# State-Space Groundwater Modelling of a Coastal Sand Dune Aquifer: Management Intervention, Coastal Retreat and Climate Impacts at Newborough Warren, Wales

*A twenty-one-year analysis of recharge, atmospheric draw and drainage across a clustered dipwell network, with intervention assessment, ecohydrological indices and critical rainfall forecasting*

M. Hollingham

ORCID: 0000-0003-0253-9301

Contact: martin.hollingham+nrg@gmail.com

## Abstract

Coastal sand dune aquifers are ecologically sensitive systems whose groundwater regimes determine the persistence of legally protected dune slack habitats. Newborough Warren, a Special Area of Conservation (SAC) on the Isle of Anglesey, Wales, is widely held to have suffered a sustained water table decline since afforestation, threatening the humid dune slack communities for which the site is designated. Its summer minima have fallen at the coastal margin and in the early part of the monitored record, while inland levels remain close to those of 1951–53. This study presents a low-cost, reproducible, data-driven analytical pipeline of 64 steps across 22 phases, requiring only monthly dipwell readings and publicly available climate data from the nearest meteorological station. The pipeline is applied to a network of 88 wells monitored over 2005–2026, integrating hierarchical clustering, Pearson affinity analysis, a three-term state-space model (SSM) parametrizing recharge, atmospheric draw and head-dependent drainage, comparative model benchmarking, water table fluctuation specific yield estimation, dune scraping and clearfell BACI analysis with three-counterfactual ANCOVA climate correction, cluster-level water balance decomposition, per-well spatial coefficient mapping, critical rainfall threshold (P_flood) forecasting, seasonal prediction equations, and a per-well scenario framework for climate and forest management projections. The approach is designed for replication at comparable coastal dune sites using minimal equipment and no specialist modelling infrastructure.

Hierarchical clustering (Ward's linkage, k = 5, selected for behavioural resolution and mechanistic interpretability, with four of five clusters confirmed by bootstrap stability analysis) identifies five hydrogeological units spanning a behavioural spectrum from shallow pan to deep sponge to canopy sponge. The eastern open dune (C1 Lake Edge, C2 Dune) responds rapidly to rainfall with high recharge sensitivity, with drainage fastest at the lake-

coupled C1 (Lumped Catchment Storage Coefficient, LCSC = 21.8%) and slower at the larger, more representative C2 Dune archetype (LCSC = 25.7%). The Western Residual (C3) acts as a deep buffer (LCSC = 26.6%). Two Forest clusters (C4 Main Forest, C5 Coastal Forest) share elevated LCSC values (40.4%, 41.5%) reflecting canopy interception of approximately 24% of rainfall (Freeman, 2008), but differ in setting: C4 sits on the elevated ridge flank (atmospheric draw coefficient $\beta_2 = 2.57$) while C5 occupies the lower coastal ground ($\beta_2 = 1.25$). Per-well analysis within the forest zone shows that elevation alone explains 98% of $\beta_2$ variance, indicating topographic position — and the substrate inferred to underlie it — as the major control on the atmospheric draw coefficient, with depth-coupling and other factors also acting. The co-temporal amplification field (Figure 75) shows C4 Main Forest to be the most climate-volatile zone on the site, with a mean spring swing 1.66 times the network mean between canonical dry and wet years. This over-response reflects two positional factors: thin saturated thickness over irregular bedrock, and the forest's occupation of the hydraulic high of the aquifer, furthest from any lateral discharge boundary and therefore without the damping that lake-edge or coastal proximity provides. Of the two forested clusters, C5's $\beta_2$ lies within the open-dune range, below C2 and C3; C4's $\beta_2$ is genuinely elevated, an identifiability diagnostic having found the drainage coefficient cleanly resolved and no $\beta_2$–$\beta_3$ degeneracy, and is best explained by position and substrate rather than canopy. Whether the canopy suppresses summer atmospheric draw is unresolved on the present evidence, and the stronger canopy evidence remains the recharge suppression captured in the lower $\beta_1$ of both forest clusters. Once corrected for interception, C4's WTF-derived specific yield falls within the open-dune range and the interpolated yield surface shows no discontinuity at the cluster margins; the C5 estimate is consistent with this but is the least reliable in the network and is treated as weakly corroborative. On this evidence the plantation's distinctive β-coefficient signature derives largely from canopy boundary conditions at the surface rather than from differences in subsurface storage architecture.

The SSM outperforms the traditional linear model in iterative forecasting (median per-well NSE improvement +0.37; 65 of 66 wells with positive NSE vs 55 of 66 for TLM), indicating that explicit parametrization of the head-dependent drainage term is essential for long-range prediction in this aquifer. The framework requires only monthly rainfall and temperature from a single climate station and is operationally transferable to other coastal dune aquifers and, in principle, to other groundwater-fed wetland settings. The water balance closes to within 1.8% at all five clusters under a displacement formulation. Seasonal prediction equations indicate that winter rainfall is the dominant predictor of winter flooding potential in the eastern open dune (C1, C2); in the Western Residual (C3) rainfall and antecedent summer minimum depth carry equal weight, and in the two forest clusters (C4, C5) antecedent summer minimum depth dominates and rainfall is not significant. The ecological significance is acute: the entire habitat gradient from SD15b wet slack to SD16 dry slack spans only 37 cm of summer minimum depth (Curreli et al., 2013), so all management interventions must be evaluated on their summer minimum effect.

Dune scraping at CEH36 produced a paired summer minimum shift of +195 mm (p = 0.004) — more than half the SD15b–SD16 ecological gradient — identified by three independent methods (raw BACI +128 mm, synthetic control +141 mm, SSM forward residual +100 mm). The benefit persisted through the post-felling period while the paired control deteriorated, and the operational zone for this intervention is the C1/C2 area where P_flood thresholds remain achievable. CEH18 and CEH21, scraped in October 2023, showed limited responses that did not survive correction for background drift — consistent with progressive coastal boundary retreat overwhelming the intervention at the more coastal sites. No off-site drawdown is resolved: the raw steps at the nearest long-record uphill well, WMC3, lie within its month-to-month variability, and no distance-decay is detectable in the drainage coefficient; the neighbour drawdown that the relief-drainage mechanism predicts remains a modelled scenario the record cannot resolve. Because scraping operates through the same physical mechanism as coastal erosion — lowering ground elevation to increase the drainage gradient — its spatial placement determines whether it complements or exacerbates the existing coastal drainage regime.

A five-tier, 17-well BACI experiment with three counterfactual controls tested the December 2017 clearfell of 4.2 ha of Corsican pine. Against the Forest control — the most direct test, comparing felled wells against unfelled forest sharing the same canopy and substrate — the single in-compartment Impact well WMC3 rose +128 mm over October–March after felling (95% CI +1 to +254 mm, p = 0.048) and not at all over June–September (−1 mm); the annual-mean step, +108 mm, is not significant on its own with autocorrelation-robust (Newey–West) errors (95% CI −21 to +237 mm, p = 0.10), and the Edge-tier step of +31 mm was not significant (p = 0.50). Summer minima at the felled wells show no clearfell signal: felled and unfelled wells tracked each other. The coefficient decomposition shows a network-mean decline in recharge sensitivity ($\beta_1$) of 4.6% across the BACI network, consistent with changing within-month rainfall distribution; it is a candidate mechanism for the observed summer minimum deterioration, but the unfelled Climate Control tier shows almost none of it, so its independence of management is not established.

The scenario framework evaluates clearfell, 50% thinning and broadleaf conversion alongside UKCP18 climate projections. The forestry scenarios are anchored to what the December 2017 felling produced at the felled well: a fall in atmospheric draw and no rise in recharge. As sustained changes in level, clearfell raises the water table by +0.038 m (Model B) to +0.066 m (Model A) at C5 and by +0.217 m at C4 (Model A); at the felled well it gives +0.060 and +0.135 m, against the +0.108 m measured there. Thinning gives at most half of each. These are the lower end of the clearfell range: the 4.2 ha compartment is narrower than the drawdown reach, so a recharge gain there would spread into the surrounding forest, and a wider felling could keep it, bounded by an open-dune end state of 0.5–0.9 m. Broadleaf conversion ranges from a slight fall (−0.037 m at C5 and −0.121 m at C4 under Model A) without an interception gain to a rise (+0.308 and +0.506 m) with one. The scenarios are evaluated well by well and cannot show whether forest management reaches the open dune

clusters, where ecological need is greatest; the modelled canopy drawdown has an e-folding length of about 220 m beyond the forest edge, an upper bound on its reach. By the 2080s the annual level at C5 falls by about 0.27 m (Model B) to 0.49 m (Model A) under UKCP18, more than the lower end of any forestry scenario raises it.

Network-wide summer minimum depths have declined at all five clusters, with C1, C2, C3 and C5 statistically significant over the full record and C4 non-significant; the decline is concentrated before 2011, and refitted from 2011 no cluster shows a significant trend. C5 Coastal Forest shows the steepest decline in both seasons, its winter decline falling just short of significance. A network-scale gradient regression of per-well summer-minimum slopes against perpendicular distance to the eroding western shoreline fits a coast-edge anomaly of −26 mm yr⁻¹ at 150 m from the shoreline over an inland reach of about 900 m. C5 sits closest to the shore of any cluster, and the gradient accounts for about half of its full-record decline (53% on the balanced basis; n = 4 in the gradient subset; Section 4.8.2), a share sensitive to which wells the cluster contains. The two other candidate mechanisms, amplified head response under thin saturated thickness at the coastal margin and progressive canopy maturation, are therefore not excluded and retain room to contribute. Averaged over each cluster’s wells, the gradient also accounts for most of the smaller decline at C3, eleven of whose twenty-two wells lie within the reach (93% on the balanced basis, 87–102% across delete-one-well refits of the gradient); C1 carries a negligible component and C2 none. C4's mean distance falls marginally inside the reach and three of its four wells lie within it; its component, −4.8 mm yr⁻¹, is 74% (58–91% across the same refits) of a decline the record does not resolve.

The deterioration is concentrated in the summer season, consistent with the observed warming trend (post-2013 period +0.94°C above baseline) and increasing atmospheric evaporative demand. Monthly decomposition of the RAF Valley PET record shows significant increases in May (+0.068 mm month⁻¹ yr⁻¹, p = 0.002), June (+0.048, p = 0.033) and July (+0.063, p = 0.010), consistent with a physical mechanism for the observed secular deepening concentrated in the early-summer period when the water table is falling from its spring peak. The decline is concentrated before 2011. Where the slack floor is surveyed, at van Willegen et al.'s quadrats, the four-year mean annual minimum is deeper at every quadrat than during Curreli et al.'s survey years (median 0.19 m), and none now lies above the SD15b community mean.

The two ecologically referenced metrics diverge sharply in their climate exposure: the five-year mean spring water level (MSL5; van Willegen et al. 2025), which describes the sustained baseline within which dune-slack vegetation establishes, is projected to move far less than the summer minimum under the 2080s trajectory in the open dune (+52 and −22 mm at C1 and C2, against summer-minimum deepening of 315 and 359 mm), and by about a third as much at C3 and a little under half as much at C5 (−150 and −141 mm against −440 and −319 mm), so the summer minimum — the binding ecological constraint — is where the

projected climate bites. On the four-year mean annual minimum, the quantity on which Curreli et al. (2013) reported the communities, the open-dune clusters stay above the forest clusters throughout; the scraping benefit is expressed on this baseline metric as well as on summer minima, while the clearfell recovery is not. An empirical comparison of five-year mean spring water levels (MSL5) between the pre-clearfell baseline (window end 2017, springs 2013–2017) and the current period (window end 2023, springs 2019–2023) shows a site-wide secular deepening of 105 mm with the largest declines at the south-western coastal margin (CEH22 −233 mm, C5 coastal members −164 to −172 mm); no spatially distinct signal attributable to either the 2015 scraping or the 2017 clearfell is resolved against the five-year window standard error of the MSL5 metric (82–181 mm by cluster; Section 4.8.3). This is consistent with the BACI finding that the clearfell rise is expressed in the winter months at the felled well rather than in five-year spring means.

Independently, MSL5 as implemented predicts the co-located Ellenberg-F vegetation moisture of van Willegen et al. (2025) between wells ($r = 0.83$, $n = 18$). An equilibrium wetness index derived from the state-space coefficients alone (independent of MSL5) is statistically indistinguishable from MSL5 as an Ellenberg-F predictor (Williams $p = 0.49$), and, calibrated onto the MSL5 scale, reconstructs observed spring levels across the open dune to 73 mm RMSE, generalizing to wells outside the vegetation-calibrated set — indicating that dynamics-only coefficients encode ecohydrological state. It is scoped to the open dune (degrading in the forest) and its practical edge is a shorter required record, not simpler construction. The modelled water table is additionally validated in space against the Sentinel-2 satellite archive: over 43 cloud-free winter scenes (2016–2026) the near-infrared wetted area of the open slack floor (306.97 ha, forest excluded) rises monotonically with the modelled median level (wet floor $R^2 = 0.77$, open water $R^2 = 0.71$), and the SSM level driven through these fitted area–level curves reproduces the observed inundation history out of sample ($n = 238$ months; wet-floor area $R^2 = 0.80$, median modelled:observed ratio 0.96) — the study's only spatial, out-of-sample validation of the water table, supporting its use across the slack floor between the wells.

The P_flood threshold equations, seasonal prediction equations and scenario framework, together with live Equilibrium Wetness Index (EWI) and five-year mean spring level (MSL5) monitoring, are implemented as interactive tools hosted on the Newborough Warren Groundwater Research project page. The monitoring evidence places the clearfell rise in the winter months at the felled well, not in the summer months or in summer minimum depth — the metric that determines dune slack viability — where felled and unfelled wells track each other; whether felling acted through a reduced atmospheric draw or through a recharge gain that dispersed beyond the small compartment, the record does not decide. The operational domain for direct hydrological intervention lies in targeted scraping in the C1/C2 zone where P_flood thresholds remain achievable, against a climate trajectory that is the binding constraint on all outcomes. The intervention window is measured in one to two decades.

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# 1. Introduction

Sand dune systems are among the most ecologically dynamic and internationally important coastal habitats in north-west Europe (Dargie, 1995). Their characteristic dune slack communities — low-lying inter-dune hollows that flood seasonally where the water table intersects the surface — support a concentration of nationally rare plant species including creeping willow (*Salix repens*), and a diverse suite of wetland-associated invertebrates. The persistence of these communities is fundamentally dependent on a shallow, responsive groundwater table that rises into the slack floor each winter.

Newborough Warren on the Isle of Anglesey (National Grid Reference SH 406 636) is one of the largest and most ecologically important coastal dune systems in Wales. Designated as a Special Area of Conservation primarily for its humid dune slack and creeping willow communities, the dune system — Newborough Forest and the open warren of the National Nature Reserve — comprises approximately 1,300 hectares of late-glacial blown sand resting on weakly permeable glacial till (Stratford et al., 2007), within the 1,869-hectare Abermenai to Aberffraw Dunes Special Area of Conservation, which also takes in the Tywyn Aberffraw dunes and of which conifer plantation makes up 38% (JNCC, 2026). The northern 700 hectares were afforested with Corsican pine (*Pinus nigra* var. *laricio*) between 1948 and 1965. The hydrological study area of this report, defined by the DEM-derived flow network, covers approximately 865 ha (Section 2).

A decline of the water table since afforestation has long been asserted for the site (Jennings, 1990), but it has not previously been measured. The one early record, Ranwell’s fortnightly readings at seventeen levelled pipes in 1951–53 (Ranwell, 1959), established a seasonal range of 70–100 cm and a baseline, not a trend. The later account of a drier phase in the 1970s rests, in the words of the guidelines that report it, on "anecdotal evidence" for a period "of less than long-term average rainfall"; the one dry spell the dipwell record itself holds is the late 1980s and early 1990s, after which "flooding of some slacks has recurred" in the wetter years since 2000 (Davy et al., 2010, p. 17).

Set against the modern network (Section 5.7.9), Ranwell’s readings support fluctuation about a near-stationary level inland of the coastal-retreat reach, rather than progressive decline, with the 1989–96 depression and its recovery (Section 5.7.8) as the one documented excursion.

Conservation management has included a December 2017 experimental clearfell of 4.2 ha within the plantation, and topographical dune scraping at selected slack sites. These interventions provide natural experiments by which the hydrological consequences of management can be assessed empirically. Despite the importance of the site and the scale of management investment, no systematic network-wide quantitative analysis has previously been undertaken.

Restoration of dune slack hydrology is widely assumed to be achievable through topographical intervention provided sufficient winter rainfall occurs. However, stationarity in the annual level does not imply stationarity in the summer minimum, and it is the minimum that governs whether a slack floods: if summer water table minima are declining at a rate that progressively increases the recharge deficit that must be overcome each winter, there exists an intervention window — a period during which topographical management can still realistically restore flooding conditions — beyond which no realistic winter rainfall total can compensate for the accumulated summer deficit. Quantifying the rate of closure of this window, and identifying the site-specific mechanisms that drive it, is therefore an urgent practical and scientific priority. This requires not only a long baseline record of groundwater behaviour but a mechanistic understanding of how recharge, evapotranspiration and drainage interact across the hydrogeologically distinct sub-units of the dune system.

This study presents such an analysis. The specific objectives are to: (1) classify the dipwell network into hydrogeologically coherent groups by hierarchical clustering and extend membership to the full monitoring network via Pearson affinity analysis; (2) characterize mechanistic groundwater signatures using a state-space model benchmarked against a traditional regression approach; (3) quantify aquifer storage — estimating specific yield by water-table-fluctuation methods with a forest-canopy interception correction — and close a cluster-level water balance; (4) map the per-well spatial structure of the recharge, atmospheric-draw and drainage coefficients across the site; (5) quantify the effects of dune scraping using a two-tier hierarchical CUSUM-BACI framework that separates localized treatment signals from background system-wide deterioration; (6) conduct a three-counterfactual ANCOVA-BACI assessment of the December 2017 plantation clearance across a 17-well five-tier experimental network, partitioning intervention signals from climate variability, scraping confounds and coastal erosion gradients; (7) quantify the water-table drawdown attributable to coastal erosion and to climate change, and characterize the changes recorded across the site over 2005–2026; (8) derive ecohydrological state indices — the five-year mean spring water level (MSL5) and a dynamics-based equilibrium wetness index (EWI) — and cross-validate them against independent Ellenberg-F vegetation-moisture observations; (9) derive operational critical rainfall threshold and seasonal prediction equations for conservation planning; (10) project groundwater responses to UKCP18 climate trajectories and forest-management scenarios (clearfell, thinning and broadleaf conversion), delivered as interactive web tools; and (11) validate the modelled water table out of sample and in space against the wetted extent of the dune-slack floor mapped from Sentinel-2 satellite imagery.

## 1.1 The Newborough Warren Analytical Architecture

To achieve a rigorous characterization of the aquifer's hydrogeological dynamics, this study employs a twelve-tiered analytical framework that scales from individual well diagnostics to network-wide predictive modelling. A further objective is to establish what the framework

requires in order to be applied at comparable dune slack systems — its input data, its record-length requirements, and the parameters that must be re-derived locally — using only manual dipwell records and publicly available climate station data, without specialist survey equipment, telemetry or laboratory analysis. Those requirements and their limits are set out in the Discussion.

- **Hydrogeological Clustering:** The 66-well reference network is partitioned into hydrogeologically distinct zones using agglomerative hierarchical clustering with Ward's minimum variance linkage (Section 3.2). The Pearson correlation distance isolates temporal hydrograph dynamics from absolute water table elevation, and the optimal cluster number (k = 5) is selected using a dual-metric approach combining Ward's merge distances and mean silhouette coefficient. The resulting five clusters provide the spatial framework for all subsequent analysis.
- **Extended Network Classification and Pattern Matching:** The primary hydrogeological classification is extended from the 66-well reference network to the full 88-well monitoring array using Pearson Affinity Analysis (Section 3.3). Unlike the mechanistic models, this phase utilizes pattern-matching on raw observed time series without model fitting. It identifies core cluster members and raises Multi-Cluster Affinity (MCA) flags where a well tracks several cluster centroids; the flag is raised for 44 of the 66 reference wells, so it describes a condition common across the network rather than one confined to transition zones. The wells in the extended network are not used in the rest of the analysis.
- **Mechanistic Modelling and Storage Characterization:** At the cluster scale (n = 5), the State-Space Model (SSM) is used to derive physical coefficients ($\beta_1$, $\beta_2$, $\beta_3$) directly from cluster-average time series (Section 3.4.1). These coefficients characterize the fundamental contrast between the "shallow pan" and "deep sponge" blocks, providing the basis for both the Lumped Catchment Storage Coefficients (LCSC; Section 3.4.1) and the operational P_flood threshold equations (Section 3.6.3) used for management forecasting. Per-well coefficients are interpolated across the site (Section 3.4.3), providing a spatially continuous expression of recharge sensitivity, atmospheric draw, and drainage rate that is independently consistent with the cluster boundaries and with the interception imprint of the plantation canopy.
- **Predictive Validation and Model Benchmarking:** The predictive skill of the SSM is formally tested through matched comparison against a Traditional Linear Model (TLM) at every individual well (n = 66) in the reference network (Section 3.4.2). This benchmarking uses iterative Nash–Sutcliffe Efficiency (NSE) and one-step $R^2$ to assess predictive skill across the network. Spatial mapping of the performance gains ($\Delta R^2$, $\Delta$NSE) indicates where the explicit parametrization of storage decay matters most for accurate multi-step forecasting.

- **Intervention analyses:** The hydrological consequences of two management interventions are quantified using complementary BACI frameworks (Sections 3.5.3–3.5.4):
  - *Dune Scraping:* A two-tier hierarchical CUSUM-BACI framework across five focal wells separates localized scraping signals from background system-wide deterioration. Tier 1 tests local control wells against a regional mean; Tier 2 differences each treatment well against its matched control. Era-specific storage decay coefficients ($-\beta_3$) are estimated independently for each management period to assess whether scraping has altered subsurface drainage mechanics.
  - *Plantation Clearfell:* A three-counterfactual ANCOVA-BACI experiment across 17 wells in a five-tier hierarchical design — Impact, Edge, Forest Control, Coastal Control and Climate Control — assesses the hydrological consequences of the December 2017 clearfell. The ANCOVA model includes a cumulative water balance covariate, a distance-weighted scraping covariate, and a coastal-drift control. The model is run against three control centroids (Forest, Climate, Combined) to bracket the clearfell effect.
- **Robustness and Confound Assessment:** Independent robustness analyses are conducted for both intervention analyses to test whether the detected signals are genuine rather than artefacts of control well limitations or climate confounds (Sections 3.5.3–3.5.4). For the scraping analysis, the reliability of CEH4 as a stable long-term control is tested using two independent methods: a synthetic control constructed as an optimally weighted combination of 11 donor wells calibrated on the pre-scraping period, and an SSM forward residual computed by calibrating the model on the pre-scraping period and running it forward iteratively. Convergence between the raw BACI estimate, the synthetic control gap and the SSM residual tests whether the scraping signal at CEH36 is detectable independently of CEH4's own trajectory of coastal retreat and warren-wide decline; an additional propagation analysis tests for distance-decay drawdown at uphill wells. For the clearfell analysis, the primary three-counterfactual ANCOVA design (Forest, Climate and Combined controls) is itself a robustness structure, with the Forest control providing the most direct test against position-matched unfelled wells. Four further supplementary analyses test the ANCOVA result: summer minima analysis with mixed-effects modelling; coefficient decomposition comparing Before and After SSM fits; SSM residual normalization against the Forest Control drift; and a synthetic control constructed from optimally weighted donor wells. A spatial transect analysis tests whether post-felling step changes show a distance-dependent gradient consistent with a localized clearfell effect.
- **Climate Trajectory Analysis:** Annual summer minima and winter maxima are extracted for all five clusters and their trends fitted over the record, and refitted from 2011, to describe the direction and rate of change of each cluster's water levels (Section 3.6.4).
- **Water Table Fluctuation Specific Yield Estimation:** Empirical upper bounds on specific yield are derived independently from the dipwell record using the water table

fluctuation method (Healy and Cook, 2002), applying an interception correction for the forest clusters (C4 and C5) following Freeman (2008). The resulting cluster-specific Sy estimates (Section 3.7.3) are consistent with broadly uniform aquifer storage across the site — an inference resting on the C4 estimate and the continuous interpolated yield surface, with the clip-constrained C5 estimate only weakly corroborative.

- **Water Balance Decomposition:** Cluster-level SSM coefficients are used to decompose the mean monthly water balance into its component fluxes — recharge, atmospheric draw, gravity drainage — expressed both as head-space quantities (m/month) and as indicative volumetric fluxes (mm/year) using assumed specific yield values. These are conservative lower bounds; the water-table-fluctuation estimates above bound them from above (Section 3.7.2). This tier quantifies the structural water balance at each cluster, independently of land cover or management history.
- **Ecohydrological State Indices and Vegetation Validation:** Two state indices are derived and cross-validated against independent Ellenberg-F vegetation-moisture observations: the five-year mean spring water level (MSL5), which places the site on the scale used in the published dune-slack literature, and a dynamics-based equilibrium wetness index (EWI) derived from the state-space coefficients themselves. This tier connects the hydrological characterization to the ecological condition it is intended to inform.
- **Independent Spatial Validation from Satellite Imagery:** The modelled water table is checked out of sample and in space against the wetted extent of the open dune-slack floor, mapped from the Sentinel-2 satellite archive (Section 3.7.7). Forty-three cloud-free winter scenes (2016–2026) are classified in the near-infrared into open water and wet floor across the warren study area, and each class’s area is related to the reference-network median level by an exponential curve; passing the state-space level through those curves tests whether the modelled water table reproduces the observed wet-area history it never saw. This is the study’s only spatial, out-of-sample validation of the water table, complementing the temporal well-by-well benchmark (Section 3.4.2) and the vegetation cross-validation above. The fitted area model additionally drives the scenario viewer’s wet-area panel and the century hindcast film.
- **Climate and Management Scenario Projection:** Critical rainfall thresholds and seasonal prediction equations are derived for conservation planning, and groundwater responses are projected under UKCP18 climate trajectories and forest-management scenarios (clearfell, thinning and broadleaf conversion). This tier is delivered as interactive web tools so that the projections can be interrogated without re-running the pipeline.

# 2. Study Site

Newborough Warren occupies the south-western tip of the Isle of Anglesey in north-west Wales (Figure 1). The dune field is underlain by weakly permeable glacial till forming the effective base of the sand aquifer. A bedrock ridge, demarcated by the 20 m AOD contour of the DEM and trending south-west to north-east, forms the northern boundary of the study area. To the south the ridge plunges beneath the dune field; its northern end reaches the coast, meeting the sea obliquely rather than face-on. The hydrological study area, defined by the extent of the DEM-derived flow network (modelled drainage lines from topographic analysis of the digital elevation model, not mapped surface watercourses; the network itself is not drawn), covers approximately 865 ha (~4.3 km × 3.9 km; the hydrological study area outlined in Figure 1). The monitoring network spans ~3.0 km × 2.1 km within this, lying entirely on the south-eastern flank of the ridge. The dune system on the north-western flank of the ridge, draining toward the Malltraeth estuary, is hydrologically separate from the main Newborough Warren aquifer; monitoring wells located on that flank do not have sufficient record length for inclusion in the clustering and state-space analyses reported here.

Lateral recharge from the rock ridge into the dune sands has been identified as a possible contribution to the aquifer's water balance (Stratford et al., 2007). The dune aquifer south-east of the ridge is divided into an eastern block, where shallower sand over till and estuarine deposits produces a more responsive, heterogeneous water table regime, and a western block, where deep sand accumulation over a broader platform permits significant groundwater storage and more buffered seasonal fluctuations (Stratford et al., 2007).

The aquifer discharges principally across the south-western foreshore (Caernarfon Bay) and the southern and eastern coast (Menai Strait) as tides recede. A small lake within the warren, Llyn Rhos-Ddu, acts as a near-fixed head body and is the dominant local drainage sink for the immediately adjacent C1 Lake Edge wells; it receives groundwater for most of the year as the regional water table maintains its head above the lake bed, and supplements aquifer recharge during the driest months (Stratford et al., 2007). The active monitoring network comprises 89 measuring points (Figure 1). Of these, 88 are dipwells forming the analytical network used in this study — 66 reference-network wells with sufficient record length and consistent monitoring for inclusion in the core analyses, plus 22 extended-network wells used in robustness checks and spatial interpolation. The Llyn Rhos-Ddu lake gauge is the 89th point. Ten further located points — nine dipwells whose records are too sparse for inclusion and one with no record — are not shown. An interactive map of the monitoring network is linked in Section 11, and the wells can also be explored in the groundwater flooding forecaster (forecaster.html).

The northern 700 hectares are managed plantation dominated by mature Corsican pine (Stratford et al., 2007). A partial clearance of approximately 4.2 ha (about 0.6% of the plantation) began in December 2017, removing trees across a defined felling zone at the

plantation–open dune transition (Figure 2). Dune scraping has been undertaken at selected western slack locations: CEH36 in April 2015, and CEH18 and CEH21 in October 2023.

The dune slack communities at Newborough Warren are classified under the National Vegetation Classification as SD15b (wet slack, *Salix repens–Calliergon cuspidatum* community) and SD16 (dry slack, *Salix repens–Holcus lanatus* community). Curreli et al. (2013) (see also Davy et al., 2006) measured the water table regimes of these communities in a site-specific vegetation survey across Newborough Warren and four other Welsh dune systems: Curreli's wet-slack (SD15b) community mean summer minimum is 0.61 m below the ground surface, and its winter maximum floods the slack, to a community mean of 0.21 m above the surface; the dry-slack (SD16) community means are a summer minimum of 0.98 m and a winter maximum within 0.25 m of the surface. These community means are levels relative to the ground at slack-floor quadrats. Where that ground is surveyed, at van Willegen et al.'s quadrats, the record is read against them directly (Section 4.8.3). Elsewhere a dipwell's depth below its own ground indicates a community only where the well stands on the slack floor, which the 2 m lidar cannot confirm (Section 5.8.2), so the means serve there as reference depths. Wells whose mean summer minimum exceeds 0.98 m lie deeper than the SD16 community-mean depth. For the purposes of assessing management options, two intermediate bands are defined: wells between 0.61 m and 0.75 m below ground are likely to be recoverable to the SD15b community-mean level by a standard dune scraping event yielding approximately 0.14 m of water-table benefit (a planning assumption; at CEH36 the BACI-measured benefit is +0.13 m, from a cut whose depth, inferred from that response, is about 0.43 m, consistent with field observation; Section 4.5); wells between 0.98 m and 1.20 m are likely to be recoverable to the SD16 community-mean level, 1.20 m reflecting a planning assumption of about 0.22 m of benefit from a deeper scrape rather than a benefit measured at any well. These "recoverable" classifications should be treated as indicative — achievable excavation depth varies with substrate, access and licence conditions, and the empirical basis is a single scraping event at one well. Wells deeper than 1.20 m are beyond the reach of a standard single scraping event under average antecedent conditions.

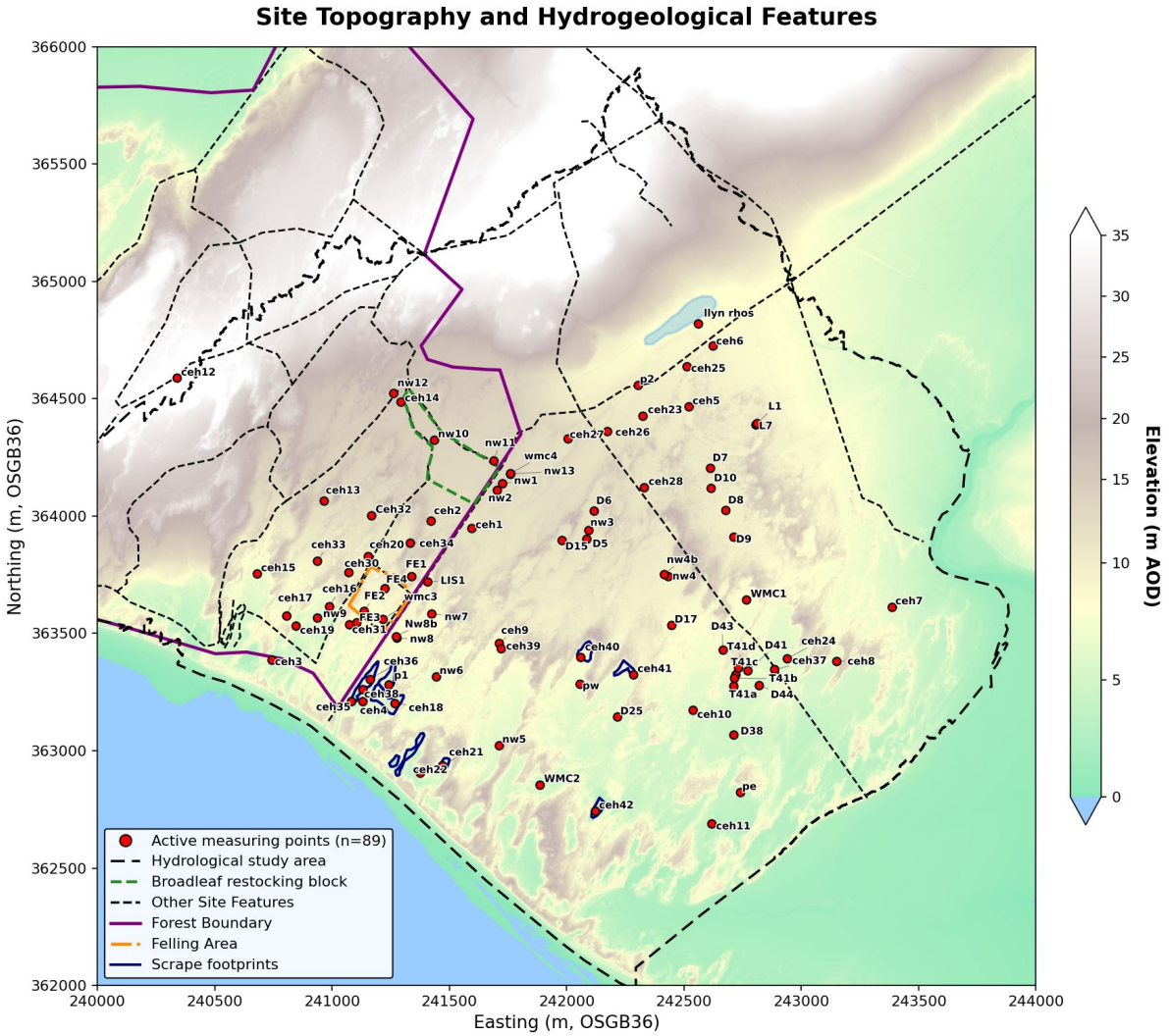



***Figure 1:*** *Site topography and hydrogeological features. Digital elevation model of Newborough Warren showing the 89 active measuring points (88 dipwells and the Llyn Rhos-Ddu lake gauge), the hydrological study area boundary, the forest boundary, the December 2017 felling area, the broadleaf restocking block, the scrape footprints, and other mapped site features (dashed). (Source: 12_01_dem_site_overview.png) (LiDAR data: © NRW & OS; see Section 11 for full attribution).*

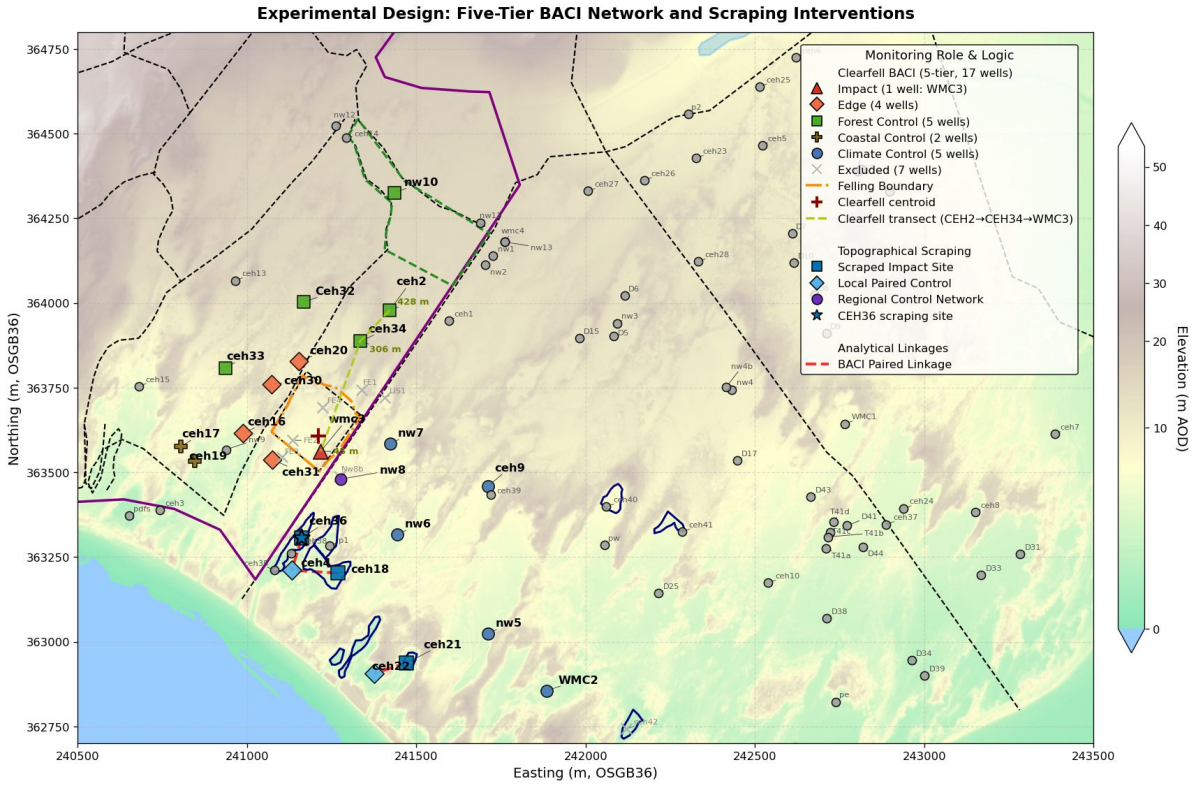



***Figure 2:*** *Hierarchical experimental design. Spatial layout of the monitoring network showing the five-tier BACI structure: Impact (WMC3), Edge (CEH31, CEH20, CEH30, CEH16), Forest Control (CEH32, CEH34, CEH33, NW10, CEH2), Coastal Control (CEH19, CEH17), Climate Control (CEH9, NW7, NW6, NW5, WMC2), scraped sites (CEH36, CEH18, CEH21) with their local controls (CEH4 for CEH36 and CEH18, CEH22 for CEH21), NW8 of the scraping regional control network (purple; the other regional controls, CEH9, NW5, NW6 and NW7, carry their clearfell tier symbol), the seven wells excluded from the clearfell BACI design (grey crosses: FE1–FE4, LIS1, NW8B, CEH42), the clearfell centroid, the CEH2 → CEH34 → WMC3 transect with distances from the centroid, and the felling boundary. (Source: 13_01_experimental_setup_map.png) (LiDAR data: © NRW & OS; see Section 11 for full attribution).*

# 3. Methods

## 3.1 Data Sources and Preparation

All analyses were implemented as a numbered Python pipeline orchestrated by run_analysis.py. The pipeline comprises 64 registered steps across 22 phases. Those steps are classified two independent ways. By tier: 44 analytical, twelve diagnostic and eight display/utility — Script 26c (MSL5 report-format figures), Script 09f (the management-effects synthesis figure), Script 09g (the mechanism-diagram figure suite), Script 27 (the greyscale figure-conversion utility), Script 43 (the Ranwell 1959 historical water-table site georeference) and Scripts 45, 46 and 47 (the Sentinel wet-area model, its public feed and the century hindcast film). By execution: 53 run in a default pass, eight (Scripts 24b, 31, 31b, 50, 51, 53, 54 and 55) only behind the --with-supplementary switch, and three (Script 14b, the year-of-crossing bootstrap withdrawn from the report; Script 27, greyscale conversion; and Script 47, the century hindcast film) only on demand behind their own flags. The two classifications each account for the same 64 steps and are not additive with one another; current values are written to outputs/pipeline_manifest.json on every run.

- **Phases 1–11** produce the analytical chain presented in this report — clustering, state-space modelling, intervention analysis, forecasting, water balance, scenario evaluation and the network-scale coastal-retreat gradient regression.
- **Phase 12** contains supplementary residual diagnostics (Scripts 22–24).
- **Phase 13** runs the van Willegen et al. (2025) mean-spring-water-level analyses — an observational five-year aggregation (Script 26), a companion UKCP18 climate projection (Section 3.7.5, Script 26b), and the report-format MSL5 figures for Sections 4.8.3 and 4.13.1 (Script 26c).
- **Phase 14** contains the cluster framework diagnostics: the C3 detrend check (Script 28, testing the aquifer-architecture framing of Section 5.1), the within-C3 variance attribution (Script 29, characterizing the spatial structure within C3), and the C4 Main Forest drainage identifiability diagnostic (Script 30, step 36, which assesses whether the C4 drainage coefficient is cleanly recovered from the record and is discussed in Section 5.2.3).
- **Phase 15** produces the observed differential-change, climate-envelope and driver-validation analyses: the secular differential water-table drift map (Script 32, Figure 73), the climate-swing amplification and dry-year floor maps (Script 33, Figures 75 and 76), the per-well climate-sensitivity coefficient (Script 35; Section 5.7.5), the absolute climate-removed secular-trend map (Script 36, Figure 74), the per-driver scale-factor validation of the modelled driver fields against that trend (Script 37; Section 3.8.1, Figure 77), and the comparative driver-footing analysis (Script 37b; Section 3.8.1, Figure 78).

- **Phase 16** runs by default the MSL5 two-window sensitivity demonstration (Script 34, Supplementary Material Note S11), the coast-to-inland transect estimate of the coastal-retreat rate (Script 38, Section 4.10.4), the out-of-sample test against the 1989–96 record (Script 39, Section 4.14.1), the measured shoreline-retreat series (Script 40, Section 4.10.6) and the canopy texture index from the dated aerial photographs (Script 41, Section 5.5), alongside a supplementary opt-in remainder — the cluster-stratified residual climatology (Script 24b), an independent k = 5 partition validation (Script 31), and the cluster separation-versus-recoverability check (Script 31b).
- **Phase 17** collects the display and historical-record steps: the management-effects synthesis figure (Script 09f), the mechanism-diagram figure suite (Script 09g), the publication-ready greyscale figure conversion (Script 27, on demand), the georeferencing of Ranwell's 1951–53 sites (Script 43) and the hindcast of his record (Script 44, Section 4.14.2).
- **Phase 18** builds the Sentinel-2 wet-area model (Script 45, Sections 3.7.7 and 4.8.5), its public feed (Script 46) and the century hindcast film (Script 47, on demand).
- **Phase 19** runs an independent cross-check of the per-well fits with the Pastas transfer-function model (Script 48, Section 3.4) and, behind --with-supplementary, the record-length analysis (Script 50).

Two further post-review scripts were inserted into earlier phases: Script 11c (P_flood achievability map, Phase 3) and Script 14b (bootstrap year-of-crossing, Phase 4). Steps 10 and 11 each invoke modular sub-script suites — the scraping intervention analysis (sub-scripts 09a–09e) and the clearfell BACI analysis (sub-scripts 10a–10n). Each script reads from and writes to a shared outputs/ directory; the public repository at https://github.com/newbroman/Newborough_Hydrology contains the full source tree and pipeline documentation, and all outputs are reproducible by running the pipeline in sequence.

### 3.1.1 Groundwater Records

Groundwater level data were obtained from the Newborough Warren dipwell network. After quality control and exclusions (described below), 88 wells entered the classified network — 66 in the reference network and 22 in the extended network. Field measurements record depth to water below the top of each standpipe as a positive number in metres; water standing above the standpipe top is recorded as a negative value. Records span April 2005 to February 2026. Raw records were quality-controlled by removing values exceeding 4.0 m below standpipe top as physically implausible for this dune system, and assigned to the appropriate calendar month by field convention: readings taken on or before the 15th of a month are assigned to the previous month (reflecting the field practice of measuring at the start of the month to characterize the

preceding month's conditions), while readings on the 16th or later are assigned to the current month. For every spring quantity in this report the consequence is that the readings taken in March, April and May carry the February, March and April labels, and it is those three months that the report calls spring, the labelling of van Willegen et al. (2025) and Curreli et al. (2013). In general most readings were taken one or two days from the month's end. Where multiple readings fall within the same month after assignment, they are averaged to produce a single monthly mean; this applies to a small number of readings in 2005–06. Gaps of a single month (one missed visit) were filled by time-based linear interpolation; longer gaps were retained as missing values. All monthly data are indexed by year and month (e.g. July 2007); where date stamps appear in tabular outputs with a day suffix (e.g. 2007-07-01), the `-01` day component is a pandas formatting convention and does not denote the first of the month — the record represents the complete calendar month, with the water-level reading taken at the end of that month. Monthly rainfall and Thornthwaite PET totals from RAF Valley were paired to the same calendar month as the water level they accompany. The monthly change in water level (Δh) is thus explained by the climate forcing that acted during that month, with drainage driven by the water table position at the end of the preceding month.

A reference network of 66 wells was selected from the 88 monitored dipwells: a well is admitted if it has at least 100 months of valid record and is still reporting in February 2026. The 100-month figure is an admission threshold, not a fitting window — admitted records are longer than it (median 192 months, range 140 to 252; Script 00). The analyses that follow deliberately use different parts of them: which record each analysis fits and evaluates is registered analysis by analysis in the pipeline's record-basis register (record_basis.csv), which is checked against each script's fitting call before every commit, and the distinction that matters most for what follows — the fixed comparison window of the per-well model against the full record of the cluster centroids — is set out in Section 3.4. Eight wells meeting the record-length criterion were excluded from the reference network prior to analysis. Wells within the clearfell management footprint (FE1, FE2, FE3, FE4 and LIS1; Section 4.6) were held out on the grounds that removal of canopy interception following the 2017 felling might have introduced a non-stationary upward drift in their water tables that the time-invariant state-space formulation is not designed to represent. These wells were subsequently excluded from the clearfell BACI analysis (Section 4.6) on the grounds of insufficient pre-felling baseline (Section 3.5.4) and contribute to the extended-network Pearson affinity classification (Section 4.3). One lake-surface well (Llyn Rhos-Ddu) was excluded because it reads a near-fixed head body rather than a water table and therefore violates the storage-decay assumptions of the state-space model. Two further wells (CEH3, CEH22) were excluded on clustering grounds described in Section 3.2.3, their correlation structure being consistent with tidal-signal contamination. Seven of these eight reference-network exclusions remain in the extended network for per-well analysis; Llyn Rhos-Ddu is

excluded from both networks, as its lake-stage signal is not interpretable as a groundwater response.

One additional well, PDFS, was excluded from both networks. Its hydrograph carries a tidal-influence signature that makes it unrepresentative of water-table behaviour — the same exclusion principle applied to CEH3 and CEH22 — and its record is too short to meet the minimum-length criterion for extended-network inclusion, so, unlike the tidally affected reference-network exclusions (CEH3, CEH22), it could not be retained for per-well analysis in the extended network. The classified network therefore comprises 88 wells: 66 in the reference network and 22 in the extended network.

The temporal completeness of the classified network is shown in Figures 3a and 3b. Each row is a single well: Figure 3a shows the 66 reference wells in cluster colour, and Figure 3b the 22 extended-network wells together with the excluded study-area dipwells and the Llyn Rhos-Ddu lake gauge as a presence-only band. Plotting the monthly observation state of every well makes the staggered installation history and the principal data gaps directly visible, and distinguishes genuine measurements from interpolated single-month bridges, recorded and inferred dry states, flooding, and intervals when a well was not found, inaccessible, or outside its record.

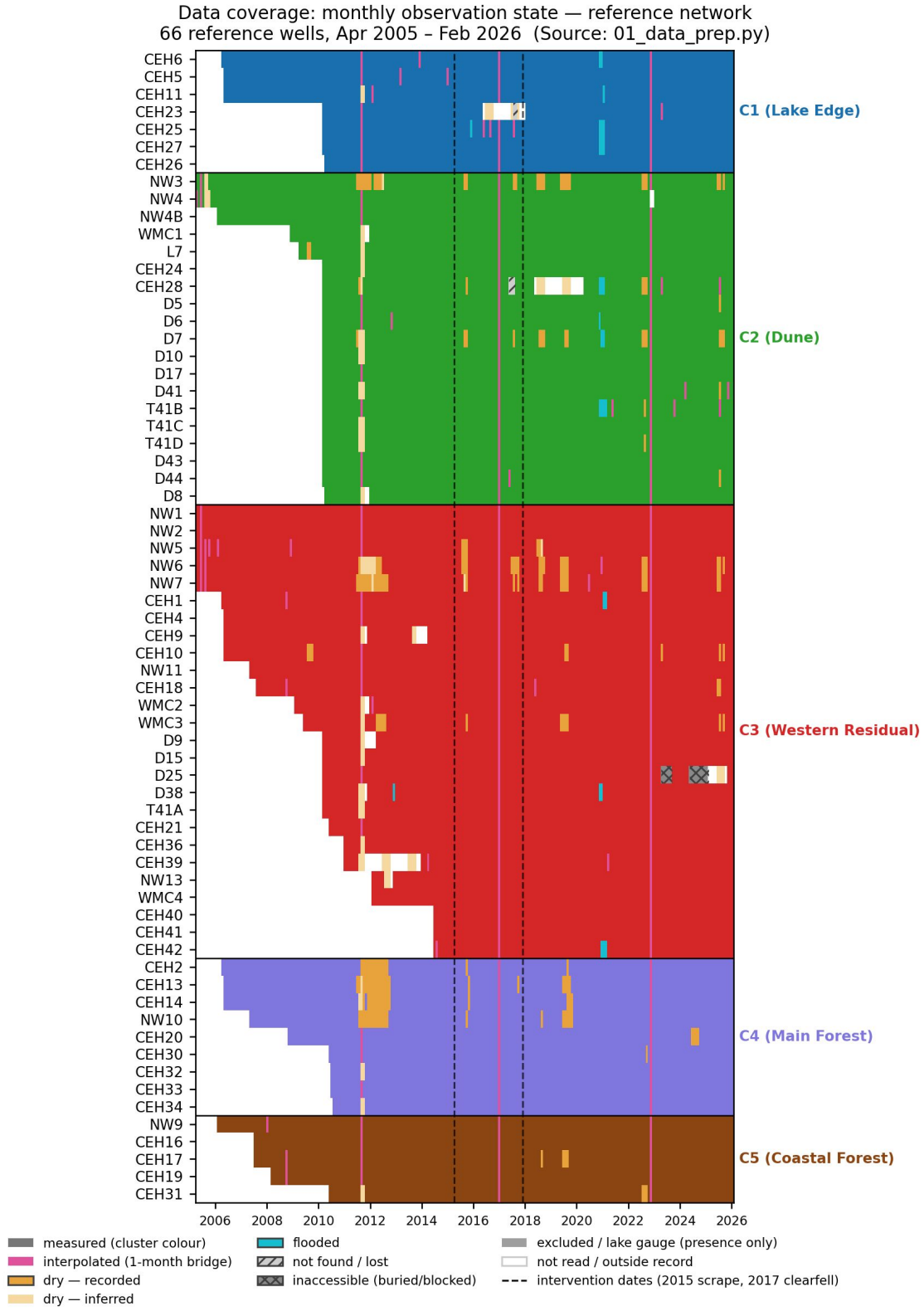



***Figure 3a:*** *Data coverage and monthly observation state of the reference network (66 wells), April 2005 – February 2026. Each row is one dipwell, grouped by cluster in cluster colour. Cell colour encodes the observation state: measured, interpolated (single-month bridge), dry (recorded or inferred), flooded, not found or lost, inaccessible, and not read or outside record. Vertical dashed lines mark the April 2015 western-slack scraping and the December 2017 clearfell. (Source: 01_coverage_states_reference.png)*

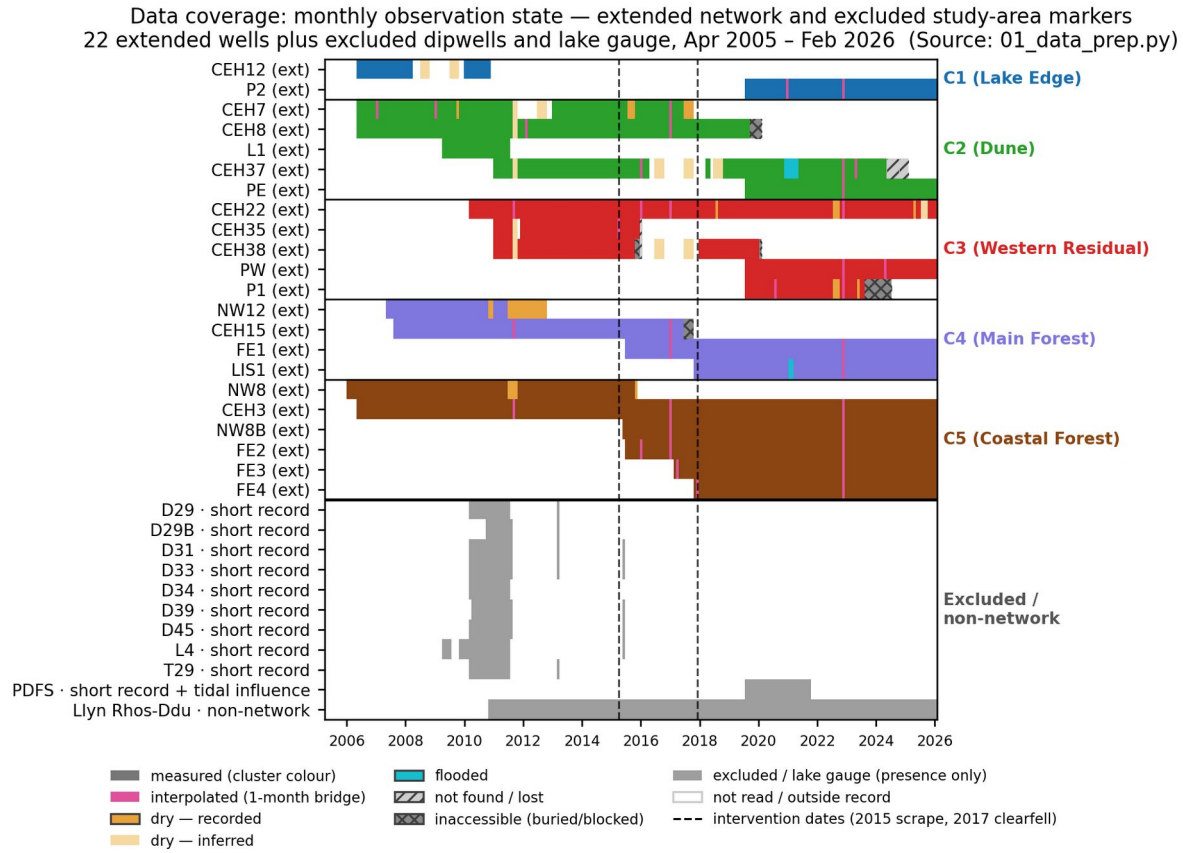



***Figure 3b**. Data coverage and monthly observation state of the extended network and excluded study-area markers, April 2005 – February 2026. The 22 extended-network wells (labelled "ext") are grouped by cluster above a grey presence-only band showing the excluded study-area dipwells and the Llyn Rhos-Ddu lake gauge, with exclusion reasons annotated. Observation-state colours and the scraping and clearfell reference lines are as in Figure 3a. (Source: 01_coverage_states_extended.png).*

CEH36, which received scraping management at a single point during the monitoring record (Section 4.5), was retained as a reference-network member on the grounds that scraping produced a one-time baseline shift rather than ongoing trend, and its post-intervention response remains consistent with its cluster mates. Wells not meeting the reference-network criteria for reasons of insufficient record length were assigned to the extended network used in the Pearson affinity analysis to extend the hydrogeological classification beyond the reference network boundary. Dipwells are positioned across a range of geomorphological contexts including dune slack floors, slack margins, and inter-slack hollows; most but not all wells are sited within slacks. Notable exceptions include L1, which is positioned adjacent to a fenced grazing experiment on slightly elevated ground; NW1, where the immediate surroundings have since been excavated; NW10, which is in an interslack area that floods when the adjacent slack overtops; and CEH31 and FE3, which sit in low dune ridges to the south of the clearfell area.

Ground surface elevations are referenced to the Natural Resources Wales LiDAR Digital Terrain Model (2 m resolution, captured March 2023; Natural Resources Wales, 2023),

obtained from the DataMapWales repository (https://datamap.gov.wales/maps/lidar-viewer/). Ground elevation at each dipwell is taken from a differential-GPS survey where one exists (78 wells; surveyed by the author and Curreli in 2010) and from the LiDAR DTM otherwise (21 wells), the two together resolving all 99 located wells; the source used at each well is recorded by Script 01 (01_well_elevations.csv), and the pipe-top elevation is the ground elevation plus the measured upstand. No check of the DTM against the survey is made here: the well metadata holds no LiDAR sample at the surveyed wells that is independent of the survey, so the two cannot be compared from it. The LiDAR-derived elevations carry the NRW stated DTM accuracy of ±0.15 m RMSE on open ground.

Field readings are dips from the pipe top, but the master workbook applies each well's surveyed upstand on export, so the modelled series is a signed height relative to the ground surface (level = upstand − dip): negative below ground, and positive where a slack is ponded. Water table elevation in metres above Ordnance Datum was therefore derived as maOD = ground elevation + level, which is the pipe-top conversion with the upstand already folded in, since maOD = pipe top − dip = (ground + upstand) − dip. Upstand heights range from −0.14 m (CEH28, a sunken casing) to +0.71 m (CEH2); they are retained for audit and reporting, but no further upstand term is applied anywhere in the analysis. The maOD head surfaces are used in the spatial groundwater analysis (Section 3.8) and the broadleaf comparison (Section 5.6.3); the temporal clustering, SSM parameterization, and BACI analyses use the ground-referenced records throughout.

For the cluster average hydrograph construction (Section 3.4.1), member-well series are averaged directly: they already share a common ground-surface datum (Section 3.1.1), so no correction is applied and between-well differences in pipe-top elevation cannot bias the cluster mean. The clustering and Pearson affinity analyses are in any case invariant to absolute depth, z-score standardization removing it. The datum convention is not, however, inert for the state-space model. A constant offset cancels in the first-difference term $\Delta h$ but not in the level term $h_{disp,prev}$, and because the model is fitted without an intercept there is nothing to absorb it. The effect falls almost entirely on $\beta_3$: across the 66 reference wells, each refitted on its comparison window, the coefficient shifts scale with each well's upstand (Pearson $r = -0.852$ for $\Delta\beta_3$), while $\beta_1$ is near-invariant. Ratios of $\beta_3$ between clusters are unaffected, so comparative results do not depend on the choice; quantities computed from $\beta_3$ alone, such as the recession half-life, are conditional on it. This is why the ground-surface frame is applied once, at source, and never re-derived downstream. Network summary statistics and well characterization figures are presented in Section 4.1 and Figure 6.

### 3.1.2 Climate Data

Monthly total precipitation and potential evapotranspiration were obtained from the RAF Valley Met Office climate station (SH 30691 75549), approximately 16 km north-west of the

site. The station record extends from December 1930 to February 2026, with a mean annual rainfall of 891 mm over the well-record period (calculated from complete calendar years 2006–2025; the partial years at the start and end of the monitoring record are excluded from this summary statistic). A single missing monthly rainfall value in the historic record (June 1941) was retained as missing rather than substituted; it falls outside the well-record period and has no influence on any analysis.

Potential evapotranspiration was calculated using the Thornthwaite and Mather (1957) method with a day-length and month-length correction, applied to mean monthly air temperature derived as the mean of daily maximum and minimum. The method proceeds in three steps. First, the heat index I is computed as the sum of twelve monthly contributions:

$$I = \sum_{m=1}^{12} \left( \frac{T_m}{5} \right)^{1.514}$$

where $T_m$ is mean monthly temperature clipped to zero for months at or below 0°C. Second, unadjusted monthly PET is calculated as:

$$PET_{unadj} = 16 \times \left( \frac{10\,T}{I} \right)^{\alpha}$$

where: $\alpha = 6.75 \times 10^{-7} I^3 - 7.71 \times 10^{-5} I^2 + 1.792 \times 10^{-2} I + 0.49239$

For months where mean temperature exceeds 26.5°C a high-temperature correction is applied $PET_{unadj} = -415.85 + 32.24\,T - 0.43\,T^2$; this threshold is not reached at RAF Valley under the historic record but is retained for formal completeness. Months with mean temperature at or below zero are assigned PET = 0. Third, the unadjusted value is scaled by a day-length and month-length correction factor:

$$PET = PET_{unadj} \times K \text{, where } K = \left( \frac{N}{12} \right) \times \left( \frac{d}{30} \right)$$

where N is the maximum possible sunshine duration for the month (hours), calculated from site latitude (53.25°N, RAF Valley) using standard solar geometry, and d is the number of days in the month. This correction accounts for the fact that the original Thornthwaite equation was calibrated for a standard 30-day month and 12-hour photoperiod.

Thornthwaite sums the twelve contributions to I over the calendar year. Here they are the twelve months ending at the month for which PET is being computed. A trailing window contains exactly one of each calendar month and is therefore a full annual heat sum, but unlike the calendar-year form it is defined for every month of a record that does not end in December, and the PET of a given month does not depend on the months that follow it. Over the well record the two forms differ by 2.1% of monthly PET on average (5th to 95th percentile −5.4% to +5.5%, 249 months) and coincide each December, when the trailing window is the calendar year. The departure from the published method is deliberate: the

monitoring record is live and is analysed part-way through a calendar year, where the calendar-year sum is undefined — for January and February 2026 it could not be calculated.

Thornthwaite PET is often reported to overestimate evaporative demand in humid maritime climates relative to physically based methods such as Penman–Monteith (Donohue et al., 2010). At RAF Valley it does not do so over the year. An FAO-56 Penman–Monteith grass-reference series, built from the station's hourly wind, humidity, temperature and radiation records (Met Office MIDAS Open; Script 50, E13b), totals within about 1% of Thornthwaite over 1957–2023 and over the years with measured radiation, 2002–2023. The two differ in their seasonal shape: Penman–Monteith is about 30% higher than Thornthwaite from January to April and about 16% lower from July to October (Section 4.1.1). Thornthwaite was selected because it requires only mean air temperature, consistent with the low-cost replication objective described in Section 3.10. Although forest recharge at Newborough is estimated to be 100–200 mm $yr^{-1}$ lower than open dune recharge (Stratford, 2006), these absolute differences do not affect the comparative conclusions of this study. The same PET series is applied to every cluster and every period, so its bias, whether a common factor or a seasonal shape, is shared by every cluster-to-cluster and before-versus-after contrast and bears on them far less than on absolute values. It does not cancel in the absolute volumetric fluxes of Section 3.7.2, which are reported as indicative for that reason.

A centred cumulative water balance series (CWB; alternatively termed the centred cumulative water balance anomaly elsewhere in this report) was computed for the well-record period by summing monthly net balance (P − PET), with a mean-centring correction applied to remove the systematic long-term offset between precipitation and Thornthwaite PET inherent to this maritime climate; the corrected series oscillates around zero and directly reflects periods of cumulative surplus and deficit relative to the long-term mean. CWB is referenced as a covariate in the BACI ANCOVA models in Section 3.5.4 and in the climate sensitivity analysis in Section 4.6.2. The centring constant, the mean monthly net balance over the well record, was 19.51 mm $month^{-1}$ (December 2004 to December 2025). The correspondence between this cumulative balance series and the unsmoothed network mean water level was assessed by linear regression at zero lag; the result (Section 4.1.1) motivates the state-space modelling framework described in Section 3.4.

Alongside the precipitation and evapotranspiration series used in the modelling, the long-term temperature record is summarized as a warming diagnostic. A mean summer maximum temperature is computed for each year of the full RAF Valley record as the mean of the June, July and August monthly maximum temperatures, retaining only years in which all three summer months are recorded, and expressed as an anomaly relative to the pre-2013 record mean. An ordinary-least-squares linear trend is fitted to the annual anomaly series to characterize the direction and rate of change across the record. The anomaly series and fitted trend are presented in Section 4.1.1.

*Baseline climate figures and well network characterization statistics (record length, mean water level, seasonal amplitude, and monthly box plots) were produced by 00_climate_summary.py, with data preparation, quality control, gap-filling and the canonical geometry derivation implemented in 01_data_prep.py.*

## 3.2 Hierarchical Clustering

To characterize the spatial structure of the aquifer and identify distinct zones of hydrological behaviour, agglomerative hierarchical clustering was applied to the continuous dipwell monitoring network, following applications of unsupervised classification to groundwater-level records elsewhere (Wu, Koch and Xu, 2021).

Why cluster at all. A partition of the network is not a substitute for the continuous covariates that describe each well — distance to the coast and to the lake, ground elevation, land cover, position — and the question of what it adds is answered rather than assumed. Regressing the per-well SSM coefficients of Section 3.4.2 on those six covariates, with and without the five cluster labels, the labels add least for recharge sensitivity ($\beta_1$: adjusted $R^2$ 0.69 without them and 0.72 with; F(4,55) p = 0.04; ΔAIC −4.0) and most for atmospheric draw and drainage ($\beta_2$: adjusted $R^2$ 0.66 → 0.77, p = $2 \times 10^{-5}$, ΔAIC −23; $\beta_3$: 0.62 → 0.69, p = 0.003, ΔAIC −11). The gain is not the forest flag in disguise: within the 50 wells outside the forest, which fall in C1–C3 alone, the labels improve all three (p = 0.045, $1 \times 10^{-5}$ and 0.0045). What the partition carries that position and land cover cannot is the east–west geological division (Section 4.2: the till-and-estuary-influenced eastern aquifer against the deeper western sand block), which sets the drainage and atmospheric-draw behaviour of a well more strongly than its recharge response. In the dendrogram that division is the second: the top Ward merge (distance about 0.99) separates the Main Forest (C4) from all other wells, and within the rest the eastern C1 and C2 divide from the western C3 and C5 at about 0.61. Beyond that, the zones serve four purposes: they pool the specific-yield estimates of Section 3.7.3 to a scale at which the water-table-fluctuation method is stable; they classify the extended network by affinity (Section 3.3); they supply the scale at which the mechanistic coefficients are fitted with the noise of individual wells suppressed (Section 3.4.1); and they give site managers a data-driven zonation rather than one drawn from land cover. One caveat governs the test: the clusters were cut from the same hydrographs the coefficients are fitted to, so it shows that the partition found structure in the record, not that the structure is geological — that reading rests on the bootstrap stability of Section 4.2 and the mapped geology. The test is emitted by Script 07 (07_05_clusters_vs_covariates.csv).

### 3.2.1 Data Pre-processing and the Pearson Correlation Distance

Because the absolute elevation of the water table varies significantly across the site due to topography, analysing raw head data would artificially separate functionally identical slacks. To isolate the temporal dynamics—the "shape" of the hydrograph—each well's monthly time

series was used directly in the correlation matrix; the correlation-based distance metric inherently removes mean and variance differences between series, making explicit z-score standardization unnecessary. (Note that explicit z-score standardization is applied at the cluster average construction stage of the subsequent Pearson affinity analysis, where it ensures that wells with different absolute ranges contribute equally to the cluster average time series.)

Pairwise dissimilarity between the time series was then computed using the Pearson correlation-based distance: $d_{i,j}$:

$$d_{i,j} = 1 - r_{i,j}$$

where $r_{i,j}$ represents the Pearson correlation coefficient calculated over contemporaneous, overlapping temporal records for wells i and j. This specific metric translates the statistical correlation directly into the physical behaviour of the aquifer:

- Synchronized Slacks (r=1): Where two dipwells rise and fall in perfect harmony, the distance metric becomes zero (d=0). The algorithm recognizes identical hydrogeological drivers and groups them tightly.
- Unrelated Slacks (r≈0): Where one well responds to flashy winter storms and another is heavily buffered, the correlation approaches zero, yielding a distance of one (d=1), effectively pushing them apart in the clustering space.
- Opposing Slacks (r=−1): Where wells exhibit inverse temporal behaviour, the distance maximizes at two (d=2).

By utilizing the Pearson correlation distance, the absolute topographic elevation of the sand dunes is stripped from the analysis. The algorithm is forced to group wells purely on the timing and shape of their temporal hydrographs, allowing deep aquifer wells to cluster together even if they are situated at vastly different physical elevations.

### 3.2.2 Agglomeration Strategy

The dissimilarity matrix was processed using agglomerative hierarchical clustering, employing Ward's minimum variance linkage **(Ward, 1963)**. At the initiation of the algorithm, each dipwell is treated as an individual cluster. The algorithm then iteratively merges the two clusters that result in the smallest possible increase in the total within-cluster variance, quantified mathematically as the Error Sum of Squares (ESS).

For any given cluster C containing n wells, the ESS is defined as the sum of the squared distances between each well's standardized time series (Xi) and the mean temporal signature (centroid, $\bar{x_C}$) of that cluster:

$$ESS_C = \sum_{i \in C} \left\| x_i - \bar{x_C} \right\|^2$$

When evaluating the potential merge of two clusters, A and B, the algorithm calculates the resulting increase in total variance (ΔESS). This merge cost is computed algebraically as:

$$\Delta ESS(A,B) = ESS_{A \cup B} - (ESS_A + ESS_B)$$

Which simplifies to the objective function:

$$\Delta ESS(A,B) = \frac{n_A \cdot n_B}{n_A + n_B} \cdot \left| \bar{x}_A - \bar{x}_B \right|^2$$

where nA and nB represent the number of wells in clusters A and B, respectively.

This specific algebraic formulation is hydrogeologically essential for mapping a continuous dune aquifer. Because the spatial groundwater gradient transitions smoothly from the bedrock ridge to the foreshore, single-linkage (nearest-neighbour) algorithms frequently suffer from "chaining"—sequentially merging adjacent wells until the entire site collapses into a single, uninterpretable cluster.

Ward's mathematical objective function actively resists this chaining effect. The $\frac{n_A n_B}{n_A + n_B}$ term acts as a strict mathematical penalty against merging disproportionately large clusters or appending single outliers to established groups. Instead, it forces the algorithm to prioritize merging smaller, tightly correlated groups first, ensuring that the hydrographic signatures within any newly formed group remain highly synchronized. By penalizing variance, Ward's linkage imposes discrete analytical boundaries onto a continuous physical system, delineating the continuous sand matrix into compact, functionally distinct hydrogeological catchment zones.

### 3.2.3 Cluster Determination and Robustness Assessment

The optimal number of clusters (k) was evaluated iteratively for k = 2 to k = 10 using three complementary metrics: the elbow method applied to Ward's merge distances, the mean silhouette coefficient (Yuan and Yang, 2019), and the Calinski–Harabasz index (Caliński and Harabasz, 1974). Silhouette and Calinski–Harabasz both peaked at k = 2, reflecting the top division of the dendrogram (Ward merge distance about 0.99), which separates the Main Forest (C4) from all other wells; within the rest, the shallower eastern clusters (C1, C2) divide from the deeper western C3 and C5 at about 0.61. However, selecting k = 2 provides insufficient resolution to distinguish the geomorphological units relevant to conservation management. Rather than selecting k on a single metric, the partition was evaluated across a range of candidate values using bootstrap stability analysis (Hennig, 2007) to quantify how sensitive cluster assignments are to the particular set of wells observed.

For each candidate k in {4, 5, 6, 7}, the 66-well reference network was resampled with replacement 1000 times. At each resample, Ward's linkage was re-applied and the resulting assignments compared with the reference (full-sample) partition. For every pair of wells, the

fraction of resamples in which both wells were assigned to the same cluster was recorded as a pairwise co-assignment probability; the median of these probabilities across a well's reference-cluster partners provides a per-well stability score. A well scoring 1.0 co-assigns with the same neighbours in every resample; lower scores indicate sensitivity to the composition of the network.

At k = 5, four of the five clusters exhibited median per-well stability of 0.98 or higher: the Main Forest cluster (stability 1.00), the Coastal Forest cluster (0.99), the Dune cluster (0.99), and the Lake Edge cluster (0.98). The fifth cluster, Western Residual, exhibited moderate stability (0.38), reflecting the landscape heterogeneity of this group rather than a failure of the partition — the two groups into which it divides at k = 6 remain the least stable in the partition (median stability 0.71 and 0.25, against 0.96 or higher for the other four), consistent with k = 5 being the finest resolution at which Ward's linkage produces behaviourally coherent groups. The final solution of k = 5 was therefore selected as the finest partition at which the majority of clusters are robust under resampling, while providing sufficient resolution to distinguish the principal geomorphological units.

The silhouette coefficient at k = 5 is 0.37. This moderate value is consistent with the physical reality of a contiguous, highly permeable sand aquifer where cluster boundaries are inherently gradational. The bootstrap stability analysis provides a more informative assessment of partition quality than the silhouette score alone, because it directly quantifies whether individual well assignments are reproducible rather than measuring only average cluster separation.

Two wells — CEH3 and CEH22 — were excluded from the reference network prior to clustering because Ward's linkage consistently identified them as singleton outliers at every k from 4 to 9, resistant to grouping with any behavioural cluster. Both are low-elevation coastal wells (ground elevation 3.2–4.4 m) whose correlation structure is consistent with tidal-signal contamination superimposed on the climate-driven response that the clustering and subsequent state-space modelling are designed to characterize. Their exclusion follows the same principle applied to the FE/LIS clearfell wells and Llyn Rhos-Ddu (Section 3.1.1): wells whose behaviour violates the stationarity or single-mechanism assumptions of the analytical framework are retained in the extended network for per-well analysis but excluded from the reference partition. Including either well distorts the Ward's tree at lower k values — CEH3 suppresses the Lake–Dune separation at k = 4, and CEH22 appears as a persistent singleton from k = 5 onward — so their exclusion is both physically motivated and algorithmically necessary.

A converse case is CEH11, which is assigned to C1 (Lake Edge) on the basis of hydrograph similarity but is geographically isolated from the other six C1 members. The remaining C1 wells occupy ground at 8.5–10.9 m AOD around Llyn Rhos-Ddu in the northern part of the site; CEH11 sits at 3.5 m AOD on the south-eastern coast near the Menai Strait, approximately 1.7 km from the nearest C1 well (CEH26). Its damped, low-amplitude

seasonal hydrograph — the signature that drives its C1 assignment — likely reflects proximity to the Menai Strait tidal boundary rather than to Llyn Rhos-Ddu: a different constant-head source producing a similar buffered water-table response. The cluster assignment is retained because the statistical grouping is hydrograph-based and CEH11's behaviour is genuinely C1-like, but the different boundary condition means its mechanistic coefficients should not be interpreted alongside the lake-edge wells without qualification. This distinction is relevant to the cluster coefficients in Section 4.2.2: CEH11 contributes nothing to the lake-proximity gradient that governs the other members.

### 3.2.4 Seasonal Amplitude as a Secondary Cluster Descriptor

The Pearson correlation distance described in Section 3.2.1 is scale-invariant: because Pearson correlation is unchanged by linear rescaling, two wells whose hydrographs rise and fall in synchrony will cluster together regardless of whether one exhibits a seasonal range of 0.5 m and the other 2.0 m. This property is desirable for identifying shared drainage mechanisms, but it means that wells within a single cluster may differ substantially in response magnitude — reflecting local differences in specific yield, proximity to hydrogeological boundaries, or depth to an impermeable layer. To characterize each cluster on this second, orthogonal axis, a suite of seasonal amplitude descriptors was computed as a supplementary characterization alongside the pattern-coherence statistics reported in Section 4.2.

The primary amplitude metric is the difference between the 90th and 10th percentiles of each well's monthly water-level series (p90 − p10), expressed in metres. This inter-decile range captures a representative seasonal swing while remaining robust to single-month outliers. It was computed for each well over three windows: the full record, the pre-2018 period (prior to the December 2017 clearfell event in the FE compartment), and the post-2018 period. Wells with fewer than 24 monthly observations in a given window were excluded from that window's calculation: two complete annual cycles is the shortest record in which each decile rests on more than one season's observations, so that a single anomalous winter or summer cannot set p10 or p90 on its own. Cluster-level statistics are reported as the median of per-well amplitudes within each cluster, preserving the visibility of within-cluster spread that would be suppressed by computing a single amplitude on the cluster-mean hydrograph.

Because the post-2018 window contains a higher proportion of drought summers than the pre-2018 window, a climate-normalized variant was also computed. Monthly observations from empirically identified drought summers (Jun–Sep rainfall total at least one standard deviation below the long-term mean of the RAF Valley climatology against the pre-monitoring baseline) were removed from both windows before recomputing p90 − p10. This normalization is reported alongside the raw values as a sensitivity check; where the two agree in direction and approximate magnitude, the raw value is taken as primary. The normalization predominantly affects the post-2018 window, since only wells whose records extend back to 2005 are modified in the pre-2018 window.

Amplitude results are presented in Section 4.2 as a secondary descriptor alongside pattern-coherence and bootstrap stability. The within-cluster range of per-well amplitude is reported alongside the median because the cluster-mean β coefficients from the state-space model (Section 3.4) represent an average response magnitude across wells whose individual magnitudes may differ meaningfully.

*Clustering, silhouette validation, and cluster assignment were implemented in 02_clustering.py. Cluster map was produced by 04_cluster_visualisations.py.*

## 3.3 Checking Cluster Allocation with Pearson Affinity Analysis

While hierarchical clustering imposes discrete boundaries, the physical reality of a highly transmissive sand aquifer is one of continuous hydraulic gradients. To evaluate the rigidity of the Ward's linkage assignments and quantify the gradational nature of the aquifer, a secondary Pearson Affinity Analysis was conducted to measure the probabilistic membership of each well to all identified clusters (Rao and Srinivas, 2006).

### 3.3.1 Cluster Membership and Affinity Scoring

First, a cluster average hydrograph time series, $\bar{X}_k(t)=\frac{1}{n_k}\sum_{i\in C_k} x_i(t)$, was generated for each of the five identified clusters (k=1,…,5; shown in Figure 9) by calculating the mean standardized response of all wells definitively assigned to that cluster by the Ward's algorithm.

The temporal affinity of every individual well i to each cluster centroid k was then quantified using the Pearson correlation coefficient (ri,k) over contemporaneous overlapping records of length N:

$$r_{i,k}=\frac{\sum_{t=1}^{N}\left(x_i(t)-\bar{x}_i\right)\cdot\left(\bar{X}_k(t)-\bar{\bar{X}}_k\right)}{\sqrt{\sum_{t=1}^{N}\left(x_i(t)-\bar{x}_i\right)^2\cdot\sum_{t=1}^{N}\left(\bar{X}_k(t)-\bar{\bar{X}}_k\right)^2}}$$

*Cluster centroid construction, per-well affinity scoring, and the four-category classification described in Section 3.3.2 are implemented in 05_pearson_affinity.py.*

### 3.3.2 Classification Criteria

To categorize the spatial transition of the groundwater dynamics, let r_{1st} represent a well's highest correlation with any cluster centroid, and r_{2nd} represent its second-highest correlation. The affinity margin, Δr = r_{1st} − r_{2nd}, serves as a mathematical index of well hydrograph behaviour. Based on these metrics, each well in the 66-well reference

network was classified into one of four structural categories. The same affinity procedure was subsequently applied to the extended network of wells with shorter or earlier records, allowing the hydrogeological classification to be extended beyond the reference network boundary (Section 3.3.3).

- **Core (Δr ≥ 0.05):** The well exhibits a dominant affinity to a single cluster, and its highest correlation matches its spatial assignment from the Ward's linkage. These represent the definitive hydrological signature of their respective geomorphological zones.
- **Fuzzy (Δr < 0.05):** The well exhibits highly competing affinities between its primary and secondary clusters. This mathematical ambivalence identifies physical ecotones — transitional zones where the groundwater regime shifts smoothly between two distinct structural blocks (e.g. the boundary between the Western Residual and the Forest margin).
- **Spy:** The well's highest temporal affinity (r_{1st}) belongs to a different cluster than the one assigned by Ward's hydrograph linkage. This divergence highlights localized hydrogeological anomalies where sub-surface stratigraphy overrides broad spatial proximity (Hannah et al., 2000).
- **Unclassified:** The temporal record is of insufficient length or overlap to generate statistically significant correlation coefficients.

The Δr ≥ 0.05 threshold distinguishing Core from Fuzzy membership is an empirical choice; lowering it to 0.03 or raising it to 0.10 reclassifies 16 and 14 wells respectively between the Core and Fuzzy confidence tiers, reflecting a concentration of affinity margins near the cut-off. This sensitivity affects only the confidence labelling within each cluster, not the partition itself: cluster assignment derives from the Ward's-linkage solution (Section 3.2), and every reference well retains its assigned cluster regardless of the Δr threshold.

A fifth diagnostic flag — Multi-Cluster Affinity (MCA) — is raised in addition to the primary classification for any well where three or more distinct cluster centroids achieved a correlation of r > 0.90. An MCA flag mathematically identifies highly generic, highly responsive boundary wells that lack a specialized local signature, acting instead as structural nodes governed by system-wide climatic forcing rather than local topography. The flag is raised for 44 of the 66 reference wells, so it describes a condition common across this aquifer rather than a rare one — which is itself the point: in a highly transmissive sand body most wells carry the shared climatic signal, and it is the minority that do not which hold a distinct local signature.

### 3.3.3 Extended Network Classification

To extend the hydrogeological classification beyond the 66-well reference network, the Pearson affinity procedure was applied to all wells with a minimum of 24 months of record that did not meet the reference network criteria. The reference network cluster averages

derived in Section 3.3.1 were used as fixed classification templates; these were not recalculated from the extended wells. Prior to correlation, all wells — both reference and extended — were z-score standardized jointly as a single combined dataset, ensuring that extended wells with shorter or earlier records were standardized relative to the same temporal distribution as the reference network rather than independently.

Extended wells were classified using the same affinity margin threshold ($\Delta r = 0.05$) and MCA criteria ($r > 0.90$ with three or more centroids) as the reference network audit. Because extended wells carry no prior Ward's cluster assignment, the Spy classification is not applicable; extended wells are assigned as either Core or Fuzzy based solely on the strength of their best-match affinity.

*Extension of the affinity classification to the full monitoring array is implemented in 06_pearson_extended.py.*

### 3.3.4 Filling Gaps in the Record from Cluster-Mates

Many wells were read for only part of the monitoring period. A large group began in March 2010, others later, and the extended wells were read irregularly. The clusters say which wells move together, and that suggests a way of filling the gaps. Over the months two wells were both read, a straight line relating one to the other is fitted, and that line then carries the donor's measured level into the target's missing months. Whether this works better than the climate model, and which donor works best, was tested before any gap was filled.

The test hides months that exist. Two windows of 54 months were hidden in turn: an early window beginning in March 2006, which holds the 2006–08 excess and is the late-starting wells' own gap, and a late window ending with the reference record. Every well read through at least 80 % of a window was a target. Each target was filled from a donor read through the window under five rules: the nearest well whose correlation with it over the overlap is at least 0.95; the best-correlated well; the nearest cluster-mate meeting the same correlation; the median of its cluster-mates; and Model A, fitted outside the window and run freely through it, as the climate-only reference. Each donor line was fitted on at least 48 months outside the window, and cluster membership never saw the hidden months: the extended wells' best match to the reference centroids was recomputed without them. Donors within 50 m of their target were treated as twins and left out of the headline. The rule adopted was fixed before the run: the one with the lowest median error in the held-out spring level over both windows or, where another rule's median lies within that rule's bootstrap 95 % interval, the simpler.

The adopted rule then fills each well's gaps, one contiguous gap at a time, by the same procedure. A donor qualifies only if it was read through at least 80 % of the gap, as in the test. A second version accepts donors read through at least 50 % of the gap, a rule the test did not examine, and only for gaps the first leaves empty. Every filled month is flagged as

reconstructed. The reconstructed levels are kept apart from the measured record and enter no other analysis in this report.

Borrowing from a cluster-mate across eras assumes the grouping held then as it does now. That was tested on the CCW wells read from 1989 to 1996. The wells with a confirmed modern counterpart were clustered on their 1989–96 record alone, by the method of Section 3.2, and the grouping was compared with their modern clusters by the adjusted Rand index against a permutation null. By the rule fixed before the run, the grouping counts as durable only if the index beats the null at $p < 0.05$ and at least two-thirds of the pairs that share a modern cluster also share a 1989–96 group. The same wells clustered on their modern record give the control, whether so few wells can recover the partition at all, and the author's own similarity charts of the CCW wells, drawn by eye at the time, were compared in the same way.

*The cross-decade test, the hold-out test and the gap filling are implemented in 51_cluster_gap_fill.py.*

## 3.4 Using Models to derive the Aquifer Architecture

Predicting how a dune aquifer responds to rainfall and evaporation over time requires a model that captures not just the immediate effect of each month's weather, but also its dependence on its own previous state. A simple approach — the Traditional Linear Model (TLM) used as a benchmark in Section 3.4.2 — treats the aquifer as a bucket that fills with rainfall and empties at a fixed background rate, regardless of how full it is (von Asmuth et al., 2002). This works reasonably well when the model is continuously corrected with observed dipwell readings, but fails when run autonomously over longer periods because it has no mechanism to slow drainage as the water table drops or accelerate it when the water table is high (Hypolite et al., 2021; Young, 2011). The state-space model (SSM) captures this physical reality by adding a head-dependent drainage term that scales drainage rate to the displacement of the current water table above a reference drainage base, consistent with the functional relationship between groundwater head and lateral discharge reported in shallow sandy aquifers (Gumuła-Kawęcka et al., 2021; Manis et al., 2022). This single addition turns the model from a statistical curve-fit into a physically constrained representation of aquifer behaviour, allowing it to simulate water table dynamics autonomously over multi-year periods with much less accumulated drift than the TLM (Knotters and van Walsum, 1997); the benchmark of Section 3.4.2 measures the difference.

The baseline formulation, working in terms of monthly water table change $\Delta h = h(t) - h(t-1)$, is:

$$\Delta h_t = \beta_1 \cdot P_t - \beta_2 \cdot PET_t - \beta_3 \cdot h_{disp,t-1}$$

where Δh(t) is the change in water table depth during month *t*; P(t) and PET(t) are monthly rainfall and potential evapotranspiration totals for month *t*; h_disp(t−1) is the displacement of the water table above the drainage datum at the end of month *t*−1; and $\beta_1$, $\beta_2$, $\beta_3$ are the recharge, atmospheric draw, and drainage coefficients respectively. The water table position is expressed in the negative-below-ground convention, so h(t−1) is negative for sub-surface levels. Displacement is defined as:

$$h_{disp,t-1} = D + h_{t-1}$$

where $z_0$ is the drainage datum depth (metres below ground surface). Working in displacement rather than raw depth ensures that h_disp is positive when the water table sits above the drainage base, so $\beta_3$ is positive (Darcy-consistent) for all clusters. Fitting on raw depth h(t−1) produces negative $\beta_3$ at two of the five clusters (the forest clusters C4 and C5, neither significantly), indicating that the depth-below-surface coordinate does not correctly represent the hydraulic gradient driving lateral drainage. $\beta_1$, $\beta_2$ and $\beta_3$ are all defined and reported as positive quantities; the negative signs in the model equation indicate that PET and drainage act as loss terms on Δh. The complete sign convention is set out in Section 3.4.1.

The drainage datum $z_0$ was selected by sweeping reference depths from 0.5 to 8.0 m in 0.1 m steps, refitting the cluster-centroid SSM at each depth and recording $\beta_3$, its p-value, $R^2$ and AIC (Burnham and Anderson, 2002) for all five clusters. The selection criterion was the minimum depth at which all five clusters simultaneously produce positive and statistically significant ($p < 0.05$) $\beta_3$. The C4 Main Forest cluster — which has the deepest mean water table in the network and is therefore closest to the drainage base — is the binding constraint: C1, C2, C3 and C5 reach significance at shallower datums, but C4 requires the deepest. The corresponding canonical value $z_0 = 3.7$ m is adopted as the network-wide datum to ensure consistent positive displacement across all clusters. The value $z_0 = 3.7$ m is set deliberately deeper than the bare empirical minimum — which under the current record is 1.6 m — so that the forest drainage term is comfortably rather than marginally identified: at the minimum the C4 $\beta_3$ is only just significant ($p = 0.035$), whereas at 3.7 m every cluster, C4 included, is comfortably significant (C4 $p = 0.001$). The coefficients are re-fitted at the adopted datum on the full current record, so 3.7 m is a conservative common reference. $R^2$ and AIC change little across the viable range — the per-cluster $R^2$ spans at most about 0.08 between the shallowest and deepest admissible datum, edging higher at C1 and C2 under a shallower base and at the forest clusters under a deeper one — so the datum choice does not materially distort model fit. Because the headline SSM carries no intercept, a shift in the datum constant does not cancel but redistributes across $\beta_1$, $\beta_2$ and $\beta_3$ together (Δh, being a difference, is unaffected); the datum thus fixes the absolute scale of the drainage coefficient and half-life, which are reported as conditional on $z_0 = 3.7$ m, while the quantities the interpretation rests on — the cluster ordering, the open-dune drainage contrasts, the mean water-balance closure and the equilibrium wetness — are robust across the swept range. The

per-well spatial distribution of optimal datum depths and the $R^2$ cost of the uniform assumption are presented in Section 4.9.1 (Figure 53a).

A related question is whether the drainage base is better represented as a fixed height above Ordnance Datum or as a uniform depth below the ground surface. The two are distinguished by a geometric test across the network's 10.9 m spread of ground elevations: each well's $R^2$-optimal datum, expressed as an elevation, is regressed on its ground elevation, a fixed-elevation base predicting a slope of 0 and a surface-following base a slope of +1. The observed slope across the 66 reference wells is +0.845, placing the effective drainage base close to surface-following, and a fixed absolute datum is in any case undefinable network-wide because no single elevation keeps every well within the swept depth range. A base that follows the dune topography supports reading $\beta_3$ as a Darcy-consistent drainage coefficient — drainage proportional to head above a nearby base — consistent with the theoretical flow paths and Darcy vectors of Section 4.9.5 (Figure 56), and it is the geometry the displacement datum is chosen to represent.

What the sweep prefers, and what is published. Two independent readings of the datum sweep agree on what the data themselves prefer. Refitting each cluster centroid across the swept range and taking the datum that minimizes AIC gives 0.7, 1.2, 2.2, 7.1 and 3.7 m at C1–C5, tracking the mean depth to water in each cluster; and Model B, in which a freed intercept absorbs the datum, implies a zero-drainage level ($z_0 - \alpha_B/\beta_3$) of 0.68, 1.19, 2.16, 7.11 and 3.71 m — the same values within 0.05 m at every cluster. The best-fitting form is therefore a mean-reversion model in which drainage is measured from each cluster's own mean level. The published form deliberately is not that: it evaluates drainage at one physical reference, a drainage base below the water-table fluctuation range, so that $\beta_3$ reads as a Darcy-consistent discharge coefficient rather than a reversion rate and one coefficient set can be compared across clusters and inverted into the threshold equations of Section 3.6.3. The cost of that choice is stated so that it need not be computed by the reader: at $z_0 = 3.7$ m the centroid fits sit 94.0, 28.7, 7.3, 0.7 and 0.00 AIC units above each cluster's own optimum, with $R^2$ lower by 0.086, 0.028, 0.005 and less than 0.001 (C1 0.818 → 0.729). The cost falls entirely where the mean water table is shallowest — C1 and the eastern C2 wells — and vanishes under the forest.

No conclusion the report draws depends on the value chosen. The ranking of the five clusters by $\beta_3$ is identical at every swept datum from 4.7 to 8.0 m, and the ranking by mean drainage flux $\beta_3 \cdot \bar{h}_disp$ from 1.2 m; at 3.7 m the flux stands at 0.97, 0.96, 0.92, 0.81 and 0.89 of its value at the deepest swept datum for C1–C5, and it stays within 80 % of that value from 1.7, 1.9, 2.5, 3.7 and 3.1 m respectively (within 90 % from 2.3, 2.7, 3.4, 4.7 and 3.9 m), which is the plateau the Darcy reading rests on. The datum is set below the fluctuation range and within the saturated thickness the four site boreholes constrain (3.65 to more than 12.8 m; Supplementary Material Table S2.1). A uniform depth below ground, rather than a fixed elevation, follows from Section 4.9.5: the head surface replicates the ground smoothed to

300 m at slope 1.01, and the slack floors on which the wells stand are cut to the water table, so a depth below ground is, to within the local relief, a depth below the local mean level (Figure 57). Each cluster's optimum, the fit cost of the shared datum and the flux plateau onsets are in 03_18_datum_invariance.csv (Script 03).

Choosing a drainage datum: a procedure for other sites. The datum question arises wherever a head-dependent drainage term is fitted without an intercept, and the steps that settle it here transfer. (1) Sweep the datum over the plausible range of the drainage base — here 0.5 to 8.0 m below ground in 0.1 m steps — refitting the model at each depth at the scale the coefficients will be published (the cluster centroids), and record $\beta_3$, its significance, $R^2$, AIC and the mean drainage flux $\beta_3 \cdot \bar{h}_disp$. (2) Find the admissible floor: the shallowest datum from which every unit's $\beta_3$ is positive and significant at every deeper datum (1.6 m here). Below it the drainage term is not identified. (3) Fit the same model with a free intercept (Model B) and compute the level at which its drainage is zero, $z_0 - \alpha_B/\beta_3$. If that level coincides with each unit's AIC-optimal datum, as it does here within 0.05 m, the fit-optimal datum is the unit's own mean level: the data prefer a mean-reversion form, and a datum chosen on fit alone would measure persistence, not drainage. (4) Locate the flux plateau: the datum from which $\beta_3 \cdot \bar{h}_disp$ stays within a stated fraction of its deepest-datum value (80 % from 1.7–3.6 m here, 90 % from 2.4–4.6 m). On the plateau the drainage flux and the drainage/evapotranspiration partition are insensitive to the datum and $\beta_3$ reads as a Darcy discharge coefficient; the half-life $\ln 2/\beta_3$ is not insensitive and must be reported as conditional on the datum. (5) Choose one datum for the network, on or beyond the plateau for the unit with the deepest water table, below the fluctuation range everywhere, and within whatever the site's boreholes say the saturated thickness is. (6) Decide the frame: regress each well's fit-optimal datum, as an elevation, on its ground elevation — a slope near 1 (0.845 here) and a head surface that follows the smoothed ground (Section 4.9.5) support a uniform depth below ground; a slope near 0 supports a fixed elevation. (7) Publish the cost: each unit's $\Delta$AIC and $\Delta R^2$ at the chosen datum against its own optimum, and the range of datums over which the rankings by $\beta_3$ and by flux are unchanged, so that a reader can see what the choice buys and what it costs without recomputing the sweep. Script 03 performs steps 1–4 and 7 and writes them to 03_18_datum_invariance.csv; step 6 is the geometric test above; step 5 is the judgement, recorded here as $z_0 = 3.7$ m.

The SSM formulation adopted here sits within an established lineage of transfer-function-noise (TFN) and linear-reservoir models for groundwater head series, in which monthly or daily head is expressed as a linear response to climatic forcing with an autoregressive storage-decay term (von Asmuth et al., 2002; Knotters and Bierkens, 2000; Peterson and Western, 2014; Peters et al., 2003). Recent open-source implementations — notably Pastas (Collenteur et al., 2019) and HydroSight (Peterson and Western, 2014) — provide impulse-response calibration at individual wells and have been applied to shallow unconfined aquifers across a range of climatic settings (Obergfell et al., 2019; Bakker and Schaars, 2019). The formulation used in this study departs from these implementations in three ways

developed in Section 3.4.1: it is fitted at cluster-average scale rather than individual-well scale; it retains the storage decay coefficient explicitly rather than absorbing it into a convolution kernel; and it constrains the intercept to zero. These choices enable the Lumped Catchment Storage Coefficient (LCSC) to be derived directly from the fitted parameters and inverted algebraically into the P_flood threshold equations reported in Section 3.6.3.

The correspondence is tested rather than asserted. Model B — the same equation with a free intercept — is term for term the exponential-response recharge model of Pastas (Collenteur et al., 2019): its response time is $-1/\ln(1-\beta_3)$ months, its steady gain $\beta_1/\beta_3$, its evaporation factor $-\beta_2/\beta_1$ and its fitted constant the level at which the fitted drainage is zero. Script 48 fits Pastas at each of the 66 reference wells on exactly the months that well's Model A and Model B were fitted on (the comparison window of Section 3.4.2) and again on the well's full record, with the monthly readings stamped at month-end and the monthly climate spread evenly over each month's days, once without a noise model and once with Pastas's AR(1) noise model, and sets the four quantities beside both SSM forms; the unit conversion is verified beforehand on noise-free wells generated from each cluster's own Model B coefficients. Model A has no Pastas counterpart, because its $\beta_3$ carries the datum. Pastas has no datum and no drainage term: it fits a free base level, as Model B does, so its agreement with Model B checks the implementation of the per-well fits rather than providing independent support for either form (48_02_pastas_agreement.csv, 48_report_numbers.csv; see "Which form is used where", Section 3.4.1). On the full record, where 63 of the 66 wells identify a response time, Pastas and Model B agree well by well — Spearman ρ 1.00 for $\beta_1$, 0.96 for $\beta_2$ and 0.98 for $\beta_3$ — with Pastas's $\beta_3$ a median 0.86 of Model B's and its response 1.19 times longer, the fitting approaches differing in what they fit (the level, against the monthly change conditional on the observed lagged level), and its base level 0.20 m lower. Against Model A, Pastas's $\beta_3$ is 1.99 times larger: the datum. On the comparison window all nine Main Forest wells fail to identify a response time, the response there being comparable with the window; on the full record six of them do, and lie on the 1:1 line.

### 3.4.1 Modelling the Cluster water levels

The state-space model defined in Section 3.4 was applied at the cluster level to derive mechanistic coefficients representative of each hydrogeological zone. For each of the five clusters, a mean monthly time series was computed by averaging the ground-referenced records of all member wells (Section 3.1.1), producing a single representative hydrograph of seasonal water table behaviour. The SSM was fitted to this average hydrograph using the full available record rather than the 100-month window applied to individual wells (Section 3.4.2), yielding a single set of coefficients ($\beta_1$, $\beta_2$, $\beta_3$) representative of the cluster as a whole. Cluster-scale fitting reduces the influence of individual well idiosyncrasies and local topographic noise, producing a parsimonious mechanistic characterization of each hydrogeological zone. Because wells were installed progressively over the early record, the cluster mean in the first years is formed from a growing subset of member wells rather than

the full complement. Each centroid was therefore also refitted from a common 2015 reference date, a recent window in which every cluster reports its full membership throughout (Section 4.2.2).

**Cluster-scale versus per-well fitting.** The framework fits the state-space model at two scales, and does so deliberately, because the two answer different questions. The mechanistic characterization (Table 3, and the water-balance and BACI analyses that draw on it) is fitted to the cluster-mean hydrograph. Averaging the member records before fitting suppresses the measurement noise, short gaps and local topographic idiosyncrasies carried by any single dipwell, and returns one internally consistent coefficient set per hydrogeological zone — a signal-to-noise gain that is the point of clustering in the first place, since the zones were defined precisely so that their members share a dynamic signature. The spatial products (the coefficient surfaces of Section 3.4.3, the water-balance residual field, and the coastal-retreat regression) are instead built from the model fitted independently at each reference well, because their purpose is to resolve within-zone structure — the gradient of recharge sensitivity across the open dune, the ridge-margin concentration of the residual, the distance-decay of the coastal signal — which a cluster mean would erase by construction.

The trade-off runs in both directions and is stated plainly here. Cluster-scale fitting buys robustness at the cost of within-zone detail, and that detail is not negligible: recharge sensitivity varies roughly two-fold within the Western Residual alone ($\beta_1$ ranging 2.3–5.5 across its member wells; Table 4), and the forest interior spans drainage half-lives from about 16 to 42 months around a cluster mean of 26, once CEH13 and CEH14 — whose drainage coefficients the SSM does not resolve — are set aside. A single cluster coefficient set represents this spread by its centroid and should not be read as a property shared uniformly by every member. Cluster-scale fitting also inherits any error in the partition itself: a well assigned to the wrong zone, or a zone that is genuinely gradational rather than discrete, will bias the mean it contributes to — a dependence the per-well fits do not carry. The two scales are therefore treated as complementary rather than interchangeable throughout: cluster coefficients for mechanism and cross-zone comparison, per-well coefficients wherever the spatial distribution within a zone is the object of interest. Where they are combined — for instance in reading a cluster's mean behaviour against the per-well range of Table 4 — the range is reported alongside the mean rather than subsumed into it.

**Sign convention.** $\beta_3$ is defined and reported as a positive quantity throughout this study. The negative sign appears explicitly in the model equation as $-\beta_3 \cdot h_{disp,t-1}$ to indicate that storage decay acts as a loss term: a higher antecedent displacement above the drainage datum generates greater drainage, tending to restore equilibrium. Where the notation $-\beta_3$ appears in subsequent figures, tables and analytical sections (including the BACI, scraping and cluster mechanistic analyses), it refers to this loss term and should be read as the magnitude of the

drainage decay coefficient. An increase in $-\beta_3$ therefore indicates accelerated drainage; a decrease indicates retarded drainage or greater storage retention.

**Intercept constraint.** The cluster regression was fitted with the intercept strictly forced to zero. This enforces a strict physical mass balance: in the absence of external climatic forcing (P = 0, PET = 0), the aquifer cannot spontaneously generate or lose water beyond the passive drainage decay dictated by its own antecedent storage state ($-\beta_3 \cdot h_{disp,t-1}$). The constraint also enables direct algebraic inversion into the P_flood threshold equations (Section 3.6.3). A companion specification with the intercept released (Model B, fitted alongside Model A on the same series) is used where a result is set by the water table's mean-reversion time rather than by the drainage datum: the intercept absorbs any constant datum shift, leaving $\beta_3$ independent of the datum, and it is the specification behind the persistence figures of Sections 4.12 and 5.7.5 and the climate projections of Section 3.7.5 (Script 03, 03_16_model_b_persistence.csv; see below).

**Which form is used where.** The two forms of the model identify different things, and each is used for what it identifies.

The no-intercept form (Model A) fixes the level to which the aquifer drains at the drainage datum and gives $\beta_3$ its reading as a drainage coefficient. Model A therefore carries the cluster coefficients, the flooding thresholds and the hindcasts. The split of the mean loss between drainage and atmospheric draw is quoted from the free-intercept form (Model B): in the mean the split follows from $\beta_1$ and $\beta_2$, and Model B's intercept absorbs any shift of the datum, so its split is the same at every datum. Model A's split, which moves with the datum, is carried beside it, with the depth-dependent model and the seasonal recession ratio, as the bracket (Section 4.2.3). In this report a coefficient, half-life or datum range given without a form is Model A at the network datum; every drainage share and every timescale names its form.

Model A's split depends on the datum, and the heads alone cannot fix the datum. It depends on it in two regimes. Close to the water table, above about 1.6 m below ground, $\beta_3$ becomes a rate of return to the mean rather than a drainage coefficient, and Model A's split swings with the datum: at 1.6 m its drainage share is 73 % at C1, 54 % at C2, 46 % at C3, 24 % at C5 and 4 % at C4. Below that depth the split settles toward the Darcy behaviour of a drainage base well below the fluctuating water table. At 3.7 m the drainage flux is 0.81 to 0.97 of its value at the deepest datum swept (8 m), and Model A's drainage share is within five points of its deep value at every cluster: 85 against 88 % at C1, 66 against 69 % at C2, 65 against 68 % at C3, 61 against 66 % at C5 and 25 against 29 % at C4. Model A's shares are quoted at 3.7 m on that understanding; the headline shares are Model B's (38 % at C1, 45 % at C2, 56 % at C3, 61 % at C5 and 29 % at C4).

A datum fixed in elevation instead of depth, each well draining to a common level above Ordnance Datum, was also tested. It can be estimated, with an optimum near +2.5 m AOD,

but it fits worse than a depth below ground at every level tried. By +6 m AOD it returns a non-positive $\beta_3$ at 18 of 79 wells, against one at the depth datum. The per-well optimum datums follow the ground surface rather than a common level. The record therefore points to a drainage base that follows the topography, each slack draining locally, and not to a regional base.

A physically anchored datum that follows the topography was then tested directly. Each well was given its own datum at its depth to mean high water (1.88 m OD), a median of 6.7 m below ground and between 1.7 and 12.5 m across the reference wells. It reproduces the water-balance partition of the 3.7 m datum: a median drainage share of 68 % against 68 %, with the spread across wells almost unchanged (43–78 % against 39–79 %, 10th to 90th percentile). Nearly every well's depth to mean high water lies in the Darcy regime, below the 1.6 m boundary, so the partition does not come from the choice of a single depth. The physical datum is nonetheless the worse model. Because Model A has no intercept, the deeper datum passes into $\beta_3$: the e-folding time of a departure roughly doubles, to 29 months against 16 at 3.7 m and 5.1 observed, and the out-of-sample efficiency falls to 0.47 forward and 0.53 in reverse, against 0.61 and 0.67. It is better than 3.7 m at fewer than one well in five, and worst at C4 (efficiency −0.6). The 3.7 m datum is therefore kept: a physically anchored datum gives the same water balance and a poorer fit to the record.

The intercept form (Model B) leaves the level free. Its $\beta_3$ is then the rate at which the water table returns to its own mean, independent of any datum. That is the mean-reversion time, the e-folding time $-1/\ln(1 - \beta_3)$ of a departure of the water table from its mean, and it sets two quantities: how far one year's level carries into the next, and in a one-reservoir model, how far a climate shift held over many years moves the level. Model B is used for the first and for forecasting within the record, for four reasons. For a sustained climate shift it is reported beside Model A as the fast-reversion bound, never alone: the one independent test of a sustained shift, the 1989–96 record (Section 4.14), favours Model A, which hindcast from fits to the full modern record is close to unbiased (median bias +0.03 m) where Model B is too wet by 0.30 m (Script 50). Climate projections are therefore given as a pair on the same wells, Model A at the 3.7 m datum with Model B beside it (Section 3.7.5).

(1) Model A's $\beta_3$ carries the datum. Its recession time changes with the datum chosen, so it is the datum's, not the aquifer's.

(2) The mean-reversion time can be measured without any model. The lag-1 autocorrelation of the observed deseasonalized monthly level gives a median e-folding time of 5.1 months across the reference wells. Model B gives 6.5 and Model A 15.8. Cluster by cluster, Model B sits with the observed value: 2.1 against 2.0 months at C1, 4.4 against 4.1 at C2, 7.4 against 6.3 at C3 and 13.3 against 12.0 at C5. Model A is three to five times longer.

(3) The year-to-year correlation of spring level is −0.11 (median across wells). Model B predicts 0.15; Model A predicts 0.44.

(4) Out of sample, Model B forecasts better. Each form was fitted on the record before 2018 and run freely over the years after. Model B's median Nash–Sutcliffe efficiency is 0.79 against Model A's 0.61, and Model B is better at 88 % of wells. In the reverse direction the figures are 0.78 against 0.67, and 77 % of wells. The advantage is clearest in the open dune (C1 to C3); at C5 the two forms are close.

Model B has a matching weakness. Its intercept and $\beta_3$ are almost interchangeable: their estimates correlate at 0.96 at the median well. The level its drainage refers to is therefore fitted rather than physical. It lies near each well's mean level, a median 1.6 m below ground, which is the shallow, mean-reversion end of the datum range above. With it, more of the mean loss is booked to atmospheric draw (a drainage share of 45 % against Model A's 68 %).

This is why Model B is not used to choose the datum. Adopting its fitted base as the datum would make Model A identical to Model B, and that base is the level the water table returns to, not where drainage stops.

Neither form identifies the absolute split between evapotranspiration and drainage from heads alone. Model A gives it on a physical assumption about the drainage base; Model B gives it on a fitted mean-reversion level. Both give the same ranking of clusters and the same dynamics. At C4 neither form is reliable: its Model B coefficient is not identified on the record, and both forms forecast poorly out of sample there (Nash–Sutcliffe efficiency about 0.1 to 0.2). C4 is not projected.

**Physical interpretation of $\beta_3$.** The storage decay coefficient provides a parsimonious empirical representation of head-dependent lateral drainage that does not require explicit knowledge of hydraulic conductivity, aquifer geometry or hydraulic gradient — parameters that are either unmeasured or spatially variable across a 700 ha dune system (Stratford et al., 2007) where borehole logs are absent across the majority of the study area. $\beta_3$ is functionally equivalent to a linearized Darcy flux under the assumption that the hydraulic gradient scales proportionally with water table height above the drainage base, which is a reasonable approximation for a shallow unconfined dune aquifer draining to a fixed boundary (Knotters and van Walsum, 1997).

**Lumped Catchment Storage Coefficient.** The LCSC is defined as the reciprocal of the recharge sensitivity, expressed as a percentage (LCSC = $100/\beta_1$). Expressing it as LCSC% converts the recharge sensitivity into physically intuitive units — the millimetres of rainfall required to raise the water table by 10 cm — and scales in the same direction as storage capacity, making it directly comparable to the empirical specific yield concept used by site managers. A higher LCSC therefore indicates a more buffered, less responsive aquifer: a shallow aquifer over impermeable till produces a large water table rise from a small rainfall event, a high $\beta_1$, and a low LCSC, whilst a deep sand aquifer absorbs rain across a thick unsaturated zone before the water table rises appreciably, producing a low $\beta_1$ and a high

LCSC. *Cluster-level SSM fits are implemented in* 03_state_space_model.py. *The five cluster-mean hydrographs are constructed and exported by the same script and become the input to all downstream cluster-scale analyses.*

### 3.4.2 State-Space vs. Traditional Linear Model Benchmark

**Model Configurations.** Both models were independently fitted to the most recent 100 months of data to predict the monthly change in water table elevation (Δh_t) using the same climatic forcing data, differing only in their structural representation of the aquifer:

1. **Traditional Linear Model (TLM)** treats the aquifer as a passive linear store driven purely by atmospheric inputs, with a fitted intercept term capturing background drift and unmeasured forcing, but no mechanism representing the aquifer's dependence on its own previous state (von Asmuth et al., 2002):

$$\Delta h_t = \beta_1 \cdot P_t - \beta_2 \cdot PET_t + c$$

2. **State-Space Model (SSM):** the physically constrained mass-balance framework defined in Section 3.4, fitted through the origin (c = 0), with antecedent displacement h_disp,t−1 included as a feedback term that causes drainage to accelerate when the water table is high and slow when it is low:

$$\Delta h_t = \beta_1 \cdot P_t - \beta_2 \cdot PET_t - \beta_3 \cdot h_{disp,t-1}$$

**Forecasting modes: diagnostic vs iterative.** Model performance was evaluated under two distinct operational modes to separate baseline statistical fit from true forecasting stability:

- **One-Step Diagnostic Fit:** The model predicts the change in head for month **t** using the *observed* field measurement of h(t−1) from the prior month. This tests the fundamental validity of the regression coefficients without compounding error.
- **Iterative Forecasting Stability:** The model predicts the change in head for month **t** using its own simulated head from the previous timestep (h_sim), relying entirely on its own simulated state over the 100-month evaluation window without periodic data assimilation; this tests resistance to cumulative drift over multi-year autonomous simulation.

**1) Performance Metrics.** Diagnostic fit (one-step) was evaluated using the Coefficient of Determination ($R^2$). Iterative forecasting stability was evaluated using $R^2$, the Root Mean Square Error (RMSE), and the Nash–Sutcliffe Efficiency (NSE) (Nash and Sutcliffe, 1970; Krause et al., 2005).

The coefficient of determination is the squared Pearson correlation between observed and simulated head:

$$R^2 = \left( \frac{\sum \left( h_{obs,t} - \bar{h}_{obs} \right)\left( h_{\mathrm{sim},t} - \bar{h}_{\mathrm{sim}} \right)}{\sqrt{\sum \left( h_{obs,t} - \bar{h}_{obs} \right)^2 \sum \left( h_{\mathrm{sim},t} - \bar{h}_{\mathrm{sim}} \right)^2}} \right)^2$$

The RMSE is mathematically defined as:

$$RMSE = \sqrt{\frac{1}{N} \sum \left( h_{obs,t} - h_{\mathrm{sim},t} \right)^2}$$

The NSE is mathematically defined as:

$$NSE = 1 - \frac{\sum \left( h_{obs,t} - h_{\mathrm{sim},t} \right)^2}{\sum \left( h_{obs,t} - \bar{h}_{obs} \right)^2}$$

where h_obs,t and h_sim,t are the observed and model-simulated water-table levels in month t; $\bar{h}$_obs and $\bar{h}$_sim are their means over the N-month evaluation window; and N is the number of months in that window (100). RMSE is reported in the units of head (m) and is zero for a perfect forecast; $R^2$ and NSE are dimensionless. NSE is the most informative metric for continuous hydrological forecasting: NSE = 1 indicates perfect predictive agreement, NSE = 0 indicates the model is only as accurate as the long-term observed mean, and NSE < 0 indicates the model performs worse than that mean.

**2) Spatial Diagnostics of Subsurface Architecture using SSM and TLM comparison.**
The comparative performance metrics are the improvement in variance explanation:

$$\Delta R^2 = R^2_{SSM} - R^2_{TLM}$$

and the improvement in iterative forecasting stability:

$$\Delta NSE = NSE_{SSM} - NSE_{TLM}$$

where the SSM and TLM subscripts denote the metric computed for the state-space model and the traditional linear model respectively, both fitted to the same 100-month record. A positive $\Delta R^2$ or ΔNSE marks a location where adding the head-dependent drainage term improves the forecast. ΔNSE serves as more than a statistical validation; it acts as a spatial diagnostic tool for the underlying geological architecture (Beven, 2012; Gupta et al., 2008). Similar multi-method approaches have been used to constrain recharge in shallow sandy aquifers, combining unsaturated-zone modelling with the water-table fluctuation method so that two independent estimates bound the same quantity (Gumuła-Kawęcka et al., 2021). Because the fundamental mathematical difference between the two models is the inclusion of the head-dependent drainage term ($-\beta_3 \cdot$h_disp,t−1), the spatial distribution of model improvement suggests where the subsurface structural boundaries of the aquifer lie:

- **High $\Delta R^2$ and ΔNSE zones:** Regions where the SSM most substantially outperforms the TLM are those where drainage is most strongly head-dependent at the monthly timescale (Fetter, 2001; Robins, Jones and Farr, 2013). In these zones the

proportional drainage rate varies considerably between high and low water table states, making the $-\beta_3$ term essential for accurate autonomous simulation. The Western Block's deep sand buffer moderates this dynamic, and canopy interception in the Forest cluster smooths it further, producing a gradient of decreasing SSM gain from east to west that mirrors the structural gradient identified by clustering.

- **Low $\Delta R^2$ and $\Delta$NSE zones:** Conversely, areas exhibiting small or marginally negative improvement between the two models reflect zones where drainage is least strongly head-dependent. A small or zero SSM gain in these zones does not indicate model failure but is consistent with the TLM's linear approximation being adequate where drainage dynamics are sufficiently buffered or smoothed (Gumuła-Kawęcka et al., 2021). The wells where the SSM scores below the TLM are identified in Section 4.4.

By mapping the success or failure of the storage decay coefficient $-\beta_3$, this comparative methodology indicates where structural boundaries between deep sand reserves, perched slacks and complex discharge zones may lie, without invasive coring; the substrate is uncored (Section 3.4.4), so these boundaries are inferred from model skill, not observed. *SSM versus traditional linear model benchmarking across all 66 reference wells was implemented in 08_model_benchmarking.py.*

### 3.4.3 Spatial Coefficient Mapping

To visualize the spatial structure of aquifer behaviour across the site, per-well SSM coefficients ($\beta_1$, $\beta_2$, $\beta_3$) were interpolated onto a regular 50 m grid (model fit quality, $R^2$, is shown at the wells alone, having no spatial structure a surface could carry) using piecewise-linear (Delaunay) interpolation between the wells and rendered over the 2 m LiDAR DEM hillshade. The per-well coefficients are derived from the displacement-formulation SSM (Section 3.4) fitted independently at each of the 66 reference wells, using the same canonical drainage datum as the cluster-centroid model. The $\beta_3$ drainage surface was plotted on a logarithmic scale to resolve the order-of-magnitude contrast between the fast-draining lake-edge wells and the near-stagnant forest interior. The resulting coefficient atlas (Figure 54a–d) shows model fit quality at the wells and maps recharge sensitivity, atmospheric draw and drainage rate across the full dune field, providing a spatially continuous expression of the mechanistic contrasts identified at cluster level in Table 3. *Spatial coefficient mapping was implemented in 07_spatial_coefficients.py.*

**Summary of SSM fitting windows.** Four distinct windows are used across the study. The cluster-centroid SSM (Section 3.4.1) is fitted to the full available record of each cluster's mean hydrograph, supplying the cluster $\beta_1/\beta_2/\beta_3$ values used in the P_flood threshold equations (Section 3.6.3). The per-well SSM is fitted to the most recent 100 months at each reference well; these per-well coefficients (written to 03_master_data.csv) underpin both the SSM-versus-TLM benchmark (Section 3.4.2) and the spatial coefficient atlas and its

interpolated surfaces (Section 3.4.3), which fits nothing itself and interpolates the per-well store. The BACI per-well SSM (Section 3.5.4) is fitted independently to the pre- and post-felling windows at each BACI-network well, supplying the era-specific $\Delta\beta_1/\Delta\beta_2/\Delta\beta_3$ decomposition (Section 4.6.6). The per-well recession constants and drainage half-lives (Section 3.4.5) are fitted on each well's full record (03_19_per_well_recession_full_record.csv).

### 3.4.4 Within-Forest Spatial Coefficient Analysis

To test whether within-forest coefficient variation tracks topographic position (a proxy for the uncored substrate) or canopy differences, per-well SSM coefficients for the 14 forest zone wells (9 in C4, 5 in C5) were regressed against three spatial predictors: DEM ground elevation, Euclidean distance from the ridge crest (E = 241,750, N = 364,500), and Easting as a proxy for distance from the south-eastern coast (Menai Strait). Single-predictor Pearson correlations were computed for each coefficient against each spatial variable, and multiple regression was used to test whether distance from the ridge added explanatory power beyond elevation alone. The C4/C5 boundary was assessed for continuity or discontinuity using t-tests on elevation, $\beta_1$, $\beta_2$ and $\beta_3$ between the two clusters. *Results are reported in Section 4.9.4 (Table 23). The analysis is implemented in 10c_forest_zone_analysis.py.*

### 3.4.5 Drainage Timescales

The drainage coefficient $\beta_3$ (Section 3.4.1) determines how rapidly the aquifer sheds excess head above the drainage datum. Its reciprocal $t_R = 1/\beta_3$ is the recession e-folding time: the time for excess displacement above the datum to decay to 37% of its initial value through drainage alone. A more intuitive expression of the same timescale is the drainage decay half-life

$$t_{\frac{1}{2}} = \frac{\ln 2}{\beta_3}$$

which is the time for excess groundwater storage above the drainage datum to drain to half its initial value. $t_{\frac{1}{2}} = 0.693 \times t_R$ and is expressed in months, the natural timestep of the monthly state-space model. A well with a short $t_{\frac{1}{2}}$ drains its stored displacement within a season; a well with a long $t_{\frac{1}{2}}$ retains excess storage across years. The per-well recession constants are fitted on each well's full record rather than the 100-month comparison window: a recession comparable with the window cannot be seen in it, and on the window none of the nine Main Forest wells identifies a response time while six do on the full record (Section 3.4). The window remains the basis on which wells are compared and ranked. The half-life is mapped across the network as a spatial field; its construction, and the two wells withheld from it, are in Section 3.8.3.

The storage-drainage index $\tau = Sy/\beta_3$ (where Sy is WTF-derived specific yield, Section 3.7.3) is a related composite that weights t_R by storage capacity. τ has the same units as t½ but is not the same quantity: two wells with identical t½ can have different τ if their Sy values differ. Where Sy varies little between wells relative to $1/\beta_3$, τ and t½ map alike; how closely they agree across the network is reported in Section 4.9.3. The physical interpretation of τ as the ratio of stored volume to drainage flux is developed in Section 5.3.2, where it provides a conceptual bridge between the drainage half-life and the volumetric water balance. Two cautions attach to that reading. τ has units of time and is a storage–drainage index, not a residence time: it is not the interval over which the aquifer replaces its water, and no result in this report is derived from reading it as one. And because $\beta_3$ carries the drainage datum under the no-intercept specification (Section 3.4.1), τ inherits that dependence — it is defined at the project datum, as t½ is, and is not comparable across studies adopting a different one.

$$\Delta NSE = NSE_{SSM} - NSE_{TLM}$$

Plotting t½ against ΔNSE (defined in Section 3.4.2) at each well, with marker area scaled to that well's specific yield, places three independently derived descriptors — drainage decay half-life (t½), drainage sensitivity (ΔNSE) and storage capacity (Sy) — in a single diagnostic space, with cluster means overlaid as summary markers. Because the three descriptors are drawn from different parts of the analysis — the water-balance fit, the model benchmarking, and the rising-limb storage estimate respectively — their joint structure provides a characterization of aquifer architecture that is independent of the hydrograph clustering of Section 3.2. The synthesis is presented in Section 4.9.3. *The storage–drainage index field and the aquifer diagnostic synthesis are implemented in 18_wtf_spatial.py.*

## 3.5 Anthropogenic Interventions

### 3.5.1 The Before-After Control-Impact (BACI) Design

To evaluate the hydrological outcomes of anthropogenic interventions at the study site, a Before-After Control-Impact (BACI) experimental design was used (Stewart-Oaten et al., 1986). This framework is designed to isolate the effect of a specific localized disturbance from broader natural variability, such as seasonal weather patterns or regional climatic drift.

The fundamental mechanism of the BACI approach involves contrasting an impact zone (where the intervention occurred) with a reference control zone (which remained undisturbed) across designated pre-intervention and post-intervention periods. Rather than simply measuring absolute changes in water levels, the model calculates the difference (Δ) between the impact and control sites. The actual effect of the intervention is subsequently quantified by measuring the shift in this difference after the event.

For highly complex environments where multiple interventions overlap, the standard BACI method was developed into a Hierarchical Nested Control design to ensure robust baseline validation. By using multiple control sites to defend against localized hydrogeological anomalies, this study employs a "Beyond BACI" approach (Underwood, 1992).

**General Before-After Control-Impact (BACI) Calculation.** The standard BACI effect size isolates the intervention signal from natural background variation. The fundamental equation calculates the difference in means (μ) between the Impact (I) and Control (C) sites across the Before (B) and After (A) periods:

$$BACI_{\text{effect}} = \left(\mu_{I,A} - \mu_{C,A}\right) - \left(\mu_{I,B} - \mu_{C,B}\right)$$

Where:

- $\mu_{I,A}$ is the mean water level at the impact site post-intervention.
- $\mu_{C,A}$ is the mean water level at the control site post-intervention.
- $\mu_{I,B}$ and $\mu_{C,B}$ represent the respective means prior to the intervention.

**Cumulative Sum (CUSUM) Statistical Verification.** To locate the temporal onset of hydrological shifts identified by the BACI framework, and to test their permanence, a Cumulative Sum (CUSUM) sequential analysis was integrated into the analytical framework (Page, 1954). While the BACI contrast quantifies the magnitude of an intervention's effect, the CUSUM statistic is used to identify the exact temporal structural break — or regime shift — within the time series data. Era-specific estimation of the storage decay coefficient $-\beta_3$ is additionally reported across treatment and control wells; changes that are confined to the intervention footprint would be consistent with a genuine drainage response, whilst changes extending uniformly to unmanipulated control wells indicate a network-wide signal attributable to climate or SSM fitting artefacts rather than the intervention itself.

**Mathematical Framework.** The CUSUM method operates by calculating a running total of the deviations of individual data points from an established baseline mean. In the context of this study, the input variables (x_i) are the sequential differences between the impact and control wells (Δ), and the baseline (μ_baseline) is the mean of these differences during the pre-intervention period. The cumulative sum at time t is formalized as:

$$C_t = \sum_{i=1}^{t} \left(x_i - \mu_{\text{baseline}}\right)$$

**Interpretation of the CUSUM Trajectory.** The resulting CUSUM trajectory provides a clear visualization of system stability:

- **Baseline Stability:** A stable relationship between the impact and control sites yields random fluctuations around the baseline mean, producing a roughly horizontal CUSUM trajectory.

- **Structural Break Detection:** A sustained shift in the hydrological relationship — such as that caused by canopy removal or slack scraping — results in directional deviations that accumulate rapidly, manifesting as a distinct change in the gradient of the CUSUM line.
- **Temporal Verification:** The exact point of inflection, where the trajectory breaks from its horizontal trend, marks the initiation of the intervention's physical effect.

By aligning these inflection points with the documented dates of the December 2017 clearfell and the 2015 and 2023 scraping interventions (the February 2013 cuts at CEH40/41/42 are unmonitored and carry no inflection to align), the analysis indicates that the observed divergence follows the site works rather than gradual environmental drift. To isolate the localized effects of topographical and silvicultural interventions from background climatic drift, all CUSUM trajectories were calculated on the BACI delta (impact minus control) rather than raw water table depth. This ensures that any identified structural breaks represent a true divergence from the undisturbed regional baseline.

Where an impact site has experienced more than one intervention over the monitoring period, or where a secondary disturbance confounds the primary signal, the record is subdivided into discrete management eras defined by the dates of known interventions. Each era is treated as a distinct analytical period with its own mean water level, inter-era BACI delta and $-\beta_3$ estimate. This allows the analysis to separate the signal of a specific intervention from preceding baseline conditions and from subsequent confounding events — for example, isolating the effect of dune scraping before the onset of a plantation clearance pulse that would otherwise inflate or mask the scraping response. Era boundaries are defined a priori from documented intervention dates rather than from the data, preventing circular reasoning in the identification of structural breaks.

### 3.5.2 Hierarchical Nested Control Application

This framework operates across two distinct analytical tiers:

- **Tier 1 (Background Drift Validation):** Local control wells are first evaluated against a broader regional mean. This step verifies that the local control (C_local) accurately tracks the broader regional mean (C_region) and is not subject to anomalous localized influences, such as unmapped coastal drawdown. The temporal difference (Δ_drift) at any time t is calculated as:

  $$\Delta_{\text{drift},t} = C_{\text{local},t} - C_{\text{region},t}$$

  (where C_local,t is the water-table depth at the local control well at time t and C_region,t is the regional mean across the C3 climate control wells at the same time. A near-zero, stable Δ_drift,t across the pre-intervention baseline is consistent with the local control tracking regional conditions rather than being subject to localized disturbance.)

A structurally stable Δ_drift,t over the baseline timeline qualifies the local control for use in Tier 2.

- **Tier 2 (Intervention Signal Isolation):** Once validated, these local controls are paired directly with specific impact wells. This pairing subtracts the regional noise and secondary intervention pulses, isolating the pure signal of the targeted experiment. The Tier 2 BACI effect calculates the net shift in the impact-minus-control gap between any two consecutive management eras:

$$\text{Net Benefit}_{E1 \to E2} = \left(\mu_{I,E2} - \mu_{C_{\text{local}},E2}\right) - \left(\mu_{I,E1} - \mu_{C_{\text{local}},E1}\right)$$

(where I is the impact well; C_local is the validated local control well; E1 is the earlier era (e.g. pre-intervention baseline) and E2 is the later era (e.g. post-intervention); and μ_{I,E} denotes the mean water-table depth at well I during era E. The outer subtraction removes whatever gap existed between impact and control before the intervention, leaving only the change in that gap — the net hydrological effect of moving from era E1 to era E2. Because C_local has been validated in Tier 1 to track the regional mean, this difference removes regional climatic noise and any secondary intervention pulses to the extent that they affect both wells equally.)

### 3.5.3 Dune Scraping Intervention Analysis

A two-tier hierarchical BACI framework was applied across five focal wells. Three wells served as scraping treatment sites — CEH36 (central Warren, scraped April 2015), CEH18 (western boundary, scraped October 2023) and CEH21 (coastal margin, scraped October 2023) — and two as local controls: CEH4 (central Warren, unmanipulated) and CEH22 (coastal margin, unmanipulated). A strip of foreshore pine about 120 m from CEH4 and 200 m from CEH36 was cleared in the same winter (between October 2014 and March 2015, dated from Landsat), so the scrape comparison carries that clearance as a confound. A regional mean was computed as the arithmetic mean of five C3 climate control wells (CEH9, NW7, NW6, NW5, WMC2) to characterize background system-wide trends. NW8 and NW8B were excluded on the grounds of data quality.

The scraping programme extended beyond these three monitored sites. Two further slacks in the western dunes (Scrape A and Scrape B) were scraped in the same 2015 campaign as CEH36 but carry no monitoring dipwells, and three sites in the eastern dunes (CEH40, CEH41 and CEH42) were scraped earlier, in February 2013, with their dipwells first recording in July 2014 — seventeen months after the works. None of these five provides a pre-scraping baseline, so the intervention analysis is necessarily restricted to CEH36, CEH18 and CEH21, the sites at which a before-and-after comparison is possible. The unmonitored and late-instrumented scrapes are represented in the combined spatial driver reconstruction

(Section 3.8.1) but cannot be evaluated for benefit; where that reconstruction requires a per-scrape response for them, the measured CEH36 response is assumed, and this assumption is stated in the relevant figure captions.

Because the three treatment wells span different management histories, each was decomposed into site-specific eras defined by intervention dates:

- **CEH36 and its paired control CEH4** were decomposed into three eras: Baseline (pre-April 2015); Pure Scraping (April 2015 to November 2017); Felling Pulse (December 2017 onwards), capturing the confounding influence of the plantation clearance.

- **CEH18 and CEH21** were decomposed into three eras: Baseline (pre-December 2017); Felling Pulse (December 2017 to October 2023); After Scraping (October 2023 onwards).

Five limitations of the control wells and treatment records bear directly on interpretation and are stated here to avoid repetition in the results.

First, the water table at CEH4 has deepened across the record, with no change in its ground surface. CEH4 lies about 90 m nearer the eroding Caernarfon Bay shore than its paired treatment well CEH36 (221 m against 313 m), and its deepening is consistent with the retreating shoreline together with the warren-wide decline; the record cannot separate the two, and no felling pulse is resolved at CEH4. Because the control sits nearer the eroding shore than the treatment, this coastal-retreat drawdown bears more heavily on CEH4 than on CEH36. Net benefit estimates at CEH36 during the Pure Scraping era should be treated as upper bounds; an additional upward bias may arise from contemporaneous foredune build-up at the site contributing to a rising water table independently of the scraping intervention. Net benefit estimates at CEH18 over the After Scraping era are similarly subject to caution given CEH4's continuing deterioration. The robustness of the CEH36 scraping signal is tested independently in part 3) below using a synthetic control constructed from donor wells calibrated on the pre-scraping period and by SSM forward residual analysis.

Second, CEH22 experienced accelerating water-table deterioration throughout the monitoring period — again a water-table effect, with no change in ground surface. This deterioration has two components. A system-wide component, evident in the network-wide $-\beta_3$ analysis, is common to the regional control mean and is therefore largely removed in the BACI differencing. A local coastal-retreat component is not: CEH22 is the most coastal of the focal wells (147 m from the eroding shore, some 85 m nearer than its paired treatment CEH21), so the retreating shoreline has drawn its water table down independently of the regional trend. It is this residual local component that prevents CEH22 from serving as a stable Tier 1 baseline; net benefits are therefore not reported for CEH21, and raw Tier 2 BACI shifts are reported for completeness only.

Third, CEH18 and CEH21 were scraped in October 2023, providing only two full post-scraping summers by the end of 2025. The short post-intervention record limits the statistical power of pre- vs post-scraping comparisons at these wells and precludes any conclusion about whether the scraping benefit observed at CEH36 over a longer timeframe will be replicated at the secondary sites.

Fourth, the pre-scraping baseline length differs substantially across the focal wells: CEH4 (paired control, installed May 2006) provides 8.9 years; CEH36 (central treatment well, installed January 2011) provides 4.2 years; CEH18 (installed September 2007) provides 7.7 years pre-2015; and CEH21 (installed July 2010) provides 4.8 years pre-2015. CEH36's shorter pre-scraping baseline does not invalidate the analysis: the scraping methodology does not require estimation of a long-baseline parametric covariate structure (in contrast to the clearfell ANCOVA, Section 3.5.4, which estimates a cumulative-water-balance slope from pre-felling data). CUSUM, Welch's t-test on era means, synthetic-control calibration and SSM forward residual all operate effectively with the available pre-event window, and the robustness triangulation in part 3) below tests the resulting BACI step against three independent methods to test for convergence.

Fifth, the scraping analysis treats the coastal-retreat drawdown of CEH4's water table as a confound documented in the era-by-era comparisons rather than as a covariate explicitly modelled in the BACI design. CEH4 sits approximately 221 m from the eroding south-western Caernarfon Bay shore, and the characteristic propagation timescale ($L_B^2Sy/(Kb)$, Section 5.4.3) for coastal-erosion effects to reach inland wells through the aquifer is multi-year to multi-decadal. At this distance a substantial part of the coastal-erosion propagation envelope falls within the analysis window, so the visible deterioration at CEH4 is consistent with a coastal-retreat drawdown — the integrated effect of historical retreat together with its continued propagation over the record — acting together with the warren-wide decline; no felling pulse is resolved at CEH4. The Tier 1 CUSUM analysis at CEH4 captures this combined drift but cannot decompose it into its component drivers; the synthetic control and SSM forward residual analyses below provide the necessary alternative counterfactuals.

### 1) Paired-control BACI design and analytical sub-procedures

Four analytical procedures are applied within the paired-control BACI framework. Each addresses a distinct research question; results are cross-referenced to the relevant sub-sections of Section 4.5.

**Paired BACI and net benefit estimation.** Tier 1 CUSUM analysis evaluated each local control well against the regional mean to verify that the controls were tracking system-wide background trends consistently. Tier 2 paired CUSUM analysis differenced each treatment well against its matched local control to isolate the scraping signal from regional climatic drift. Net benefit was computed as the BACI shift in the treatment-minus-control difference

between consecutive eras, as defined in Section 3.5.2. Results are presented in Section 4.5.1 (background drift) and Section 4.5.2 (paired BACI shifts).

**Scraping propagation analysis.** A separate analysis tested whether the CEH36 scraping event — which created a topographic low acting as a local drain — produced a detectable drawdown at wells uphill of the scraped site. Groundwater flow toward the new drain would steepen the local hydraulic gradient, lowering the water table at neighbouring wells. The displacement-formulation SSM was fitted independently to pre-scraping (start of record to March 2015) and post-scraping (April 2015 to November 2017) windows for each of 10 wells located north and north-west of CEH36, positioned along the expected groundwater flow path into the forest interior. Seven distant C3 wells (NW1, NW2, NW11, NW13, WMC4, D25, WMC2), located 850–1070 m from CEH36 and well beyond any expected scraping influence, served as BACI controls. Per-well coefficient shifts ($\Delta\beta_1$, $\Delta\beta_2$, $\Delta\beta_3$) were corrected by subtracting the control centroid shift, isolating any scraping-specific perturbation from background climate drift. The BACI-corrected $\Delta\beta_3$ was examined as a function of distance from CEH36, to see whether it diminished with distance as would be expected if the scraping drain were pulling water from the uphill aquifer. Results are presented in Section 4.5.5.

**Summer minima analysis.** The annual summer minimum depth (June–September) was computed for each of the five focal wells. For each well, the gap between the well's summer minimum and both the climate control centroid and the paired control summer minimum was computed annually, and the pre- vs post-scraping shift tested by Welch's t-test. This mirrors the methodology applied to the clearfell intervention (Section 3.5.4) and addresses the ecological question directly: does scraping improve the critical summer low that determines dune slack habitat viability? Results are presented in Section 4.5.4.

**Scenario comparison.** To contextualize the observed scraping benefit at CEH36, five alternative scenarios were computed at the same well using its own SSM coefficients and its own per-well WTF-derived specific yield. Three forestry scenarios — clearfell, 50% thinning, and broadleaf conversion — are hypothetical, asking what the response would be if CEH36 had pine canopy and was then managed. These keep the pine-canopy rainfall, with no interception recharge term because the felled well’s recharge did not rise after felling (Section 4.13.2), and apply the change in atmospheric draw measured at the felled well (Section 3.5.4) to CEH36's SSM. Two climate scenarios apply the UKCP18 RCP8.5 dry and wet 2050s multipliers, not the central estimate of Section 3.8.4. The equilibrium volumetric water table change (mm water equivalent per month) was computed from the SSM equilibrium response under two forcings: annual-mean rainfall and PET, and summer (July–September) rainfall and PET. Both figures report the same volumetric quantity, so all scenarios are directly comparable; the summer figure is the equilibrium response to summer forcing and not a summer minimum, which the equilibrium framework cannot resolve. For scraping, the on-site bar is the observed paired-BACI response converted to a volumetric

equivalent, and a second bar gives the modelled off-site neighbour drawdown at 100 m (with a 250 m reference line) from the drain-cone model of Section 4.11. Results are presented in Section 4.5.6. *These four sub-procedures are implemented in 09a_paired_baci.py, 09b_scraping_propagation.py, 09c_summer_minima.py and 09d_scenario_comparison.py respectively.*

**2) Era-specific $\beta_3$ analysis on the drainage residual**

Beyond the per-era CUSUM and BACI shift tests, the scraping intervention may have altered the storage decay coefficient $\beta_3$ at the treatment wells. To diagnose this, a two-step procedure is applied separately at each treatment well and its paired control. The procedure isolates per-era variation in $\beta_3$ from the confounding influence of inter-era differences in climate forcing.

First, $\beta_1$ and $\beta_2$ are estimated by fitting the full displacement-formulation SSM (Section 3.4) to the complete well record. The climate forcing contribution is then removed from each monthly observation to yield a drainage residual series, representing the component of water table change attributable to head-dependent drainage after removing the estimated rainfall and evapotranspiration contributions:

$$D_{residual} = \Delta h_t - \beta_1 \cdot P_t - \beta_2 \cdot \left(-PET_t\right)$$

Second, for each management era separately, the era-specific drainage coefficient is estimated by regressing this drainage residual against minus the displacement at the end of the previous month, with a constant term:

$$D_{residual} = -\beta_3 \cdot \left(-h_{disp,t-1}\right) + c + \epsilon$$

with 95% confidence intervals computed by OLS on each era's data subset. Era-specific $\beta_3$ values are compared across the Baseline, Pure Scraping (or Felling Pulse) and where applicable After Scraping eras, with non-overlapping confidence intervals interpreted as evidence of an era-specific shift in drainage behaviour.

This procedure reduces the influence of differential climate forcing between eras, but does not eliminate all sources of artefact: where the water table undergoes a sustained directional shift between eras — as occurs following both clearfell and scraping — the h_disp,prev term itself tracks a non-stationary trajectory, and the model may interpret systematic head changes as an elevated drainage signal. Era-specific $\beta_3$ estimates should therefore be interpreted cautiously, and spatial extent is used as the primary diagnostic: changes confined to the intervention footprint are consistent with a genuine drainage response, whilst changes extending uniformly to unmanipulated control wells point to a network-wide artefact rather than an intervention effect. Results are presented in Section 4.5.3. *This procedure is implemented in 09a_paired_baci.py as part of the paired-control BACI workflow.*

**3) Robustness triangulation of the CEH36 step**

The CEH36 Pure Scraping era step change is the central scraping result. Because the water table at the paired control well CEH4 has itself deepened — consistent with the retreating shoreline (CEH4 sits about 90 m nearer the coast than CEH36) together with the warren-wide decline, which the record cannot separate — the magnitude of the CEH36 step estimated by paired BACI alone may carry a CEH4-specific bias. Three independent estimation methods are applied at CEH36 to test whether the step is robust under alternative counterfactuals.

The first method is the **paired BACI** estimate itself, set out in part 1) above and presented here for comparison.

The second method is a **synthetic control** estimate. A weighted composite of donor wells, drawn from the C3 climate control set and from other reference-network wells with strong pre-scraping correlation to CEH36, is constructed to match CEH36's pre-scraping trajectory. The donor weights are estimated on the pre-scraping period only, ensuring that the synthetic control is calibrated independently of the post-intervention data it is then used to evaluate. The scraping step is computed as the difference between observed CEH36 and the synthetic-control prediction over the Pure Scraping era.

The third method is an **SSM forward residual** estimate. The displacement-formulation SSM is fitted to the CEH36 pre-scraping record alone, and the fitted model is then forward-simulated through the Pure Scraping era using observed climate forcing as input. The simulated water table represents the predicted CEH36 trajectory in the counterfactual where no scraping had occurred. The scraping step is computed as the difference between observed and predicted CEH36 over the Pure Scraping era.

Convergence between the three methods — paired BACI, synthetic control, and SSM forward residual — supports reading the estimated step as an intervention effect rather than an artefact of CEH4's own deterioration. The three estimates and their convergence are reported in Section 4.5.2. *This triangulation is implemented in 09e_robustness.py.*

### 3.5.4 Clearfell BACI experiment

A five-tier hierarchical BACI experiment was conducted across 17 wells to assess the hydrological consequences of the December 2017 clearfell. The dates of the three management interventions used throughout this analysis — the April 2015 and October 2023 dune scrapes and the December 2017 clearfell — are the author's recollection, and are used here as given; config.py additionally records CEH36 as the documented April 2015 dune-scrape site. The clearfell date is corroborated independently of any written record by the dated aerial series (Section 4.6.8): the canopy texture index across the felled compartment falls from 1.243 in the frame of 22 April 2017 to 0.130 in that of 28 June 2018, while the unfelled conifer control holds at 1.005. The scrape sites are identifiable in the same imagery, so the scraping dates are bracketable in principle by the same means; the frames currently registered do not do it, because April 2015 falls inside a gap in the series between May 2012

and March 2017, and October 2023 is bracketed only between April 2021 and March 2026. Narrowing those windows would require frames not yet in the series, and a documentary cross-check against Natural Resources Wales or forestry operations records would settle them outright; both are identified as outstanding work rather than claimed here. Wells were assigned to tiers defined a priori from spatial proximity to the felled compartment and cluster membership, with all wells required to have a minimum of seven years' pre-felling baseline to ensure stable coefficient estimation:

- **Impact:** WMC3 (45 m from felling compartment centroid E=241210, N=363607, inside the cleared area; 8.4 years pre-felling record).
- **Edge:** CEH31, CEH20, CEH30, CEH16 (152–229 m, adjacent to the felling boundary; 7.4–10.3 years pre-felling).
- **Forest control:** CEH32, CEH34, CEH33, NW10, CEH2 (C4 Main Forest interior, under unfelled pine canopy; 306–752 m; 7.3–11.6 years pre-felling).
- **Coastal control:** CEH19, CEH17 (C5 Coastal Forest; separated from C4 on the basis of a distinct $\beta_2$ regime and non-overlapping elevation range — Section 4.2; 9.7–10.3 years pre-felling).
- **Climate control:** CEH9, NW7, NW6, NW5, WMC2 (C3 Western Residual, no forest canopy; 215–1010 m; 8.7–12.7 years pre-felling).

Wells with insufficient pre-felling baseline were excluded from the primary analysis. The candidate-well criterion adopted here was monitoring continuity from the 2010 network expansion onwards, providing a minimum of seven years of pre-felling record and at least four years of pre-scraping record before the April 2015 scraping. This baseline length is required for two reasons. First, the April 2015 scraping at CEH36 (Section 3.5.3) introduced a level shift in the BACI gap at the same date, requiring a scraping covariate in the ANCOVA set out in part 1) below to avoid biasing the clearfell estimate; reliable estimation of this covariate requires that included wells have monitoring data spanning both pre- and post-scraping periods so that the April 2015 step is identifiable separately from the December 2017 felling step. Second, multiple full annual cycles are needed to constrain the seasonal cycle and the cumulative water balance slope independently of the felling step. FE1, FE2 (both installed August 2015) and NW8B (July 2015) were installed after the 2010 expansion and within four months of the scraping; their entire pre-felling baseline sits within the post-scraping era. FE3 (April 2017), FE4 (November 2017) and LIS1 (November 2017) provide nine months or fewer of pre-felling record. Following general BACI guidance that pre-intervention monitoring should span sufficient years to characterize inter-annual variability and resolve overlapping perturbations (Smokorowski and Randall, 2017), these wells were excluded. CEH36 was excluded as the scraping treatment site itself: its post-scraping water table record does not represent undisturbed pre-intervention conditions.

The clearfell BACI opens its common pre-felling window in January 2011 — the first full calendar year clear of the 2010 network installation, when every clearfell-suite well reports

continuously, and the start that aligns the monthly frame with the four-zone summer panel (Section 4.6). CEH34, in the Forest control tier, came online in August 2010, one month after the other Forest control wells, so only a single hindcast month (July 2010) is needed to align it with their common start and keep the five-well Forest control centroid balanced from its earliest point. That value is estimated by donor regression from CEH9 (Climate control), fitted by ordinary least squares on their overlapping record (August 2010 to November 2017; $R^2 = 0.91$); CEH9 was chosen in preference to the more strongly correlated CEH2 because CEH9 is a Climate control well, independent of the Forest control set, whereas CEH2 is itself a Forest control member and using it would have partially double-counted its baseline contribution. Under the January-2011 window the alignment does not affect the headline BACI, since CEH34's observed record already spans the whole window; the hindcast (extended back over 2006–2010 for cross-version reproducibility) bears only on rows before the window. The hindcast is implemented in `10i_ceh34_hindcast.py`.

The separation of controls into three counterfactual pools — Forest, Climate and Combined — allows the clearfell question to be asked three ways: (a) did the felled well (WMC3) diverge from unfelled C4 wells sharing the same canopy, substrate and elevation range, the most position-matched comparison; (b) did it diverge from the climate trajectory defined by unforested C3 Western Residual wells; and (c) against a blended counterfactual pooling all 12 control wells (C4 Forest, C5 Coastal Forest and C3 Climate). The Coastal control tier (C5: CEH19, CEH17) is not used as a primary counterfactual in the three-pool ANCOVA — its distinct $\beta_2$ regime and non-overlapping elevation range from C4 mean it is not position-matched to the felled well — but it appears in the Combined pool and is analysed separately as a coastal-erosion diagnostic in the summer minima and coefficient decomposition analyses (Sections 4.6.4 and 4.6.6).

The clearfell effect is quantified through three complementary analyses, presented in turn below: (i) the network structure and ANCOVA model, which estimates the felling step from the BACI gap against each counterfactual pool, with covariates for the April 2015 scraping and the east–west coastal-retreat gradient; (ii) a summer-window analysis with a climate-collinearity sensitivity variant, isolating the response in the ecologically critical low-water season; and (iii) a per-era SSM coefficient decomposition of the clearfell effect, attributing the change to the recharge, atmospheric-draw and drainage terms.

**1) Network structure and ANCOVA model**

The BACI gap for each target tier (Impact, Edge) was calculated as the target centroid mean water level minus the control centroid mean, producing a monthly time series of (target − control) values for each of the three control definitions. This monthly BACI gap is distinct from the SSM displacement (Section 3.4), which is the head above the drainage datum used internally as a predictor in the SSM $\beta_3$ term; the BACI gap is the dependent variable in the clearfell ANCOVA.

To separate the intervention signal from background climate variability and from the prior scraping event, an ANCOVA-BACI model (Underwood, 1992; Gotelli & Ellison, 2004; Osenberg et al., 1994) was fitted to each monthly BACI gap series:

$$BACI_{gap} = \alpha + \delta \cdot CWB + \sigma \cdot D_{scrape}(t) + \phi \cdot D_{fell}(t) + \eta \cdot (CWB \times D_{fell}) + \psi \cdot Easting(t) + \epsilon$$

where α_B is the intercept — the mean BACI gap (target centroid minus control) over the pre-intervention baseline, when CWB = 0 and D_scrape = D_fell = 0, absorbing the constant offset between the two centroid levels rather than being an effect of interest; δ_cwb is the coefficient on CWB, the centred cumulative water balance anomaly (cumulated P − PET departure from the long-run mean), and measures the climate sensitivity of the BACI gap; σ is the coefficient on D_scrape(t), a distance-weighted scraping covariate applying an exponential decay from CEH36 with length scale L_s = 300 m, activated at the April 2015 scraping date; φ is the coefficient on D_fell(t), a step function for the December 2017 clearfell, and is the felling step of primary interest; CWB × D_fell is an interaction term (coefficient η) testing whether climate sensitivity changed after felling; and ξ is the coefficient on Drift(t), a differential-drift covariate carrying the progressive divergence between the target wells and their controls — the drift that coastal retreat at the western margin would otherwise be attributed to φ. Which covariate is fitted is a documented choice of the analysis rather than part of the model. The published fits use the coastal design, in which Drift(t) is the fitted coastal drawdown profile averaged over each tier's wells, each evaluated at its own distance from the shore (not at the tier's mean distance, which would credit a tier straddling the reach with too little drift), differenced between target and control and multiplied by elapsed months; ξ is then the fraction of that field the contrast feels, and ξ = 1 is exactly the amplitude fitted network-wide. An easting × time form is retained as an option and, with ξ free, is the same fit re-parameterized — identical step, p-value and AIC (Section 4.6.3). The coefficient symbols follow the same convention as the four-zone model below: δ_cwb, subscripted by its covariate, for climate sensitivity; σ for the scraping step; and φ for the felling step. Only δ_cwb carries a subscript, because it is the one symbol of the three that names more than one quantity in this report; σ and φ are defined where they are introduced.

The BACI displacement series (target centroid minus control centroid) contains several overlapping signals: the scraping step, the clearfell step, climate-driven fluctuations (absorbed by CWB), and a differential drift between target and control wells arising from the coastal margin. If that drift is not modelled the regression distributes it across whichever coefficients best reduce the residual, biasing all estimates including the clearfell step. Coastal retreat is a sustained forcing rather than a recent perturbation: the shoreline has been retreating since at least 1899 (Section 4.10.6). Over a forcing of that duration the diffusive reach $2\sqrt{(Dt)}$, with $D = Kb/Sy \approx 101$ m² day⁻¹, exceeds the extent of the monitoring network, so the head field at the western margin is better described as quasi-steady than as a front still travelling inland. Two consequences follow. The fitted reach of about 900 m is the distance

over which the rate of change still varies detectably with distance, not the extent of the influence. And no tier of the BACI design is coastal-free: the differential the covariate carries is a contrast between positions on an established gradient, not the arrival of a signal at one tier and not another. That differential is therefore expected within the observation window, and is what ξ measures, because the independently fitted retreat gradient reaches across the network (Section 4.10.2); its value against each control tier is reported in Section 4.6.3 and discussed in Section 5.5.

An additional step term for the October 2023 re-scraping at CEH36, distance-weighted from that well as the April 2015 covariate is, was tested in each model and retained where it improved fit (ΔAIC < −2). The scraping distance-decay length scale was fixed at L_s = 300 m on physical grounds (Section 3.5.3); sensitivity to L_s = 200 m and L_s = 500 m was tested and reported. The inclusion of the scraping covariate is empirically motivated: a significant level shift in the BACI gap at the April 2015 scraping date was detected independently in the scraping analysis (Section 3.5.3). Whether the scrape perturbed the drainage coefficient at uphill wells, or the level at WMC3, is tested in Sections 4.5.5 and 4.6.3. The covariate is retained on the strength of the CEH36 level shift and as a precaution, so that any scraping drain the record cannot resolve is kept out of the fitted clearfell step.

A non-linear extension of this model was also fitted as a sensitivity test of the climate-buffering hypothesis. The linear CWB × D_fell interaction tests whether climate sensitivity changed after felling under the assumption that the BACI gap responds linearly to cumulative water balance; a buffering effect concentrated at climate extremes would, however, be largely averaged out by a single linear slope. The extension adds a centred $CWB^2$ term and a $CWB^2$ × D_fell interaction to the Forest-control Impact and Edge specifications, allowing the post-felling climate response to take a curved form. It is fitted as a reported sensitivity variant only — the headline clearfell step and the linear model above are unaffected — and its results are given in Section 4.6.2.

This three-counterfactual design produces six primary ANCOVA results (three controls × two target tiers), each with its own clearfell step estimate, confidence interval, and model diagnostics. Because the monthly residuals are strongly autocorrelated, inference on the clearfell step uses autocorrelation-robust (Newey–West) errors with 12 monthly lags (4 for the June–September fit and 6 for the October–March fit); ordinary least-squares errors would overstate its significance. *The primary three-counterfactual ANCOVA-BACI analysis is implemented in 10a_ancova_baci.py.*

**Four-zone pooled-panel cross-check**

The three-counterfactual ANCOVA above fits the model six times — once for each pairing of a control definition with a target tier — and each fit has its own dependent variable, because the BACI gap is measured against a different control centroid each time. The six step estimates are consequently not directly comparable with one another: a contrast such as

Impact relative to Edge cannot be read off by differencing two of them. As a cross-check in which every zone-to-zone contrast comes from a single consistent fit, a pooled-panel model was fitted alongside the principal ANCOVA.

The pooled-panel model is a single regression, fitted once to one combined dataset. Rather than analysing each zone in a separate model, the monthly water-table records of all wells in four zones are stacked into one long table — one row per well per month — and zone membership is carried as a categorical variable within that table. One of the four zones, the Forest control, is designated the reference level; the model then estimates how each of the other three zones behaves relative to that reference. A single fit therefore yields three felling steps at once — C3/Warren, Edge and Impact, each expressed as a difference from the Forest control — and because all three rest on the same baseline and the same fitted error structure, any further contrast between two zones (Impact relative to Edge, for example) follows by exact subtraction. This is the structural advantage over the six separate ANCOVAs: there, each fit has its own baseline and the steps cannot be differenced; here, one fit supplies a single internally consistent set of contrasts.

The four zones are the Forest control (the five unfelled C4 interior wells, the reference level); the Edge and Impact tiers as defined above; and a fourth zone of four open western-dune wells (CEH1, NW1, NW2, NW11), here termed C3/Warren. The C3/Warren zone is included as a second control: it lies more than 500 m from the felled compartment, on the western side away from the direction in which forest-management effects propagate, and is therefore expected to show no felling response — a zero result there is a check on the design rather than a finding. The Coastal control wells are excluded from this model because they carry the coastal-erosion confound.

With the Forest control as the reference zone, the fitted model is:

$$h_{\mathrm{it}} = \alpha + \delta_F \cdot CWB_t$$

*(1) Baseline water level, common to all zones. h_it is the monthly water-table depth at well i in month t; α_B is the grand intercept; δ_F is the climate sensitivity of the Forest reference zone to the centred cumulative water-balance anomaly CWB_t.*

$$+ \delta_W \cdot CWB_t \cdot Warren_i + \delta_E \cdot CWB_t \cdot Edge_i + \delta_I \cdot CWB_t \cdot Impact_i$$

*(2) Zone-differential climate sensitivity. δ_W, δ_E and δ_I are the differential sensitivities of the C3/Warren, Edge and Impact zones relative to the Forest control. Warren_i, Edge_i and Impact_i are binary zone indicators — equal to 1 if well i belongs to that zone, 0 otherwise. Only one switch is ever active for a given well; Forest control wells have all three equal to zero.*

$$+ \sigma_F \cdot Scr_t$$

*(3) Scraping step for the Forest control zone. Scr_t = 0 before April 2015, 1 from April 2015 onward. σ_F is the mean water-level change in the reference zone after scraping.*

$$+ \sigma_W \cdot Scr_t \cdot Warren_i + \sigma_E \cdot Scr_t \cdot Edge_i + \sigma_I \cdot Scr_t \cdot Impact_i$$

*(4) Additional scraping response in each non-Forest zone relative to the Forest control. The absolute scraping step in zone z is σ_F + σ_z.*

$$+\phi_F \cdot Post_t$$

*(5) Clearfell step for the Forest control zone. Post_t = 0 before December 2017, 1 from December 2017 onward. φ_F is the mean water-level change in the reference zone after felling.*

$$+\phi_W \cdot Post_t \cdot Warren_i + \phi_E \cdot Post_t \cdot Edge_i + \phi_I \cdot Post_t \cdot Impact_i$$

*(6) Additional clearfell response in each non-Forest zone relative to the Forest control. The absolute felling step in zone z is φ_F + φ_z. The contrast between any two non-Forest zones follows by exact subtraction: φ_z1 − φ_z2.*

$$+\psi \cdot \left(Easting_i \times Time_t\right) + \mu_i + \epsilon_{\mathrm{it}}$$

*(7) Three housekeeping terms: ψ·(Easting_i × Time_t) absorbs the easting-correlated coastal-retreat trend; μ_i is a per-well fixed effect removing time-invariant depth differences between wells; ε_it is the residual error, with standard errors clustered on the well.*

The scraping and felling indicators enter the model twice — once as reference-zone main effects (σ_F·Scr_t and φ_F·Post_t) and once interacted with each non-reference zone indicator — so the absolute step in any zone z is σ_F + σ_z for scraping and φ_F + φ_z for felling, and a contrast between two non-reference zones follows by exact subtraction of their interaction coefficients. The ψ·(Easting_i × Time_t) covariate absorbs the easting-correlated coastal-retreat trend quantified in Section 4.10.2, preventing it from being attributed to the felling steps. Standard errors are clustered on the well to allow for serial correlation between repeated monthly readings at the same well.

Because each felling step is a fitted regression coefficient, its confidence interval and p-value follow directly from the OLS output. A contrast between two non-reference zones — such as the Impact step relative to the Edge step — is not itself a fitted coefficient but the difference of two; its point estimate is exact, but its standard error must be computed from the model's coefficient covariance matrix rather than by subtracting the two separate standard errors. Such derived contrasts can acquire misleadingly narrow intervals when the two zones are strongly correlated, so their p-values are not treated as independent evidence of an effect; the Impact-to-Edge contrast is instead corroborated against the separate two-zone estimator of Script 10j. This pooled-panel model tests whether the felling signal survives estimation in one jointly fitted panel; it supplements, and does not replace, the principal ANCOVA.

The same four-zone structure is also applied to the annual June–September summer-minimum series, to provide the equivalent cross-check at the summer extreme; at that resolution the scraping term is dropped. *The monthly four-zone model is implemented in 10k_four_zone_baci.py and the summer-minimum companion in 10l_four_zone_summer_minima.py.*

**Spatial step-change mapping.** The post-felling step at each reference well — relative to its pre-felling mean and corrected by subtracting the appropriate control centroid step — was computed and rendered as climate-corrected spatial maps for the felling and scraping eras (Figures 36 and 27). These maps provide a visual diagnostic of whether the step changes show a spatial gradient consistent with a localized intervention effect. *Spatial step-change mapping is implemented in 10b_spatial_step_maps.py*.

**Within-forest position-matching.** The within-forest spatial coefficient analysis (Section 3.4.4, results in Section 4.9.4) underpins the C4-only composition of the Forest control: it indicates that elevation is the dominant control on within-forest $\beta_2$ (Section 4.9.4), so position-matched controls rather than canopy-matched controls are the appropriate counterfactual for testing canopy management effects. *The within-forest spatial coefficient analysis is implemented in 10c_forest_zone_analysis.py.*

**2) Summer-window analysis and climate-collinearity sensitivity**

The annual BACI step estimated from the ANCOVA above is computed over all monthly observations from the start of record to the analysis cutoff. Because the ecological consequences of any water table change concentrate in the summer months when slack inundation determines habitat viability, a parallel summer-window ANCOVA is fitted directly to June–September observations only, using the same model specification, control definitions, and covariates as the annual analysis. This directly-fitted summer ANCOVA provides a monthly-resolution estimate of the clearfell step over the ecologically critical period, without the seasonal scaling assumption implicit in extrapolating an annual estimate.

A sensitivity variant of the summer-window analysis is additionally fitted with the CWB covariate and the CWB × D_fell interaction dropped from the design matrix. Within the restricted June–September window, the CWB covariate covaries more closely with the clearfell step indicator than it does in the annual window, because the summer-only data captures the part of the seasonal cycle in which the climate-balance trajectory and the post-2017 felling step share the most temporal overlap. This collinearity inflates the standard error on the clearfell step coefficient in the full-specification fit and may absorb a portion of the felling signal into the CWB term. The sensitivity variant tests how much the summer step estimate depends on the CWB specification; the interpretation of the two estimates and the mechanistic argument for the collinearity behaviour are presented in Section 4.6.3.

A separate annual summer-minimum analysis is also computed. For each of the 22 wells (the 17 wells of the BACI network and the five far-field control wells), the annual summer minimum depth (June–September) was computed and the gap between the well's summer minimum and the control centroid summer minimum was tracked annually. The pre- versus post-felling shift in this gap was tested by Welch's t-test, and tier-mean shifts were computed for both forest and climate control centroids. A mixed-effects model with random intercept

per well and fixed effects for post-felling and post-scraping indicators was fitted to each tier to provide a pooled clearfell step estimate with proper uncertainty:

$$gap = \alpha + \phi \cdot D_{fell} + \sigma \cdot D_{scrape} + u_{well} + \epsilon$$

where gap is the annual difference between the well's summer-minimum depth and the control-centroid summer minimum; α_B is the fixed intercept, the mean gap over the pre-felling baseline; φ is the coefficient on D_fell, the pooled clearfell step and the quantity of interest; σ is the coefficient on D_scrape, the scraping step; D_fell and D_scrape are post-intervention step indicators, equal to zero before and one from the December 2017 clearfell and April 2015 scraping dates respectively; u_well is a random intercept for each well, absorbing time-invariant between-well differences in mean gap; and ϵ is the residual error. As elsewhere in this section, φ denotes the felling step and σ the scraping step.

The directly-fitted summer ANCOVA, the CWB-dropped sensitivity, and the annual summer-minimum mixed-effects model are each reported in Section 4.6.3 to provide three complementary perspectives on the summer-specific response of the clearfell-affected wells.

*The directly-fitted summer-window ANCOVA and the CWB-dropped sensitivity variant are produced by the same script as the annual analysis (10a_ancova_baci.py); the annual summer-minimum mixed-effects model is implemented in 10d_summer_minima.py.*

### 3) Per-era SSM coefficient decomposition of the clearfell effect

The clearfell intervention may have altered the recharge, atmospheric draw and drainage responses of the affected aquifer (Komatsu et al., 2011; Sun et al., 2015). To diagnose these changes mechanistically, the SSM coefficients $\beta_1$, $\beta_2$ and $\beta_3$ are estimated separately for the pre-intervention (Before) and post-intervention (After) eras at each well in the clearfell BACI network, and the differences $\Delta\beta_1 = \beta_{1,\text{After}} - \beta_{1,\text{Before}}$, $\Delta\beta_2 = \beta_{2,\text{After}} - \beta_{2,\text{Before}}$ and $\Delta\beta_3 = \beta_{3,\text{After}} - \beta_{3,\text{Before}}$ are interpreted as the recharge, atmospheric draw and drainage shifts associated with the intervention.

Both era-specific SSMs are fitted on the displacement formulation defined in Section 3.4, so $\beta_3$ is the displacement-storage coefficient (positive, Darcy-consistent) in both eras. The Before era is the record-length-balanced pre-intervention window, with an additional binary dummy in the design matrix at wells affected by the April 2015 scraping pulse to absorb that earlier intervention. The After era extends from the intervention date to the end of record. Both fits use the canonical no-intercept SSM (Model A), consistent with Script 03.

The shifts are reported per well and per tier by sign and magnitude, as a direction-and-pattern diagnostic of which pathway moved after felling. The decomposition is not a second estimator of the BACI step and is not compared against it: the ANCOVA step of part 1) is a control-differenced centroid coefficient, with regional climate forcing removed by

construction, whereas a per-era, per-well fit carries no control subtraction, so the two cannot be reconciled term by term. The clearfell magnitude remains the ANCOVA step of part 1); the decomposition is presented in Section 4.6.6.

$$\Delta h_{predicted} = \Delta\beta_1 \cdot \bar{P}_{after} - \Delta\beta_2 \cdot \bar{PET}_{after} - \Delta\beta_3 \cdot \bar{h}_{disp,after}$$

where bar P_after, bar PET_after and bar h_disp,after are the post-intervention era means of rainfall, PET and displacement above the drainage datum. This expression isolates the change in the climate-driven and drainage-driven response of the aquifer from the change in baseline level, by holding the climate forcing and head position at their post-intervention values and varying only the coefficients.

Δh_predicted is compared to the observed BACI step Δh_observed estimated from the ANCOVA model in part 1) above. The Δh_predicted formula captures only the β·X component of the era shift; the intercept difference Δα_B = α_After − α_Before absorbs the era-mean residual, which at most wells is the dominant component of the observed step. The predicted-versus-observed comparison is therefore informative about the mechanistic decomposition ($\Delta\beta_1$ recharge, $\Delta\beta_2$ atmospheric draw, $\Delta\beta_3$ drainage) but should not be read as a complete reconstruction of the BACI step. The interpretation of the Δα residual — whether it represents genuinely non-mechanistic baseline drift, mis-specification of one of the β terms, or a structural state shift not captured by the time-invariant SSM — is taken up in Section 4.6.7.

Two additional supplementary analyses extend the mechanistic decomposition. The SSM residual normalization calibrates the SSM on the pre-felling record for each well, runs it forward iteratively using observed climate forcing, and normalizes the post-felling residuals (observed minus predicted) by subtracting the forest control mean residual; this provides a climate-independent test of clearfell-specific displacement and is presented in Section 4.6.7. A synthetic counterfactual for the Impact and Edge tiers is constructed from an optimally weighted combination of six donor wells calibrated on the pre-intervention period, with the gap between observed and synthetic post-felling providing an estimate of the intervention effect that does not rely on the ANCOVA parametric assumptions; this is also presented in Section 4.6.7. A clearfell transect analysis tests whether the post-felling response shows a spatial gradient consistent with a localized clearfell effect, comparing era-mean depths across a transect from the plantation interior to the felling core. Script 10g also tests whether the broadleaf-canopy well NW10 diverges from the pine composite over time, and provides rolling-window SSM coefficient tracking and drainage diagnostic plots; the limitations of the NW10 comparison are discussed in Section 6.5.

A separate synthetic-extension analysis tests whether the FE1 and FE2 wells, sited inside or at the immediate edge of the clearfell compartment but excluded from the primary ANCOVA on the grounds of insufficient pre-scraping baseline, corroborate the WMC3 Impact result. Their records were extended backwards using OLS donor regression onto the

three Forest control wells CEH34, CEH2, and CEH33, selected on the basis of proximity, shared canopy conditions, and pre-clearfell correlation ($r > 0.99$ for both FE wells). The regression was calibrated on the pre-clearfell overlap window (July 2015 to November 2017, $n = 29$ months) and applied to hindcast FE well levels back to July 2010, providing a synthetic pre-scraping baseline. The synthetic hindcast was spliced with the actual FE observations to create a stable-composition impact centroid spanning all three eras (pre-scraping, post-scraping, post-felling). Three impact centroid variants were tested: (A) WMC3 + FE1 + FE2; (B) WMC3 + FE2; and (C) WMC3 alone. Each was run through the same ANCOVA framework as the primary analysis, against the Forest, Climate, and Combined control definitions. FE1 sits approximately 46 m outside the clearfell boundary, in standing forest, and is caveated accordingly. Results are presented in Section 4.6.7.

*The per-era SSM coefficient decomposition is implemented in 10e_coefficient_decomposition.py. The SSM residual normalization and synthetic-control analyses are implemented in 10f_robustness.py; the transect, NW10 broadleaf trend, and rolling-window diagnostics in 10g_diagnostics.py; and the synthetic FE well extension in 10h_synthetic_impact_baci.py. Shared data loading and spatial functions are in clearfell_common.py. The clearfell BACI experiment as a whole is orchestrated by run_10_clearfell.py.*

### 3.5.5 Temporal decomposition, drift-treatment robustness, and the canopy setting

The clearfell ANCOVA-BACI step (Section 3.5.4) is decomposed in time by splitting the post-felling record at January 2021 into an early era (December 2017 to December 2020) and a late era (January 2021 onward), each entered as its own dummy on the shared BACI frame so that every era carries its own step against the Forest control. To establish whether an early-to-late change is a real relaxation rather than an artefact of the coastal-drift correction, each contrast is refitted under four drift treatments: the coastal differential pinned to the independently measured Script 25 field (the published convention); no drift term; a single free drift slope over the record; and a free slope fitted separately in each era. The last is the direct test of an accelerating coastal retreat — were the differential drift to have steepened after 2021, a per-era slope would absorb it, so a decay that survives this treatment cannot be a drift artefact. The era split is implemented in 10a_ancova_baci.py and written to 10a_13_era_split.csv.

The sensitivity of the step to the composition of the Forest control is examined by refitting it across control subsets: all five Forest-control wells; the two conifer-canopy wells outside any 1998 replant block (CEH34 and CEH2 — NW10 also lies outside, at 240 m, but carries a broadleaf canopy); the two lying inside one (CEH32 and CEH33); and single-well drops. The subsets listed are illustrative; the full set is in 10a_12_control_subset_sensitivity.csv. The proximity of each monitored well to the 1998 replant blocks is recorded by Script 01 as

an in-block flag and a distance, together with an exposure index summing exp(−d/λ) over the three blocks, where λ is the drawdown reach defined in Section 3.8.2 and read live from the Script 20 outputs, so that the developing-canopy setting of the treatment and control wells can be quantified rather than asserted (10a_11_replant_proximity.csv, 10a_12_control_subset_sensitivity.csv).

The canopy setting of the felled compartment and its controls is characterized from the dated aerial photographic series by the texture index of Script 41, measured over the felled compartment, the three 1998 replant blocks, the broadleaf restock, a pooled four-patch untouched-forest control drawn around the Forest-control wells, and a vegetated open-warren reference, each normalized in-frame to the unfelled mature-conifer canopy. Aerial frames on which cloud or shadow falls across a region are withheld from that region's series by inspection (the 2020-03-31 frame over the felled compartment and 1998 blocks 1 and 2, and the 2020-04-24 frame over 1998 block 3); the index is reported both to the last clean pre-2026 frame and including the isolated 2026 frame, the developing trend being present either way.

## 3.6 Threshold Forecasting

To translate the observed spatial dynamics into predictive management tools, operational forecasting equations were derived using a multi-tiered mathematical framework. This approach combines continuous mass-balance modelling with seasonal prediction equations to quantify the aquifer's climatic vulnerability.

### 3.6.1 Cluster-Level Mechanistic State-Space Equations

The cluster-level SSM coefficients derived in Section 3.4.1 were used to construct operational P_flood threshold equations for each of the five principal clusters. The model was fitted with the intercept strictly forced to zero, ensuring that the coefficients represent a strict physical mass balance and providing the basis for algebraic inversion in Section 3.6.3.

### 3.6.2 Seasonal Prediction Equations

While the SSM operates on a continuous monthly basis, conservation management typically requires inter-seasonal risk forecasting — specifically, predicting winter flood peaks and summer drought minima from antecedent conditions. To address this, autoregressive prediction equations were fitted at a hydrological-year resolution for each of the five clusters.

Two equations are fitted for each cluster. The winter prediction equation predicts the peak winter water table (h_peak) from cumulative winter precipitation and the antecedent summer minimum:

$$h_{peak} = a \cdot P_{winter} + b \cdot h_{min} + c$$

The summer prediction equation predicts the summer minimum water table (h_min) from cumulative summer precipitation and the antecedent winter peak:

$$h_{min} = a \cdot P_{summer} + b \cdot h_{max,winter} + c$$

where P_winter is cumulative precipitation over October–March, P_summer is cumulative precipitation over April–September, h_min is the preceding summer minimum water table depth, h_{max,winter} is the preceding winter maximum water table depth, *a* is the fitted regression coefficient on the seasonal precipitation predictor, *b* is the fitted coefficient on the antecedent-state predictor, and *c* is a fitted intercept. The seasonal equations are AR(1)-style empirical regressions at hydrological-year resolution, providing operational forecasting at coarser temporal resolution than the monthly SSM described in Section 3.4. Data are organized by hydrological year commencing 1 October.

Head-dependent drainage (the $-\beta_3 \cdot h(t-1)$ term in the SSM) is not fitted as a separate parameter at the seasonal scale: with one observation per hydrological year, drainage and the antecedent-state term are aliased, so drainage is absorbed implicitly into *b* and the fitted intercept *c* rather than appearing as a separate coefficient.

The antecedent-state terms (h_min in the winter equation, h_max in the summer equation) explicitly formalize the carry-over between seasons in the dune aquifer system — the degree to which a winter flood peak is conditioned by the depth of the preceding summer drought, and vice versa. This parameterization allows managers to construct simple conditional forecasts: given a known or projected winter rainfall total and the current summer water table position, the expected winter flood maximum can be estimated directly from the fitted equation without running the full SSM.

### 3.6.3 Algebraic Derivation of the Critical Rainfall Threshold (P_flood)

The most operationally significant metric for dune slack management is the critical rainfall threshold (P_flood) — the cumulative winter rainfall depth required to raise the water table from its summer minimum to ground level at the well (h_target, where depth to water table equals zero).

This threshold was derived by iterating the cluster-level state-space equation (Section 3.4.1) forward from the September summer minimum ($h_0$) over the months from October to each cluster's historical peak-of-record month. The horizon is therefore cluster-specific and reflects the observed timing of winter peak water table: four months (October–January) for C1 and C2, and five months (October–February) for C3, C4 and C5.

The monthly recurrence equation — applied once per step from October to the peak month — is:

$$h_t = (1 - \beta_3) \cdot h_{t-1} + \beta_1 \cdot P_t - \beta_2 \cdot E_t - \beta_3 \cdot D$$

*(1) h_t is the water level at the end of month t. The term (1 − $\beta_3$)·h_{t−1} is how much of last month's displacement survives into this month — close to 1 for sluggish aquifers such as C4, lower for fast-draining ones such as C1. $\beta_1$·P_t is the recharge contribution from this month's rainfall; $\beta_2$·E_t is the atmospheric draw-down from PET; $\beta_3$·$z_0$ is the drainage loss proportional to the fixed datum displacement. Monthly precipitation is expressed as P_t = m_P·$\bar{P}$_t, where $\bar{P}$_t is the long-term monthly mean rainfall from the RAF Valley reference climatology and m_P is a multiplier representing the departure of a given winter from the long-term mean. Monthly PET is taken directly from the climatology (E_t = $\bar{E}$_t) without scaling, reflecting the fact that winter PET varies considerably less year-to-year than winter rainfall at this latitude.*

Letting ρ_h = 1 − $\beta_3$, iterating the recurrence forward n months from $h_0$ yields the closed form:

$$h_n = h_0 \cdot \alpha^n + \beta_1 \cdot \sum_{i=1}^{n} \alpha^{n-i} \cdot \lambda \cdot \bar{P}_i - \beta_2 \cdot \sum_{i=1}^{n} \alpha^{n-i} \cdot \bar{E}_i - D \cdot \left(1 - \alpha^n\right)$$

*(2) h_n is the water level after n months of winter recharge. Rather than stepping through the equation n times, this single expression accumulates the effect of all n monthly steps at once. The term $h_0$·ρ_h$^n$ is how much of the starting summer deficit survives n months of drainage — for C2, with ρ_h = 0.937 from the cluster-level fit, roughly 77% of the September deficit persists to January. The two summation terms accumulate the rainfall benefit and PET cost month by month, with the weighting ρ_h^{n−i} giving greater influence to earlier months because they have more remaining winter months to amplify their contribution. The final term $z_0$·(1−ρ_h$^n$) is the cumulative drainage loss over the horizon, where $z_0$ is the drainage datum defined in Section 3.4.1 — a fixed reference depth of 3.7 m below ground surface.*

Setting h_n = h_target and solving for m_P:

$$\lambda = \frac{h_{target} - h_0 \cdot \alpha^n + \beta_2 \cdot S_E + D \cdot \left(1 - \alpha^n\right)}{\beta_1 \cdot S_P}$$

*(3) m_P is the rainfall multiplier needed to raise the water table to the slack floor from a given summer minimum $h_0$. The numerator is everything the aquifer must overcome: the slack-floor target level (h_target), the starting deficit (−$h_0$·ρ_h$^n$), the cumulative PET demand ($\beta_2$·S_E), and the drainage losses ($z_0$·(1−ρ_h$^n$)). The denominator is the drainage-weighted rainfall the winter can supply at climatological rates ($\beta_1$·S_P). Here S_P = Σρ_h^{n−i}·$\bar{P}$_i and S_E = Σρ_h^{n−i}·$\bar{E}$_i are the drainage-weighted sums of climatological rainfall and PET over the horizon. An m_P of 1.0 means an exactly average winter suffices; m_P = 1.5 means a winter 50% wetter than average is needed.*

The critical rainfall threshold then follows directly:

$$P_{flood} = \lambda \cdot \sum_{i=1}^{n} \bar{P}_i$$

*(4) P_flood is m_P converted from a dimensionless multiplier into a depth in millimetres — the number a site manager can compare directly against a seasonal rainfall forecast or a historical percentile. For example, a C2 well with a September minimum of 0.5 m below ground requires m_P ≈ 0.99, giving P_flood ≈ 397 mm — an average October–January total. A well requiring m_P = 2.5 has never received sufficient winter rainfall in the 95-year RAF Valley record.*

Because the weighting ρ_h^{n−i} places greater emphasis on early-winter rainfall than on late-winter rainfall, the formulation encodes the physical fact that October recharge has more months to contribute to the peak than March recharge does, modulated by the cluster-specific drainage coefficient $\beta_3$. Three natural classes of well emerge from this framework: those whose required m_P is less than unity (P_flood reachable under sub-average winters), those requiring 1.0 ≤ m_P < 2.5 (reachable only in wetter-than-average winters of varying rarity), and those for which m_P ≥ 2.5 (effectively unreachable under the present climate). This classification is applied spatially across the dipwell network in Section 4.

This algebraic inversion provides site managers with a dynamic, site-specific forecasting tool. By entering a measured summer minimum dipwell reading ($h_0$) into the cluster's equation, a manager obtains both the cumulative winter rainfall (mm) required to raise that well to the slack-floor threshold, and its expression as a multiple of climatological winter rainfall (m_P) — a direct measure of ecological flooding feasibility in the subsequent recharge season.

### 3.6.4 Long-Term Climate Trajectory Analysis

To describe the long-term trajectory of the water table, annual summer minima and winter maxima were extracted for all five clusters from a fixed-effects cluster series: each cluster's reference wells are fitted as h(w,t) = a_w + g_t, a level for each well and a common monthly term, and the series g_t is placed at the mean level of all members, so that a well joining the network late does not shift the cluster. The plain cluster mean behind the model coefficients rests, before 2010, on a few early wells, which an annual extreme, unlike a fitted coefficient, does not average out. C4 and C5, both forested clusters, are included for completeness to characterize the full network trajectory and to support future analysis of forest management effects on seasonal extremes. The hydrological year was defined as 1 October to 30 September, consistent with Section 3.6.2, and a minimum of three valid monthly observations per season was required for a year to be included.

For summer minima, ordinary least squares linear regression was fitted to the annual series for each cluster over the hydrological years of record (2005–2025; C1, C4 and C5 from

2006) and extrapolated to 2035, with 95% confidence intervals derived analytically. Every trend is refitted from the 2011 hydrological year, the first full year clear of the 2010 install ramp, as a robustness basis.

For winter maxima, linear trends are fitted to the observed annual maxima of each cluster (reported in Section 4.8.1) but not projected: transfer function projection via the seasonal models of Section 3.6.2 extrapolates beyond the fitted range for the eastern clusters and gives physically implausible trajectories there. The winter analysis therefore presents the observed record only. Flooding is measured at each well, as the proportion of winters in which its winter maximum reached the ground at that well (Section 3.6.5).

*Critical rainfall threshold equations, peak flood prediction equations, and summer drought prediction equations were implemented in 11_forecasting_thresholds.py. Climate trajectory projections and threshold exceedance analysis are implemented in 14_climate_projections.py.*

### 3.6.5 Spatial Distribution of Threshold Metrics

The cluster-level threshold framework of Sections 3.6.3 and 3.6.4 is complemented by a set of per-well spatial fields that render the eco-hydrological status of every dipwell across the classified network. Four metrics are mapped. The mean summer-minimum depth is computed at each well from its annual minimum above-Ordnance-Datum water level over the August–September window, requiring at least one valid observation per year, and converted to depth below ground by subtracting the level from the LiDAR ground elevation. The mean winter-maximum depth is computed analogously from the annual maximum over the October-to-March window, requiring at least three valid observations in that window for a year to be admitted, and converted to depth below ground on the same convention. The critical rainfall threshold P_flood is evaluated at each well by applying the iterated closed-form solution of Section 3.6.3 to that well's observed mean summer minimum, using per-well state-space coefficients at the reference wells and cluster-centroid coefficients at the extended wells assigned through the Pearson affinity audit (Section 3.3). The winter flooding frequency is computed at each well having at least five years of winter record as the percentage of hydrological years in which the winter-maximum water table reached or exceeded the local ground surface, taking the LiDAR ground elevation as the surface datum.

Aggregating on above-Ordnance-Datum elevations renders the summer-minimum and winter-maximum metrics invariant to the ground-surface lowering produced by scraping at treatment wells; the surface change is carried instead by the depth-below-ground conversion, which applies the post-intervention ground elevation. Each per-well field is interpolated to the standard site grid by piecewise-linear (Delaunay) interpolation and masked across the dune ridges where the elevation model exceeds the

interpolated water surface, on the convention described in Section 3.8. The summer-minimum, winter-maximum and flooding-frequency surfaces are referenced to the SD15b / SD16 community-mean levels of Curreli et al. (2013). A well's class is its depth below its own ground, and indicates a community only where the well stands on the slack floor. The spatial fields are presented in Section 4.7. *The per-well threshold metrics and their spatial fields are implemented in 11b_spatial_thresholds.py.*

## 3.7 Characterizing Water Table Losses and Long-Term Trajectories

### 3.7.1 Depth-Dependent Evapotranspiration Model

To test whether the atmospheric draw coefficient varies significantly with water table depth — and whether a depth-dependent formulation is warranted over the standard fixed-$\beta_2$ SSM — an extended model was implemented in which $\beta_2$ was replaced by the depth-dependent term $\beta_2 \cdot \exp(-d_w / L_E)$, where d_w is the depth of the water table below ground surface (metres) and L_E is an attenuation length (m) controlling the distance over which evapotranspiration influence declines as the capillary fringe retreats from the root zone. Depth below ground surface was computed at each timestep as $d_w = \max(-h, 0)$, where h is the water table position (negative convention). No upstand term enters: h is already referenced to the ground surface (Section 3.1.1). The floor at zero keeps d_w a physical distance below ground when a slack is ponded. Because the modified predictor $-\exp(-d_w / L_E) \cdot PET$ remains linear in $\beta_2$ for any fixed L_E, the three coefficients $\beta_1$, $\beta_2$ and $\beta_3$ were recovered by no-intercept OLS at each value of L_E. A systematic grid search over the rate $1/L_E \in [0, 6]$ $m^{-1}$ at intervals of 0.05 $m^{-1}$ was conducted independently for each cluster, with the optimal 1/L_E selected by maximizing iterative simulation NSE. Setting 1/L_E = 0 (L_E unbounded) recovers the standard SSM exactly, providing a direct baseline for comparison.

Despite the statistical improvement that the grid search produces (Section 4.4), the depth-dependent formulation introduces a nonlinearity that would propagate through the threshold equations (Section 4.7), forestry scenarios (Section 4.13.2), and forecasting tools. The standard SSM's fixed $\beta_2$ provides an effective depth-averaged parameterization that preserves the closed-form analytical framework on which these operational outputs depend. The depth-dependent formulation was therefore not adopted for the primary analysis; the standard fixed-$\beta_2$ SSM is retained throughout. The fixed $\beta_2$ should be understood as absorbing a real depth-coupling effect whose magnitude varies across clusters; its diagnostic significance is discussed further in Section 5.6.1. *The depth-dependent PET grid search and benchmark comparison are implemented in 15_depth_dependent_pet.py.*

### 3.7.2 Mean Monthly Water Balance Decomposition

Water balance decomposition of the kind applied here — partitioning monthly head change into recharge, evaporative draw, drainage and residual — has a long history in the coastal dune hydrology literature, particularly in the Dutch and North Sea coastal systems where managed water tables and dune slack conservation drive analogous restoration questions (Stuyfzand, 1993; Grootjans et al., 2002, 2017). The decomposition applied in this study derives the component fluxes directly from the fitted SSM coefficients, allowing direct comparison of fluxes across clusters subject to identical climate forcing using only the dipwell and climate records already available for the site.

To quantify and compare the hydrological inputs and losses operating on each hydrogeological cluster, a mean monthly water balance decomposition was applied to the five principal clusters (C1–C5) using cluster-level SSM coefficients derived in Section 3.4.1. All terms are expressed in metres of water table change per month (m/month), the natural unit of the state-space model and directly comparable across clusters receiving the same climate forcing.

The decomposition isolates three components from the canonical state-space equation (Section 3.4):

$$\Delta h = \beta_1 \cdot P - \beta_2 \cdot PET - \beta_3 \cdot h_{disp}$$

Mean monthly recharge was computed as $\beta_1 \times \bar{P}$ , where $\bar{P}$ is the mean monthly precipitation over the monitoring period from the RAF Valley record. Atmospheric draw was computed as $\beta_2 \times \overline{PET}$ , where $\overline{PET}$ is the mean monthly Thornthwaite PET. Gravity drainage was computed as $\beta_3 \times \bar{h}$ _disp, where $\bar{h}$ _disp is the mean displacement above the drainage datum ($\bar{h}$ _disp = $z_0 + \bar{h}$ , with $z_0$ = 3.7 m; see Section 3.4) for each cluster derived from the cluster-average hydrograph. Because the drainage term scales with displacement above the datum, the drainage/ET partition depends on the datum regime: a datum set within the water-table fluctuation range reduces the drainage term to mean-reversion of anomalies and attributes almost no steady flux to drainage, whereas at depths beyond the fluctuation range the term represents Darcy outflow above a fixed base and the partition is insensitive to the exact choice. The 3.7 m datum sits within this stable Darcy regime (Section 3.4; the supporting datum-regime diagnostic is presented in Supplementary Material Note S9). For the Main Forest cluster (C4), where this datum sensitivity is greatest, an identifiability diagnostic indicates that the low drainage coefficient is cleanly recovered at the cluster-centroid scale — C4 has among the lowest predictor collinearity in the network (VIF ≈ 1.1, second-lowest of the five clusters) — so the drainage/ET partition is well determined there rather than a fitting artefact (Section 5.2.3). Canopy interception is not included as a separate loss term — it is implicitly captured in the forested clusters' (C4 and C5) lower $\beta_1$ and lower $\beta_2$, and adding it explicitly would double-count losses already embedded in the fitted coefficients. Total losses are therefore the sum of atmospheric draw and drainage only. The net monthly

balance — recharge minus total losses — quantifies the water balance residual — the flux that would be required from some unmodelled source to maintain long-run water table equilibrium.

A companion volumetric partition is presented alongside the primary head-space decomposition (Table 7; Figure 11b), expressing the recharge, evapotranspiration and drainage terms as annual depths (mm $yr^{-1}$). The partition is obtained from the head-space fractions applied to the net climate input (precipitation less canopy interception), without a specific-yield conversion; the split between evapotranspiration and drainage is bracketed by two independent estimates — the SSM head-space ratios and the observed seasonal recession rates — and the range between them is shown as a hatched band at the evapotranspiration/drainage boundary in Figure 11b. For the forested clusters (C4 and C5), canopy interception is shown as an explicit segment on the inputs bar of Figure 11b, computed as $0.24 \times \bar{P}$ following Freeman (2008; see also Betson and Scholefield, 2004), reducing the net input rather than being additive. The mean monthly water balance decomposition, volumetric partition and recession partition were produced by 16_water_bal.py (outputs: 16_water_bal_table.csv, 16_water_bal_vol_table.csv, 16_water_bal_rec_table.csv).

The second of those two estimates — the seasonal recession analysis — is computed from the observed cluster-mean series and is independent of the state-space model. Months in which the cluster mean water table fell ($\Delta h < 0$) are separated into a winter window (November–February) and a summer window (June–September), March being excluded as transitional, and the mean month-on-month decline is taken over each. Winter potential evapotranspiration at this latitude is small relative to summer, so a winter decline is approximately drainage alone while a summer decline carries drainage plus evapotranspiration; the ratio of the two mean declines therefore estimates the drainage share of summer losses, with the evapotranspiration share as its complement. Both terms are head declines in the same units, so specific yield cancels in the ratio and the estimate is independent of Sy — which is why the volumetric partition can be reported without the water table fluctuation specific yields of Section 3.7.3. The two methods share a name but not a construction: the WTF method reads rising limbs and returns a magnitude, whereas the recession analysis reads falling limbs and returns a dimensionless ratio. Two biases are acknowledged. Winter potential evapotranspiration is small but not zero, so the winter decline slightly overstates drainage; and drainage is head-dependent, so it is faster in winter when heads are high than in summer when they are low. Both act in the same direction, inflating the drainage fraction, so the recession estimate is read as leaning high and the bracket between the two estimators — rather than either one alone — is reported as the partition uncertainty. The per-cluster recession and summer decline rates, the month counts behind each and both drainage fractions are given in Table 6; their mid-point is carried into Table 7 and Figure 11b.

### 3.7.3 Specific Yield Estimation: Water Table Fluctuation Method

To characterize the substrate storage capacity independently of the head-space water balance — and to compare against the literature specific-yield range for unconfined dune sand (C1 = 0.08 over glacial till, C2–C5 = 0.12; Fetter, 2001) — an independent estimation of the specific yield (Sy) was performed using the water table fluctuation (WTF) method of Healy and Cook (2002):

$$S_y = \frac{R}{\Delta h}$$

where R is the net recharge reaching the water table (m) and Δh is the corresponding water table rise (m).

Three complementary derivations of Sy are reported. Approach A (through-origin OLS) fits a linear regression of corrected water-table rise against net recharge for each cluster over the winter months (November–March, PET < 25 mm/month), with the regression forced through the origin so that the slope is the cluster-mean Sy. The water-table rise is corrected for storage decay by adding the drainage component $\beta_3 \cdot |h_prev|$ derived from the SSM, isolating the recharge-driven rise from the contemporaneous drainage loss. Approach B (event median) computes a per-month Sy = R / Δh for each rising-limb event in the record (months where Δh > 5 mm and net recharge P − PET > 10 mm), with event-level estimates further constrained to the physically plausible range 0.01 < Sy < 0.50 to exclude months where the suppressed or negative Δh signal produced implausibly high individual Sy values; the cluster-mean Sy is reported as the median of the event distribution with interquartile range as the uncertainty bound. Approach C (rapid recharge events, after Crosbie et al., 2005) is a third, methodologically independent estimator: rather than correcting for drainage (as Approach A does through $\beta_3$) or ignoring it (as Approach B does), it selects short, sharply-rising episodes in which drainage during the rise is negligible by construction. A candidate episode requires at least two consecutive prior months of falling or static water table (Δh ≤ 0), begins on the first rising month, extends for at most two further months while the rise continues, and must accumulate a cumulative rise of at least 50 mm; the per-episode Sy is the accumulated net recharge over the episode divided by the cumulative rise, filtered to the same 0.01–0.50 plausibility range as Approach B, with the cluster estimate taken as the median of qualifying episodes and a 1000-sample bootstrap 95% confidence interval. Forest clusters carry the same Freeman (2008) interception correction as Approach B. Approach B is the canonical method, and it is computed at two granularities that are not interchangeable. Script 17 (17_wtf_specific_yield.py) estimates it at cluster level, writing Sy_event_median to 17_wtf_01_sy_estimates.csv alongside the Approach A and C estimates and the interception-corrected forest variants; those are the values reported in Section 4.2.4 and the ones entering the WTF-Sy volumetric water balance, and outside Script 17 they are read only by the scraping-propagation analysis (Script 09b). Script 18 (18_wtf_spatial.py) estimates it at each reference well, writing Sy_median to 18_wtf_01_well_sy_estimates.csv;

that is the table the downstream consumers read — Scripts 09d, 19, 19b, 20, 29, 31 and 37b — and the cluster median of its per-well values is the specific yield behind the forest-drawdown reach λ. The two are close but not identical, and a value quoted without its file is ambiguous, so every Approach B figure in this report names the file it comes from. Approaches A and C are reported alongside as independent methods cross-checks and neither propagates to any downstream calculation. Agreement among the three approaches on the cluster ordering is taken as evidence that the specific-yield gradient is robust to the choice of estimation strategy, while divergences at individual clusters are interpreted as diagnostic of the estimators' differing sensitivities rather than as a ranking between them.

All three approaches are applied to the cluster-mean hydrographs derived in Section 3.4.1. Monthly data cannot isolate individual storm events, and the WTF method at monthly resolution is known to overestimate effective Sy because it cannot separate true gravity drainage from capillary fringe release (Healy and Cook, 2002; Scanlon et al., 2002). The resulting estimates should therefore be interpreted as upper bounds on Sy rather than point estimates of the long-term effective value.

For the Forest clusters (C4 and C5), a correction was applied for canopy interception prior to computing net recharge:

$$R_{eff} = (1 - 0.24) \cdot P - PET$$

following the site-specific interception fraction of Freeman (2008). Only the precipitation term is reduced. The PET term is retained unchanged because Thornthwaite PET is an energy-based atmospheric demand derived from temperature and day length, independent of land cover; it represents the capacity of the atmosphere to receive water vapour rather than the realized evaporative flux from any particular surface. Reducing PET to account for canopy-evaporated intercepted water would conflate atmospheric demand with actual evapotranspiration and remove the same quantity from both sides of the water balance. Without the interception correction, months in which canopy suppression of Δh is most pronounced produce individual Sy estimates above the plausibility threshold and are excluded from the event pool; applying the correction recovers these months, increases the event count, and lowers the event-median Sy at both forest clusters by repopulating the event pool with below-median values (see Section 4.2.4 and Table 8). Approach A interception-corrected variants were tested but found to be unstable in this dataset, and are not reported; the Approach B forest values cited in Section 4.2.4 use the corrected event pool.

Results of the cluster-level and well-level WTF analyses are presented in Section 4.2.4. *The cluster-level analysis was implemented in 17_wtf_specific_yield.py and the well-level spatial analysis in 18_wtf_spatial.py.*

### 3.7.4 Network-Scale Coastal-Retreat Gradient Regression

To partition each cluster's observed summer-minimum decline into a climate-driven component and a coastal-retreat-driven component, a non-linear panel regression was fitted to the full 21-year monthly record. Each well's perpendicular distance to the eroding Caernarfon Bay High Water Mark was computed within the pipeline from a committed OpenStreetMap mean-high-water coastline (EPSG:27700) restricted to the 7.8 km west-facing Newborough shoreline; the Menai Strait and Llanddwyn Island were excluded as non-eroding boundaries.

Two functional forms were fitted: a Dupuit–Forchheimer steady-state strip-aquifer form,

$$\delta(d) = max\left(\delta_0 \cdot (1 - d/L), 0\right) + c$$

and an exponential decay form,

$$\delta(d) = \delta_0 \cdot \exp(-d/L) + c$$

where d is distance from the coast, δ_0 is the coast-edge water-table deepening rate above climate, L_cg is the inland reach, and c_far is the far-field climate background. Both were fitted by profile non-linear least squares, with well and month fixed effects and a cumulative-water-balance covariate absorbed by within-well demeaning. The two functional forms were compared by the Akaike Information Criterion, which favours the exponential; the linear-with-cutoff form is retained on physical grounds. Three nested specifications were fitted to test robustness to forest-cover inclusion: the full network (67 wells, all clusters minus the BACI clearfell-zone wells and the bedrock-ridge well CEH12), a forest-free specification dropping the wells under canopy on the land-cover flag (61 wells), and a C3-only specification (26 wells: the reference wells of C3 and the extended wells assigned to it by the Pearson affinity audit, with the clearfell Impact well removed). The forest-free linear-with-cutoff fit is taken as the headline. A fourth fit controls forest cover explicitly rather than by exclusion — the full network with a canopy × time term on the wells' land-cover flag — and returns an essentially unchanged coast-edge amplitude alongside a separate −9.0 ± 2.4 mm yr⁻¹ deepening absorbed under pine, so the gradient does not depend on how forest cover is handled.

The headline fit is then applied to the per-cluster decomposition of the long-term summer-minimum slope into gradient and climate components (Section 4.10.2). The coastal term of each cluster is the mean over its wells of the capped gradient, each evaluated at the well's own distance from the shore, not the gradient at the cluster's mean distance, so a cluster that straddles the inland reach carries the share of its wells that lie inside it. *The network-scale coastal-retreat gradient regression is implemented in 25_coastal_gradient.py.*

### 3.7.5 Five-Year Mean Spring Water Level (MSL5) and Associated Forecasting Tools

The seasonal prediction equations of Section 3.6.2 and the critical rainfall threshold of Section 3.6.3 operate in summer-minimum and winter-peak space. Van Willegen et al. (2025) revisited the eco-hydrological calibration for the Newborough dune slacks using a larger vegetation dataset and a wider set of candidate hydrological metrics, and identified the unweighted five-year mean spring water level as the best-performing predictor of community-mean vegetation moisture response. This metric, here denoted MSL5, is computed across the monitoring network as a complement to the summer-minimum framework. Three analytical components are described in turn: the MSL5 metric itself, an empirical transfer function that forecasts the next spring's mean level, and a climate-scenario projection of the metric.

**The MSL5 metric.** For each well, the annual mean spring water level for hydrological year $y$ is the unweighted mean of that year's three spring values — the readings dated March, April and May, which under the convention of Section 3.1.1 carry the February–April labels. To place the entire antecedent forcing window within a single hydrological year, the van Willegen hydrological year is defined as 1 June ($y-1$) to 31 May ($y$), the readings so dated, which under the same convention carry the May–April labels — distinct from the October-commencing hydrological year used for the seasonal equations of Section 3.6.2, and adopted here so that the spring response window and its winter and spring antecedents share one annual unit. The five-year mean spring water level MSL5, indexed by its closing year, is the unweighted mean of five consecutive annual spring means. Two completeness rules are applied: all three spring months must be present in the quality-controlled record for an annual spring mean to be admitted — presence includes the isolated single-month gap-fills of Section 3.1.1, which are admitted and tracked per window rather than excluded — and all five constituent annual means must be valid for a five-year mean to be admitted; windows containing any excluded annual mean are themselves excluded rather than bridged. Across the study record the gap-fill provision is marginal: 9 of 1272 admitted annual spring means (0.7%) and 29 of 866 admitted five-year windows (3.3%) contain an interpolated month, never more than one; the per-window counts are carried in the committed Script 26 outputs. The metric is computed at every well with sufficient record and aggregated to cluster level as a fixed-effects series of the member-well values: each well is held at its own level, so a well joining the network does not shift its cluster, and the series equals the unweighted mean wherever every member reports. Levels are reported below ground surface, the convention of the source study, which the ground-referenced records of Section 3.1.1 already provide without further correction.

The Curreli et al. (2013) community levels are means of the four-year average annual minimum water level over the same 1 June–31 May hydrological year (their Table 4). The like-for-like quantity is therefore computed alongside MSL5 (Script 26): the minimum of the monthly level over each hydrological year, admitted only where every June–September

month is present in the quality-controlled record and the well reports the year's closing level, and its unweighted mean over the four years ending in each window year, every year required. A five-year window is carried beside it as a sensitivity. The community levels are applied to this series at the van Willegen quadrats, each on its own ground (the pipeline level less the quadrat's datum offset; Section 4.8.3), and not to MSL5. The cluster series of the same quantity is reported as a trajectory.

**The spring MSL transfer function.** To allow the coming spring's mean level to be forecast before the season closes, an empirical transfer function was fitted independently for each of the five clusters, predicting the annual spring mean from antecedent winter and spring conditions:

$$MSL_{y} = \alpha \cdot h_{max,\,winter} + b \cdot P_{win-spr} + \gamma \cdot PET_{win-spr} + c$$

where $MSL_{y}$ is the cluster-centroid mean of the spring water levels in hydrological year *y*; $h_{max,\,winter}$ is the cluster-centroid maximum head over October (*y*−1) to February (*y*); $P_{win-spr}$ and $PET_{win-spr}$ are cumulative precipitation and Thornthwaite PET over October (*y*−1) to April (*y*), the month of the last spring level; α_W, b and γ are the fitted coefficients on the winter peak, the rainfall total and the PET total; and *c* is a fitted intercept. The coefficients are fitted by ordinary least squares with an intercept term, using the cluster-centroid monthly series — the same baseline against which the cluster state-space coefficients of Section 3.4.1 are fitted, ensuring internal consistency between the empirical transfer function and the mechanistic model. The winter peak enters as the antecedent-state term and carries the antecedent state between seasons, in the same role as the antecedent terms in the Section 3.6.2 seasonal equations. The transfer function gives managers a forecast of the coming spring mean from readings already in hand by late winter, which can be combined with the rolling record of recent observed annual means to update the MSL5 statistic ahead of the end-of-spring readings.

**Climate-scenario projection.** To place the MSL5 metric in the context of projected climate change, the water-table response to UKCP18 RCP8.5 central-estimate scenarios for the 2050s and 2080s was computed as a sustained change in level. A scenario changes each month's forcing by

$$\Delta f(t) = \beta_1 \cdot P(t) \cdot (s_P(m) - 1) - \beta_2 \cdot PET(t) \cdot (s_{PET}(m) - 1)$$

where $s_P(m)$ and $s_{PET}(m)$ are the seasonal UKCP18 multipliers for the calendar month m of t, and, because the state-space model is linear, the difference d between a run under the scenario and a run under the observed climate obeys

$$d(t) = (1 - \beta_3) \cdot d(t - 1) + \Delta f(t),$$

in which the intercept and the drainage datum cancel. The response therefore accumulates through the drainage term: a shift held year after year reaches a level about $1/\beta_3$ times one month's change. The size of the response therefore depends on which $\beta_3$ carries it, and the

projection is given under both forms as a pair on the same wells, never averaged: Model A at the 3.7 m datum, the form the 1989–96 record supports for a sustained shift (Section 3.4.1), with the intercept form (Model B), whose $\beta_3$ is datum-free, beside it as the fast-reversion bound. Model A is projected only where Model B is identified on the record (the rule of Section 3.4.1, Script 48), so C4 is withheld under both forms. Because Model A's $\beta_3$ carries its datum, the datum the 1989–96 record supports, 2.9 m (Script 50), is reported as the projection's sensitivity: on comparison-window fits at the same 57 wells the median 2080s summer fall is 0.51 m under Model A at 3.7 m, 0.44 m under Model A at 2.9 m and 0.36 m under Model B. The scenario run is started from the sustained state and carried over the observed 2005–2026 climate, and each MSL5 window is shifted by the mean of d over its springs. The projection is a climatological response under the observed sequence of years, not a forecast of the level in any future year. The 2050s scenario changes annual P − PET by about as much as the 1989–96 dry stretch (Section 4.14); the 2080s change lies beyond anything in the record.

*The MSL5 metric and its forecasting and projection companions sit alongside, and do not replace, the summer-minimum framework of Sections 3.6 and 4.8; the two measure the same multi-year hydrological state through different seasonal windows. The MSL5 aggregation is implemented in 26_van_willegen_msl.py; the spring transfer function in Section 5 of 11_forecasting_thresholds.py; and the UKCP18 projection in 26b_van_willegen_msl_projections.py.*

### 3.7.6 Equilibrium Wetness Index and Vegetation Cross-Validation

The MSL5 metric of Section 3.7.5 measures spring wetness directly from the water-level record; the state-space model of Section 3.4.1 offers a structural counterpart computed from the fitted coefficients. Section 3.7.5 sets aside the equilibrium-displacement form of the model in favour of a seasonal perturbation for climate projection; here that same equilibrium form is used not for projection but to define a static wetness index, calibrated against MSL5 and cross-validated against independent vegetation data.

**The equilibrium wetness index.** Setting the mean monthly change to zero in the state-space model (Section 3.4.1) and solving the head-dependent drainage term for the steady-state displacement gives $h_{disp,eq} = (\beta_1 \cdot \bar{P} - \beta_2 \cdot \overline{PET})/\beta_3$, whence the equilibrium wetness index $EWI = h_{disp,eq} - z_0$, reported below ground surface to match the MSL5 convention, the records being ground-referenced (Section 3.1.1). Here $\beta_1$, $\beta_2$, $\beta_3$ are the per-well recharge, atmospheric-draw and drainage coefficients; $z_0$ is the drainage datum (3.7 m); and $\bar{P}$, $\overline{PET}$ are the full-record monthly-mean precipitation and Thornthwaite PET — the same long-term climatology used as the projection baseline in Section 3.7.5. Because the forcing is the long-term normal rather than a specific observation window, the index is climate-window-independent, estimating a well's intrinsic wetness from its response coefficients alone and computable from a shorter record than MSL5 requires. Two network tiers are distinguished: reference wells take the quality-controlled coefficients of Section 3.4.1, while extended wells

are fitted with the same shared state-space routine as a single pass without that control, and flagged accordingly. Wells whose drainage coefficient is near zero or negative (CEH13, CEH14) are excluded, that coefficient being the denominator of the equilibrium expression.

**Uncertainty in the index.** The equilibrium expression is a ratio, so uncertainty in the fitted coefficients propagates into the index. Because the drainage datum $z_0$ is a constant, subtracting it shifts the equilibrium displacement without changing its uncertainty; the propagation is therefore anchored on h_disp,eq itself rather than on the datum-referenced index value, which would understate the error at wells whose equilibrium level lies close to the datum. Two variants are computed. The first propagates the drainage coefficient alone: since h_disp,eq is inversely proportional to $\beta_3$, the relative error in $\beta_3$ passes directly into the index, and this dominant term is the variant quoted below. The second is a first-order propagation over all three coefficients treated as independent — a close approximation rather than an exact joint interval, given the mild collinearity between the recharge and atmospheric-draw terms. Standard errors for reference wells are recovered from the committed coefficient, its two-sided p-value and the sample size, the t statistic inverting exactly for the no-intercept three-predictor model; extended-tier fits return standard errors directly from the shared fitting routine.

**Calibration onto the spring-level scale.** The index sits on a systematically deeper scale than the observed spring level and is calibrated onto the MSL5 scale by ordinary least squares, MSL5 = a + b·EWI, across the wells carrying both. Consistent with MSL5's open-dune character (Section 4.8.3) and with the forest coefficients being the least constrained on the site (Sections 3.4.4, 4.9.2), the calibration is scoped to the open-dune network (clusters C1–C3); forest wells (C4, C5) are predicted but reported out of scope. Cluster membership for reference wells follows Section 3.4.1, and for extended wells the Pearson affinity analysis of Section 3.3.

**Vegetation cross-validation.** To test both metrics against the ecological target directly, the co-located Ellenberg-F moisture indicator values (the wetness indicator, scale 1–12; Hill et al., 1999) of van Willegen et al. (2025) were obtained from the associated open dataset (van Willegen et al., 2024) and aggregated to a mean per piezometer. Observed MSL5 and the equilibrium index were each regressed on mean Ellenberg-F between wells, and the difference between the two correlations — dependent, since they share the response variable — was tested by Williams' test (Williams, 1959). This cross-validation draws on an external dataset and is computed once rather than as part of the recurring pipeline.

*The equilibrium wetness index and its MSL5 comparison are implemented in 26_van_willegen_msl.py; the vegetation cross-validation uses the van Willegen open dataset.*

### 3.7.7 Wet Area from Sentinel-2

*An independent, satellite-derived measure of slack-floor wetness tests the modelled water table in space (Results, Section 4.8.5). Sentinel-2 L2A scenes over the warren (2016–2026) are read at 10 m from the Earth Search archive; scenes with tile cloud above 25 % or with less than 95 % of the warren cloud-free by the scene-classification layer are discarded, leaving 43 winter scenes each with a dipwell reading in the same month. On the open dune-slack floor of the warren — a fixed 306.97 ha mask of 30,697 cells, the planted forest excluded — each scene's near-infrared reflectance (Band 8) is normalized by that scene's own clear-floor median and split at two fixed fractions: at or below 0.50, open water; between 0.50 and 0.80, wet floor. No manual classification enters.*

*Each class's area is regressed on the reference-network median water level as area = a·exp(b·h), and the fitted curves are driven by the modelled level to test the wetted extent out of sample (Section 4.8.5). The scene selection, the two-class near-infrared classification and the per-cell switching levels are those of tools/sentinel_wet_floor.py and Scripts 45–47; this section gives the outline only. The result is an illustration of the area–level relationship, not a flood map; which individual slacks carry the water is not resolved. The state-space model driving these curves is fitted to 2005–February 2026; the wet-area analysis uses winter scenes through March 2026, that month’s water level drawn from continued monitoring held separately from the model inputs.*

## 3.8 Spatial Analysis

### 3.8.1 Baseline Fields, Combined Drivers and Network Change

This subsection presents the spatial fields that characterize the aquifer’s baseline state and its change over the monitoring period, in three groups: two baseline products derived directly from the per-well SSM parameterization; a group of maps that combine or difference the active drivers; and a group that characterizes observed network change directly, independently of the driver model. All are rendered on the standard 50 m British National Grid (EPSG:27700) site grid and site-polygon clip described in Section 3.8.4. The two baseline products are (1) the mean annual water-table surface, from which the groundwater flow-direction field is derived; and (2) the state-space water-balance residual field, which maps how completely the three-term model closes the long-term balance at each well. The water-table surface is built by kriging from the dipwells (Script 01b), the geostatistical approach generally preferred for mapping groundwater levels from sparse wells over deterministic interpolators such as inverse-distance weighting or triangulation (Varouchakis and Hristopulos, 2013; Rao et al., 2022), and one in which the ground surface can be tested as an external drift (Desbarats et al., 2002). Each well's mean head above Ordnance Datum is kriged, with a spherical variogram, to a 20 m grid over the study area drawn on the LiDAR catchments, for three states of the network: the wettest and driest tenth of months by the network-median level, and all months (the mean state, which is the baseline product). A drift

on the ground surface was tested rather than assumed: the raw DEM, the DEM smoothed to 300 m and a slack-floor envelope were each compared with ordinary kriging by leave-one-well-out error and by a physical test, whether the surface stands above the ground where Sentinel-2 never recorded water; ordinary kriging was selected. The edges of the surface are anchored rather than extrapolated: the sea along the high-water mark at a coastal head fitted to the coastal wells between mean tide level and mean high water (UKHO, 2026); and three landward parts, each kept only if it did not worsen the leave-one-out error at the wells within 1 km of it — the Afon Braint at its channel elevation, Llyn Rhos-Ddu at its gauged level, and the ridge well CEH12 at its measured depth below ground. All three were kept; a boundary along the whole ridge divide was tested and dropped because it over-predicted the forest-edge wells. No elevation-model mask is applied: the water table need not sit below the ground everywhere, and in the slacks it periodically does not. How closely the surface follows the ground, and at what scale, is measured rather than assumed: the mean surface is regressed on the DEM at each smoothing width in HEAD_DEM_SMOOTHING_M over the grid nodes inside the reference-network hull, and the reference wells' mean head on their surveyed ground elevation; Script 01b emits both (01b_14_head_vs_dem.csv, 01b_report_numbers.csv) and Section 4.9 reports them. From the interpolated head surface, groundwater flow-direction vectors are obtained as the negative head gradient,

$$Q = -K \cdot b \cdot \nabla h$$

(Bear, 1972; Freeze and Cherry, 1979), evaluated by finite differences on the kriged grid and drawn one per 100 m square, each arrow sized and coloured by the head gradient, so that steep and slack gradients are distinguished rather than normalized away. Neither a hydraulic conductivity nor an aquifer thickness enters the field, and no thickness surface is constructed. Arrows more than 500 m from any dipwell are drawn faint: there a boundary, not a measurement, sets them. The same kriged surface is the water table under the residual, drawdown and scrape maps of this section, the modern level in the Ranwell comparison (Section 5.7.9) and the head and depth views of the scenario viewer, so one water table is used throughout.

The state-space water-balance residual field expresses, at each reference well, the degree to which the three-term model closes the long-term balance. Setting the monthly change to zero at long-run equilibrium, the residual is the excess of modelled losses over modelled recharge,

$$\alpha = \left(\beta_2 \cdot \overline{PET} + \beta_3 \cdot \overline{h_{disp}}\right) - \beta_1 \cdot \overline{P}$$

where the overbars denote long-term monthly means, overline h_disp is the mean displacement above the drainage datum (Section 3.4), overline PET is Thornthwaite potential evaporation — above-canopy at the forested clusters and open-dune elsewhere — and overline P is gross rainfall; both are taken uncorrected at every cluster, including the forested ones (Section 3.8). A positive residual marks a location where the rainfall-driven

recharge captured by the model does not account for the observed head — an inadequacy in the three-term model at that location, or in principle an unmodelled input; a negative residual marks slight over-prediction. On the present evidence the diagnostic reading is the supported one (Sections 4.9.6, 5.2.1). The residual is computed from the fitted β coefficients alone — specific yield does not enter — and shown at the wells alone: no interpolation between wells predicts a withheld well's residual better than the network mean, so no surface is drawn. The head surface, flow field and residual field are presented in Section 4.9.

The remaining figures extend the analysis beyond the baseline state. The first group combines or differences the active drivers: a measured clearfell step-change map, a parametric net state map combining all five active drivers, a 2005→2025 driver-change map, and an empirical MSL5 change map. The second group characterizes observed network change directly, independently of the driver model.

The clearfell step-change map (Figure 69) is derived from the climate-corrected difference in mean water-table level between the pre-felling scraping era and the post-felling era at each well (fell_step_cc, Script 10b; Section 3.5.4), converted to millimetres. Values are interpolated to the site grid by piecewise-linear (Delaunay) interpolation, without distance weighting or smoothing, and masked to the site polygon. Well markers use the same discrete banded diverging colourmap as the surface (±150 mm symmetric scale, 14 bands) to ensure self-consistency between data points and the interpolated field. Seventy-one wells pass the quality filter (minimum six post-felling monthly observations required to estimate a stable post-felling era mean; wells with data-quality flags excluded). This map shares its underlying per-well values with the climate-corrected clearfell map of the BACI analysis (Figure 36; Section 4.6.5); the two renders differ in convention rather than data — Figure 36 uses a continuous, data-scaled colour range on a finer 40 m interpolation grid with cluster-coded markers, serving the BACI evidence, whereas the fixed banded scale here places the felling step on the common footing of the driver-map group. Implemented in *20_spatial_figures.py*.

**The parametric net state map** (Figure 70) superimposes five annual-mean driver fields on the site grid. Each field uses an exponential decay $H_0 \times \exp(-d/\lambda)$ from the relevant source geometry, where λ is the forest-drawdown reach of Section 3.8.2, ≈ 220 m, built from the C3 open-dune values given there rather than re-stated here. Its uncertainty is set out there too; the drainage datum contributes less than the conductivity or the thickness, $\beta_3$ varying by about a quarter over a plausible datum range and λ with its square root. The results below are stated to the nearest tens of metres, and the conclusions they support depend only on the reach being of order 200 m rather than of order 20 m or 2 km. The five components are: (i) forest canopy drawdown — loss, $H_0$ = 150 mm, decaying from the forest boundary, zeroed over the felling polygon; (ii) clearfell gain — positive, same $H_0$ and λ, decaying from the felling polygon boundary; (iii) scrape drain — loss, $H_0$ = 128 mm anchored to the measured CEH36 response, same λ; (iv) coastal erosion drawdown — loss, $h_0$ = 81 mm at the dune toe

from the 6 m Storm Brendan exemplar retreat and the live Script 25 fit; (v) sea-level-rise head gain — positive, +20 mm over five years at the shore. Net = SLR + clearfell − forest − erosion − scrape. A first-order superposition of single-mechanism illustrative annual-mean fields; not a closed water balance. Implemented in *20_spatial_figures.py*.

**The 2005→2025 driver-change map** (Figure 71) re-bases the net-state construction to the monitoring record, differencing each modelled driver field between its 2005 and 2025 states so that the map shows what the drivers imply should have changed over the study window rather than the accumulated state since the forest was planted. Because the standing pine canopy is at equilibrium at both endpoints, its drawdown cancels in the difference and the mature forest interior returns to near zero, as expected of an epoch difference rather than a re-drawing of Figure 70. Five fields change over the window and are summed: the clearfell gain, on the net-state construction but scaled to the measured clearfell BACI step; the dune-scrape response, with all cuts within the window contributing; a broadleaf-restock field new to this map, computed on the same exponential-decay construction but scaled by the broadleaf interception fraction (0.15, against 0.24 for pine; Komatsu et al., 2011) and by the 2005→2025 canopy-establishment increment, giving of order 56 mm of added drawdown at source and appearing as a loss patch of the opposite sign to the clearfell gain; the chronic coastal drawdown over the full twenty-year window, twenty years of the observed coast-edge decline rate $\delta_0$ fitted in Section 4.10.2, applied at the dune toe; and the sea-level-rise head gain of Section 3.8.2, which keeps its five-year window. The map is drawn on a symmetric logarithmic colour scale so that the deep coastal field and the shallower management fields both read. As a first-order superposition the map is illustrative and not a closed water balance; the broadleaf and coastal fields in particular are modelled quantities the dipwell record cannot fully resolve, and are presented as indicative. It is the modelled companion to the absolute climate-removed trend map (Figure 74) over the same window. *Implemented in 20_spatial_figures.py.*

**The empirical MSL5 change map** (Figure 72) is the observed counterpart of the modelled driver maps. For each well with a valid five-year mean spring water level (Section 3.7.5) at both window ends, it differences the current MSL5 window ending 2023 (springs 2019–2023) against the last complete pre-clearfell window ending 2017 (springs 2013–2017), taking the raw below-ground difference with no common-mode normalization: the five-year spring mean already averages inter-annual climate noise, and the slower storage decay of wet years at slow-draining wells is not removable by a common-mode subtraction. The 2023 current window is the canonical choice, supported by the three independent checks of Section 5.7.5. The second of these, reported with the results (Section 4.12), correlates the per-well 2017→2023 change with the per-well long-run summer-minimum slope of Script 25 (Pearson and Spearman, across the wells carrying both), a metric that shares no window with the MSL5 change; the 2012 drought year falls outside both windows. Values are drawn from the Script 26 per-well MSL5 series with the MSL5 well exclusions honoured, and interpolated to the site grid by IDW (power 2) with a one-cell Gaussian blur applied within

the site polygon; changes below ±25 mm — a display threshold, well inside the five-year window standard error of the metric (82–181 mm by cluster; Section 4.8.4) — are left uncoloured. Implemented in *20_spatial_figures.py*.

The four following network-change maps are constructed from the same per-well spring series (the readings dated March–May) and differ only in climate handling and frame of reference; Supplementary Material Note S10 gives a comparison table and construction schematic. Two of the maps remove the common climate signal by subtracting the network-mean spring: the differential-movement map (Figure 73) asks which wells are gaining or losing ground against the site as a whole (mm $yr^{-1}$, relative to the site mean), and the amplification field (Figure 75) asks how strongly each well swings between dry and wet springs (a ratio to the network-mean swing). Two maps work in each well's absolute frame: the absolute climate-removed trend map (Figure 74) asks how fast each well itself is drifting once the shared climate signal is removed (mm $yr^{-1}$, per well), and the dry-year spring depth map (Figure 76) asks how close a dry spring already sits to the SD15b / SD16 community-mean levels (metres below ground). Neither of the two accompanying analyses adds a further map: the per-well climate-sensitivity coefficient (Script 35) is a discrete companion to the amplification field (Figure 75), extending that metric to short-record wells, and the two-window sensitivity demonstration (Script 34) is presented in Supplementary Material Note S11.

**The differential-movement map** (Figure 73) isolates the secular component of spring water-table change that the two-window MSL5 comparison cannot resolve. For each well a spring value is formed for every year as the mean of the available readings dated March–May; the well's anomaly is its spring level minus the site-mean spring level in the same year, the subtraction removing the common climate signal so that what remains is movement relative to the network as a whole. The site-mean reference panel is restricted to the wells present in a fixed majority fraction of the period's springs, so the reference does not drift with coverage. The per-well metric is the ordinary-least-squares slope of this anomaly against year, in millimetres per year, fitted over two periods — 2011–2025 (primary) and 2005–2025 (robustness). Significance is assessed by a lag-1 autoregression-corrected t-test on the effective sample size and cross-checked against a moving-block bootstrap confidence interval; significant wells are drawn solid and non-significant wells hollow. The Llyn Rhos-Ddu lake gauge is excluded throughout; CEH13 and CEH14, whose drainage coefficients are near zero or negative, are excluded from the mapped per-well trends and retained only in the site-mean reference panel. The per-well slopes are interpolated to the 50 m grid by linear interpolation within the convex hull of the wells, extended 100 m beyond the hull, and clipped to the site outline. Because the anomaly is referenced to the network mean, this map shows relative movement — which well gains or loses ground against the site as a whole — rather than the absolute rate of change at each well, which is mapped separately (Figure 74). *Implemented in 32_differential_movement.py.*

**The absolute climate-removed trend map** (Figure 74) complements the differential map by giving each well's secular rate of change on its own terms, with the shared climate signal removed but without re-referencing to the network mean. Where the differential map subtracts the site-mean spring level — which over a wet-spring-lifted window can invert the reading, so that a well holding its position against a rising mean appears to fall — the absolute map removes only the variance attributable to an external climate index, leaving the real site-wide recharge decline and coastal drying in the result. Spring values are formed as for the differential map. The external index is the spring climatic water balance, the sum of precipitation minus Thornthwaite PET over March–May, taken from the RAF Valley record contemporaneously with the spring level and without lag. For each well a single joint regression is fitted, $h(t) = a + b \cdot CWB(t) + c \cdot t$, and the secular trend is the fitted time coefficient c in millimetres per year; fitting the climate index and the linear trend together orthogonalizes the two, so the climate term cannot absorb the secular signal. A two-step alternative — regressing level on the climate index and then fitting a trend to the residual — was rejected because over a climatically non-stationary window the climate index itself trends and absorbs the real drying before the slope is measured. A coverage filter admits a well to a window only where it carries at least one spring observation before 2011 and its observed record spans at least 80% of the window, excluding short-record wells whose trend would otherwise be fitted to a truncated drought-recovery trajectory. Two periods are fitted — 2005–2025 (primary) and 2011–2025 (robustness); the longer window is primary here, in contrast to the differential map, because the shorter window is dominated by post-2011 short-record wells. Significance, exclusions and marker convention follow the differential map. The per-well slopes are interpolated to the 50 m grid by linear interpolation within the convex hull of the wells, extended 100 m beyond the hull to reach the coastal margin, and clipped to the site polygon. The per-cluster trends are reported in Section 4.12. I*mplemented in 36_absolute_climate_trend.py.*

**The climate-response envelope** (Figure 75) characterizes the range the spring water table moves between its dry and wet extremes and yields two window-independent products. Spring values are formed as for the differential map. Dry and wet extreme states are taken as the per-well mean spring level over fixed sets of extreme years, selected by site-mean spring extremity and antecedent-rainfall consistency — dry {2011, 2012, 2019} and wet {2014, 2016, 2021, 2024}; the year 2006 is excluded as a wet extreme because its 2004–2005 antecedent was the driest in the record and the slow-draining forest wells had not refilled by that spring. Each well's swing is its wet state minus its dry state, and its amplification coefficient is that swing divided by the network-mean swing, with the common-mode swing removed, so that values above one magnify the shared climate forcing and below one damp it. The drought-floor surface (Figure 76) is the dry-state depth to water, retained in absolute depth below ground and contoured against the Curreli et al. (2013) SD15b (0.61 m) and SD16 (0.98 m) community-mean levels, which are absolute distances to surface rather than relative quantities. As in the differential map, the lake gauge is excluded; unlike it, CEH13

and CEH14 are included, because the coefficient is observational. The amplification field is interpolated to the 50 m grid by linear interpolation within the convex hull of the wells, extended 100 m beyond the hull; the drought-floor surface is likewise linearly interpolated with a ridge mask applied to suppress extrapolation onto the bedrock ridge, and the single raised inter-slack well CEH10 is shown as a distinct slack-edge marker but excluded from the interpolated slack-floor surface, which would otherwise smear its raised-ground depth across the neighbouring slacks. *Implemented in 33_envelope_amplification.py.*

The short-coverage wells that the amplification field (Figure 75) cannot interpolate are not dropped but shown on it as filled triangles, each carrying the same co-temporal climate-sensitivity coefficient (utils/envelope_metric.py) used for the surface: the well's dry-to-wet spring swing normalized against a reference core of full-coverage wells, with the core swing recomputed over that well's own extreme years, so that wells measured on different subsets of extreme springs remain comparable. Each coefficient carries a confidence tier (A: at least two dry and two wet years — the wells that build the surface; B: at least one of each; C: a single year on one side) and a 90% interval from refitting with each extreme year withheld in turn. Script 35 develops this coefficient as a standalone per-well product for Paper 1 and cross-calibrates it against the independently fitted SSM response — it correlates with the per-well $\beta_2$ and $\beta_3$ coefficients, and is reported as consistent with, rather than confirming, that response, since at the short-record wells that are its main use case the β coefficients are not available for direct cross-check. Computed across the whole network the coefficient reproduces the earlier matched-window amplification ($r \approx 0.97$), indicating that the field is not an artefact of the window choice. *Implemented in 35_per_well_amplification.py.*

**The two-window sensitivity demonstration** (Script 34) tests directly whether the two-window MSL5 change measures secular decline or merely the choice of windows. Per-well MSL5 for a window ending in year Y is the mean spring level over years Y−4 to Y (from the committed van Willegen aggregation, 26_msl_annual_per_well.csv), and a well qualifies for a window only if all five spring-years are present. For each ordered pair of windows the site-mean change is the mean, over the common panel of wells qualifying in both windows, of the per-well change, with the panel held fixed across the pair so composition change cannot inflate the result; admissible pairs are those whose common panel reaches at least forty wells, which excludes the thin early baselines (every window ending 2014 or earlier). Placing the single 2017→2023 comparison inside the envelope of every admissible pair shows the site-mean change reverses sign and spans −161 to +229 mm, so the −105 mm headline is one interior cell of a wide distribution and the two-window method characterizes window choice as much as secular change. The full envelope and its figure are given in Supplementary Material Note S11 (Figure S11.1).

**Driver validation (Script 37).** The 2005→2025 driver-change map described above presents what the modelled fields imply should have changed; Script 37 tests that implication against the observed change directly. Per-well climate-corrected endpoint

differences (the absolute climate-removed trend of Figure 74) are regressed on the same unit driver fields at their modelled amplitudes, per window, with a free spatially-uniform intercept, using ordinary least squares with HC3 heteroskedasticity-robust standard errors. Three windows are fitted: a pre-intervention window (2006–2012, coastal field only), a post-clearfell window (2018–2025, coastal and clearfell fields), and the full record (2005–2025). The intercept absorbs the site-wide recharge decline, so a uniform background drying cannot be attributed to the coastal gradient; the resulting scale factors are dimensionless multipliers on the modelled amplitude, unity meaning the aquifer feels exactly the modelled field. C2 Dune, carrying no driver field, serves as a driver-free negative control, and the sluice-controlled C1 Lake Edge wells are excluded from the fit as management-influenced. This is a validation, not a fitting step: it sets no amplitude used elsewhere. Its outcome is bounded: in the window where each driver acts, its scale factor is distinguishable from zero but not from unity, so the network detects the modelled fields without resolving their amplitude, and the common-mode intercept remains the larger term in every window (Section 4.12.1). The modelled coastal amplitude in particular cannot be independently recovered from the monitoring record (Section 5.7); the coastal shape it tests is moreover fitted from the same network (Section 3.7.4), making the validation partly self-confirming, with only an expanding-window trajectory of the implied coastal rate serving as an independent check (Figure 77). *Implemented in 37_driver_validation.py.*

**Comparative driver footing (Script 37b).** The same modelled fields are placed on a common footing over the 2005→2025 window, with each driver entered as its gain and loss components (forest as clearfell and broadleaf restock, scraping as on-site rise and off-site drain cone, coast as sea-level rise and erosion drawdown). In this report the drivers are compared on three common currencies — peak local head change (mm), area-integrated volume, and crossings of the SD15b / SD16 community-mean levels — shown as grouped bars in Figure 78, with amplitudes taken from the observed BACI anchors and the modelled Script 20 fields, not the Script 37 scale factors; the peak-and-reach view of the same drivers, as depth-against-distance profiles, is the reach panel of Figure 83. The area-integrated volumetric construction that underlies the net scraping-versus-coast ratio reported in Section 4.12 is that of Script 37b (37b_driver_footing.csv): each field is integrated over the site mask and converted to a volume through a representative specific yield; fields are superposed linearly, a first-order construction that is an upper bound in overlap zones. *Implemented in 37b_driver_footing.py.*

### 3.8.2 Drawdown-Field visualization

Six spatial fields visualize how localized perturbations to the water table propagate through the dune aquifer: the drawdown imposed by canopy interception within the forest block, the drawdown imposed by shoreline retreat at the western dune margin, the head gain produced by sea-level rise at the coastal boundary, the cell-by-cell difference of the latter two, the

drawdown imposed by the eight mapped dune scrapes, and the combined drawdown imposed on the clearfell pre-fell baseline. All six are rendered as filled or line contours over the 2 m LiDAR digital elevation model hillshade and clipped to the study-area outline, so that the forest, erosion, sea-level-rise and net maps are mutually comparable. They are single-mechanism illustrative constructions: each isolates one process to show its spatial reach, and is not a closed water-balance solution.

The forest drawdown is propagated outward from the forest boundary as an exponential decay, $\Delta h(d) = H_0 \cdot \exp(-d/\lambda)$, with edge magnitude $H_0$ = 150 mm taken from the canopy-interception literature (Freeman, 2008) and characteristic decay length $\lambda = \sqrt{(D/\beta_3)} \approx 220$ m. The hydraulic diffusivity D = Kb/Sy and the SSM drainage coefficient $\beta_3$ are sourced from the open-dune medium *through which* the drawdown propagates (the C3 Western Residual cluster) rather than from the forest interior: K = 6.0 m day$^{-1}$, b = 5 m, Sy = 0.298 (the C3 per-well median, 18_wtf_01_well_sy_estimates.csv) and $\beta_3$ = 0.0620 month$^{-1}$. The reach is a scaling estimate rather than a calibrated length — across the conductivity range reported for dune sand (2–20 m day$^{-1}$) λ spans roughly 130–410 m, and across an aquifer thickness of 3–8 m it spans 170–280 m — and the live value is carried in 20_report_numbers.csv, quoted here to the nearest 10 m. Distance from the forest edge is computed as a flow-direction-weighted cost-distance over the DEM, so that the drawdown follows hydraulically plausible paths rather than straight-line Euclidean distance.

The coastal-erosion field visualizes the drawdown produced by a single discrete shoreline-retreat event of 6 m, a Storm-Brendan-scale exemplar of the episodic, storm-driven retreat process at the western Newborough margin (Pye and Blott, 2024; Forgrave, 2020). The coast-edge anomaly fitted in Section 3.7.4 is a deepening per year. Dividing it by the shoreline retreat rate over the same years converts it to a deepening per metre of shoreline lost, and multiplying by the 6 m event gives the edge drawdown. The retreat rate is E_c = 2.32 m yr$^{-1}$, measured from shoreline positions digitized from dated aerial imagery over 2006–2026. Both quantities in that ratio are measured by this study: the anomaly from the water-table record, the retreat rate from shoreline position. They come from different data, so the ratio is not circular; but neither term is now external to the study, and the edge drawdown inherits the uncertainty of both. The field decays inland from the dune-toe front following the forest-free linear-capped coastal-retreat gradient of that fit, $\Delta h(d) = h_0 \cdot \max(1 - d/L_{cg}, 0)$, with $\delta_0$ and L_cg read live from it. The dune-toe front is derived by extracting the DEM contour at the waterline and offsetting it 100 m inland to where the aquifer outcrops; the seaward side is set to zero. The field references the eroding Caernarfon Bay (south-western) shore only.

The retreat rate is measured between dune-edge lines throughout: the dune edge surveyed on the 1899 Ordnance Survey sheet (Anglesey XXV.NW, revised 1899) and four lines digitized from dated aerial imagery (2006, 2017, 2021 and 2026). The indicator is therefore consistent across the series, but the 1899 line alone derives from a field survey rather than from

imagery, and it was georeferenced to OSGB36 from 27 ground control points at an RMSE of 5.95 m (3.27 m excluding two outliers). That error applies to one end of the century comparison only. Across the 107 years separating the 1899 and 2006 lines it is equivalent to 0.06 m yr⁻¹, so the long-run rate is 0.64 ± 0.06 m yr⁻¹ against 2.32 m yr⁻¹ for 2006–2026, and the contrast between the two is not sensitive to it; over the shorter modern intervals a positional error of that size would dominate, which is why the 1899 line does not enter them. Registration accuracy for the imagery-derived lines is not independently recorded; a blind repeat tracing of the 2006 frame, with the first line not loaded, returns a median offset of 2.34 m and a 95th percentile of 5.53 m. Within the modern series the three intervals return 1.81, 2.37 and 2.14 m yr⁻¹ (2006–2017, 2017–2021 and 2021–2026, on the frontage all four lines share): a steady rate carrying noise rather than a monotonic climb. The acceleration the record resolves is between the century mean and the modern window, and a mean across 107 years is not the like-for-like comparator of a 20-year one: retreat at this frontage is episodic, so a two-decade window is short enough for its own storm history to set its rate. The century contrast is consistent with an increase in retreat rate, but does not on its own establish when that increase occurred, or whether it was progressive.

The sea-level-rise field models the head gain at the coast as a finite-window boundary perturbation diffusing inland, $\Delta h(d) = \Delta SLR \cdot \mathrm{erfc}(d/(2\sqrt{(Dt)}))$, with $\Delta SLR = 0.02$ m accumulated over a 5-year window (≈ 4 mm yr⁻¹, the north-Wales near-term UKCP18 central estimate). The diffusivity $D = Kb/Sy$ uses the same C3 specific yield as the forest and erosion fields; the diffusive length-scale $2\sqrt{(Dt)} \approx 860$ m sets the inland reach. The boundary is referenced to mean sea level (0 m AOD), distinct from the dune-toe front of the erosion field — the water table is pinned to mean sea level at the coast, not at the dune toe.

The net coastal field is the cell-by-cell difference between the sea-level-rise head gain and the erosion drawdown over a matched 5-year window. It reuses the two component fields and is subject to the same single-shore, single-mechanism assumptions; it is a difference of two illustrative fields, not a mass-balance solution.

The dune-scrape drawdown field renders the April 2015 CEH36 scrape as a steady-state topographic drain, the same leaky-aquifer construction as the forest field. The edge magnitude $H_0$ is set not to an assumed excavation depth but to the directly measured CEH36 response (the paired-BACI pure-scraping step, Section 4.5), read live; the unsurveyed excavation depth is recovered as the inferred quantity $d_{ex} = H_0 / Sy$ through the open-dune specific yield. The field decays from the excavation along a flow-weighted cost-distance with the same characteristic length $\lambda \approx 220$ m as the forest field, sourced from the C3 open-dune medium through which it propagates. The excavation interior is masked: the scraped slack itself rises (Section 5.4.1), so the rendered field is the surrounding drawdown — the dipole counterpart of the slack rise — not a drawdown at the scraped site.

The clearfell pre-fell baseline field combines the two non-felling drawdown drivers bearing on the December 2017 clearfell compartment — the 2015 CEH36 scrape and a single Storm-

Brendan-scale shoreline-retreat event — as the cell-by-cell sum of the scrape-drain and coastal-erosion fields. Both are head losses, so the combined field is rendered on the same sequential drawdown scale as its components rather than on a diverging scale; sea-level rise is excluded by construction, the field being the footprint of the two confounders the clearfell BACI must separate from the felling signal (Section 4.6) rather than a coastal mass balance. It is a first-order superposition of two illustrative fields, not a closed water-balance solution. *The drawdown-field visualizations are implemented in 20_spatial_figures.py.*

### 3.8.3 Drainage Half-life Field

The drainage decay half-life t½ = ln 2/$\beta_3$, defined in Section 3.4.5, is computed at every reference well for which a positive $\beta_3$ is available. Two wells are withheld: CEH14 (negative $\beta_3$, $t_{\frac{1}{2}}$ undefined) and CEH13 (near-zero $\beta_3$, $t_{\frac{1}{2}}$ approximately 108 months on the full record, more than twice the next-longest in the network, and not identified by either code in Section 3.4), both ridge-flank settings where the displacement formulation does not apply. The per-well values are interpolated to the standard site grid by piecewise-linear (Delaunay) interpolation and rendered over the LiDAR hillshade on the same convention as the coefficient atlas (Section 3.4.3).

### 3.8.4 Scenario Framework and Interactive Viewer

A well-level scenario framework was implemented to translate the per-well SSM parameterization into site-wide spatial summaries of head change (Δh) under prescribed climate and forest management perturbations. For each well, the scenario head change relative to baseline was computed directly from the per-well SSM equation as:

$$\Delta h = \left(\beta_1 \cdot P_{eff,sc} - \beta_{2,sc} \cdot PET_{sc} - \beta_3 \cdot h_{disp}\right) - \left(\beta_1 \cdot P_{eff,0} - \beta_2 \cdot PET_0 - \beta_3 \cdot h_{disp}\right)$$

where P_eff applies the canopy interception factor at forest wells (C4 and C5; 24% under the Corsican pine baseline following Freeman (2008)) and is left unchanged under the clearfell, thinning and broadleaf scenarios, because the felled well's recharge did not rise after felling (Section 4.13.2); returning the intercepted rain (0% interception under full clearfell; 15% annual mean under broadleaf, following Komatsu et al. (2011)) gives the upper end of each range. $\beta_2$,sc applies a scenario-specific multiplier at forest wells. For full clearfell it is the felled well's (WMC3) post-felling $\beta_2$ ratio net of the climate-control drift ($\beta_2$ × 0.952; Section 4.13.2). The thinning multiplier is set as half the clearfell perturbation, an upper bound (Gebhardt et al., 2014).

Under broadleaf conversion, $\beta_2$ is scaled seasonally with phenological windows: a leaf-on (May–October) multiplier applied above the Corsican pine baseline, and a leaf-off (November–April) multiplier applied below it, reflecting the seasonally asymmetric transpiration signature of deciduous canopy compared to the year-round pine. The two phenological multipliers are estimated from the BACI Edge-tier monthly $\beta_2$ profile under the

assumption that broadleaf canopy approaches the leaf-off canopy state in winter and a moderate-transpiration state in summer. Canonical values and their derivation are presented in Section 4.13.2. The same clearfell multiplier is applied at both C4 and C5 forest wells; no felled C5 wells were available in the BACI design, so the C5 application represents extrapolation from the felled well, a C3 well at the C4/C5 transition, rather than direct measurement.

The $\beta_3 \cdot h_disp$ term is identical in the scenario and baseline expressions and cancels in the difference. Specific yield Sy does not appear in the equation because the fitted β coefficients already embed the aquifer storage response through the SSM fit to observed head data; the per-well WTF-derived specific yield values (Section 3.7.3) are retained as supplementary display information in the interactive scenario viewer, not as a divisor on Δh.

Six scenarios were evaluated: baseline; UKCP18 2050s (winter P +10%, summer P −15%, winter PET +5%, summer PET +20%, RCP8.5 central estimate for Wales, informed by the UKCP18 Regional 12 km ensemble of Met Office (2018) and the CHESS-SCAPE bias-corrected projections of Robinson et al., 2023); UKCP18 2080s (winter P +20%, summer P −30%, winter PET +10%, summer PET +35%, same sources); full clearfell; 50% thinning; and broadleaf conversion. The two climate scenarios are evaluated as the sustained level response of Section 3.7.5, per well under both forms, as a pair on the same wells, Model A at the 3.7 m datum with Model B beside it (extended wells on their cluster's), and reported as the mean over the winter, summer or all months; the forestry scenarios are likewise carried from the one-month forcing change of the expression above to their sustained level response (Section 3.7.5), reported under Model B and, beside it, Model A, under which the Main Forest cluster withheld by Model B also appears.

The per-well Δh field was interpolated to a site-wide 50 m grid on the British National Grid (EPSG:27700) using Delaunay triangulation with linear barycentric weighting (scipy.interpolate.griddata, method="linear"), masked to the rectangular sea-boundary extent; the resulting field is rendered in the interactive scenario viewer described below rather than as a static map. Two wells were excluded from the interpolation: CEH12 (bedrock-anchored, geologically distinct from the dune aquifer) and CEH15 (forest slack edge, transitional between Forest and Western Residual clusters). The hydraulic conductivity reported for the site (K = 6.0 m/day; Betson et al., 2002) is retained as context for the flow arrows of Figure 56 and does not enter the Δh calculation.

The accompanying interactive scenario viewer (Hollingham, 2026; scenario_viewer.html) renders the per-well Δh field by inverse-distance weighting (power 1, eight nearest neighbours) for responsiveness under slider-driven re-interpolation. Its head and depth-below-surface views use the kriged water table instead: the kriging weights are computed once in Python and embedded, and the browser applies them to each well's scenario head, which reproduces the kriged surface exactly because kriging is linear in the data. Depth is the DEM minus that head, with inter-dune ridge cells masked (DEM more than 1.0 m above

the level interpolated from the wells' ground elevations). The Δh map and the head and depth views therefore use different interpolators, and the viewer states which.

A two-dimensional steady-state groundwater flow model in the Helmholtz form, with Dirichlet head at the sea boundaries and no flow at the ridge (Bear, 1972; Freeze and Cherry, 1979), was explored during development as an alternative spatial framework and not adopted:

$$\nabla \cdot (T \cdot \nabla h) - \beta_3 \cdot h = -(\beta_1 \cdot P_{eff} - \beta_2 \cdot PET)$$

The form solves directly for the equilibrium head field h(x, y) in level rather than displacement terms, the constant datum offset being absorbed into the boundary specification. It requires a transmissivity field T = K·b, and the site constrains K by a single tracer test (Betson et al., 2002) and b by four borehole logs across 700 ha (Bristow, 2003, Volume 4; Supplementary Material S2.1); a field built from those constraints claims spatial flow dynamics the monitoring data cannot support, which is the difficulty that also limited the MODFLOW model of Betson et al. (2002) at this site. The model is not part of the pipeline and no result from it is reported. The well-level interpolation framework described above was adopted instead, on the basis that it retains the physical content the SSM parameterization supports without asserting a flow field the data do not constrain.

All scenario outputs are provided through a self-contained interactive viewer allowing site-wide and cluster-level Δh responses to be explored under any combination of the six defined scenarios and their underlying parameter sliders (scenario_viewer.html; available at https://newbroman.github.io/Newborough_Hydrology/scenario_viewer.html).

*The spatial groundwater analysis and scenario mapping are implemented in 19_spatial_groundwater.py. Publication-quality spatial figures are produced by 20_spatial_figures.py. The forestry scenario equilibrium framework and the broadleaf seasonal $\beta_2$ derivation are implemented in 21_forestry_scenarios.py.*

## 3.9 Computational Implementation

### 3.9.1 Data Pre-processing and Time-Series Management

All data pre-processing, statistical modelling, and spatial mapping were executed within a Python programming environment. Time-series data manipulation, interpolation of missing temporal records, and standardization were performed using the pandas and NumPy libraries.

*Data preparation including quality control, gap-filling and the canonical geometry derivation is implemented in 01_data_prep.py; baseline climate figures and network summary statistics are produced by 00_climate_summary.py.*

### 3.9.2 Statistical and Predictive Modelling

The core parameterization of the aquifer mechanics was executed using the statsmodels module. For the continuous state-space modelling (Section 3.4), the Ordinary Least Squares (OLS) algorithm was constrained to fit through the origin (omitting the intercept constant) to enforce physical mass balance. Conversely, seasonal prediction equations predicting peak flood and summer drought minima (Section 3.6.2) used unconstrained OLS regressions. The statsmodels architecture was further used to extract 95% confidence intervals and p-values for all SSM and ANCOVA coefficients, including the era-specific $\beta_3$ regressions of Section 3.5.3 and the ANCOVA-BACI $\beta$ coefficients of Section 3.5.4. Iterative forecasting metrics, including the Nash–Sutcliffe Efficiency (NSE) and Root Mean Square Error (RMSE), were computed via NumPy array operations to benchmark the state-space model against traditional regression approaches (Section 3.4.2).

*State-space model fitting and the shared SSM utility functions used by all downstream scripts are implemented in utils/model_utils.py. Shared pipeline constants — including DRAINAGE_DATUM, HEADLINE_LAG, FOREST_INTERCEPTION, and the canonical cluster labels and colours — are defined in utils/config.py.*

### 3.9.3 Geospatial Processing and Cartographic Visualization

The integration of the topographical data with the monitoring network was managed through a dedicated spatial stack. GeoPandas, through the shared reader utils/kml_io.py, was used to parse external KML feature boundaries (e.g. the clearfell perimeter and hydrological boundaries) and standardize all spatial geometries to the British National Grid coordinate reference system (EPSG:27700).

The high-resolution Digital Elevation Model (DEM) was ingested and masked using the rasterio library. To visualize the localized dune topography effectively, the DEM arrays were rendered using a dual-slope normalization technique (TwoSlopeNorm) provided by matplotlib, anchoring the visual transition at specific elevation thresholds to highlight critical hydrological features. Contextual basemaps were fetched via contextily when necessary, and label placement and occlusion avoidance were handled using the adjustText library.

*Spatial coefficient mapping and the SSM spatial fits (Section 3.4.3) are implemented in 07_spatial_coefficients.py. Shared geospatial utilities used across the spatial analyses are in utils/paths.py (output path conventions) and utils/map_utils.py (map rendering helpers).*

## 3.10 Data requirements and Replicability

A deliberate objective of this study was to demonstrate the hydrological insight achievable from minimal, low-cost monitoring infrastructure. All primary data inputs consist of manual dipwell measurements, recorded with a standard electrical contact meter or tape measure, and publicly available monthly climate summaries from the Met Office historic station dataset. No specialist survey equipment, telemetry, or laboratory analysis was required. Potential evapotranspiration was estimated using the Thornthwaite and Mather (1957)

method specifically because it requires only mean monthly air temperature — a variable recorded at every Met Office station and at the majority of automatic weather stations worldwide. Methods requiring wind speed, humidity, or radiation data were deliberately avoided as these would limit replicability to sites with more comprehensive climate instrumentation.

The analytical framework presented here could be deployed at any comparable lowland wetland, dune slack, or shallow unconfined aquifer system with a dipwell network and access to a nearby climate station. The principal constraint on analysis quality is record length rather than measurement sophistication; the state-space modelling and BACI analyses both benefit substantially from the long baseline available here, but meaningful cluster separation and mechanistic coefficient estimation are likely achievable with records of eight to ten years or more.

# 4. Results

## 4.1 Climate and Well Network Characterization

### 4.1.1 Climate Record

The RAF Valley climate station provides a continuous monthly record from December 1930 to February 2026 (Section 3.1.2). Over the well-record period (April 2005 to February 2026), mean annual precipitation was 891 mm and mean annual Thornthwaite PET was 653 mm, giving a mean P/PET ratio of 1.37 (Table 1). The wettest year on record within this period was 2008 (1,040 mm), followed by 2012 (1,022 mm); the driest complete year was 2010 (736 mm). Two years recorded a P/PET ratio below 1.2 — indicating near-balance between precipitation and evaporative demand — namely 2011 and 2022, with 2022 recording the lowest ratio of the monitoring period at 1.11; 2010 (1.20) and 2007 (1.22) sat close to that line. There is no meaningful difference in mean annual precipitation between the pre-felling (2006–2017: 887 mm) and post-felling (2018–2025: 896 mm) periods.

PET shows strong seasonality and a small positive inter-annual trend across the record, with the 12-month rolling mean remaining stable at approximately 49–58 mm/month throughout (Figure 4b).

The levelled cumulative water balance series (Figure 4c), expressed relative to the study-period mean of 19.51 mm/month, oscillates with a clear seasonal signal — rising sharply each winter as precipitation exceeds PET and falling each summer — while tracking the multi-year rhythm of wetter and drier phases. No sustained multi-year drift away from the baseline is evident. The water table tracks this movement (Figure 4d). Linear regression of the cumulative balance against the unsmoothed network mean water level at zero lag yielded $R^2 = 0.528$. This close correspondence between cumulative climate forcing and water table behaviour provided the empirical motivation for the state-space modelling framework described in Section 3.4.

Mean monthly rainfall peaks in October–December (around 101–109 mm/month) and falls to its minimum in April (around 46 mm/month), whilst Thornthwaite (1957) PET peaks in July (around 107 mm/month) and is negligible from November to February (Figure 4b). Over the monitoring period, mean annual winter rainfall (October–March) was 518 mm and mean annual summer rainfall (April–September) was 373 mm, together summing to the 891 mm annual mean.

Thornthwaite PET was compared with an FAO-56 Penman–Monteith grass-reference series built from RAF Valley's hourly records (Script 50, E13b). Over 1957–2023 the annual totals agree within about 1% (Thornthwaite 640 mm, Penman–Monteith 633 mm), and they agree as closely over 2002–2023, when the station measured radiation. Month by month they do not. Penman–Monteith rises earlier in spring and peaks lower: it is about 30% above

Thornthwaite from January to April and about 16% below it from July to October. Over 2002–2023 Penman–Monteith rose by about 25 mm per decade, against about 7 mm for Thornthwaite. The comparison cannot be carried further back, because the wind and dew-point records before 2002 contain instrument changes. Refitting the model with Penman–Monteith in place of Thornthwaite did not improve it: forward forecasts were poorer at the clusters that carry the 2006–08 excess, and the month-by-month pattern in the residual (Section 5.2.1) grew larger, not smaller.

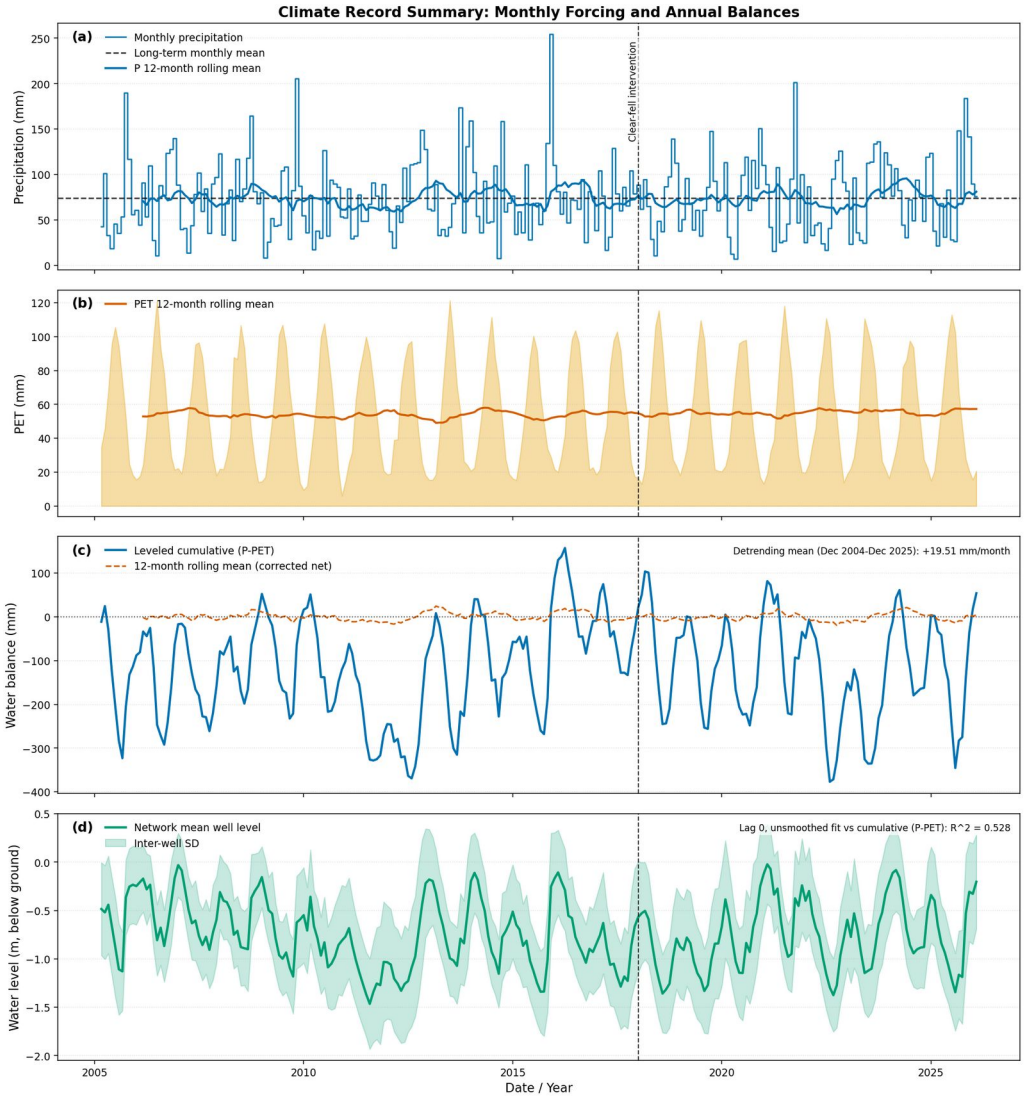



***Figure 4:*** *Climate record summary for the well-monitoring period (April 2005 – February 2026), derived from RAF Valley Met Office station. Panel (a): monthly precipitation (mm) with long-term monthly mean indicated by dashed line and 12-month rolling mean as solid line. Panel (b): monthly Thornthwaite PET (mm) as filled area with 12-month rolling mean overlaid. Panel (c): levelled cumulative water balance (P − PET, mm), corrected by subtracting the long-term mean monthly net balance to centre the series on zero; the dashed orange line shows the 12-month rolling mean of the corrected net balance. The detrending mean is shown inset. Panel (d): network mean monthly water level (m below ground) across the 66-well reference network, with inter-well standard deviation as shaded envelope; the R² inset gives the coefficient of determination from linear regression of the cumulative balance against the unsmoothed network mean at zero lag. The vertical dashed line in all four panels marks the December 2017 clearfell. (Source: 00_01_climate_timeseries_short.png)*

| Year | Annual P mm | Annual PET mm | Months complete | P PET ratio |
|---|---|---|---|---|
| 2005 | 693.1 | 612.0 | 10 | — |
| 2006 | 954.0 | 669.8 | 12 | 1.42 |
| 2007 | 773.7 | 636.1 | 12 | 1.22 |
| 2008 | 1039.7 | 639.1 | 12 | 1.63 |
| 2009 | 871.3 | 653.2 | 12 | 1.33 |
| 2010 | 736.2 | 612.4 | 12 | 1.20 |
| 2011 | 760.6 | 673.8 | 12 | 1.13 |
| 2012 | 1021.7 | 616.0 | 12 | 1.66 |
| 2013 | 844.0 | 653.0 | 12 | 1.29 |
| 2014 | 943.0 | 667.8 | 12 | 1.41 |
| 2015 | 974.0 | 623.0 | 12 | 1.56 |
| 2016 | 858.4 | 653.0 | 12 | 1.31 |
| 2017 | 871.0 | 658.7 | 12 | 1.32 |
| 2018 | 873.2 | 655.8 | 12 | 1.33 |
| 2019 | 832.0 | 648.1 | 12 | 1.28 |
| 2020 | 944.2 | 658.2 | 12 | 1.43 |
| 2021 | 906.6 | 660.5 | 12 | 1.37 |
| 2022 | 747.2 | 671.7 | 12 | 1.11 |
| 2023 | 986.2 | 678.0 | 12 | 1.45 |
| 2024 | 911.2 | 643.1 | 12 | 1.42 |
| 2025 | 969.2 | 686.5 | 12 | 1.41 |
| 2026 | 165.6 | 36.1 | 2 | — |
| Long-term mean | 890.9 | 652.9 | | 1.37 |

***Table 1:*** *Annual climate summary for RAF Valley station over the well-monitoring period. Annual precipitation (P) and Thornthwaite PET are in mm. P/PET ratio below 1.0 indicates evaporative demand exceeds precipitation. Months_complete gives the number of months with valid data in that year. Partial years (2005: March–December only, ten months; 2026: January–February only) are included for completeness but no annual P/PET ratio is calculated for these years (shown as —). Long-term mean row is calculated from complete years 2006–2025 only. (Source: 00_01_annual_climate_summary_short.csv).*

The RAF Valley record also shows a statistically significant warming trend in mean summer maximum temperature over the full 95-year record (slope +0.014°C yr⁻¹, $R^2 = 0.142$, $p < 0.001$; Figure 5). The post-2013 period has run 0.94°C above the pre-2013 baseline mean of 17.89°C, with six of the thirteen summers between 2013 and 2025 exceeding this baseline by more than 1°C (2013, 2018, 2021, 2022, 2023 and 2025). The largest post-2013 anomalies occurred in 2018 (+2.04°C), 2025 (+1.84°C) and 2022 (+1.77°C). The year 2022 combined a +1.77°C temperature anomaly with only 747 mm annual rainfall — the most severe dual-stress year in the monitoring period. Annual rainfall shows no significant trend over the full record (1931–2025, $p = 0.57$).

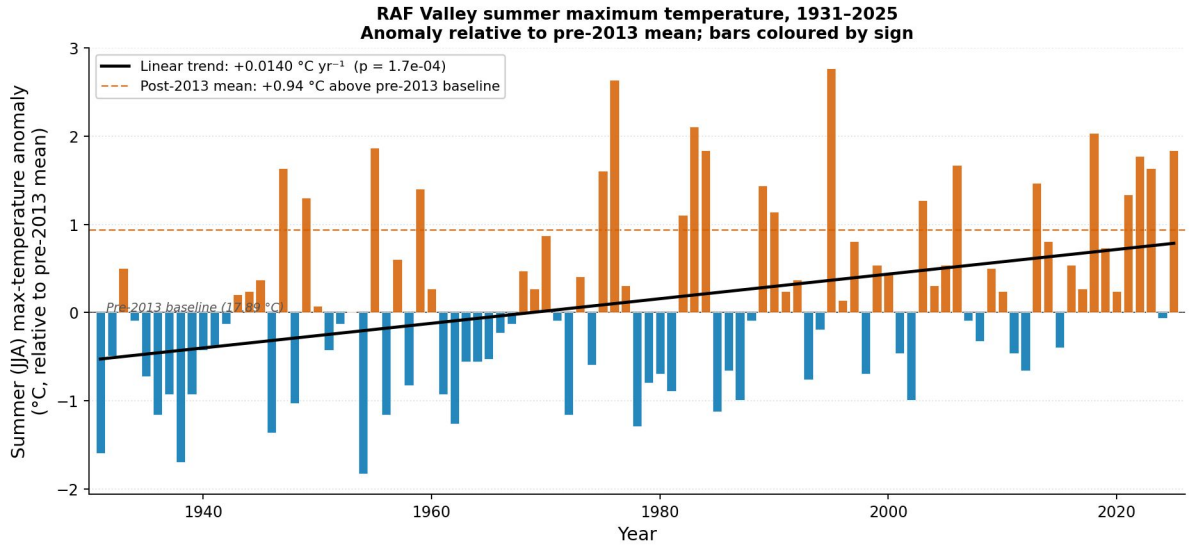



***Figure 5:*** *Summer maximum temperature anomaly at RAF Valley, 1931–2025, expressed as departure from the pre-2013 mean (17.89°C). Bars coloured by sign: orange = above baseline, blue = below. Solid black line: OLS linear trend (+0.014°C yr⁻¹, $p < 0.001$). Dashed orange line: post-2013 mean anomaly (+0.94°C above baseline). (Source: 00_03_summer_warming_trend.png)*

## Well Network

The 66-well reference network (Section 3.1.1) spans a median record length of 192 months (16.0 years), with individual cleaned series ranging from 140 to 252 months (Figure 6, upper left); all wells satisfy the 100-month minimum-observation threshold for admission to the reference network. The network mean water level across all wells and months was 0.77 m below ground — water levels throughout are reported as depth below the ground surface (negative below ground, positive when ponded above ground) — with individual well means ranging from 0.26 m to 2.07 m below, reflecting the considerable topographic and hydrogeological diversity of the site (Figure 6, upper right). The median seasonal amplitude — defined as the difference between mean August and mean February water levels — was 0.68 m, with the water table deepest in late summer and shallowest in late winter, consistent with the expected seasonal recharge cycle for a maritime dune system (Figure 6, lower

right). The network mean time series (Figure 6, lower left) shows a clear seasonal cycle throughout the record, with the inter-well standard deviation envelope widest around 2011–2013 and not visibly wider after the December 2017 clearfell; the clearfell response of the impact zone wells is explored in Section 4.6.

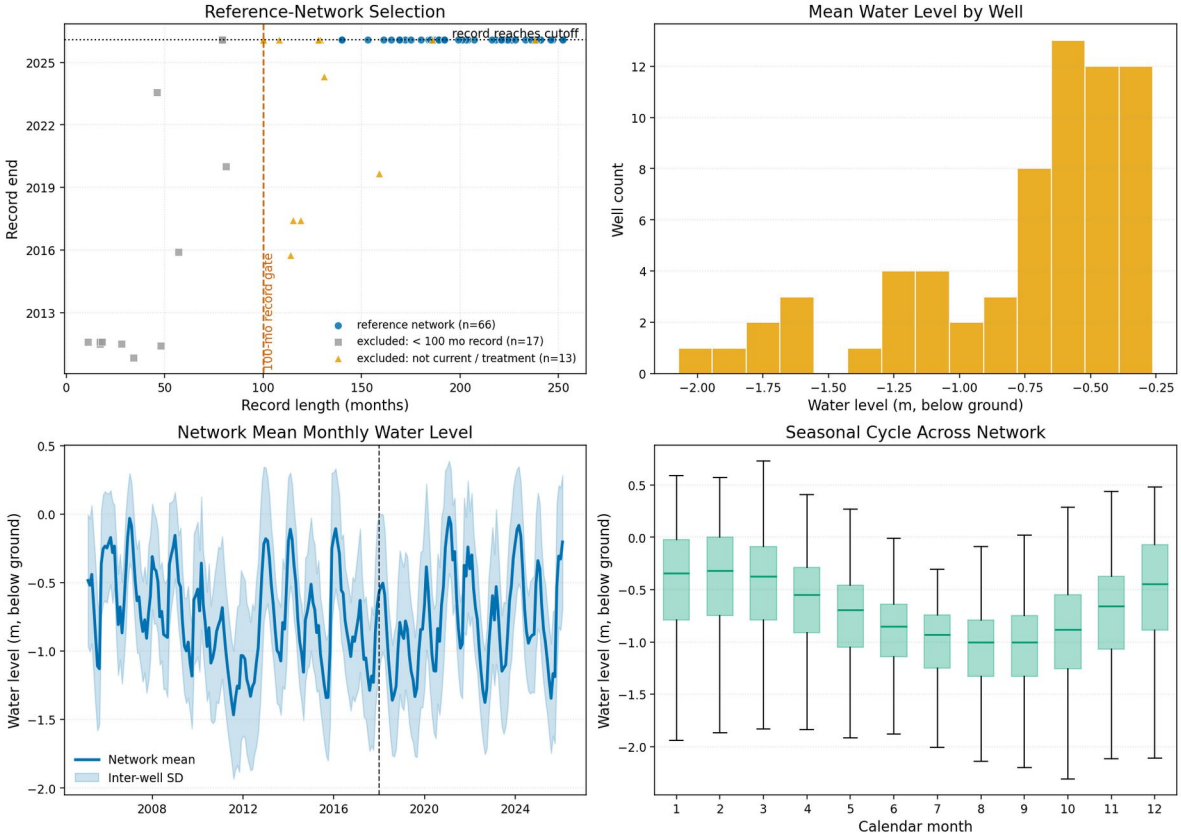



***Figure 6:*** *Well network characterization for the 66-well reference network. Upper left: reference-network selection — record length (months) against record end for every monitored well, with the 100-month record-length gate (orange dashed) and the record-extends-to-cutoff recency gate (dotted); the 66 reference wells (blue) clear both gates and are reference rather than treatment or extended wells, while the 17 wells failing the length gate (grey) and the 13 excluded on recency or as treatment/extended wells (orange) are shown for context. Upper right: frequency distribution of mean water level across the 66 reference wells (m below ground; positive values indicate water ponded above ground). Lower left: mean monthly water level across the 66 reference wells (m below ground) with inter-well standard deviation as shaded envelope; vertical dashed line marks the December 2017 clearfell intervention. Lower right: seasonal cycle across the 66 reference wells shown as monthly box plots pooling all reference well-month observations; boxes show interquartile range, horizontal line shows median, whiskers extend to 1.5 × IQR, outliers suppressed. (Source: 00_02_well_network_summary_short.png)*

Set against the SD15b / SD16 community-mean depths of Curreli et al. (2013), the network-wide seasonal extremes lie largely on the dry side of both; a well's depth indicates a community only where it stands on the slack floor (Section 4.8.3). The mean annual summer minimum across the 66-well reference network was −1.18 m, ranging from −0.71 m to −2.32 m across individual wells. No well maintains a mean summer minimum shallower than the

SD15b community mean of −0.61 m; 27 wells lie between the SD15b and SD16 community means (−0.61 to −0.98 m); and 39 wells record mean summer minima at or deeper than the SD16 community mean of −0.98 m, of which 21 lie at or below −1.20 m, beyond the reach of a single scraping event under average conditions on the planning assumption of about 0.22 m of benefit from a deeper scrape. For winter maxima, no well reaches the SD15b community-mean winter maximum of +0.21 m (standing water above the ground surface; the highest mean winter maximum is +0.20 m, at CEH6), 11 record mean winter maxima above ground, and 39 reach the SD16 community-mean winter maximum of −0.25 m. Against the nine-year guideline levels of the Dune Managers Handbook v2 (Denning et al., 2024), the summer picture barely moves (25 wells between the two levels and 41 at or deeper than its SD16 level of −0.96 m), but the winter one does: five wells reach its SD15b winter level of +0.09 m (CEH6, CEH24, CEH25, D15 and T41a) and 43 its SD16 winter level of −0.34 m. The winter comparison is therefore sensitive to the guideline source; the two sets of values are averaged over different periods (four years and nine) and are not interchangeable. These network-level baselines provide the reference against which cluster-specific dynamics, intervention effects, and climate trajectory are assessed in subsequent sections.

## 4.2 Network Clustering and Spatial Architecture

The clustering analysis (Section 3.2) identified k = 5 as the preferred cluster solution (silhouette coefficient = 0.37; bootstrap stability analysis confirming four of five clusters with median per-well co-assignment ≥ 0.98). The five clusters are spatially coherent and hydrogeologically interpretable (Table 3; Figures 7–9); their mechanistic characterization including LCSC values is presented in Section 4.2.2.

The top merge of the dendrogram, at a Ward linkage distance of approximately 0.99, separates C4 Main Forest from all other wells. Within the remaining 57 wells the east–west division follows at approximately 0.61, separating the eastern wells (C1 Lake Edge, C2 Dune) from the western wells (C3 Western Residual, C5 Coastal Forest). The two forest clusters therefore do not separate together: C4 stands apart from the whole network, while C5 separates from adjacent C3 wells only lower in the tree, despite their geographical proximity. The C4 separation also reflects the position of the C4 wells in the northern part of the study area, closest to the rock ridge, at the highest topographic positions in the dune system. C5 (Coastal Forest) occupies the coastal edge of the plantation where canopy and coastal influences combine.

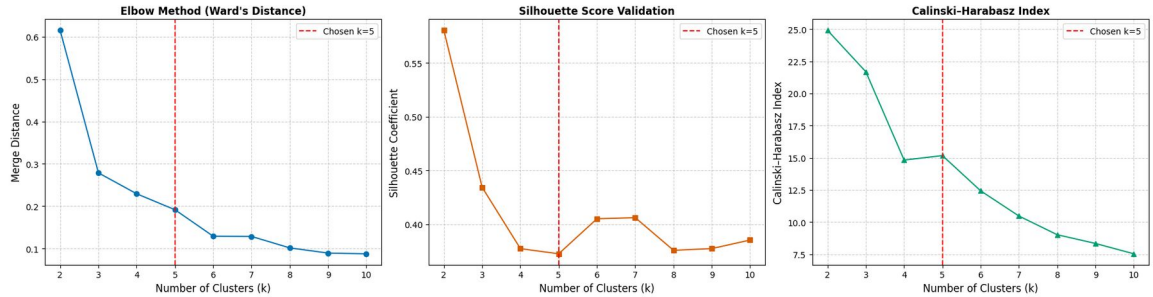



***Figure 7:*** *Cluster validation plots for k = 2–10: Ward's merge distance (elbow), silhouette coefficient and Calinski–Harabasz index, with chosen k = 5 indicated. The silhouette coefficient and the Calinski–Harabasz index both peak at k = 2, the top merge of the dendrogram, which separates the Main Forest cluster (C4) from the rest of the network; k = 5 was selected as the finest partition at which bootstrap stability analysis confirms four of the five clusters (median per-well co-assignment ≥ 0.98). (Source: 02_02_validation_plots.png).*

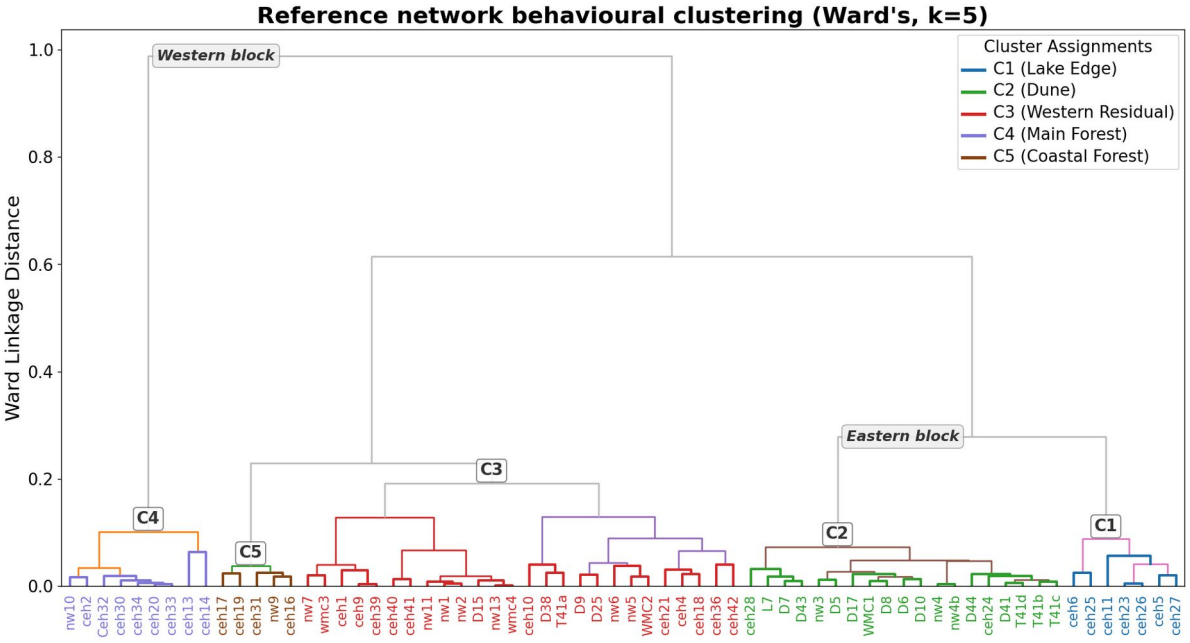



***Figure 8:*** *Clustering results: Ward's hierarchical clustering dendrogram for the 66-well reference network. Wells are labelled by identifier and coloured by assigned cluster. The top merge, at a Ward linkage distance of approximately 0.99, separates the Main Forest cluster (C4) from all other wells; within the rest, the east–west division at approximately 0.61 separates the eastern till-influenced clusters (C1, C2) from the western sand clusters (C3, C5). C5 Coastal Forest, which also carries pine canopy, joins the western group rather than C4, so the C4 separation is not canopy alone; it is consistent with the ridge-flank position of the C4 wells as well as their plantation cover. (Source: 02_01_dendrogram.png).*

### 4.2.1 Cluster Hydrographs and Seasonal Amplitudes

The five cluster average hydrographs (Figure 9b) reveal a clear stratification of groundwater behaviour that reflects the hydrogeological architecture identified by the clustering analysis. C1 (Lake Edge) shows the shallowest mean depth (−0.38 m). The one exception within C1 is CEH11, assigned to the cluster on hydrograph similarity but geographically isolated on the south-eastern coast, so its mechanistic coefficients are not interpreted alongside the lake-edge wells (Section 3.2.3). C2 (Dune) tracks C1 closely but sits systematically deeper (mean −0.50 m). The fundamental geological contrast between the Eastern and Western Blocks is visible as a persistent vertical offset between C1/C2 and C3 (Western Residual), with C3 running approximately 0.3 m deeper on average (mean −0.77 m) despite occupying a broadly similar topographic position. C4 (Main Forest) is the most distinctive cluster, sitting 0.6 m deeper than adjacent C3 wells throughout the record (mean −1.34 m). The C4 cluster mean hydrograph should be interpreted cautiously prior to 2010, when only three of the nine cluster wells were operational (CEH2, CEH13 and NW10); these wells are disproportionately located at higher elevations on the ridge margin so the apparent high mean heads in 2006–2008 reflect an unrepresentative spatial sample rather than a genuine network-wide signal. Full nine-well coverage was not achieved until mid-2010. C5 (Coastal Forest) sits at comparable depth to C4 (mean −1.19 m) but with a damped seasonal signal. Across all five clusters the average hydrographs (Figure 9b) follow the levelled cumulative water balance (Figure 9a) throughout the record. Per-well seasonal amplitudes and their temporal evolution are analysed below.

The network mean water table elevation was 7.45 m AOD across the 66-well reference network (elevations computed as water level above each well’s NRW 2 m LiDAR-derived ground surface; Section 3.1.1), ranging from 3.02 m AOD (CEH11, C1 Lake Edge) to 13.33 m AOD at the ridge flank (CEH14, C4 Main Forest). CEH11's assignment to C1 is discussed in Section 3.2.3: its low elevation and coastal position place it near the Menai Strait tidal boundary rather than Llyn Rhos-Ddu, and it represents the low-elevation extreme of the reference network rather than a typical lake-edge well. Excluding CEH11, the C1 cluster-mean elevation rises from 8.46 m AOD to 9.37 m AOD, and the network minimum shifts to CEH21 (C3, 3.10 m AOD). The Main Forest cluster (C4) occupies the highest mean water table elevation in the network at 9.25 m AOD, compared with 8.46 m AOD in the Lake Edge cluster (C1), 7.78 m AOD in the Dune cluster (C2), and 6.90 m AOD in the Western Residual (C3). The Coastal Forest cluster (C5) sits at 4.39 m AOD, the lowest cluster-mean elevation in the network.

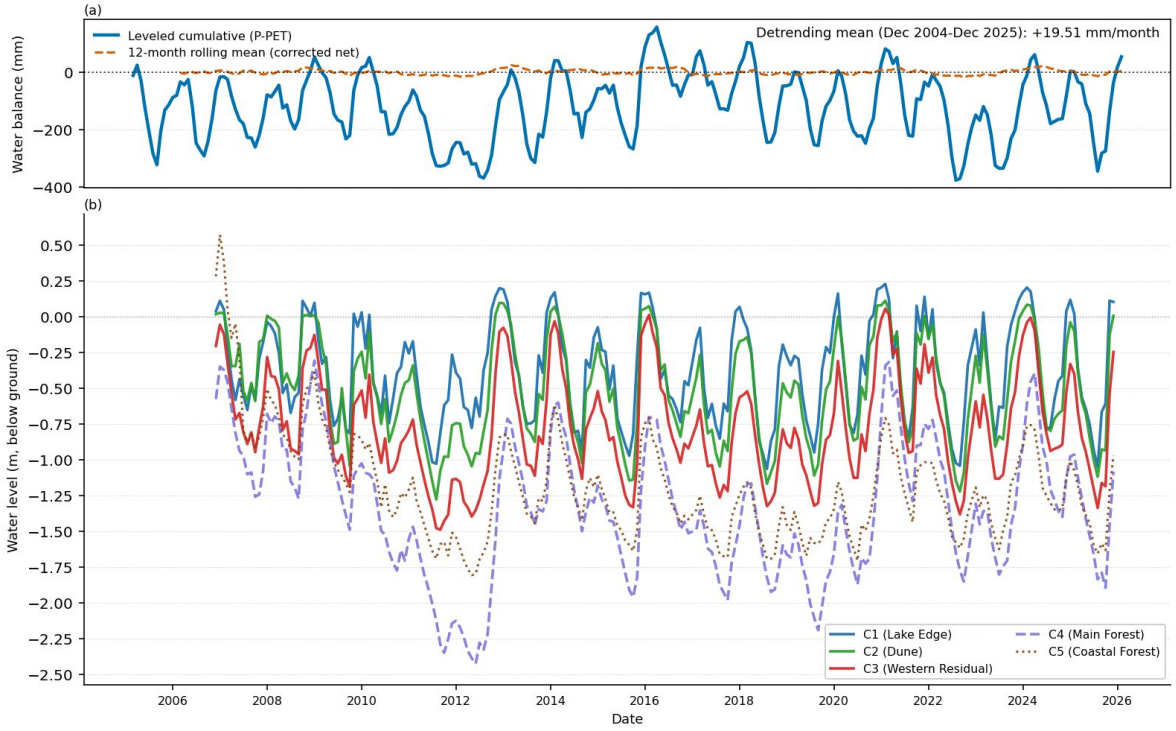



***Figure 9:*** *Cluster-mean relative water level hydrographs for the five hydrogeological clusters, December 2006 to February 2026. (a) Levelled cumulative water balance (P − PET), detrended at the study-period mean of +19.51 mm/month (December 2004 to December 2025); the 12-month rolling mean of the corrected net balance (orange dashed) shows no long-run directional trend in climate forcing over the monitoring period. (b) Monthly cluster-mean water level relative to pipe-top datum (m), where zero approximates ground surface, computed from all wells assigned to each cluster in the 66-well reference network. Cluster labels and colours follow those used in Figure 8. (Source: 02_03_cluster_hydrographs_wb.png)*

Separating the reference wells into one panel per cluster (Figure 10) shows the within-cluster spread that the cluster means in Figure 9 necessarily collapse. On the shared vertical axis the between-cluster stratification is evident: the two forested clusters (C4 Main Forest, C5 Coastal Forest) sit at consistently lower relative water levels than the lake-edge and dune clusters, and all clusters share the same seasonal winter-recharge and summer-recession cycle. C1 (Lake Edge) is the most internally coherent, while C2 (Dune) contains one well (L7) sitting persistently below the cluster mean; the individual traces nonetheless broadly track their cluster's seasonal cycle throughout the record.

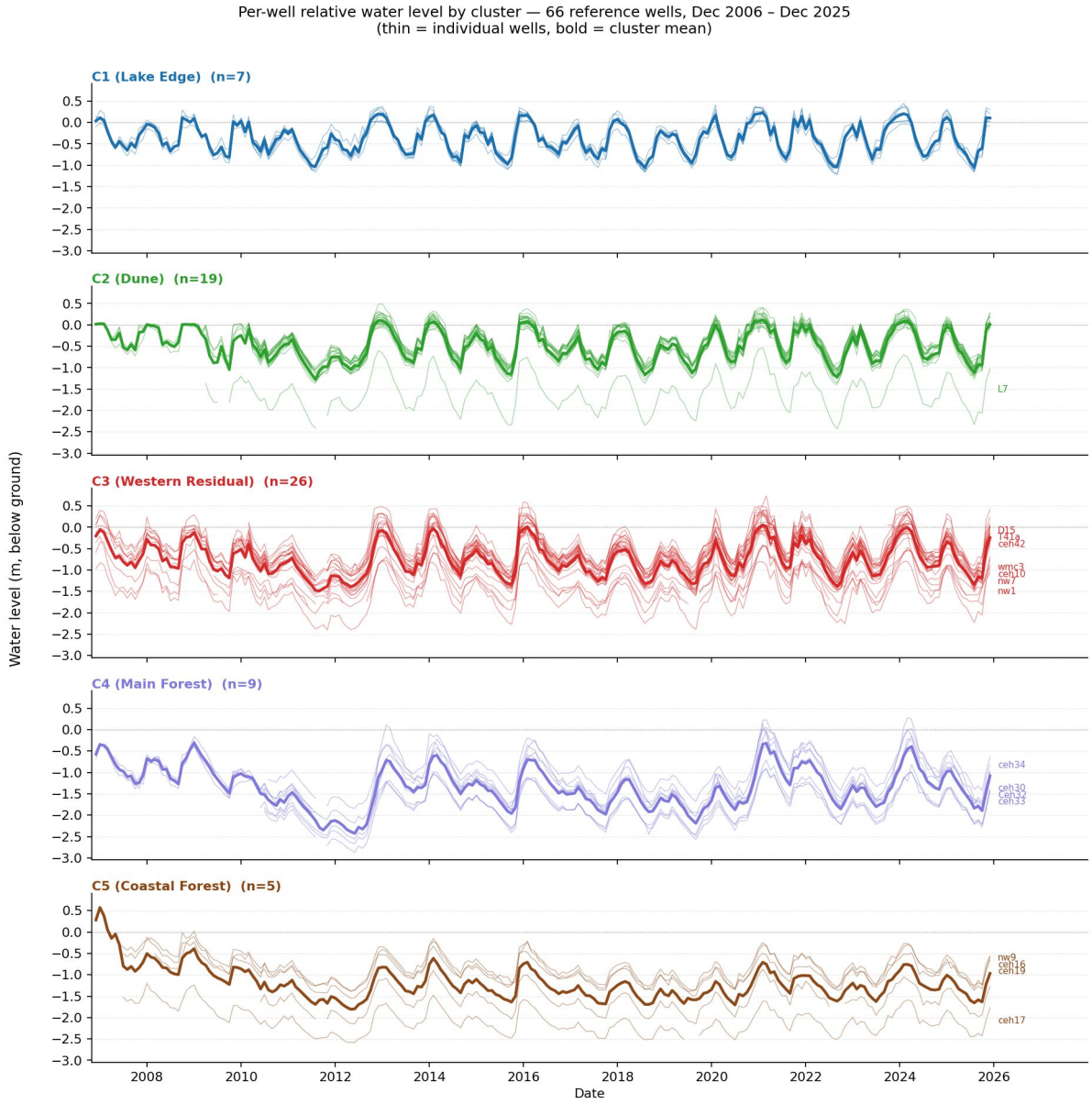



***Figure 10:*** *Per-well relative water level for the 66 reference wells, shown as one stacked panel per hydrogeological cluster, December 2006 to December 2025. Within each panel the thin lines are individual wells and the bold line is the cluster mean. All panels share a common vertical axis: monthly water level relative to pipe-top datum (m), where zero approximates ground surface, as in Figure 9b. Cluster labels and colours follow Figure 9; well counts per cluster (n) are given in each panel title. (Source: 02_03b_cluster_spaghetti.png)*

**Per-Well Seasonal Amplitude.** Per-well inter-decile amplitudes (p90 − p10) computed across three windows (full record, pre-2018, post-2018) agree with and refine the cluster-mean amplitude readings (Section 4.2.1; Figure 9). Cluster medians of per-well amplitudes are summarized in Table 2.

| Cluster | n | Pre-2018 (m) | Post-2018 (m) | Δ pre→post (%) | Climate-normalized Δ (%) |
|---|---|---|---|---|---|
| C1 Lake Edge | 7 | 0.83 | 1.00 | +20% | +15% |
| C2 Dune | 19 | 0.99 | 0.97 | −3% | −8% |
| C3 Western Residual | 26 | 1.09 | 1.09 | −0% | −8% |
| C4 Main Forest | 9 | 1.16 | 1.09 | −6% | −6% |
| C5 Coastal Forest | 5 | 0.89 | 0.73 | −18% | −21% |

***Table 2:*** *Negative values indicate amplitude reduction (damping); positive values indicate amplitude amplification. Climate-normalized values exclude empirically-identified drought summers (2005, 2018, 2022) where Jun–Sep rainfall fell ≥1σ below the 1931–2017 baseline. (Source: 02_09_cluster_amplitude_summary.csv)*

Three patterns emerge. First, the open dune clusters (C2, C3) show stable raw amplitude but modest climate-normalized damping (≈8%), consistent with the post-2018 drought summers temporarily inflating apparent variability while the underlying seasonal range has slightly contracted. Second, **C5 Coastal Forest shows a pronounced amplitude collapse** of 18% (raw) to 21% (climate-normalized), the largest reduction in the network: recharge peaks no longer reach their pre-2018 heights even after correcting for drought summers (Section 5.7.2). Third, **C1 Lake Edge shows the opposite pattern**: amplitude has increased by 15–20%. The C3 cluster shows the highest within-cluster spread (post-2018 ratio max:min ≈ 2.1), approaching the amplitude-heterogeneity flagging threshold (Section 3.2.4).

The amplitude analysis shows that the cluster-mean coefficients reported in Section 4.2.2 represent stable averages across wells whose individual amplitude responses differ within most clusters by less than a factor of two, but with sufficient within-cluster variation in C3 and C4 to motivate the per-well coefficient mapping presented in Section 4.9.

### 4.2.2 Cluster Mechanistic Characterization

The five clusters are characterized mechanistically in Table 3 using a state-space model fitted to the average hydrograph for each cluster (Section 3.4.1); Table 4 gives the range of the per-well fits on the 100-month comparison window, against which each centroid can be read, so the spread behind every coefficient below is visible alongside it. $\beta_1$ is the recharge sensitivity — the water table rise (mm) per mm of rainfall; $\beta_2$ is the atmospheric draw — the water table decline (mm) per mm of PET; and $-\beta_3$ is the proportional drainage decay rate — the fraction of antecedent displacement above the drainage datum (Section 3.4) dissipated by

lateral drainage each month. All five cluster-centroid coefficient sets reported in Table 3 are statistically significant on the full record ($p < 0.01$); within-cluster, individual per-well coefficients vary in significance, most notably the near-zero C4 drainage coefficient and the small C1 atmospheric-draw term, as discussed below and in Section 4.9. Because wells were installed progressively over the early record, each centroid was also refitted from a common 2015 reference date, when every cluster reports its full membership (Section 3.4.1): the coefficients change by a few per cent at the open-dune clusters and by about 9% at C4 Main Forest, where the coefficient is smallest in absolute terms, and the between-cluster ordering is unchanged.

The recharge sensitivity coefficient $\beta_1$ is highest in the Lake Edge cluster (C1: 4.58), indicating that each millimetre of rainfall drives a proportionally larger water table rise in the shallow-till aquifer than in the Dune cluster (C2: 3.99) or Western Residual (C3: 3.81). The two forested clusters return near-identical and substantially lower recharge sensitivities (C4: 2.48, C5: 2.46). The atmospheric draw coefficient $\beta_2$ is highest at C4 (2.57), followed by C3 (1.83) and C2 (1.84), with C5 (1.34) and C1 (0.94) lowest. The C4 value of $\beta_2$ slightly exceeds its $\beta_1$ (2.56 against 2.50), meaning that the Main Forest cluster loses as much water table per mm of PET as it gains per mm of rainfall. Because β coefficients fold together vegetation effects, specific yield and soil moisture dynamics, they cannot be decomposed into crop coefficients without independent Sy estimates; this decomposition is developed in Section 5.6.1. C5 (Coastal Forest) has low $\beta_2$ despite also carrying Corsican pine canopy (Section 5.6.1). C1's $\beta_2$ (0.94) is the lowest in the network; the depth-dependent PET diagnostic (Section 3.7.1) returns the strongest residual depth-coupling at C1 (L_E = 0.43 m). The drainage decay coefficient $-\beta_3$ is highest at C1 (0.088), similar at C2 (0.062) and C3 (0.062), lower at C5 (0.045), and lowest at C4 (0.019). The C4 value is the smallest in the network, but it is well resolved at the cluster scale: the drainage and climate predictors are nearly uncorrelated at C4 (variance inflation factor ≈ 1.1) and the cluster carries the network's largest displacement variation, so the centroid coefficient is significant ($p = 0.001$; Section 5.2.3). Two ridge-flank wells, CEH13 and CEH14, return drainage coefficients indistinguishable from zero over their full records; excluding them from the cluster fit raises the coefficient to 0.029 month$^{-1}$, reported as a sensitivity in Section 5.2.3 rather than adopted as the cluster value. The LCSC ($= 100/\beta_1$) expresses these contrasts as a directly interpretable management metric: 22–25 mm of rainfall is required to raise Lake Edge and Dune water tables by 10 cm, compared with 26 mm in the Western Residual and 40–41 mm beneath the plantation.

The C1 cluster-mean coefficients in Table 3 include CEH11, which is geographically separated from the other six C1 members (Section 3.2.3). CEH11 sits at 3.5 m AOD on the south-eastern coast, approximately 1.7 km from the nearest lake-edge well, with a damped hydrograph (Section 5.1.1). Its $\beta_2$ value of 0.67 lies within the C1 per-well range (Table 4). CEH11's inclusion extends the C1 elevation range from 8.5–10.9 m down to 3.5 m. Excluding CEH6 as well leaves five wells in a 9.1–10.9 m band. CEH6's small negative $\beta_2$

(−0.40, not significant at p = 0.26) is the only negative $\beta_2$ in the network and is why the C1 $\beta_2$ range in Table 4 crosses zero; the only other range that does is $\beta_3$ at C4 (Section 5.2.3). The cluster-mean coefficients are not materially affected by CEH11's inclusion — its $\beta_1$ (4.46) and $\beta_3$ (0.093) are close to the C1 mean — but per-well spatial analyses within C1 should be interpreted with awareness that CEH11 responds to a different boundary condition.

An extended form of the state-space model incorporating a depth-dependent evapotranspiration coefficient (Section 3.7.1) was evaluated at cluster level; as the results of this analysis bear primarily on the mechanistic interpretation of the Forest cluster $\beta_2$ coefficient, they are discussed in Section 5.6.1 rather than reported here. The within-cluster coefficient variation is substantial, particularly in the forest zone where per-well $\beta_2$ ranges from 0.776 to 3.880 across the 14 C4 and C5 wells. Whether this variation tracks topographic position (a proxy for the uncored substrate) or canopy differences is examined through per-well spatial regression in Section 4.9.4.

The elevated C4 atmospheric-draw coefficient ($\beta_2$ = 2.56, the largest in the network) is accompanied by the network's lowest drainage coefficient ($\beta_3$ = 0.018 month$^{-1}$), so that at the cluster centroid the C4 loss budget is atmospheric-draw-dominated under both forms: drainage takes 29% of the loss under Model B, the form the split is quoted from, and 25% under Model A. The identifiability of this low $\beta_3$ is examined in Section 5.2.3; the partition remains dependent on the drainage datum and substrate assumption, and is flagged as modelled and unconfirmed wherever it enters downstream analysis.

The identifiability of the C4 drainage coefficient was tested directly (Script 30). The variance inflation factor on the displacement term at the C4 centroid is 1.09 and the correlation between PET and displacement −0.04, against 1.82 and −0.62 at C1, 1.38 and −0.41 at C2, 1.31 and −0.35 at C3, and 1.07 and −0.17 at C5; C4 carries the network's largest displacement variation (standard deviation 0.48 m), and the centroid coefficient is significant (p = 0.001). Per well, the median variance inflation factor at C4 is 1.20 on the 100-month window, the lowest in the network; the per-well correlation between $\beta_2$ and $\beta_3$ is −0.77 across all nine wells, −0.26 with CEH14 removed and −0.04 with CEH13 and CEH14 both removed. The four wells whose $\beta_3$ is significant on the window (NW10, CEH20, CEH30, CEH33) span 0.023–0.043 month$^{-1}$. At the fitted centroid coefficient of 0.018 month$^{-1}$ the steady-state water balance closes to a residual of 0.13% of the C4 loss budget (Section 4.2.3), and swept across $\beta_3$ that residual is minimized at 0.019 month$^{-1}$. Excluding the two ridge-flank wells raises the centroid from 0.018 to 0.029 month$^{-1}$ and shortens Model A's recession time $1/\beta_3$ from 54 to 34 months. The reading of these diagnostics is in Section 5.2.3.

| Cluster | Basis | n | $\beta_1$ | $\beta_2$ | $-\beta_3$ | p($\beta_3$) | LCSC (%) | $R^2$ |
|---|---|---|---|---|---|---|---|---|
| C1 Lake Edge | full record (238 mo) | 7 | 4.577 | 0.942 | 0.088 | <0.001 | 21.849 | 0.729 |
| | 100-month window (100 mo) | 7 | 5.008 | 0.524 | 0.103 | <0.001 | 19.968 | 0.784 |
| C2 Dune | full record (251 mo) | 19 | 3.989 | 1.842 | 0.062 | <0.001 | 25.072 | 0.764 |
| | 100-month window (100 mo) | 19 | 4.180 | 1.624 | 0.069 | <0.001 | 23.923 | 0.805 |
| C3 Western Residual | full record (251 mo) | 26 | 3.808 | 1.833 | 0.062 | <0.001 | 26.263 | 0.827 |
| | 100-month window (100 mo) | 26 | 3.836 | 1.713 | 0.063 | <0.001 | 26.071 | 0.854 |
| C4 Main Forest | full record (238 mo) | 9 | 2.502 | 2.564 | 0.019 | <0.001 | 39.975 | 0.733 |
| | 100-month window (100 mo) | 9 | 2.418 | 2.625 | 0.013 | 0.176 | 41.360 | 0.759 |
| C5 Coastal Forest | full record (240 mo) | 5 | 2.462 | 1.336 | 0.045 | <0.001 | 40.621 | 0.706 |
| | 100-month window (100 mo) | 5 | 2.341 | 1.158 | 0.045 | <0.001 | 42.718 | 0.765 |

*Table 3: Cluster mechanistic characterization derived from the displacement-formulation SSM (Section 3.4). $\beta_1$ = recharge sensitivity (mm water table rise per mm rainfall); $\beta_2$ = atmospheric draw (mm water table decline per mm PET); $-\beta_3$ = proportional drainage decay rate (fraction of displacement above the 3.7 m drainage datum dissipated per month). LCSC = 100/$\beta_1$; values express the millimetres of rainfall required to raise the cluster average water table by 10 cm. $R^2$ is one-step diagnostic fit. Each cluster is shown on both fitting bases: its full record, which is the basis for every cluster coefficient used elsewhere in this report, and the 100-month comparison window used by the per-well analyses (Section 3.4.3). On the full record all coefficients are significant at $p < 0.01$ or better. On the comparison window C4's drainage term is not significant ($p = 0.176$) — its standard error is unremarkable, but the coefficient is 3.6 to 8.2 times smaller than any other cluster's, so the shorter record cannot resolve it from zero. C1's atmospheric-draw term is the only other coefficient that is not significant on the window ($p = 0.103$). (Source: 03_03_cluster_mechanistic_coefficients.csv; window basis from 03_14_centroid_window_sensitivity.csv).*

### 4.2.3 Recharge and Losses: A Cluster-Level Water Balance

| Cluster | n wells | $\beta_1$ range | $\beta_2$ range | $\beta_3$ range |
|---|---|---|---|---|
| C1 Lake Edge | 7 | 4.267–6.237 | −0.400–1.251 | 0.088–0.134 |
| C2 Dune | 19 | 3.425–4.876 | 1.039–2.531 | 0.057–0.094 |
| C3 Western Residual | 26 | 2.332–5.514 | 0.884–2.185 | 0.034–0.109 |
| C4 Main Forest | 9 | 2.101–3.493 | 2.084–3.880 | −0.024–0.039 |
| C5 Coastal Forest | 5 | 2.151–2.752 | 0.776–1.359 | 0.039–0.061 |

***Table 4:*** *Within-cluster range of the per-well SSM coefficients, reference network (Script 07). Each range is the minimum and maximum across that cluster's member wells, from the per-well fits on the 100-month comparison window (Section 3.4.3), not the cluster-average hydrographs to which the centroid coefficients of Table 3 are fitted; n wells is the number of reference wells the range spans, so the narrow C5 range rests on five wells and the wide C3 range on twenty-six. Two ranges cross zero and are discussed where they matter: $\beta_2$*

*at C1 (Section 5.2.3, the CEH6 coefficient) and $\beta_3$ at C4 (Section 4.2.2, drainage identifiability). The $\beta_3$ ranges at C1 and C4 do not overlap. (Source: 07_coeff_05_cluster_ranges.csv)*

Water balance decomposition applied to all five clusters (Section 3.7.2) reveals markedly different recharge sensitivities across clusters receiving nearly identical climate forcing. The head-space decomposition is presented in Table 5 and Figure 11; an indicative volumetric conversion is presented in Table 7 and Figure 11b. The derivation and defence of the specific yield values is addressed in Section 4.2.4.

Recharge into the water table ($\beta_1 \cdot Pl$) declines systematically from the Lake Edge cluster (C1: 0.341 m/month) through the Dune (C2: 0.296 m/month) and Western Residual (C3: 0.282 m/month) to the forested clusters (C5: 0.183 m/month; C4: 0.186 m/month). Total losses (atmospheric draw plus drainage) track recharge closely in every cluster. Residuals are less than 1.7% of total losses in all clusters. The split varies markedly across clusters. Under Model B, the form it is quoted from, drainage takes 38% of losses at C1, 45% at C2, 56% at C3, 61% at C5 and 29% at C4, where atmospheric draw dominates. Model A at the network datum gives drainage more at the dune clusters (85, 66 and 65%) and about the same under the forest (61 and 25%).

| Cluster | Label | LCSC (%) | Recharge (m/month) | Atm. Draw (m/month) | Drainage (m/month) | Total Loss (m/month) | Residual (m/month) |
|---|---|---|---|---|---|---|---|
| C1 | Lake Edge | 21.8 | 0.341 | 0.051 | 0.293 | 0.345 | −0.004 |
| C2 | Dune | 25.1 | 0.296 | 0.100 | 0.198 | 0.298 | −0.003 |
| C3 | Western Residual | 26.3 | 0.282 | 0.100 | 0.182 | 0.282 | +0.000 |
| C4 | Main Forest | 40.0 | 0.186 | 0.140 | 0.046 | 0.186 | +0.000 |
| C5 | Coastal Forest | 40.6 | 0.183 | 0.073 | 0.114 | 0.186 | −0.003 |

***Table 5:*** *Mean monthly head-space water balance decomposition by hydrogeological cluster, Newborough Warren 2005–2026. Recharge ($\beta_1 \cdot Pc$), atmospheric draw ($\beta_2 \cdot PETc$) and gravity drainage ($\beta_3 \cdot hc_disp$) are expressed as metres of water table change per month (m/month), where hc_disp is the mean displacement above the 3.7 m drainage datum (Section 3.4). Coefficients are cluster-centroid fits from the no-intercept form (Model A, Table 3); the split quoted in the text is Model B's (Table 6, Figure 11a). Residuals are less than 1.7% of total losses in all clusters, consistent with the three-term formulation closing the water balance without requiring an unmeasured boundary contribution. All clusters receive nearly identical climate forcing: Pc = 74.4 mm/month, PETc = 54.5 mm/month. Drainage and ET percentages shown below. (Source: 16_water_bal_table.csv).*

Model A's head-space decomposition of Table 5 is one way to split the losses into evapotranspiration and drainage, and it inherits whatever uncertainty attaches to $\beta_2$ and $\beta_3$ separately. A second, model-independent estimate comes from the observed seasonal recession of the cluster means themselves (Section 3.7.2): the mean month-on-month decline over winter months in which the water table fell (November–February, when potential evapotranspiration is small) as a share of the same quantity in summer (June–September)

estimates the drainage fraction of summer losses. Both terms are head declines, so specific yield cancels in the ratio and the estimate is independent of the WTF specific yields of Section 4.2.4. Model A and the recession ratio are compared in Table 6, beside Model B. Across the four estimators of Table 6 the spread is 10 percentage points at C3, where Model A and the recession ratio sit closest (recession 0.66, Model A 0.65), and 12 points at C5 (recession 0.73, Model A 0.61), widening to 47 and 32 points at C1, C2 and C4. C4 Main Forest reverses the ordering: Model A's head-space ratio makes the forest interior much the most evapotranspiration-dominated cluster in the network (drainage fraction 0.25) while the recession ratio places it second-lowest of the five clusters (0.51). Two acknowledged biases in the recession estimator — winter potential evapotranspiration is small but not zero, and head-dependent drainage is faster in winter when heads are high — both act to inflate its drainage fraction, so it is read as leaning high rather than as a point estimate. Model B's fraction, the one the split is quoted from (D-283), is 0.38 at C1, 0.45 at C2, 0.56 at C3, 0.29 at C4 and 0.61 at C5. The range of the four estimators (Model A, Model B, the depth-dependent model and the recession ratio), not any one alone, is the partition uncertainty carried into Table 7 and drawn as the hatched band of Figure 11b.

| Cluster | Winter decline (mm/month) | Summer decline (mm/month) | ET = summer − winter (mm/month) | n months (winter/summer) | Drainage fraction (recession) | Drainage fraction: Model B; Model A (admissible datum range) |
|---|---|---|---|---|---|---|
| C1 Lake Edge | 98.0 | 164.0 | 66.0 | 27 / 50 | 0.60 | 0.38; 0.85 (0.73–0.88) |
| C2 Dune | 60.1 | 172.4 | 112.3 | 19 / 60 | 0.35 | 0.45; 0.66 (0.54–0.69) |
| C3 Western Residual | 92.5 | 139.3 | 46.9 | 17 / 63 | 0.66 | 0.56; 0.65 (0.46–0.68) |
| C4 Main Forest | 64.8 | 127.8 | 63.0 | 15 / 72 | 0.51 | 0.29; 0.25 (0.04–0.29) |
| C5 Coastal Forest | 67.0 | 92.3 | 25.3 | 21 / 69 | 0.73 | 0.61; 0.61 (0.24–0.66) |

***Table 6:*** *Seasonal-recession partition of water-table losses by cluster (Section 3.7.2). Winter and summer figures are the mean month-on-month decline of the cluster-mean water table over months in which it fell (Δh < 0), in the windows November–February and June–September; March is excluded as transitional. n months is the number of falling months behind each mean — note the asymmetry, 15 to 27 winter months against 50 to 72 summer months. The recession drainage fraction is the winter mean divided by the summer mean; the evapotranspiration fraction is its complement. Both terms are head declines, so specific yield cancels in the ratio and the partition is Sy-free. The last column gives Model B's fraction, the one the split is quoted from (D-283), then Model A's, the head-space fraction implied by Table 5, with its value at the shallowest and deepest admissible datum (1.6 and 8 m) in brackets; the range of the four estimators (Model A, Model B, the depth-dependent model and the recession ratio) is the partition uncertainty in Table 7 and the hatched band of Figure*

*11b. (Source: 16_water_bal_rec_table.csv).*

An indicative volumetric expression of the water balance is presented in Table 7 and Figure 11b. Each cluster receives 889–893 mm/yr of rainfall over its own fitted months; for the forested clusters (C4, C5) canopy interception removes 214 mm/yr (24%; Freeman, 2008), leaving net rainfall of 678–679 mm/yr reaching the ground surface. The ET/drainage split is Model B's (D-283), and its range spans the four estimators of Table 6: Model A, Model B, the depth-dependent model and the recession ratio. At steady state, total losses equal total inputs (P or P_net for forest clusters) and the hatched band in Figure 11b spans that range. The ET column counts only what evaporates after the canopy has taken its share. Interception is itself evaporation, so a forest's total evaporative loss is I + ET: 695 mm/yr at C4, close to its PET (655 mm/yr), against 488 mm/yr at C2. At C5 the total (478 mm/yr) is similar to C2's. Across the four estimators, drainage takes 38–85% of the loss at C1, 35–66% at C2, 56–66% at C3, 19–51% at C4 and 61–73% at C5.

| Cluster | Label | P (mm/yr) | I (mm/yr) | P_net (mm/yr) | ET, Model B (mm/yr) | I + ET, Model B (mm/yr) | Drainage, Model B (mm/yr) |
|---|---|---|---|---|---|---|---|
| C1 | Lake Edge | 893 | 0 | 893 | 553 | 553 | 340 |
| C2 | Dune | 889 | 0 | 889 | 488 | 488 | 401 |
| C3 | Western Residual | 889 | 0 | 889 | 387 | 387 | 502 |
| C4 | Main Forest | 893 | 214 | 679 | 481 | 695 | 198 |
| C5 | Coastal Forest | 892 | 214 | 678 | 264 | 478 | 414 |

***Table 7:*** *Indicative annual volumetric water balance by cluster, 2005–2026. At steady state, total losses (ET + drainage) equal total inputs (P for open dune clusters; P − I for forested clusters). I = canopy interception (24% of P; Freeman, 2008), applied to C4 and C5 only. ET is the evaporative loss after interception, from net rainfall and the water table; because interception is itself evaporation, the total evaporative loss is I + ET, computed before rounding, so it can differ by one from the sum of the rounded columns. ET and drainage are split by Model B, the free-intercept form (D-283); the range across Model A, Model B, the depth-dependent model and the seasonal recession ratio is shown graphically in Figure 11b as a hatched band. Interception appears identically on both input and loss bars of Figure 11b and cancels in the net surplus. (Source: 16_water_bal_vol_table.csv)*

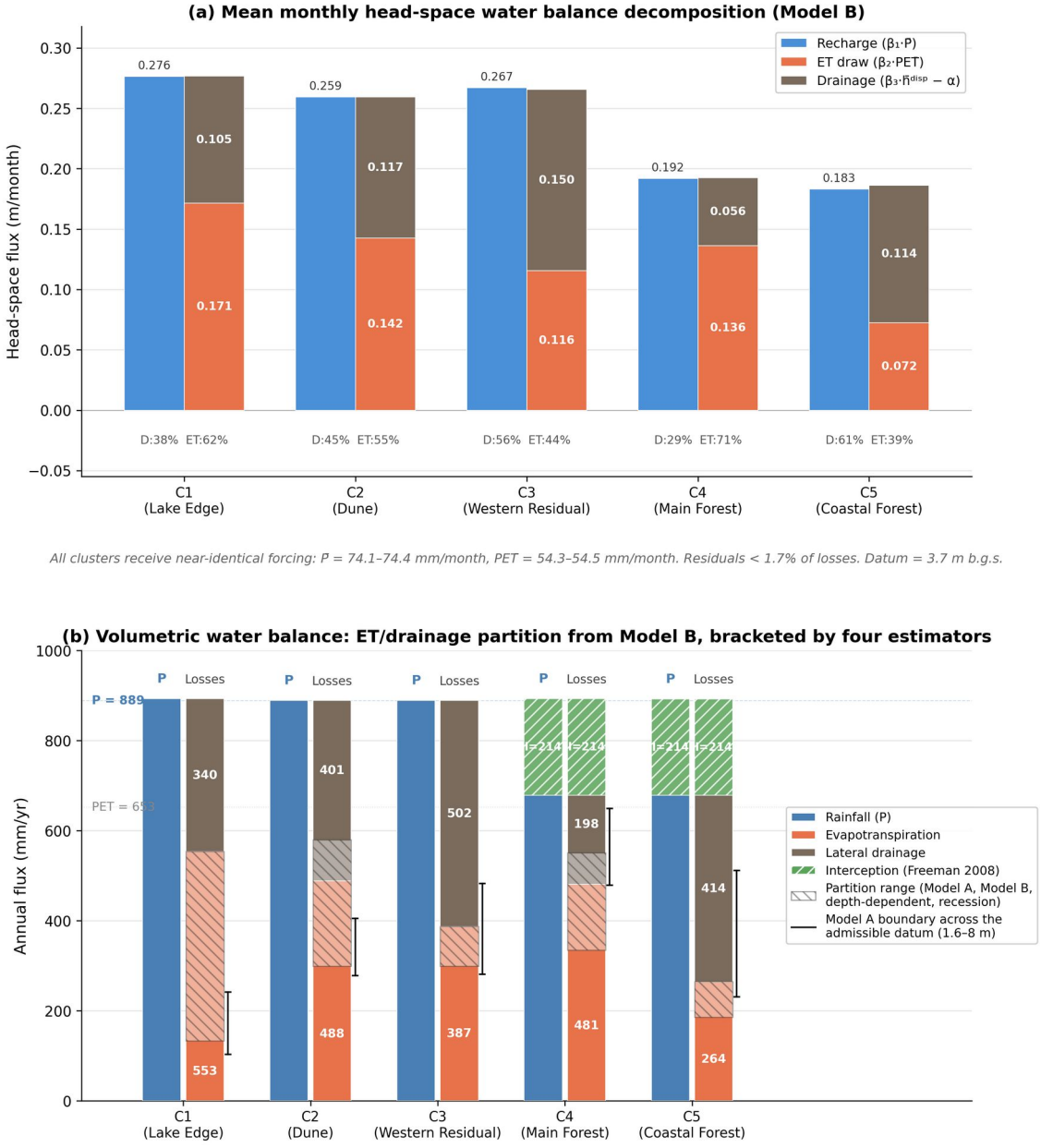



***Figure 11:*** *Water balance decomposition by hydrogeological cluster, Newborough Warren 2005–2026. (a) Model B's mean monthly head-space decomposition: for each cluster, the left bar shows recharge ($\beta_1 \cdot \bar{P}$) and the right bar shows losses decomposed into drainage ($\beta_3 \cdot \bar{h}_disp - \alpha_B$) and atmospheric draw ($\beta_2 \cdot P\bar{E}T$). Drainage and ET percentages are shown below each cluster label. All clusters receive near-identical climate forcing: $\bar{P}$ = 74.1–74.4 mm/month, $P\bar{E}T$ = 54.4–54.5 mm/month. Drainage datum = 3.7 m b.g.s. (b) Indicative annual volumetric water balance. All clusters receive P ≈ 890 mm/yr (889–893 by cluster); for forested clusters (C4, C5) canopy interception (I = 214 mm/yr, 24% of P; Freeman, 2008) appears identically on both input and loss bars and cancels in the net surplus. The bars split the losses by Model B; the hatched band spans the four estimators (Model A, Model B, the depth-dependent model and the seasonal recession ratio). The black bracket beside each loss bar spans Model A's ET/drainage boundary across the admissible datum (1.6–8 m). (Source: 16_water_bal_bar_ms.png).*

### 4.2.4 Specific Yield Estimation

WTF event-based median Sy estimates for all five clusters, derived from 36–64 qualifying monthly rising-limb events per cluster, are presented in Table 8, and the full event-level distribution behind each cluster median in Figure 12. A month qualifies when net recharge exceeds 10 mm, the cluster-mean water table rises by more than 5 mm, and the resulting event-level Sy falls within a plausibility window of 0.01–0.50. The spread in event counts is not a sampling imbalance: the net-recharge criterion is met in 141–148 months at every cluster, as it must be when all clusters share one climate record, and the rise criterion in 92–113, fewer beneath the canopy where recharge is suppressed. What separates the clusters is the plausibility ceiling, which discards 39–60% of the months clearing both criteria — 60% at C5, which is why its event count is the lowest in the network; the 0.01 floor removes no events at any cluster. Because the ceiling censors only the upper tail, every event-median is biased low by an amount that scales with the censored fraction: removing the clip raises the medians by +0.08 (C1) to +0.25 (C5), into a range of 0.29–0.61 that exceeds the drainable porosity of dune sand. The ceiling is therefore doing real work rather than trimming noise, but the correction it applies is not uniform across clusters, and at C5 the unclipped median (0.611) sits above the ceiling itself, so that cluster's estimate is set by the limit rather than by its own distribution. Uncorrected median values range from 0.210 (C1 Lake Edge) to 0.364 (C5 Coastal Forest), with broad interquartile ranges reflecting the inherent noise in monthly cluster-mean data. The two forested clusters return uncorrected medians (C4: 0.321, C5: 0.364) that sit at or above the open dune range (0.210–0.312): the WTF method attributes the full rainfall to recharge when 24% is intercepted, inflating the apparent Sy. After applying the interception correction of 24% of P (Freeman, 2008), the forested cluster medians fall to 0.256 (C4) and 0.316 (C5): C4 falls cleanly within the open dune range (0.210–0.312), while C5 sits just above its upper edge. The C5 estimate is the least reliable in the network — a majority of its uncorrected rising-limb events (60%, against 36% after the interception correction) return event-level Sy at or above the 0.50 plausibility ceiling (Section 3.7.3), so its median is constrained by that limit rather than freely estimated, and its modest apparent response to the interception correction is partly an artefact of the clip; it is therefore reported as a weak corroborator. Within this corrected dataset, a secondary pattern is visible: the two clusters with the lowest corrected Sy — C1 (Lake Edge; 0.210) and C2 (Dune; 0.270) — return lower values than C3 (Western Residual; 0.312) and C5 (Coastal Forest; 0.316), with C4 (Main Forest; 0.256) intermediate (Section 5.1). The pattern recurs in the per-well $\beta_2$ regression (Section 4.9.4) where elevation — reflecting both vadose zone thickness and proximity to buried bedrock — explains 98% of within-forest $\beta_2$ variance (97% leave-one-out).

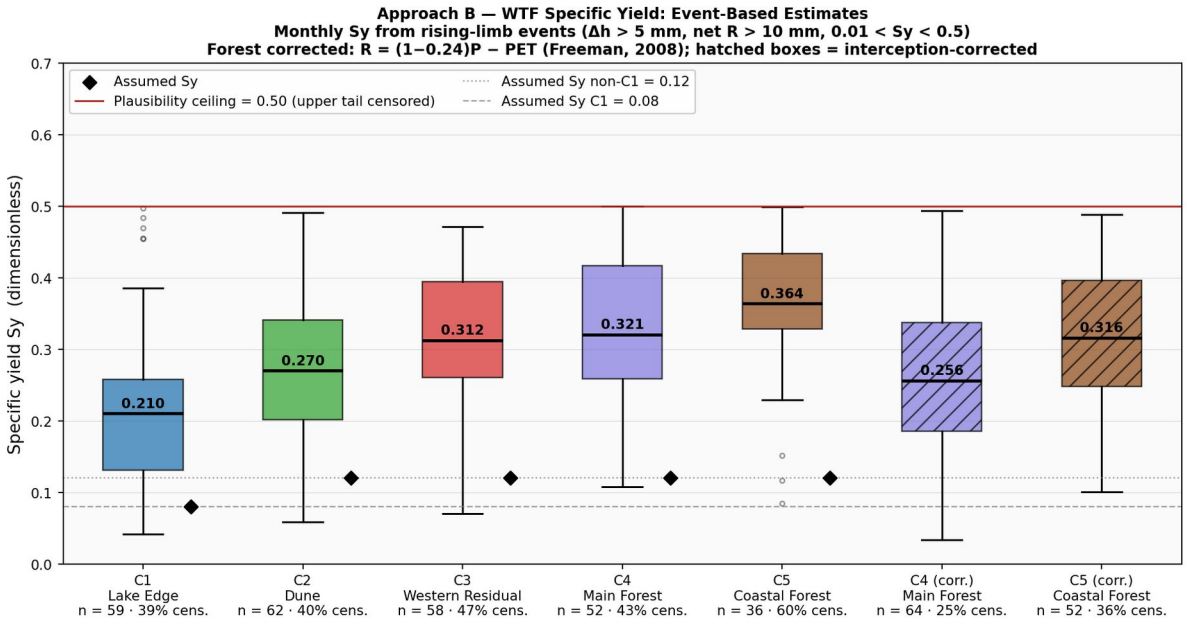



***Figure 12:*** *Distribution of event-level specific yield (Sy) from monthly rising-limb events, by cluster, with the interception-corrected forest variants hatched. Qualifying events satisfy Δh > 5 mm, net recharge > 10 mm and the plausibility window 0.01 < Sy < 0.50; the number of qualifying events (n) and the fraction removed by the 0.50 ceiling are given beneath each cluster. The ceiling censors the upper tail only, so each median is biased low by an amount that scales with the censored fraction — from 39% at C1 to 60% at C5, whose distribution is truncated hard against the limit. (Source: 17_wtf_03_event_boxplot.png).*

An independent cross-check on these estimates is available from a second WTF method: an ordinary-least-squares regression of winter water-table rise on net recharge, which returns a per-cluster Sy from the regression slope (Table 8, OLS-winter column). The OLS estimates run higher than the event-medians at four of the five clusters (mean ratio 1.22), consistent with the documented tendency of WTF methods to bias high on absolute magnitude; the OLS values, like the event-medians, should be read as upper bounds rather than point estimates. The two methods agree on the principal spatial pattern — both place the Coastal Forest (C5) cluster at the high-storage end of the network, and the event median also places the deep-sand Western Residual (C3) next to it. They diverge at two clusters, and the divergence is itself informative. At C1 (Lake Edge), the OLS regression returns the weakest fit of the five clusters ($R^2$ = 0.60, with C2 next-weakest at 0.62 and C3–C5 at 0.85–0.87), so the OLS estimate is least reliable where it disagrees; the event-median's placement of C1 as the low-storage end-member is retained. At C4 (Main Forest), the uncorrected OLS estimate ranks the cluster lowest in the network; the interception-corrected event-median is the estimate carried forward. Across both methods, the event-based median is retained as the primary specific-yield estimate for each cluster; the OLS regression serves as a corroborating check on the spatial pattern.

A third, independent WTF estimator — the rapid-recharge-event method of Crosbie et al. (2005; Section 3.7.3) — provides a further check that shares the assumptions of neither the

event-median nor the winter-OLS estimate. Selecting only short, sharply-rising episodes in which drainage during the rise is negligible, it returns cluster median Sy values of 0.180 (C1, 95% CI [0.101, 0.220], n = 19), 0.274 (C2, [0.213, 0.334], n = 21), 0.314 (C3, [0.243, 0.389], n = 17), 0.306 (C4, interception-corrected, [0.236, 0.361], n = 13) and 0.311 (C5, interception-corrected, [0.281, 0.337], n = 13). At C3 (Western Residual) all three methods agree within 0.037 (0.312, 0.314 and 0.349), so the high-storage placement of the deep-sand cluster is robust to estimator choice. At C1 (Lake Edge) the rapid-event estimate (0.180) sits with the event-median (0.210) well below the winter-OLS value (0.346); the low-storage placement of C1 is retained on the agreement of the two drainage-insensitive estimators. The coarse specific-yield gradient across the network is therefore consistent across three methodologically distinct estimators, while the fine ordering at the high-storage end (C3 versus C5) remains method-dependent — the winter-OLS and the event-median placing C5 highest, the rapid-event method placing C3 highest — and all three, as noted, are upper bounds on effective storage rather than point estimates.

Individual well Sy values underlying the spatial mapping of Figure 13 are considered separately below. All five WTF medians exceed the literature specific yields for unconfined dune sand and, at C1, glacial till (C1: 0.08; C2–C5: 0.12; Fetter, 2001) — benchmark values, not inputs to the Sy-free water balance of Section 4.2.3 — by a factor of two to three, consistent with the known tendency of monthly WTF to conflate gravity drainage with capillary fringe release (Healy and Cook, 2002; Scanlon et al., 2002). The WTF estimates should therefore be interpreted as upper bounds on effective storage. Direct field measurement by slug test or pumping test at representative wells per cluster is identified as a priority for future work.

| Cluster | Sy assumed | WTF event-median | n events | OLS-winter | n months | Rapid-event median | n episodes |
|---|---|---|---|---|---|---|---|
| C1 Lake Edge | 0.08 | 0.210 | 59 | 0.346 | 41 | 0.180 | 19 |
| C2 Dune | 0.12 | 0.270 | 62 | 0.332 | 50 | 0.274 | 20 |
| C3 Western Residual | 0.12 | 0.312 | 58 | 0.349 | 55 | 0.314 | 17 |
| C4 Main Forest | 0.12 | 0.321 | 52 | 0.306 | 49 | 0.306 | 13 |
| C5 Coastal Forest | 0.12 | 0.364 | 36 | 0.418 | 51 | 0.311 | 14 |
| C4 Main Forest (corrected) | 0.12 | 0.256 | 64 | — | — | — | — |
| C5 Coastal Forest (corrected) | 0.12 | 0.316 | 52 | — | — | — | — |

***Table 8:*** *Specific yield (Sy) estimates by hydrogeological cluster derived from the water table fluctuation (WTF) method (Healy and Cook, 2002). Event-based median Sy computed from monthly rising-limb events satisfying Δh > 5 mm, net recharge P − PET > 10 mm, and an event-level Sy within the plausibility window 0.01–0.50, over the monitoring period April 2005 to February 2026. For the forested clusters (C4, C5), two estimates are reported: uncorrected (using gross rainfall) and corrected (net recharge adjusted for 24% canopy interception following Freeman, 2008). The corrected C4 median (0.256) falls within the range of the open dune clusters (0.210–0.312) and the C5 median (0.316) sits just above its upper edge; the C5 estimate is constrained by the 0.50 plausibility clip and is treated as weakly corroborative. The C4 result, together with the continuous interpolated yield surface (Figure 13), is consistent with broadly uniform aquifer storage across the site. Monthly WTF estimates are known to overestimate effective Sy through conflation of gravity drainage with capillary fringe release (Healy and Cook, 2002; Scanlon et al., 2002) and should be interpreted as upper bounds rather than point estimates of long-term effective storage. Two further, independent WTF estimators are shown alongside: the drainage-corrected winter OLS regression (Sy OLS-winter, Approach A) and the rapid-recharge-event median of Crosbie et al. (2005) (Approach C). Neither is fitted to the separately tabulated interception-corrected forest series, shown as '—'; the rapid-event estimate is itself interception-corrected at C4 and C5. Each estimate is followed by the number of observations it rests on — qualifying rising-limb events (n events), winter months in the regression (n months) and rapid-recharge episodes (n episodes). Sy assumed is the literature specific yield for unconfined dune sand and, at C1, glacial till (Fetter, 2001), shown as a comparison benchmark; it is not an input to the Sy-free water balance of Section 4.2.3. The event-median remains the primary estimate (see text). (Source: 17_wtf_01_sy_estimates.csv).*

Individual well WTF Sy estimates were derived from 27–73 qualifying rising-limb events per well (mean 52 across the 66-well reference network). Per-well cluster-mean values — C1: 0.210, C2: 0.264, C3: 0.305, C4: 0.246, C5: 0.310 — are consistent with the cluster-aggregate estimates of Table 8. These per-well means are by-products of the spatial interpolation: they describe the inputs to Figure 13 and preserve within-cluster variability, but each per-well Sy is noisier than the cluster-aggregate estimate of Table 8, which fits a single median to the smoothed cluster-mean hydrograph and is therefore more robust as an estimate of cluster storage. The well-level analysis is presented here for its spatial content rather than as a competing storage estimate. The spatial distribution of Sy, shown in Figure

13, is broadly uniform across the site, with no abrupt discontinuities coinciding with the cluster margins.

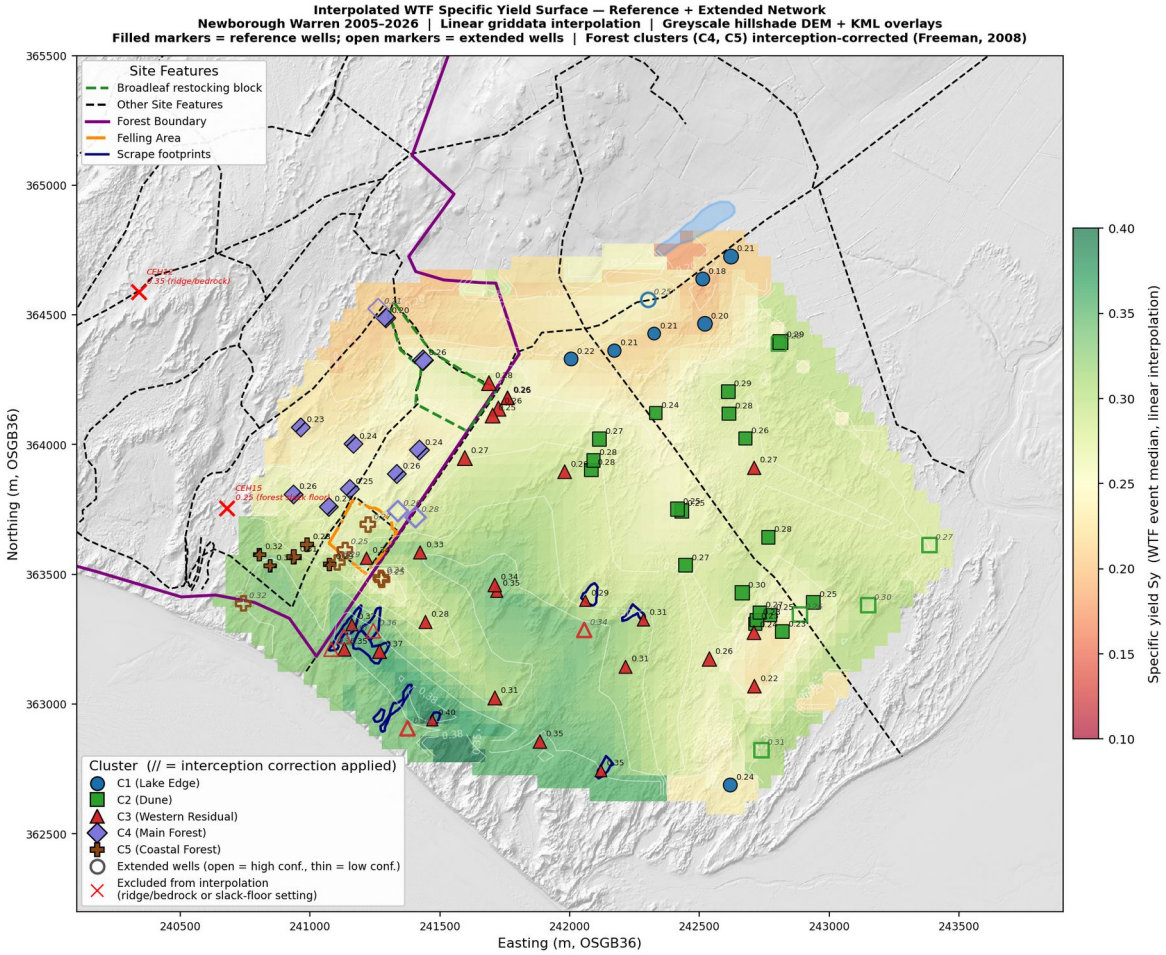



***Figure 13:*** *Interpolated WTF specific yield surface for the combined reference and extended networks, Newborough Warren 2005–2026. Per-well event-median Sy interpolated by piecewise-linear (Delaunay) interpolation over 2 m LiDAR DEM hillshade. Filled markers denote reference-network wells; open markers denote extended-network wells classified by Pearson affinity (Section 4.3, best-match cluster). Forest cluster (C4, C5) wells are interception-corrected following Freeman (2008). Two wells are excluded from interpolation: CEH12 (ridge/bedrock setting) and CEH15 (forest slack floor), both of which produce anomalous Sy values inconsistent with the surrounding aquifer. Per-well cluster-mean Sy values — C1: 0.210, C2: 0.264, C3: 0.305, C4: 0.246, C5: 0.310 — are consistent with the cluster-aggregate estimates of Table 8. The spatial distribution is broadly uniform across the open dune (Sy 0.22–0.31), with no abrupt discontinuities coinciding with cluster margins. (Source: 18_wtf_04_sy_contour_extended.png) LiDAR data: © NRW & OS*

## 4.3 Checking the Clusters: Pearson Affinity Analysis

Of the 66 wells in the reference network, 60 retained their primary cluster membership under the Pearson Affinity Analysis (Section 3.3). Membership is characterized by three distinct tiers based on the correlation delta (Δr) between assigned and neighbouring cluster centroids: Core members (19 wells) with a clear dominant affinity to their assigned cluster (Δr > 0.05), representing stable hydrogeological anchors within their designated geomorphological zone; Fuzzy members (41 wells) exhibiting closely competing affinities between their primary and secondary clusters (Δr < 0.05), identifying areas where the groundwater regime transitions smoothly between adjacent structural blocks; and six Spy classifications where the well's highest temporal affinity belongs to a different cluster than Ward's assignment. CEH21, CEH4 and WMC3 (all assigned C3 Western Residual) show strongest affinity with C5 Coastal Forest, consistent with their coastal- and forest-margin positions where the Western Residual and Coastal Forest signatures overlap. D38, T41a and CEH40 (assigned C3) show strongest affinity with C2 Dune; D38 and T41a are two of the five boundary wells that moved from C2 to C3 in the final revision of the partition. All six Spy deltas are small (|Δr| < 0.015): borderline assignments at gradational boundaries.

Classification confidence was further characterized by the Multi-Cluster Affinity flag, raised where three or more cluster centroids achieved a correlation of r > 0.90. This criterion was met at 44 of 66 wells, reflecting the high mutual correlation among the five cluster centroids in a contiguous, highly transmissive sand aquifer. Of these, 22 are concentrated within the gradational C1/C2/C3 transitional zone of the mid-eastern dune sector, 13 lie along the C2/C3/C5 boundary where the Dune, Western Residual and Coastal Forest signatures converge, 8 lie along the C3/C4/C5 forest–dune boundary, and 1 lies along the C2/C3/C4 boundary. The count measures how much of each hydrograph the clusters share (the rainfall signal every well carries), not how well they are separated. Separation rests instead on the shape of the response once that shared signal is removed: this is what the correlation distance of Section 3.2.1 measures and what the bootstrap stability of Section 4.2 tests, and the covariate test of Section 3.2 shows that the cluster labels carry drainage and atmospheric-draw structure that position and land cover do not. The MCA flags identify the locations of greatest structural ambiguity within the network (Figure 15a).

The extended network analysis (Section 3.3.3) classified all 22 extended wells against the reference centroids, yielding 3 Ext_Core and 19 Ext_Fuzzy assignments (Figure 15b). With less temporal overlap against the centroid time series, affinity margins are compressed, which is why Fuzzy classifications predominate.

Within the extended network, the clearfell treatment wells showed mixed affinities across the C3/C4/C5 boundary, with all wells carrying Multi-Cluster Affinity flags (r > 0.90 with three or more centroids). FE1 and LIS1 returned strongest affinity with C4 Main Forest (r = 0.989 and 0.973 respectively), while FE2 and FE4 showed marginally strongest affinity with C5 Coastal Forest (r = 0.975 and 0.971) with C4 within 0.006. FE3 showed strongest affinity

with C5 (r = 0.985) followed by C3 (r = 0.968). WMC3, in the reference network, showed strongest affinity with C5 (r = 0.961) with C3, its assigned cluster, a close second (r = 0.954). The affinity profiles of the extended-network clearfell wells (FE1–FE4, LIS1) and the other selected wells are plotted in Figure 14; WMC3 is not among them. The small deltas (Δr = 0.003–0.037) place the clearfell zone at the gradational boundary between the Western Residual, Main Forest and Coastal Forest clusters. Forest-range affinities (C4 or C5) persist at these wells despite the December 2017 clearfell (Section 5.1.1).

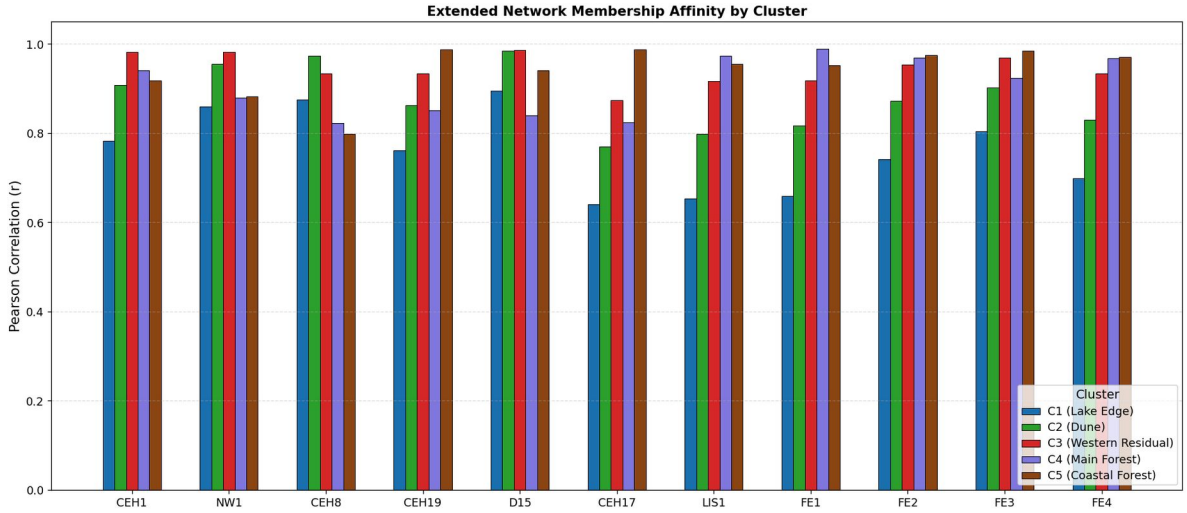



***Figure 14:*** *Pearson affinity analysis of selected reference and extended network wells, showing the correlation coefficient (r) between each well's z-scored time series and each of the five cluster centroids. Wells are selected to illustrate the range of affinity confidence from strong single-cluster dominance (Core members) to near-equal competing affinities (Fuzzy members), along with wells in or close to the clearfell experiment. (Source: 06_pear_01_affinity_chart_extended.png)*

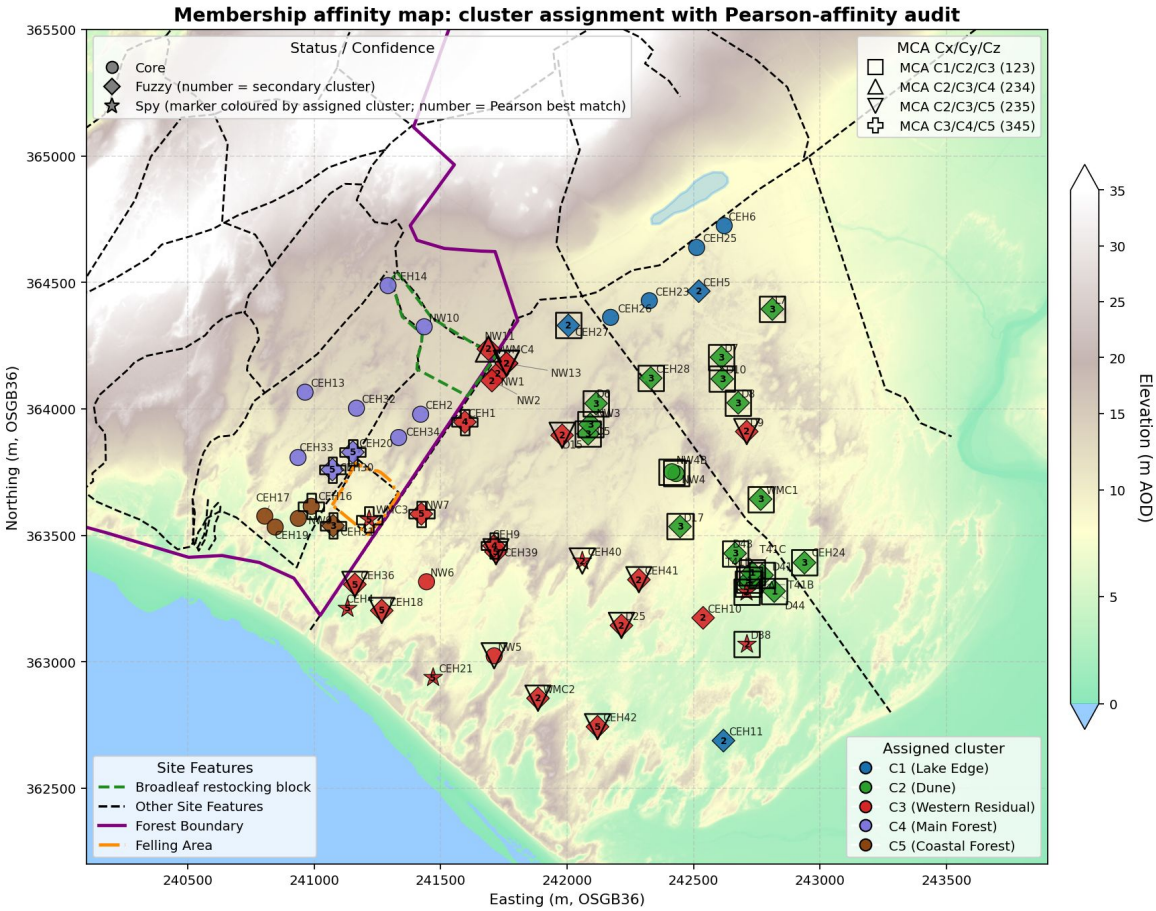



***Figure 15a**: Spatial distribution of the five primary hydrogeological clusters checked by Pearson affinity analysis, reference network (66 wells, records ≥ 100 months to February 2026). Points coloured by Ward's cluster assignment. Symbology denotes Pearson affinity confidence: circles = Core members (Δr > 0.05), diamonds = Fuzzy members (Δr < 0.05; number inside diamond = secondary cluster), stars = Spy classifications (Pearson best-match ≠ Ward's-assigned cluster; number inside star = best-match cluster). Large hollow overlays identify Multi-Cluster Affinity (MCA) wells: convergence r > 0.90 with three or more cluster centroids; overlay shape encodes the specific MCA combination. LiDAR: © NRW & OS (Section 11). (Source: 05_pear_01_spatial_confidence_map.png)*

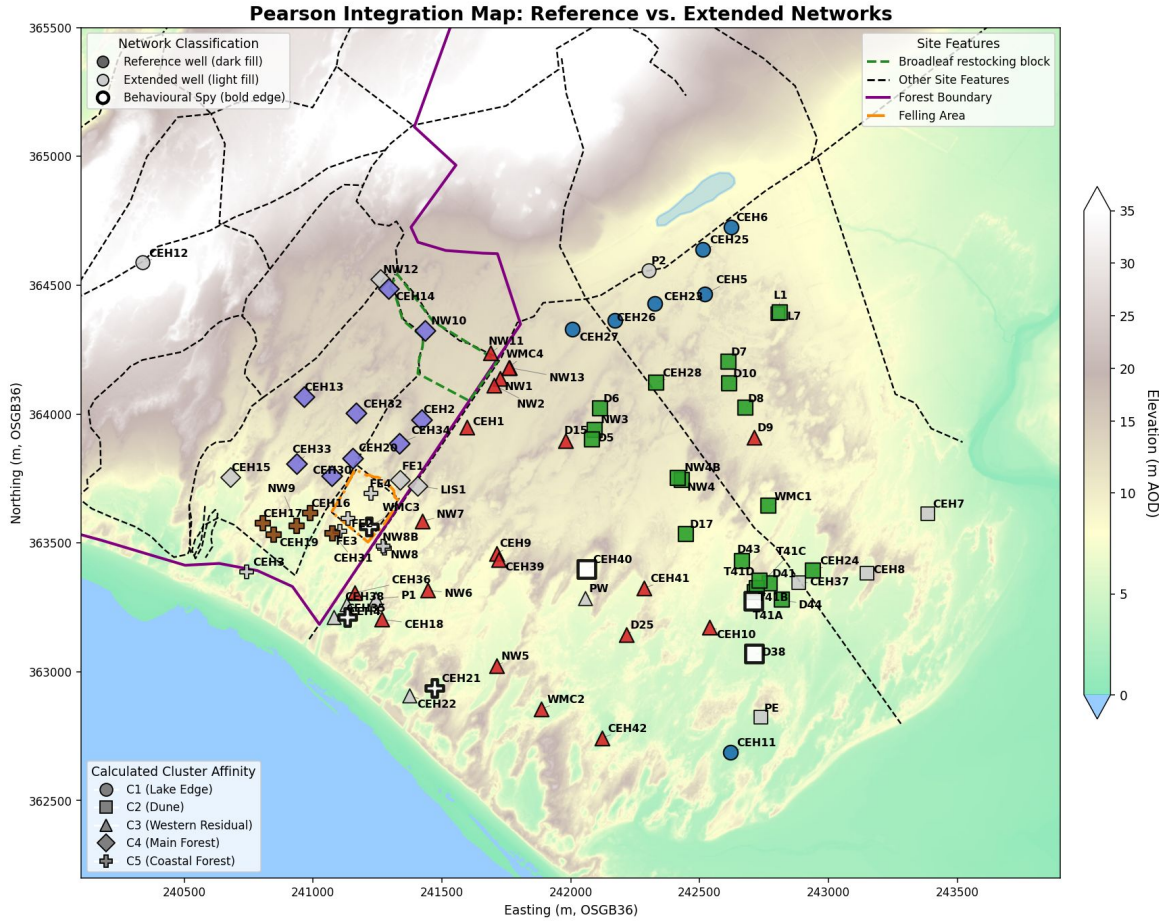



***Figure 15b.*** *Pearson affinity integration map, full 88 well network. Cluster identity encoded by marker shape; fill distinguishes network membership (reference = dark, extended = light grey, Spy = white with bold edge). LiDAR: © NRW & OS (Section 11). (Source: 06_pear_02_integration_map.png)*

Three further checks test the partition against evidence the Pearson analysis does not use (Scripts 28, 29, 31 and 31b). The first asks whether the C3 Western Residual cluster is the C2 Dune cluster with a western-margin coastal-erosion drift superimposed. For each C3 well with coast-distance metadata, the monthly hydrograph was de-trended against the network-scale coastal-retreat gradient model (Section 4.10.2; forest-free linear-capped fit, $\delta_0 = -31.3$ mm yr$^{-1}$, L_cg ≈ 900 m) and re-classified against the un-de-trended cluster centroids with the correlation-distance metric used by Ward's clustering. Of twenty-one testable C3 wells, fifteen were best matched to C3 before de-trending and all fifteen stayed in C3; after de-trending eighteen are best matched to C3, two to C2 and one to C4, and those three (D38, T41a and NW7) were already best matched to C2, C2 and C4 before it. All fifteen also stay in C3 under each of the five sensitivity variants (full-fit versus forest-free $\delta_0$, three values of L_cg, and summer-only versus monthly-uniform de-trending), and all 19 C2 wells keep their assignment under the same procedure.

The second examines the variance within C3. A panel regression of nine per-well behavioural metrics against five spatial and hydrogeological predictors (the Section 4.10.2 exponential coastal predictor, distance to CEH36, distance to the forest polygon edge, ground elevation and depth to water) returns $R^2 = 0.77$ for $\beta_1$ recharge, 0.69 for $\beta_3$ drainage and 0.64 for the recession time $1/\beta_3$. The coefficient on the exponential coastal predictor against per-well slopes is +0.87 on its own and −0.21 in the five-predictor model, and the strongest unique predictor across the three SSM coefficients and the recession time $1/\beta_3$ is distance to CEH36. Along the south-west to north-east axis, $\beta_1$ and $\beta_3$ rise inland ($r = +0.74$ and +0.55) and WTF-derived specific yield falls inland from 0.40 at the south-west margin to 0.22 at the north-east ($r = −0.80$, $p < 0.001$; the 21 C3 wells with a coastal distance). At $k = 6$, Ward's correlation-distance clustering splits C3 into two groups of thirteen wells and leaves C2 intact, and C3 is the least stable cluster under bootstrap resampling of wells (median co-assignment 0.38).

The third asks whether the partition is recoverable from evidence the clustering never used (Script 31). The five clusters are spatially compact: a mean within-cluster separation of 707 m, a same-cluster join-count of $z = 16.0$, and Moran's I of 0.45 to 0.60 for the five cluster indicators, each at $p \leq 0.0001$. They separate the wells on external descriptors the clustering did not see: distance to coast ($\eta^2 = 0.58$), ground elevation ($\eta^2 = 0.22$), and the canopy polygon, which recovers the two forest clusters with $\kappa = 0.91$ (0.96 with edge wells excluded). Hydrograph descriptors that do not depend on the correlation metric separate them as well (mean depth to water $\eta^2 = 0.52$, summer minimum 0.49, dry depth 0.40, seasonal amplitude 0.28), and the SSM coefficients and LCSC, estimated after the clustering, separate them most strongly ($\eta^2 = 0.60$ to 0.69; WTF specific yield 0.50). Re-clustering the same 66 wells under average and complete linkage, and under Spearman and dynamic-time-warping distance, reproduces the partition in part (adjusted Rand index 0.48, 0.44, 0.45 and 0.61 against the canonical Ward's solution). Script 31b asks the converse, whether any single descriptor could have produced the partition on its own. None does: the descriptors that separate the clusters most strongly, easting ($\eta^2 = 0.72$) and the forest flag ($\eta^2 = 0.85$), recover only about a quarter of the partition when the wells are clustered on them alone (ARI 0.23 and 0.26), and separation and recoverability rise together from ground elevation ($\eta^2 = 0.22$, ARI 0.01) without any one variable reaching ARI 0.3.

**Filling gaps from cluster-mates.** The cluster-mate rule was adopted by the test (Section 3.3.4). Pooled over both windows, the nearest cluster-mate restored the held-out spring level (Figure 16a) with a median error of 0.041 m, against 0.139 m for Model A. In the early window, which holds the 2006–08 excess the climate model cannot reproduce, the errors were 0.065 m and 0.149 m; in the late window, 0.037 m and 0.132 m. The other donor rules were close behind: the nearest well at 0.041 m and the best-correlated well at 0.043 m, with the cluster median poorer at 0.059 m. Distance matters as well as the cluster: the error rises with the distance between donor and target (Figure 16b; Spearman $\rho = +0.61$).

Filled by that rule, 69 wells gain 4,432 reconstructed months, and 37 gaps have no qualifying donor and stay empty. Accepting donors read through half the gap brings the total to 75 wells and 4,903 months, and leaves 24 gaps empty. In the median filled well, none of the filled months needed the donor outside the range its line was fitted over.

The grouping did not carry back to 1989–96 by the rule. The 12 CCW wells clustered on their early record (Figure 16c) agree with their modern clusters at an adjusted Rand index of 0.03 (p = 0.45), and only 28 % of the pairs that share a modern cluster were grouped together then. Clustered on their modern record, the same wells reproduce the modern partition at 0.75, so the number of wells is not what prevents agreement. The disagreement has a pattern. In 1989–96 the three seaward warren wells of the second transect (NW6, NW5 and WMC2, now in C3) grouped with the inland warren wells NW3 and NW4 (now in C2), and the forest and border wells NW10, NW11, NW13 and NW9 grouped together, although they now fall in three clusters. The author's charts saw the same thing: 2 of the 3 he marked “similar” drew their wells from one modern cluster. Many early readings sit at the foot of the pipe; keeping them raises the index to 0.20, and using only the 10 least-affected wells gives 0.17. Neither reaches the rule. Durability is therefore not shown, which is not the same as shown to differ: twelve wells over seven years is a weak test.

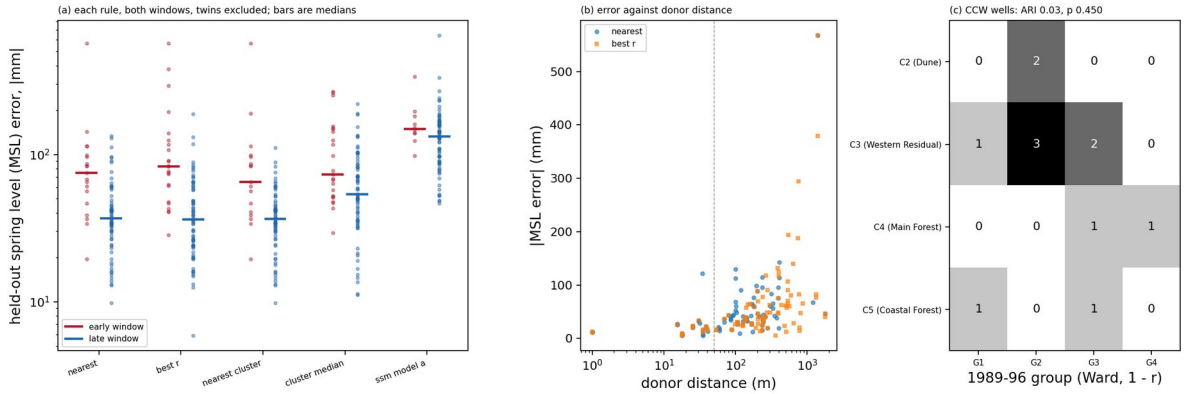



Figure 16: *Filling gaps from cluster-mates. (a) The error in the held-out spring level (MSL) for each donor rule, in the early window (from March 2006, red) and the late window (to the end of the reference record, blue), with twins excluded; bars are medians, on a log scale, and Model A is the climate-only reference. (b) The same error against the distance between donor and target, for the nearest-well and best-correlated rules; the dashed line marks the twin distance. (c) The twelve confirmed CCW wells: their modern cluster (rows) against their group when clustered on 1989–96 alone (columns). (Source: 51_07_cluster_gap_fill.png)*

## 4.4 Model Benchmarking: State-Space Model Comparison with Traditional Linear Models

The SSM was compared to a TLM as outlined in Section 3.4.2. In one-step diagnostic mode — where each prediction is corrected with the observed water level from the previous month — both models track the observed record almost identically (median $R^2$ 0.92 and 0.92 respectively; Table 9). The difference emerges under autonomous simulation. Over 100 months without observational correction, the TLM diverges rapidly at wells with strong head-dependent drainage (e.g. CEH6, C1 Lake Edge: TLM NSE = −0.35, SSM NSE = +0.71; Figure 17), while the SSM maintains physical consistency through its storage decay term. Across the full 66-well reference network, the SSM achieves a median iterative NSE of 0.77 compared with 0.43 for the TLM — a median gain of +0.34 NSE units — and returns a positive iterative NSE at 63 of 66 wells compared with 56 of 66 for the TLM (Table 9). The improvement varies systematically across the five clusters (Figure 18a, 17b): C1 Lake Edge shows the largest median gain (+0.455); C2 (+0.379), C3 (+0.386) and C5 (+0.175) show substantial gains; and C4 Main Forest shows the smallest (+0.121), where $\beta_3$ is lowest (0.019; Section 5.2.3). The sole well where the SSM fails is CEH14 (C4, iterative NSE = −12.87), a ridge-flank well (Section 5.2.3). Spatial visualization (Figure 18) shows the same pattern geographically, with the largest gains in the C1 lake-buffer sector and the eastern dune and the smallest at the forest-boundary C4 wells.

The depth-dependent PET grid search (Section 3.7.1) provides an independent diagnostic of how completely the standard fixed-$\beta_2$ formulation captures the atmospheric draw signal. The grid search improved iterative NSE at all five clusters, from negligibly at C4 to substantially at C1, where L_E (m) is the attenuation length of evaporative draw, the water table depth over which the PET signal falls by a factor of e: C1 (ΔNSE = +0.147, L_E = 0.43 m), C2 (+0.071, L_E = 1.05 m), C3 (+0.026, L_E = 2.22 m), C5 (+0.012, L_E = 2.50 m) and C4 (+0.002, no attenuation detected: the fitted rate of 0.05 $m^{-1}$ is one grid step above the canonical model). The PET draw at C1 therefore falls by a factor of e for every 0.43 m of water table depth; at C4 the fitted rate implies effectively no attenuation over the working depth range; their interpretation is developed in Sections 5.2.2 and 5.6.1.

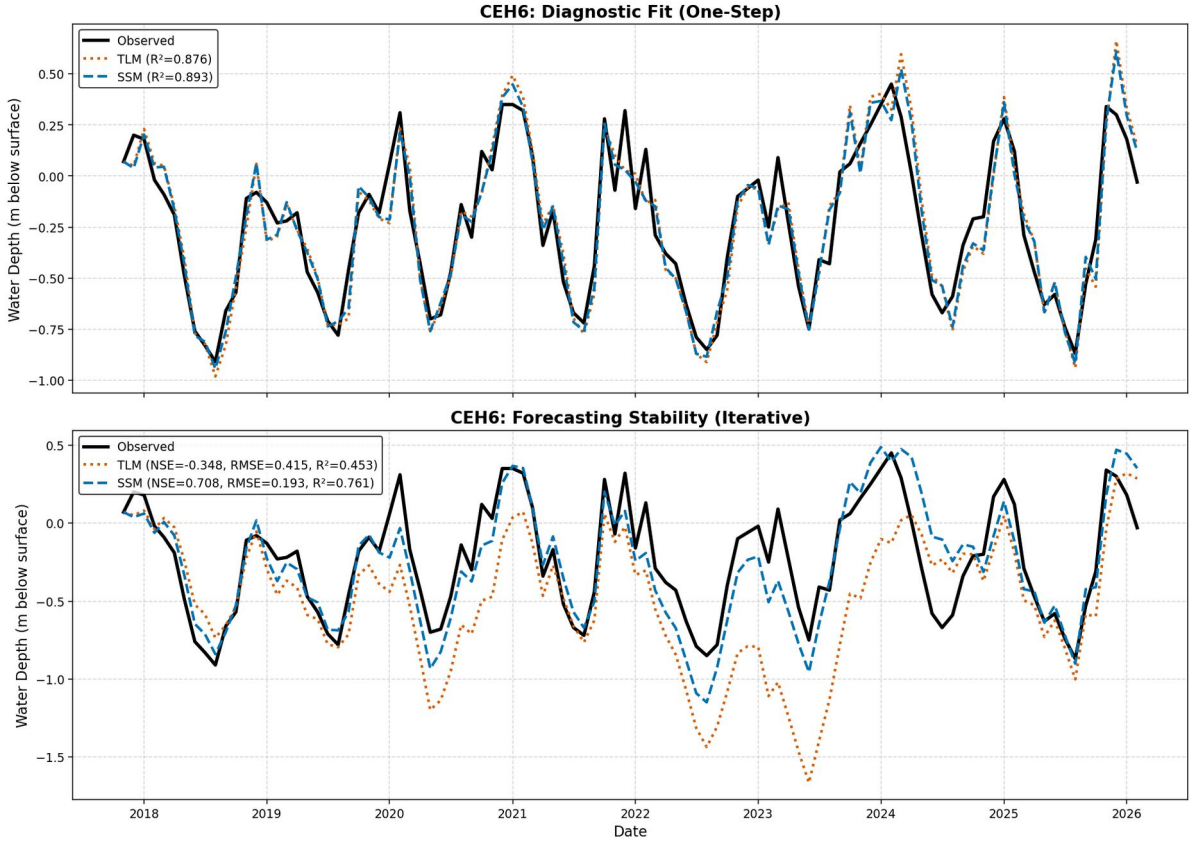



***Figure 17:*** *Dual-model performance comparison at Lake Edge well CEH6. Upper panel: one-step diagnostic fit — both TLM and SSM track observed water levels closely when corrected with real observations at each step. Lower panel: iterative 100-month autonomous simulation without observational correction — the TLM (NSE = −0.35) diverges rapidly, while the SSM (NSE = +0.71) maintains physical consistency through its storage decay term. CEH6 records the eighth-largest NSE improvement (+1.06) in the 66-well reference network. (Source: 08_lcsc_01_ceh6_showdown.png)*

| Metric | TLM | SSM | Improvement Δ |
|---|---|---|---|
| Median one-step R² | 0.920 | 0.924 | 0.0026 |
| Median iterative R² | 0.631 | 0.788 | 0.1552 |
| Median iterative NSE | 0.433 | 0.770 | 0.3379 |
| Wells with iterative NSE > 0 | 56 / 64 | 63 / 64 | — |
| Max NSE improvement (D7) | −5.499 | 0.212 | 5.7107 |
| CEH6 iterative NSE | −0.348 | 0.708 | 1.0561 |
| CEH14 iterative NSE | −0.040 | −12.874 | −12.8340 |

***Table 9:*** *Model benchmarking summary comparing the Traditional Linear Model (TLM) and State-Space Model (SSM) across the 64-well reference network. Performance evaluated over the most recent 100 months at each well. One-step R² uses observed h(t−1) at each step; iterative NSE uses simulated h(t−1) over the full 100-month evaluation window without observational correction. For the median rows, Δ is the median of the per-well improvements (not the difference of the column medians). D7 is the well recording the largest single*

*NSE improvement; CEH6 is the illustrative C1 Lake Edge well shown in Figure 17; CEH14 is the sole well returning a negative SSM NSE, and with CEH13 one of the two where the SSM scores below the TLM. (Source: 08_lcsc_04_table3_benchmark_summary.csv).*

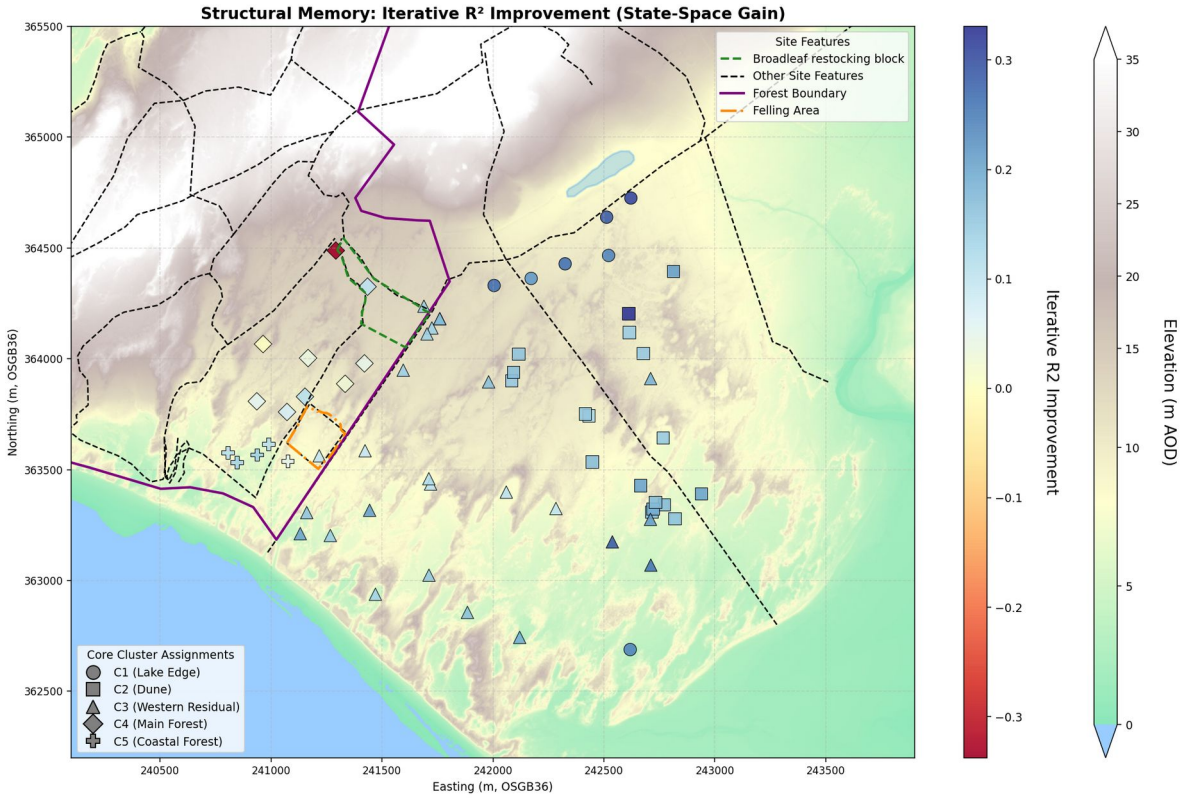



***Figure 18a:*** *SSM gain over TLM across all 66 reference wells. Iterative R² improvement — largest gains in the Lake Edge sector and eastern dune, smallest in the Main Forest cluster where the TLM already performs adequately; the one strongly negative point is CEH14, the negative-$\beta_3$ well (SSM NSE −7.54; Section 4.9.3). (Source: 08_lcsc_02_r2_improvement_map.png) LiDAR data: © NRW & OS; see Section 11 for full attribution.*

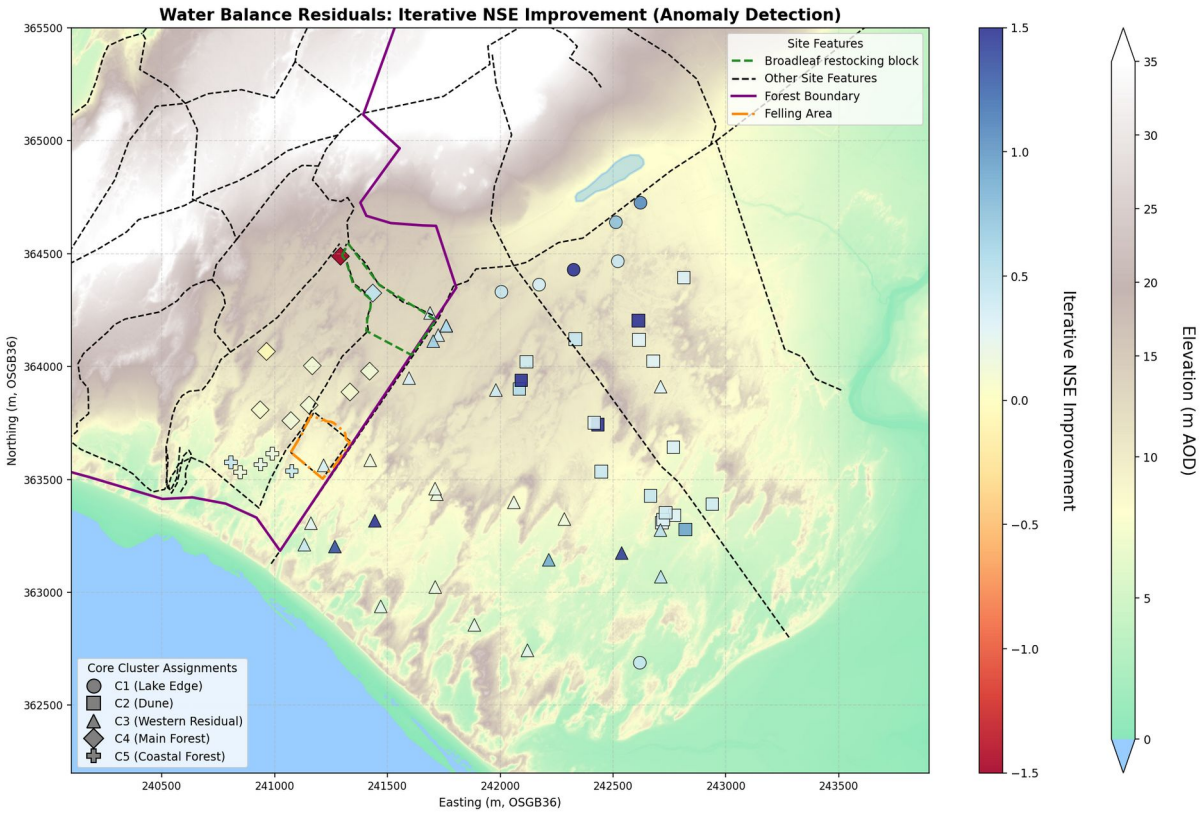



***Figure 18b:*** *SSM gain over TLM across all 66 reference wells. NSE improvement — corresponding spatial pattern with the largest gains concentrated in the C1 Lake Edge cluster. The spatial pattern of improvement suggests where subsurface structural boundaries lie, without invasive coring. (Source: 08_lcsc_03_nse_improvement_map.png) LiDAR data: © NRW & OS; see Section 11 for full attribution.*

## 4.5 Dune Scraping Intervention Analysis

### 4.5.1 Background Environmental Drift

Three dipwells within the dune slack system — CEH36, CEH18 and CEH21 — were subject to ground surface lowering by mechanical scraping, with CEH36 scraped in April 2015 and CEH18 and CEH21 scraped in October 2023. The analysis uses a two-tier CUSUM-BACI framework (Section 3.5) to isolate the scraping signal from background environmental drift, with CEH4 and CEH22 serving as paired controls. Tier 1 analyses at CEH4 and CEH22 both show accelerating negative CUSUM trajectories across the monitoring period, reaching −10.61 m and −19.46 m respectively by early 2026 (Section 3.5.3; Figure 19). Both controls — CEH22 and CEH4 near the coast — deteriorate independently of any management action (Section 5.4.3). CEH22's trajectory is the more severe, accelerating markedly from 2021 onwards.

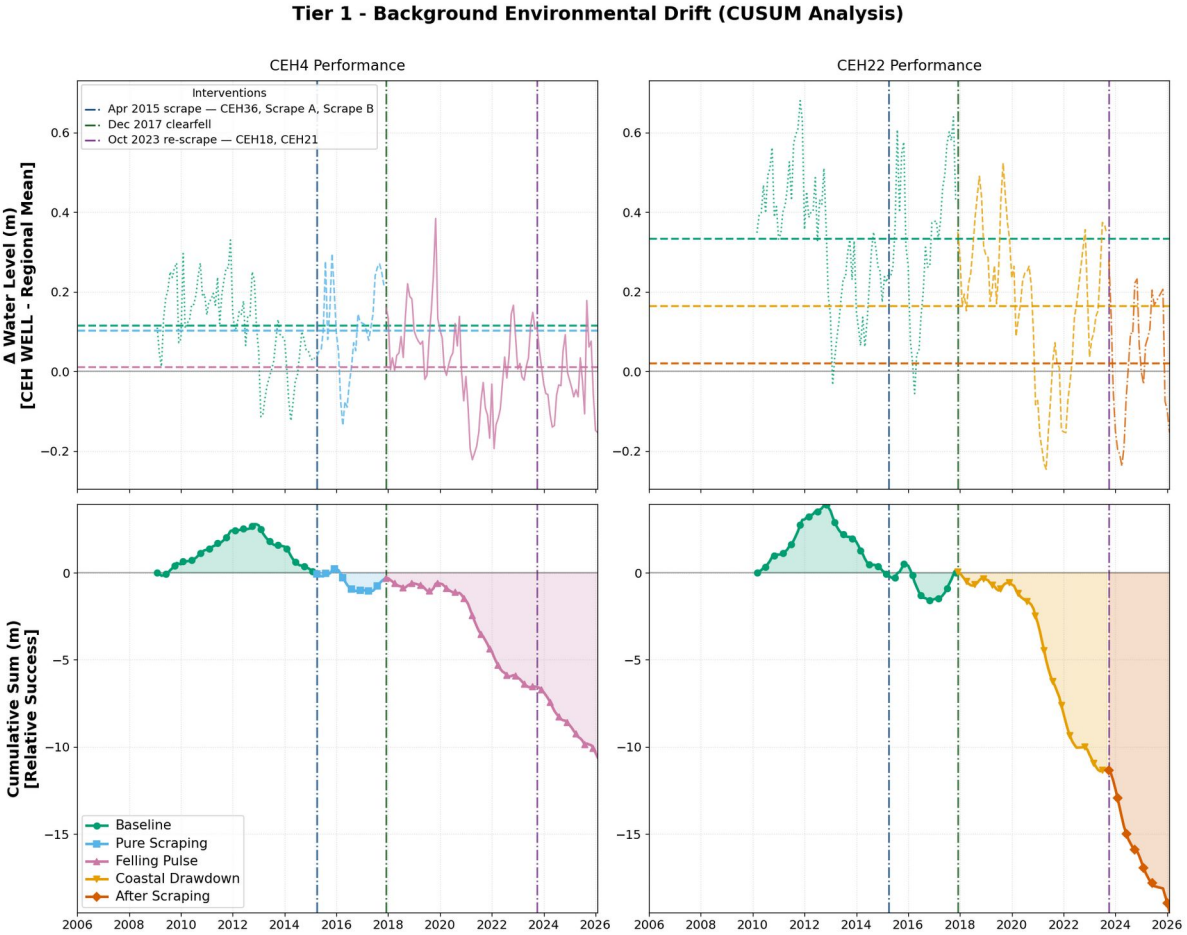



***Figure 19:*** *Tier 1 background environmental drift analysis. CUSUM time series for control wells CEH4 (left) and CEH22 (right) decomposed by management era relative to the regional mean. Upper panels show monthly Δ water level (control well minus regional mean) with era mean levels indicated by dashed horizontal lines. Lower panels show cumulative CUSUM trajectories. A rising CUSUM indicates the well is performing above the regional mean (shallower water table); a falling CUSUM indicates below-average performance (deeper water table). The accelerating negative CUSUM at both control wells is consistent with a system-wide deterioration trend independent of local management interventions. (Source: 09_scrape_05_tier1_background_drift.png)*

CEH4 sits on the eroding coastal margin; its deepening is consistent with coastal retreat together with the warren-wide decline, the record cannot separate the two, and no felling pulse is resolved at CEH4. These Tier 1 trajectories establish that both local controls are experiencing substantial background decline that must be accounted for before any scraping benefit can be assessed.

### 4.5.2 Paired BACI and Robustness

In this context, Tier 2 paired analysis shows a modest positive signal at CEH36 during the Pure Scraping era (April 2015 to December 2017), the only well and era combination to show a net benefit after Tier 1 correction. Three independent methods all return a positive scraping benefit: the two control-differenced methods agree closely — raw BACI vs CEH4 (+0.128 m) and synthetic control from 11 donor wells (+0.141 m) — while the SSM forward residual, an autonomous-forecast estimator and the most conservative of the three, is lower at +0.100 m (Figure 21c). The cumulative CUSUM for CEH36 rises from April 2015 and more steeply from 2018 onwards, as CEH4 deteriorates (Figure 20); the synthetic control separates the scraping signal from CEH4's own trajectory (Figure 21). CEH18 sits consistently negative relative to CEH4 throughout the full record including the pre-scraping baseline; only a modest recovery is visible after the October 2023 scraping. CEH21's rising CUSUM from 2022 onwards is read against a control, CEH22, that is itself deteriorating (Section 4.5.1), which inflates the apparent relative improvement; these results are noted for completeness only.

After Tier 1 correction for background drift, the net benefit at CEH36 against the CEH21 coastal benchmark reversed to −0.055 m during the Felling Pulse era (against CEH4 the paired step fell to +0.023 m but did not reverse), as the deepening at CEH4 (consistent with coastal retreat together with the warren-wide decline) progressively eroded the scraping gain. The raw Tier 2 CUSUM tells a different story, with CEH36 rising strongly relative to CEH4 through this period — but as established above, this reflects CEH4's own deterioration rather than genuine improvement at CEH36. At CEH18 and CEH21, scraped in October 2023, no scraping-specific response survived Tier 1 correction for background drift. CEH18's corrected net benefit was −0.075 m after scraping; CEH21's paired control (CEH22) is itself on a coastal erosion trajectory, so a net benefit cannot be reliably estimated. The monitoring window for both wells is short (two post-scraping summers) and the analysis is likely underpowered to detect benefits of the magnitude observed at CEH36.

The annual summer minimum records for all three treatment wells, partitioned by management era and plotted against the Curreli et al. (2013) SD15b / SD16 community-mean levels, are presented in Figure 23. The deepening of the CEH4 control well after the Pure Scraping era (era mean summer minima of 1.075 m at baseline, 1.053 m during the Pure Scraping era and 1.148 m during the Felling Pulse) is consistent with coastal retreat at this eroding-margin well together with the warren-wide decline documented in Section 4.8.1; the record cannot separate the two, and no felling pulse is resolved at CEH4. It indicates that management benefit must be assessed against this background trajectory rather than in absolute terms.

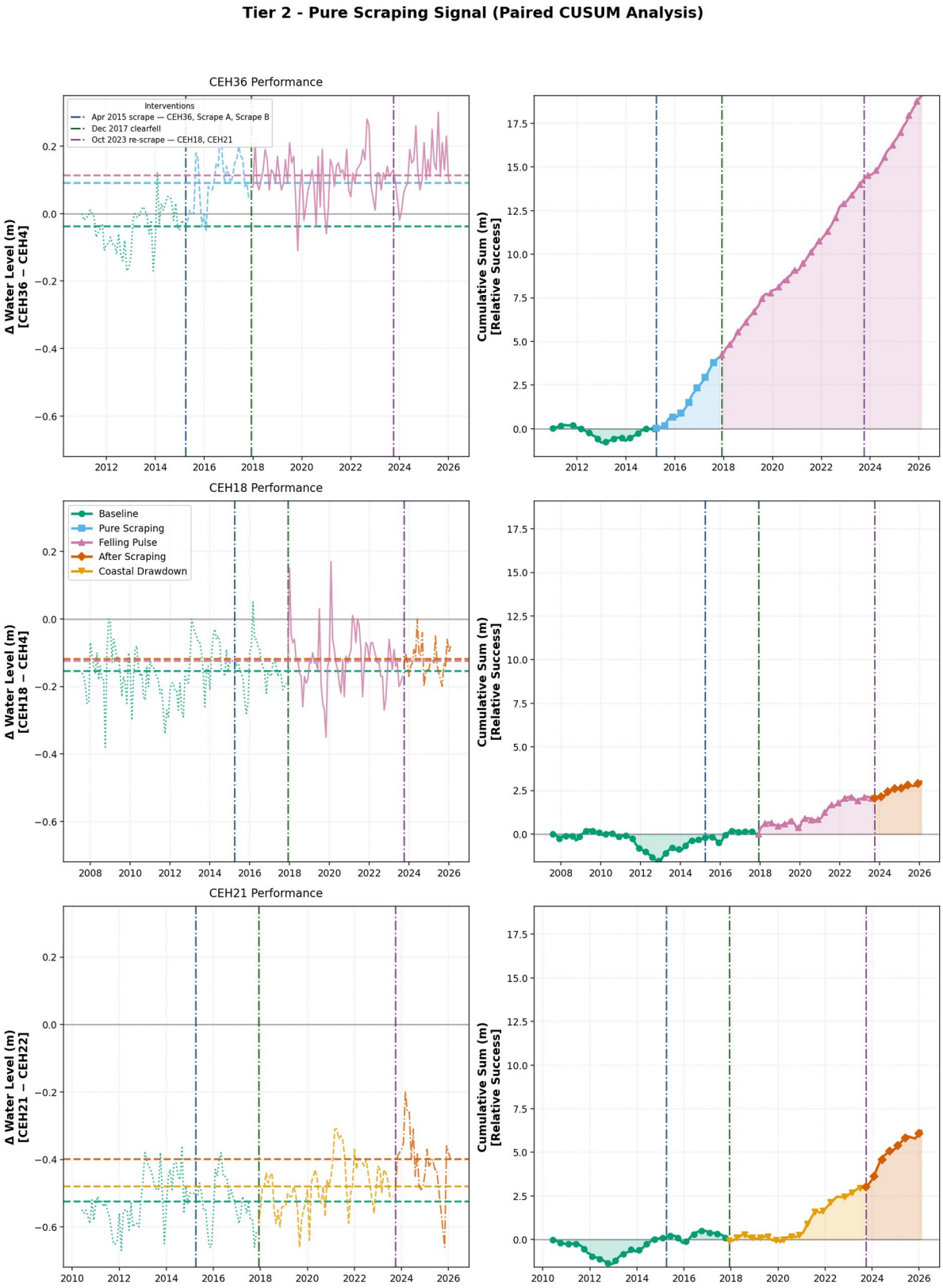



***Figure 20:*** *Tier 2 paired CUSUM analysis at treatment wells CEH36 (top), CEH18 (centre) and CEH21 (bottom), differenced against matched controls (CEH36 and CEH18 vs CEH4; CEH21 vs CEH22). Left panels: monthly Δ water level relative to control with era means as dashed lines. Right panels: cumulative BACI CUSUM — rising indicates the treatment well becoming shallower relative to control, falling indicates deepening. CEH36 shows a rising CUSUM from April 2015, steepening from 2018, though this partly reflects the progressive deterioration of CEH4 rather than genuine improvement at CEH36. Neither CEH18 nor CEH21 shows a scraping response that survives Tier 1 correction; CEH21's apparent post-2022 rise reflects deterioration of its control (CEH22) through coastal erosion rather than a scraping benefit. (Source: 09_scrape_06_tier2_scraping_signal.png).*

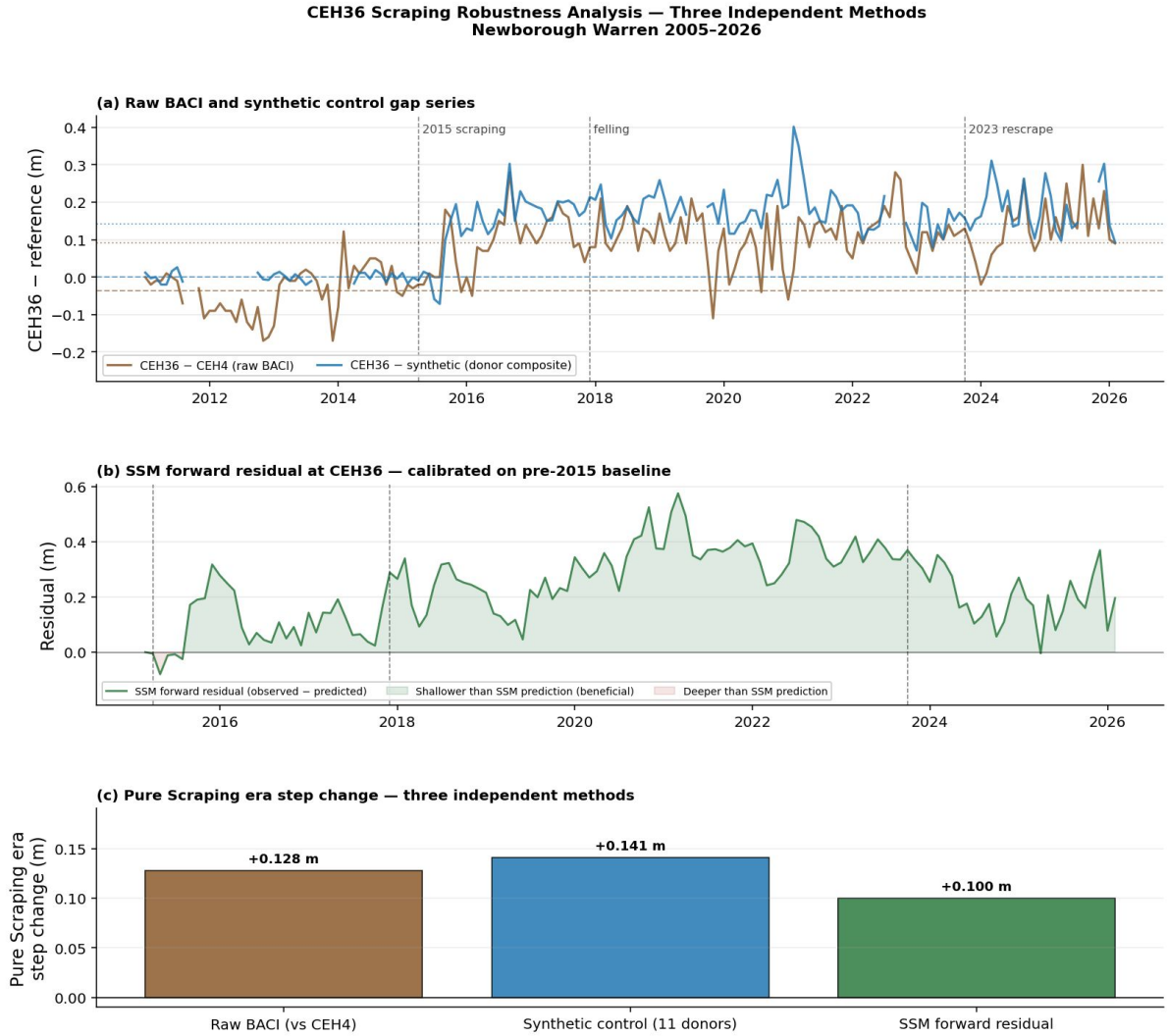



***Figure 21:*** *Three-method robustness assessment of the scraping signal at CEH36. (a) Raw BACI gap series (CEH36 minus CEH4, brown) and synthetic control gap series (CEH36 minus donor composite, blue), with intervention dates marked. The persistent positive offset after April 2015 is consistent with the scraping step change, and the two gap series stay close together through the post-2018 record. (b) SSM forward residual at CEH36, computed by calibrating the model on the pre-scraping period and running forward iteratively. Green shading indicates periods where CEH36 is shallower than the SSM prediction. The residual is persistently positive throughout the post-scraping record, indicating that the water table sits systematically above the level predicted by the pre-intervention model — consistent with a permanent structural benefit from scraping. The residual increases further post-clearfell (2018 onwards), suggesting the combined effect of scraping plus canopy removal exceeds the pre-intervention SSM baseline by 0.3–0.5 m in peak winters. (c) Pure Scraping era step changes from three independent methods: Raw BACI vs CEH4 (+0.128 m), synthetic control (+0.141 m), and SSM residual (+0.100 m). All three return a positive benefit, with the two control-differenced methods near 0.13 m. (Source: 09_scrape_08_ceh36_robustness.png)*

### 4.5.3 Drainage Coefficient Response

Era-specific $\beta_3$ estimates show directional increases at both treatment and control wells (Table 10): CEH36 increased from 0.084 at baseline to 0.139 during the Pure Scraping era; CEH4, its unmanipulated control, showed a comparable increase from 0.070 to 0.162 over the same period. CEH18 rises from 0.075 at baseline through 0.098 to 0.146 after the October 2023 scraping, whereas CEH21 falls from 0.103 to 0.052 and recovers to 0.095 (not significant) against a flat CEH22 (0.140, 0.140, 0.155) (Figure 22). The $\beta_3$ increase is shared by the unmanipulated CEH4 but not by CEH22. The per-well era estimates are reported in Table 10; their reading is in Section 5.4.3.

| Well | Role | Era | $\beta_3$ | 95% CI | p-value | Sig |
|---|---|---|---|---|---|---|
| CEH36 | Treatment | Baseline | 0.084 | [0.004, 0.163] | 0.041 | * |
| CEH36 | Treatment | Pure Scraping | 0.139 | [0.057, 0.220] | 0.002 | ** |
| CEH36 | Treatment | Felling Pulse | 0.110 | [0.053, 0.168] | <0.001 | *** |
| CEH18 | Treatment | Baseline | 0.075 | [0.037, 0.114] | <0.001 | *** |
| CEH18 | Treatment | Felling Pulse | 0.098 | [0.020, 0.176] | 0.014 | * |
| CEH18 | Treatment | After Scraping | 0.146 | [0.012, 0.279] | 0.034 | * |
| CEH21 | Treatment | Baseline | 0.103 | [0.052, 0.153] | <0.001 | *** |
| CEH21 | Treatment | Coastal Drawdown | 0.052 | [−0.006, 0.109] | 0.080 | ns |
| CEH21 | Treatment | After Scraping | 0.095 | [−0.018, 0.208] | 0.095 | ns |
| CEH4 | Control | Baseline | 0.070 | [0.026, 0.114] | 0.002 | ** |
| CEH4 | Control | Pure Scraping | 0.162 | [0.079, 0.244] | <0.001 | *** |
| CEH4 | Control | Felling Pulse | 0.113 | [0.064, 0.162] | <0.001 | *** |
| CEH22 | Control | Baseline | 0.140 | [0.088, 0.192] | <0.001 | *** |
| CEH22 | Control | Coastal Drawdown | 0.140 | [0.062, 0.217] | <0.001 | *** |
| CEH22 | Control | After Scraping | 0.155 | [0.007, 0.302] | 0.040 | * |

***Table 10:*** *Era-specific $\beta_3$ storage decay coefficients at scraping treatment and control wells with 95% confidence intervals. Treatment wells (CEH36, CEH18, CEH21) shaded mid-grey; control wells (CEH4, CEH22) shaded light grey. The $\beta_3$ changes at the control wells are not read as a scraping effect: CEH4 rises in the Pure Scraping era, before the December 2017 clearfell, and falls again in the Felling Pulse, and CEH22 is flat within its confidence intervals. Treatment-well era-specific $\beta_3$ estimates are plotted in Figure 22. Significance: *** p < 0.001, ** p < 0.01, * p < 0.05. (Source: 09_scrape_04b_beta3_era_summary.csv).*

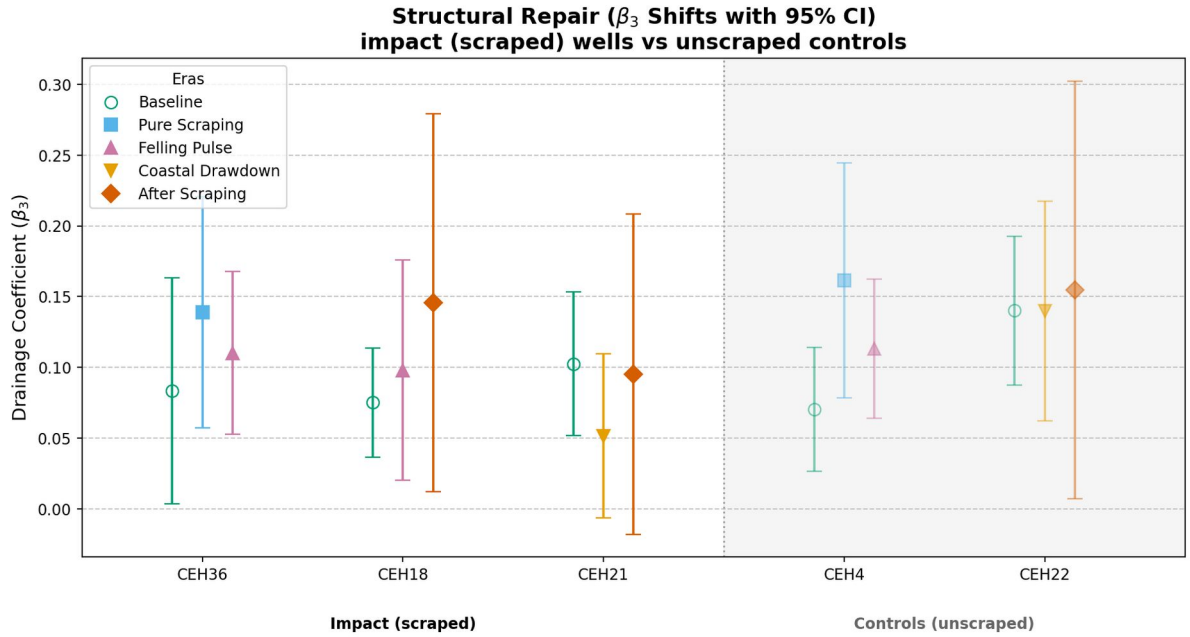



***Figure 22:*** *Era-specific $\beta_3$ storage decay coefficients with 95% confidence intervals at scraping treatment wells (CEH36, CEH18, CEH21) and their unscraped controls (CEH4, CEH22). Each symbol represents a management era: Baseline (hollow circle), Pure Scraping (filled square), Felling Pulse (triangle), After Scraping (diamond) and Coastal Drawdown (inverted triangle). Where confidence intervals overlap between eras, $\beta_3$ changes are not statistically distinguishable. CEH36 rises from Baseline (0.084) to Pure Scraping (0.139) and eases to 0.110 in the Felling Pulse, though the confidence intervals overlap across all three eras. CEH18 rises from Baseline (0.075) through Felling Pulse (0.098) to 0.146 after the October 2023 scraping; that estimate differs from zero (p = 0.034), but its wide confidence interval [0.012, 0.279] overlaps both earlier eras. CEH21's Coastal Drawdown estimate (0.052, ns) and After Scraping estimate (0.095, ns) are not significant. Control wells (CEH4, CEH22) are drawn faded, to the right of the dashed separator; CEH4's $\beta_3$ rises in the Pure Scraping era and falls in the Felling Pulse as CEH36's does, while CEH22's is flat, which is why the cross-era changes are not read as a scraping effect (full control values in Table 10). (Source: 09_scrape_07_beta3_confidence.png).*

### 4.5.4 Summer Minima

The annual summer minimum depth records for the three scraping treatment wells and the CEH4 control well, partitioned by management era, are shown in Figure 23. CEH36 shows a clear reduction in summer minimum depth during the Pure Scraping era (April 2015 to December 2017), with the era mean shallowing from −1.115 m (pre-scraping, n = 4 summers) to −0.917 m (2015–17, n = 3 summers) — a raw improvement of 198 mm (Figure 23). The era mean is −0.986 m in the post-felling era. Neither CEH18 nor CEH21 shows a clearly established summer minimum response to the October 2023 scraping — the post-scraping record comprises only two summers with high inter-annual variability, and CEH21's apparent improvement does not survive background correction against CEH22. The CEH4 control well deepens after the Pure Scraping era (−1.075 m pre-scraping, −1.053 m in 2015–17, −1.147 m during the Felling Pulse).

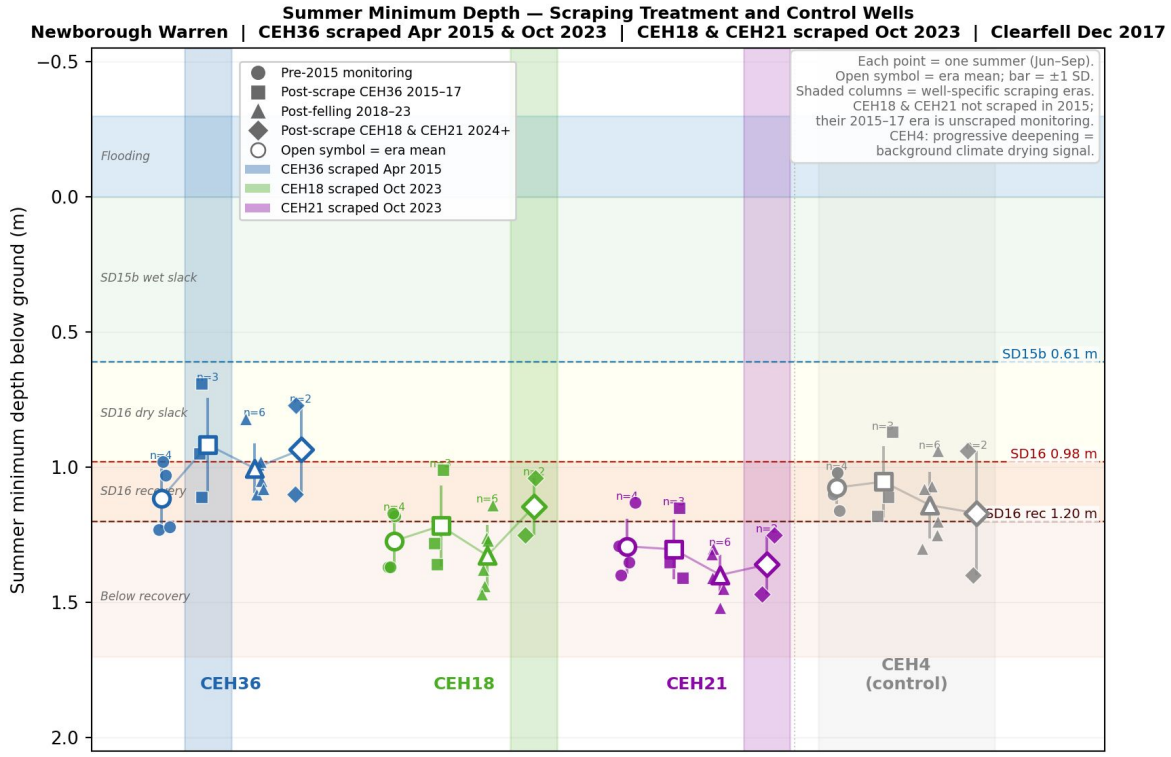



***Figure 23:*** *Annual summer minimum depth below ground surface at scraping treatment wells (CEH36, CEH18, CEH21) and control well (CEH4), partitioned by management era. Each point represents the mean annual summer minimum depth for one year (June–September). Open symbols show era means; vertical bars = ±1 SD. CEH36 was scraped in April 2015 (blue band, 2015–17 Pure Scraping era) and re-scraped in October 2023, to a depth not recorded; CEH18 and CEH21 were scraped in October 2023 (green and purple bands, post-scraping era). Post-scraping era means for CEH18 and CEH21 are based on two summers (2024 and 2025) only. CEH18 and CEH21 were not scraped in 2015 — their 2015–17 records represent unscraped monitoring. CEH4 (grey) is the unscraped control, plotted uncorrected; it sits on the eroding coastal margin, and its deepening in the Felling Pulse era is consistent with coastal retreat together with the warren-wide decline; the record cannot separate the two, and no felling pulse is resolved at CEH4. Zone colouring is bounded by the SD15b / SD16 community-mean levels of Curreli et al. (2013), 0.61 m and 0.98 m below ground, with a further band to 1.20 m marking the planning depth reachable by a deeper scrape: SD15b wet slack (0–0.61 m), SD16 dry slack (0.61–0.98 m), SD16 recovery (0.98–1.20 m). (Source: 21_forestry_03_scraping_eras.png).*

The paired BACI summer minima analysis (09c) shows the scraping benefit at CEH36 to be statistically significant against the paired control CEH4 (+195 mm, p = 0.004) and also significant against the climate control centroid (+161 mm, p = 0.006; Figures 24 and 25). Neither CEH18 (+56 mm, p = 0.20) nor CEH21 (−8 mm, p = 0.89) showed significant paired shifts. CEH4 showed a non-significant change against the climate centroid (−34 mm, p = 0.33); the background drying trajectory is evidenced instead by the Tier-1 CUSUM decline and the era-mean progression (Section 4.5.1; Figures 19 and 23) and, in climate-corrected anomaly terms, by the divergence of the two wells after April 2015 (Figure 26). The distributional view (Figure 35) is consistent with this pattern: Impact and Edge tier summer minima track the Forest Control trajectory

without detectable divergence after felling, while the Coastal Control summer minima show no progressive deepening across the record.

The same paired comparison on the five-year mean spring water level (MSL5; Section 4.8.3) corroborates the benefit on the baseline vegetation metric. MSL5 at CEH36 shallows from −0.64 m at window-end 2015 to between −0.42 and −0.55 m (mean −0.49) across window-ends 2017–2023, opening a sustained separation of +0.03 to +0.09 m (mean +0.07) from the paired control CEH4, which sits between −0.45 and −0.62 m over the same window-ends and recovers slightly in the two latest. Because MSL5 is a rolling five-year mean, the scraping signal enters the metric gradually as post-scrape springs populate the window rather than as a discrete step, and part of the 2015–2017 shallowing is shared with CEH4 as the wet springs of that period enter both wells' windows; the sustained CEH36-minus-CEH4 separation that persists through the post-scrape record is, however, consistent in direction, though smaller in magnitude, with the +195 mm paired summer-minimum benefit (Figure 25) and the +128 mm BACI-corrected monthly benefit (Section 4.5.2). The benefit is thus present on all three metrics — the monthly mean, the summer minimum and the five-year spring baseline (Section 5.4.1).

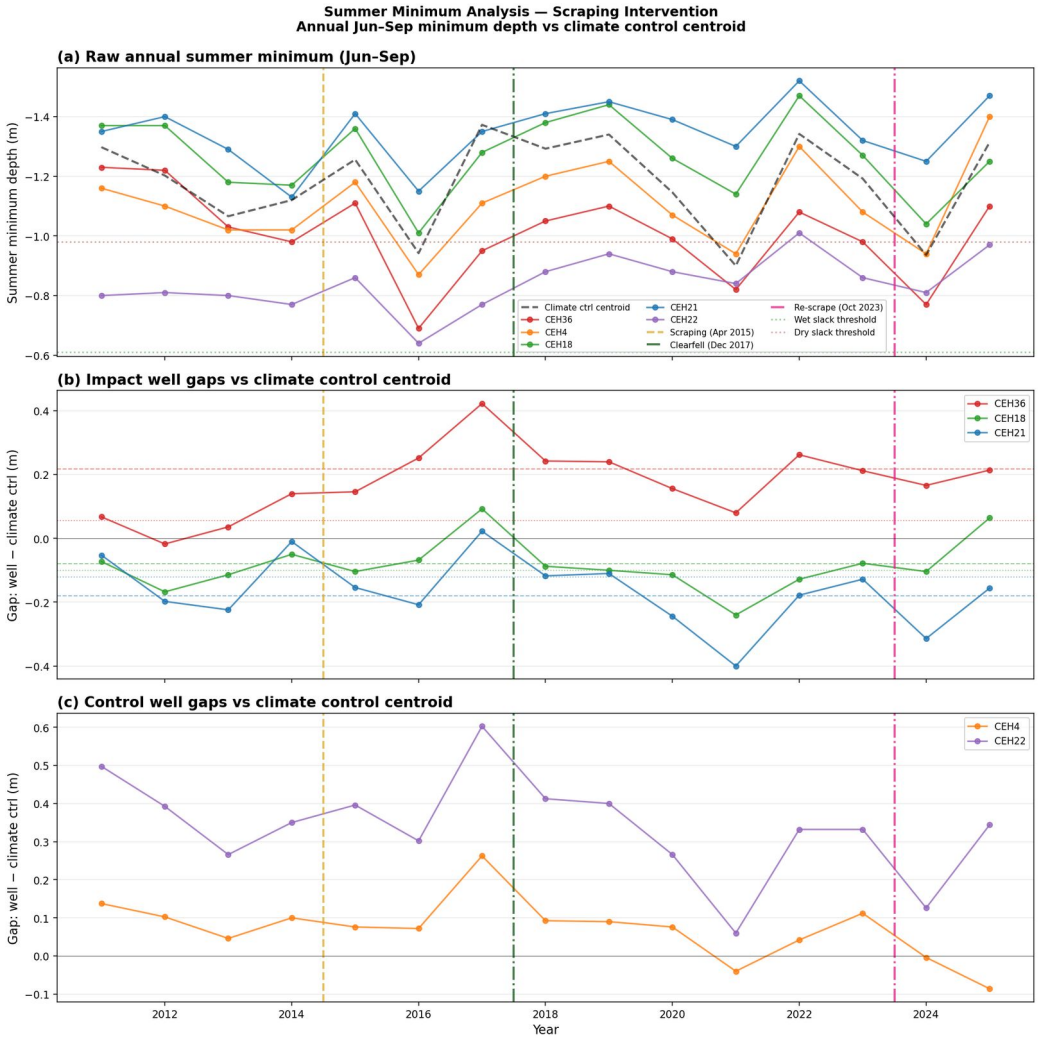



***Figure 24:*** *Summer minima analysis for the scraping intervention against the climate control centroid. (a) Raw annual summer minimum (June–September) for treatment and control wells. (b) Impact well gaps vs climate control centroid, pre/post scraping. (c) Control well gaps vs climate control centroid (diagnostic). (Source: 09c_03_summer_minima_climate_ctrl.png)*

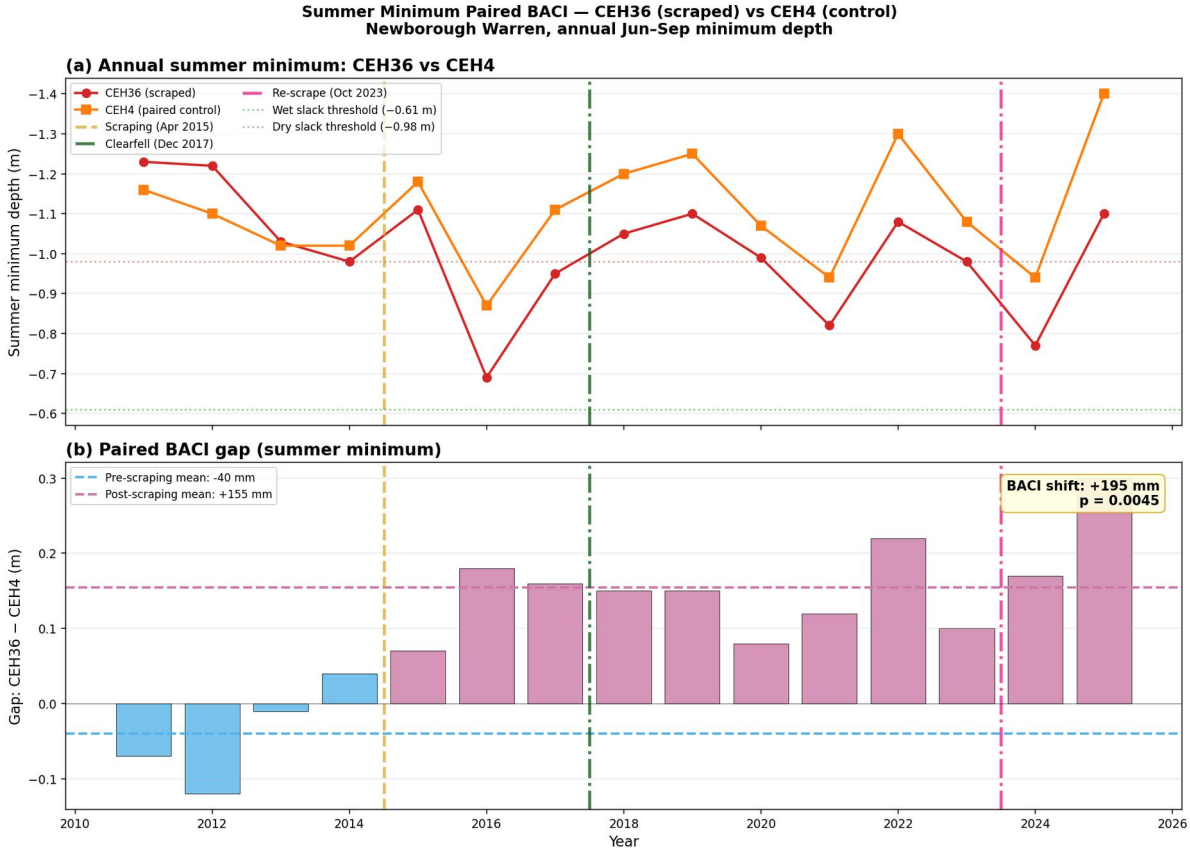



***Figure 25:*** *Paired BACI summer minimum analysis: CEH36 (scraped) vs CEH4 (control). (a) Annual summer minimum depth with the SD15b / SD16 community-mean levels of Curreli et al. (2013). (b) BACI gap (CEH36 minus CEH4) by year, with pre- and post-scraping means. BACI shift: +195 mm, p = 0.004. (Source: 09c_04_summer_minima_paired.png).*

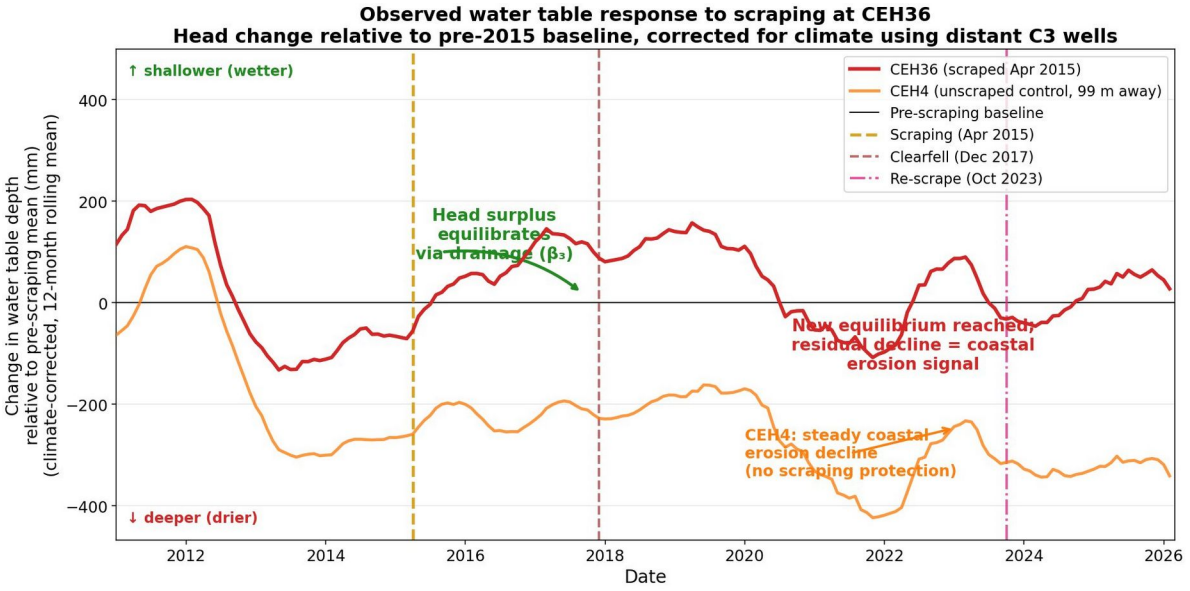



***Figure 26:*** *Climate-corrected anomaly trajectory at CEH36 (scraped) vs CEH4 (control), showing the divergence following April 2015 scraping and the protection of CEH36 against the deepening visible in CEH4. The two later vertical lines mark the December 2017 clearfell and the October 2023 re-scrape. (Source: 09b_03_ceh36_equilibration.jpg)*

### 4.5.5 Scraping Propagation into the Surrounding Aquifer

A supplementary analysis tested whether the CEH36 scraping signal propagated uphill into the surrounding aquifer, comparing split-window SSM fits at ten wells between 250 m and 780 m from CEH36, with BACI correction against seven distant C3 control wells. Under a full three-coefficient refit, seven of the eight wells within 575 m returned a positive BACI-corrected change in the drainage coefficient (mean $\Delta\beta_3 = +2.9 \times 10^{-3}$; one-sample $t = 2.47$, $p = 0.043$). Its interpretation is in Section 5.4.3.

The uphill wells traverse pre-existing spatial gradients in the drainage coefficient. The wells nearer CEH36 lie mostly within the C3 Western Residual cluster (the nearest, CEH31, is C5), which carries a significant internal $\beta_3$ gradient along the south-west to north-east axis (Section 4.9.2; $\beta_3$ increasing inland, $r = +0.55$, $p = 0.009$); the higher uphill wells lie mostly within the C4 forest zone (CEH9 and CEH1 are C3), where $\beta_3$ shows a significant elevation dependence (Section 4.9.4; $r = -0.836$, $p < 0.001$). The BACI-corrected $\Delta\beta_3$ shows no decay with distance from CEH36.

The aquifer diffusivity timescale L_B²Sy/(Kb) exceeds seven years at a separation of 570 m (Fetter, 2001), against a 31-month post-scraping record. The inferred ground surface lowering at CEH36 is approximately 0.43 m (back-calculated as d_ex = $H_0$/Sy from the measured BACI step $H_0 \approx 0.13$ m and open-dune Sy ≈ 0.30), consistent with field observation of the cut. The water-table shallowing partially re-equilibrates over approximately two years, retaining +128 mm as the BACI-corrected benefit during the pure scraping era (Section 4.5.2). The SSM forward residual (Figure 21b) stays positive through

this period (+0.100 m over the Pure Scraping era; Figure 21c), consistent with a lasting lowering of the ground surface.

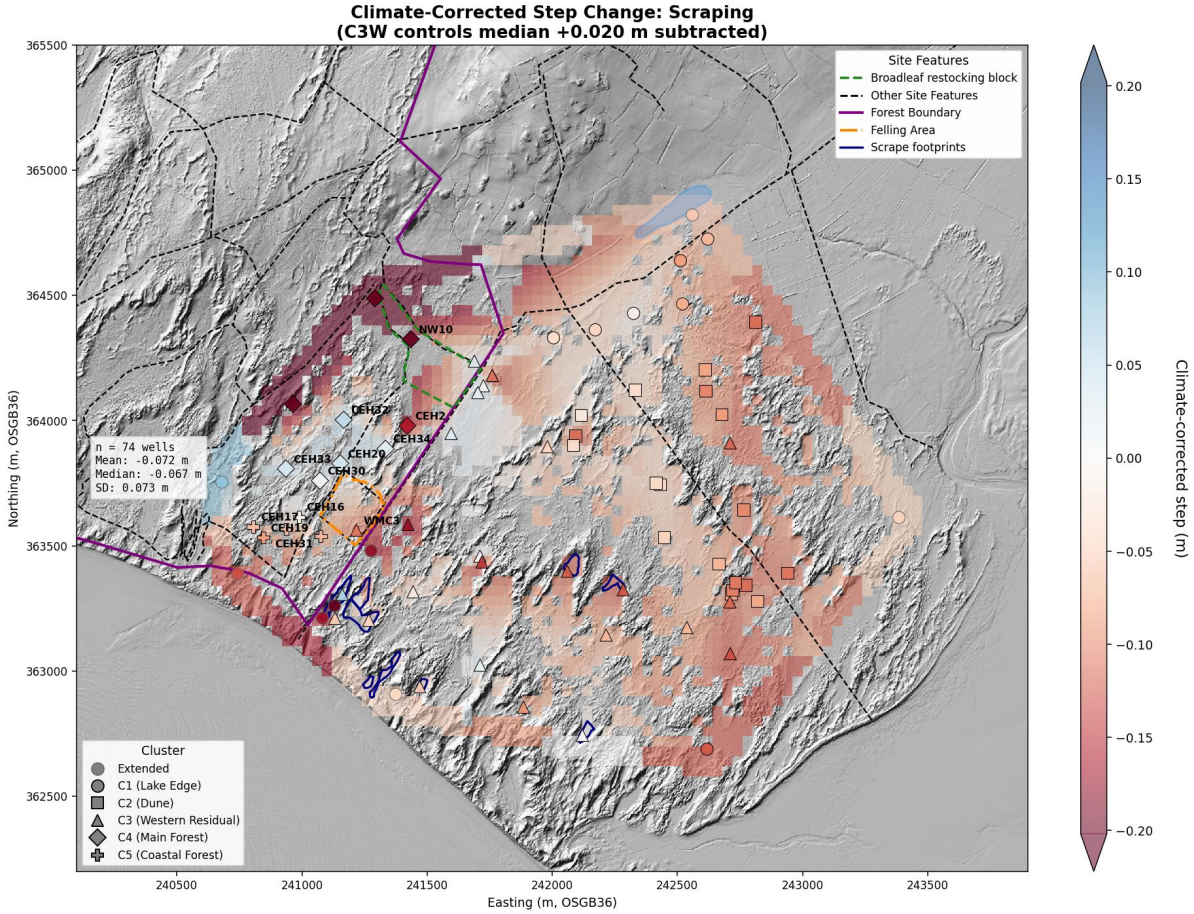



***Figure 27:*** *Climate-corrected spatial step-change map for the scraping era (April 2015 – November 2017). Each well's step relative to its pre-scraping mean, corrected by subtracting the C3 control centroid step. The localized response at CEH36 and absence of a spatial gradient uphill at well distances of 250–780 m within the 31-month observation window are consistent with the scraping effect being either spatially confined or propagating on timescales longer than the available record (Section 5.4.3). (Source: 10b_spatial_scrape_corrected.png)*

### 4.5.6 Scenario Contextualization

To contextualize the observed scraping benefit at CEH36, five alternative interventions were compared at the same well under two climate forcings (09d). The forestry alternatives take the felled well's measured change in atmospheric draw, with no interception recharge. Under annual-mean forcing (Figure 28), the observed scraping benefit (+46.9 mm w.e./month) exceeds each alternative: clearfell +0.8, thinning +0.4, broadleaf −1.3, and climate wet +8.1 mm/month, against −14.3 mm/month under climate dry. Evaluated under summer forcing (Figure 29) scraping still exceeds every alternative and climate wet (+8.0) exceeds each forestry scenario; the forestry responses are marginally larger (clearfell +1.4, thinning +0.7, broadleaf −2.2), and the climate-dry penalty is marginally larger (−14.7); both figures report the same equilibrium volumetric quantity and are directly comparable. Both figures also show the modelled off-site neighbour drawdown the scrape imposes on the surrounding water table (−31.3 mm w.e./month at 100 m, decaying to −15.9 at 250 m): scraping raises the slack locally but draws its neighbours down; the canopy-management scenarios carry no off-site term. The raw level steps at WMC3, the nearest long-record uphill well, are examined in Section 4.6.3, where none is resolved against its month-to-month variability. (For reference, the observed paired-BACI summer-minimum shift at CEH36 was +195 mm; the equilibrium framework does not resolve summer minima, so it is not plotted.)

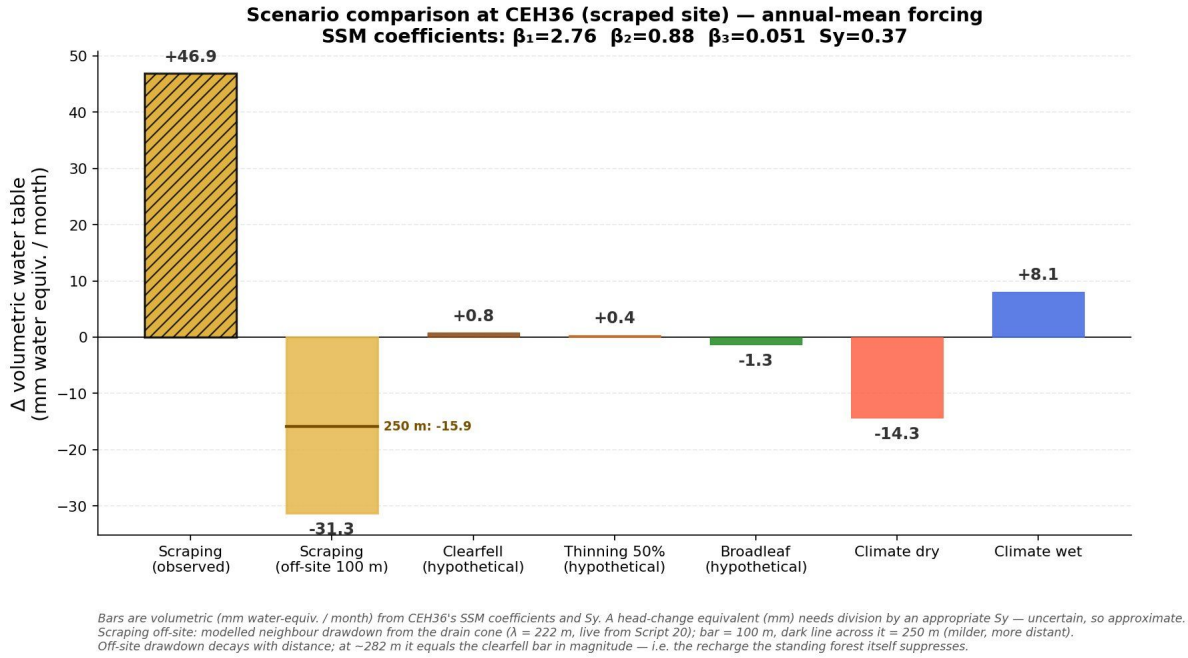



***Figure 28:*** *Scenario comparison at CEH36 (scraped site) — annual-mean forcing. Equilibrium volumetric water table change (mm water-equivalent per month) under each scenario. Observed scraping benefit (+46.9) compared with hypothetical forestry scenarios (clearfell +0.8, 50% thinning +0.4, broadleaf −1.3 — 'what if CEH36 had pine canopy?') and UKCP18 RCP8.5 climate projections (wet +8.1, dry −14.3). A second scraping bar shows the modelled off-site neighbour drawdown at 100 m (−31.3), with a reference line marking the milder drawdown at 250 m (−15.9), from the drain-cone model of Section 4.11. The forestry scenarios take the felled well's measured change in draw with no interception recharge; broadleaf applies its seasonal $\beta_2$ profile. Converting a volumetric bar to a head change requires dividing by an appropriate, uncertain specific yield and is only approximate. All computed using CEH36's own SSM coefficients ($\beta_1$ = 2.76, $\beta_2$ = 0.88, $\beta_3$ = 0.051) and Sy = 0.37. (Source: 09d_01_scenario_comparison.jpg)*

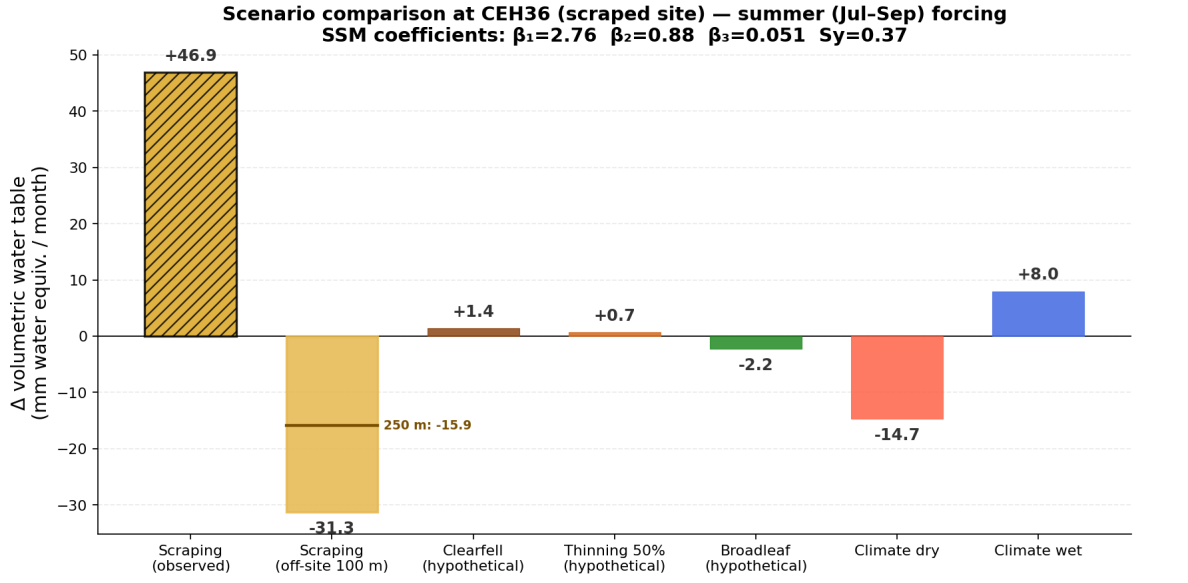


***Figure 29:*** *Scenario comparison at CEH36 (scraped site) — summer (July–September) forcing. As Figure 28 but with the SSM equilibrium evaluated under summer P and PET; bars are the same volumetric quantity (mm water-equivalent per month) and directly comparable to Figure 28. Observed scraping +46.9 (on-site) and −31.3 off-site at 100 m (reference line at 250 m, −15.9); clearfell +1.4, thinning +0.7, broadleaf −2.2, climate dry −14.7, wet +8.0. This is the equilibrium response to summer forcing, not a summer minimum, which the equilibrium framework cannot resolve; for reference, the observed paired-BACI summer-minimum shift at CEH36 was +195 mm. Broadleaf applies the seasonal $\beta_2$ adjustment of Section 4.13.2. As Figure 28, converting to a head change requires an appropriate specific yield and is approximate. (Source: 09d_02_summer_scenario_comparison.png)*

## 4.6 The December 2017 Experimental Clearfell: Monitoring and Retrospective Analysis

### 4.6.1 Interpretive Context

Three features of the monitoring record bear on interpretation of the clearfell results. First, wells within or at the edge of the clearfell footprint (FE1–FE4, LIS1) and the scraping treatment well CEH36 were excluded from the BACI design on the grounds of insufficient or confounded pre-felling baselines (Section 3.5.4). FE1 and FE2, which have 29 months of pre-felling record but no pre-scraping baseline, were excluded from the primary BACI design because their late start dates create an unstable impact centroid when pooled with WMC3 — the changing well composition introduces spurious level shifts that the ANCOVA conflates with the intervention steps. These wells were instead included as a robustness check via donor-regression synthetic extension, which provides a stable-composition centroid across all three eras (Section 4.6.7). Second, all clearfell analyses use the period from January 2011 onwards as the effective baseline, because only three of nine C4 wells were operational before mid-2010 and these are unrepresentative of the cluster (Section 4.2.1). Third, the April 2015 scraping event at CEH36 produced a significant level shift 32 months before the clearfell; the ANCOVA's distance-weighted scraping covariate absorbs this, but the post-scraping, pre-felling window (2015–17) encompasses only three summer minima, limiting baseline power.

### 4.6.2 Climate Sensitivity

The CWB × clearfell interaction was non-significant in the Forest control models (Impact p = 0.58, Edge p = 0.26), indicating that the rate at which the BACI gap responds to cumulative water balance deficit did not detectably change after felling when that response is modelled as a single linear slope. The relationship is shown in Figure 30: the fitted pre-felling and post-felling climate-response lines are effectively parallel, offset by the clearfell step (Section 4.6.3).

A non-linear extension of the headline model tested whether the response is concentrated at the tails of the water-balance range (the buffering hypothesis, Section 5.5.1), adding a centred $CWB^2$ term and a $CWB^2$ × clearfell interaction to the full ANCOVA specification. The extension does not support the hypothesis. The $CWB^2$ × clearfell interaction is negative at both zones — the direction a dry-tail buffering effect would produce — but reaches significance at neither (Impact −14.4 mm per (100 mm $CWB)^2$, p = 0.22; Edge −8.4 mm per (100 mm $CWB)^2$, p = 0.36). The joint F-test of the two added curvature terms is non-significant at Impact (p = 0.20) and at Edge (p = 0.079), and the curvature model is not preferred over the linear one by AIC at Impact (ΔAIC +0.7); at Edge it is, narrowly (ΔAIC −1.4). The $CWB^2$ main effect is non-significant at both zones. The negative interaction sign means the post-felling response is concave in cumulative water balance: the felling-

attributable rise in the water table is larger in dry, low-water-balance conditions than in wet ones.

This is reported as a preliminary finding. It rests on a single interacted quadratic term and one felling event at one site, and it is a sensitivity variant only: the headline clearfell step and the linear ANCOVA model of Section 4.6.3 are unchanged. The point estimates place the felling-attributable rise in dry periods; the alternative explanations and the buffering reading are in Section 5.5.1.

The Climate-control models — which contrast the felled wells against the unforested Climate control (CEH9, NW5, NW6, NW7, WMC2, all C3 Western Residual wells) — return non-significant CWB × clearfell interactions (Impact p = 0.47, Edge p = 0.22). The Forest-control comparison is the position-matched test and its results, linear and non-linear, are the headline findings; the Climate-control interactions are not interpreted as a clearfell-specific change in canopy water-balance physics.

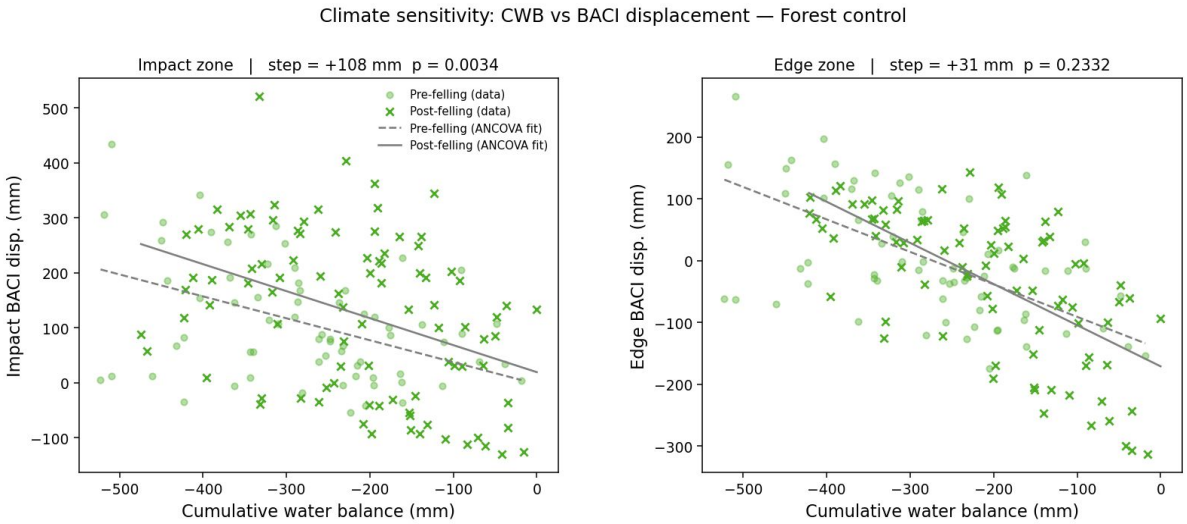



***Figure 30:*** *Climate sensitivity: cumulative water balance (CWB) vs BACI displacement against the Forest control, pre-felling (circles) and post-felling (crosses), for Impact (left) and Edge (right) zones. Solid and dashed lines are the ANCOVA-fitted climate-response lines for the post- and pre-felling eras respectively; their near-parallel form reflects the non-significant linear CWB × clearfell interaction. (Source: 10a_06_climate_sensitivity.png).*

### 4.6.3 ANCOVA-BACI Results

The ANCOVA-BACI model (Section 3.5.4) was run three times with different control centroids (Table 11).

| Control | Zone | Clearfell step (m) | 95% CI (m), robust | p, robust | Sig | Scraping step (m) | Scraping p (OLS) | $R^2$ | n |
|---|---|---|---|---|---|---|---|---|---|
| Forest | Impact | +0.108 | [−0.021, +0.237] | 0.100 | ns | +0.316 | 0.005 | 0.232 | 162 |
| Forest | Edge | +0.031 | [−0.059, +0.121] | 0.501 | ns | +0.562 | <0.001 | 0.455 | 156 |
| Climate | Impact | −0.019 | [−0.097, +0.058] | 0.624 | ns | +0.014 | 0.914 | 0.189 | 140 |
| Climate | Edge | −0.106 | [−0.208, −0.004] | 0.042 | * | +0.273 | 0.513 | 0.203 | 136 |
| Combined | Impact | +0.066 | [+0.012, +0.119] | 0.016 | * | +0.070 | 0.376 | 0.259 | 138 |
| Combined | Edge | −0.021 | [−0.068, +0.026] | 0.391 | ns | +0.300 | 0.019 | 0.451 | 134 |

***Table 11:*** *Three-counterfactual ANCOVA-BACI results for the December 2017 clearfell. Each row reports the clearfell step estimate from a separate ANCOVA model fitted to the monthly BACI gap (target tier centroid minus control centroid). Covariates: centred cumulative water balance (CWB), distance-weighted scraping dummy (L_s = 300 m), CWB × clearfell interaction, and coastal-drift term. The Forest control is the most direct test of the clearfell hypothesis: it compares felled wells against unfelled forest sharing the same canopy and substrate, isolating the felling response from the coastal erosion gradient. All comparisons use a record-length-balanced window beginning January 2011 — the date from which every well in the five-tier comparison network has continuous record. The control centroid is computed on a fixed-membership basis: a month enters the BACI series only when every control-roster well has a reading, so the centroid is not silently re-weighted by control wells going offline mid-record (clearfell_common v1.7.0). Sample sizes therefore differ between comparisons. (Source: 10a_01_ancova_comparison_table.csv)*

Against the Forest control, the clearfell step was +108 mm for Impact and +31 mm for Edge (Figure 32, Figure 33). The monthly residuals are strongly autocorrelated (lag-1 0.85), so inference uses autocorrelation-robust (Newey–West) errors with 12 monthly lags: the Impact step has a 95% CI of −21 to +237 mm (p = 0.10) and the Edge step −59 to +121 mm (p = 0.50), so neither is significant on its own. Ordinary least-squares errors, which ignore the autocorrelation and overstate significance, give p = 0.003 (95% CI +37 to +180 mm) for Impact and p = 0.23 for Edge. Against the Climate control, the clearfell step was −19 mm for Impact (95% CI −97 to +58 mm, p = 0.62) and −106 mm for Edge (95% CI −208 to −4 mm, p = 0.042). Against the Combined control, the clearfell step was +66 mm for Impact (95% CI +12 to +119 mm, p = 0.016) and −21 mm for Edge (95% CI −68 to +26 mm, p = 0.39). The interpretation of the contrasting Forest, Climate and Combined results is taken up in Section 5.

The scraping covariate was significant in both Forest-control models: +316 mm at Impact (p = 0.019) and +562 mm at Edge (p = 0.002), with autocorrelation-robust errors. Sensitivity analysis across three decay length scales (L_s = 200, 300, 500 m) produced consistent clearfell coefficients, so the scraping and clearfell terms are temporally well-separated. These covariate coefficients index the temporal separation of the two interventions in the model; they are not a physical estimate of the scraping effect on the impact well. In the raw, uncorrected BACI gap the era mean falls at each scraping event, whereas the climate-

corrected covariate carries the opposite sign because it is estimated jointly with the cumulative-water-balance and coastal-drift terms; neither raw scraping step is distinguishable from the gap's month-to-month variability (below). Figure 31 shows the raw gap and its per-era steps directly.

The raw before-and-after evidence for all three interventions is shown in Figure 31. In the uncorrected WMC3-minus-forest-control gap, the December 2017 clearfell registers a +40 mm difference-in-differences step — the raw counterpart to the +108 mm climate-corrected ANCOVA step above, the two differing because the raw step retains the drier-than-average pre-felling climate the ANCOVA removes. The gap's era mean also falls at each scraping event (−46 mm in 2015, −53 mm in 2023). None of the three raw steps is distinguishable from the gap's month-to-month variability. Against autocorrelation-robust standard errors each lies within one standard error of zero (−46 ± 63 mm at the 2015 scraping, +40 ± 57 mm at the clearfell, −53 ± 68 mm at the 2023 re-scraping), and sliding the break date across the record with the same era lengths returns a step at least as large in 73%, 78% and 65% of positions respectively. The 2015 step also depends on the baseline era, whose mean is set largely by the 2010–2012 excursion visible in Figure 31b. The raw steps therefore do not resolve an off-site scraping effect at WMC3, 262 m from CEH36; this agrees with the essentially zero BACI-corrected $\Delta\beta_3$ WMC3 returned in Section 4.5.5 and with the absence of any distance-decay in the whole-array step map (Figure 27). The clearfell rests on the climate-corrected ANCOVA, not on the raw step. WMC3 is retained as the sole impact well, while CEH36 and the FE wells are excluded, because it alone has a pre-2015 record against which the scraping era can be separated from the pre-felling baseline (Section 4.6.1).

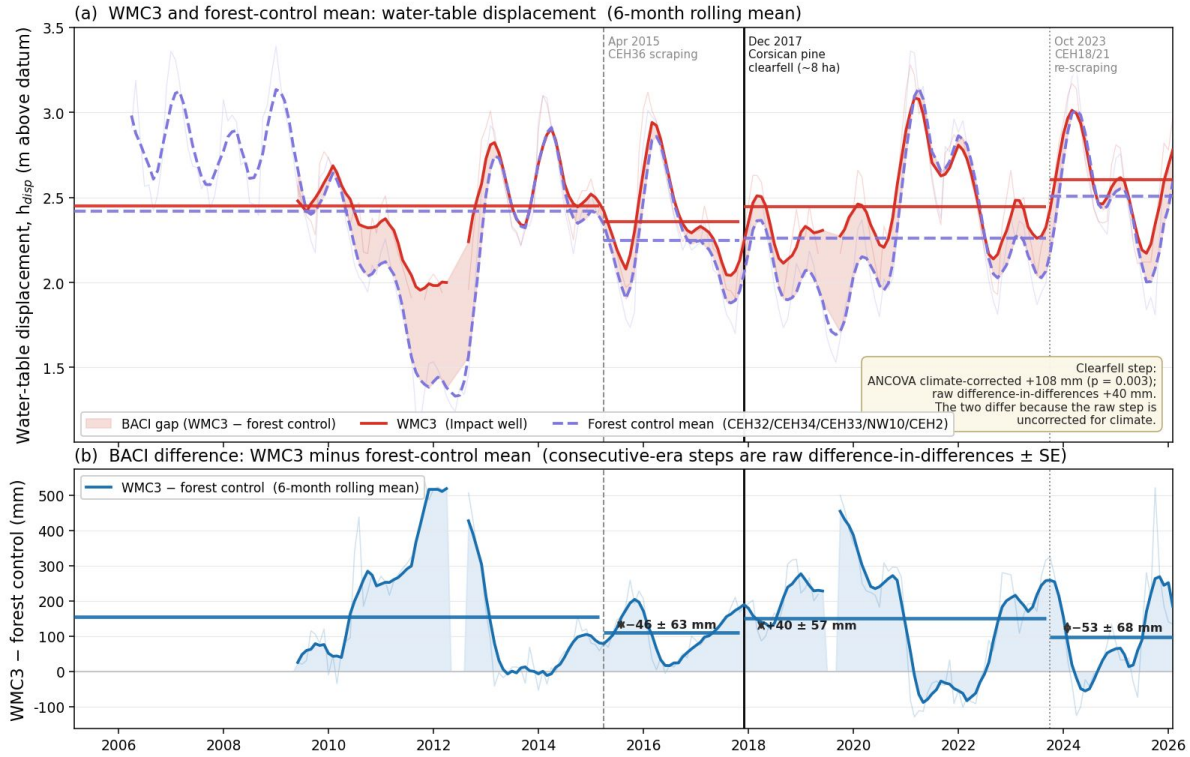



***Figure 31:*** *WMC3 (Impact well) and the forest-control mean across the three interventions, in water-table displacement (h_disp = datum + depth). (a) Six-month rolling means (bold) over faint monthly values, with the WMC3-minus-forest-control gap shaded and per-era mean bars; vertical lines mark April 2015 scraping (grey dashed), December 2017 clearfell (black solid) and October 2023 re-scraping (grey dotted). The boxed note reconciles the raw and climate-corrected clearfell steps. (b) The BACI gap (WMC3 minus forest control) as a six-month rolling mean over faint monthly values, with per-era mean bars; annotations give the consecutive-era difference-in-differences steps with their autocorrelation-robust standard errors (−46 ± 63 mm at the 2015 scraping, +40 ± 57 mm at the clearfell, −53 ± 68 mm at the 2023 re-scraping). None is distinguishable from the gap's month-to-month variability, so the scraping steps are not evidence of an off-site drawdown (Section 4.6.3). These raw difference-in-differences steps are uncorrected for climate and are not a substitute for the climate-corrected ANCOVA step of Section 4.6.3. (Source: 10m_02_wmc3_baci_dual.png)*

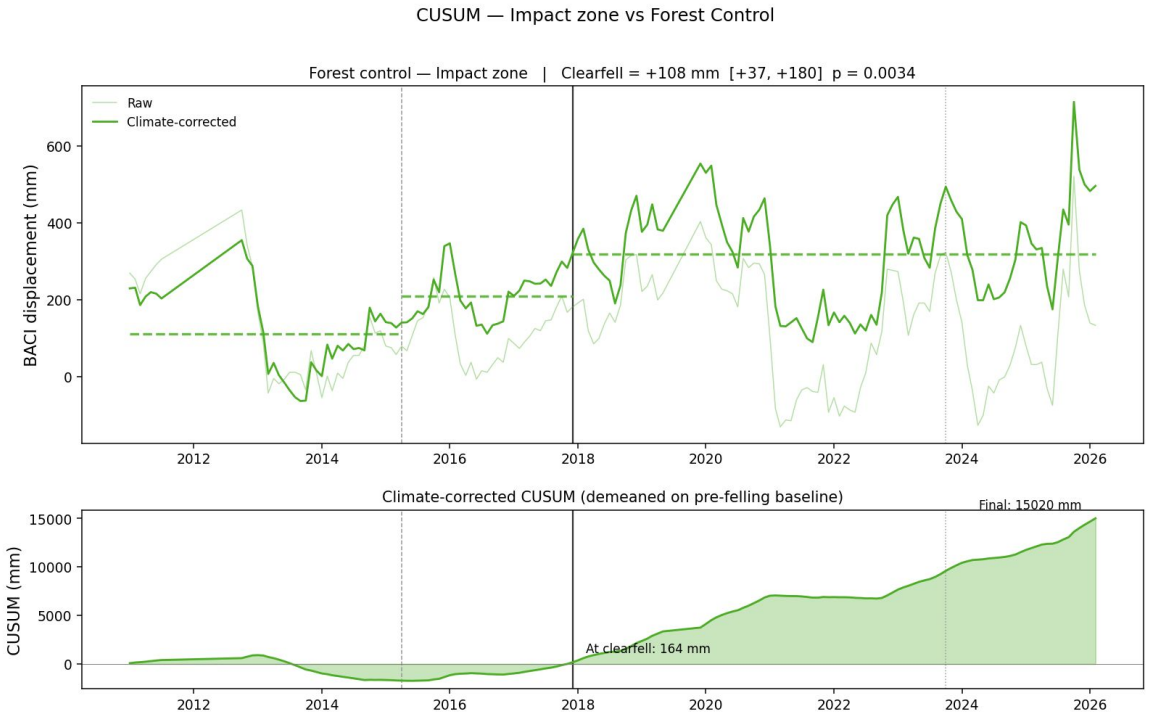



***Figure 32:*** *Forest control BACI timeseries and CUSUM for the Impact tier (WMC3). Upper panel: monthly BACI gap (WMC3 minus Forest control centroid) with climate-corrected series and era means. Lower panel: cumulative sum of the climate-corrected series demeaned on the pre-felling baseline; sustained positive accumulation after December 2017 is consistent with the clearfell step. Vertical lines: April 2015 scraping (grey dashed), December 2017 clearfell (black solid), October 2023 re-scraping (grey dotted). (Source: 10a_07_cusum_impact.png).*

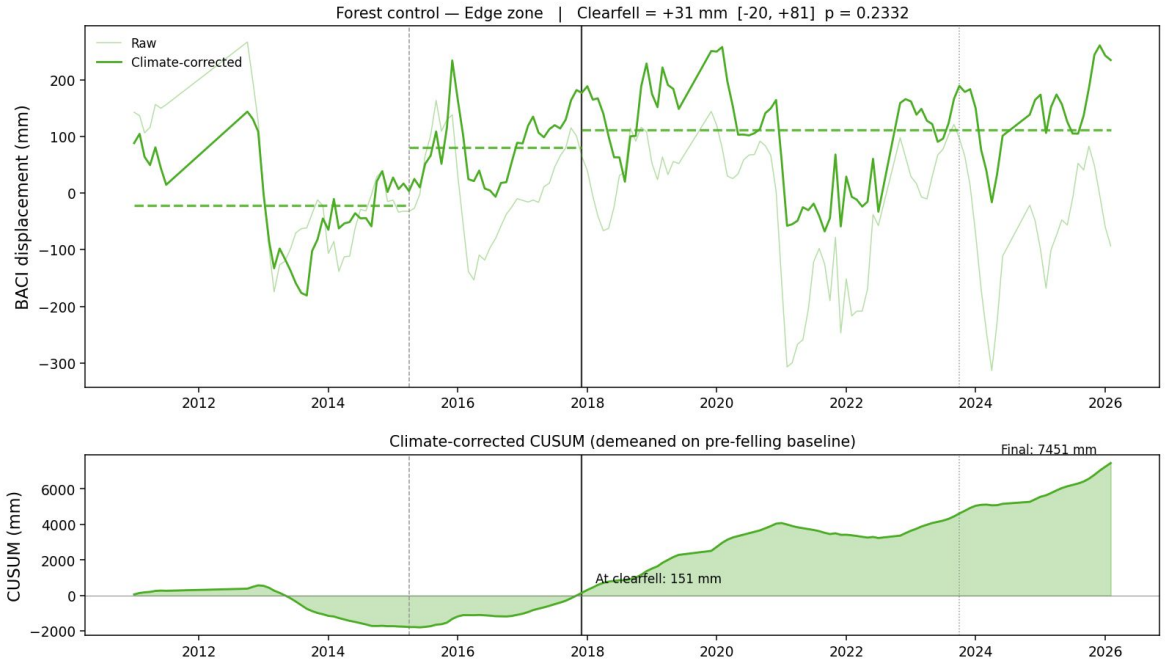



***Figure 33:*** *Forest control BACI timeseries and CUSUM for the Edge tier (CEH31, CEH20, CEH30, CEH16). Upper panel: monthly BACI gap (Edge centroid minus Forest control centroid) with climate-corrected series and era means. Lower panel: cumulative sum of the climate-corrected series demeaned on the pre-felling baseline; the positive accumulation after December 2017 is consistent with the Edge-tier clearfell step of +31 mm [−59, +121] (p = 0.50, not significant; Table 11). Vertical lines: April 2015 scraping (grey dashed), December 2017 clearfell (black solid), October 2023 re-scraping (grey dotted). (Source: 10a_08_cusum_edge.png)*

The coastal-drift covariate in Table 11's ANCOVA design absorbs progressive divergence between impact and control wells across the post-felling era, and Table 12 reports the differential each contrast absorbs. Table 12 also carries the far-field control tier (NW4b, WMC1, CEH5, L7, CEH6), which Script 10a fits beside the three counterfactuals of Table 11, and has no Combined rows. The covariate is the fitted coastal profile of Section 4.10.2, averaged over each tier's wells at their own distances from the shore (not evaluated at the tier's mean distance) and differenced between target and control, so its coefficient ξ is the fraction of that field the contrast feels and ξ = 1 is exactly the amplitude fitted network-wide. What the covariate does is keep that drift from being attributed to the felling step, which is the use made of it in Section 4.6.3; the residual coastal-retreat decline it absorbs is discussed in Section 5.5.

| Control tier | Impact zone | Coastal differential (mm yr⁻¹) | ξ | SE | p | Absorbed drift (mm yr⁻¹) |
|---|---|---|---|---|---|---|
| Forest | Impact | −10.68 | +1.513 | 0.447 | <0.001 | −16.1 ± 4.8 |
| Forest | Edge | −9.48 | +1.183 | 0.365 | 0.001 | −11.2 ± 3.5 |
| Climate | Impact | −0.93 | −8.713 | 3.521 | 0.015 | +8.1 ± 3.3 |
| Climate | Edge | +0.26 | +44.204 | 16.210 | 0.007 | +11.4 ± 4.2 |
| Far-field | Impact | −13.21 | −1.519 | 0.614 | 0.014 | +20.1 ± 8.1 |
| Far-field | Edge | −12.02 | −1.741 | 0.742 | 0.020 | +20.9 ± 8.9 |

***Table 12:*** *The coastal-drift term fitted in each ANCOVA contrast (Section 4.6.3). ξ is the amplitude of the network-scale coastal gradient that each target-minus-control contrast feels, and is dimensionless: ξ = 1 is exactly the field fitted independently in Section 4.10.2. The absorbed differential is ξ × the tier's coastal differential (mm yr⁻¹), with ±1 SE propagated from the coefficient SE. Because the covariate is the fitted gradient field itself, the comparison with the model prediction is an internal consistency check rather than independent corroboration: it asks, in effect, whether ξ = 1. It is a drift control, not a measurement of a spatial gradient. (Source: 25_04_baci_corroboration.csv)*

**Four-zone pooled-panel cross-check.** The three-counterfactual ANCOVA above fits the felling step six times — once per (control, zone) pair — and each fit has its own dependent variable, because the BACI gap is the target centroid minus a control centroid and the control centroid differs between the Forest, Climate and Combined configurations. The six steps therefore do not share a covariance matrix and are not subtractable. As a cross-check, a four-zone pooled-panel model (Script 10k) was fitted: the Forest control, a second western-dune control (C3/Warren: CEH1, NW1, NW2 and NW11, open-dune C3 Western Residual wells all at least 370 m from the edge of the felled compartment), the Edge tier and the

Impact well are stacked into one long-form panel, and every zone contrast is read off a single internally-consistent joint fit. This supplements, and does not replace, the three-counterfactual ANCOVA, which remains the principal clearfell analysis.

The joint fit returns three primary zone-versus-Forest monthly contrasts: Impact−Forest +0.027 m (95% CI [+0.003, +0.051], p = 0.025), Edge−Forest −0.040 m (p = 0.055, ns), and C3/Warren−Forest +0.008 m (p = 0.526, ns); joint model $R^2$ = 0.848, N = 2492 well-months. The four-zone result is consistent with the principal analysis. The joint-fit Impact−Forest step (+0.027 m) is smaller in magnitude than the three-counterfactual Forest × Impact step (+0.108 m); the two estimators partition climate variance differently — the joint panel estimates climate sensitivity from the full cross-zone monthly record, so a portion of the monthly variance the separately-fitted ANCOVA attributes to the felling step is attributed to climate in the joint fit. The felling response is positive under both estimators, though the three-counterfactual Impact step is not significant on its own once autocorrelation is allowed for (p = 0.10); the magnitude is sensitive to the partitioning choice. Dropping the easting × time covariate moves the joint-fit Impact step only from +0.027 m to +0.029 m, significance unchanged.

The Impact−Edge difference of +0.067 m is an exact arithmetic combination of two primary contrasts; its standard error is covariance-dependent and not comparable to a primary contrast, so it is not cited with a p-value here. It is instead corroborated independently by the two-zone Impact-versus-Edge estimator (Script 10j), which returns a monthly Impact−Edge step of +0.070 m (p = 0.0001) — the close agreement between the two estimators is the warrant for the contrast.

### 4.6.4 Summer Minima

The annual summer minimum depth (June–September) — the quantity whose four-year mean defines the SD15b and SD16 community-mean levels (Curreli et al. 2013) — was analysed for each of the 22 wells — the 17-well BACI network and the five far-field control wells. Four complementary analyses are reported: per-well Welch t-tests on the annual summer-minimum gap (Table 13); a mixed-effects model pooling within-tier (Table 14); a directly-fitted Jun–Sep ANCOVA on monthly observations, using the same three-counterfactual specification as the annual analysis (Section 3.5.4) restricted to the June–September window; and a four-zone pooled-panel cross-check fitted on the annual summer-minimum series (Section 3.5.4). The two ANCOVA-based analyses differ in estimator and in unit of observation: the Jun–Sep ANCOVA estimates the felling step from monthly observations as a differenced centroid, while the four-zone model estimates every zone contrast from a single joint fit on annual summer-minimum well-years — the annual deepest point rather than the summer-window mean.

**Per-well summer-minimum shifts.** Pre- vs post-felling Welch t-tests on the annual summer-minimum gap (well minus Forest control centroid) found no significant clearfell

effect at any well in the Impact or Edge tier (Table 13). The Impact well WMC3 showed a shift of −7 mm (p = 0.91). Individual Edge well shifts ranged from −9 mm at CEH30 to −97 mm at CEH16, all non-significant.

| Well | Tier | Control | n pre | n post | Pre gap (m) | Post gap (m) | Shift (mm) | p | Sig |
|---|---|---|---|---|---|---|---|---|---|
| WMC3 | Impact | Forest | 6 | 7 | +0.127 | +0.120 | −7 | 0.905 | ns |
| CEH31 | Edge | Forest | 7 | 8 | −0.075 | −0.132 | −56 | 0.581 | ns |
| CEH20 | Edge | Forest | 7 | 7 | +0.122 | +0.111 | −10 | 0.739 | ns |
| CEH30 | Edge | Forest | 7 | 8 | −0.311 | −0.320 | −9 | 0.859 | ns |
| CEH16 | Edge | Forest | 7 | 8 | +0.603 | +0.506 | −97 | 0.297 | ns |

***Table 13:*** *Per-well pre- vs post-felling shifts in annual summer minimum depth gap (well minus Forest control centroid), with Welch t-test. (Source: 10d_02_summer_minima_shifts.csv)*

**Pooled mixed-effects model.** A mixed-effects model with random intercept per well and fixed effects for post-felling and post-scraping indicators, fitted to the same Forest-control centroid gap series (Table 14), produced non-significant clearfell-step estimates across all five tiers. The Forest × Edge pooled estimate was −64 mm (p = 0.12); the Forest × Coastal Control estimate of −176 mm (p = 0.06) sits just outside conventional significance.

| Control | Tier | Model | Clearfell coef (m) | SE | p | n obs |
|---|---|---|---|---|---|---|
| Forest | Impact | OLS (single well) | −0.001 | 0.078 | 0.991 | 13 |
| Forest | Edge | Mixed-effects | −0.064 | 0.042 | 0.123 | 59 |
| Forest | Forest Ctrl | Mixed-effects | −0.008 | 0.016 | 0.618 | 72 |
| Forest | Coastal Ctrl | Mixed-effects | −0.176 | 0.093 | 0.059 | 29 |
| Forest | Climate Ctrl | Mixed-effects | −0.046 | 0.058 | 0.425 | 69 |

***Table 14:*** *Mixed-effects model results: pooled clearfell step by tier against Forest control, with random intercept per well, except the single-well Impact tier which is fitted by OLS. (Source: 10d_03_mixed_model_results.csv).*

**Directly-fitted Jun–Sep ANCOVA.** A directly-fitted summer-window ANCOVA was run on the same five-tier specification as the annual analysis (Section 3.5), restricted to June–September observations (n = 50 months). Against the Forest control, the Jun–Sep clearfell step at the Impact tier was −1 mm (95% CI [−118, +116], p = 0.99), with $R^2 = 0.34$. A sensitivity variant dropping the CWB covariate and the CWB × clearfell interaction from the

design matrix produced an estimate of +91 mm (95% CI [−35, +218], p = 0.16, ordinary errors), with $R^2 = 0.07$. The contrast between the full-specification and CWB-dropped estimates is discussed in Section 5. The same specification fitted to October–March observations (n = 84 months) gives a winter step of +128 mm (95% CI [+1, +254], p = 0.048, with autocorrelation-robust errors and 6 monthly lags), with $R^2 = 0.34$; the winter-minus-summer difference of +129 mm is not itself significant (p = 0.059).

**Distributional comparison.** Figure 35 shows the distribution of summer minima across the five BACI tiers across three phases: pre-scraping baseline (2011 to April 2015), the 2015–17 era, and post-felling 2018 onwards. The Forest Control group shows stable summer minima throughout. The Impact and Edge tiers track the Forest Control trajectory across the three phases. The Coastal Control tier (CEH19, CEH17) shows progressive deepening across all three phases. The two coastal controls stand 120–180 m from a strip of foreshore pine cleared between October 2014 and March 2015, inside the before period, and the comparison carries that clearance as a confound.

**Four-zone pooled-panel cross-check (summer minimum).** Whereas the directly-fitted Jun–Sep ANCOVA above estimates the felling step from monthly observations in the Jun–Sep window as a differenced centroid, the four-zone pooled-panel model (Section 3.5.4) estimates every zone contrast from a single joint fit on the annual summer-minimum series (Script 10l) — the annual deepest point rather than the summer-window mean. The three primary zone-versus-Forest contrasts were Impact−Forest −7 mm (95% CI [−23, +8], p = 0.36), Edge−Forest −29 mm (p = 0.16) and C3/Warren−Forest +12 mm (p = 0.26); joint model $R^2 = 0.724$, N = 181 well-years. No zone contrast resolves a significant differential felling response at the summer minimum.

The four summer-window analyses converge: the per-well Welch tests, the pooled mixed-effects model, the directly-fitted Jun–Sep ANCOVA and the four-zone pooled-panel cross-check all return non-significant clearfell steps at the Impact and Edge tiers. The felling was followed by a winter rise in the felled compartment's water table and a monthly-mean rise that is not significant on its own once autocorrelation is allowed for (Section 4.6.3), but no detectable change at the summer minimum — the driest point of the year, and the quantity whose four-year mean defines the Curreli SD15b and SD16 community-mean levels.

Five-year mean spring water level. The same absence of a discrete felling response is seen in the five-year mean spring water level (MSL5; Section 4.8.3), the baseline metric most directly tied to dune-slack vegetation (van Willegen et al. 2025). At the Impact well WMC3, MSL5 carried no step at the December 2017 felling: the window-end values deepen through the post-felling years (−1.01 m at window-end 2018, −1.14 m at 2019, between −1.02 and −1.12 m through 2023) before recovering to −0.89 m at window-end 2024. The four Edge wells (CEH16, CEH20, CEH30, CEH31) show the same pattern — deepening by 0.14–0.17 m at window-end 2019 and staying deeper than their 2018 values through 2023, then a recovery first appearing at window-end 2024. The 2024 recovery is not clearfell-attributable:

it appears synchronously across the felled wells, the unfelled Forest control, and every other cluster in the network (Section 4.8.3, Figure 45), six years after the intervention, and reflects the 2023–24 hydrological year entering the rolling five-year window rather than a delayed canopy-removal response. On the baseline metric, as on the summer minimum, the clearfell produced no detectable step against the background climate signal (Figure 34 shows the summer-minimum result by tier). The mean-monthly rise estimated by the ANCOVA (Section 4.6.3) is thus specific to the monthly-mean level; it is not expressed in either of the two ecologically referenced metrics — the summer minimum or the five-year spring baseline — to which the dune-slack community levels are tied.

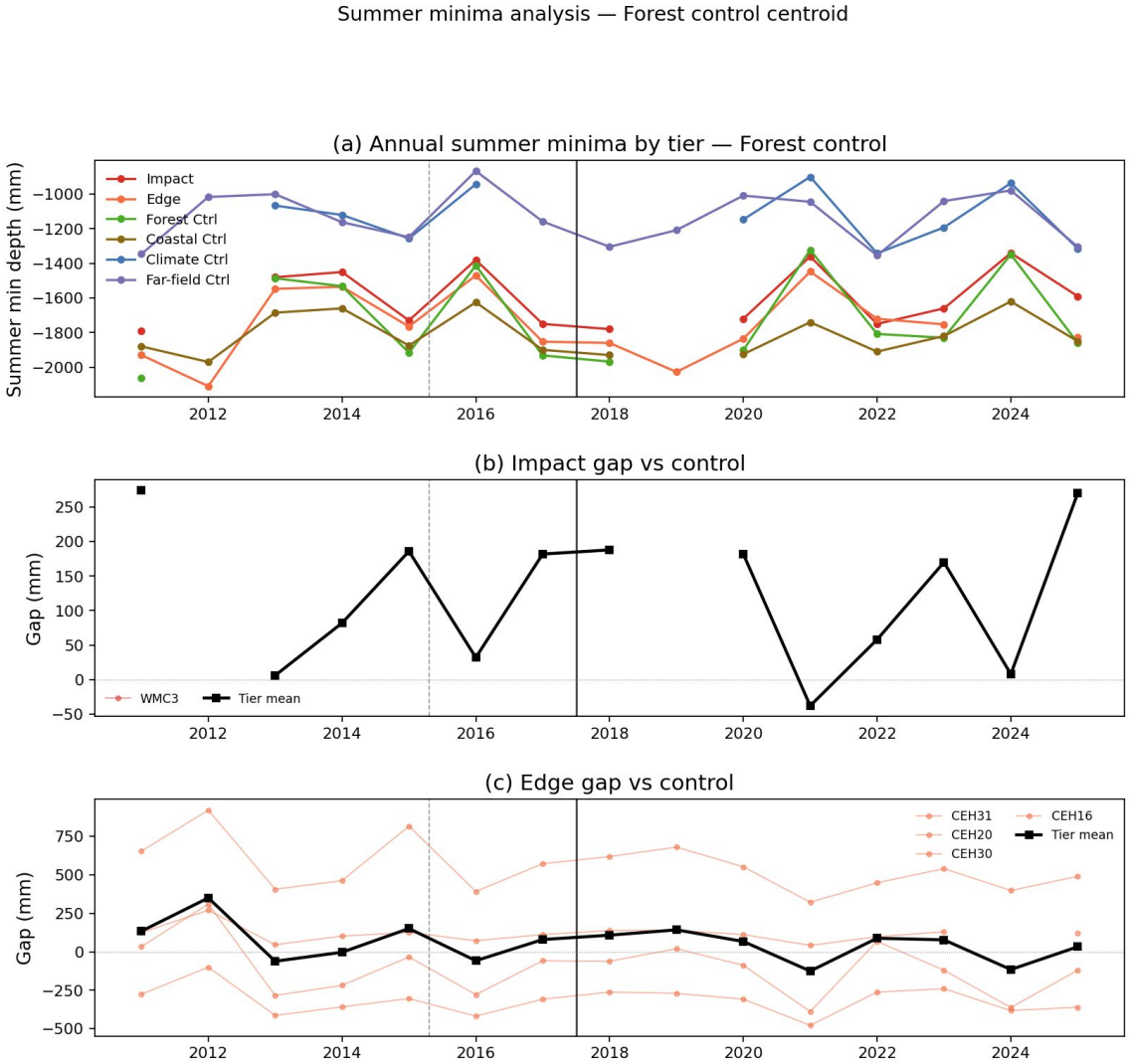



***Figure 34:*** *Summer minima analysis against the Forest control. (a) Raw annual summer minimum depth (June–September) by tier. (b) Impact well gap (WMC3 minus Forest control centroid) by year, before and after felling; era means are not drawn. (c) Edge well gaps vs Forest control centroid. No significant clearfell effect on summer minima was detected at the Impact or Edge tier; the mixed-effects Edge estimate (−64 mm, p = 0.12) is non-significant. (Source: 10d_04_summer_minima_forest_ctrl.png).*

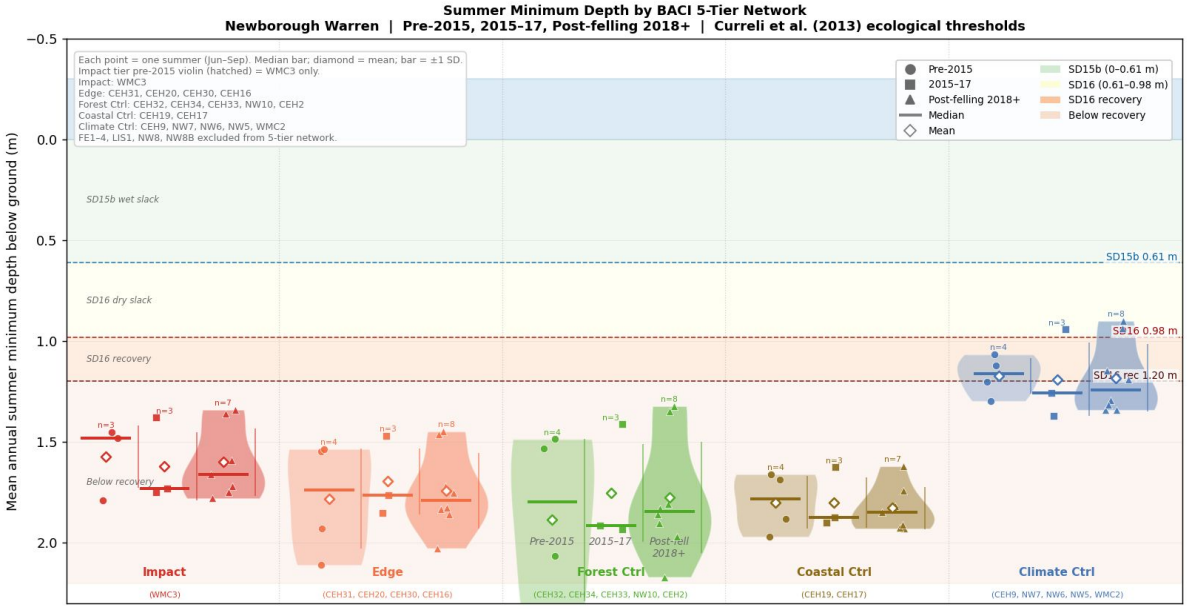



***Figure 35:*** *Summer minimum depth distributions by BACI tier, Newborough Warren 2011–2025. Three phases: pre-scraping baseline, 2015–17 era, post-felling 2018+. Violin width indicates frequency; diamond = mean; bar = median. The Forest Control group shows stable summer minima throughout. Both Impact and Edge tiers track the Forest Control trajectory, with no detectable divergence in summer extremes after felling despite the positive shift in the monthly ANCOVA (+108 mm; +128 mm over October–March, p = 0.048) (Section 4.6.3). The Coastal Control tier shows progressive deepening consistent with coastal erosion. Reference levels are the Curreli et al. (2013) SD15b and SD16 community means. (Source: 21_forestry_04_baci_zone_violin.png).*

### 4.6.5 Spatial Step-Change Maps

Climate-corrected step-change maps for the clearfell era are shown in Figure 36. Each well's step is computed relative to its own pre-intervention mean, corrected by subtracting the median step of four western C3 Western Residual wells (NW5, NW6, NW7, CEH1) to remove the background climate signal. This correction removes shared climate forcing but does not account for the coastal erosion gradient, which produces a systematic east–west trend in the corrected steps. The map shows no clear distance-decay from the felling compartment (Figure 36). The transect analysis does not detect a distance gradient either (Figure 38). The ANCOVA (Section 4.6.3), which explicitly models the coastal-retreat gradient, provides a cleaner estimate of the clearfell-specific effect.

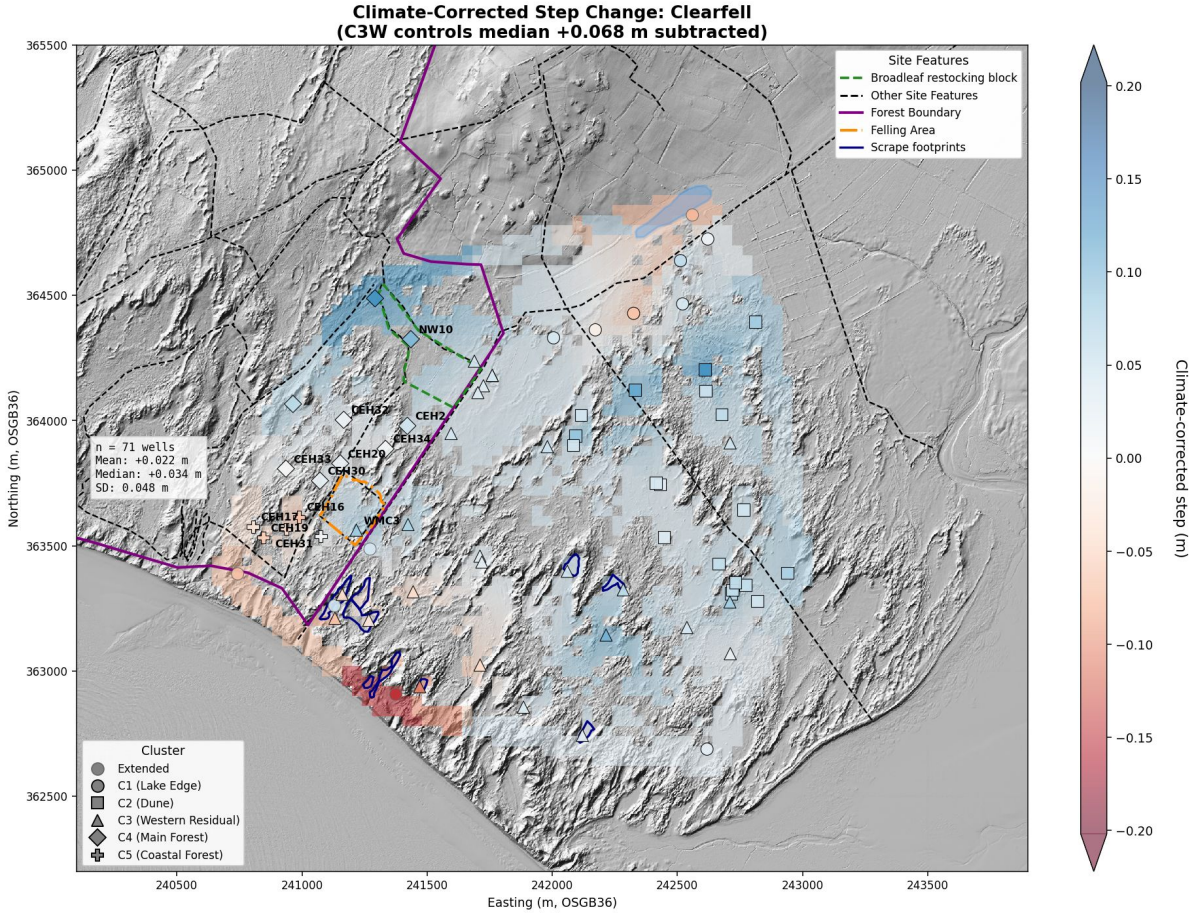



***Figure 36:*** *Climate-corrected spatial step-change map for the clearfell era (December 2017 – end of record). Each well's step is computed relative to its pre-felling mean, corrected by subtracting the median step of four western C3 Western Residual wells (NW5, NW6, NW7, CEH1) to remove the background climate signal. Step changes show no gradient consistent with a localized felling response; the most negative steps lie along the south-western coastal margin. The absence of a distance-decay from the felling compartment centroid is supported by the transect analysis, which detects no gradient (Figure 38). Source: 10b_spatial_fell_corrected.png.*

### 4.6.6 Coefficient Decomposition

The SSM was fitted independently to Before (record-length-balanced pre-felling window) and After (felling to end of record) windows for each well in the 17-well network, yielding per-well coefficient shifts $\Delta\beta_1$, $\Delta\beta_2$ and $\Delta\beta_3$ (Script 10e). This decomposition is a mechanistic diagnostic: it characterizes *which* SSM pathway — recharge, atmospheric draw or drainage — shifted between eras and in *what direction*. It is not an estimate of the clearfell magnitude; the clearfell magnitude is the BACI step reported in Section 4.6.3. The per-era fits use the canonical no-intercept SSM (Section 3.5), consistent with Script 03 and the master coefficient table.

$\beta_1$ (recharge) declined at most wells across all five tiers: mean $\Delta\beta_1 = -0.05$ at Impact, −0.29 at Edge ($p = 0.047$), −0.09 at Forest Control, −0.26 at Coastal Control and −0.01 at Climate

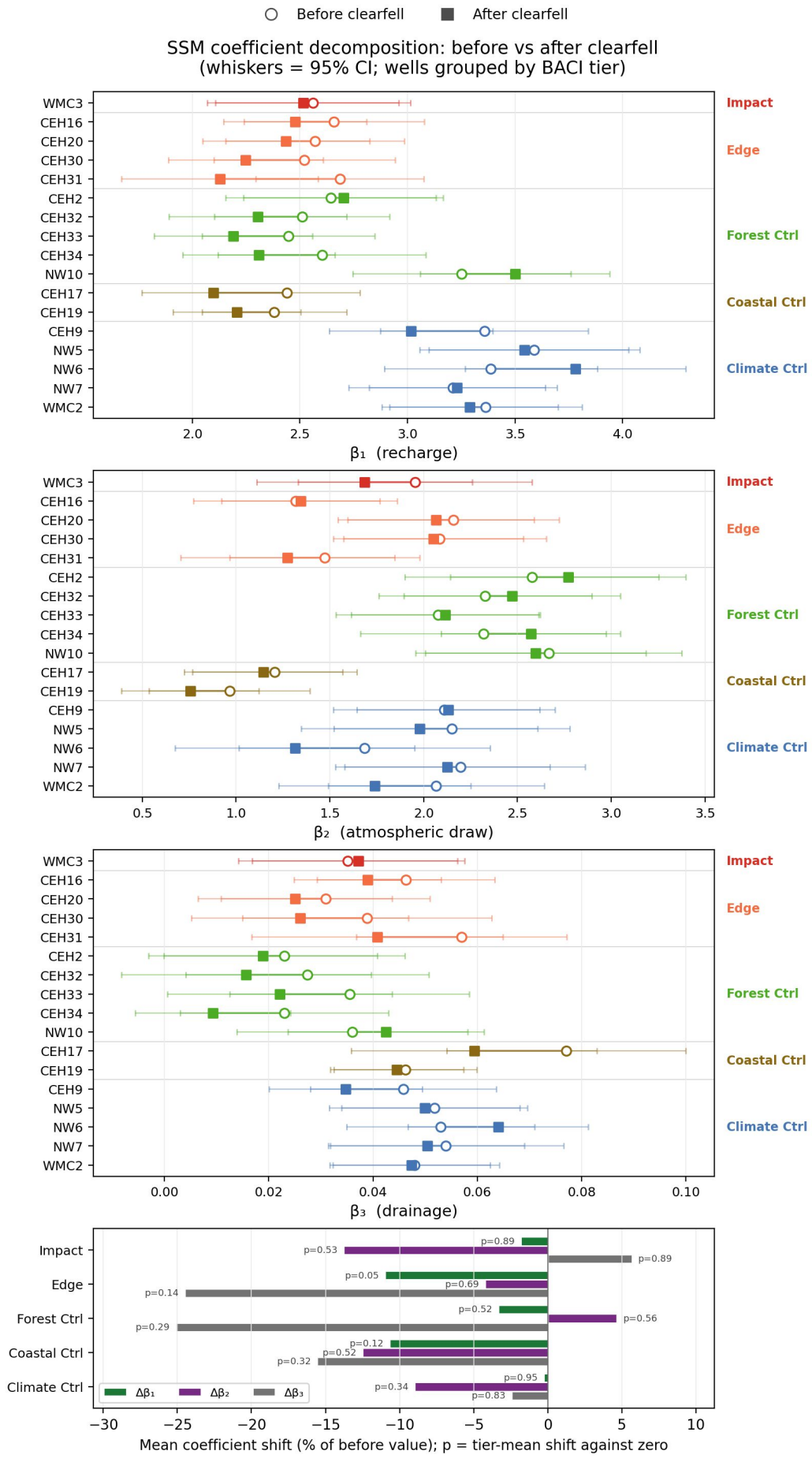
Control (Table 15). The network-mean shift across the seventeen-well BACI network is $\Delta\beta_1 = -0.13$, a 4.6% reduction against a pre-felling network mean of 2.83. The decline is present at the unfelled Forest Control tier as well as at the treatment tiers, but is negligible at the unfelled Climate Control tier (−0.3%, three of five wells); its candidacy as a mechanism for the network-wide summer-minimum decline is taken up in Section 5.5.

$\beta_2$ (atmospheric draw) increased at four of the five unfelled Forest-Control wells (mean $\Delta\beta_2 = +0.11$; NW10 fell by 0.07) with no comparable rise at the non-forest tiers. Refitting $\beta_2$ on a common PET support across the two eras leaves the Forest-Control increase positive while the Climate-Control tier remains negative (Section 5.5). The Impact-tier $\beta_2$ shift ($\Delta\beta_2 = -0.27$) is a single-well estimate lying within one standard error of zero and is not significant on its own; it is nonetheless the only measured response of a felled well at this site, and the forest-management scenarios take it, net of the climate controls, as their anchor, with the caveats stated in Section 4.13.2; Edge-tier shifts were near zero in the mean (−0.07) and negative at three of the four wells.

$\beta_3$ (drainage) shifts were small across all tiers (mean $\Delta\beta_3$ between −0.01 and 0.00 at every tier) but negative at fourteen of the seventeen wells.

Individual wells departed from these tier means. Within the Forest Control, NW10 and CEH2 showed $\beta_1$ increases ($\Delta\beta_1 = +0.25$ and +0.06) against a network of general decline. These departures are retained: the per-well era coefficients carry wide confidence intervals (Figure 37) and exclusion would not be justified on physical grounds.

The decomposition therefore identifies a recharge decline across the BACI network, negligible at the Climate Control tier, and an atmospheric-draw rise at the forest controls, consistent with canopy maturation or, in part, with a fitting artefact, but does not by itself isolate a clearfell-specific coefficient signature; the clearfell effect is quantified by the BACI analysis of Section 4.6.3.


Before clearfell
After clearfell
SSM coefficient decomposition: before vs after clearfell
(whiskers = 95% CI; wells grouped by BACI tier)
WMC3
CEH16
CEH20
CEH30
CEH31
CEH2
CEH32
CEH33
CEH34
NW10
CEH17
CEH19
CEH9
NW5
NW6
NW7
WMC2
Impact
Edge
Forest Ctrl
Coastal Ctrl
Climate Ctrl
2.0
2.5
3.0
3.5
4.0
β₁ (recharge)
0.5
1.0
1.5
2.0
2.5
3.0
3.5
β₂ (atmospheric draw)
0.00
0.02
0.04
0.06
0.08
0.10
β₃ (drainage)
p=0.89
p=0.53
p=0.89
p=0.05
p=0.69
p=0.14
p=0.52
p=0.56
p=0.29
p=0.12
p=0.52
p=0.32
p=0.95
p=0.34
p=0.83
Δβ₁
Δβ₂
Δβ₃
−30
−25
−20
−15
−10
−5
0
5
10
Mean coefficient shift (% of before value); p = tier-mean shift against zero

confider

***Figure 37:*** *Before- and after-clearfell SSM coefficient estimates with 95% CI for all 17 wells in the BACI network, grouped by tier. (a) $\beta_1$ recharge: declined at most wells, negligibly at the Climate Control tier. (b) $\beta_2$ atmospheric draw: increased at Forest Control wells, consistent with canopy maturation. (c) $\beta_3$ drainage: small, spatially incoherent shifts. (d) Per-tier mean coefficient shift $\Delta\beta_1$, $\Delta\beta_2$ and $\Delta\beta_3$, as a percentage of the before-clearfell value, with the p-value of each tier-mean shift against zero; most shifts are not significantly different from zero. (Source: 10e_03_coefficient_shifts.png).*

| Well | Tier | $\beta_1$ before | $\beta_1$ after | $\Delta\beta_1$ | $\beta_2$ before | $\beta_2$ after | $\Delta\beta_2$ | $\beta_3$ before | $\beta_3$ after | $\Delta\beta_3$ |
|---|---|---|---|---|---|---|---|---|---|---|
| WMC3 | Impact | 2.562 | 2.515 | −0.047 | 1.955 | 1.685 | −0.270 | 0.0352 | 0.0372 | +0.0020 |
| CEH31 | Edge | 2.686 | 2.128 | −0.558 | 1.473 | 1.275 | −0.199 | 0.0570 | 0.0408 | −0.0163 |
| CEH20 | Edge | 2.570 | 2.436 | −0.134 | 2.159 | 2.066 | −0.093 | 0.0309 | 0.0251 | −0.0058 |
| CEH30 | Edge | 2.521 | 2.249 | −0.273 | 2.087 | 2.052 | −0.034 | 0.0389 | 0.0260 | −0.0129 |
| CEH16 | Edge | 2.659 | 2.477 | −0.182 | 1.317 | 1.344 | +0.027 | 0.0463 | 0.0390 | −0.0074 |
| CEH32 | Forest Ctrl | 2.510 | 2.305 | −0.206 | 2.329 | 2.473 | +0.144 | 0.0274 | 0.0157 | −0.0116 |
| CEH34 | Forest Ctrl | 2.604 | 2.310 | −0.294 | 2.319 | 2.571 | +0.253 | 0.0230 | 0.0093 | −0.0137 |
| CEH33 | Forest Ctrl | 2.447 | 2.190 | −0.257 | 2.076 | 2.114 | +0.038 | 0.0355 | 0.0221 | −0.0134 |
| NW10 | Forest Ctrl | 3.252 | 3.500 | +0.248 | 2.667 | 2.597 | −0.070 | 0.0360 | 0.0425 | +0.0065 |
| CEH2 | Forest Ctrl | 2.644 | 2.702 | +0.059 | 2.577 | 2.771 | +0.194 | 0.0230 | 0.0189 | −0.0042 |
| CEH19 | Coastal Ctrl | 2.381 | 2.208 | −0.173 | 0.966 | 0.756 | −0.210 | 0.0462 | 0.0446 | −0.0016 |
| CEH17 | Coastal Ctrl | 2.440 | 2.099 | −0.341 | 1.207 | 1.145 | −0.061 | 0.0771 | 0.0594 | −0.0176 |
| CEH9 | Climate Ctrl | 3.357 | 3.016 | −0.341 | 2.108 | 2.131 | +0.023 | 0.0458 | 0.0348 | −0.0110 |
| NW7 | Climate Ctrl | 3.211 | 3.231 | +0.020 | 2.196 | 2.125 | −0.070 | 0.0540 | 0.0504 | −0.0036 |
| NW6 | Climate Ctrl | 3.388 | 3.781 | +0.394 | 1.684 | 1.313 | −0.371 | 0.0530 | 0.0640 | +0.0109 |
| NW5 | Climate Ctrl | 3.589 | 3.542 | −0.047 | 2.151 | 1.978 | −0.173 | 0.0518 | 0.0499 | −0.0019 |
| WMC2 | Climate Ctrl | 3.364 | 3.291 | −0.073 | 2.067 | 1.740 | −0.327 | 0.0480 | 0.0474 | −0.0006 |

***Table 15:*** *Before- and after-clearfell SSM coefficient estimates for all 17 wells in the BACI network, with per-well shifts Δβ. Before = record-length-balanced pre-felling window; After = felling to end of record. Coefficients are from the canonical no-intercept SSM (Section 3.5). (Source: 10e_01_coefficient_shifts.csv)*

### 4.6.7 Robustness

Four methods were applied to test whether a clearfell-specific signal could be detected once the ANCOVA assumptions were relaxed.

**SSM residual normalization.** The SSM was calibrated on the pre-felling record for each well and run forward iteratively into the post-felling era. The Impact well WMC3 showed a +81 mm normalized step — its post-felling water table 81 mm above the pre-felling SSM prediction. The Forest Control wells drifted upward by a mean of +53 mm over the same period, so against the Forest-control drift the Impact-specific component is approximately +27 mm: about two-thirds of the Impact well's upward residual is shared with the unfelled forest controls, and the method attaches no uncertainty to the remainder.

**Synthetic control.** A synthetic counterfactual for WMC3 constructed from six optimally weighted donor wells showed an Impact step of +101 mm (p = 0.002 with autocorrelation-robust errors); the corresponding Edge synthetic-control step was +44 mm (p = 0.27), non-significant. The synthetic-control and SSM-residual methods agree that WMC3's gross post-felling step is of order +80 to +100 mm; they differ on how much survives normalization against the unfelled forest controls, and neither, after that normalization, resolves a clearfell-specific step distinguishable from zero.

**Clearfell transect.** Step changes across a transect from the plantation interior to the felling core showed no significant spatial gradient (−0.5 mm per 100 m, p = 0.14). A distance-decay response would be strongest at WMC3 and diminish outward; the observed pattern is statistically uniform (Figure 38). The post-felling spread in the transect hydrographs is wider than in the 2015–17 scraping era, though no wider than in the 2012 trough (Figure 38, panel a); WMC3 sits above the other transect wells in the post-felling troughs but not at the winter peaks (Section 5.5).

**Synthetic FE-well extension.** FE1 and FE2 — which sit inside or at the immediate edge of the clearfell compartment but lack pre-scraping baselines — were extended backwards using OLS donor regression calibrated on Forest Control wells (calibration $R^2 > 0.99$, RMSE 18–25 mm; Section 3.5.4). Against the Forest control, the centroid of WMC3 + FE2 produced a clearfell step of +94 mm (95% CI [−24, +212], p = 0.12, $R^2$ = 0.30), consistent with the WMC3-only ANCOVA estimate of +108 mm (Section 4.6.3). FE2's post-felling divergence from its own donor counterfactual was +29 mm ($p < 0.001$, ordinary errors), consistent with a within-compartment response. Extending the centroid to WMC3 + FE1 + FE2 returned a Forest-control clearfell step of +78 mm (95% CI [−16, +172], p = 0.10). FE1 sits approximately 46 m outside the felling boundary in standing forest, so including it attenuates the step magnitude — from +108 mm (WMC3 alone) through +94 mm (WMC3 + FE2) to +78 mm — and none of the three centroids resolves a step that is significant on its own once autocorrelation is allowed for.

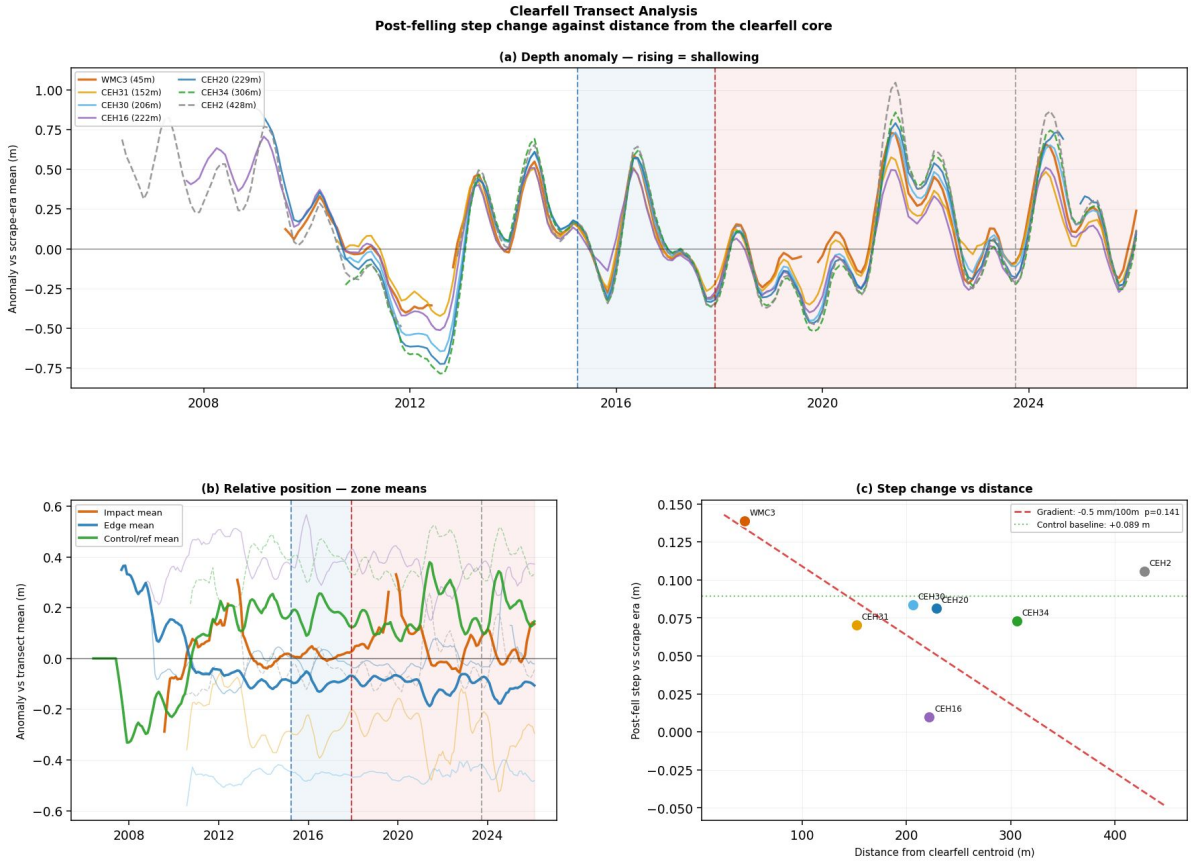



***Figure 38:*** *Clearfell transect. (a) Monthly depth anomaly against each well's scrape-era mean (rising = shallowing) for the seven transect wells, WMC3 at 45 m to CEH2 at 428 m from the felling-compartment centroid; the shaded bands are the 2015–17 scraping era and the post-felling era, the vertical lines the April 2015 scrape, the December 2017 clearfell and the October 2023 re-scrape. (b) The same anomalies as zone means (Impact, Edge, control/reference) relative to the transect mean, individual wells faint. (c) Post-felling step change against distance from the centroid: gradient −0.5 mm/100 m (p = 0.14, ns) against the control baseline of +0.089 m. No spatial decay consistent with a localized clearfell response; the uniform pattern indicates a common climate signal. (Source: 10g_02_clearfell_transect.png)*

### 4.6.8 Temporal decay of the clearfell step and the canopy setting

The Forest-control clearfell step is not stationary across the record. Split at January 2021, the Impact-well recovery against the Forest control falls from +152 mm in the early era (December 2017 to December 2020) to +39 mm in the late era (January 2021 onward), no longer distinguishable from zero. The Edge-tier contrast falls similarly, from +66 mm to −6 mm over the same split. The Impact well nonetheless remains significantly elevated relative to the Edge tier in both eras (+87 mm early, +57 mm late). The relaxation is not an artefact of the coastal-drift correction: refitting each contrast under the four drift treatments of Section 3.5.5 leaves the early step above the late step in every case (Impact +152 to +39 with the Script 25 field, +120 to −36 with no drift, and +120 to −37 with a free per-era slope), and

the per-era treatment, the direct test of an accelerating retreat, widens the gap rather than closing it (Figure 40; 10a_13_era_split.csv).

The step is robust to the composition of the Forest control. Estimated against only the two control wells lying outside any 1998 replant block (CEH34 and CEH2) it is +110 mm, essentially the +108 mm headline, so the two controls lying inside a replant block do not inflate it; a contrast between the in-block pair (CEH32, CEH33) and the outside pair is flat and non-significant in both eras (−21 mm early, −9 mm late) (10a_12_control_subset_sensitivity.csv for the +110 mm; 10a_13_era_split.csv for the in-block contrast).

The canopy trajectories give the setting (Figure 39; Section 3.5.5). Across the felling itself the index falls from 1.243 in the frame of 22 April 2017 to 0.130 in that of 28 June 2018 over the felled compartment, and from 1.248 to 0.123 over the felling experiment, while the unfelled conifer control holds at 1.005. The felled compartment falls from a mature-conifer texture to about 0.13 of the reference at felling and is held there by grazing, reaching only 0.14 by 2020 and 0.09 by 2026: it does not regenerate. The three 1998 replant blocks rise from near-bare in 2006 (0.15, 0.36 and −0.05 relative to the mature-conifer reference) to 0.77 to 0.90 by 2020 and 1.01 to 1.30 by 2026 — a developing canopy across the study window, on a trend present whether or not the isolated 2026 frame is included. The pooled untouched-forest control drifts gently from 1.47 in 2006 to about 1.15 and is flat across the BACI window, remaining dense and above the reference: the control forest was neither felled nor heavily thinned. The broadleaf restock grew in by about 2012, and the vegetated warren sits near zero throughout, the open-dune end of the scale. The replant reads above the reference by 2026 because it is unthinned whereas the reference stand has been thinned.

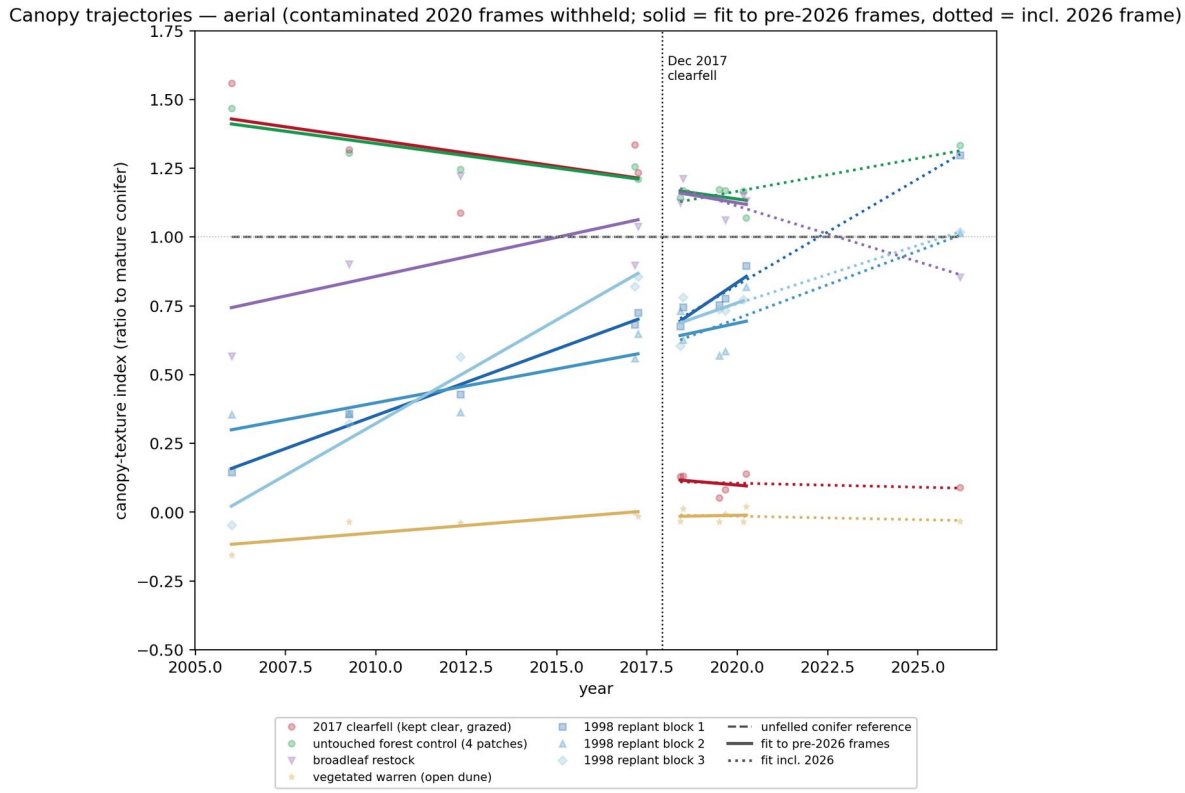



***Figure 39:*** *Canopy-texture trajectories from the dated aerial series, 2006 to 2026, relative to the unfelled mature-conifer canopy. The 2017 clearfell compartment is held clear by grazing; the three 1998 replant blocks develop across the study window; the pooled untouched-forest control stays dense and flat; the broadleaf restock is grown in; and the vegetated warren marks the open-dune end. Solid lines fit the frames before 2026, dotted lines include the isolated 2026 frame; cloud- and shadow-contaminated 2020 frames are withheld. (Source: 41_05_canopy_trajectory.png)*

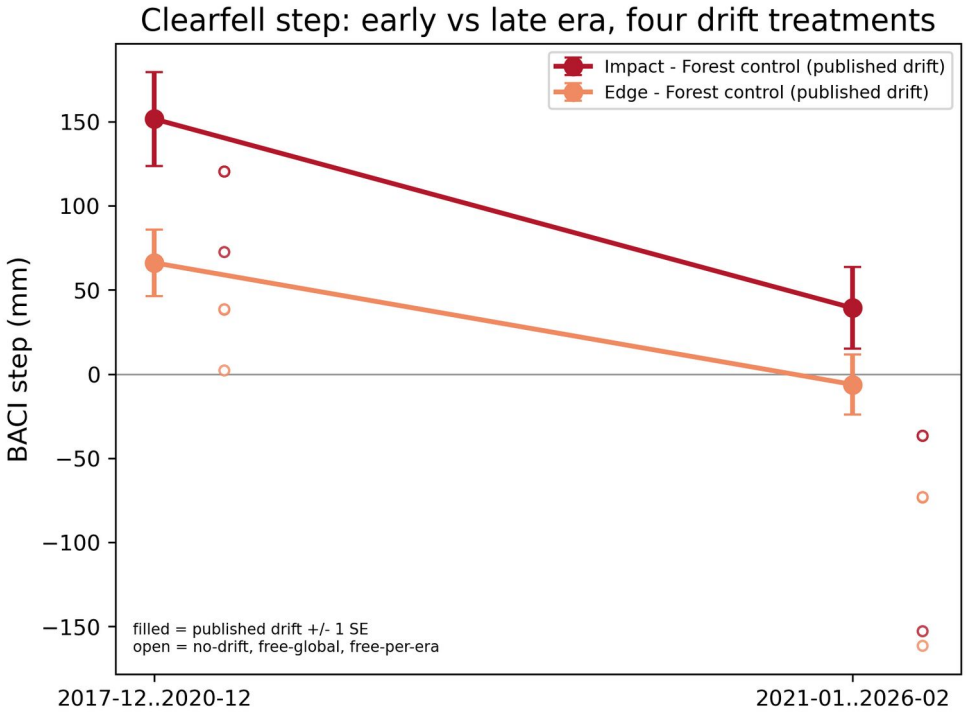





***Figure 40:*** *Early versus late clearfell BACI step against the Forest control, split at January 2021, for the Impact and Edge tiers under four coastal-drift treatments. Filled markers are the published (Script 25) drift with one standard error; open markers are the no-drift, free-global and free-per-era treatments. The step decays in every treatment. (Source: 10a_14_era_split_decay.png)*

## 4.7 Critical Rainfall Threshold Forecasting: Predicting Winter Flooding and Summer Droughts

### 4.7.1 Mechanistic State-Space Equations

The cluster-level state-space equations (Section 3.6.1) fitted without intercept produced statistically significant coefficients at all five clusters (all $p < 0.001$ except the C4 drainage coefficient $\beta_3$, where $p = 0.001$). Model fit ranged from $R^2 = 0.827$ at Western Residual (C3) to $R^2 = 0.706$ at Coastal Forest (C5), with C5 the least well fitted and C4 Main Forest second-lowest ($R^2 = 0.733$; Section 5.2.4). The full coefficient values are reported in Section 4.2.2; note that the β coefficients used in the forecasting tool are expressed in m/mm rainfall, one thousand times smaller than the m/m values in Table 3.

### 4.7.2 Seasonal Prediction Equations

The prediction equations were fitted at hydrological-year resolution, one equation per cluster under the k = 5 partition (Table 16). The two predictor terms — winter rainfall total (P_winter) and antecedent summer minimum (h_min) — show a clear spatial partitioning of dominance across the site.

In the Eastern Block, winter rainfall is the dominant predictor. P_winter is significant at Lake Edge ($p = 0.002$) and Dune ($p = 0.011$), while h_min is non-significant in both ($p = 0.823$ and 0.398 respectively).

In the forest clusters the pattern reverses: the antecedent summer minimum is the dominant predictor at Main Forest ($p < 0.001$, $R^2 = 0.872$) and Coastal Forest ($p < 0.001$, $R^2 = 0.863$), where winter rainfall is non-significant ($p = 0.903$ and 0.388). At Western Residual ($R^2 = 0.613$) the two carry equal weight, each just significant (summer minimum $p = 0.060$, winter rainfall $p = 0.051$). The intercepts are significant at Forest ($p = 0.028$) and Coastal Forest ($p = 0.007$). The reading of both patterns is in Section 5.2.5.

Two individual coefficients require comment. The negative h_min coefficient at Lake Edge (−0.047) is not mechanistically meaningful given its non-significance; it is an artefact of the small sample (n = 19) and the dominance of P_winter in that cluster. The near-zero P_winter coefficient in the Forest cluster (−0.00007, $p = 0.903$) reflects multicollinearity with h_min and should not be interpreted mechanistically.

| Block | Equation | $R^2$ | n | p(P_winter) | p(h_min) |
|---|---|---|---|---|---|
| Lake Edge (C1) | h_peak = 0.00108·P_winter − 0.047·h_min − 0.550 | 0.48 | 19 | 0.002 | 0.823 |
| Dune (C2) | h_peak = 0.00134·P_winter + 0.171·h_min − 0.621 | 0.46 | 20 | 0.011 | 0.398 |
| Western Residual (C3) | h_peak = 0.00149·P_winter + 0.695·h_min − 0.302 | 0.61 | 20 | 0.051 | 0.060 |
| Main Forest (C4) | h_peak = − 0.00007·P_winter + 1.254·h_min + 1.204 | 0.87 | 19 | 0.903 | <0.001 |
| Coastal Forest (C5) | h_peak = 0.00042·P_winter + 1.565·h_min + 1.253 | 0.86 | 19 | 0.388 | <0.001 |

***Table 16:*** *Winter peak prediction equations fitted at hydrological-year resolution (Section 3.6.2), one equation per cluster under the k = 5 partition. P_winter = total winter rainfall (mm) accumulated over the cluster-specific recharge horizon (October–January for C1 and C2; October–February for C3, C4 and C5). h_min = preceding summer water table minimum (m relative to ground surface, negative below). (Source: 11_forecast_winter_transfer_functions.csv)*

### 4.7.3 Summer Drought Prediction Equations

The summer drought prediction equations performed better than the winter peak functions at four of the five blocks (Lake Edge is the exception) (Section 3.6.2). Both predictors — antecedent winter peak (h_max_winter) and cumulative summer precipitation (P_summer) — were significant in all five blocks ($p < 0.05$), except summer precipitation at Coastal Forest, which is marginal ($p = 0.018$). Model fit was highest for the Forest ($R^2 = 0.930$) and Coastal Forest ($R^2 = 0.900$) blocks, high for the Western Residual ($R^2 = 0.689$), moderate-to-high for the Dune ($R^2 = 0.537$), and moderate for the Lake Edge cluster ($R^2 = 0.419$). The equations are given in Table 17.

| Block | Equation | $R^2$ | p(P_summer) | p(h_max) |
|---|---|---|---|---|
| Lake Edge (C1) | h_min = 0.00139·P_summer + 0.427·h_max_winter − 1.399 | 0.419 | 0.005 | 0.040 |
| Dune (C2) | h_min = 0.00215·P_summer + 0.696·h_max_winter − 1.693 | 0.537 | 0.003 | 0.001 |
| Western Residual (C3) | h_min = 0.00134·P_summer + 0.529·h_max_winter − 1.491 | 0.689 | 0.006 | <0.001 |
| Forest (C4) | h_min = 0.00140·P_summer + 0.763·h_max_winter − 1.487 | 0.930 | 0.002 | <0.001 |
| Coastal Forest (C5) | h_min = 0.00082·P_summer + 0.534·h_max_winter − 1.337 | 0.900 | 0.018 | <0.001 |

***Table 17:*** *Summer drought prediction equations (Section 3.6.2). P_summer = total April–September rainfall (mm); h_max_winter = preceding winter water table peak (m relative to ground surface, negative below). All coefficients significant at p < 0.05 except the Coastal Forest P_summer term (p = 0.018). (Source: 11_forecast_summer_transfer_functions.csv).*

### 4.7.4 Spatial Distribution of P_flood and Flooding Status Across the Well Network

Application of the cluster-level P_flood equations derived in Section 3.6.3 to each well in the 88-well classified network (66 reference wells with per-well SSM coefficients; 22 extended wells assigned cluster-level coefficients via the Pearson affinity audit of Section 3.3), using that well's observed mean summer minimum as $h_0$, produces a per-well critical rainfall threshold expressed as either millimetres of cumulative winter rainfall or as a dimensionless multiplier m_P = P_flood / Σ P_clim, where Σ P_clim is the long-term mean rainfall over the cluster-specific recharge horizon (400 mm for C1/C2 over Oct–Jan; 461 mm for C3/C4/C5 over Oct–Feb). A well with m_P = 1.0 would flood under an average winter; m_P > 1.0 indicates that above-average rainfall is required. The network-wide picture is summarized in Figures 41–44.

The summer minimum depth map (Figure 41; Section 3.6.5) classifies each well's depth into the five bands set at the Curreli et al. (2013) community-mean depths and the two scrape-recovery limits (Section 2); a well's band is its depth below its own ground: SD15b wet slack (<0.61 m), SD15b-recoverable (0.61–0.75 m), SD16 dry slack (0.75–0.98 m), SD16-recoverable (0.98–1.20 m), and beyond-recoverable (>1.20 m). The two forest clusters (C4 and C5) show the deepest mean summer minima across the network, consistently below 0.98 m. All 24 C4 and C5 wells fall in the SD16-recoverable (6 wells) or beyond-recoverable (18 wells) bands. By contrast, none of the nine C1 Lake Edge wells is within the SD15b wet slack band: two lie in the SD15b-recoverable band, six in the SD16 dry slack band and one is beyond recoverable. Of the 88 wells in the classified network, 48 have mean summer minima (August–September, below the DEM ground surface) at or deeper than 0.98 m — the SD16 community-mean level — of which 28 are at or deeper than the SD16 recovery limit of 1.20 m.

The P_flood map (Figure 45) quantifies the rainfall demand implied by these summer minima and reveals a progressive, cluster-ordered closure of the operational window (Table 18). Every well in the classified network requires winter rainfall above its climatological mean to reach the slack floor: all 88 wells have m_P > 1.0, the smallest (m_P = 1.06, at CEH6) flooding only just above an average winter. Under mildly wet winters (1.0 ≤ m_P < 1.5), 60 wells (68%) become reachable; a further 23 (26%) require wet winters of varying rarity (1.5 ≤ m_P < 2.5). The remaining 5 wells (6%) have m_P ≥ 2.5 — they require at least 2.5 times the climatological mean winter rainfall, a total well beyond the wettest winter in the ~95-year RAF Valley record. For these wells the slack floor is not reachable under any winter rainfall yet observed: the wettest October–February on record (701 mm, in 1959/60, 1.52× the climatological mean over the same months) delivers only 61% of the m_P = 2.5 demand.

| Cluster | n | P_flood range (mm) | Median (mm) | m_P |
|---|---|---|---|---|
| C1 Lake Edge | 9 | 423–574 | 433 | 1.08 |
| C2 Dune | 24 | 498–823 | 535 | 1.16 |
| C3 Western Residual | 31 | 491–853 | 576 | 1.25 |
| C4 Main Forest | 13 | 681–1,205 | 820 | 1.78 |
| C5 Coastal Forest | 11 | 712–1,490 | 1,086 | 2.36 |

***Table 18:*** *Per-cluster P_flood summary across the 88-well classified network. m_P = P_flood / Σ P_clim; values above 1.0 indicate rainfall above the climatological mean is required. (Source: 11b_06_pflood_cluster_summary.csv; m_P is the cluster median).*

The winter maximum depth map (Figure 43) shows that Eastern Block wells achieve the shallowest winter maxima. Western Residual and Forest wells rarely approach the ground surface (0 m), let alone the SD15b community-mean winter level of +0.21 m. Against the handbook's SD15b winter level (+0.09 m; Denning et al., 2024) six wells meet it on this basis (CEH6, CEH24, CEH25, CEH27, D15 and T41a), against none at +0.21 m, and four more (NW2, NW11, CEH4 and CEH40) meet its SD16 winter level of −0.34 m. The flooding frequency map (Figure 44) summarizes the record as a frequency: C1 wells show the highest flooding frequency across the monitoring period; C3 wells flood less often than C1 and C2; C4 and C5 Forest wells show essentially zero, with the water table failing to reach the ground surface in any monitored year.

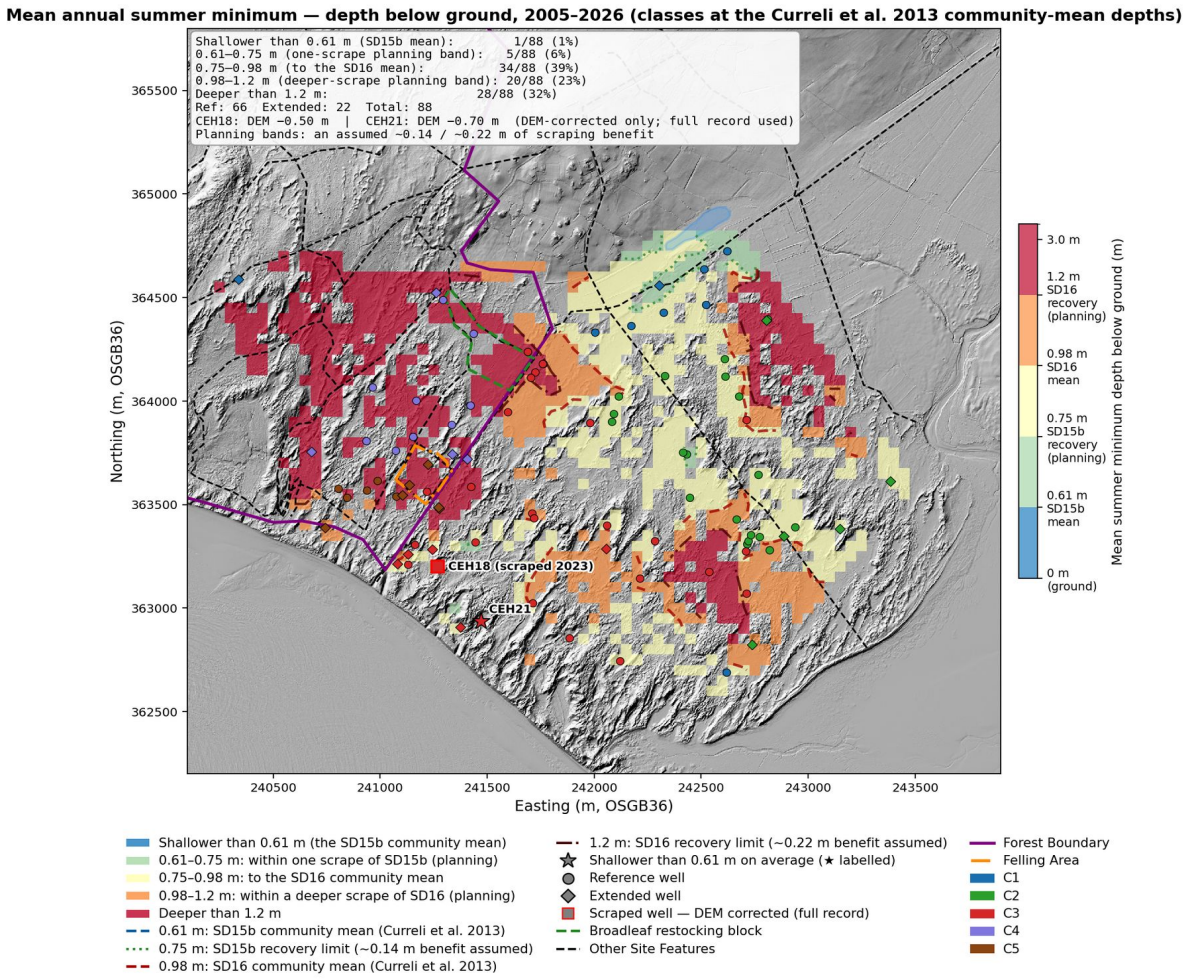



***Figure 41:*** *Spatial distribution of mean annual summer minimum (August–September) water table depth below ground across the full dipwell network (66 reference wells and 22 extended wells), Newborough Warren 2005–2026. Depth is expressed as metres below the LiDAR ground surface (DEM_Ground_Elev − mean maOD), with positive values indicating the water table lies below the surface. The continuous piecewise-linear (Delaunay) surface is masked across dune ridges (cells where the DEM raster exceeds the interpolated well surface by more than 1 m). Zone boundaries are the SD15b and SD16 community-mean levels of Curreli et al. (2013) and the recovery limits set from them: SD15b wet slack (0.61 m); SD15b recovery limit (0.75 m); SD16 dry slack (0.98 m); SD16 recovery limit (1.20 m). Recovery limits are indicative, a planning assumption of about 0.14 m (one scrape) and 0.22 m (a deeper scrape) of water-table benefit; the measured benefit at CEH36 is +0.128 m (Section 4.5); deeper scraping or repeat interventions may achieve greater uplift. Wells CEH18 and CEH21 use their full record, with DEM corrections of −0.50 m and −0.70 m respectively to account for material removed during scraping. Well symbols are coloured by cluster; circles indicate reference wells, diamonds indicate extended network wells. (LiDAR data: © NRW & OS; see Section 11 for full attribution.) A well's class is its depth below its own ground; it indicates a community only where the well stands on the slack floor. A ridge-masked variant of this surface is available in the Newborough Warren Hydrogeological Modelling Suite scenario viewer (Hollingham, 2026). (Source: 11b_01_summer_minima_depth.png).*

On these bands the Eastern Block (C1 and C2) lies within the reach of current scraping practice, C3 Western Residual at its margin, and the two forest clusters (C4 and C5) beyond it (Section 5.8.1).

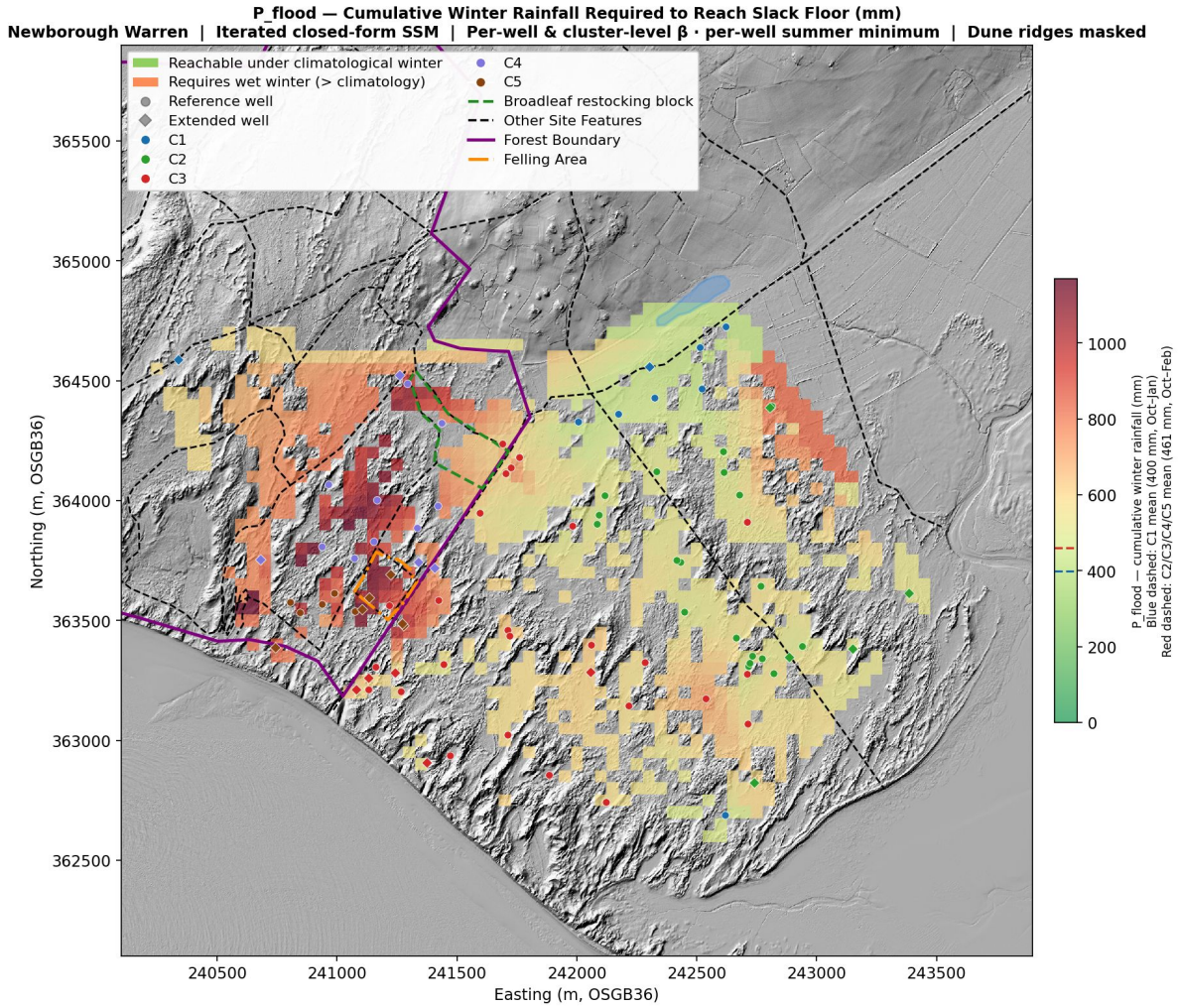



***Figure 42:*** *Spatial distribution of P_flood — the minimum cumulative winter rainfall (mm) required to raise the water table from its observed mean summer minimum to the local slack floor (h_target = 0 m) — across the 88-well classified network, Newborough Warren. P_flood is computed at each well using per-well SSM coefficients where available (66 reference wells) or cluster-level centroid coefficients assigned via Pearson affinity (22 extended wells), applying the iterated closed-form threshold equation of Section 3.6.3 to that well's observed mean summer minimum. The piecewise-linear (Delaunay) surface is masked across dune ridges. Dashed reference lines on the colourbar indicate the cluster-specific climatological rainfall totals: 400 mm for C1/C2 (Oct–Jan, blue) and 461 mm for C3–C5 (Oct–Feb, red). Wells whose P_flood falls below their cluster's reference line are reachable under average winter rainfall. (LiDAR data: © NRW & OS; see Section 11 for full attribution.) (Source: 11b_03_pflood.png).*

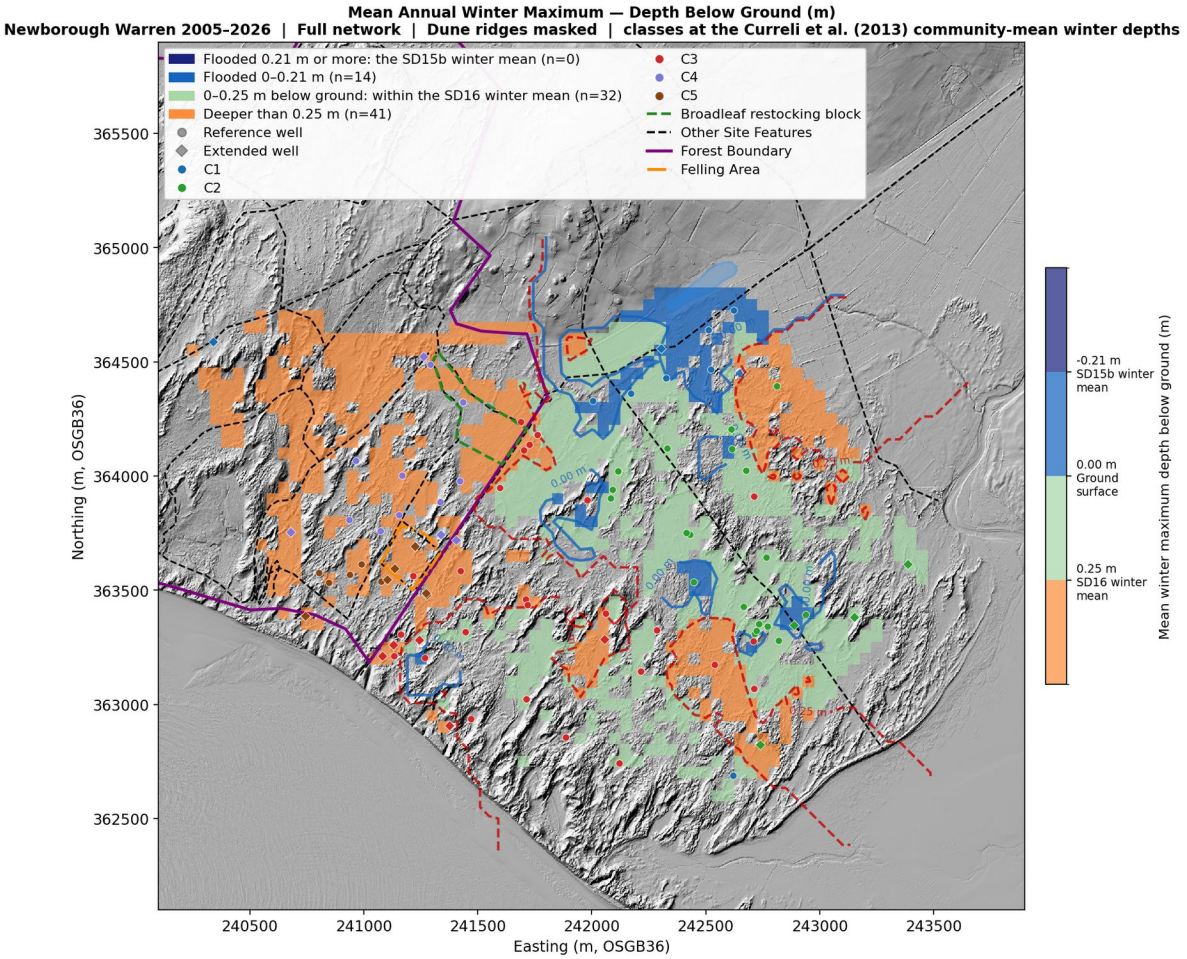



***Figure 43:*** *Spatial distribution of mean annual winter maximum water table depth below ground across the full dipwell network (66 reference wells and 22 extended wells), Newborough Warren 2005–2026. Winter maximum is defined as the highest monthly maOD value in October–March of each hydrological year (October start); the mean is taken across all years with at least three October–March readings. Depth is expressed using the same DEM-referenced convention as Figure 41. The piecewise-linear (Delaunay) surface is masked across dune ridges. Contours mark the Curreli et al. (2013) community-mean winter levels: 0.21 m above ground (the SD15b wet slack winter level, the community-mean winter maximum); 0 m (the ground surface); 0.25 m below ground (the SD16 dry slack winter level). No well reaches the SD15b level. A well's class is its depth below its own ground; it indicates a community only where the well stands on the slack floor. The spatial contrast between the C1 Lake Edge — where winter maxima frequently reach or approach the surface — and the C3 Western Residual, C4 Main Forest and C5 Coastal Forest, where winter maxima rarely approach 0 m, reflects the contrasting aquifer storage architectures identified in Section 4.2. (LiDAR data: © NRW & OS; see Section 11 for full attribution.) (Source: 11b_02_winter_maxima_depth.png).*

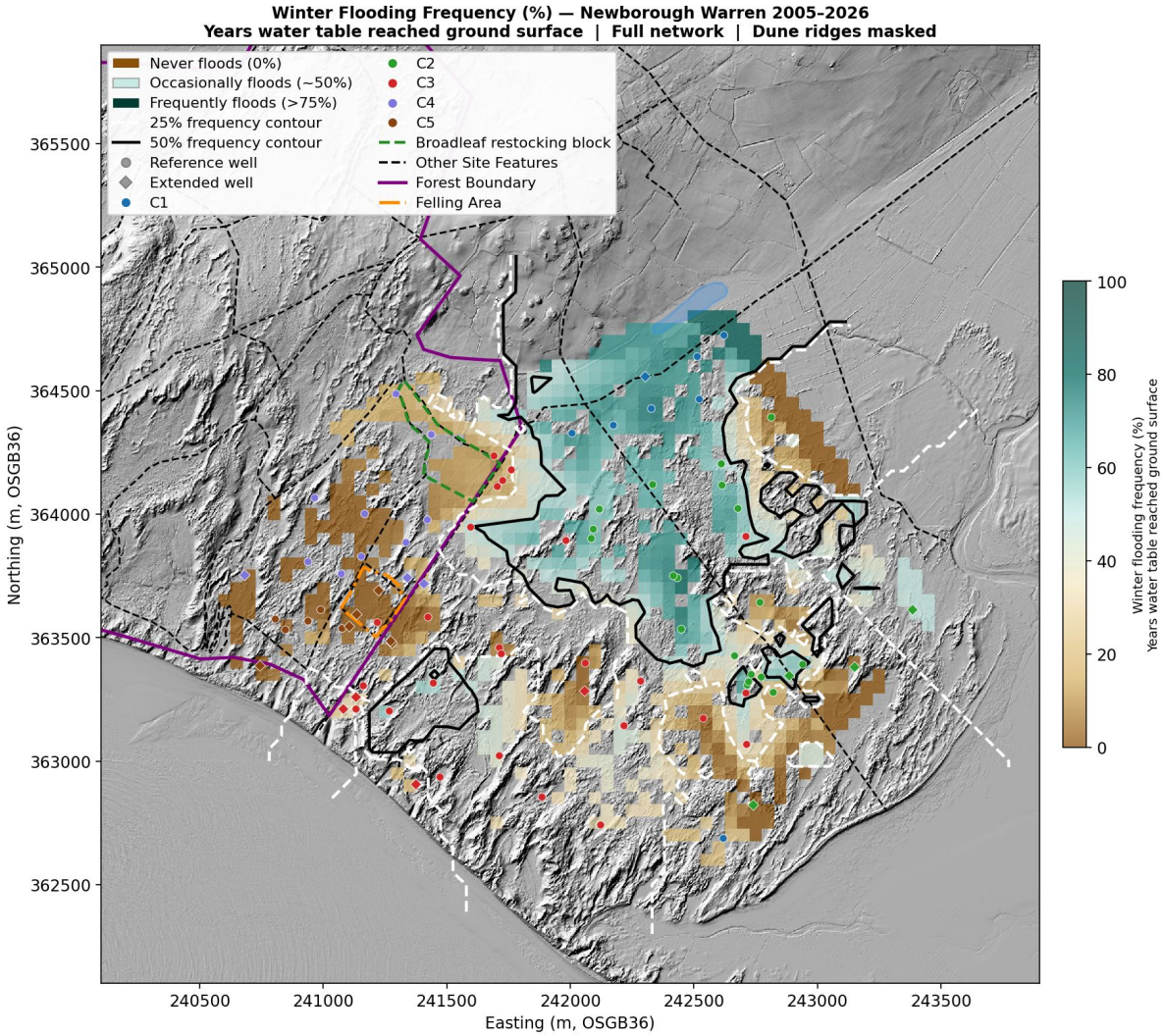



***Figure 44:*** *Winter flooding frequency across the full dipwell network (66 reference wells and 22 extended wells), Newborough Warren 2005–2026, expressed as the percentage of monitored hydrological years in which the winter maximum water table reached or exceeded the local ground surface (DEM ground elevation used as ground surface proxy at each dipwell location). Only wells with at least five years of winter data are included. The piecewise-linear (Delaunay) surface is masked across dune ridges. Contours mark the 25% and 50% frequency thresholds. The C1 Lake Edge shows the highest flooding frequency, consistent with the flashy, fill-and-spill dynamics of the shallow till substrate. C3 Western Residual wells flood less often than C1 and C2, and C4 and C5 Forest wells record essentially zero — the water table fails to reach the ground surface in any monitored year. (LiDAR data: © NRW & OS; see Section 11 for full attribution.) (Source: 11b_04_flood_frequency.png).*

### 4.7.5 Critical Rainfall Threshold Equations (P_flood)

The structural difference between clusters is immediately apparent from the threshold equations (Table 19; Section 3.6.3). The coefficients from Table 3, together with the cluster-

specific recharge horizons from October to each cluster's historical peak month (four months for C1 and C2, five months for C3, C4 and C5) and the long-term monthly RAF Valley rainfall and PET climatology, propagate through the iterated closed form of Section 3.6.3 to produce a linear relationship between observed summer-minimum depth and required winter rainfall. The C4 recharge horizon endpoint is marginal — the cluster's historical mean peak falls between February and March depending on the treatment of partial monitoring years — and the shorter February horizon is adopted as the conservative choice; the C1, C2, C3 and C5 horizons are unambiguous.

The iterated derivation reduces to the form P_flood = A · d_w + B for each cluster, where d_w is the observed summer-minimum depth below ground (m, positive) and P_flood is the cumulative winter rainfall required over the cluster's recharge horizon (mm). The slope A represents the marginal rainfall demand per additional metre of summer drawdown, and the intercept B represents the rainfall demand required even for a well already at the surface — non-zero because accumulated winter PET continues to draw down the water table through the recharge season. Both terms express the cluster gradient directly: A rises from 174 mm per metre at C1 through 210 (C2) and 220 (C3) to 358 (C5) and 379 (C4), and B ranges from 279 mm (C4) to 411 mm (C5).

As a worked example, at a typical C3 well with a mean summer minimum of 1.0 m below ground, P_flood = 219.57 × 1.0 + 368.94 = 588.5 mm of cumulative winter rainfall over the October–February recharge horizon, corresponding to a rainfall multiplier m_P = 589/461 = 1.28 times the long-term climatological total. The same depth at a C2 well produces P_flood = 209.78 × 1.0 + 351.31 = 561.1 mm (m_P = 561/461 = 1.22), an 27 mm difference that comes almost entirely from the intercept B (369 mm at C3 against 351 mm at C2); the slopes A are nearly equal (220 and 210 mm per metre). Both wells lie within the mildly wet band (1.0 ≤ m_P < 1.5); the C3 well needs more rainfall, accumulated over a longer recharge horizon.

| Cluster | Label | Horizon | P_flood equation (mm) | Σ P_clim (mm) |
|---|---|---|---|---|
| C1 | Lake Edge | Oct–Jan (4 mo) | 174.10·d_w + 311.31 | 400 |
| C2 | Dune | Oct–Feb (5 mo) | 209.78·d_w + 351.31 | 461 |
| C3 | Western Residual | Oct–Feb (5 mo) | 219.57·d_w + 368.94 | 461 |
| C4 | Main Forest | Oct–Feb (5 mo) | 378.79·d_w + 278.83 | 461 |
| C5 | Coastal Forest | Oct–Feb (5 mo) | 357.68·d_w + 411.04 | 461 |

***Table 19:*** *Cluster-specific linear forms derived from the iterated closed-form P_flood solution of Section 3.6.3, reduced by substituting fitted SSM coefficients ($\beta_1$, $\beta_2$, $\beta_3$), cluster recharge horizons (n = 4 for C1/C2, 5 for C3/C4/C5), and RAF Valley long-term monthly rainfall and PET climatology (2005–2026). Input variable d_w is the observed mean summer minimum water table depth below ground in metres, entered as a positive number (e.g. d_w = 1.26 for a water table 1.26 m below the surface). Output P_flood is the cumulative rainfall (mm) required over the cluster's recharge horizon (October to the historical mean peak month) to raise the water table to the slack floor (h_target = 0 m), assuming monthly rainfall scales uniformly by a multiplier m_P and monthly PET follows climatological means. The rainfall multiplier is m_P = P_flood / Σ $\bar{P}_i$; values above 1.0 indicate rainfall above the climatological mean is required. (Source: 11_forecast_pflood_summary.csv).*

## 4.8 Climate Trajectory

The spatial distribution of mean annual summer minima and winter maxima across the full dipwell network is presented in Figures 41 and 43. This section follows each cluster's water levels through the 21-year monitoring period: the direction and rate of change of the summer trough, the winter peak and the five-year spring baseline, and the order of the clusters on each. The clusters group wells by how they respond to rainfall, evaporation and drainage, not by where they stand in the dune topography, so the trajectories are read as change in level and as differences between clusters. How they bear on the dune-slack community water levels is discussed in Section 5.8.2.

### 4.8.1 Summer and Winter Trajectories

Summer minimum trends were fitted for all five clusters over the monitoring period (Section 3.6.4; Figure 47, panel a). All clusters show declining summer minima on the full record. C1 Lake Edge declined at −0.0115 m yr⁻¹ ($R^2$ = 0.251, p = 0.024, n = 20), C2 Dune at −0.0143 m yr⁻¹ ($R^2$ = 0.217, p = 0.033, n = 21), C3 Western Residual at −0.0175 m yr⁻¹ ($R^2$ = 0.231, p = 0.028, n = 21) and C5 Coastal Forest at −0.0291 m yr⁻¹ ($R^2$ = 0.385, p = 0.004, n = 20) — the four clusters reaching statistical significance at the 5% level; C4 Main Forest (−0.0115 m yr⁻¹, p = 0.362) was non-significant. The decline is concentrated in the early record: refitted from 2011, no cluster shows a significant trend (C1 −0.0046 m yr⁻¹, p = 0.56; C2 +0.0070, p = 0.48; C3 +0.0087, p = 0.44; C4 +0.0200, p = 0.30; C5 +0.0026, p = 0.79), so the drying the full record shows took place largely before 2011 and is not detectable in the fifteen years since. The cluster series is a fixed-effects centroid, each well held at its own level, so that the few wells recording before 2010 do not set the early cluster level (Section 3.6.4).

The trend is a shift in the trough of an ordinary year, set against large year-to-year variability: C1's summer minimum ranges from −0.64 m (2007) to −1.06 m (2018) over its 20 years. Through that variability the order of the clusters holds: in every one of the 20 years in which all five clusters are recorded, the forest clusters (C4, C5) lie deeper than every open-dune cluster, and C1 or C2 is the shallowest.

**Winter maximum trends** were fitted for all five clusters. C1 showed a positive but non-significant trend (+0.0069 m $yr^{-1}$, p = 0.226); C2 (+0.0058, p = 0.449), C3 (−0.0061, p = 0.616) and C4 (−0.0010, p = 0.955) were all non-significant. C5 showed the steepest winter decline (−0.0271 m $yr^{-1}$), falling just short of significance (p = 0.056); no cluster has a significant winter trend.

**The C5 Coastal Forest trajectory anomaly** is the outlier. At −0.029 m $yr^{-1}$ in summer and −0.027 m $yr^{-1}$ in winter, C5's decline rate is 2.5 times the summer rate of C1 and 1.7 times that of C3, the next steepest; its winter decline, alone among the clusters, approaches significance (p = 0.056). These are full-record rates, and like every cluster's the C5 decline is concentrated before 2011: refitted from 2011, its summer trend is +0.003 m $yr^{-1}$ (p = 0.79) and its winter trend +0.007 m $yr^{-1}$ (p = 0.60), so the anomaly describes the early record rather than a departure that has continued since. The candidate explanations are weighed in Section 5.7.2.

The post-2018 collapse in C5 seasonal amplitude (−18% raw, −21% climate-normalized; Section 4.2.1) is the largest amplitude reduction in the network: recharge peaks are no longer reaching their pre-2018 heights even after correcting for drought summers. The C5 CUSUM trajectory does not stabilize through to the end of the monitoring record.

### 4.8.2 Observed Winter Maximum Flooding Record

The observed winter flooding record shows a strong spatial contrast between structural domains (Figure 47, panel b), and it is read at each well, where standing water means a winter maximum at or above the ground at that well (Figure 44). Flooding is common in the open dune and almost absent under the forest. 7 of the 8 C1 wells flood in at least half their winters, as do 16 of the 23 C2 wells; in C3, 17 of 30 flood in at least a quarter of their winters but only 6 in half. Under the forest, 6 of the 12 C4 wells have flooded in some winter and 1 of the 11 C5 wells, none in as many as a quarter. The cluster winter-maximum series carry no significant trend (Section 4.8.1), so the contrast is between places, not a change through time.

### 4.8.3 Five-Year Mean Spring Water Level (MSL5)

The cluster-level summer-minimum trajectories examined in Sections 4.8.1–4.8.2 describe stress events — single annual values capturing the deepest water-table excursion of each year. A vegetation community's response is not driven by stress events alone; it integrates the long-term baseline water level over which it grows. The 5-year mean spring water level

(MSL5), as defined in Section 3.7.5 and implemented per van Willegen et al. (2025), is the metric that captures this baseline. It is the rolling 5-year mean of the cluster-mean March–May water table, taken at the end of each hydrological year. Where the summer-minimum metric is sensitive to single dry years, MSL5 is sensitive to sustained hydrological state — the regime within which dune-slack vegetation establishes and persists.

The cluster MSL5 trajectories for window-ends 2014–2025 are shown in Figure 45(a). Figure 45(b) shows, over the same window-ends, the cluster four-year mean annual minimum water level, the summer-trough counterpart of MSL5 and the quantity on which Curreli et al. (2013) reported their slack communities (Section 5.8.2), and Table 20 gives both at window-end 2025.

On the four-year mean annual minimum the open-dune clusters stand above the forest clusters at every window-end from 2014 to 2025, as on the annual summer minimum (Table 20): C1 Lake Edge between −0.83 and −0.91 m, C2 Dune between −0.96 and −1.06 m, C3 Western Residual (−1.10 to −1.26 m), C4 Main Forest (−1.48 to −1.86 m) and C5 Coastal Forest (−1.65 to −1.89 m). Both metrics describe the trough of the year.

Where the slack floor is surveyed the record can be read against the communities directly. Van Willegen et al.'s quadrats lie on the slack floor beside their piezometers, and each quadrat's ground is known from its datum offset (Supplementary Table S7.2), so each quadrat's four-year mean annual minimum can be placed on its own ground. Over Curreli et al.'s own window (window-end 2010), 11 of the 64 quadrats at 12 piezometers lay above the SD15b community mean, 42 between SD15b and SD16 and 11 below SD16. At window-end 2025 none of the 72 quadrats at 16 piezometers lies above SD15b, 48 lie between the two and 24 below SD16.

On the 47 quadrats (eight piezometers) read in both windows, every one is deeper at 2025, by a median of 0.19 m (0.10 to 0.47 m). Ten of them lay above SD15b in 2007–10 and none does now, and the number below SD16 has risen from 5 to 20. The deepening holds against every later window-end, by medians of 0.19 to 0.42 m. Between later window-ends the sign of the change depends on the windows chosen, as it does for MSL5 (Section 5.7.5), so the result is a difference from the survey years rather than a steady trend since.

A change in MSL5 between window-ends needs no floor height, but its size and sign depend on the window-ends chosen, on the five-year mean and on the nine-year MSL9 alike (Section 5.7.5; Supplementary Material Note S11).

MSL5 orders the clusters as the annual minimum does — C1 wettest, C5 driest — and the spread between them is of the same size on both metrics: 1.02 m between C1 and C5 on MSL5 at window-end 2025 and 0.81 m on the four-year mean annual minimum. C3 Western Residual sits 0.19 m below C2 on MSL5 at 2025 (−0.383 m against −0.189 m), and its two most recent window-ends are the two containing the wet 2023–24 hydrological year.

The forest clusters sit apart from the open dune on both metrics. On MSL5 at window-end 2025 C4 Main Forest is at −0.781 m and C5 Coastal Forest at −1.149 m, 0.65 and 1.02 m below C1; on the four-year mean annual minimum the deficits are 0.60 and 0.81 m. MSL5 is, in the van Willegen framework, primarily an open-dune metric (Section 3.7.5). The C5 trajectory rests on ten wells at window-end 2025 (five reference and five extended-network) located in the south-western Corsican-pine block; the cluster is among the smallest in the classified network and the result should be read with that in mind, but the wells are spatially coherent and the trajectory is clear within that reach.

The spatial pattern of the MSL5 surface across the network is shown in Figure 46. The figure is a piecewise-linear (Delaunay) interpolation across the 83 mapped wells — the 85 that carry at least one valid MSL5 window, less CEH13 and CEH14, which are excluded for unreliable SSM drainage coefficients; for 75 of these the latest valid window ends in 2025, while for eight, recent gaps in spring measurements mean the latest valid window ends earlier (2015–2024), and these appear at their latest available value. The map is distinct from the cluster-mean trajectories of Figure 45 in that it shows the full spatial reach of the per-well MSL5 surface rather than the cluster-identity-based aggregation. The two figures show complementary things — Figure 45 is cluster-mean MSL5 over time, Figure 46 is the per-well MSL5 surface at the most recent window. The map shows the deepest MSL5 concentrated in the south-western corner of the site — the C5 Coastal Forest block — with the surrounding open-dune network markedly shallower. The Western Residual cluster's western margin appears as the transition zone on the map's western edge, consistent with C3's position below C2 in the trajectory analysis.

The most recent window-ends carry a synchronous upward step. All five clusters jump wetter between window-end 2023 and window-end 2024, with the magnitude varying by cluster (Figure 45). The step reflects the 5-year window incorporating the 2023–24 hydrological year, which carried significantly higher annual rainfall than the 2018–22 range it displaces. The magnitude varies because the displacement-sensitive clusters (C4 +205 mm, C3 +157 mm, C5 +140 mm) respond more strongly than the near-surface ones (C1 +84 mm) to a single wet year entering the rolling window. Because the metric is a 5-year rolling mean, the step's persistence depends on subsequent years: a return to the recent dry hydrological pattern would erode it as the wet year ages out of the window. The 2025 window-end holds the step at every cluster (C5 −1.165 m at 2024, −1.149 m at 2025). The step is the entry of one wet year into the window, not a trend.

The MSL5 metric requires a five-year record of valid spring measurements. The earliest valid window-ends fall in 2010–2011 depending on cluster, but the early window-ends (2011–2014) carry between one and ten wells per cluster (C1 3, C2 3–5, C3 9–10, C4 1–2, C5 3–6), and membership rises to 7, 20, 18, 6 and 7 wells at window-end 2015 as the 2010 installations complete their first five-year window. With within-cluster well-to-well differences of 0.3 m at C2, 0.3 m at C3, 0.4 m at C4 and 0.4 m at C5 (window-end 2025), a

plain mean would move each time a well joined; the cluster trajectories are therefore fixed-effects series, each well held at its own level (Section 3.6.4), which equal the plain mean wherever every member reports. The trajectory plot starts at window-end 2014. Earlier window-ends are computed and stored in the supporting outputs for completeness, but are not plotted or discussed here.

The headline reading of Figure 45 is therefore twofold. MSL5 orders the clusters from the lake edge to the coastal forest and holds that order through the recorded window-ends, with the forest clusters 0.6–1.0 m below C1 at window-end 2025. The four-year mean annual minimum keeps the open-dune clusters above the forest clusters throughout. The 2024 step is climatic and is held at 2025. Implications for the climate trajectory are taken up in Section 4.13.1; the relation between this reading and the summer-minimum reading of Section 4.8.1 is returned to in Section 5.1.

| Cluster | n wells (2025) | MSL5 (m) | 4-yr mean annual minimum (m) |
|---|---|---|---|
| C1 Lake Edge | 8 | −0.133 | −0.885 |
| C2 Dune | 17 | −0.189 | −0.964 |
| C3 Western Residual | 24 | −0.383 | −1.097 |
| C4 Main Forest | 10 | −0.781 | −1.482 |
| C5 Coastal Forest | 9 | −1.149 | −1.692 |

***Table 20:*** *Cluster five-year mean spring water level (MSL5) and four-year mean annual minimum water level at window-end 2025 (fixed-effects cluster series, Section 3.6.4). n is the number of wells with a valid four-year window at 2025; MSL5's own well count is carried in the source file. (Source: 26_curreli_min_cluster_threshold_summary.csv).*

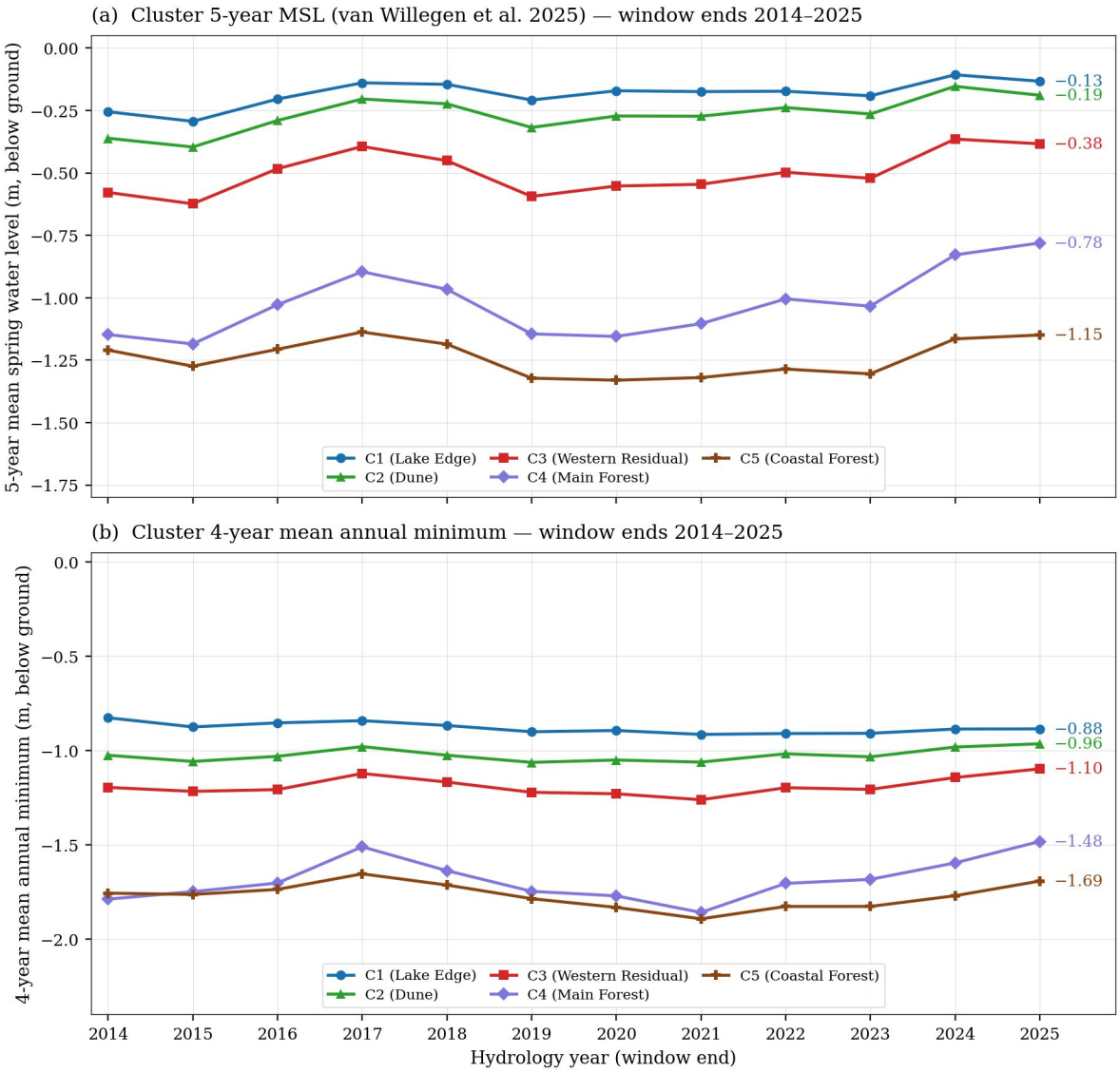



***Figure 45:*** *Cluster five-year mean spring water level (MSL5), window-ends 2014–2025, for the five clusters of the classified network (a), and the cluster four-year mean annual minimum water level over the same window-ends (b). Both are fixed-effects cluster series (Section 3.6.4); no community level is drawn (Section 5.8.2). The open-dune clusters stand above the forest clusters on both metrics throughout. The synchronous upward step between 2023 and 2024 in both panels is the 2023–24 hydrological year entering the windows. (Source: Script 26c output fig_msl5_trajectory_report.png).*

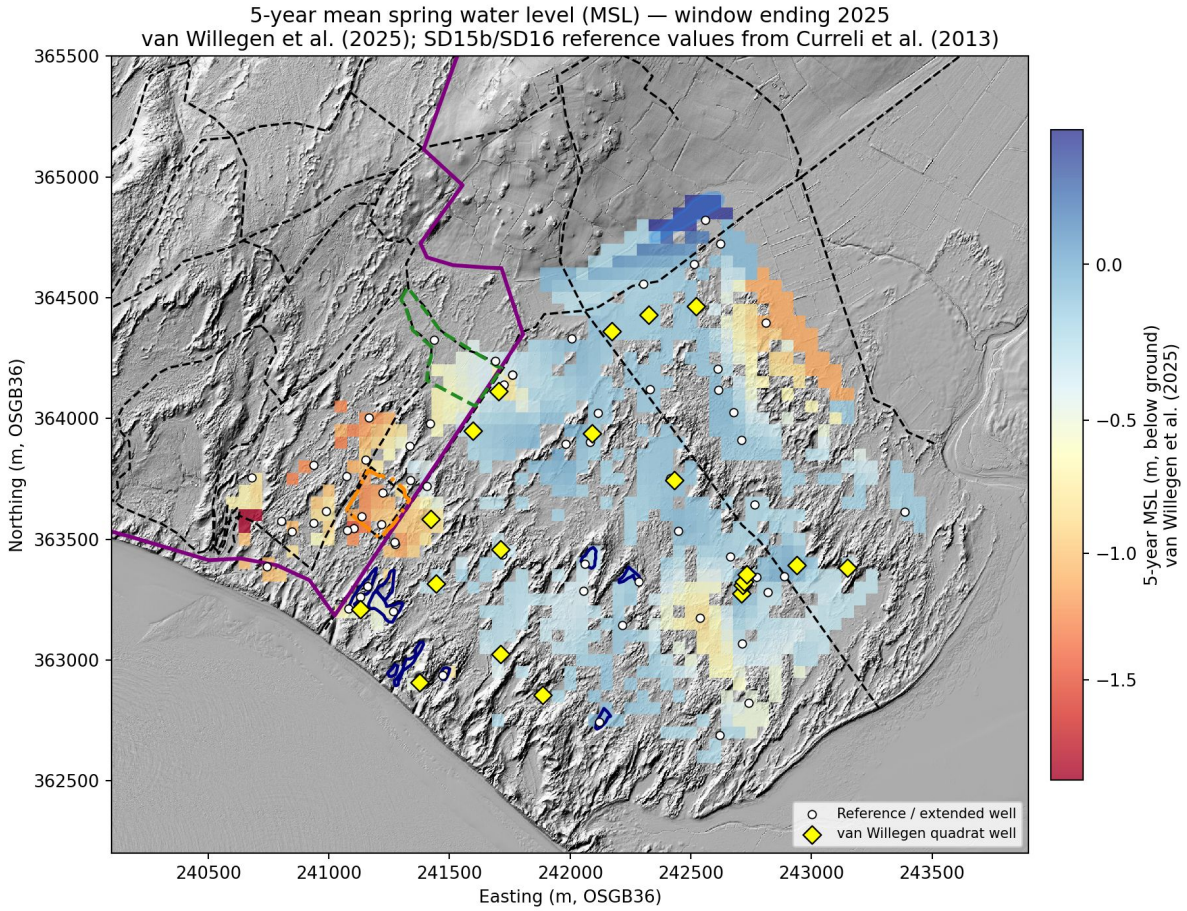



***Figure 46:*** *Spatial pattern of the latest available 5-year mean spring water level (MSL5) per well, interpolated piecewise-linearly (Delaunay) across the 83 mapped wells — the 85 wells of the 88-well classified network carrying at least one valid MSL5 window, less CEH13 and CEH14, which are excluded for unreliable SSM drainage coefficients (75 wells at window-end 2025, eight at earlier latest-valid windows). White circles mark reference and extended-network wells; yellow diamonds mark the twenty van Willegen quadrat-calibrated wells at their seventeen stations where vegetation community has been empirically calibrated against MSL5. The colour scale is centred on the Curreli et al. (2013) SD15b value (−0.61 m) and marks SD16 (−0.98 m) as reference levels only: both are community means of the four-year mean annual minimum, not of spring levels, so a well's MSL5 against them is not a threshold crossing (Section 4.8.3). The orange-red zone in the south-western corner covers the C5 Coastal Forest and C4 Main Forest clusters, the deepest MSL5 on the site; the open dune to the east is shallowest, and the Western Residual block at the western margin lies between them. (Source: 26_msl_5yr_map.png)*

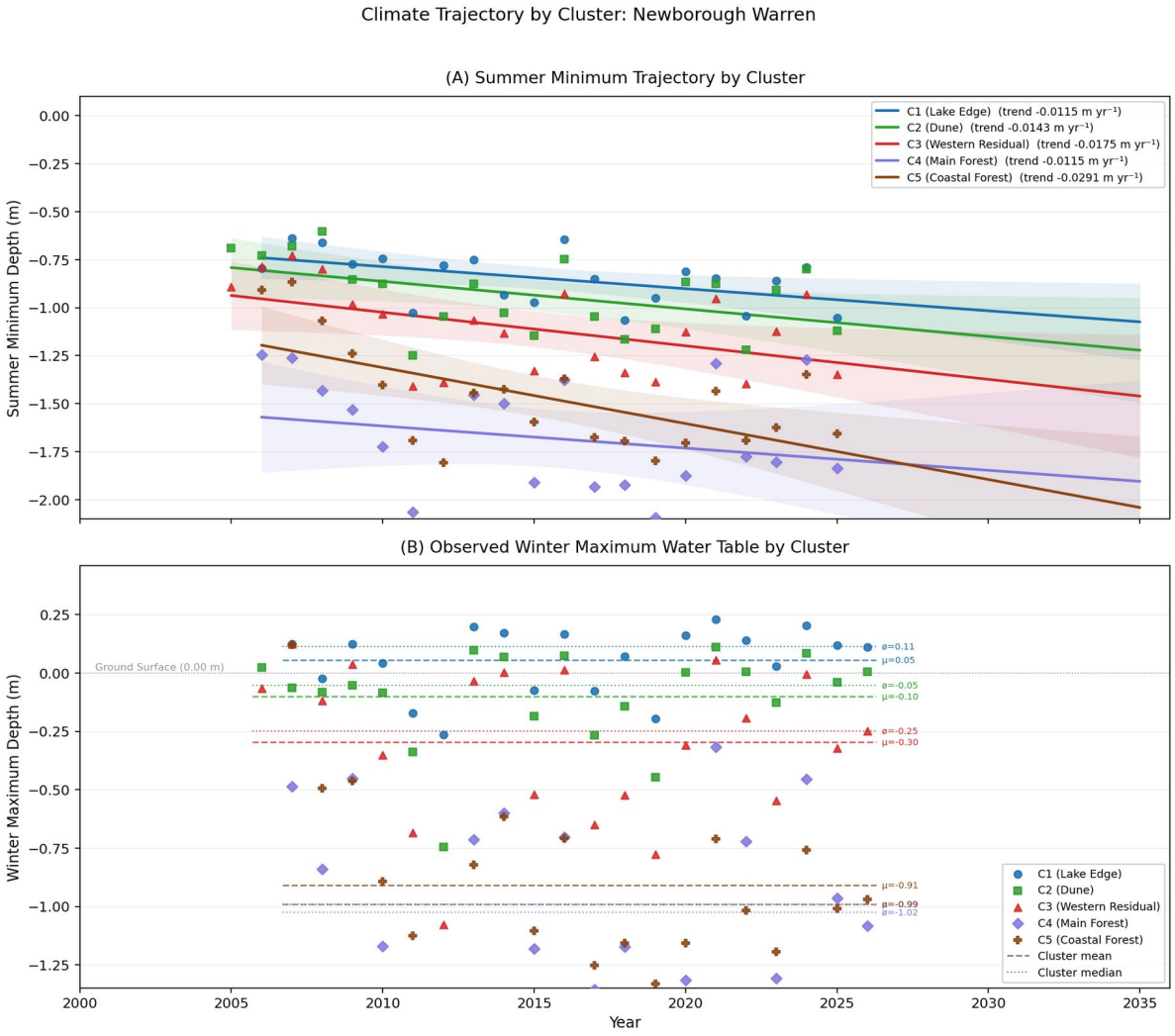



***Figure 47:*** *Climate trajectory by cluster for Newborough Warren. Panel (a): observed annual summer minimum water table depth for all five clusters, 2005–2025, with OLS linear trend lines and 95% confidence intervals projected to 2035. Panel (b): observed annual winter maximum water table depth for all five clusters, with each cluster's mean and median. (Source: 14_climate_trajectory_stacked.png)*

### 4.8.4 Equilibrium Wetness Index and Vegetation Cross-Validation

MSL5 (Section 4.8.3; defined Section 3.7.5) follows van Willegen et al. (2025), who identified it as the strongest single hydrological correlate of dune-slack vegetation moisture at Newborough. Van Willegen et al.'s published per-quadrat spring levels are this network's own readings re-referenced to each quadrat's ground surface, so recomputing them from the pipeline checks that the processing matches theirs; it is not an independent validation. After one constant offset per quadrat the two agree to within millimetres (Supplementary Table S7.2). The offsets span several decimetres, so a level in this report cannot be compared with one referenced to quadrat ground without applying that quadrat's offset. Between wells, MSL5 predicts mean Ellenberg-F — the moisture indicator value (wetness scale 1–12; Hill et al., 1999) — strongly ($r = +0.83$, $n = 18$). These wells lie in the lake-edge, open-dune and western-residual clusters, van Willegen having placed no quadrats in the forest interior.

The state-space coefficients afford a structurally derived estimate of the same wetness. Setting the mean monthly change to zero in the fitted model (Section 3.4) and inverting the head-dependent drainage term yields an equilibrium water-table level, $h_{eq} = (\beta_1 \bar{P} - \beta_2 PE\bar{T})/\beta_3 - z_0$ ($z_0 = 3.7$ m), from each well's coefficients and the long-term annual-mean climate. Tested directly against Ellenberg-F this equilibrium index performs as well as observed MSL5: the two are statistically indistinguishable as vegetation predictors (Figure 48; Table 21), their correlation confidence intervals overlapping almost entirely and a Williams test for the difference between the two dependent correlations returning $p = 0.59$. The equivalence holds not only in aggregate but well by well: binned by prediction accuracy, the two metrics are broadly comparable across match bands, the equilibrium index placing marginally more wells in the closest band but also more in the poorest (Table 22). That coefficients calibrated purely on month-to-month dynamics predict vegetation moisture as well as the direct spring-level metric is the result; eighteen wells give limited statistical power, so this is equivalence within the data rather than proven identity. Because the equilibrium index is computed from the coefficients and climate alone, this comparison is independent of MSL5 — each metric is regressed on Ellenberg-F separately, so the equivalence tests two independent predictors rather than restating the spring-level metric. No single state-space coefficient predicts Ellenberg-F nearly as well (best $|r| = 0.38$, versus 0.80 for the equilibrium index), the vegetation signal residing in the equilibrium level rather than any individual rate or storage term.

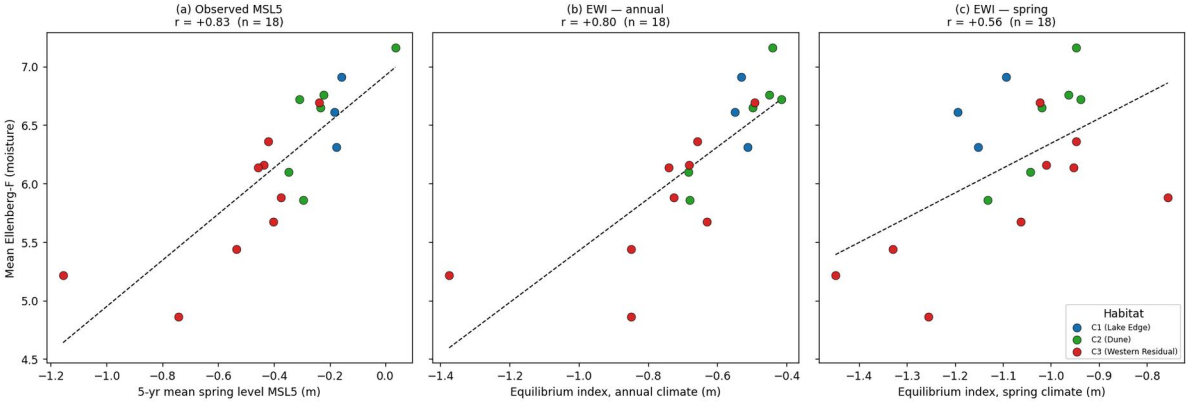



***Figure 48:*** *Between-well prediction of mean Ellenberg-F moisture by three water-table metrics, across the eighteen van Willegen et al. (2025) piezometers carrying a valid state-space fit. (a) Observed five-year mean spring water level (MSL5); (b) equilibrium wetness index on the annual-climate basis; (c) equilibrium wetness index on the spring-climate basis. Points are coloured by hydrogeological cluster (C1 lake-edge, C2 open-dune, C3 western-residual); the dashed line is the ordinary-least-squares fit, with Pearson r and sample size annotated on each panel. The annual-basis index (b, r = +0.80) predicts vegetation moisture almost as well as observed MSL5 (a, r = +0.83) and more strongly than the spring basis (c, r = +0.56), motivating the annual climatology adopted in Section 3.7.6. Ellenberg-F from van Willegen et al. (2024); per-well β from Script 03. Source: 26_ebf_prediction_scatter.png.*

| | MSL5 | Equilibrium wetness index |
|---|---|---|
| r [95% CI] | +0.83 [0.59, 0.93] | +0.80 [0.53, 0.92] |
| RMSE (Ellenberg-F units) | 0.337 [0.20, 0.39] | 0.361 [0.18, 0.46] |

***Table 21:*** *Between-well prediction of mean Ellenberg-F moisture by observed MSL5 and by the equilibrium wetness index, across the van Willegen et al. (2025) piezometers with a valid state-space fit (n = 18: of the 20 wells at their seventeen stations, T41 counted as its four transect wells, the 18 that carry a valid state-space fit). Pearson r and residual root-mean-square error (in Ellenberg-F units) are given with 95% confidence intervals. The two predictors are statistically indistinguishable (Williams' test for dependent correlations, p = 0.59; Williams, 1959). Ellenberg-F from van Willegen et al. (2024). Source: 26_ebf_prediction_summary.csv.*

| Match band | Absolute error (Ellenberg-F units) | MSL5 (wells) | EWI (wells) |
|---|---|---|---|
| A — excellent | ≤ 0.15 | 4 | 7 |
| B — good | 0.15–0.30 | 7 | 5 |
| C — moderate | 0.30–0.50 | 5 | 2 |
| D — poor | > 0.50 | 2 | 4 |

***Table 22:*** *Ellenberg-F prediction accuracy by match band, MSL5 versus the equilibrium wetness index (n = 18 wells). Each band counts wells by the absolute prediction error \|predicted − observed\| in Ellenberg-F units. The two metrics distribute similarly across bands, corroborating the aggregate equivalence of Table 21. Source: 26_ebf_band_summary.csv.*

Beyond predicting vegetation, the equilibrium index can be calibrated onto the MSL5 scale itself by ordinary least squares. Consistent with MSL5's own open-dune character (Section

4.8.3) and with the forest coefficients being the least constrained on the site (Section 4.9.2), this calibration is scoped to the open-dune network (clusters C1–C3, reference and extended wells): MSL5 = 0.401 + 0.973·EWI (n = 61, r = 0.96, RMSE = 73 mm). Every open-dune well is reconstructed within 234 mm, and the relationship carries beyond the vegetation-calibrated subset with some loss of accuracy: a root-mean-square error of 79 mm across the 41 open-dune wells outside van Willegen's set, against 60 mm at the 20 wells of their seventeen stations (their T41 station is the four-well T41a–d transect, each of its quadrats read from its own well). The C4/C5 forest wells are predicted but held out of scope (n = 21, RMSE 167 mm): there the coefficients are least constrained and the estimate is too coarse to place a well against the SD15b / SD16 community-mean levels. Calibrated instead on the nine-year mean spring level of the Dune Managers Handbook (MSL9; Supplementary Material Note S11), the fit is unchanged: MSL9 = 0.374 + 1.048·EWI (n = 57, r = 0.97, RMSE = 73 mm), against r = 0.97 and RMSE = 69 mm for MSL5 on the same 57 wells; and MSL9 predicts Ellenberg-F no better than MSL5 (r = +0.80 against +0.83, n = 18). MSL5 therefore remains the calibration target, as van Willegen et al.'s metric. The full per-well reconstruction, with the out-of-scope forest wells flagged, is given in Table S7.1 of the Supplementary Material.

**Precision of the two indices**

The two indices differ in what limits them. MSL5 averages five spring observations, so its precision is set by how far spring levels swing from year to year: the interannual standard deviation rises from about 182 mm at C1 through 252 mm at C2 and 324 mm at C3 to 291 mm at C5 and 404 mm at C4, giving a five-year window mean a standard error of about 82, 113, 145, 130 and 181 mm respectively. The equilibrium index carries no window, but it divides by $\beta_3$, so its precision is set by how well that coefficient is identified. Propagating each well's $\beta_3$ uncertainty gives a standard error of about 303 mm at C1, 390 mm at C2, 403 mm at C3, 415 mm at C5 and 1014 mm at C4.

The equilibrium index is therefore the less precise of the two throughout the network, by a factor of between 2.8 and 5.6 at the cluster medians, and it is more precise than a five-year MSL5 mean at none of the 65 wells where both can be computed. Its advantages lie elsewhere: it requires no five-year spring window, so it can be computed at wells whose spring record is too broken for MSL5, and its precision shows no measurable dependence on record length ($\beta_3$ relative standard error against number of observations, Spearman −0.09, p = 0.71 across the extended network). The two indices also degrade at the same wells for opposite reasons — MSL5 where spring levels swing furthest, the equilibrium index where $\beta_3$ is weakly identified — and the forest clusters are adverse on both counts at once. Under closed canopy both carry a precision caveat, and the Ellenberg-F equivalence established above rests on eighteen open-dune piezometers, van Willegen having placed no quadrats in the forest interior.

### 4.8.5 Slack-floor wet area: an independent spatial check on the modelled water table

The vegetation cross-validation of Section 4.8.4 tests the modelled water table against an independent ecological record at the wells and quadrats; it does not test whether the modelled level predicts the wetted extent of the slack floor between them. A wholly independent check on that spatial reach is available from the Sentinel-2 satellite archive (Script 45), which images the warren every few days at 10 m resolution in the near-infrared, where standing water and saturated ground read dark. Over 43 cloud-free winter scenes (2016–2026) the open dune-slack floor of the warren — a 306.97 ha study area of 30,697 cells, the planted forest excluded — was classified into two near-infrared wetness classes: open water (the darkest cells) and wet floor (dark but not ponded). This is an area measure derived entirely from satellite reflectance; it shares no wells, no data and no model coefficients with the SSM water table.

Closed canopy is excluded from the classification because Band 8 there measures the canopy and not the floor. Scored as a control, the 1998 compartments and the leafless broadleaf block read dark at every water-table level, and the 2017 clearfell, the one open floor under the plantation, reads bright at every level, its regrowth indistinguishable from a dry floor; the classification is therefore not extended into the forest. Sentinel-1 C-band radar was tried on the same polygons and resolves neither the floors nor the canopy; flooding beneath the canopy would need L-band. The rides are one 10 m cell wide and edge-contaminated, and are counted but not classified.

Across the 43 scenes each class area rises with the contemporaneous median water level as area = a·exp(b·h), where h is the reference-network median level relative to the slack floor. Open water grows steeply (a = 24.9 ha, b = 4.43; Spearman ρ = +0.80, $R^2$ = 0.71) and wet floor more gradually (a = 56.1 ha, b = 0.95; ρ = +0.85, $R^2$ = 0.77) over the observed level range −0.80 to +0.03 m (Figure 49). The darkened floor (wet floor plus open water) comprises 11,703 cells (117.0 ha) with a fitted switching level (seen in at least ten scenes), of which 4,483 (44.8 ha) are open-water cells. The fits indicate that satellite-observed slack wetness is a monotonic, well-behaved function of the modelled level — the precondition for using it as a spatial test.

That test is whether the modelled water table, driven through these fitted curves, reproduces the observed wet-area history it never saw. Passing the SSM monthly level (Section 4.4) through the two curves yields a modelled open-water and wet-floor area for every month of record; comparing these to the satellite-observed areas over the months with adequate well coverage (n = 238 well-months, ≥ 20 reporting wells) tests the model out of sample and in space (Figure 50). Wet-floor area is reproduced closely ($R^2$ = 0.78 under the recharge-referenced level, 0.80 under the calibrated level; median modelled:observed area ratio 0.96 in both modes), open water more approximately ($R^2$ = 0.62 and 0.64; median ratio 0.84 and 0.81, the model running slightly dry at the ponded extreme).

This is the report's only spatial, out-of-sample test of the modelled water table. The well hydrographs (Section 4.4) test it through time at points, and the vegetation record (Section 4.8.4) tests it against ecology at points; neither tests whether the modelled level predicts the wetted extent of the slack floor between the wells. The satellite-observed extent is recovered from the model, most strongly for the wet-floor class that carries the dune-slack habitat signal (Section 5.7.6). The check is an aggregate one over the whole warren floor, not a per-slack map (the per-slack aerial comparison was withdrawn as unresolvable), and it is confined to the open warren; the planted forest is not modelled and is excluded. The wet-area model is presented as an illustration of the area–level relationship, not as a flood map. Its implications for the dune-slack habitat area at stake under the climate trajectory are taken up in Section 5.

The same scene stack, read cell by cell, gives the wetted extent a spatial expression the aggregate curves cannot (Figure 51). For each 10 m cell of the slack floor the scenes are ordered by the reference-network median level at their dates, and the cell's switching levels are the level at which it first reads wet floor and, separately, the level at which it first reads open water — each a ground-referenced height of the water table at that cell, resolved at 10 m and independent of the wells. Of the 30,697 floor cells 11,703 have read wet floor and 4,483 open water within the observed range of −0.80 to +0.03 m; the remainder have not wetted in the record. The maps are the floor's own answer to where the water table reaches the ground first, and they carry two cautions. They are not the prediction of a digital elevation model plus a uniformly rising water table: the kriged surface of Section 4.9.5 carries relief only at the scale of the well spacing, and a slack’s floor is finer than that. And the floor lags the wells on the way down: at the same median level a scene taken while the water table is rising carries 2.2 times the open water of one taken while it is falling (wet floor 1.15 times; the phase term is significant at p = 0.003 and 0.04). Rainfall in the scene month adds no positive term once the level is in the model. The aggregate fits of Figure 49 are level-only by design and absorb this hysteresis into their scatter (Script 45: 45_03_switching_levels_map.png, 45_04_phase_hysteresis.csv).

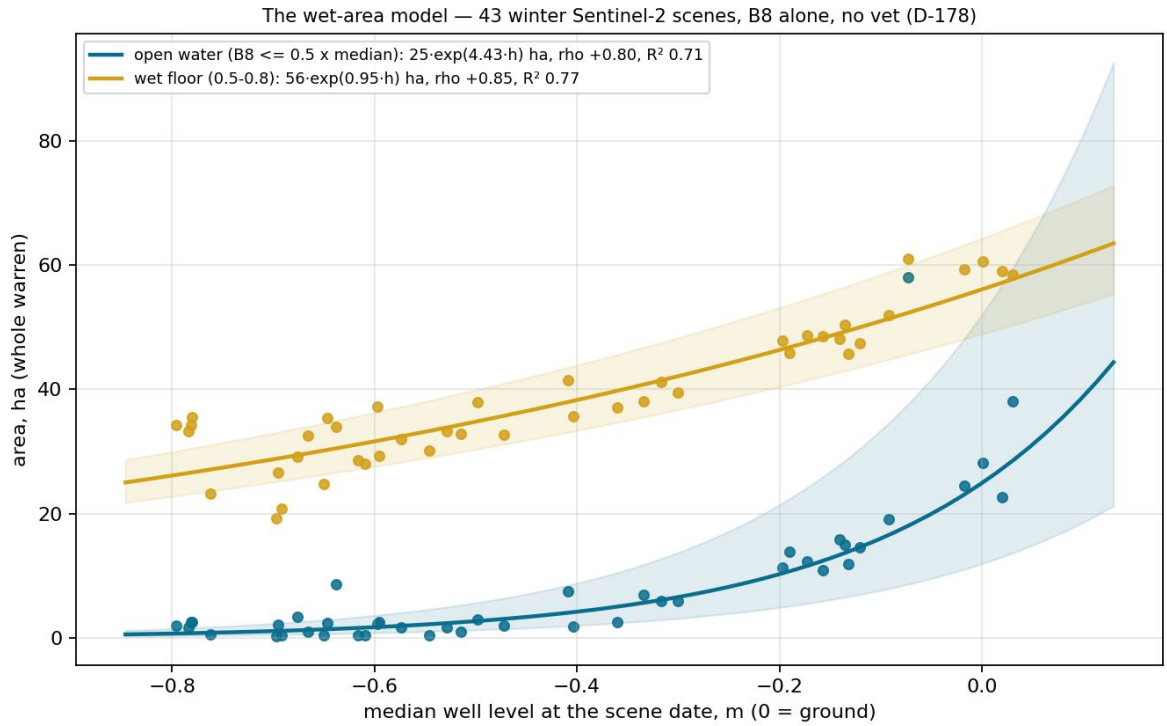



***Figure 49:*** *Slack-floor wet area against the modelled water table. Near-infrared wet area of the open dune-slack floor of Newborough Warren (306.97 ha study area, forest excluded), from 43 cloud-free winter Sentinel-2 scenes (2016–2026), classified into open water (darkest cells) and wet floor (dark, unponded) and plotted against the reference-network median water level. Fitted curves area = a·exp(b·h): open water a = 24.9 ha, b = 4.43 (Spearman ρ = +0.80, R² = 0.71); wet floor a = 56.1 ha, b = 0.95 (ρ = +0.85, R² = 0.77), over the observed range −0.80 to +0.03 m. (Source: 45_01_wet_area_model.png). Sentinel-2 imagery: © Copernicus.*

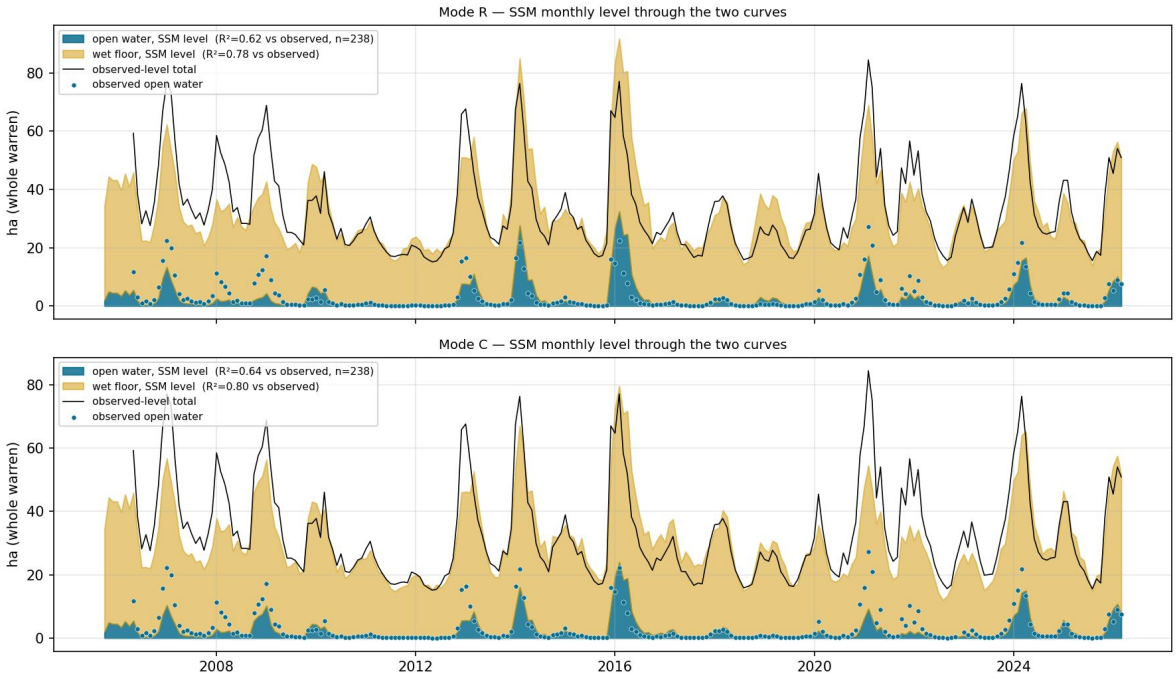



***Figure 50:*** *Modelled versus satellite-observed slack-floor wet area — an out-of-sample spatial test of the water table. The SSM monthly level (Section 4.4) driven through the fitted area–level curves of Figure 49 (modelled), against the Sentinel-2 observed area over 238 well-months with ≥ 20 reporting wells, under the recharge-referenced (mode R) and calibrated (mode C) level. Wet floor: R² = 0.78 (R) / 0.80 (C), median area ratio 0.96. Open water: R² = 0.62 (R) / 0.64 (C), median ratio 0.84 / 0.81. (Source: 45_02_ssm_through_nir_curves.png). Sentinel-2 imagery: © Copernicus.*

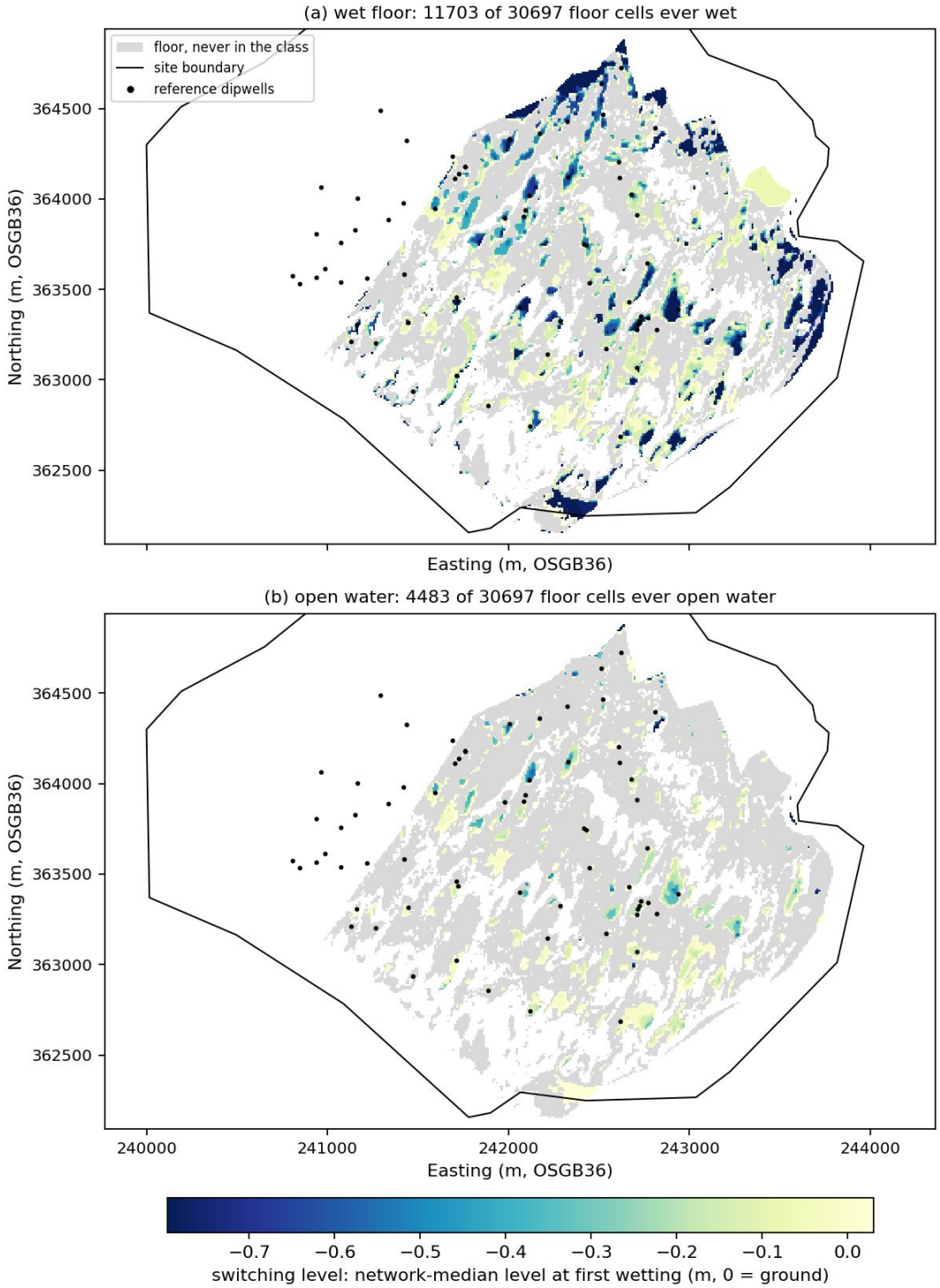



Figure 51: *Per-cell switching levels of the Newborough Warren slack floor from 43 winter Sentinel-2 scenes (2016–2026). For each 10 m floor cell, the reference-network median water level (m relative to the slack floor, 0 = ground) at which the cell first reads (a) wet floor and (b) open water; floor cells that have not entered the class within the observed range (−0.80 to +0.03 m) in light grey, non-floor cells blank. 11,703 of 30,697 floor cells have read wet floor and 4,483 open water. Black points: the 66 reference dipwells; black line: the study-area boundary. The planted forest is excluded (near-infrared darkness under canopy measures the canopy, not the floor). Switching levels are censored to the observed range of the scene series and are a ground-referenced water-table height at 10 m, independent of the wells; they are not a prediction from the elevation model, and at a given level the floor holds more water on a rising than on a falling water table (Section 4.8.5). Generated by 45_wet_area_model.py. (Source: 45_03_switching_levels_map.png; data/sentinel/cell_thresholds.npz). Sentinel-2 imagery: © Copernicus.*

Read the other way, the switching levels test the kriged water table cell by cell (Script 01b; Figure 52). Each wetted floor cell's switching level implies a water-table height there: the floor, plus the wet-state network level, less the switching level, less the cell-to-well offset measured at the wells (0.32 m). That height is compared with the kriged wet-state surface. Of 11,616 tested cells, 56% agree with the surface to within 0.5 m; 23% have no dipwell within 300 m, so the surface there is set by its boundaries and they are not tested; 17% lie between 0.5 and 1.0 m from it; and 4.2% (488 cells) imply a water table more than 1.0 m above the surface. Most of those 488 are accounted for. 95 lie on north-facing dune faces in cast shadow at the lowest winter sun, where the near-infrared reads dark without water. 180 lie at floors that the nearest well's own wet-state level already reaches. 51 are steep floor-edge pixels mixing dune and floor. The remaining 162 cells (1.4% of those tested) are unexplained. They are treated as error in the check rather than as evidence of perched water, which neither the wells nor the field record support. The shadow class carries one consequence for the switching levels themselves: north-facing faces can read wet early in low-sun scenes, which biases their switching levels low (Script 45 is unchanged).

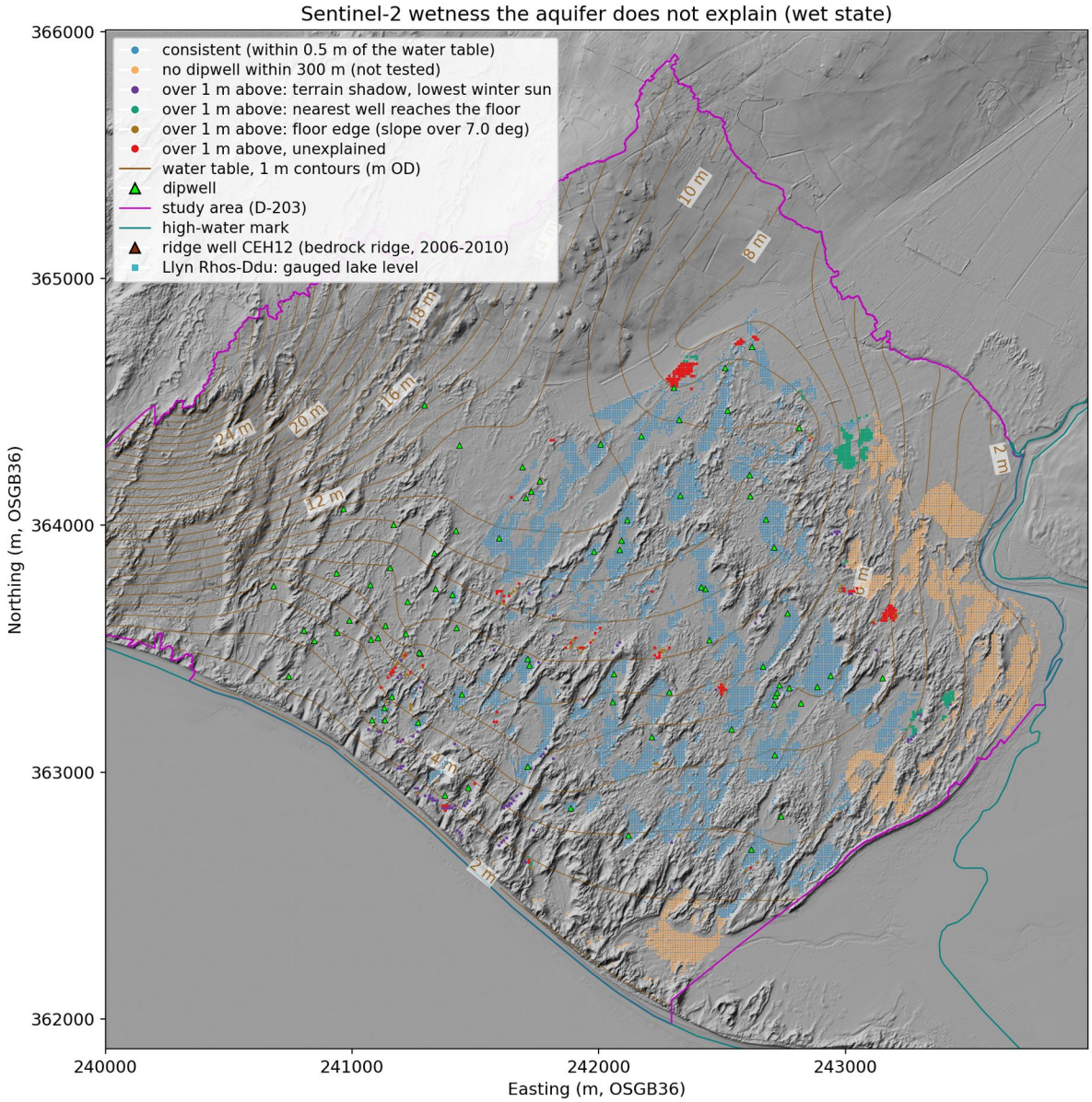



***Figure 52:*** *Sentinel-2 switching levels tested against the kriged water table. Each wetted slack-floor cell is coloured by class: consistent (within 0.5 m of the wet-state surface); untested (no dipwell within 300 m); more than 1.0 m above the surface and explained as terrain shadow at the lowest-sun winter scene, as reached by the nearest well's wet-state level, or as a floor-edge pixel; or unexplained. Cells between 0.5 and 1.0 m are not drawn. Black-edged triangles: dipwells; magenta line: study area; brown lines: wet-state water table, 1 m contours. Generated by 01b_water_table.py. (Source: 01b_04_unexplained_wetness.png; 01b_04_sentinel_check.csv). Sentinel-2 imagery: © Copernicus.*

## 4.9 Spatial Groundwater Analysis

The per-well SSM reparametrization provides a spatially resolved view of aquifer behaviour that complements the cluster-level summaries of Sections 4.2–4.4. This section presents the spatial coefficient structure across the 88-well classified network (66 reference wells with per-well SSM coefficients; 22 extended wells assigned cluster-level coefficients via the

Pearson affinity audit of Section 3.3), the baseline head surface and groundwater flow field, and the SSM water balance residual; the equilibrium scenario projections under prescribed climate and forest management perturbations that these fields feed are presented in Section 4.13. An interactive scenario viewer (Hollingham, 2026) integrates these outputs with the Curreli et al. (2013) SD15b / SD16 community-mean levels and is available as a companion tool to this report.

### 4.9.1 Datum Sensitivity

Per-well drainage datum optimization (Figure 53a), obtained by independently optimizing the displacement datum $z_0$ at each well to maximize SSM fit (Section 3.4), shows that the $R^2$-maximizing datum varies systematically across the site: the cluster medians are 0.8 m at C1 Lake Edge, 0.9 m at C2 Dune, 1.8 m at C3 Western Residual, 2.0 m at C5 Coastal Forest and 3.0 m at C4 Main Forest (network median 1.55 m), and the per-well value correlates with mean depth to water ($r = 0.60$, $n = 66$). The $R^2$ gain from per-well optimization over the uniform 3.7 m datum has a median of 0.013 and exceeds 0.05 at eight wells, reaching 0.080 at C1 and 0.074 at the eastern C2 wells where the effective drainage base is shallowest (Figure 53b); it is below 0.01 at every C4 well and at most 0.030 at C5. The uniform datum is not the fit-optimal choice at the shallow-water-table wells; Section 3.4.1 states its cost and why it is retained.

Per-well SSM coefficients interpolated across the site (Section 3.4.3) reveal spatially variable patterns whose broad structure aligns with the cluster boundaries (Figure 54a–d), although within-cluster heterogeneity is substantial. The three coefficient surfaces are interpolated by piecewise-linear (Delaunay) interpolation to a 50 m grid with dune ridge cells masked where the DEM raster exceeds the interpolated surface by more than 1 m (see Section 3.8). $R^2$ (Figure 54a) is shown at the wells alone: no interpolation between wells predicts a withheld well's $R^2$ better than the network mean, so a surface would draw a pattern the wells do not support.

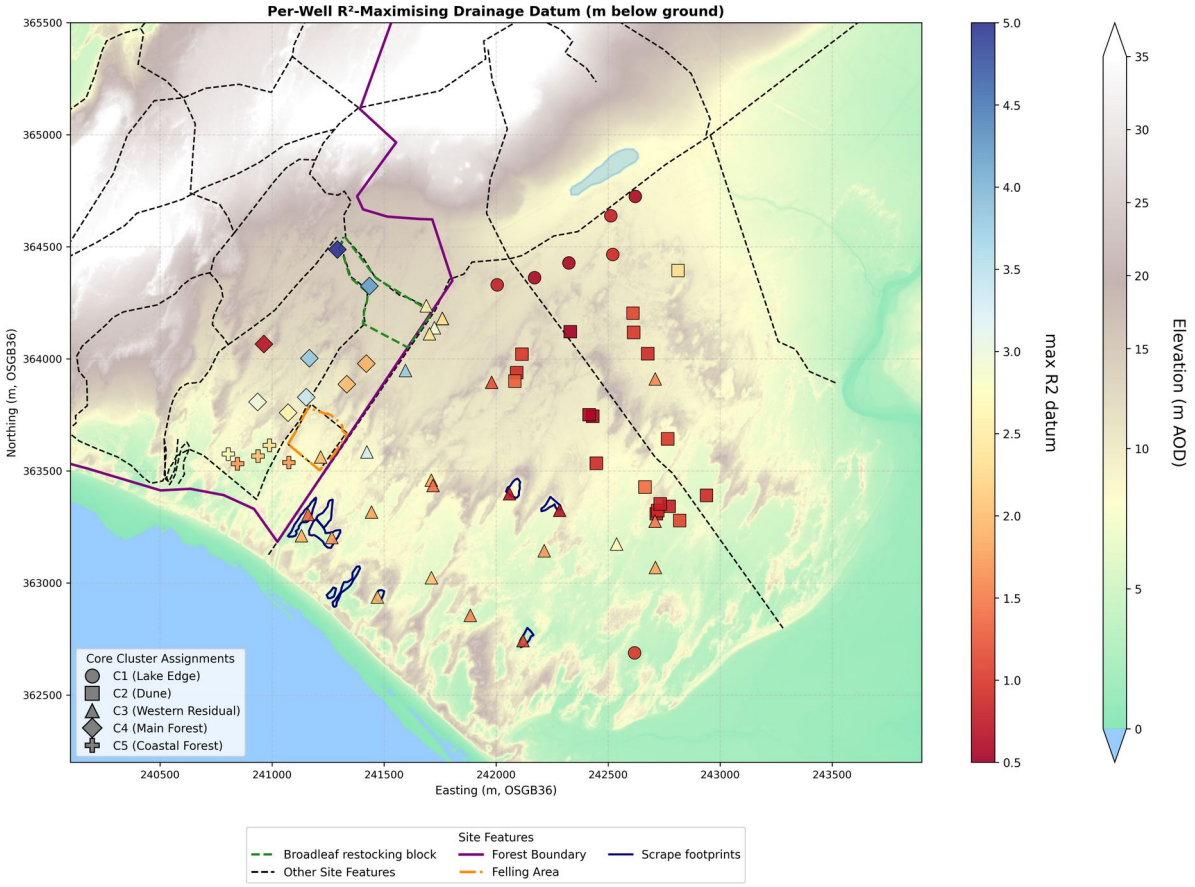



***Figure 53a:*** *Per-well optimal SSM drainage datum across the 66-well reference network, Newborough Warren. Each well's displacement datum $z_0$ was optimized independently by sweeping from 0.5 to 8.0 m in 0.1 m steps and selecting the value maximizing one-step R² (Section 3.4). (a) R²-maximizing datum depth (m below ground surface; the colour scale is capped at 5.0 m). The gradient from shallow at the C1 Lake Edge and eastern dune wells (cluster medians 0.8 and 0.9 m), through the western block (1.8 m), to deepest under the forest (C5 2.0 m, C4 3.0 m, spanning 0.5 to 8.0 m) reflects the depth at which the SSM drainage term best captures lateral discharge at each well location; the per-well values are in 03_09_well_optimal_datums.csv and the cluster summaries in 03_18_datum_invariance.csv. (Source: 03_10_well_datum_r2max_map.png) Well markers over 2 m LiDAR DEM hillshade; no surface is interpolated for this map, with ridge and non-aquifer areas masked. Marker fill shows the value; marker shape shows cluster assignment. LiDAR data: © NRW & OS; see Section 11 for full attribution.*

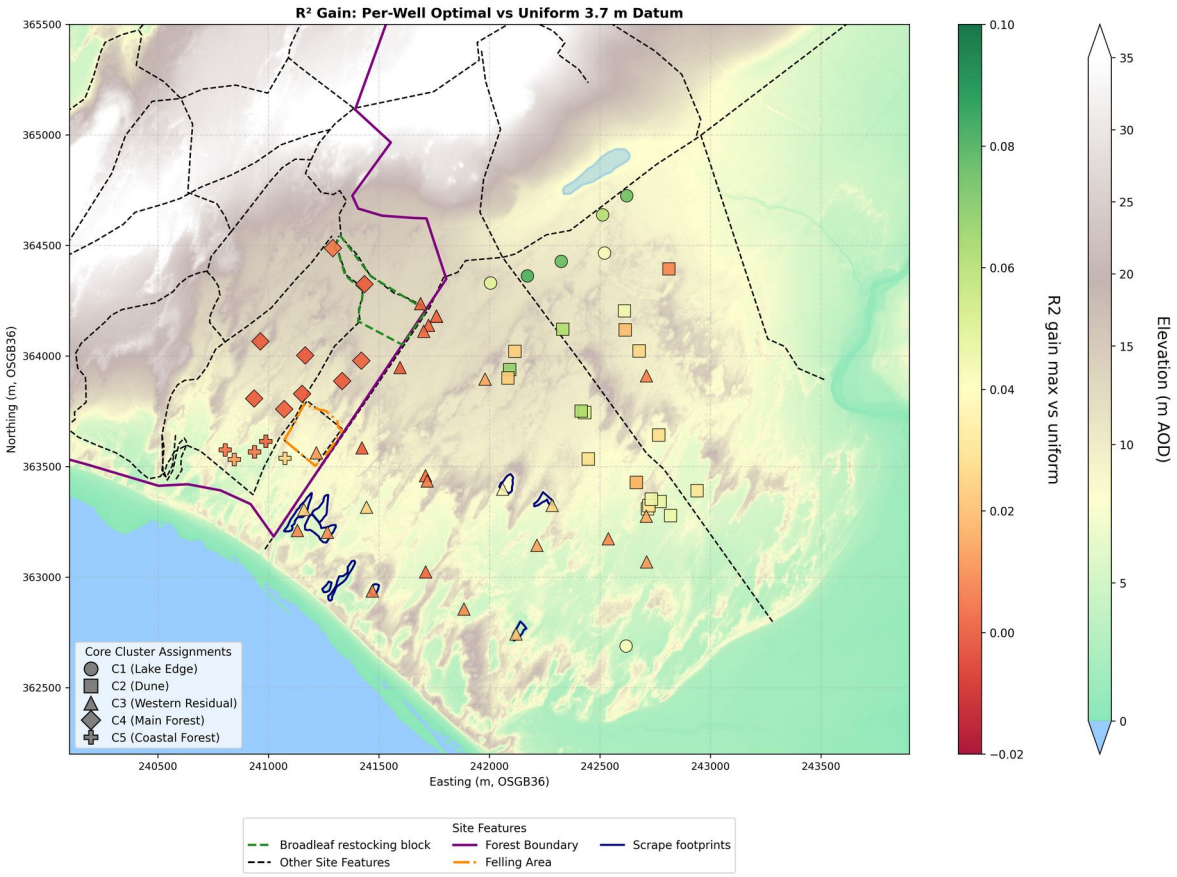



***Figure 53b.*** *Per-well optimal SSM drainage datum across the 66-well reference network, Newborough Warren. Each well's displacement datum $z_0$ was optimized independently by sweeping from 0.5 to 8.0 m in 0.1 m steps and selecting the value maximizing one-step R² (Section 3.4). R² gain from per-well optimal datum vs the uniform 3.7 m network datum. The gain has a median of 0.013 over the network, is below 0.01 at every C4 well and at most 0.030 at C5, and is largest at the C1 Lake Edge (up to 0.080) and eastern dune wells (up to 0.074), where the effective drainage base is shallowest: the cost of the uniform datum is concentrated where the fit-optimal datum is shallowest. Its size per well is in 03_09_well_optimal_datums.csv; the cluster maxima are in 03_18_datum_invariance.csv. (Source: 03_10_well_r2_gain_map.png). Well markers over 2 m LiDAR DEM hillshade; no surface is interpolated for this map, with ridge and non-aquifer areas masked. Marker fill shows the value; marker shape shows cluster assignment. LiDAR data: © NRW & OS; see Section 11 for full attribution.*

### 4.9.2 Spatial Coefficient Structure

The R² map (Figure 54a) shows each well's SSM fit at the project drainage datum, well by well; R² carries no spatial skill, so no surface is interpolated between the wells. The model explains 70–81% of monthly water-table variance at most open-dune wells (10th–90th percentile of per-well R², 52 wells), declining toward the forest (C4 and C5 mean 0.70) and at some peripheral south-eastern wells. Fitting each well at its own optimal datum would raise these values appreciably only at the lake-edge and eastern dune wells (Figure 53b).

The three coefficient surfaces (Figure 54b–d) reveal a coherent spatial story about how rainfall enters, is consumed, and drains from the aquifer. Recharge sensitivity ($\beta_1$) is highest in the north-east around the C1 Lake Edge wells closest to Llyn Rhos-Ddu (>5 mm/mm), declines across the open dune (3–5 mm/mm), and drops to 2–3 mm/mm west of the forest boundary. Atmospheric draw ($\beta_2$) shows a different geography: the highest values occur in two zones — the C4 Main Forest interior and a band across the northern open dune adjacent to the ridge — while C1 Lake Edge wells show notably low values (mean 0.50 mm/mm). C5 Coastal Forest shows moderate $\beta_2$ (mean 1.14), lower than C4 despite also carrying pine canopy. Drainage rate ($\beta_3$, logarithmic scale) shows the strongest spatial contrast of any coefficient: the seven C1 wells, seven C3 wells and two C2 wells drain fastest (>8% of displacement per month), the forest interior shows the lowest rates in the network (C4 mean 1.5%, with individual wells below 1%), and C5 falls between the two (mean 4.7%). One C4 well (CEH14) records a negative $\beta_3$ (−2.4%), although the estimate is not significantly different from zero ($p = 0.14$), and five of the nine C4 wells likewise carry a non-significant $\beta_3$ on the 100-month per-well window from which these surfaces are built. That weakness is largely a property of the window rather than of the cluster: refitting each well on its full record leaves seven of the nine significant, with only CEH13 and CEH14 failing. The mapped C4 coefficients are therefore best read as the least precisely resolved in the network, rather than as evidence that forest-interior drainage is unresolvable.

Within the C3 Western Residual cluster, the coefficient surfaces resolve into a coherent internal gradient aligned with the south-west to north-east axis — the same axis as distance inland from the eroding south-west coast. Against distance from the coast, across the 21 C3 wells that carry one (Script 29), recharge sensitivity $\beta_1$ increases inland ($r = +0.74$, $p < 0.001$), drainage rate $\beta_3$ increases inland ($r = +0.55$, $p = 0.009$), and WTF-derived specific yield declines inland, from 0.40 at the south-west margin (CEH21) to 0.22 in the north-east (D38) ($r = −0.80$, $p < 0.001$). The interpretation of this gradient is developed in Section 5.1.1.

The combined picture is of an aquifer whose eastern margin fills quickly, loses water quickly, and resets each season, while its western and forested interior fills slowly, holds water against evaporative demand, and drains sluggishly — the "shallow pan" versus "deep sponge" contrast identified at cluster level in Section 4.2.

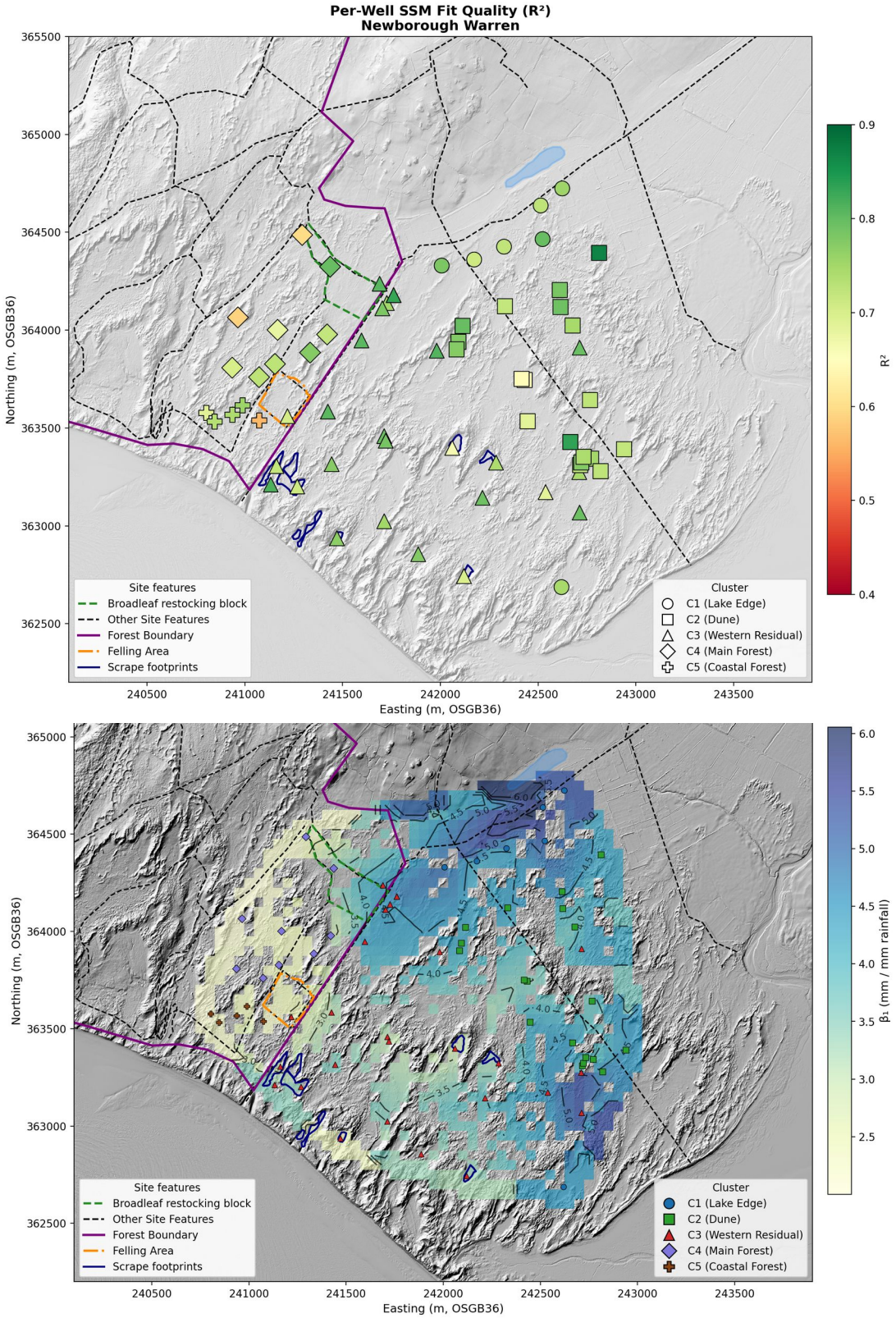



***Figure 54:*** *Spatial SSM coefficient atlas for the 66-well reference network, Newborough Warren. Per-well coefficients interpolated by piecewise-linear (Delaunay) interpolation over 2 m LiDAR DEM hillshade, with ridge and non-aquifer areas masked. Well markers coloured by cluster assignment in panels (b)–(d). (a) R² one-step model fit quality, shown at the wells alone (marker fill = R², marker shape = cluster), since R² has no spatial structure a surface could carry — the SSM explains 64–87% of monthly water table variance across the open dune, declining to 58–79% in the forest interior (C4). (Source: 07_coeff_04_r2_quality.png). (b) $\beta_1$ recharge sensitivity (mm water-table rise per mm rainfall) — highest at the Lake Edge (C1, 4–6 mm/mm), with a sharp drop at the forest boundary consistent with 24% canopy interception. Generated by 07_spatial_coefficients.py. (07_coeff_01–04_*.png) LiDAR data: © NRW & OS; see Section 11 for full attribution.*

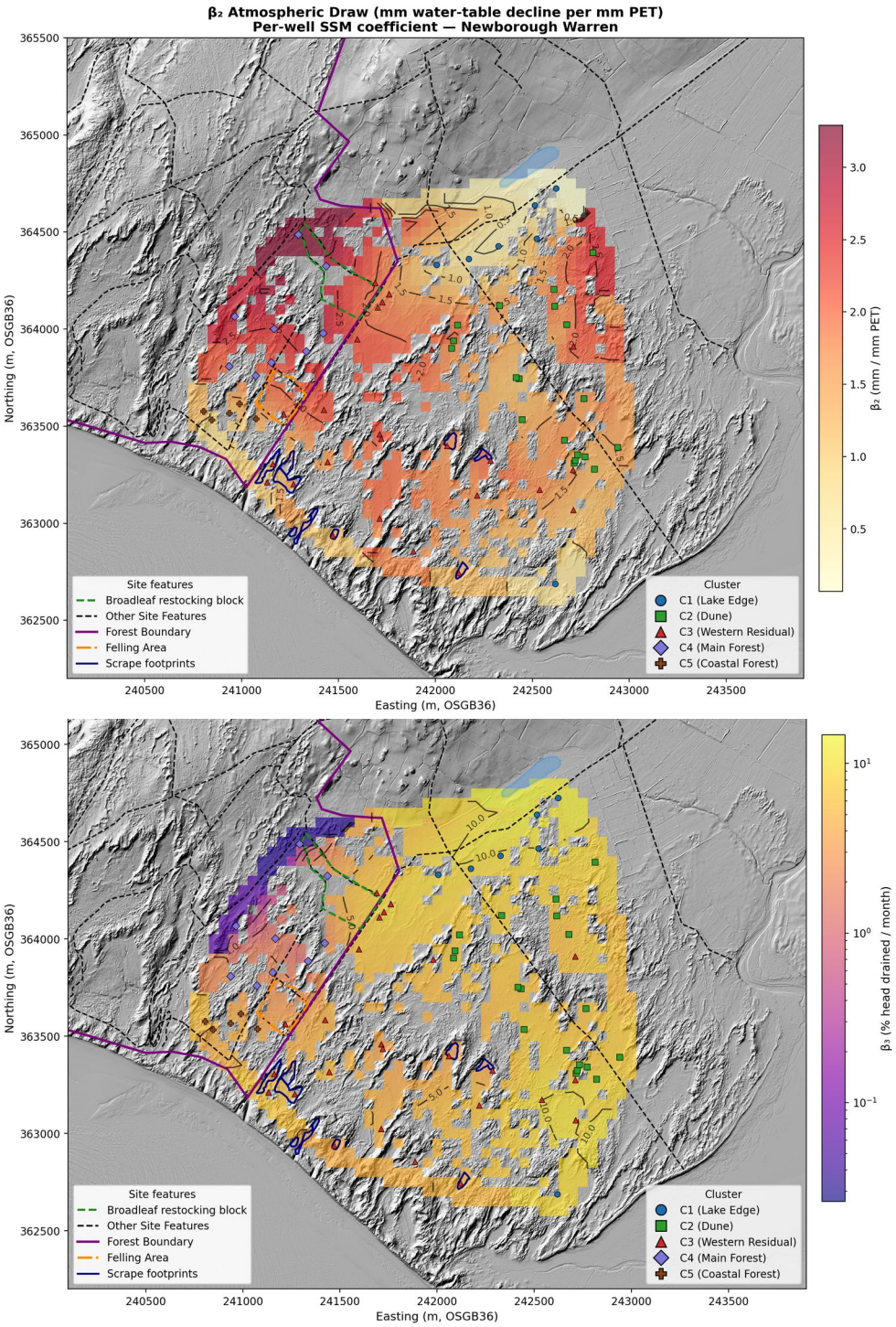



***Figure 54 (continued).*** *Spatial SSM coefficient atlas for the 66-well reference network, Newborough Warren. Per-well coefficients interpolated by piecewise-linear (Delaunay) interpolation over 2 m LiDAR DEM hillshade, with ridge and non-aquifer areas masked. Well markers coloured by cluster assignment. (c) $\beta_2$ atmospheric draw (mm water-table decline per mm PET) — highest in the forest interior (C4, 2.4–3.9 mm/mm), lowest at the Lake Edge (C1) and Coastal Forest (C5). (d) $\beta_3$ drainage rate (% displacement drained per month, logarithmic scale) — a sevenfold contrast in cluster-mean drainage rate, from the fast-draining Lake Edge (C1, 9–13% per month, mean 10%) to the near-stagnant forest interior (C4, mean 1.5% and no well*

*above 3.9% per month). CEH14, the sole well with negative $\beta_3$, sits at the ridge flank. Generated by 07_spatial_coefficients.py. (07_coeff_01–04_*.png) LiDAR data: © NRW & OS; see Section 11 for full attribution.*

### 4.9.3 Drainage Decay Half-Life

The drainage decay half-life $t½ = \ln(2)/\beta_3$ (Figure 55; Section 3.4.5) maps the time for excess groundwater storage above the drainage datum to drain to half its initial value at each reference well. It is derived from the SSM drainage coefficient $\beta_3$ alone, without requiring an independent storage estimate. The eastern open dune shows the shortest mean half-lives in the network: C1 Lake Edge t½ = 5.9–8.9 months (mean 7.8 months), meaning excess storage is largely dissipated within a single season; C2 Dune t½ = 8.3–13.0 months (mean 10.6 months). The western dune shows intermediate mean values over a wide range: C3 Western Residual t½ = 6.0–19.9 months (mean 12.2 months). The forest clusters show markedly longer half-lives: C5 Coastal Forest t½ = 11.3–15.7 months (mean 14.0 months), comparable to C3 despite carrying pine canopy (Section 4.9.4); and C4 Main Forest t½ = 16.8–45.4 months (mean 26.7 months), the longest half-lives in the network. At the upper end of the C4 range, excess groundwater storage persists for some three and a half years before draining to half its initial level. The per-well values are on each well's full record (Section 3.4.5): on the 100-month comparison window the C4 recessions are comparable with the window itself and are not identified there (Section 3.4), so the window values overstate them — 18–74 months, mean 36, at C4. The pronounced C4 hotspot lies on the elevated plateau, where $\beta_3$ is lowest (Section 5.2.3). The wide within-C4 range (17–45 months on the full record) shows the forest block to be hydraulically non-uniform. CEH14 (negative $\beta_3$) and CEH13 (near-zero $\beta_3$) are excluded as ridge-flank wells where $\beta_3$ is not reliably identified; $t½ = \ln(2)/\beta_3$ cannot meaningfully be formed from a negative or near-zero coefficient. Across the reference network Sy and $1/\beta_3$ are only weakly correlated ($r = 0.19$, $p = 0.13$, $n = 64$), and the storage–drainage index $\tau = Sy/\beta_3$ follows the half-life closely ($r = 0.95$ between the per-well values), so the two maps carry much the same pattern.

The model benchmarking diagnostic (Section 4.4; Figure 18) complements the half-life map. The iterative NSE improvement from SSM over TLM (ΔNSE) correlates positively with $\beta_3$ ($r = 0.46$, $p < 0.001$) and negatively with $\beta_2$ ($r = −0.39$, $p = 0.001$). The TLM fails most severely at C1 Lake Edge (median ΔNSE = +0.45, median TLM NSE = +0.31), where $\beta_3$ is highest, and least severely at C4 Main Forest (median ΔNSE = +0.12, median TLM NSE = +0.54), where $\beta_3$ is lowest (Section 5.2.4).

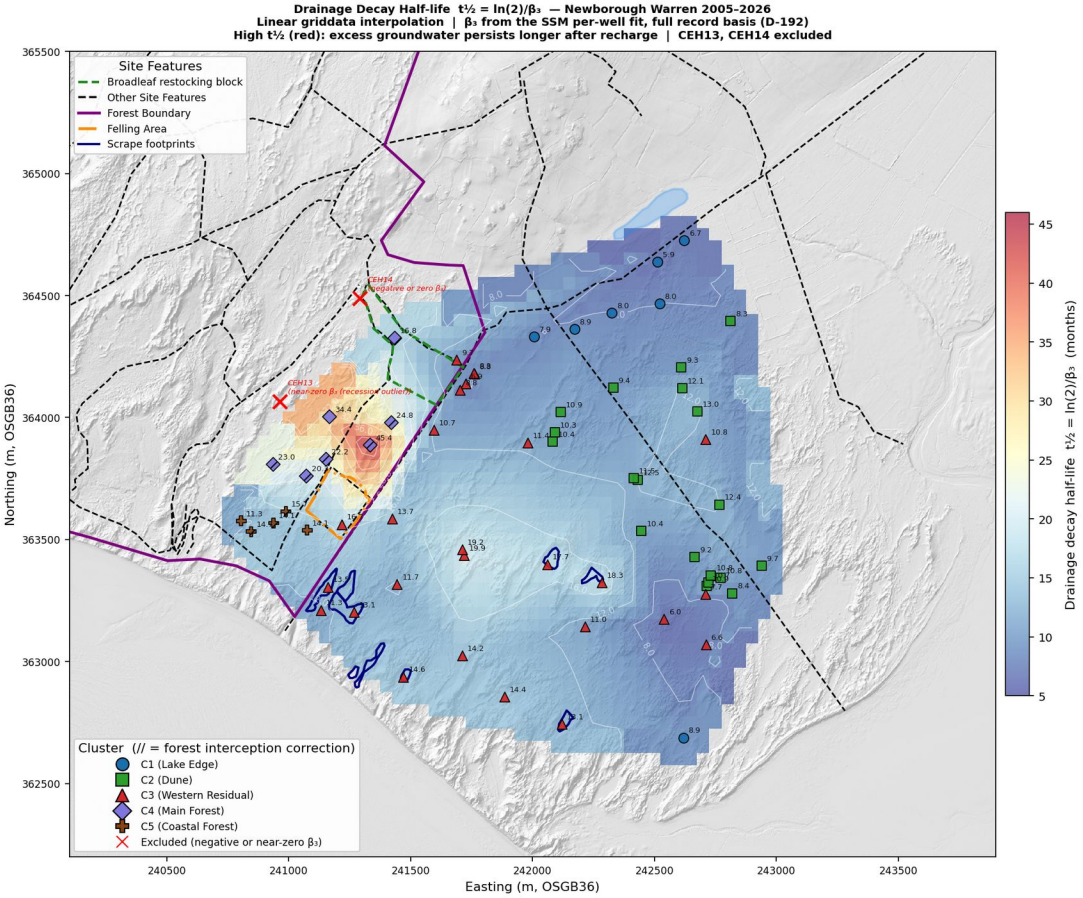



***Figure 55:*** *Model A drainage decay half-life ($t½ = ln(2)/β_3$, months, at the network datum) across the 66-well reference network (CEH 13 and 14 excluded), Newborough Warren 2005–2026. The half-life is the time for excess groundwater storage above the drainage datum (3.7 m below ground level; Section 3.4.1) to drain to half its initial value through natural discharge alone, derived from the per-well SSM drainage coefficient $β_3$ fitted on each well's full record (Section 3.4.5). Low values (blue) indicate rapid aquifer turnover; high values (red) indicate persistent storage following recharge. For formal comparisons, the mathematically equivalent recession e-folding time is $t_R = 1/β_3 = t½ / ln(2)$. Under Model A at the network datum, the eastern open dune contrasts sharply with the forest interior: C1 Lake Edge (t½ = 6–9 months) drains within a season; C4 Main Forest (t½ = 17–45 months, mean 27 months) retains excess storage for months to years, with a pronounced hotspot on the elevated plateau where $β_3$ is lowest. The wide within-C4 range reflects the progressive thinning of the sandy aquifer over the underlying bedrock ridge. Under Model A, C5 Coastal Forest (t½ = 11–16 months) resembles C3 Western Residual (t½ = 6–20 months) despite carrying pine canopy, consistent with an elevation and substrate control (Section 4.9.4). Two wells are excluded: CEH14 (negative $β_3$, t½ undefined) and CEH13 ($β_3$ not significantly different from zero on the full record), both ridge-flank settings where the displacement formulation does not apply; shown as red crosses. Piecewise-linear (Delaunay) interpolation over 2 m LiDAR DEM hillshade, site boundary clipped. Well markers coloured by cluster; hatched markers indicate forest interception correction applied to Sy. LiDAR data: NRW and OS; see Section 11 for full attribution. (Source: 18_wtf_05_halflife_map.png)*

### 4.9.4 Per-Well Forest Zone Analysis

The spatial coefficient atlas (Figure 54b–d) shows that the C4 and C5 forest wells span nearly the full site-wide range of $\beta_2$ values despite sharing the same Corsican pine canopy. To test whether this within-forest variation tracks topographic position (a proxy for the uncored substrate) or canopy differences, per-well SSM coefficients for the 14 forest-zone wells (9 in C4, 5 in C5) were regressed against elevation, distance from the ridge and easting (Section 3.4.4).

Elevation is overwhelmingly the dominant spatial predictor (Table 23). The atmospheric draw coefficient $\beta_2$ shows the strongest relationship ($r = 0.989$, $p < 0.001$; $R^2 = 0.979$; leave-one-out predicted $R^2 = 0.970$, the out-of-sample check for $n = 14$), with higher-elevation wells on the ridge returning systematically higher $\beta_2$ values. Adding distance from the ridge contributes virtually no additional explanatory power ($R^2 = 0.979$). The drainage coefficient $\beta_3$ shows a significant but weaker elevation dependence ($r = -0.836$, $p < 0.001$; $R^2 = 0.699$), with distance from the ridge adding meaningful information (combined $R^2 = 0.784$): wells further from the ridge and at lower elevation drain faster. The recharge coefficient $\beta_1$ shows no significant relationship with elevation ($r = 0.130$, $p = 0.657$); Easting is the best single predictor ($r = 0.529$, $p = 0.052$) but explains only 28% of variance. This elevation–drainage relationship is visible directly in the Model A drainage decay half-life map (Figure 55), where C4 wells reach t½ values of up to 45 months compared with 11–16 months at C5.

C4 and C5 form two clearly distinct groups rather than a continuum. They separate almost entirely on the $\beta_2$ axis: C4 wells cluster at $\beta_2 = 2.1–3.9$ while C5 wells sit at $\beta_2 = 0.8–1.4$, with zero overlap. Mean elevations differ significantly (C4: 10.6 m, C5: 5.6 m; $p < 0.001$), with a clean gap between 6.3 m and 8.6 m containing no wells (Source: 10c_forest_zone_cluster_summary.csv). This corresponds to the transition from the elevated ridge and plateau terrain to the low-lying coastal fringe. Both zones carry Corsican pine canopy; C4 sits on the flank of the bedrock ridge (which outcrops within the forest), C5 on the lower coastal sand (Section 5.6.1).

Two individual wells merit comment. CEH14, at the highest elevation in the forest zone (14.4 m), has the highest $\beta_2$ in the network (3.880), on the elevation regression line. Its negative $\beta_3$ (−0.021) and low $R^2$ (0.596) are not well identified at this ridge-proximal position. The estimate is not significantly different from zero ($p = 0.14$), and five of the nine C4 wells likewise carry a non-significant $\beta_3$ (Section 5.2.3). NW10, the broadleaf block well, has the highest $\beta_1$ in the forest zone (3.481, 2.4 SD above the C4 mean) and the highest $\beta_3$ among C4 wells (0.039). Its reading is in Section 5.6.3.

| Coefficient | vs Elevation r (p) | vs Dist. ridge r (p) | vs Easting r (p) |
|---|---|---|---|
| $\beta_1$ (recharge) | 0.130 (0.657) | −0.452 (0.104) | 0.529 (0.052) |
| $\beta_2$ (atm. draw) | 0.989 (<0.001) | −0.909 (<0.001) | 0.743 (0.002) |
| $\beta_3$ (drainage) | −0.836 (<0.001) | 0.657 (0.011) | −0.487 (0.077) |

***Table 23:*** *Per-well spatial predictors of SSM coefficient variation within the forest zone (C4 Main Forest and C5 Coastal Forest, n = 14). Pearson correlation coefficients and p-values for three spatial variables. Elevation dominates $\beta_2$ ($R^2$ = 0.979); $\beta_1$ shows no significant elevation dependence. (Source: 10c_forest_zone_correlations.csv).*

The per-well analysis places the C4/C5 partition of the Ward's solution at an elevation discontinuity; its mechanistic interpretation — and its implications for forest management — is developed in Section 5.6.1.

### 4.9.5 Mean Water Table Surface and Groundwater Flow Direction

The water table was kriged from the dipwells in a wet, a mean and a dry state of the network (Figure 56; Section 3.8). The drift test chose ordinary kriging, with no ground-surface drift. Left out one at a time, the 83 wells are predicted with a median absolute error of 0.14 m in the mean state and almost no bias (−0.005 m). One well in ten misses by more than 0.68 m. The largest miss is CEH15, at the forest-slack edge (2.26 m). The coastal head fitted to the coastal wells is 1.87 m OD, at the top of its range and within 0.01 m of mean high water. All three landward boundary parts were kept: the Afon Braint, Llyn Rhos-Ddu and CEH12. Without CEH12, the surface under-predicts the ridge well by 25 m. The surface is a single mound. It stands highest at CEH14, 14.1 m OD in the wet state, beside the bedrock ridge. It falls to the coastal head along Caernarfon Bay and to the river and lake levels on the landward side. Flow follows the mound outward: west of the network the slacks drain south and south-west to Caernarfon Bay; in the centre mainly south, but a northern group drains north and north-east towards Llyn Rhos-Ddu; in the east, east and south-east towards the Afon Braint. The pattern hardly changes with the state of the aquifer. The median gradient falls from 0.0071 in the wet state to 0.0062 in the dry. The median slack turns 4.7° between the two states, and only 7 of 116 slacks turn more than 20°. Of the choices tested, only the drift moves the pattern. Setting the coastal head anywhere from 0 m to mean high water changes the median direction by less than 1°. Removing every landward boundary changes it by 2°. The runner-up drift, the DEM smoothed to 300 m, changes it by 19° and turns a fifth of the slacks.

The transects (Figure 58) cut the mound along four coast-normal lines and a fan of radials from CEH14. Together they carry 45 of the 83 wells, and they show the water table in all three states against the ground. In the south-west, the wet-to-dry range narrows towards the sea: 0.70 m at CEH22, 145 m from the high-water mark, against 1.42 m at CEH14, 1.4 km

inland (Spearman ρ = 0.80; ρ = 0.62 with ground height held constant). No such damping is seen in the east, where the nearest well stands 524 m from the shore.

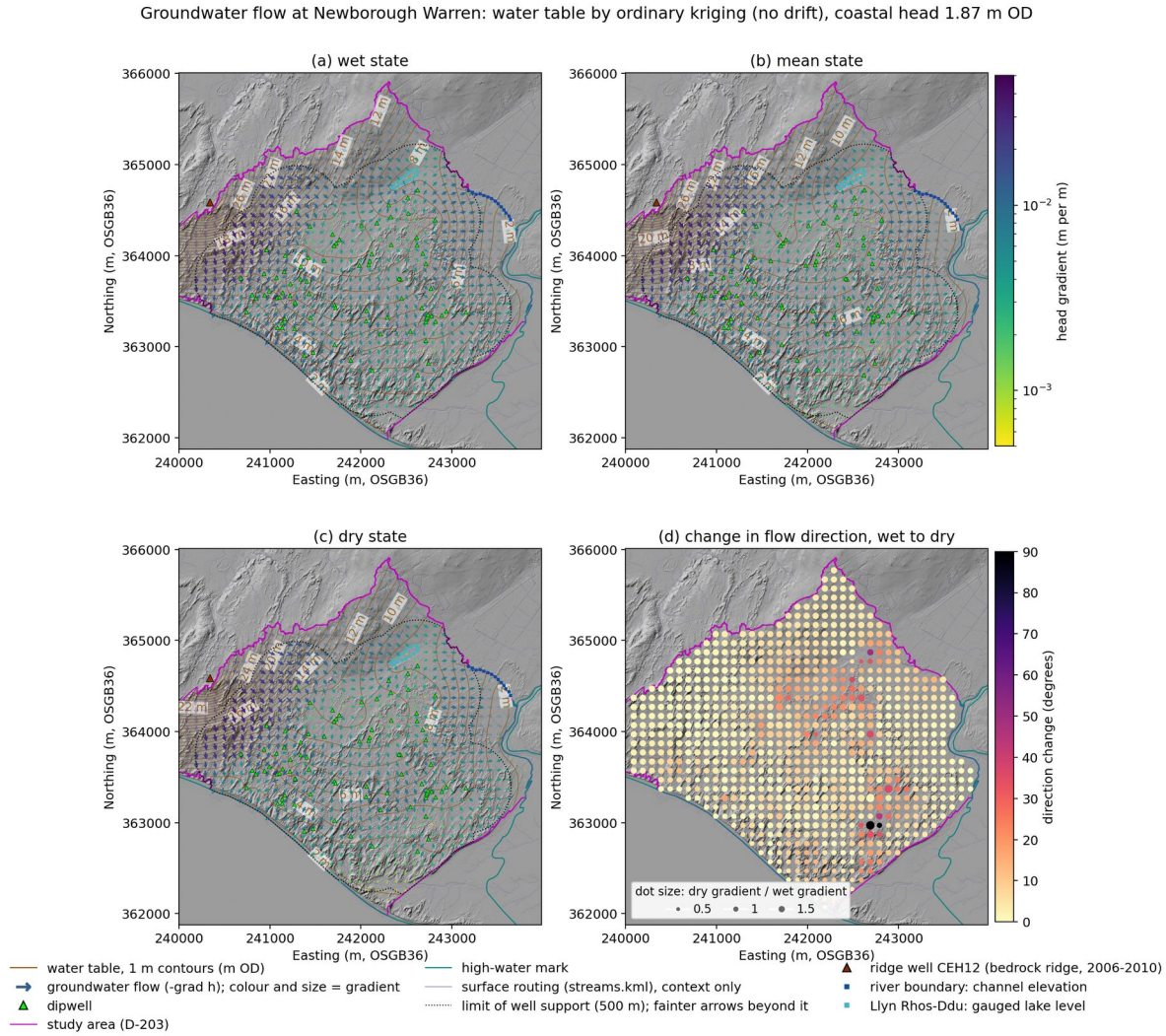



***Figure 56:*** *The water table of Newborough Warren kriged from the dipwells in (a) the wet state (the wettest tenth of months by the network-median level), (b) the mean state (all months) and (c) the dry state (the driest tenth), with 1 m contours (m OD). Ordinary kriging with a spherical variogram, chosen by leave-one-well-out error and a physical test over three ground-surface drifts; sea anchors along the high-water mark at the fitted coastal head (1.87 m OD), and the Afon Braint at channel elevation, Llyn Rhos-Ddu at its gauged level and the ridge well CEH12 at its measured depth as landward boundaries. Arrows are the negative head gradient, one per 100 m square, sized and coloured by the gradient; arrows more than 500 m from a dipwell (dotted limit) are faint, because there the boundaries rather than the wells set them. No hydraulic conductivity or aquifer thickness enters the arrows. (d) The change in flow direction from the wet to the dry state; dot size is the ratio of dry to wet gradient. Surface routing (streams.kml) is drawn faintly as context only. Background: LiDAR hillshade (LiDAR data: © NRW & OS; see Section 11 for full attribution). Generated by 01b_water_table.py. (Source: 01b_01_water_table_flow.png; 01b_report_numbers.csv)*

How closely the water table follows the ground, and at what scale, is measured directly (Figure 57). Regressed on the DEM at native resolution over the 7,939 grid nodes (20 m) inside the reference-network hull, the mean surface has slope 0.66 and r = 0.76. Smoothed to 300 m, the ground gives slope 1.01 and r = 0.92; 300 m is the first width at which the slope

reaches 1. The water table is therefore a subdued replica of the topography. It follows the dune mound almost 1:1, but not the ridge-and-slack relief finer than about 300 m. That scale is also near the finest relief the network can resolve (median nearest-well spacing 123 m), so the measurement bounds it from above. At the 66 reference wells, the mean water table tracks the surveyed ground at slope 0.98 (r = 0.99). This is a different relation. Dipwells stand on slack floors, and slack floors are deflation surfaces cut down to the water table (Ranwell, 1959). At the wells, therefore, the ground follows the water table rather than the reverse.

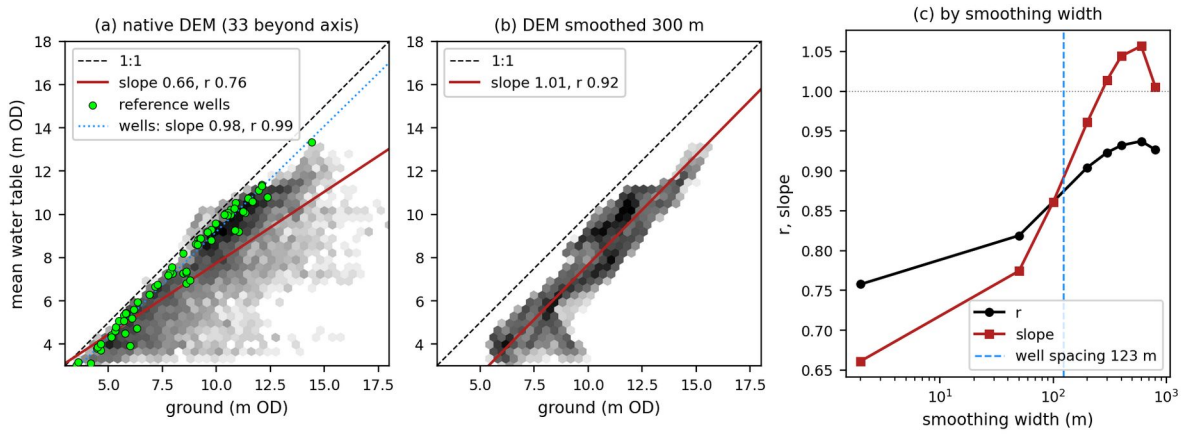



***Figure 57:*** *Mean water table against the ground surface, by DEM smoothing width. (a) The kriged mean-state water table (Figure 56b) against the LiDAR DEM at native resolution, at the 7,939 20 m grid nodes inside the convex hull of the reference network (hexagonal bins, log count; 33 nodes lie beyond the axis limit and are included in the fit); the dashed line is 1:1 and the red line the fitted relation (slope 0.66, r = 0.76). Points are the 66 reference wells, mean water table against surveyed ground, with their own fit (dotted; slope 0.98, r = 0.99): the slack floors on which the wells stand are cut to the water table, so this is the slack-formation relation, not topographic control. (b) The same surface against the DEM smoothed by a 300 m mean filter (slope 1.01, r = 0.92). (c) r and slope against smoothing width; the dashed vertical line is the median nearest-well spacing (123 m), the finest relief the surface can carry. Generated by 01b_water_table.py. (Source: 01b_06_head_vs_dem.png; 01b_14_head_vs_dem.csv; 01b_report_numbers.csv)*

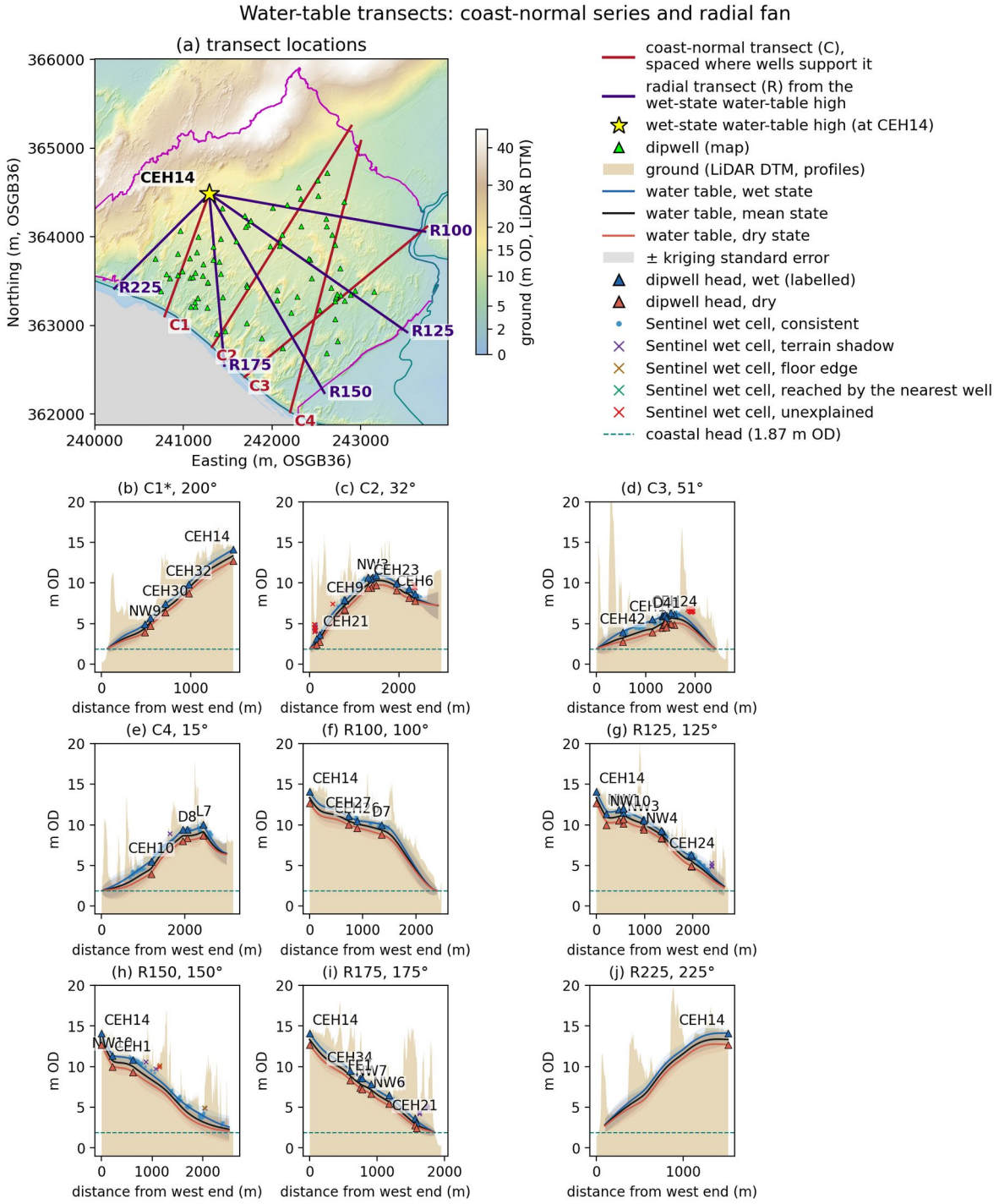



***Figure 58:*** *Water-table transects across Newborough Warren. (a) Four coast-normal transects (C1–C4, red), spaced where the dipwells support them, and a fan of radials (R, purple) from the wet-state water-table high at CEH14; C1 merges one radial (C1*). (b–j) Profiles along each line, measured from its west end: ground (LiDAR DTM, shaded), the kriged water table in the wet (blue), mean (black) and dry (red) states with ± one kriging standard error, dipwell heads in the wet and dry states (triangles), Sentinel-2 wet cells beside the line by class, and the fitted coastal head (1.87 m OD, dashed). 45 of the 83 wells lie on a transect. Generated by 01b_water_table.py. (Source: 01b_02_transects.png; 01b_06_transects.csv; 01b_07_transect_profiles.csv)*

### 4.9.6 Water Balance Residual Field

The SSM water balance residual field (Figure 59; Section 3.8.1) quantifies the degree to which the three-term SSM closes the long-term water balance at each well. The residual at each reference well is computed as $\alpha_B = (\beta_2 \cdot \overline{PET} + \beta_3 \cdot \bar{h}_disp) - \beta_1 \cdot \bar{P}$, where overbars denote long-term monthly means, $\bar{h}_disp$ is mean displacement above the 3.7 m drainage datum, and $\bar{P}$ is gross rainfall. The averaging window is taken from the mean-head record itself (March 2005 to February 2026, n = 252 months; $\bar{P}$ = 74.11 mm/month, $\overline{PET}$ = 54.39 mm/month), so that the climate means and the mean heads describe the same period.

Gross rather than interception-corrected rainfall is used at forest wells. Script 03 fits the SSM on gross rainfall and above-canopy Thornthwaite PET, so the canopy interception loss is already carried inside the fitted $\beta_2 \cdot \overline{PET}$ term; reducing $\bar{P}$ as well would subtract it twice. Interception is a partition of the atmospheric energy budget rather than a term additive to it (Section 3.8).

Positive values indicate locations where modelled losses exceed modelled recharge, implying an unmodelled input; negative values indicate the converse. The field closes tightly and carries no coherent spatial structure, with a small systematic negative offset. Residuals span −0.0079 to +0.0089 m/month about a median of −0.0040 m/month (mean −0.0026), with 66 of the 66 reference wells falling within ±0.01 m/month and 52 negative. Four of the five cluster medians are negative, from −0.0065 m/month at C4 Main Forest to −0.0010 m/month at C5 Coastal Forest, and C1 Lake Edge is slightly positive (+0.0006 m/month), all small relative to the coefficient uncertainties that generate them. No DEM-height mask is applied. Residual magnitude is uncorrelated with position on either axis (Spearman $\rho$ = −0.20 against Easting, p = 0.10; $\rho$ = +0.14 against Northing, p = 0.28). The signed residual shows no gradient on either axis ($\rho$ = +0.10 against Easting, p = 0.42; $\rho$ = −0.09 against Northing, p = 0.49).

No well exceeds +0.02 m/month, and only fifteen of the sixty-six carry a positive residual at all. The largest positive values are NW3 (+0.0089) in C2 Dune, CEH23 (+0.0053) in C1 Lake Edge and NW6 (+0.0058) in C3 Western Residual, with CEH28 (+0.0036) in C2 and NW7 (+0.0027) in C3 next — all in the open dune or at the lake edge rather than at the ridge. No C4 Main Forest well is positive. The largest negative residual occurs at the ridge-flank well CEH14 (−0.0105 m/month), with CEH32 (−0.0079 m/month) and NW11 (−0.0071 m/month) next; at these, modelled recharge marginally exceeds modelled losses.

The residual field is read in Section 5.2.1. Across all five clusters the three-term SSM closes the long-term water balance without requiring an additional input term, and the corrected field agrees with the independent water balance of Script 16, which computes the same closure on gross rainfall over each cluster's fitted months and agrees to within ±6.7 mm/month at every cluster, the largest discrepancy falling at C4 Main Forest. The kriged water table (Section 4.9.5, Figure 56) is a single mound with flow running outward from the high point beside the bedrock ridge — south and south-west to Caernarfon Bay across the

west and centre, north to Llyn Rhos-Ddu from the slacks beside the lake, and east and south-east to the Afon Braint in the east; it describes the site's flow geometry and is not evidence bearing on the residual field, which shows no ridge-proximal signal.

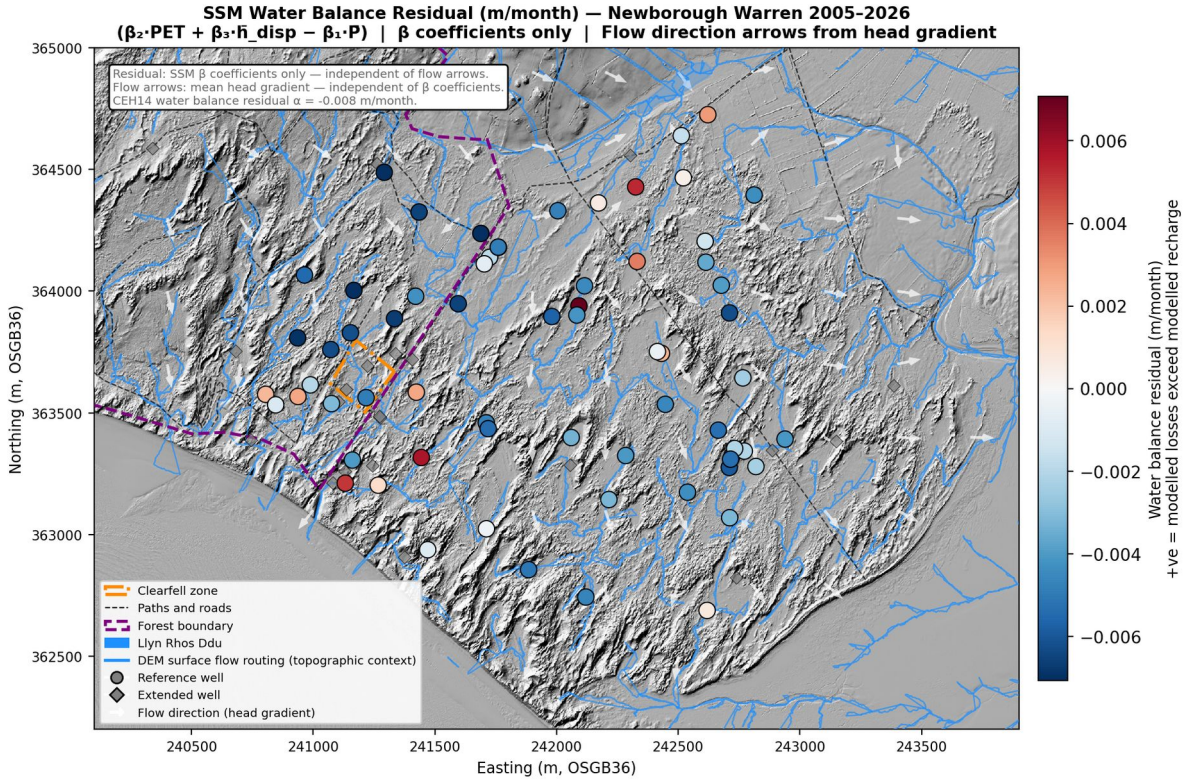



***Figure 59**: SSM water balance residual across the Newborough Warren dune aquifer, 2005–2026. The residual at each reference well is computed from β coefficients only as ($\beta_2 \cdot PE\bar{T} + \beta_3 \cdot \bar{h}_disp) - \beta_1 \cdot \bar{P}$, using gross rainfall, and shown at the wells alone: no interpolation between wells predicts a withheld well's residual better than the network mean, so no surface is drawn. Climate means are taken over the same period as the mean heads (March 2005 to February 2026, n = 252 months). Positive values (red) indicate locations where modelled losses exceed modelled recharge; negative values (blue) the converse. The field closes tightly and shows no coherent spatial structure, with a small systematic negative offset: residuals span −0.0077 to +0.0089 m/month about a median of −0.0040 m/month, with 66 of the 66 reference wells within ±0.01 m/month and 52 negative. No well exceeds +0.02 m/month; the largest positive value is NW3 (+0.0089 m/month) in C2 Dune. The largest negative residual is at the ridge-flank well CEH14 (−0.0105 m/month), with CEH32 (−0.0079 m/month) next. White arrows show normalized Darcy flow direction vectors derived from the mean head gradient field (independent of the β coefficient residual calculation). Well symbols are coloured by residual value; circles indicate reference wells, diamonds indicate extended wells. Background: LiDAR greyscale hillshade. (LiDAR data: © NRW & OS; see Section 11 for full attribution.) Generated by 20_spatial_figures.py (Source: 20_residual_ssm.png).*

## 4.10 Coastal erosion and the cross-shore section

The spatial analysis of Section 4.9 asks how quantities vary in plan view, over easting and northing, and answers by interpolation. This section asks a different question of the same network: how does change vary along a single coordinate, the perpendicular distance to the eroding shoreline, and it answers by fitting a physical decay law. Same wells, same records,

different geometry. The change of coordinate is what makes a far field definable at all, and with it the identification problem set out below. This is the boundary-referenced view of the network, not a transect cut through the maps of Section 4.9.

### 4.10.1 The cross-shore coordinate and what it replaces

Every quantity in this section is indexed by d, the perpendicular distance from a well to the eroding Caernarfon Bay high-water mark. The coordinate is computed within the pipeline from a committed OpenStreetMap mean-high-water polyline (EPSG:27700), restricted to the west-facing Newborough frontage; the Menai Strait and Llanddwyn Island are excluded as non-eroding boundaries, so a well's distance is to the shoreline that is actually retreating and not to the nearest water.

That restriction is the substantive part of the definition. An unrestricted distance-to-water would place the eastern lake-edge wells within a few hundred metres of a boundary, and the gradient fitted here would be a fit to the wrong coastline.

The coordinate replaces easting as the spatial variable for questions about coastal forcing. Easting serves that role in the BACI design of Section 4.6, where it absorbs west-to-east divergence between treatment and control wells, and its coefficient is not a measurement of coastal retreat (Section 4.6.3). The reason is visible once the coordinate is stated: easting is a proxy for distance to the shore only to the extent that the shoreline runs north–south, and within the fifteen wells of the clearfell panel that correspondence fails entirely. Distance to the eroding boundary is the variable the physics is written in; easting is a nuisance covariate that happens to correlate with it across the network and not within a compartment of it.

### 4.10.2 Network-scale partition of the summer-minimum decline

The third candidate mechanism for the C5 anomaly — progressive coastal boundary retreat — can be tested quantitatively at network scale. A non-linear panel regression was fitted to the full 21-year monthly record (11,982 observations across 67 wells; FE wells, the scraped well CEH36, the BACI clearfell-zone wells and the bedrock-ridge well CEH12 excluded) to test whether the C5 anomaly is quantitatively consistent with coastal boundary retreat operating site-wide, against the cross-shore distance d of Section 4.10.1. The headline fit is the forest-free panel (the wells under canopy on the land-cover flag dropped; 61 wells, 10,909 observations), giving a Dupuit–Forchheimer steady-state strip-aquifer form. The headline coast-edge trend is quoted at the 150 m reference distance, which sits inside the observed range of well positions: −26.4 ± 3.5 mm yr⁻¹ (95 % CI −33.2 to −19.5, well basis). The fitted decay amplitude at the shoreline, $\delta_0$ = −31.3 ± 4.8 mm yr⁻¹, is reported as a model parameter and not as the headline: no well sits at the shoreline, so d = 0 lies beyond the network and is the one distance at which the linear-capped and exponential forms disagree materially — a spread of 9.3 mm yr⁻¹ there against 1.6 mm yr⁻¹ at 150 m. The two forms also imply inland reaches of 902 ± 144 m for the linear-capped strip and a decay length of 498 ± 225 m for the exponential, which the record does not separate firmly. The amplitude

carries one well's leverage: withholding CEH3, the second-nearest well to the shore at 176 m, moves $\delta_0$ by 3.1 mm yr$^{-1}$ and the 150 m rate by 2.4 mm yr$^{-1}$, 1.9 times the next-largest single-well shift (each well withheld in turn over the 61-well panel; 25_16_delta0_leave_one_out.csv). Uncertainties on the coastal gradient are quoted on the well basis: the standard error is estimated from those refits, because consecutive monthly readings within a well are not independent observations, and the fitted standard errors from the monthly rows remain in 25_01_panel_fit_parameters.csv as the nominal precision of the model and are not quoted as uncertainties. On that basis $\delta_0$ is more than six standard errors from zero, and the leverage of CEH3 lies inside the error — a property of a network with few wells inside 300 m rather than an instability of the fit. The fit gives an inland reach L_cg of about 900 m (± 144 m, well basis) beyond which the gradient effect vanishes, and a far-field asymptote c_far of −0.3 ± 1.5 mm yr$^{-1}$ (well basis), which is not a climate rate and is not quoted as one (Section 4.10.3). Model selection by AIC favours the exponential form ($\Delta$AIC = −6.0); the linear-with-cutoff form is retained on physical grounds, as the Dupuit–Forchheimer steady-state solution for a strip aquifer with a fixed inland reach, and because the exponential’s far-field asymptote takes the opposite sign to the observed far-field trend — not on the AIC margin, which points the other way. The fit is robust to forest-cover exclusion: it is itself the forest-free panel, and the full-network and C3-only fits shift no parameter beyond sampling uncertainty (Script 25; Figure 60).

Applied to C5's four wells, each at its own distance from the shore, and averaged, the model predicts a gradient contribution of −17.1 mm yr$^{-1}$; all four lie within the fitted reach, so this equals the gradient at the cluster’s mean distance. The decomposition is computed against a balanced observed basis — the slope of the annual cross-well mean of the per-well summer minima, −32.0 mm yr$^{-1}$ — rather than against the fixed-effects cluster slope of −29.1 mm yr$^{-1}$, because a mean taken over per-well fits reports the composition of record lengths as though it were a rate: within C5 the two wells with twenty years of record average −32.4 mm yr$^{-1}$ while the two ten- and eleven-year records average −25.3. On that basis the gradient accounts for about half of C5's decline (53 %), leaving −17.6 mm yr$^{-1}$ unexplained. The attribution is sensitive to which wells the cluster contains: before the current partition, when the extended well NW8B was assigned to C3 rather than C5, the same decomposition left −0.7 mm yr$^{-1}$ unexplained. The panel's climate term is +2.9 mm yr$^{-1}$, acting against the decline rather than adding to it, and the far-field offset is −0.3 mm yr$^{-1}$; the two are not separately identified and only their sum, +2.6 mm yr$^{-1}$, is recovered. This attribution should be read against the sampling base of the forest-free fit. The forest-free panel that anchors $\delta_0$ and L_cg excludes most C4 and C5 wells; of the C5 wells, only three of the five reference wells and the extended well NW8B meet the per-well summer-minimum slope criterion that enters the cluster’s coastal term. The observed decline it is set against is the balanced annual mean over the same four wells (Table 24), so the direction of the C5 result is clear — C5 is the most coastal-affected cluster on this fit — but the precise magnitude remains sensitive to which C5 wells enter the forest-free filter. The coastal term is evaluated at each well’s own

distance and averaged over the cluster, so a cluster that straddles the fitted inland reach L_cg carries the share of its wells inside it. C2 has no well inside the reach and carries no coastal term, and C1 has one (CEH11, 783 m) and carries −0.6 mm yr⁻¹. C3's mean distance, 906 m, falls just beyond the 902 m reach, but eleven of its twenty-two wells lie inside it, including the four nearest the shore, and it carries −7.0 mm yr⁻¹, most of its balanced decline (93 %; 87–102 % across delete-one-well refits of the headline gradient, which move $\delta_0$ and L_cg together); C4 has three of its four wells inside the reach and carries −4.8 mm yr⁻¹, 74 % (58–91 %) of a decline the record does not resolve. The model leaves more than the whole of the observed decline unexplained at C1 and C2, about two-fifths of it at C3 (−3.2 mm yr⁻¹), two-thirds at C4 (−4.3 mm yr⁻¹) and about half at C5 — and what is left is not the same at each. Taking the balanced observed decline less the modelled coastal gradient, the three open-dune clusters return −10.4 (C1), −14.6 (C2) and −0.5 mm yr⁻¹ (C3), across seven, twenty-two and twenty-two wells on different substrates and at different distances from the sea: a mean of −8.5 mm yr⁻¹ with a spread of 14.1 mm yr⁻¹. The independent site-mean trend of Section 5.7.5 gives −13.0 mm yr⁻¹ over the same record. At that spread, with the residual largest at C2 and close to zero at C3, the decomposition indicates that the residual decline is not spatially uniform, and it does not resolve its rate either, because the clusters share the same year-to-year weather and their errors are therefore common-mode rather than independent, leaving −8.5 inside the ±28.5 mm yr⁻¹ minimum detectable rate. The mean is carried forward as the unexplained residual decline, applied uniformly by assumption, a central estimate rather than a climate rate (Section 5.7.5). The per-cluster decomposition is tabulated in Table 24 (and visualized in Figure 60) and itemizes each cluster's observed decline on the balanced basis, partitioned into the climate and far-field term, the coastal-retreat gradient contribution averaged over the cluster’s wells at their own distances (not at the cluster's mean coast-proximity), and the unattributed residual. The weighing of this result against the other candidate mechanisms for the C5 anomaly is in Section 5.7.2.

| Cluster | n in fit | Mean dist. to coast | Observed, balanced (mm/yr) | Climate + far-field (mm/yr) | Coastal (mm/yr) | Unexplained (mm/yr) |
|---|---|---|---|---|---|---|
| C1 (Lake Edge) | 7 | 1819 m | −10.97 | +2.65 | −0.59 | −13.03 |
| C2 (Dune) | 22 | 1565 m | −14.56 | +2.65 | 0.00 | −17.21 |
| C3 (Western Residual) | 22 | 906 m | −7.53 | +2.65 | −7.03 | −3.15 |
| C4 (Main Forest) | 4 | 889 m | −6.51 | +2.65 | −4.82 | −4.34 |
| C5 (Coastal Forest) | 4 | 410 m | −31.98 | +2.65 | −17.07 | −17.56 |

***Table 24:*** *Per-cluster decomposition of the observed summer-minimum decline under the forest-free linear-capped panel regression. The observed column is the balanced basis — the slope of the annual cross-well mean*

*of the per-well summer minima — and not the cluster-centroid slope, because a mean taken over per-well fits reports the composition of record lengths as though it were a rate. The climate + far-field column is the identified sum of the panel's cumulative-water-balance trend contribution and its fitted constant, which trade off against one another and are not separately identified. The coastal term is the mean over the cluster's wells of $\delta_0 \cdot \max(0, 1 - d_i/L_cg)$, each evaluated at the well's own perpendicular distance $d_i$ to the eroding Caernarfon Bay shoreline, so a cluster that straddles the reach carries the share of its wells inside it; the distance column is the cluster's mean, given for orientation. Unexplained is the balanced basis minus the modelled total. All slopes in mm yr⁻¹. Negative = declining water table. The model under-explains the decline at every cluster, by 3 to 18 mm yr⁻¹: C2 has no well within the fitted reach and carries no coastal term, C1 has one (CEH11), C3 eleven of twenty-two and C4 three of four, and at C5, all four of whose wells lie within it and which lies deepest in the gradient zone, the coastal term accounts for about half of the observed decline. (Source: 25_03_cluster_partition.csv).*

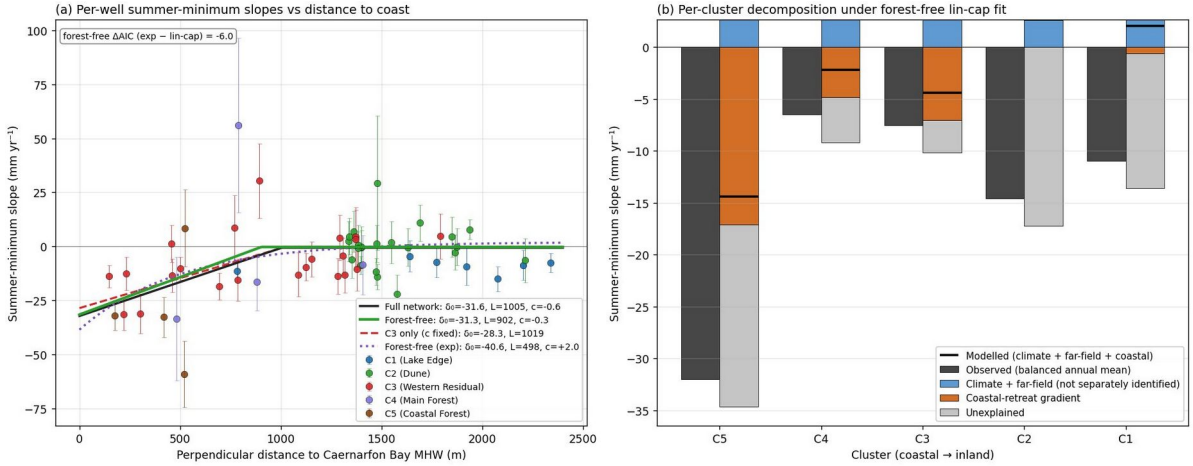



***Figure 60:*** *Network-scale coastal-retreat gradient. (a) Per-well summer-minimum slope (mm yr⁻¹, error bars = ±1 SE) plotted against perpendicular distance to the Caernarfon Bay High Water Mark, with three linear-with-cutoff fits overlaid: full-network panel (black), forest-free panel after dropping the wells under canopy (green, headline), and C3-only with climate background held to the forest-free value (red, dashed), and a fourth, exponential, forest-free fit (purple, dotted) as a comparator — the boxed ΔAIC is the exponential form's AIC less the linear-with-cutoff form's on the same panel (negative favours the exponential; the linear-with-cutoff form is retained on physical grounds, as set out in the text). The three linear-with-cutoff fits are statistically indistinguishable on δ_0 and L_cg. C5 wells (brown) sit closest to the coast; C2 lies wholly beyond the fitted reach L_cg ≈ 900 m; C1 has one well inside it, C3 half its wells and C4 three of its four. (b) The observed summer-minimum slope of each cluster (dark bar, balanced annual mean; clusters ordered coastal to inland) beside its decomposition under the forest-free fit: the coastal-retreat gradient term at the cluster's wells, each at its own distance to the coast, averaged (orange), the climate and far-field term (blue — their sum is identified, the parts are not; Section 4.10.3) and the unexplained remainder (grey). Positive terms stack upward from zero and negative terms downward, so each is drawn at its own size; the black tick is the modelled total, their algebraic sum. (Source: 25_05_fit_diagnostic.jpg)*

### 4.10.3 What the fit cannot resolve: the far-field level

The decay model carries a third parameter, the constant c_far, to which the profile decays inland. On the committed fit it is −0.30 mm yr⁻¹ (SE 1.46, well basis) — small, and indistinguishable from zero. That value should not be read as a measurement of the site-wide

background rate, and the reason is worth setting out, because it is the same limitation Section 5.7.5 reaches from a different direction.

The constant is separately identified but not stably estimated. Within any one fitting window it is cleanly separable from the cumulative-water-balance covariate the panel uses to absorb spatially uniform climate forcing: its variance inflation factor against that covariate is 1.01 after within-well and month demeaning. Before those fixed effects are removed the same two covariates are very nearly collinear (variance inflation about 149), because a cumulative sum of a near-constant-mean series is almost a straight line in time; the small figure is a property of the demeaned design and should not be quoted without it. What moves the constant is the choice of window. Holding the well set (61 wells, 60 from August 2011) and the end of the fit fixed and advancing only the first month of the fit, the constant runs from −0.30 to +23.96 mm yr⁻¹ while the coast-edge amplitude stays negative throughout, but the reach falls from about 900 m to about 520 m once the window starts after August 2007 (455 to 904 m across the sweep). The instability is not sampling error: across fixed-length rolling windows the constant moves seven to sixteen times its own within-window standard error.

The constant and the climate covariate compensate. Sliding a window of constant length along the record, the constant and the covariate's trend contribution correlate at −0.59 to −0.82 depending on window length: when the covariate is apportioned more of the decline, the constant takes less, and the reverse. Their sum is better determined than either part, and the spread of both narrows sharply as the window lengthens (Figure 61) — at eighteen years the constant settles to −0.39 mm yr⁻¹ with a standard deviation of 4.56, against 13.14 at ten years.

No far-field rate is therefore quoted anywhere in this work, and the site-wide component of the decline is treated under the recharge-efficiency and seasonal-redistribution account of Section 5 rather than here.

Over the well record potential evapotranspiration at RAF Valley rises by 1.96 mm yr⁻¹ (t = 2.67). That short-window figure should be read against the long record rather than as a secular rate: PET rises by 0.32 mm yr⁻¹ over 1931–2025 (t = 4.51) and by 0.59 over 1960–2025 (t = 5.03), so the trend is real, of long standing, and steepening. It is also the only climate term at this site that separates from zero, and the reason is variability rather than magnitude — PET varies by about 3% about its mean where winter rainfall varies by about 19%, so PET's rise over the well record is close to two standard deviations of its own year-to-year scatter while winter rainfall's is under one. The winter trend is the larger of the two in millimetres and the smaller relative to its own noise.

What does not change is the balance. The summer (June–September) water balance trends at −4.19 mm yr⁻¹ and the winter (October–March) balance at +3.74 mm yr⁻¹. Neither is individually distinguishable from zero (t = −1.28 and +0.84), and neither is claimed to be; what matters is that they are of opposite sign and comparable size, so that their sum —

which is what an annual term sees — is +1.10 mm yr⁻¹ with t = 0.27, which is flat. The far-field constant is fitted against a covariate built from that annual balance (Section 5.7.7).

The rolling-window output should not be read as a climate history. A ten-year water-balance trend at this site is dominated by which extreme years fall inside the window, and the constant inherits that window sensitivity, which is why it is not quoted and why it settles only at window lengths long enough that no single year can dominate.

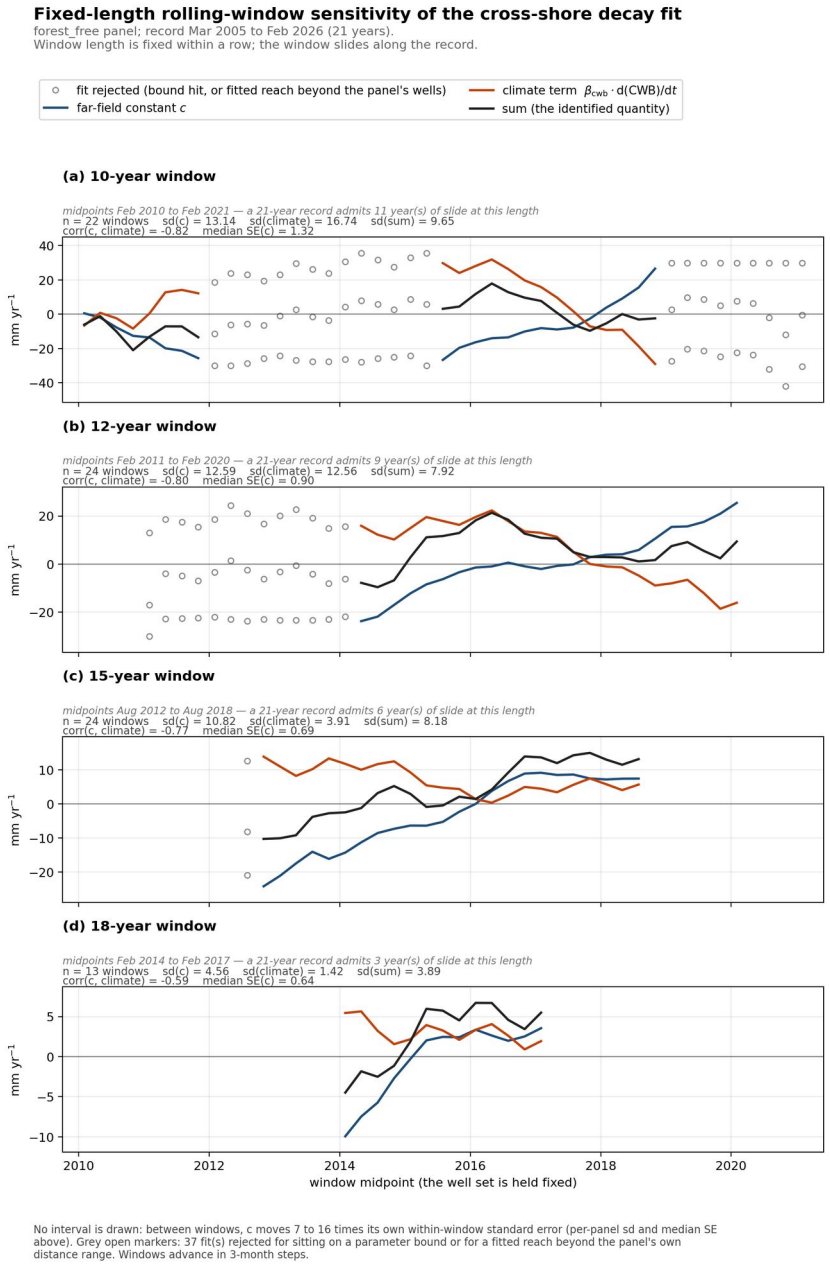



***Figure 61:*** *Fixed-length rolling-window sensitivity of the cross-shore decay fit. One panel per window length; within a panel the window length is constant and the whole window slides along the record, so the span of midpoints each panel admits is fixed by arithmetic — a 21-year record allows only three years of slide at an 18-year length. Lines break where fits were rejected for sitting on a parameter bound or returning a reach beyond the panel's own wells; rejected fits are drawn as open markers rather than dropped. The far-field constant and the climate term compensate as the window moves, and their spread narrows as the window lengthens. Source: 25_13_rolling_window.png.*

### 4.10.4 Independent transect estimate of the coastal drawdown rate

The $\delta_0$ of the panel regression above is cross-checked directly, and without a model, along a single coast-to-inland transect. Taking MAM spring means from the most coastal well (CEH22, at the dune toe) and an inland anchor beyond the fitted reach L_cg (NW4, ≈1.5 km inland), the coast-minus-inland head difference deepens progressively across 2010–2023 at −27.8 mm yr⁻¹ (95 % CI −36.1 to −19.3; n = 14; AR(1)-corrected; Figure 62) — within error of the network-scale $\delta_0$ of −31.3 mm yr⁻¹, from an entirely independent construction, and robust to any single year (leave-one-out slopes −32.2 to −25.5 mm yr⁻¹). The divergence is shared between its two ends: CEH22 falls at −14.6 mm yr⁻¹ in absolute terms while the inland anchor rises at +13.2 mm yr⁻¹ over the same springs, so the growing gradient combines the interior refilling through the wet springs of the window with the coast dropping. The observed offset trends downward (anchor-referenced coastal-end ordering Spearman ρ = −0.79, against −0.21 for the raw levels), and spring climate water balance does not trend over the window; what the growing offset distinguishes is set out in Section 5.7.3. The estimate remains a single coast-to-inland line whose rate rests on two wells (CEH22 and NW4) over a window opening after drawdown was already underway, so it distinguishes a growing (erosion-like) gradient from a static one rather than resolving erosion from a time-varying substrate effect (Section 5.7.3).

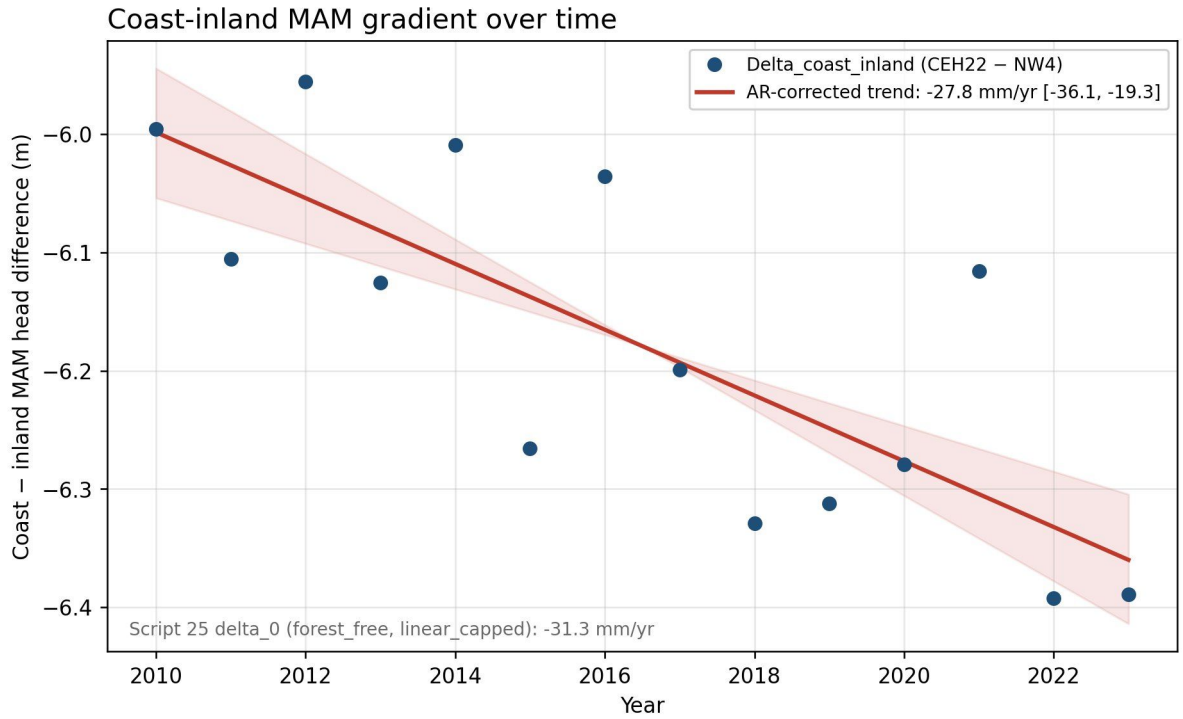



***Figure 62:*** *Independent, model-free estimate of the coastal drawdown rate. The coast-minus-inland MAM head difference (CEH22 at the dune toe minus the inland anchor NW4, ≈1.5 km inland and beyond the fitted reach L_cg) plotted per year over 2010–2023, with the AR(1)-corrected trend (red) and its bootstrap 95% confidence band. The difference deepens at −27.8 mm yr⁻¹ (95% CI −36.1 to −19.3; n = 14), within error of the network-scale $\delta_0$ of −31.3 mm yr⁻¹ from the independent gradient regression (grey annotation). A progressively deepening difference is consistent with a retreating coastal boundary; a static substrate geometry would hold it constant. (Source: 38_coast_inland_difference.jpg.)*

### 4.10.5 Coastal-Margin Processes and the Easting Gradient

The forest drawdown propagation map (Figure 66) raises a natural comparison: how does the spatial reach of canopy-driven drawdown compare with the coastal-margin processes acting on the same aquifer? Two such processes operate at the western boundary, and because they act in opposing directions, neither can be assessed in isolation. Shoreline retreat lowers the ground surface relative to the water table, raising the displacement head and enhancing seaward drainage — the same mechanism, in reverse sign of intent, as the dune scraping of Section 4.5. Sea-level rise acts oppositely, raising the water table at the coastal boundary. Mapped as companion fields on the same hillshade base as the forest drawdown figure, constructed as set out in Section 3.8.2, these processes are shown in Figure 63 (erosion drawdown from a single shoreline-retreat event), Figure 64 (the sea-level-rise head response as a finite-window transient), and their cell-by-cell difference in Figure 65.

The net map (Figure 65) shows that the two processes do not cancel but partition spatially. An erosion-dominated band, reaching about 900 m inland to the fitted reach, lies along the dune-toe margin, coincident with the western Coastal-Forest (C5) and western Western-Residual (C3) wells; this is flanked by a sea-level-rise–dominated rise at the immediate shore and near-neutral conditions across the dune interior. Over a matched five-year window the erosion contribution is the larger (an edge drawdown of 81 mm from a single retreat event, against a sea-level-rise gain of at most +20 mm), so the western-margin signal is the residual of two opposing processes in which erosion dominates everywhere but the immediate shore.

The coastal-drift covariate carried in the clearfell BACI analysis (Section 4.6.3) absorbs this west-to-east deepening gradient; it carries the net field and does not decompose it into its constituent processes (Section 5.5).

These three figures are illustrative constructions on the eroding Caernarfon Bay shore only; they combine a single storm-magnitude retreat event with gradual sea-level rise, and depend on assumed retreat-rate, hydraulic-diffusivity, datum, and window parameters. They illustrate the relative spatial reach of the two processes, not a deterministic forecast or a closed water balance.

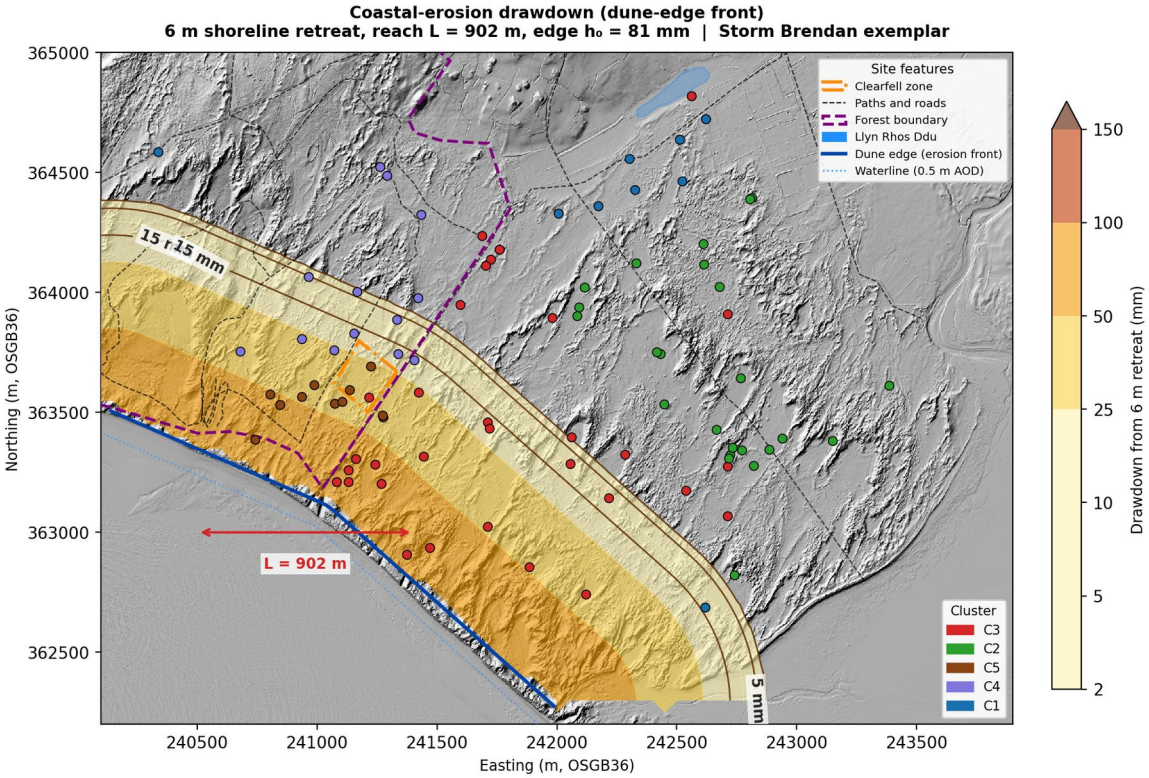



***Figure 63:*** *Coastal-erosion drawdown at the western dune margin, Newborough Warren, on the DEM hillshade base. The field shows the water-table drawdown produced by a single shoreline-retreat event, decaying inland from the dune-toe front (the DEM mean-waterline contour offset 100 m landward) following the forest-free linear-capped coastal-retreat gradient fitted in Section 4.10.2. Edge magnitude is set by a single 6 m retreat event (the Storm Brendan exemplar); $\delta_0$ and the characteristic length L_cg are read live from the Script 25 fit. The field is clipped to the study-area outline and references the eroding Caernarfon Bay (south-western) shore only. (Source: 20_coastal_erosion.png).*

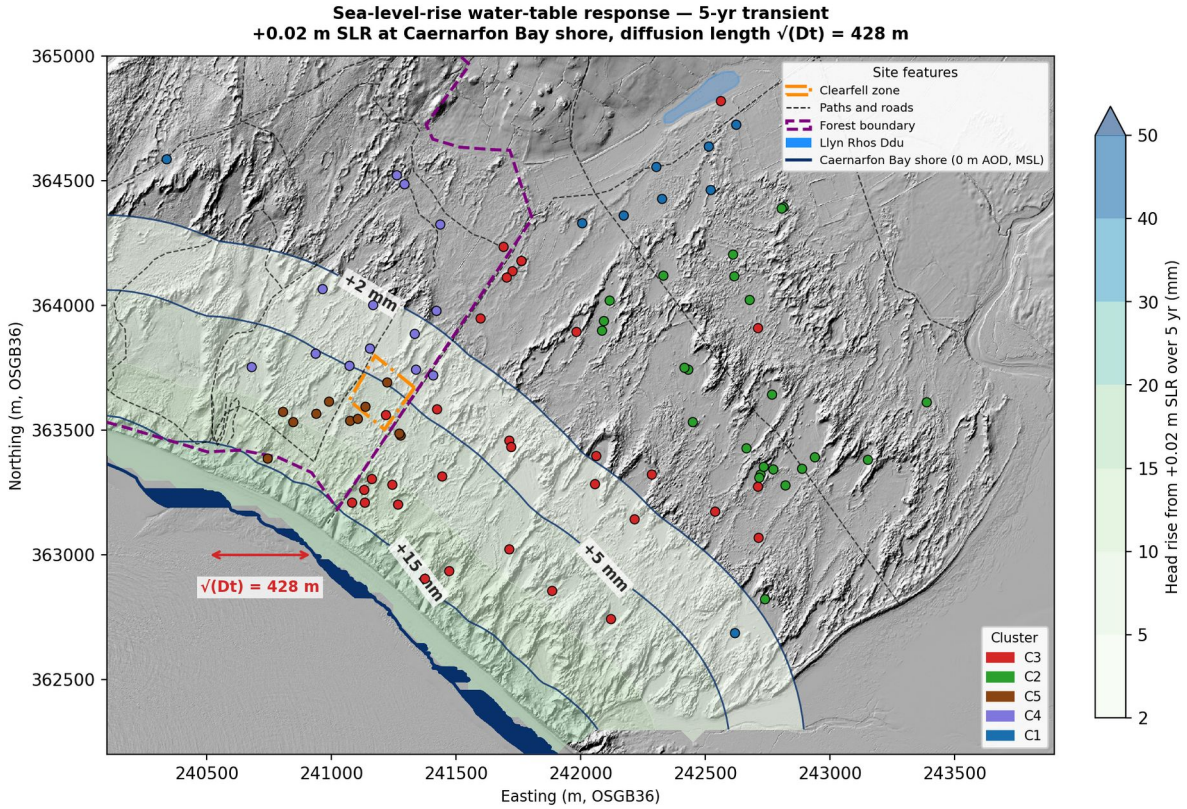



***Figure 64:*** *Sea-level-rise head response at the western dune margin, on the same hillshade base and clip as Figure 63. The field shows the water-table rise produced by gradual sea-level rise over a five-year window, modelled as a boundary perturbation diffusing inland, Δh(d) = ΔSLR · erfc(d / (2√(Dt))), with hydraulic diffusivity D = Kb/Sy (K and b from literature; Sy the C3-cluster median from the water-table-fluctuation analysis, Section 3.7.3). The boundary is referenced to mean sea level (0 m AOD), distinct from the dune-toe front of the erosion field. (Source: 20_slr_response.png).*

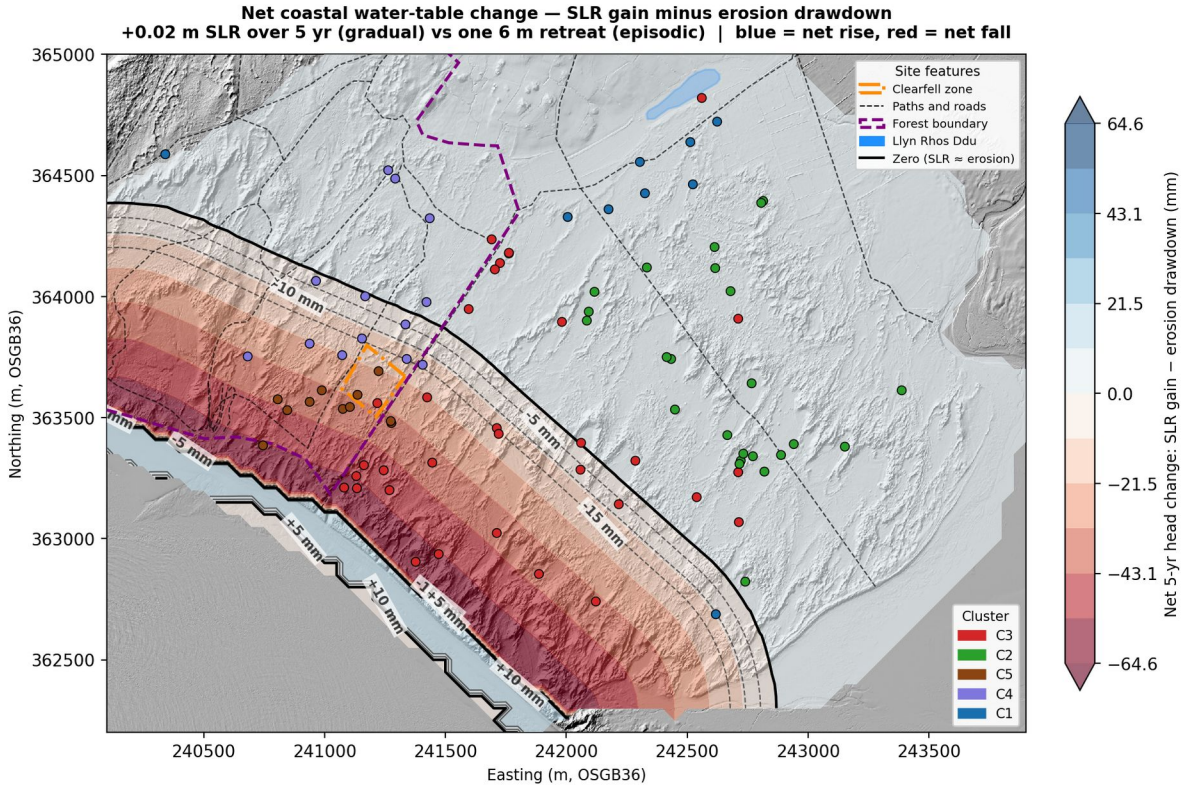



***Figure 65:*** *Net coastal-margin change at the western dune margin: the cell-by-cell difference between the sea-level-rise head gain (Figure 64) and the erosion drawdown (Figure 63) over a matched five-year window, on a diverging scale with the zero contour emphasized. The two processes partition spatially rather than cancelling: an erosion-dominated band (net deepening, reaching about 900 m inland) along the dune-toe margin, coincident with the western Coastal-Forest (C5) and Western-Residual (C3) wells, flanked by a sea-level-rise–dominated rise at the immediate shore and near-neutral conditions across the dune interior. This spatial partition provides a physical interpretation of the drift absorbed by the coastal-drift covariate in the clearfell BACI analysis (Section 4.6.3). The figure is an illustrative construction on the Caernarfon Bay shore only, combining an episodic retreat event with gradual sea-level rise under assumed retreat-rate, diffusivity, datum, and window parameters; it depicts relative spatial reach, not a deterministic forecast or a closed water balance. (Source: Script 20 output 20_coastal_net_effect.png).*

### 4.10.6 Episodicity, retreat rates and sea-level counteraction

The gradient fitted above is a rate averaged over twenty-one years, and the process producing it is not steady. Shoreline retreat at this frontage is episodic: the operative constant in the spatial construction is a single-event retreat of 6 m against the measured 2006–2026 retreat rate $E_c = 2.32$ m yr$^{-1}$ (Script 40; Section 3.8.2). Whether the hydraulic boundary has been displaced further than the dune toe over the study window — the two are not obliged to move together — is not resolved by this record: the per-driver scale-factor regression returns a coastal scale factor indistinguishable from unity in every window examined (distinguishable from zero in the pre-intervention window) (Section 4.12.1).

Nor is the retreat uniform along the frontage. Pye and Blott (2024) record a north-positive to south-negative gradient, with progradation in the north and chronic retreat concentrated at the southern Twyni Penrhos frontage. A single $\delta_0$ fitted across the network is therefore a frontage average, and the wells nearest the most active section of shoreline are not necessarily those with the smallest d.

Acting against retreat is sea-level rise, which raises the head at the seaward boundary rather than lowering the ground beside it. The two are opposite in sign and differ in inland reach, which is why the fields above partition spatially rather than cancelling: an erosion-dominated band from the dune toe, flanked by a sea-level-rise-dominated rise at the immediate shore.

None of this changes the fitted profile. It bounds how it should be read: as a frontage-averaged, time-averaged rate, whose local and episodic realization the monitoring network is not designed to resolve.

## 4.11 Management influence on the water table

The spatial reach of forest management perturbations is shown directly by the drawdown propagation analysis (Figure 66), which models the steady-state head perturbation from forest canopy interception as an exponential decay with flow-weighted cost-distance from the forest boundary ($\lambda \approx 220$ m). Estimated drawdown exceeds 50 mm out to about 240 m of flow-weighted cost-distance from the forest edge, decays to 24 mm at CEH23 and 8 mm at CEH6, and is negligible (≤2 mm) at the eastern dune and lake-edge wells CEH24, CEH10 and CEH11. These are steady-state values; the timescale to reach equilibrium is governed by the Model A drainage decay half-life values mapped in Figure 55 (at the network datum), ranging from months in the eastern dune (t½ approximately 6–13 months) to years in the forest interior (C4 mean t½ approximately 27 months) (Section 5.4.3). Forest management perturbations are therefore concentrated in the forest zone and its margins and decline steeply eastwards across the six eastern wells of the lake-edge cluster, from 72 mm at CEH27 nearest the forest to 8 mm at CEH6 furthest east (CEH26 42 mm, CEH23 24 mm, CEH25 and CEH5 11 mm). The seventh lake-edge well, CEH11, lies 1.7–2.0 km south of that group rather than further east, 1600 m from the forest, and carries 0.3 mm — the smallest modelled drawdown in the cluster.

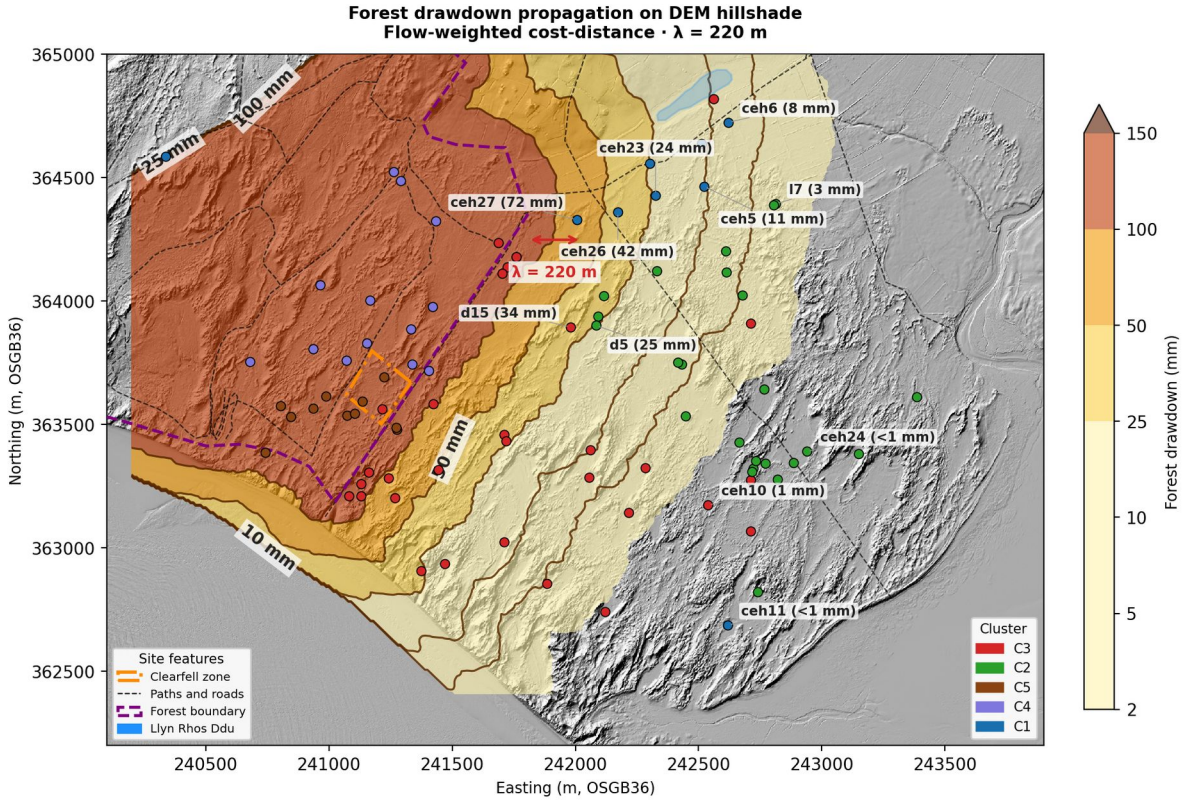



***Figure 66:*** *Forest drawdown propagation across the Newborough Warren dune aquifer, showing the estimated steady-state head perturbation (mm) attributable to Corsican pine canopy interception. Drawdown at each well is modelled as an exponential decay with flow-weighted cost-distance from the forest boundary, using a characteristic length scale λ ≈ 220 m. The background is the DEM hillshade; the coloured bands are the modelled drawdown (mm) by flow-weighted cost-distance from the forest edge, with the forest boundary, the clearfell zone, paths and Llyn Rhos-Ddu overlaid. Labelled wells show estimated drawdown in parentheses: the signal exceeds 50 mm out to about 240 m of flow-weighted cost-distance from the forest edge (CEH27: 72 mm, CEH26: 43 mm), falls to 34 mm at D15, 24 mm at CEH23 and 8 mm at CEH6, and is ≤2 mm at CEH24, CEH10 and CEH11. With a 150 mm forest-edge deficit decaying at λ ≈ 220 m, the 50 mm contour lies at roughly 240 m of cost-distance from the forest boundary and the 10 mm contour at roughly 600 m. Forest management perturbations are concentrated in the forest zone and its margins and decline steeply eastwards across the six eastern C1 Lake Edge wells, from 72 mm at CEH27 to 8 mm at CEH6; the seventh, CEH11, lies 1.7–2.0 km south of that group rather than further east, 1600 m from the forest, and carries 0.3 mm. Well markers coloured by cluster. LiDAR data: © NRW & OS; see Section 11 for full attribution. (Source: 20_drawdown_propagation_nohead.png).*

The dune scraping of Section 4.5 can be placed on the same spatial footing as the forest and coastal fields. Figure 67 renders the April 2015 CEH36 scrape as a steady-state drain, with edge magnitude anchored to the measured CEH36 response (128 mm) and the same leaky-aquifer decay length as the forest field (λ ≈ 220 m). On this anchoring the unsurveyed excavation depth is recovered as an inferred ≈ 0.43 m. Modelled drawdown exceeds 50 mm out to about 200 m from the CEH36 excavation on its own and about 470 m where the fields of all the mapped cuts combine, falling to 10 mm at about 550 m and 810 m respectively. As with the forest field these are steady-state values: the equilibration time at these distances is

years to decades (Section 5.4.3), which is why the uphill propagation is not resolved as a distance-decay signal within the 31-month BACI window (Section 4.5).

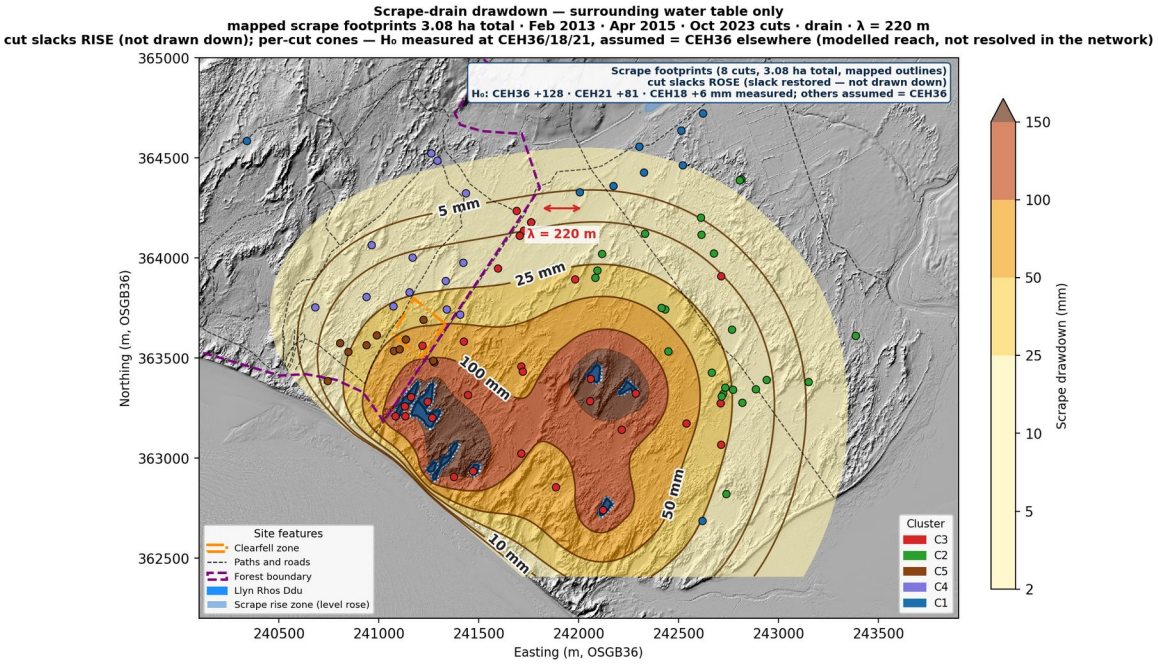



***Figure 67:*** *Dune-scrape drawdown across all eight mapped cuts, Newborough Warren (Feb 2013, Apr 2015, Oct 2023), on the DEM hillshade base. All cuts are treated as permanent co-active features; the displayed field is the long-term equilibrium superposition $\Sigma_i H_{0,i} \cdot [\exp(-d_i/\lambda) - \exp(-d_i'/\lambda)]$ where $d_i$ is Euclidean distance to cut i's footprint, $d_i'$ is distance to its mirror image reflected across the OS High Water Mark coastline (method of images; enforces zero drawdown at the fixed-head sea boundary), and $\lambda \approx 220$ m is the leaky-aquifer decay length (live C3 Sy and $\beta_3$; K=6 m d−1, Betson & Bristow 2002; b=5 m). Edge magnitude H_0,i is the measured BACI response for three monitored cuts (CEH36 +128 mm, CEH21 +81 mm, CEH18 +6 mm; Section 4.5) and assumed equal to the CEH36 value for the five cuts with no record from before they were cut. Scrape interiors are masked — each cut slack rises (Section 5.4.1) — so the field shown is the surrounding drawdown, the dipole counterpart of the slack rise. The modelled drawdown reach is not independently resolved: across the ten uphill wells the BACI-corrected change in $\beta_3$ shows no decay with distance ($r = -0.28$, $p = 0.43$; Script 09b), although the eight within 575 m rise on average (7 of 8 positive, $p = 0.043$). Steady-state construction; equilibration over years to decades (Section 5.4.3). Clipped to the study-area outline; well markers coloured by cluster. (Source: 20_scrape_drawdown_nohead.png).*

## 4.12 Combined Driver Assessment and Observed Network Change

Figure 68 combines the two non-felling drawdown drivers acting on the clearfell compartment — the February 2013 and April 2015 scrapes and a Storm-Brendan-scale retreat event — as a single head-loss field on the pre-fell baseline. The combined drawdown reaches ≈ 250 mm between the 2013 cuts and tapers through the clearfell compartment, with the coastal-erosion term adding a south-western band; the felling step estimated by the BACI ANCOVA (Section 4.6) is the residual once both are accounted for. This is the spatial form of the gradient that the coastal-drift covariate of Section 4.6.3 absorbs: the covariate absorbs

the net west-to-east gradient that this figure decomposes into the scrape drain and the coastal retreat.

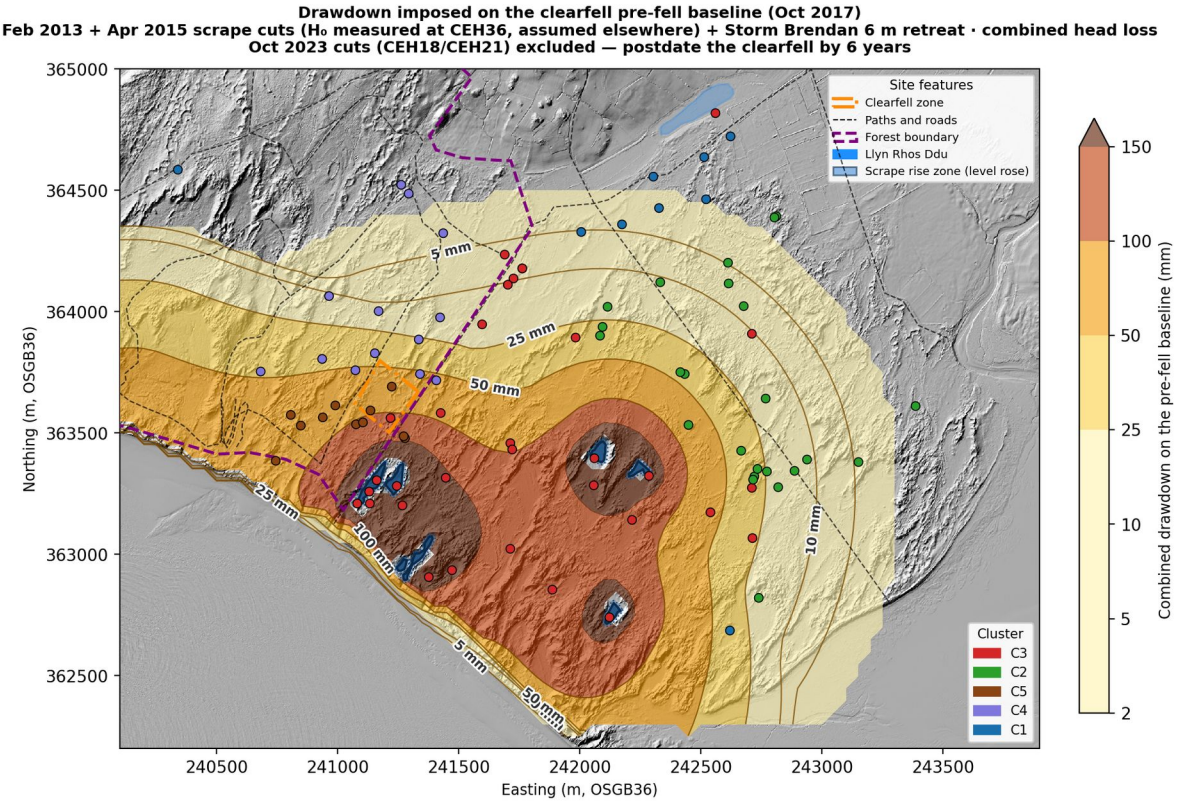



***Figure 68:*** *Combined drawdown imposed on the clearfell pre-fell baseline (Oct 2017): the cell-by-cell sum of the Feb 2013 and Apr 2015 dune-scrape cuts (five cuts; $H_0$ measured at CEH36 +128 mm, assumed = CEH36 for the four cuts with no pre-cut record; Oct 2023 cuts excluded — they postdate the clearfell by six years) and a single Storm-Brendan-scale coastal-retreat event (Figure 63), both head losses, on the shared sequential drawdown scale. The scrape component uses leaky-aquifer superposition with method-of-images coastal boundary correction (see Figure 67); the coastal-retreat component is illustrative at the Storm Brendan scale. Sea-level rise is excluded; the field is the spatial footprint of the two non-felling drivers the clearfell BACI must separate from the felling signal (Sections 4.6, 5.5), not a coastal mass balance. A first-order superposition, not a closed water-balance solution. Clipped to the study-area outline; well markers coloured by cluster. (Source: 20_clearfell_baseline_drawdown.png).*

Four further spatial figures are derived from the framework developed in Sections 4.10.5 and 4.11: a measured clearfell step-change map (Figure 69), a parametric net state map superimposing all five active drivers (Figure 70), a modelled 2005–2025 driver-change map (Figure 71) and an empirical MSL5 change map comparing the pre-clearfell and current MSL5 windows (Figure 72). Methods for all four are in Section 3.8.1; interpretation of the comparison between Figures 69 and 70 is in Section 5.7.5.

### *Measured clearfell spatial response*

The clearfell step-change map (Figure 69) shows the difference in mean water-table level between the pre-felling scraping era and the post-felling era at each well (*fell_step_cc*, Script

10b; positive = shallower after felling). The map shows a broadly positive response across the open dune interior (+25 to +100 mm), with the largest gains at the C4 Forest Control wells immediately adjacent to the felled compartment. A zone of −25 to −100 mm appears along the south-western coastal margin at the C5 Coastal Forest and southern C3 wells. The coastal-margin zone coincides with the coastal-retreat gradient of Section 4.10.2 and with the Coastal Control tier of Section 4.6.4; the positive zone coincides with the C4 and immediately adjacent C3 interior. As in Section 4.6.5, the map shows no distance-decay from the felled compartment and resolves no spatial felling footprint; the felled-well signal is the winter rise at WMC3 (+128 mm over October–March, p = 0.048; Section 4.6.3), and the map is read in Section 5.7.5.

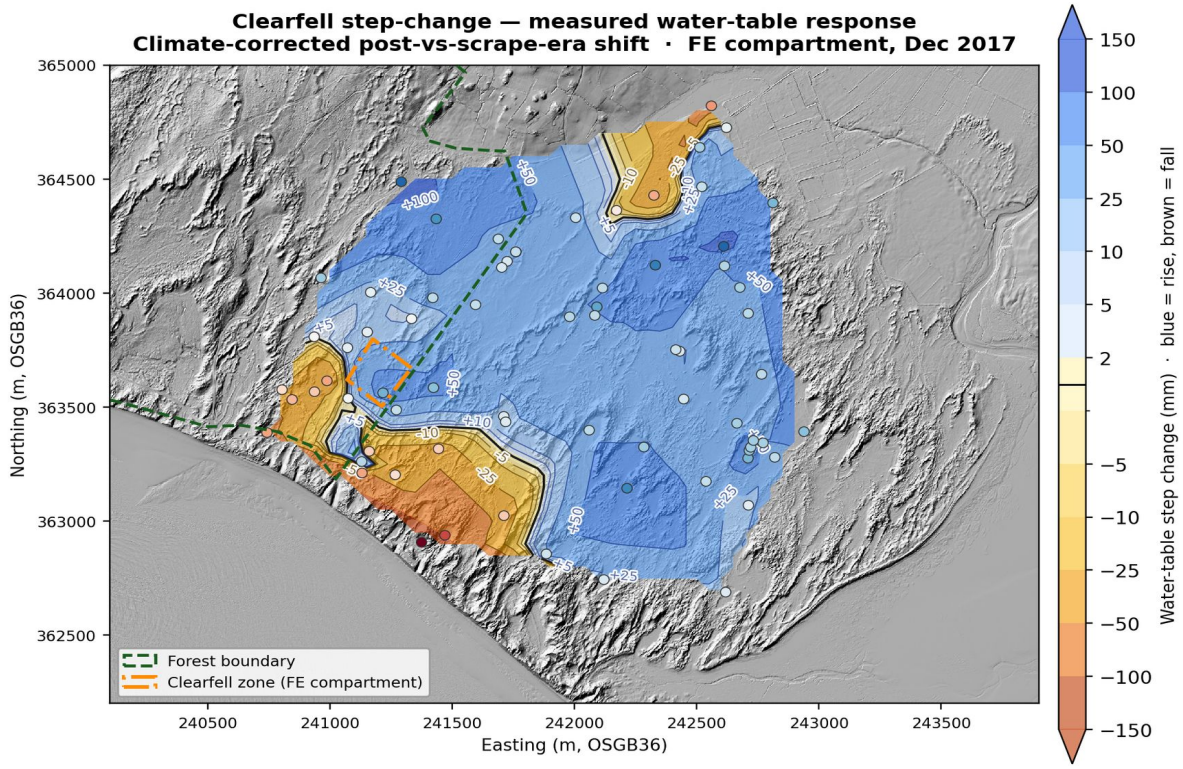



***Figure 69:*** *Clearfell step-change map — climate-corrected mean water-table shift, pre-felling (scraping era) versus post-felling, at each well. Derived from column fell_step_cc (Script 10b; Section 3.5.4). The same per-well values as Figure 36, re-rendered on the driver-group common scale for band-for-band comparison. Positive values (blue) = shallower after felling. Wells excluded where the post-felling record is too short to estimate a stable era mean (<6 post-felling monthly observations); n = 71 wells retained. Both interpolated surface (piecewise-linear Delaunay interpolation, no distance weighting or smoothing) and well markers use the same discrete diverging colourmap. Forest boundary (green dashed), clearfell zone (orange dash-dot). Generated by 20_spatial_figures.py (Source: 20_clearfell_gain.png).*

### *Parametric net state map*

Figure 70 (method: Section 3.8.1) superimposes all five annual-mean driver fields on the site grid. The forest interior carries a large net loss (−50 to over −150 mm) where the standing canopy drawdown dominates. A small net-gain zone sits over the felled compartment where the clearfell interception recovery locally exceeds the surrounding forest loss. The open dune is near-neutral: neither the erosion drawdown nor the SLR head gain reaches the eastern dune interior at meaningful magnitude. Along the southern shoreline, a narrow net-gain band reflects SLR marginally exceeding the Storm Brendan erosion drawdown at the immediate dune toe.

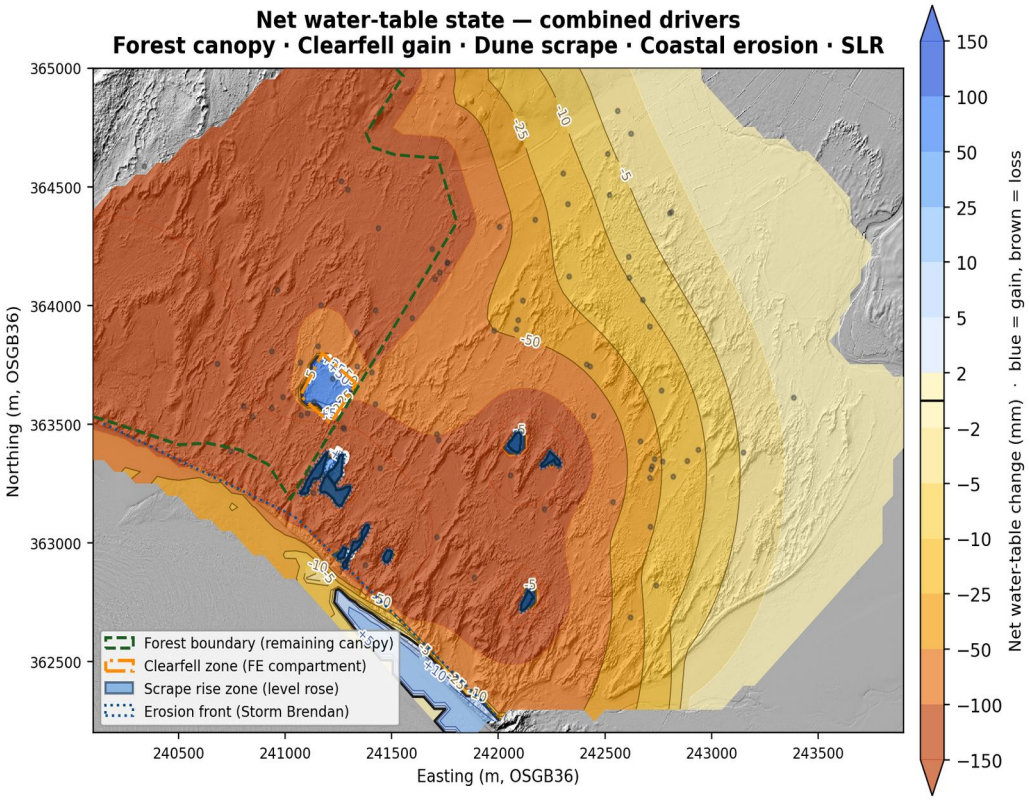



***Figure 70:*** *Net water-table state — parametric combination of five simultaneous annual-mean drivers: forest canopy drawdown ($H_0$ = 150 mm, λ ≈ 220 m), clearfell gain (same $H_0$, λ), dune-scrape contribution (all eight cuts; $H_0$ measured at CEH36 +128 mm, CEH21 +81 mm, CEH18 +6 mm; others assumed = CEH36; leaky-aquifer superposition with method-of-images coastal correction; scrape footprints contribute +$H_0$ rise, surrounding area contributes drawdown), coastal erosion ($h_0$ = 81 mm), SLR (+20 mm over 5 yr). Net = SLR gain + clearfell gain + scrape rise − forest loss − erosion loss − scrape drawdown. Symmetric ±150 mm diverging scale; bold zero contour. First-order superposition of annual-mean fields; not a closed water balance (caveats: Section 4.10.5). (Source: 20_net_state_map.png).*

### *The 2005→2025 driver-change map*

The 2005→2025 driver-change map (Figure 71) re-bases the parametric net-state construction to the study record, differencing each modelled driver field between its 2005 and 2025 states so that the map shows what the drivers imply should have changed over the monitoring window rather than the accumulated state since the forest was planted. Because the standing pine canopy is at equilibrium at both endpoints, its drawdown cancels in the difference and the mature forest interior returns to near zero — the check that the map is an epoch difference and not a re-drawing of Figure 70. Five fields change over the window and are summed: the clearfell gain, plotted at the observed BACI step (+108 mm; Section 4.6.3); the dune-scrape response, with all cuts within the window contributing; the chronic coastal drawdown accumulated over the window; sea-level-rise head gain (+20 mm, on its five-year horizon); and a broadleaf-restock field new to this map. The broadleaf-restock block, felled in 1993 and restocked in 1995, was already near closed canopy at the 2005 baseline and reaches full canopy by 2025; only the increment of interception deficit that develops over 2005→2025 enters the map, computed on the same exponential-decay construction as the forest field but scaled by the broadleaf interception fraction (0.15 for broadleaf, Komatsu et al., 2011; 0.24 for pine, Freeman, 2008) and by the canopy-establishment increment, giving of order 56 mm of added drawdown at source. It appears as a drawdown patch of the opposite sign to the clearfell gain — the atmospheric-draw signature of a canopy returning where the clearfell removed one. The coastal-retreat field is the chronic drawdown accumulated over the twenty-year window at the observed coast-edge decline rate (from the live Script 25 forest-free fit, $\delta_0 = 31.3$ mm yr$^{-1}$), giving $20 \times \delta_0 \approx 626$ mm at the dune toe and decaying to zero at the reach $L_{cg} \approx 900$ m — much the strongest of the five fields at the western coastal margin, and independent of the measured shoreline-retreat rate because it is built from $\delta_0$ alone. A five-year companion map (Script 20, 20_driver_change_2005_2025.png) carries the coastal field at $5 \times \delta_0 \approx 156$ mm, the horizon of the sea-level-rise field, for a legible comparison of the management drivers. As a first-order superposition of single-mechanism fields the map is illustrative and not a closed water balance; the broadleaf and coastal fields in particular are modelled quantities the dipwell record cannot fully resolve, and are presented as indicative. The map is the modelled companion to the observed climate-removed trend map (Figure 74, Section 3.8) over the same 2005→2025 window: the two are built from different data by different methods and are expected to agree in broad direction rather than in detail, so their agreement where it occurs supports the attribution and their divergence flags a driver the model does not capture.

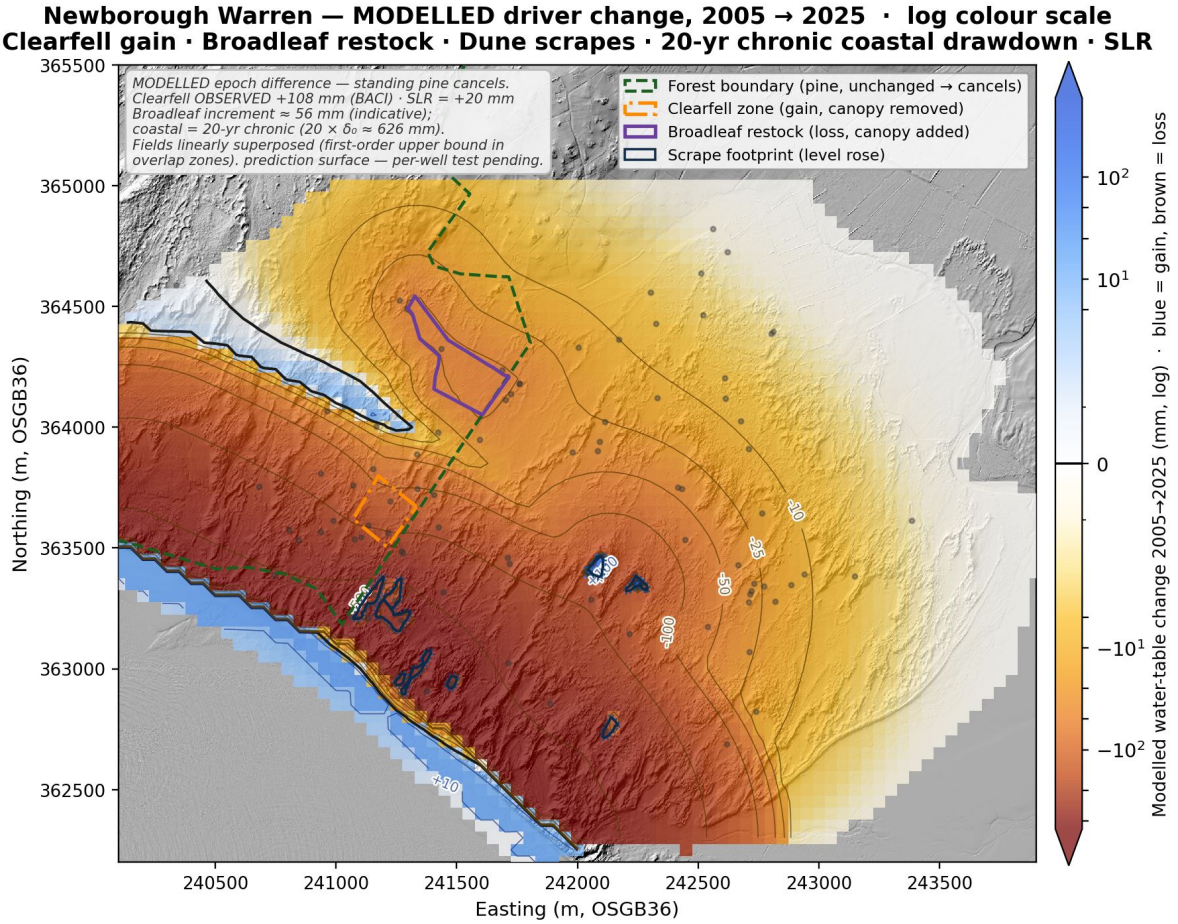



***Figure 71:*** *Modelled water-table change across Newborough Warren, 2005–2025 — full-window (20-year) coastal horizon, logarithmic scale. The companion to the five-year driver-change map, showing the same modelled drivers but with the coastal field accumulated over the full study window. The map is the epoch difference of the modelled driver fields (2025 minus 2005): the standing pine canopy cancels between the epochs, so the forest interior shows only the coastal drawdown and the broadleaf increment, not the standing-forest drawdown. The drivers that change over the window are the December 2017 clearfell (interception deficit removed → head gain, plotted at the observed BACI step of +108 mm), the broadleaf restock (canopy interception added → head loss, an indicative +56 mm increment as the block approached closed canopy), the 2013, 2015 and 2023 dune scrapes (slack rises with surrounding drawdown cones), sea-level rise (+20 mm), and the coastal chronic decline. The coastal field here is the full-window accumulation of the fitted coast-edge trend (20 × $\delta_0$ ≈ 626 mm at the dune toe, decaying to zero at the reach L_cg ≈ 900 m); the five-year companion instead shows the near-term chronic value (≈ 156 mm) for a legible comparison of the management drivers. Because the coastal drawdown (of order several hundred millimetres) is far larger than the management effects (of order one hundred millimetres), the colour scale is logarithmic so both read on one map: deep red marks the strongest coastal drying at the toe, fading through amber to white where change is negligible, and blue marks head gain. The fields are combined by linear superposition; where they overlap the map shows their sum, which is a first-order upper bound and may over-state the true combined drawdown. The map is a modelled prediction surface — at each well it predicts a 2005–2025 change that can be tested against the observed record (a per-well validation is reported separately); the coastal and broadleaf fields are the least record-constrained and are flagged as modelled. Inland propagation of the coastal signal toward the reach is diffusive and develops over years to decades; the magnitudes shown are the accumulated end-of-window values, not the present-day resolved field. (Source: 20_driver_change_20yr.png)*

## *Observed network change*

Figure 72 presents the conventional two-window reading of observed network change --- checked on its own terms (Section 5.7.5) but, as set out below, limited in what a two-window difference can establish about absolute change.

The MSL5 change map (Figure 72; method: Section 3.8.1) compares window end 2017 (springs 2013–2017, pre-clearfell, network mean −467 mm below ground) with window end 2023 (springs 2019–2023, network mean −572 mm). The raw 2023-minus-2017 difference exceeds the ±25 mm display threshold at 58 of 59 wells, every one of them deepening; no well shallows. Set against the five-year window standard error (82–181 mm by cluster, above), no single well's difference is resolved; it is the coherence of sign across the network that carries the signal. The map shows a broadly deeper aquifer in the current window, the deepening intensifying from the eastern dune interior toward the forest and the western half of the site. Because the figure differences two five-year windows, every time-invariant driver cancels — in particular the constant drawdown of the 60-year standing forest, which is present in both windows and contributes nothing to the difference — so the surviving pattern reflects only drivers that progressed across the study period (Section 5.7.5).

The per-well pattern is not an artefact of the window pair. Across the 46 wells carrying both measures, the raw 2017 → 2023 change correlates with the per-well long-run summer-minimum slope fitted over the full record (Section 4.10.2; Pearson $r = 0.43$, $p = 0.003$; Spearman $\rho = 0.40$, $p = 0.006$). The two metrics are independent: the slope is a per-well linear regression over 14 to 21 years (median 16) of annual April–September minima (hydrological-year indexed), the change a difference of two five-year means of the readings dated March–May.

The eastern C1/C2 dune wells deepen least (CEH25 −11, D10 −29, L7 −31 mm), CEH25 alone within the ±25 mm near-neutral band. The clearfell Impact well WMC3 (−90 mm) and the scrape well CEH36 (−97 mm) deepen broadly in line with the −105 mm site-wide mean, showing no intervention-specific departure from the surrounding network. The most strongly negative zone is the south-western coastal margin: CEH22 (−233 mm), CEH21 (−213 mm), CEH18 (−145 mm), and the C5 coastal members CEH19, CEH17, NW9 (−164 to −172 mm). These wells lie nearest the eroding coast (Section 4.10.2).

The C4 Forest Control wells deepen by −138 mm (cluster mean), more than the −105 mm site-wide average. These are unfelled controls, and the map applies no climate correction; the excess is read in Section 5.7.5 against the climate-response envelope of Figure 75.

The interannual standard deviation of annual spring level differs widely across the network, from about 182 mm at C1 to 404 mm at C4 (Section 4.8.4). Controlling for the drainage recession time, this standard deviation correlates with the atmospheric-draw coefficient $\beta_2$ at partial rank $r = +0.58$ ($p < 0.0001$); controlling for $\beta_2$, its association with the recession time is not significant (partial $r = +0.10$, $p = 0.42$). The two parameters are themselves correlated across wells (Spearman +0.36). Across the 65 wells with at least twelve valid spring years,

the lag-1 autocorrelation of annual spring level is approximately zero at every cluster, with cluster means between −0.16 and +0.09, against an expectation, from the datum-free drainage coefficient of the intercept form (Model B), fitted on the same record as the published coefficients, of 0.01 at C1 to 0.27 at C5 (median 0.15 across wells; C4's coefficient is not identified). Read from the published form's $\beta_3$, which carries the drainage datum, the expectation would be 0.27 to 0.76; that is the datum's recession constant, not the mean-reversion time (Section 3.4.1). The observed autocorrelation is uncorrelated with the recession time (Spearman +0.16, p = 0.21), and the site-mean spring series over 20 years gives −0.03.

These results are robust within the two-window method: the spring-gap and independent-metric checks of Section 5.7.5 indicate the 2017→2023 deepening is not an artefact of window length or spring-record gaps. Their limitation is comparative rather than internal. Because the amplitude of interannual variation differs so widely across the network, the absolute magnitude of a two-window difference — and at some wells its sign — depends on which two windows are compared, even between windows matched for antecedent rainfall; the matched-window sensitivity analysis of Section 5.7.5 yields site-mean changes spanning −0.16 to +0.23 m and changing sign across all admissible window pairs (17 of 55 falling, 38 rising) — the −105 mm comparison being one interior point and the wet-2024 windows the +0.23 m extreme — which is why absolute site-wide change is not rested on this two-window method (Script 34). The −105 mm site-wide figure is therefore specific to the 2017/2023 comparison, not a robust estimate of secular change. The two figures that follow remove this dependence: Figure 73 maps the differential movement with the common climate signal removed, and Figure 75 the climate-response envelope — the latter quantifying directly the forested clusters’ climate response.

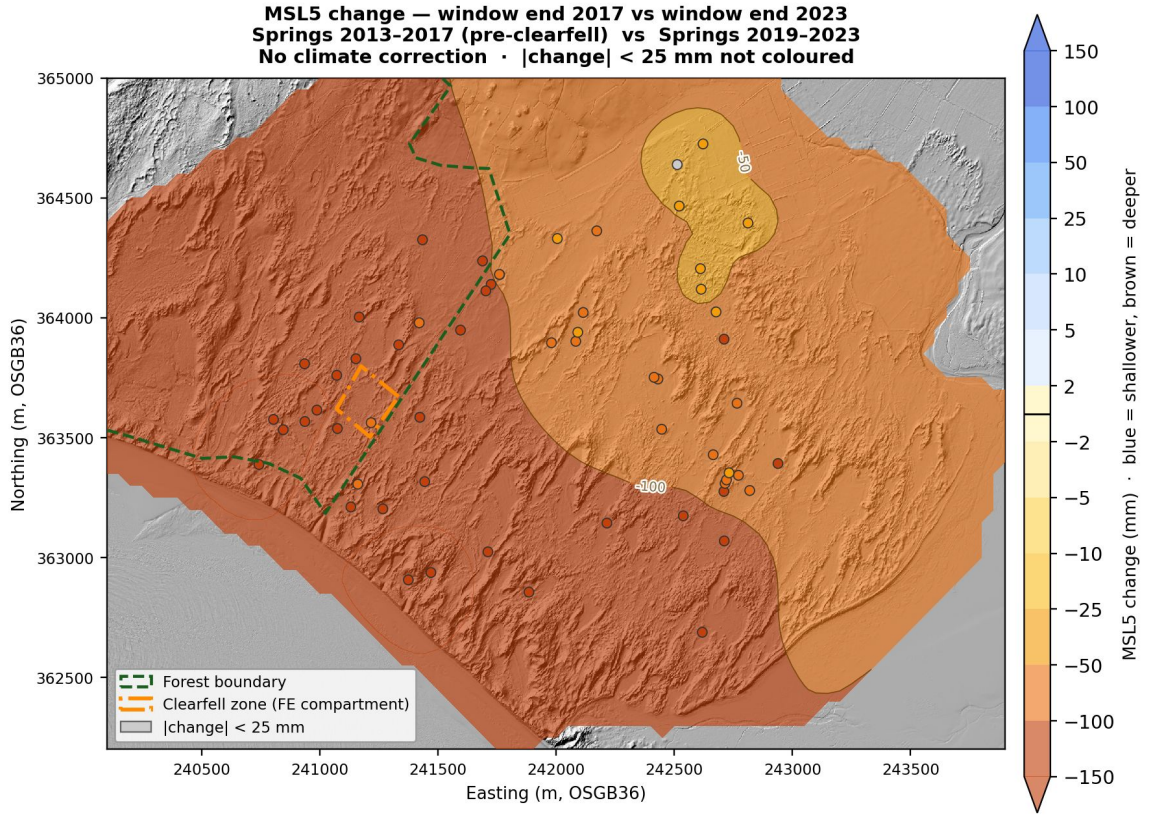



***Figure 72:*** *Observed MSL5 change: window end 2017 (springs 2013–2017, pre-clearfell) vs window end 2023 (springs 2019–2023). Raw difference in mm (positive = shallower); no climate correction or common-mode normalization. Network mean −467 mm (2017) to −572 mm (2023), below ground. Changes < ±25 mm (display threshold) not coloured; 58 of 59 wells exceed it and all deepen; the five-year window standard error is 82–181 mm by cluster (Section 4.8.4). n = 59 wells. IDW power 2.0, Gaussian σ = 50 m; well markers use same discrete banded colourmap as surface. Window end 2025 (springs 2021–2025) was examined but not adopted: it contains the two wettest springs of the record, and the resulting Forest Control excess is not separable from window composition (Section 5.7.5). The two C4 wells CEH13 and CEH14 are excluded separately, on a near-zero and a negative SSM drainage coefficient respectively, which makes their five-year window values unreliable. As a window-to-window difference, the map cancels time-invariant drivers — including the standing-forest drawdown of Figure 66; the forested zone's appearance is a climate-response artefact, not a canopy signal (Section 4.12). (Source: 20_msl5_change_2017_2023.png).*

### *Secular differential movement*

Figure 73 shows the secular differential movement of the spring water table across Newborough Warren: with the common climate signal removed, the forest mound holds its position while the lake-edge and coastal margins decline relative to the site as a whole. The quantity mapped is, for each well, the linear trend through its spring anomaly — the well's mean-spring level minus the site-mean spring level in the same year — over 2011–2025. Subtracting the site mean each year removes the shared climate forcing that dominates the raw record, so what remains is the rate at which each well is gaining or losing ground against the rest of the Warren. Because it is built from a per-well trend rather than a two-window

difference, the map is insensitive to the window-selection problem of Figure 72; it agrees closely with a two-window difference taken across the same 2011–2025 span, but not with the pre- and post-clearfell windows of Figure 72, whose fragility it therefore does not inherit.

The differential signal is spatially coherent but statistically uneven. Of the 74 wells mapped over 2011–2025, 13 reach individual significance under an AR(1)-corrected trend test cross-checked by a moving-block bootstrap; these are eight sinkers (CEH22, CEH21, CEH19, CEH17, CEH18, CEH4 and NW9 on the coastal margin, and CEH11 at the lake edge) and five risers in the western residual block (NW1, NW2, NW11, NW13 and WMC4). The forest-interior rise is coherent across neighbouring wells but individually sub-significant — each forest well is rising slowly against a noisy background — and remains individually sub-significant over the longer 2005–2025 window, in which none of the 12 significant wells of 77 is a C4 well. No C5 coastal-forest well holds up in the same way: all five decline against the site (−14.0 to −5.0 mm $yr^{-1}$, least at CEH16), so the inland C4 forest mound holds while the C5 wells nearer the eroding coast decline with their open-dune neighbours (Section 5.7.5).

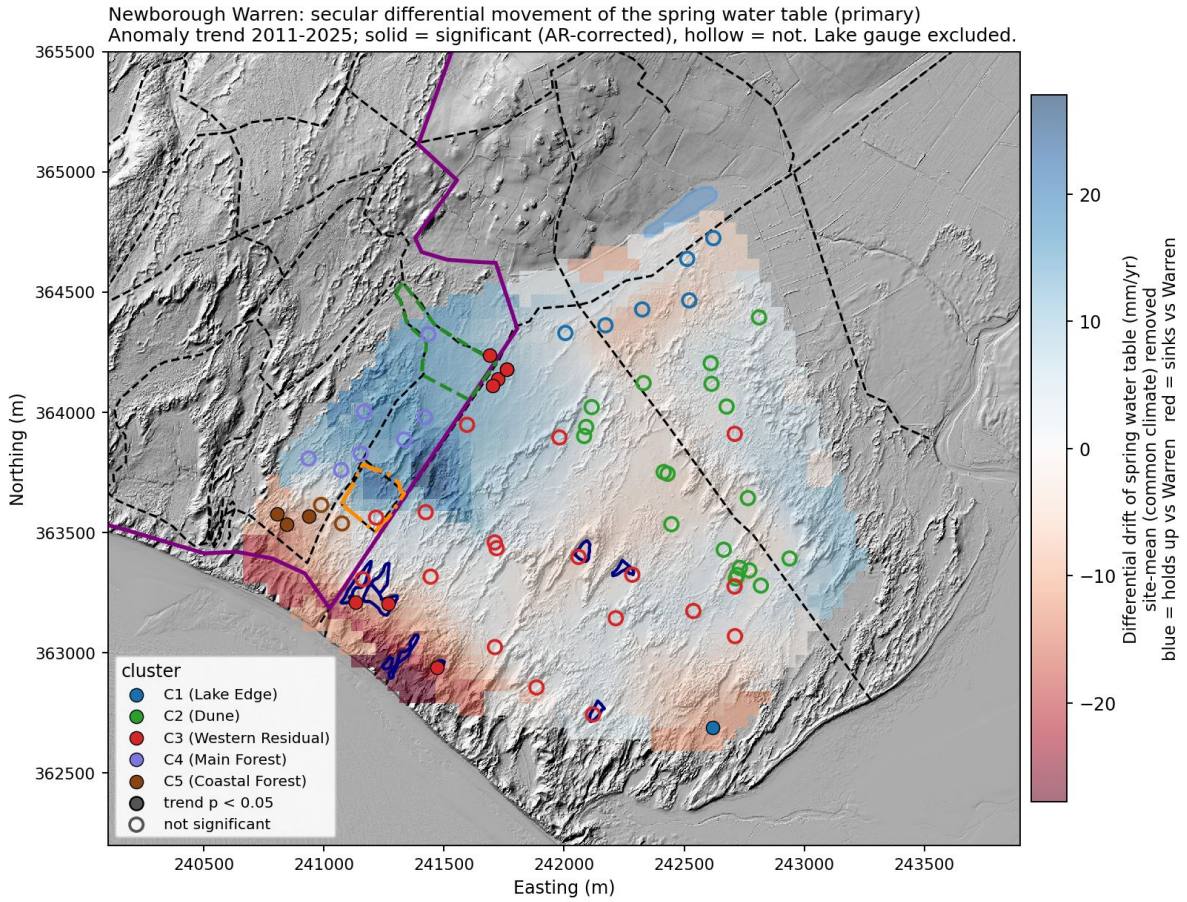



***Figure 73:*** *Secular differential movement of the spring (March, April, May) water table, 2011–2025. For each well the map shows the linear trend (mm yr $^{-1}$) of its spring anomaly, defined as the well's mean-spring level minus the site-mean spring level in the same year, so that the common climate signal is removed and only movement relative to the site remains. Read as a management diagnostic, it shows the direction and rate at which the spring water table is moving across the site over the study period — which areas are holding or gaining ground relative to the Warren as a whole, and which are losing it — the differential complement to the site-wide drying trend. Solid markers are significant under an AR(1)-corrected trend test corroborated by a moving-block bootstrap; hollow markers are not individually significant. Cool (blue) colours hold up relative to the site, warm (red) colours sink. CEH13 and CEH14 (SSM outliers) and the Llyn Rhos-Ddu lake gauge are excluded. (Source: 32_differential_movement_2011_2025.png)*

### *Absolute movement with the climate signal removed*

Figure 73 measures each well against the network mean, and over a wet-spring-lifted window that referencing can invert the plain reading: a well that is holding steady while the site mean rises appears to decline, and a slow-draining well reads as rising while falling more slowly than the rest. Figure 74 removes that ambiguity by mapping the absolute secular trend at each well — the shared climate signal taken out by regression against an external climatic-water-balance index, but with no re-referencing to the network mean, so red marks ground where the water table is falling once the weather is removed and blue where it is holding or rising. The construction and window are described in Section 3.8.1; the trend is fitted over 2005–2025, the longer window being primary here because the shorter window is dominated by wells whose records begin only after 2011.

The absolute map is more spatially specific than the differential one, although no well's trend reaches individual significance over 2005–2025 (none of the 30 mapped wells; Script 36), so its pattern rests on coherence across neighbouring wells. Once the weather is removed, most of the Warren is close to holding its position — the lake-edge, dune and main-forest clusters all sit within a millimetre per year of zero (cluster means C1 −0.3, C2 +0.2, C4 −0.6 mm $yr^{-1}$) — and the weather-independent losses are confined to the western and coastal margins. The coastal-forest cluster carries much the most negative trend (C5 ≈ −16.7 mm $yr^{-1}$), and the western residual cluster is also negative (C3 ≈ −6.9 mm $yr^{-1}$); the strongest individual driers are the western and coastal wells (NW9 −27.0, CEH3 −26.9, CEH4 −20.9 mm $yr^{-1}$). This is, in absolute terms, the coastal-margin decline that Figure 73 identifies as the fastest-losing part of the network: the forest interior is stable rather than recovering. Figure 73 answers which part of the Warren is gaining or losing ground against the rest and Figure 74 where the water table is rising or falling; the two agree on the coastal margin (Section 5.7.5).

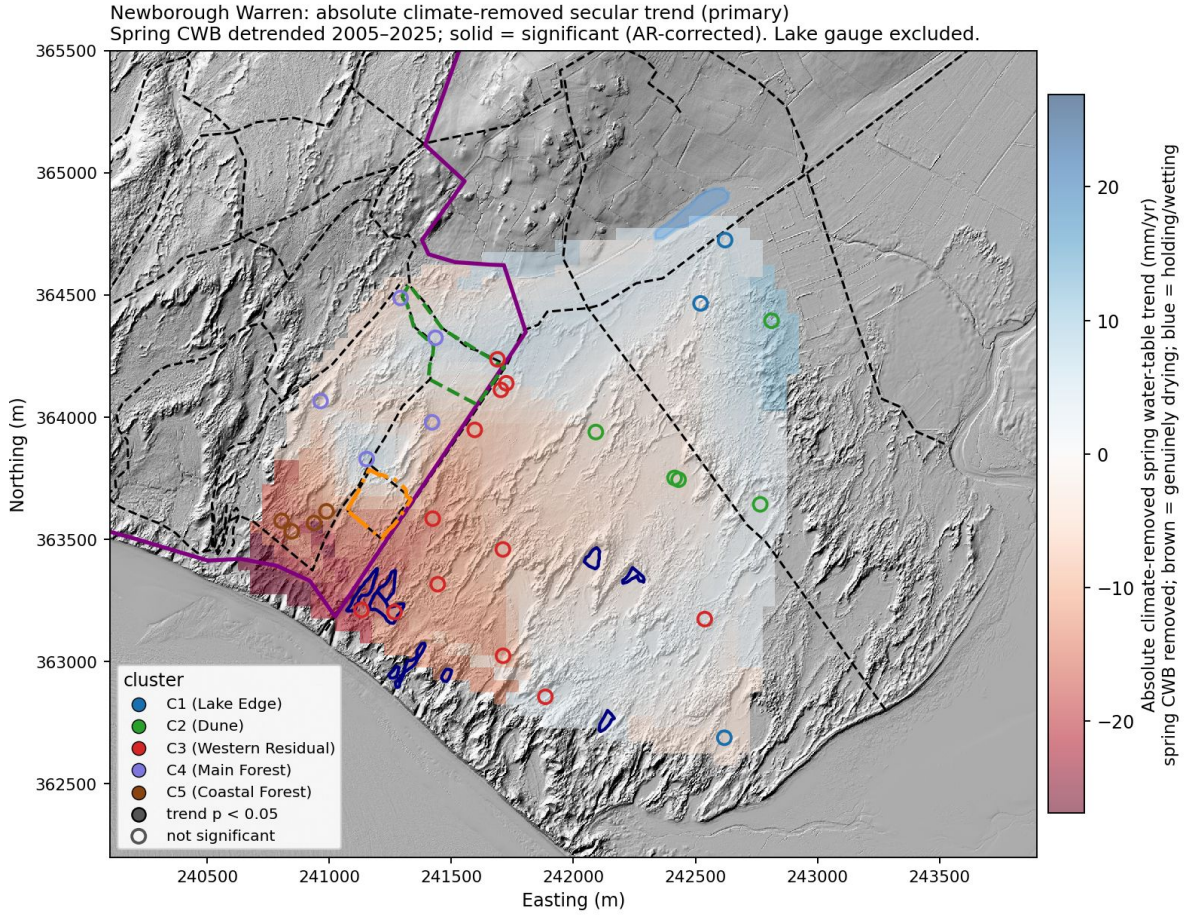



***Figure 74:*** *Absolute climate-removed secular trend of the spring (March, April, May) water table, 2005–2025. For each well the spring level is regressed jointly on an external spring climatic-water-balance index (RAF Valley) and a linear time term, and the fitted time coefficient is the secular trend in mm yr⁻¹, with the climate signal removed but without re-referencing to the network mean. Blue = rising or holding; red = falling. Solid markers would mark individual significance (AR(1)-corrected, bootstrap cross-checked); none of the 30 mapped wells reaches it, so all are hollow. Wells lacking a genuine pre-window record are excluded (Section 3.8.1). The coastal and western margins carry the largest secular losses; the interior, including the forest, holds once the weather is removed. (Source: 36_absolute_climate_trend_2005_2025.png.)*

### *The climate-response envelope*

Where Figure 73 measures slow secular change, Figure 75 measures climate sensitivity: how far each part of the network moves between the driest and wettest springs the record contains, and Figure 76 how close the dry-year water table already lies to the SD15b / SD16 community-mean levels. The amplification map (Figure 75) addresses the first. Between its antecedent-dry and antecedent-wet spring extremes the site-mean water table swings approximately 0.77 m; expressing each well's swing as a ratio to that network-mean swing, with the common-mode removed, gives a relative amplification field independent of window choice, because it compares true end-members rather than adjacent windows. The forest interior amplifies the shared swing to about 1.7×, while the lake edge damps it to about 0.6×, with the open dune close to unity. This is the behaviour behind the C4 deepening of Figure 72 (Section 5.7.5). The amplification correlates most strongly with the atmospheric-draw coefficient $\beta_2$ ($r = +0.79$, $n = 66$), which survives control for elevation and exclusion of the forest clusters.

The second map (Figure 76) maps the dry-year spring water-table depth — the mean spring level across the dry extreme years (2011, 2012, 2019). Spring is used in preference to the summer minimum because the spring record is near-complete whereas summer-minimum readings are sparse; the choice is deliberately conservative, since spring is the seasonal high and therefore the shallowest the water table reaches in those years. On this map the Curreli et al. (2013) SD15b (wet slack, 0.61 m) and SD16 (dry slack, 0.98 m) community-mean levels are contoured as reference levels only: they are means of the four-year mean annual minimum, not of a spring level, so a well beyond them is not a threshold crossing. Because the summer minimum lies deeper still than the spring high, where the forest interior and western dune sit beyond SD16 even in spring the water table lies deeper than the dry-slack community mean for the whole of those dry years. Together the two maps separate the two questions that the single two-window map of Figure 72 conflated — how sensitive each area is to climate, and how close it already sits to the slack community-mean levels in dry years — neither of which depends on an arbitrary choice of comparison window.

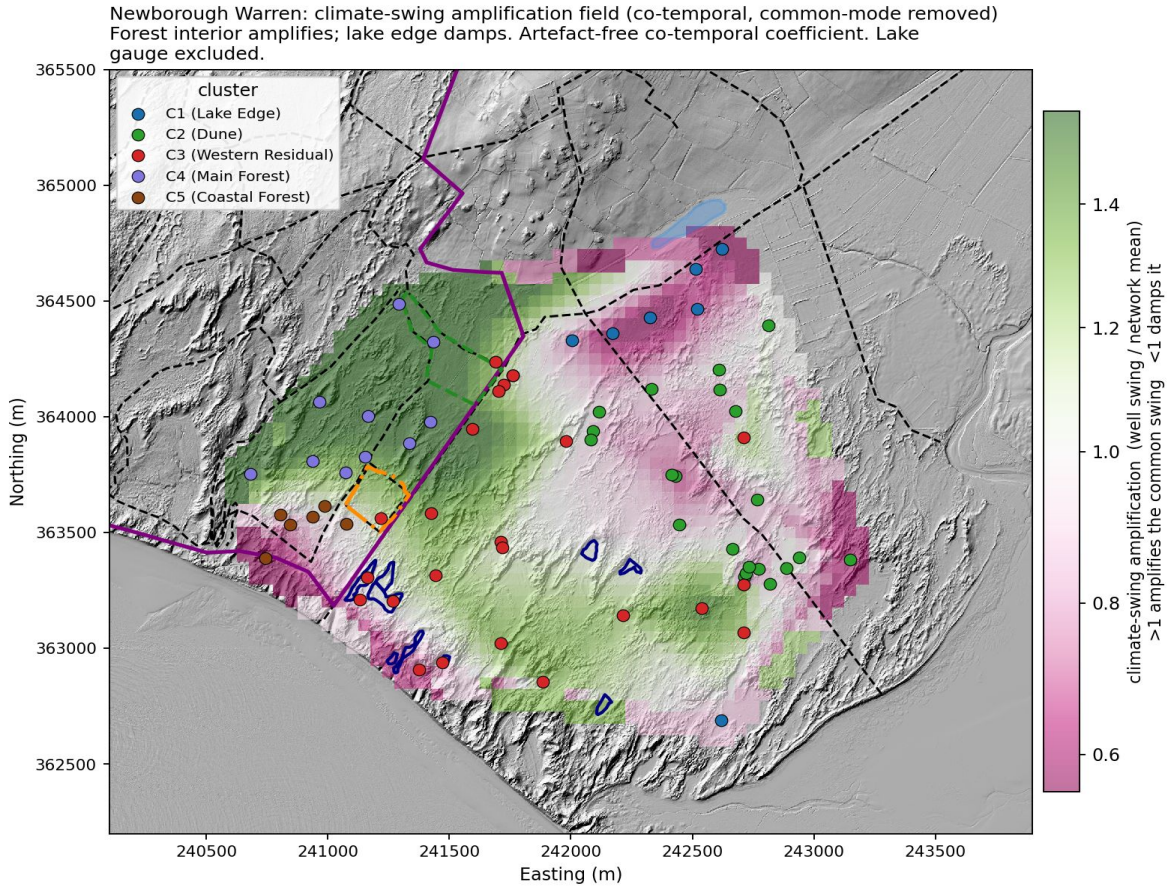



***Figure 75:*** *The climate-response envelope of the spring water table. Relative amplification field — each well's dry-to-wet spring swing as a ratio to the network-mean swing (≈0.77 m), with the common-mode removed; values above one amplify the shared climate swing, below one damp it. Wells with full dry-extreme coverage build the interpolated surface; a well below the confidence threshold would be drawn off-surface as an open triangle, and none is in this partition. The Llyn Rhos-Ddu gauge excluded. (Source: 33_amplification_field.png).*

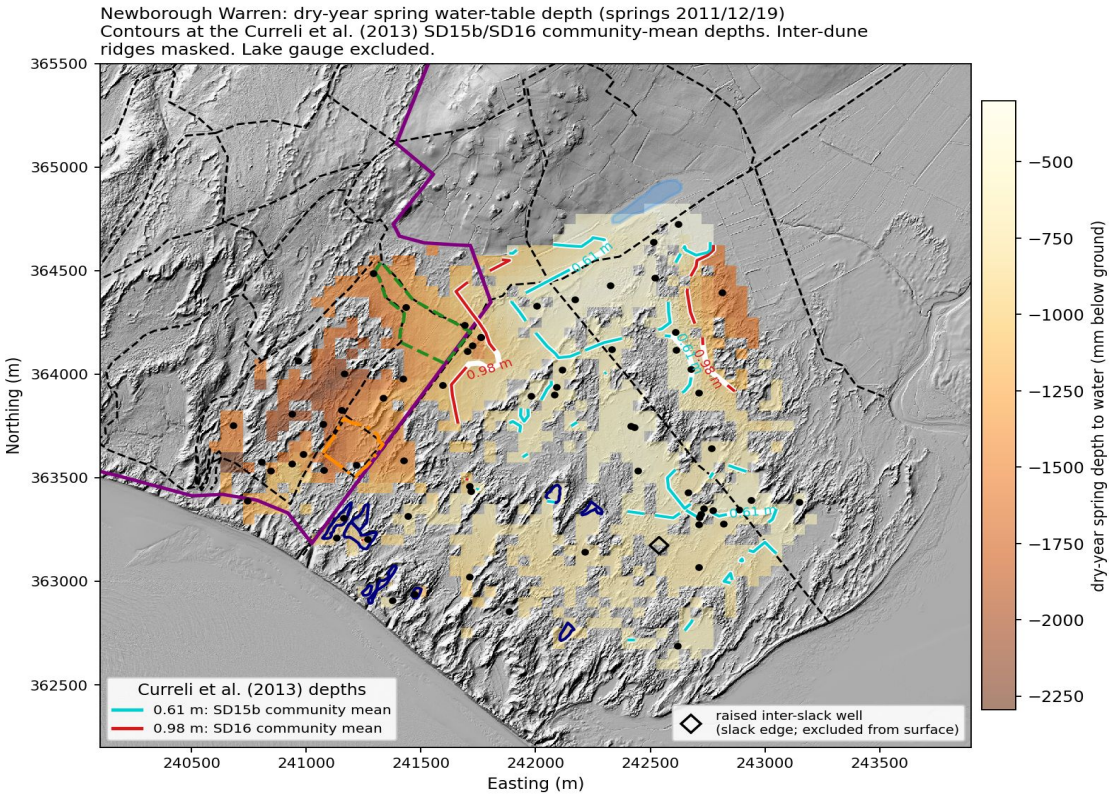



***Figure 76:*** *The climate-response envelope of the spring water table: Dry-year spring water-table depth — mean spring level across the dry extreme years (2011, 2012, 2019), with the Curreli et al. (2013) SD15b (0.61 m, wet slack) and SD16 (0.98 m, dry slack) community-mean levels contoured for reference. CEH13, CEH14 and the Llyn Rhos-Ddu gauge excluded. The single raised inter-slack well (CEH10), sited on high ground between slacks and flooding only when the neighbouring slack overtops, registers the deep slack-edge drawdown directly; it is shown as a slack-edge marker but excluded from the slack-floor surface, which would otherwise smear its raised ground depth across the adjacent slacks. (Source: 33_dry_spring_depth.png).*

## Summary of pattern

Figure 70 predicts a large net loss across the forest interior under the parametric annual-mean drivers, and the 2017/2023 window comparison (Figure 72) shows a broadly deeper aquifer with the largest declines at the south-western coastal margin; but the absolute site-wide deepening is specific to that window pair (Section 5.7.5). Figure 73 shows, with the common climate signal removed, the coastal-margin decline as the largest secular movement and the forest interior holding its position relative to the rest of the Warren. The per-cluster rates quantify this split: C4 Main Forest rises at a cluster mean of +13.7 mm yr $^{-1}$ relative to the site (range +6.9 to +18.9 mm yr $^{-1}$ across the seven mapped C4 wells; none individually significant after AR(1) correction); C1 Lake Edge declines at −5.6 mm yr $^{-1}$ (CEH11 significant, p = 0.048); C5 Coastal Forest at −8.5 mm yr $^{-1}$ (NW9, CEH17 and CEH19 significant, p = 0.017, 0.046 and 0.020); and C2 Dune is approximately neutral (−0.6 mm yr $^{-1}$). CEH22, at the south-western coastal margin, is the most rapidly declining well in the

network at −27.8 mm yr⁻¹ ($p < 0.001$). The site-mean spring trend over 2011–2025 is +19.8 mm yr⁻¹ (AR-corrected $p = 0.284$; bootstrap CI −3.4 to +43.2 mm yr⁻¹), with the exceptionally wet springs of 2021 and 2024 in the window; Figure 73 shows the forest rising relative to the network, not in absolute terms.

Figure 75 shows the forest amplifying the dry-to-wet climate swing more strongly than any other zone (C4 mean 1.66×, range 1.31–2.32×, the upper end being CEH14; 1.56× with CEH13 and CEH14 excluded; C1 lake edge 0.64×; open dune near unity). The amplification correlates positively with $\beta_2$ ($r = +0.79$, $n = 66$); the per-well climate-sensitivity coefficient of Script 35, on a co-temporal normalization, gives $r = +0.65$ against $\beta_2$ and −0.41 against $\beta_3$ ($p < 0.001$, $n = 64$) (Section 5.7.5).

The felled well FE2 (inside the December 2017 clearfell boundary) registers a spring differential trend of +8.8 mm yr⁻¹ (Figure 73) and a co-temporal amplification of 1.21× (Figure 75). Both values lie notably below the unfelled C4 interior mean (+13.7 mm yr⁻¹; 1.66×). FE2's wet-year spring depth (−1.00 m) rises less than the surrounding unfelled C4 reference wells (mean wet-year depth −0.54 m), and its swing (763 mm) is substantially smaller than the C4 mean (1,135 mm). The spring drought floor at FE2 (−1.77 m in dry years) is slightly deeper than the C4 reference mean (−1.67 m). FE1, approximately 46 m outside the felled boundary, matches the unfelled interior on the differential trend (+15.3 mm yr⁻¹) and sits just below it on amplification (1.54×). Their reading against the clearfell result of Section 4.6 is in Section 5.5.1.

The reconciliation of the parametric prediction with the observed change — the window-sensitivity of the absolute deepening, the differential movement of Figure 73, and the climate amplification of Figure 75 — is taken up in Section 5.7.5.

### 4.12.1 Driver validation and comparative footing

The differential pattern above identifies where the network is moving, but not how much of that movement each modelled driver accounts for. Two diagnostics address this, and their shared conclusion is cautious: the network can bound the drivers' relative footing, but cannot independently confirm the coastal-retreat amplitude from the monitoring record alone.

A per-driver scale-factor regression (Section 3.8.1) fits the climate-corrected per-well change against the modelled coastal and clearfell fields as separate regressors, together with a free spatially uniform intercept that absorbs the site-wide background decline present even at unfelled control wells. The scale factors are not uniformly indeterminate. In the window where each driver acts, its factor is distinguishable from zero but not from unity. The coastal factor is 2.95 (95% CI 0.18 to 5.71) over the pre-intervention 2006–2012 window, and the clearfell factor is 2.28 (0.65 to 3.90) over the post-clearfell 2018–2025 window. Over the full 2005–2025 record, by contrast, the coastal factor is 0.48 (−0.02 to 0.99), spanning zero narrowly, and the clearfell factor, −0.47 (−3.34 to 2.41), is unresolved. The uniform intercept is consistently well-determined and large — a decline of 520 mm over 2006–2012 (95% CI

−589 to −451) and a rise of 269 mm over 2018–2025 (223 to 316) — so the dominant secular movement in the network remains common-mode rather than spatially structured by any single modelled driver. The regression explains a modest fraction of the between-well variance ($R^2$ 0.09 and 0.11 in the two shorter windows, 0.35 over the full record, and 0.25 in the broadleaf-covariate variant). In that variant the broadleaf-restock covariate is −3.52 (−6.23 to −0.80), and the clearfell factor, 1.84 (0.82 to 2.85), keeps its footing when the covariate is carried. The full-record window is fitted on 20 wells against 40 in the two shorter windows, because it requires a well to carry a trend across the whole span.

The intervals are wide — a factor of three in the coastal case — and the coastal signature is collinear with easting (r = +0.67 to +0.73 across the three windows); what limits the test is discussed in Section 5.7.5.

A complementary analysis (Section 3.8.1) places the drivers on a common footing over the 2005–2025 horizon, using observed BACI anchors where available (clearfell +108.2 mm; on-site scraping +128.1 mm) and modelled fields elsewhere. The off-site scrape peak, in particular, is taken from the modelled drain cone at the distance of the nearest uphill long-record well, because the raw WMC3 step there is not resolved (Section 4.6.3). The unexplained residual decline of Section 4.10.2 is carried on the same footing at −8.5 mm $yr^{-1}$, applied uniformly to every well by assumption and as a central estimate the record does not resolve: the open-dune residuals it averages span 14.1 mm $yr^{-1}$, from −0.5 (C3) to −14.6 mm $yr^{-1}$ (C2), so on this decomposition it is not spatially uniform. The panel's fitted constant is not used for it: that quantity is not separately identified from the cumulative-water-balance covariate, and only their sum is recovered. Expressed as a local water-table depth change, the coastal driver is the largest of the spatially structured terms: a lowering of up to approximately 626 mm at the dune toe over the twenty-year horizon, tapering inland over a reach of roughly 900 m; its depth-against-distance profile, alongside those of the forest and scraping drivers, is shown in Figure 83 — which draws only terms that vary with distance. On the area-integrated volume and the SD15b / SD16 crossing measures (Figure 78) the ordering inverts: the unexplained residual decline is the largest single term, because it is applied across the whole site by assumption rather than to a fringe, while the coastal drawdown — deeper than anything else but confined to the western dune field — is second. The inversion rests on a central estimate for the residual term, and on its being applied uniformly, not on a resolved rate, and is stated on that footing. Dune scraping produces a net site-wide loss because its off-site drainage cone draws down the water table across the area surrounding each cut by more than the on-site slack rise recovers; integrated across the site, the net scraping effect is of order 25% of the coastal drawdown in magnitude (Section 5.4.3). Sea-level rise, evaluated on the near-term five-year projection rather than the twenty-year horizon of the other terms, is indicative only and small by comparison. Both diagnostics are first-order and do not close a water budget; the modelled coastal drawdown remains a projection anchored on the observed $\delta_0$ and unconfirmed spatially by the network, while the off-site scrape drawdown, and with it the net scraping loss, is modelled throughout: no off-

site effect of scraping is resolved in the record (Section 4.6.3). The ranking on the two footings is taken up in Section 5.8.2. The expanding-window trajectory of the implied coastal decline rate — the one genuinely independent test of the coastal signal — is shown in Figure 77, and the comparative footing of all drivers in Figure 78.

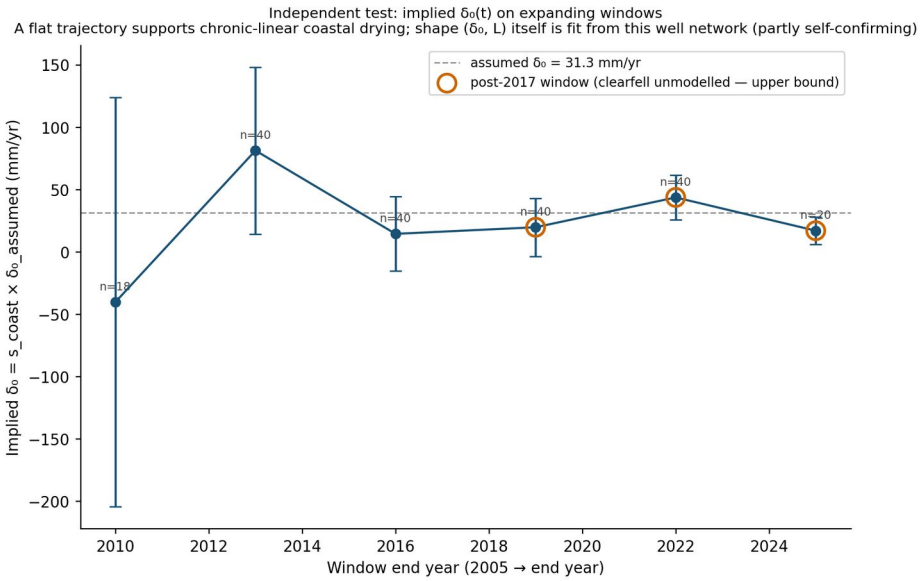



***Figure 77:*** *Independent test of the chronic coastal-drying assumption: implied coastal decline rate $\delta_0$ from the coast-only scale-factor regression re-run on expanding windows (2005–2010 through 2005–2025). For each window the implied $\delta_0$ is the fitted coastal scale factor multiplied by the assumed rate (the Section 4.10.2 headline $\delta_0$, dashed line). A trajectory that stays above zero throughout is the signature a steadily propagating erosional drawdown would produce — the coast-to-inland gradient steepening year on year; the value relative to the 31.3 mm yr⁻¹ line then indicates whether that drawdown is proceeding at, above, or below the assumed chronic rate. A dip to zero indicates no detectable coastal signal on this line (a flat head difference, consistent with static substrate geometry rather than a moving erosional boundary), and a value below zero would indicate a reversing gradient that refutes chronic coastal drawdown — though in the short early windows a near-zero value more likely reflects insufficient statistical power to resolve the signal than a genuine absence of it. The three fully pre-clearfell windows (ending 2010, 2013, 2016) are the only clean coast-only reads but are individually too noisy — the earliest falls within the 2006–2012 drought — to establish a stable rate; the windows ending 2019 or later, ringed in orange, span the December 2017 clearfell without a term to absorb its recovery signal, so their implied $\delta_0$ is an upper bound on the coastal contribution rather than a direct estimate. The test is therefore inconclusive on the amplitude: it neither confirms nor refutes the modelled coastal rate, whereas the windowed scale-factor regression in the text separates the coastal field from zero in the pre-intervention window without resolving its size. (Source: 37_implied_delta0_trajectory.png)*

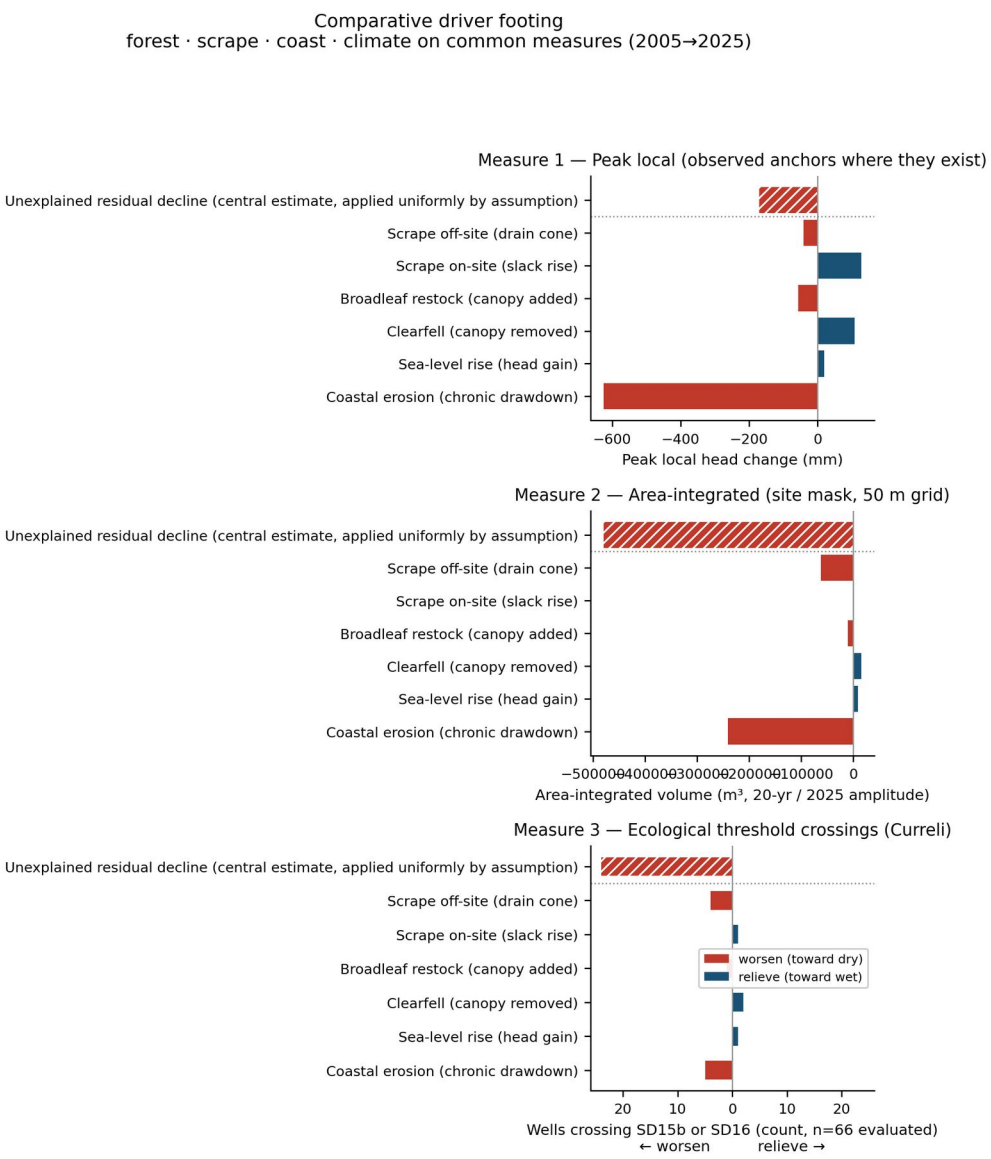



***Figure 78:*** *Comparative footing of the drivers over the 2005–2025 horizon, on three common measures — peak local head change (top, mm), area-integrated volume (middle, m³; the head field integrated over each footprint with a representative specific yield), and crossings of the slack community-mean levels (bottom; wells moved across the Curreli et al. (2013) SD15b wet-slack and SD16 dry-slack community-mean levels). Amplitudes are the observed BACI anchors (clearfell and on-site scraping), the modelled Script 20 fields (coastal drawdown and sea-level rise, the off-site scrape cone, whose peak is read at the nearest uphill well's distance, and broadleaf restock), and the unexplained residual decline, applied uniformly by assumption and carried at the −8.5 mm yr⁻¹ central estimate rather than a resolved rate. Coastal drawdown is modelled and spatially unconfirmed by the network; the off-site scrape drain is modelled throughout (Section 4.12.1), the raw step at WMC3 being unresolved (Section 4.6.3). The residual term is shown apart from the spatially structured drivers (hatched, above the separator) as the background decline not attributed to a single mechanism. Fields are superposed linearly, an upper bound in overlap zones that does not close a water budget. (Source: 37b_driver_footing.png)*

| Driver | Equivalent depth (site-mean, 20 yr) | Relative to coastal retreat | Character |
|---|---|---|---|
| Unexplained (uniform) | −170 mm | 199% | assumed uniform |
| Coastal retreat | −85 mm | 100% | progressive |
| Dune scrape (net) | −21 mm | 25% | redistributive |
| Clearfell (relief) | +6 mm | +7% | step |
| Broadleaf restock | −4 mm | 5% | progressive |

*Table 25: Site-wide footing of each water-table driver. Equivalent depth averaged over the 949 ha site at a common twenty-year horizon, expressed as a percentage of the coastal-retreat term — the reference because it is the largest term the record resolves, not because it is the largest. The unexplained residual row is the balanced observed decline less the modelled coastal gradient, averaged over the three open-dune clusters, at −8.5 mm yr⁻¹ over the horizon and applied uniformly to every well by assumption; it is a central estimate and not a resolved rate, and the percentage against it should be read as an order of magnitude rather than a ratio. Character is the mechanism type carried in the source: assumed uniform, progressive (accumulating), step (settling to an equilibrium) or redistributive (an on-site gain against an off-site loss). Of the terms ranked, the coastal drawdown and the off-site scrape drawdown are modelled, the clearfell step and the on-site scrape rise are observed, and the residual term is applied uniformly by assumption. (Source: 37b_driver_footing.csv)*

## 4.13 Scenario Analysis

The per-well SSM coefficients derived in Section 4.9 were used to project equilibrium water table responses under prescribed climate and forest management perturbations, following the framework of Section 3.8. Six scenarios were evaluated: baseline; UKCP18 2050s and 2080s climate trajectories; full clearfell; 50% thinning; and broadleaf conversion. Scenario difference maps were computed as Δh = h(scenario) − h(baseline) for each well, season and cluster. The full per-scenario, per-season, per-cluster Δh output is available through the Newborough Warren Hydrogeological Modelling Suite (Hollingham, 2026), an interactive web-based scenario viewer that renders the per-well Δh field across the site using IDW interpolation, with a complementary depth-below-surface view masked to hydrologically meaningful slack positions and overlaid with the Curreli et al. (2013) SD15b / SD16 community-mean levels. The cluster-level results are summarized below.

### 4.13.1 Climate Scenario Projections

Climate scenario responses are reported as the sustained change in water level (Section 3.7.5). Under UKCP18 2050s forcing the site-wide mean annual level falls by 0.163 m, from 0.088 m at C1 to 0.201 m at C3; under the 2080s forcing by 0.293 m, from 0.159 m at C1 to 0.362 m at C3. C4 is not projected: its drainage coefficient is not identified on the record. The response is dominated by summer: site-wide summer levels fall by 0.216 m (2050s) and 0.394 m (2080s), reaching 0.447 m at C3. Winter levels fall too, by 0.107 and 0.188 m site-wide, everywhere except C1. The extra winter rain under RCP8.5 raises the winter forcing, but the summer deficit, carried forward by the slow return of the water table to its mean,

outlasts it except where the mean-reversion time is shortest. The 2080s change in the annual water balance lies beyond the observed record; the 2050s change is about the 1989–96 dry stretch (Section 4.14). The summer-dominated signature is discussed with the observed record in Section 5.7.1.

The same projection, applied to the five-year mean spring water level (MSL5; Section 4.8.3), gives a smaller response than the summer level wherever it is projected, by a margin that depends on the mean-reversion time. In the open dune it is small: +23 and −23 mm at C1 and C2 by the 2050s, +51 and −32 mm by the 2080s, against summer falls of 171–204 and 315–371 mm. At C3 and C5, where the mean-reversion time is longer, a summer deficit reaches the following spring: −92 and −80 mm by the 2050s, −154 and −137 mm by the 2080s, about a third of the summer response at C3 and a little under half at C5 (Figure 79). No reference value is defined for a spring mean (Section 4.8.3), so the projected MSL5 shifts are read against the metric's own structure. In the open dune they are smaller than the spacing between clusters on MSL5; at C3 and C5 by the 2080s they are comparable to it. The summer-level deterioration is read against the reference values directly (Section 4.8.1). The implication for the binding ecological constraint is taken up in Section 5.2.5.

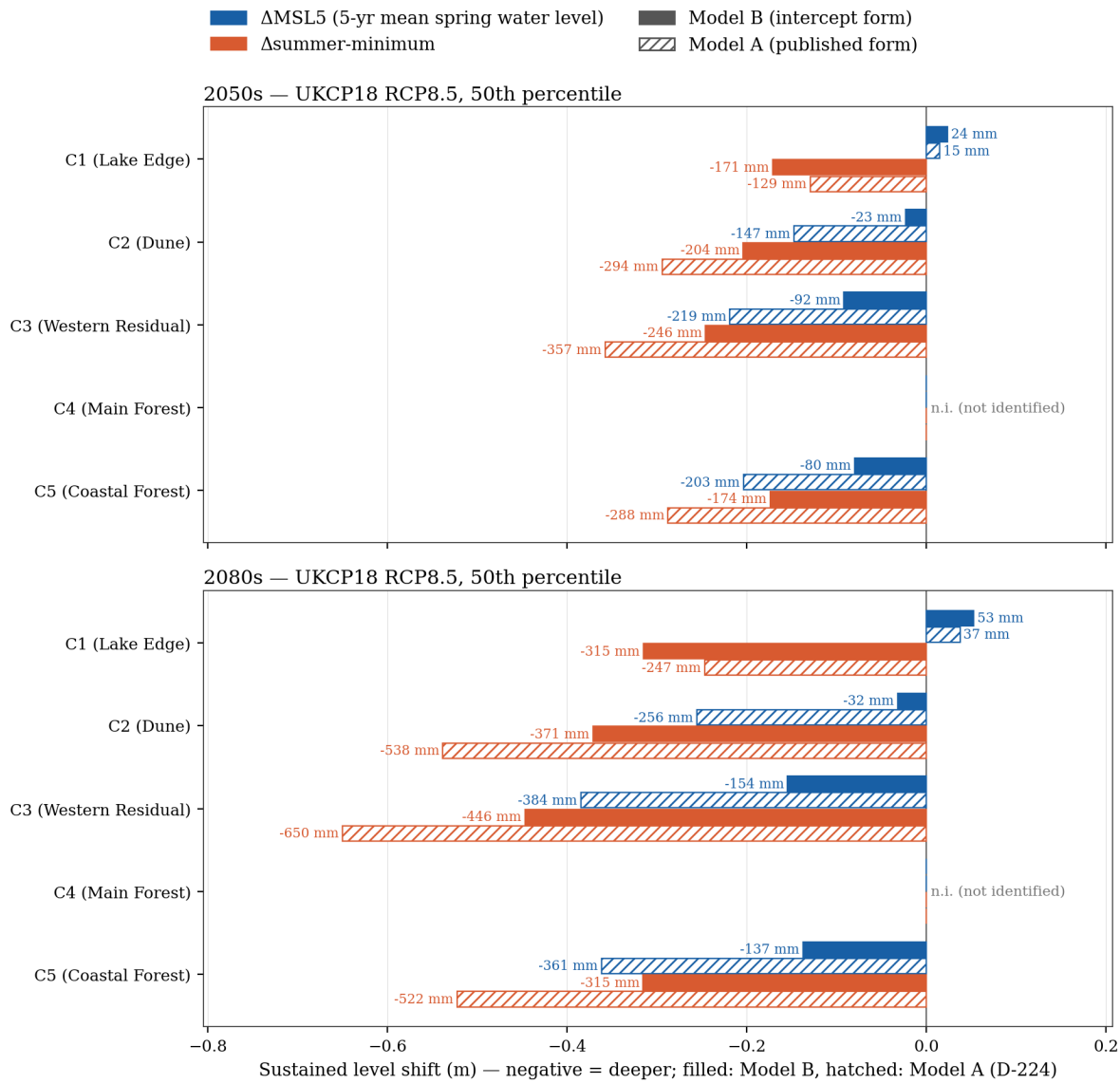



***Figure 79:*** *Projected sustained shift in the five-year mean spring water level (ΔMSL5, blue) and in the summer water level (orange), by cluster, under UKCP18 RCP8.5 forcing: 2050s (upper panel) and 2080s (lower panel). Both are the mean, over the same wells, of the per-well sustained level response, under Model B (filled; the intercept form, Section 3.7.5) and Model A (hatched; the published form); C4 is not projected under either (n.i.: its intercept-form drainage coefficient is not identified). Negative is deeper. Under Model B the spring metric moves far less than the summer level in the open dune (C1, C2), and about a third as much at C3 and a little under half at C5, where the longer mean-reversion time carries a summer deficit into the following spring. (Source: fig_msl5_vs_summer_min_projection.png)*

### 4.13.2 Forest Management Scenarios

Forest management scenarios act on the two forested clusters (C4 and C5) and on the two C3 wells under canopy, NW11 and WMC3, and are anchored to the one felling the record observed, the December 2017 clearfell at WMC3. The felled well's own fits before and after the felling (Script 10e) show no rise in recharge sensitivity ($\beta_1$ 2.56 → 2.52) and a fall in

atmospheric draw ($\beta_2$ 1.95 → 1.69), 4.8% beyond the fall at the climate-control wells over the same years. Full clearfell is therefore modelled as that change in draw ($\beta_2 \times 0.952$ at forest wells) with no interception recharge term, and broadleaf conversion as the seasonal $\beta_2$ profile of Section 3.8, lower in the leafless months and higher in full leaf, again with no interception term. Fifty percent thinning is modelled as half the clearfell change. That is an upper bound, and the record cannot test it: after thinning the remaining trees and the understorey take up part of the water the felled trees used, so the saving, though large in the first year, shrinks as the stand recovers (Gebhardt et al., 2014). Each scenario is reported as the sustained change in water level (Section 3.7.5), under Model B and Model A, as the climate projections are.

At WMC3 the sustained clearfell rise is +0.061 m under Model B and +0.101 m under Model A, either side of the +0.108 m step the BACI comparison measures against the Forest control (95% CI −0.021 to +0.237 m with autocorrelation-robust errors; Section 4.6.3), which already carries the 2015 scrape. Clearfell raises the water table by +0.038 m (Model B) and +0.065 m (Model A) at C5, and by +0.211 m under Model A at C4 (the mean of the per-well responses; +0.346 m on the cluster-centroid coefficients, Script 21), where Model B's drainage coefficient is not identified on the record. Thinning gives at most half of each. Site-wide means are small, because only 14 of the 66 reference wells lie in the two forest clusters. The scenarios are evaluated well by well and carry no flow between wells, so they cannot show whether a change in the forest reaches the open dune. The drawdown the canopy imposes on its surroundings has an e-folding length of about 220 m beyond the forest edge (Section 3.8.2), which sets the scale over which such a change could travel.

Broadleaf conversion is reported as a range, because no conversion has been observed here and the felling cannot test a broadleaf canopy's interception. Without an interception gain, the higher draw of a canopy in full leaf in summer outweighs its lower draw in the leafless months, and the water table falls: by 0.037 m at C5 and 0.118 m at C4 under Model A, and by 0.021 m at C5 under Model B. If the lower interception of a broadleaf canopy (15% of rainfall, against 24% under pine) were returned as recharge, it would rise instead: by 0.125 m (Model B) and 0.308 m (Model A) at C5, and by 0.498 m at C4 (Model A). Catchment modelling finds that replacing conifers with broadleaves raises recharge and groundwater levels, most of all on sandy soils (Sonnenborg et al., 2017), which places the outcome toward the upper end of the range; neither end is tested on this record.

Earlier versions of this report returned the 24% of rainfall the pine intercepts (Freeman, 2008) as recharge when the canopy was removed. Run as a sustained change, that form would raise WMC3 by +0.475 m under Model B and +1.021 m under Model A, three to ten times what the felling produced. The felled well's recharge did not rise: its $\beta_1$ after felling was 0.98 of its value before, net of the climate controls (standard error 0.13), where a full local return of the intercepted rain would raise it to 1.32. The size of the felling accounts for this. The compartment covers 4.2 ha, a disc of radius 115 m, narrower than the drawdown

reach, so extra recharge inside it spreads into the forest around it. At WMC3 a disc of that size keeps about 15% of a full local gain (7–28% across the range of conductivity), a rise in $\beta_1$ of about 5% that the record cannot distinguish from none (Script 20). The Edge wells beside the compartment would receive about half that share, and they rose +31 mm, not significantly (Section 4.6.3). The record therefore rules out a full local gain but is consistent with a gain that has dispersed. Grazing kept the felled compartment open, so regrowth does not explain the difference. And if felling had exposed the ground to drying, draw in the canopy-on months (May–October) should have risen after felling; at WMC3 it changed by −1.7% net of the climate controls (Script 10e). On one felled well, the record does not support a loss of the canopy's shielding.

A felling wider than the drawdown reach would keep the gain the compartment lost, so the interception form marks the upper end of the clearfell range rather than a result to discard. Returned in full at the scale of a cluster, the intercepted rain would raise C5 by 0.427 m (Model B) and 0.984 m (Model A), and C4 by 1.853 m (Model A). Under Model A these exceed the open-dune end state set out below, which bounds what removing the canopy can do, so that end state is the tighter upper limit. Field studies of clearcutting report rises of the order of the measured step: 0.13 m on wet mineral soils, where interception fell from 31% to 10% of rainfall (Roy et al., 2000), and 0.29 m in pine flatwood uplands when the whole area was harvested, with no significant upland rise when only the wetlands within it were cut (Sun et al., 2000). The +0.108 m at WMC3 lies within that range; the interception form under Model A lies well beyond it.

Felling has not undone the plantation's long-run effect. If each forest cluster took the coefficients of the open dune it resembles, C4 those of C2 (the high ground) and C5 those of C3 (the low coastal ground), its equilibrium level would stand +0.898 m (C4) and +0.498 m (C5) higher under Model B, and +0.785 m and +0.490 m under Model A. This is a bound, not a prediction: a cluster's equilibrium carries its position and topography as well as its canopy. It is of the same order as the 0.4–0.7 m by which forest floors stand further above the modern water table than open-dune floors (Section 5.7.9), a contrast that has not reversed at the wells felled in 1993. Eight years after the 2017 felling the recovery at the felled well is about a tenth of a metre; the drawdown the plantation imposed is several times that and, on this record, returns slowly if at all.

## 4.14 Out-of-sample tests against earlier records

Every result in this study is fitted to the 2005–2026 record and tested inside it. Two earlier observations of this aquifer exist: a dipwell record held by Natural Resources Wales for 1989–96 at nine wells of the present network, and Ranwell's 1951–53 readings. Both are run as hindcasts from the modern per-well coefficients with no refitting (Scripts 39 and 44); what each can and cannot establish is set out in Sections 5.7.8 and 5.7.9.

### 4.14.1 The 1989–96 record

The Natural Resources Wales record covers nine wells of the present network between May 1989 and April 1996, eighty-two monthly observations per well, ending nine years before the modern record opens. Data provenance and permission: the records were supplied for this study; access is covered by the Environmental Information Regulations 2004, and re-use and republication are governed by the Open Government Licence v3.0, under which Natural Resources Wales has confirmed these records may be reproduced with the attribution “Contains Natural Resources Wales information © Natural Resources Wales and database right. All rights reserved.” The derived results below are this study's own; the underlying records are not.

The test uses each well's own coefficients from the modern per-well fit, driven by the RAF Valley climate record for 1989–96, with no refitting and no parameter free to absorb the difference between the epochs. Historic depths are reduced to the present ground surface through each well's measured upstand, so the two epochs sit on one datum. Simulation begins from an equilibrium head and the initial condition is immaterial: the spin-up is long enough that varying the starting level across the probed range changes no well's result at four decimal places. Of the thirteen historic well codes, twelve are matched to present wells and one remains unidentified and is excluded; at one well, fifteen of eighty-two months sit at the pipe base and are treated as left-censored.

The reduction can be checked against a published figure that the analysis did not use. The annual range of water-table elevation at Newborough over 1989–95 has been reported independently as 0.75 m, with a standard deviation of 0.27 m (Davy et al., 2010); the series reduced here returns 0.725 m with a standard deviation of 0.255 m across fifty-one well-years, an agreement to 0.025 m on a quantity neither study derived from the other.

Across the nine wells the correlation between observed and predicted monthly levels has a median of 0.848 and runs from 0.652 to 0.943. Nash–Sutcliffe efficiency is negative at five of the nine wells while the bias-removed efficiency is 0.19 to 0.86 at eight wells and −0.10 at the well under forest in both epochs. The mean residual — observed level less predicted — is −0.035 m across the nine wells and +0.109 m across the six cleanest, so the climate-only prediction sits within about 0.11 m of the observations in both directions.

The same nine wells test whether a correction fitted on the modern record carries to the earlier decade. At each well a quantile map was fitted between the hindcast and the well's own 2005–2026 readings, then applied unchanged to 1989–96 (Script 50, E18). By a rule fixed before the run, the correction transfers if the median RMSE falls by at least 0.05 m, the map improves a majority of wells, and the median absolute bias does not grow. On each well's own coefficients the median RMSE falls from 0.28 to 0.26 m, but the map improves only 4 of the 9 wells: it helps where the hindcast is furthest off and hurts where it is already close. On cluster coefficients it improves 6 wells, but the median RMSE falls by only 0.04

m. By that rule the correction does not transfer. The months at the pipe base are left out, so the test does not reach the driest levels.

The epoch shift compares each well's mean over 1989–96 with its mean over the modern record. All nine are negative, from −0.135 m to −0.835 m with a median of −0.589 m: the water table of the early 1990s stood lower than it stands today at every well. Six of the nine wells were open ground in 1989 and are open ground now, and they carry the cleanest signal: a median correlation of 0.903, a median bias-removed efficiency of 0.742, and a median epoch shift of −0.347 m. The remaining three have all changed canopy state or sat under canopy throughout, and both their fits and their shifts are worse; the well that has been forest across both epochs is simultaneously the poorest fit in the set and the largest shift.

The comparison window is the driest sustained stretch in the ninety-five-year RAF Valley record: rainfall averages 0.770 m yr⁻¹ over 1989–96 against 0.889 m yr⁻¹ over the modern record, and the surplus of rainfall over potential evapotranspiration is +0.127 m yr⁻¹ against +0.237 m yr⁻¹. What is left over at the six open-ground wells is +0.109 m against a contrast of −0.347 m. The reading of the epoch shift, and what it does not resolve, is in Section 5.7.8.

### 4.14.2 The 1951–53 record

The only observation of this aquifer older than the 1989–96 record is Ranwell's. Between February 1951 and August 1953 he read seventeen galvanized pipes across the open slacks of the Warren at least fortnightly, levelled each to Ordnance Datum in 1951 and re-surveyed the mobile-dune sites in 1953, and published the result as a seasonal range of 70–100 cm and a set of figures (Ranwell, 1959). He printed no table. Three of the series survive as plotted points (his Sites 1, 4 and 8) and seven as monthly ranges (Sites 1, 4, 11, 12, 13, 16 and 18; Sites 1 and 4 are in both forms, so eight sites are covered), and these were recovered from the published figures to about a centimetre; the two figures are independent digitizations of the same readings and agree at the sites they share. The pipes stood in the Penlon and Clwt Gwlyb slacks and in the coastal slack south of them; none stood under the plantation, which was then being planted, and none stood in the slack of his Sites 2–15 for which he printed no series. Where the pipes stood is known to the slack and, on sloping ground, to a few metres: his site sketch is not metric, but his levelled heights agree with the modern terrain model to within the scatter of the placement (median offset 0.16 m, spread 0.14 m), and every site falls in the terrain basin the sketch draws it in.

Two tests follow. The first is the 1989–96 test taken thirty-eight years further back. Each of Ranwell's plotted sites is paired with the modern wells that share its terrain basin and lie within 300 m of it, and each paired well's own coefficients from the modern fit are driven by the RAF Valley record from 1930, with no refitting. Because the pipe and the well are not at the same point, the comparison is made after removing the mean level difference between them, and because a flooded slack reads at the ground, the modelled level is capped at the surface. What remains is a test of the dynamics alone (Script 44; Figure 80): at the nearest

paired well the correlation between Ranwell's monthly means and the hindcast is 0.74, 0.75 and 0.91 at the three sites, the seasonal range is reproduced to within a tenth of a metre at each, and the month of the 1951 minimum is the same in the observations and the model at all three. Across every same-basin pairing the correlation runs from 0.70 to 0.91 and the bias-removed efficiency from 0.32 to 0.80, the lower values belonging to the Penlon site, where the modelled 1951–52 winter rise overshoots readings that the pipe could not follow above the ground. The forcing can be checked over exactly these years, because Ranwell also published the rainfall at the Parc Mawr nursery: over his reading span it totals 0.97 of the RAF Valley figure, with a monthly correlation of 0.92.

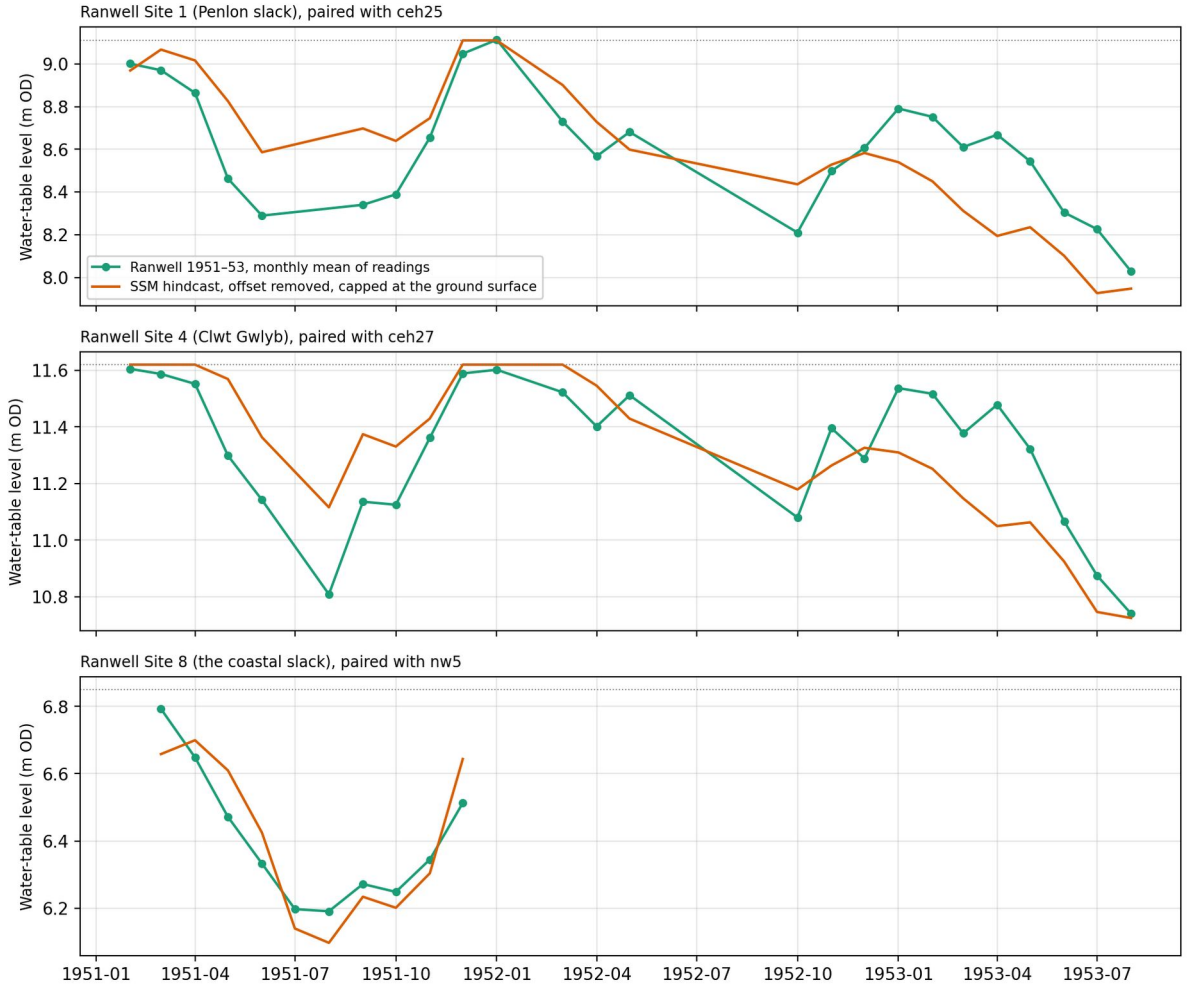



*Figure 80: Ranwell's 1951–53 water-table readings against the SSM hindcast, at the three sites whose series he published. Points are the monthly means of his fortnightly readings at Sites 1 (Penlon slack), 4 (Clwt Gwlyb) and 8 (the coastal slack); the line is the modern per-well model at the nearest dipwell in the same terrain basin (CEH25, CEH27, NW5), driven by RAF Valley climate from 1930 with the coefficients fitted to 2005–2026, shifted by the mean level difference between pipe and well and capped at the ground surface (dotted), which is where a flooded slack reads. Correlations and bias-removed efficiencies are given in the text (Script 44). (Source: 44_07b_hindcast_report.png)*

The second test asks whether the level has moved. Ranwell's mean level at each of the eight printed sites is taken in metres above Ordnance Datum, through his own levelling. It is set against the modern mean water table kriged to that point (Section 4.9.5), less the difference the climate of the two periods would produce on its own: the hindcast's own 1951–53 mean against its modern mean at the paired well, which is within 0.08 m of zero at every site. The comparison is not point to point, and its error budget says so. Four terms are combined in quadrature: the kriged surface's leave-one-out error at the six modern wells nearest the site; the surface gradient times the positional uncertainty of the pipe; the difference between a reading mean and a mid-range mean; and the spread of Ranwell's datum against the terrain model. The per-site uncertainties are 0.18 to 0.31 m in the Clwt Gwlyb and Penlon slacks, 0.37 and 0.45 m at the two coastal-slack sites, and 0.78 m at Penlon Site 1, where the surface misses its nearby wells by 0.75 m and the site is treated as unconstrained. Seven of the eight sites stand within 0.3 m of their 1951–53 level, none of them more than 1.3 standard errors from it. Site 8 stands 1.05 ± 0.37 m lower (2.8 standard errors). It lies at the seaward end of the coastal slack, 489 m from the eroding shoreline, and it is one of only two sites inside the reach of the coastal-retreat gradient fitted in Section 4.10 (902 m). Carried across the interval, that gradient's modern rate would lower the water table there by 0.95 m, and by 0.51 m at Site 18, which stands 0.26 ± 0.45 m lower. This carries a 2005–2026 rate across sixty-six years that the fit does not cover, and the shoreline's retreat since 1899 need not have been steady; Site 8's fall is therefore consistent with coastal retreat, but not shown to be caused by it. Weighted by their uncertainties, the eight sites give a change of −0.15 ± 0.09 m over the sixty-six years between the midpoints of the two records, and the seven with a constrained modern surface give −0.16 ± 0.09 m. The six sites beyond the coastal reach give −0.09 ± 0.10 m, with a scatter well inside the error budget. No combined estimate differs from zero at two standard errors. The reading of these bounds is in Section 5.7.9.

The slack floors can be read against the water table directly, each reference well against the lowest ground within 100 m of it, the definition that best reproduces Ranwell's levelled heights, and Ranwell's pipes against his own floors (Figure 81). Across the 38 open-dune wells beyond the coastal reach the floor stands a median 0.21 m above the spring water table, 0.46 m above the record mean and 0.93 m (median absolute deviation 0.11 m) above the annual minimum. Ranwell's 1951–53 readings give the same relation from his own floors: mean depths of 0.34–0.50 m and deepest readings of 0.78–1.08 m at Sites 1, 4 and 8, and with each site's modern floor taken as the lowest ground within 100 m of its placed position, the definition that reproduces his levelled heights best (Script 43), the six inland sites show a floor a median 0.07 m lower than his (median absolute deviation 0.12 m) and a mean water table a median 0.04 m deeper below it (0.21 m), the scatter of a comparison that cannot place a pipe closer than a slack. Of the fourteen open-dune wells within the coastal reach, four stand 0.3–0.7 m further above the annual minimum than the inland baseline, more than twice the standard error of the 1951–53 comparison (0.19 m), and ten do not. Under the forest the floors stand further above the water table: a median 0.94 m above the spring level and 1.59

m above the annual minimum at C4, and 0.87 and 1.22 m at C5, an excess over the open-dune baseline of 0.66 m at C4 and 0.29 m at C5 on the annual minimum (Script 44).

Under the plantation the floors stand further above the water table than the open-dune baseline and the modelled retreat response together account for. At the nine C4 wells the residual is 0.41–1.29 m (median 0.61 m), or 0.41 m once the litter that lies on the mineral floor under the canopy, up to 0.20 m, is allowed for; the modelled canopy drawdown at every forest well is 0.15 m, four times smaller. Across the forest wells of both clusters, the residual does not differ between the wells clearfelled in 1993 and replanted in 1998 (median 0.60 m over four, −0.53 to 0.79 m) and those under unfelled canopy (0.44 m over eight, −0.76 to 0.66 m). The one drainage channel in the plantation, which follows the road between CEH14 and CEH13, has its bed above the floor at every forest well but CEH14, and the two wells on ground recorded as ploughed before planting, CEH2 and CEH16, lie within the spread of their clusters. At C5 the retreat response exceeds the excess (median −0.53 m). Nor does the climate of the planting years account for it: the wells' own fitted responses driven by the RAF Valley record put the water table of 1953–1960 within 0.1 m of its level over the modern record at the C4 wells (annual minimum −0.01 m, mean +0.08 m, medians), and the shoreline has retreated 79–127 m since then depending on whether the 1899–2006 loss is spread evenly over its years or the modern rate is carried back, which leaves a median 0.66 or 0.61 m at C4 unaccounted for on the two bases, or 0.2 m less once the litter is allowed for. Five forest wells are hindcast with their cluster's coefficients because their own drainage term is not identified (Script 44).

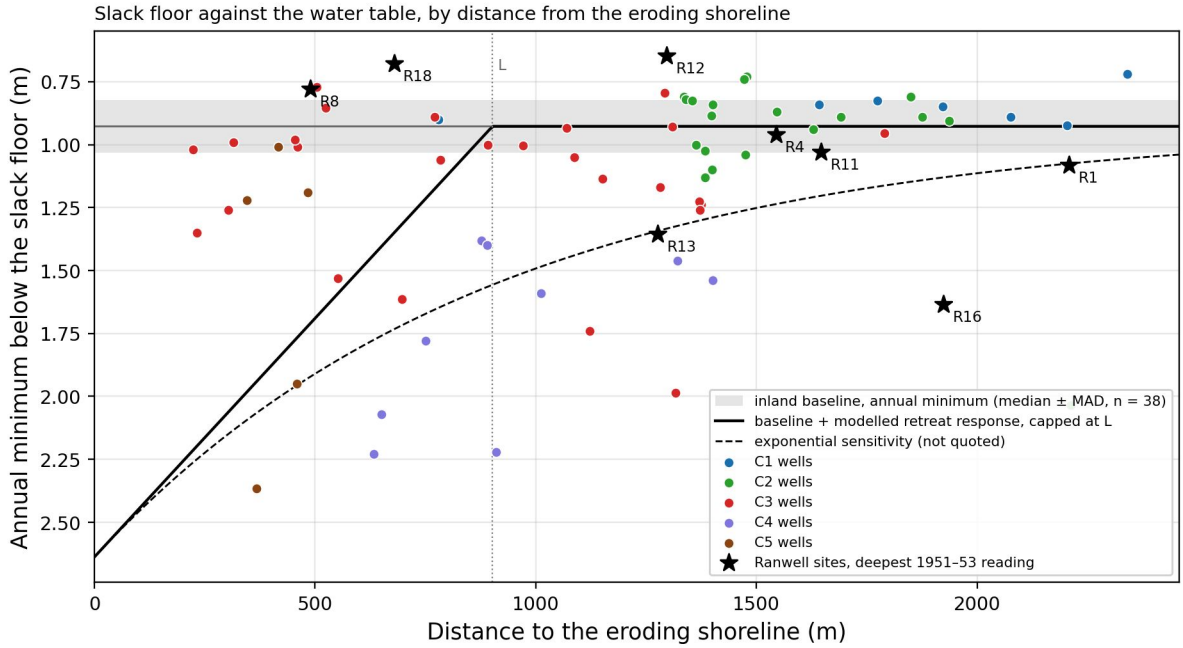



*Figure 81: The slack floor against the water table, by distance from the eroding shoreline. For each reference well, the median annual minimum below its slack floor (positive down) against its distance to the eroding shoreline, coloured by cluster. The grey band is the inland baseline: the median and median absolute deviation over the open-dune wells (C1–C3) at or beyond the coastal reach L_cg (dotted line). The solid line adds to that baseline the modelled response to the 1899–2026 shoreline retreat, the coastal response per metre of Section 4.10 shaped by the same capped form as the coastal expectation of the level comparison; the dashed line is an exponential sensitivity, not quoted. Stars are Ranwell's 1951–53 sites at the deepest reading below his own floor (Script 44). (Source: 44_09_slack_floor_datum.png)*

# 5. Discussion

The analytical pipeline applied to 21 years of monitoring data across 88 wells yields a consistent picture of a coastal dune aquifer. Its water balance is governed by the interplay of rainfall recharge, atmospheric evaporative demand and head-dependent drainage, with land cover acting as a surface boundary condition on a sand body whose effective storage, once interception is corrected, is broadly uniform. Five hydrogeological clusters emerge from the data, each with distinct mechanistic coefficients but broadly convergent specific yield once canopy interception is corrected — convergence carried by C4 and the continuous yield surface, with the clip-constrained C5 estimate only weakly corroborative. The dune slack vegetation gradient spans a narrow 37 cm window of summer minimum water table depth between the SD15b and SD16 community-mean levels (Curreli et al., 2013; Section 4.7.4), and the predictors of winter flooding potential divide across the site: winter rainfall dominates in the eastern clusters (C1, C2), winter rainfall and antecedent summer minimum depth carry equal weight in the Western Residual (C3), and antecedent summer minimum depth dominates in the two forest clusters (C4, C5), where rainfall is not significant (Table 16). This seasonal asymmetry is the organizing principle of the discussion that follows.

The cluster structure links groundwater behaviour to geology and land cover, establishing the physical contrasts that determine how each part of the aquifer responds to climate forcing (Section 5.1). The SSM framework and its benchmarking against the traditional linear model indicate that the head-dependent drainage term is essential for long-range prediction (Section 5.2), and the water balance decomposition shows how recharge, atmospheric draw and drainage partition differently across the five clusters under the displacement formulation (Section 5.3). These foundations set the context for evaluating the two management interventions in the monitoring record.

Dune scraping produced measurable improvements in summer minimum depth at CEH36, with a benefit that persisted through the post-felling period; the durability of these gains is set by the background climate trajectory and by interactions with adjacent interventions (Section 5.4). After the December 2017 clearfell the Impact well rose against the Forest control over October–March (+128 mm, 95% CI +1 to +254 mm, $p = 0.048$) and not at all over June–September (−1 mm); the annual-mean step, +108 mm, is consistent with a recovery but not significant on its own with autocorrelation-robust (Newey–West) errors (95% CI −21 to +237 mm, $p = 0.10$; Section 5.5). The apparent response at the surrounding Edge wells did not survive a balanced analysis. The recovery is concentrated in the mean monthly signal: summer minima at the felled wells tracked the unfelled Forest controls, consistent with the canopy acting simultaneously as a hydrological sink (recharge suppression through interception) and as a microclimatic buffer on atmospheric demand at the water table (Section 5.6). The kriged flow field indicates that any forest management perturbation would propagate south-westward toward Caernarfon Bay rather than south-

eastward into the C1 and C2 clusters (Sections 5.7, 5.8); the BACI record itself resolves no spatial felling footprint (Section 5.5).

The discussion brings these threads together into a management synthesis (Section 5.9): the operational domain for direct hydrological intervention, the role of canopy management in retaining the plantation's buffering function while partially recovering the recharge penalty, and the climate trajectory that sets the binding constraint on all outcomes.

## 5.1 Linking Groundwater Behaviour to Geology and Land Cover

The analytical framework rests on two foundational datasets: a 95-year climate record from RAF Valley and a 21-year dipwell network spanning 88 wells across the full geomorphological range of the dune system. The climate record establishes that Newborough Warren operates under a persistent annual surplus (mean P/PET = 1.37 over the well-record period; Table 1) in which inter-annual water balance variability is driven almost entirely by precipitation rather than by evaporative demand — PET shows strong seasonality and a small positive inter-annual trend (Section 4.1.1, Figure 4). This surplus is not seasonally uniform: winter rainfall exceeds PET by a factor of three or more, while summer PET exceeds rainfall, concentrating atmospheric demand on the aquifer during the period of lowest recharge input. The seasonal opposition is the primary climatic driver of dune slack hydrology, and the post-2013 period has run substantially warmer than the pre-2013 summer temperature baseline (Section 4.1.1, Figure 5), intensifying the summer deficit. The monitoring network itself is unusually dense for a dune site — 66 reference wells with a median record length of 15.9 years, supplemented by 22 extended-network wells — providing the spatial resolution needed to distinguish cluster-scale responses from site-wide climate forcing. The network mean water level explains approximately half the variance from cumulative climate forcing alone (Figure 4), with the residual variance reflecting the spatial heterogeneity, boundary subsidies and land cover effects that the subsequent analyses decompose.

The hierarchical clustering solution (Section 3.2; Section 4.2, Figures 7 and 8) identifies five spatially coherent and physically interpretable hydrogeological units that collectively explain the principal axes of groundwater variability at Newborough Warren. In the dendrogram the top Ward merge (≈0.99) separates C4 Main Forest from all other wells; within the rest, the east–west division between the responsive eastern open dune (C1, C2) and the buffered west (C3 Western Residual with C5 Coastal Forest) follows at ≈0.61. That east–west contrast reflects a well-documented difference in sub-surface stratigraphy, which the mechanistic coefficients resolve as a gradient rather than a hard binary. Shallower sand over till and estuarine deposits in the east constrains storage and drives rapid, flashy responses — fastest at the lake-coupled C1 Lake Edge, and at the larger, more representative C2 Dune archetype still rainfall-responsive but with slower drainage and longer turnover — while a deeper sand column toward the south-west supports the more buffered C3 Western Residual (Stratford et

al., 2007; Grootjans et al., 2004). Whether this contrast reflects two distinct substrate bodies or a single estuarine-clay base that crops out on the higher eastern and southern ground and dips toward Caernarfon Bay is examined in Section 5.1.1. The LCSC values (Table 3) quantify this physical contrast: the Eastern Block clusters show low LCSC values reflecting their constrained storage, the Western Residual is intermediate, and the Forest clusters show the highest values reflecting the combined effect of their inferred deep sand substrate and canopy interception.

LCSC is closely related to specific yield but is not identical to it: the SSM's $\beta_1$ coefficient absorbs canopy interception losses alongside true storage response, so that at the forested clusters the elevated LCSC partly reflects the 24% rainfall interception by the Corsican pine canopy (Freeman, 2008) rather than a difference in aquifer drainability. After applying canopy interception correction, the WTF-derived specific yield for C4 falls within the range of the open dune clusters and C5's sits at its upper edge (clip-constrained, weakly corroborative; Section 4.2.4, Table 8), indicating that the forest clusters' elevated LCSC reflects canopy boundary conditions at the surface rather than differences in subsurface storage architecture. The LCSC framework translates a qualitative geological description into a management-relevant metric expressing the precipitation depth required to achieve a unit rise in the water table: open-dune wells respond to rainfall more efficiently than forested wells, the latter reflecting canopy interception rather than storage capacity (Table 3).

The plantation-covered forest clusters (C4 and C5) are inferred to share the deep sandy substrate of the Western Block, yet segregate into distinct hydrogeological units (Figure 15a). Land cover therefore modifies the hydrological signature of the underlying geology sufficiently to produce a distinct cluster signature. C4 occupies the highest mean water table elevation in the network yet shows the weakest drainage decay coefficient in the network (Table 3), while C5 shows an intermediate drainage rate consistent with its position between the forest interior and the coastal margin. C4's combination of high topographic position and slow drainage reflects two reinforcing mechanisms: canopy interception dampens the recharge pulse so that the water table never rises sufficiently above its suppressed equilibrium to generate the strong head-dependent drainage response seen in the open dune clusters, and weak lateral discharge produces the within-cluster heterogeneity in $\beta_3$ (Section 4.9.4), consistent with the irregular bedrock base that outcrops within the forest and on which C4 sits; its detailed geometry and depth beneath the cluster are not constrained without coring. The displacement formulation (Section 3.4.1), which references drainage to a datum 3.7 m below the ground surface, resolves C4's $\beta_3$ to a clearly significant value (Table 3), indicating that head-dependent drainage operates at all five clusters when the hydraulic reference level is correctly specified.

The cluster-level contrasts are reflected in the state-space model parameters (Section 4.2.2, Table 3). The Western Block's lower recharge sensitivity reflects the deep unsaturated sand buffer that attenuates the rainfall-to-water-table transfer. The Eastern Block's higher $\beta_1$ and

$\beta_3$ values are consistent with a fill-and-spill mechanism in which shallow till promotes surface ponding and rapid lateral overflow rather than slow drainage through the aquifer matrix. The structural divide is evident in the winter flooding record: C1 and C2 reached the dry slack winter level in most years, C3 in just over half, C4 in none and C5 in a single year, while the wet slack winter level was reached only by C1, in a single year of the record (Section 4.8.2, Figure 47). The consequences of this asymmetry for summer minimum trajectory and forecasting are developed in Section 5.2; the anomalous C5 decline is discussed in Section 5.7.2.

The implication for conservation management is that Eastern Block slacks respond sharply to individual rainfall events but retain little water between events, making them sensitive to both exceptional winter rainfall and prolonged summer drought in ways that the buffered Western Residual is not. The post-2018 amplification of C1 seasonal amplitude (Section 4.2.1) is consistent with this sensitivity: the post-2018 period contained two of the three drought summers identified in the record (2018, 2022), producing deeper summer minima at the lake-buffered cluster. Because the entire ecological gradient of dune slack vegetation is compressed into a 37 cm window of summer minimum depth (Curreli et al., 2013; Section 4.7.4), any management intervention must be evaluated primarily on its effect on summer minima rather than on annual-mean water balance metrics.

### 5.1.1 Cluster Classification and Validation

Two diagnostic checks support the aquifer-architecture framing of Section 5.1 (Section 4.3; 28_c3_detrend_check.py and 29_c3_within_variance_check.py). Removing a linear coastal-erosion drift from the C3 hydrographs moves only two of twenty-one wells to C2, so the C2/C3 behavioural distinction cannot be reproduced as a coastal-drift artefact. This is consistent with a genuine aquifer-architecture control — whether two distinct substrate bodies or a single estuarine-clay base dipping toward the bay (Section 5.1, and below) — rather than a coastal-drift artefact (shallow sand over till in the east, deeper sand in the south-west; Stratford et al., 2007; Grootjans et al., 2004).

Within C3 the variance follows a coherent spatial axis (Section 4.3). The panel regression explains most of the per-well variation in $\beta_1$, $\beta_3$ and t½, and its strongest unique predictor is distance to CEH36, a hydrogeological axis within C3 anchored near the south-western interior of the warren and distinct from the network's coastal-margin gradient. The coastal predictor's coefficient against per-well slopes is +0.87 on its own and −0.21 in the five-predictor model; the univariate value is independently consistent with the Section 4.10.2 functional form. Because C3 divides into two at k = 6 and is the least stable cluster under bootstrap resampling, it can no longer be called "structured but not strongly enough to split": the within-cluster axis is real and informative for spatial interpolation and per-well diagnostics, and C3 is the weakest-bounded unit of the k = 5 partition the report adopts throughout.

These two strands — a genuine, non-coastal C2/C3 distinction, and a coherent within-C3 gradient along the south-west to north-east axis — admit a parsimonious reading. Along that axis recharge sensitivity $\beta_1$ and drainage rate $\beta_3$ rise inland ($r = +0.74$ and $+0.55$), and WTF-derived specific yield falls inland from 0.40 at the south-west margin to 0.22 at the north-east ($r = -0.80$, $p < 0.001$; Script 29, the 21 C3 wells with a coastal distance). A single estuarine-clay base that crops out on the higher eastern and southern ground, where a thin overlying sand column produces the flashy Dune response, and dips beneath the surface toward Caernarfon Bay, where a thickening sand column produces the buffered C3 response, would make C2 and C3 successive positions on one dipping system rather than two discrete blocks — C3 grading toward the open dune at its inland edge and toward the forest and coast at its margins, consistent with its diffuse affinity signature (Section 4.3). The geometry is interpretation, not observation: the clay base is not cored, and coastal position, plantation proximity and the inferred clay dip covary along the same axis, so two distinct depositional bodies fit the same coefficient evidence equally well. A cored or geophysical south-west to north-east transect carrying the Caernarfon Bay coastal-retreat rate as a covariate would be required to separate them (Section 5.7.3). Within C3 the $\beta_1$ and $\beta_3$ gradients are consistent with a single control, a spatial trend in near-surface specific yield, rather than a spatially varying recharge input: because $\beta_1$ is recharge divided by Sy, lower Sy at the inland margin produces both a larger water-table rise per unit recharge and a faster drainage response per unit head.

The partition is also recoverable from evidence the clustering never used (Section 4.3; Scripts 31 and 31b). The clusters are spatially compact, they separate on external descriptors, and they survive a change of linkage or distance metric in part, yet no single descriptor reproduces them. The $k = 5$ partition is therefore a multivariate structure that its descriptors corroborate but do not reproduce.

The gradational nature of the aquifer between these structural units is shown by the Pearson affinity analysis (Section 3.3; Section 4.3). Of the 66 reference wells, the great majority confirmed their primary cluster membership; six Spy wells showed strongest temporal affinity with a neighbouring cluster at borderline correlation deltas, consistent with gradational boundaries rather than misclassification. The Multi-Cluster Affinity flag was met at a substantial fraction of wells, reflecting the high mutual correlation among the five centroids in a contiguous, highly transmissive sand aquifer (Rao and Srinivas, 2006). The spatial concentration of MCA flags along the C1/C2/C3 transitional zone, the C2/C3/C5 boundary where the Dune, Western Residual and Coastal Forest signatures converge, and the C3/C4/C5 forest–dune boundary (Figure 15a) identifies the areas where geological and vegetative controls overlap. Hydrogeologically, these are candidate locations for rapid state change in response to sustained conservation management or climatic intervention (Grootjans et al., 2004).

The persistence of Forest cluster affinity at the clearfell treatment wells (Section 4.3) admits at least three interpretations: residual suppression by the surrounding intact canopy extending beyond the felling boundary; incomplete hydrological recovery within the felled footprint itself; or, most fundamentally, that the Forest cluster signature at these wells reflects high topographic elevation and its inferred deep sandy substrate rather than canopy influence alone — in which case convergence toward open dune behaviour may not be achievable regardless of management intervention. Distinguishing between these mechanisms requires continued monitoring beyond the current record.

The clusters also have a practical use: a well's cluster-mate carries information about it that the climate model does not (Section 4.3). Restoring a hidden spring level, the nearest cluster-mate erred by less than half as much as Model A in both windows, including the early one, where the 2006–08 excess lies outside what the model reproduces.

The excess has resisted explanation. Each of the following was tested by a rule fixed before the run, and none accounts for it: a second, slower store responding to the same climate, which at every cluster reduced to one store; a store drawn down by the plantation, tested with the open-dune coefficients the ground had before the canopy closed; how the winter rain fell, in long events or short; other estimates of evaporation, of which Hargreaves, Priestley–Taylor, Makkink and Penman–Monteith each remove less than half of it; a higher rainfall coefficient before 2008–11, resolved at 7 western wells but short of its rule; and a dated change in level, which earns its place only at the coastal-forest cluster. The excess is largest beside the 1998 fellings and fades with distance from them, but it had drained before the felled ground's canopy recovered, so the felling is consistent with it in place and unresolved in time (Script 50). A measured donor carries the excess whatever its cause, because its target experienced the same, which is why a gap filled from a cluster-mate can describe a reference period the climate model cannot.

The limit is the one the cross-decade test exposes. The modern partition was not recovered from the 1989–96 record. The difference lies in the company the seaward warren wells keep: in 1989–96 they moved with the inland warren, and today they group with wells on their landward side, WMC3, NW11 and NW13, which then stood under or at the edge of the forest canopy. That is consistent with a change between the two eras at those neighbours, as much as at the seaward wells themselves, NW5 and NW6 being the same pipes in both; it is equally consistent with a short early record read partly at the pipe base. The gap filling is therefore applied within the modern network, where it was tested, and not used to carry the modern clusters back to the CCW era.

## 5.2 Boundary Conditions, Model Limitations and Forecasting

The SSM is a lumped mass-balance model fitted independently at each well. It does not resolve lateral groundwater flow, and its three-term structure — recharge ($\beta_1 \cdot P$), atmospheric

draw ($\beta_2 \cdot$PET) and head-dependent drainage ($\beta_3 \cdot$h_disp) — necessarily omits any flux that does not covary with rainfall or PET at monthly resolution. Where the three-term balance does not close, the residual appears as a structural mismatch between predicted and observed water levels.

### 5.2.1 Water Balance Residuals as a Model-Adequacy Diagnostic

The water balance residual field (Section 4.9.6, Figure 59) makes this mismatch spatially explicit, and its result is largely negative: across the reference network the three-term balance closes without requiring an additional flux. No well exceeds +0.02 m/month, the fifteen positive residuals all fall below +0.01 m/month, and four of the five cluster medians are negative, C1 being slightly positive (Section 4.9.6). Residual magnitude is uncorrelated with position on either axis (Spearman $\rho = -0.14$ against Easting, $p = 0.27$; $\rho = +0.16$ against Northing, $p = 0.20$). The signed residual shows no gradient on either axis ($\rho = +0.02$ against Easting, $p = 0.86$; $\rho = -0.10$ against Northing, $p = 0.40$), including the ridge-to-dune axis on which a ridge-derived input would express itself.

The field is therefore a diagnostic of where the SSM's three terms are sufficient rather than a map of lateral inflow. It should not be read as the latter in any case, because the residual also absorbs nonlinear soil-moisture storage dynamics, β coefficient sensitivity and any systematic bias in the Thornthwaite PET estimate. On the present evidence no unmeasured boundary flux need be invoked to close the balance at any cluster, and the independent water balance of Script 16 agrees with the field to within ±6.7 mm/month at every cluster. The structure of the residuals themselves — their lag structure, the ridge-recharge hypothesis and their seasonality (Scripts 22, 23 and 24) and the cluster-stratified residual climatology (Script 24b) — is a diagnostic of the fit rather than of the site and is not repeated here.

The kriged water table (Section 4.9.5, Figure 56) is a single mound. Flow runs outward from the high point beside the bedrock ridge: south and south-west to Caernarfon Bay across the west and centre, north to Llyn Rhos-Ddu from the slacks beside the lake, and east and south-east to the Afon Braint in the east. It describes the site's flow geometry and, with no ridge-proximal residual signal, does not bear on the residual field.

Two independent diagnostic tests on the SSM time-series residuals return null results and are reported in full in Supplementary Notes S5 and S6: a cross-correlation lag analysis testing for distance-dependent transport from the ridge (Spearman $\rho = -0.028$, $p = 0.85$, $n = 49$) and a seasonal climatology analysis testing for unmodelled summer evaporative demand (network mean $r = -0.035$ against sunshine hours). Note that these operate on the monthly SSM regression residual $\varepsilon(t)$, a different quantity from the scalar water-balance closure residual mapped in Figure 59.

The seasonal residual diagnostic (Script 24) provides a further constraint. The dominant phase of the SSM residual peaks in autumn through early spring, with no wells of the 63 analysed peaking in summer. This is not the signature of unmodelled summer

evapotranspiration — if Thornthwaite PET were systematically underestimating summer losses, the residual would peak in summer months.

### 5.2.2 Displacement Formulation and Depth-Dependent PET

The displacement formulation (Section 3.4.1) — which references drainage to a datum 3.7 m below the ground surface — captures head-dependent drainage at all five clusters, with $\beta_3$ significant at $p < 0.01$ or better throughout the network (Table 3). Under the pre-displacement specification, C4's drainage term was borderline; the displacement formulation resolves this by correctly specifying the hydraulic reference level against which drainage operates (Section 5.1). The per-well datum sensitivity analysis (Section 4.9.1, Figure 53a) shows that the uniform 3.7 m datum produces negligible $R^2$ loss relative to per-well optimized datums across most of the network, with gains above 0.05 confined to four C1 Lake Edge and four C2 Dune wells, where the effective drainage base is shallowest.

A depth-dependent extension of the SSM, in which the $\beta_2$ coefficient declines exponentially with water table depth according to $\beta_2(d_w) = \beta_2 \cdot \exp(-d_w/L_E)$, was evaluated at cluster level (Section 3.7.1). The depth-dependent formulation improves iterative NSE at all five clusters (ΔNSE +0.002 to +0.148), consistent with capillary connectivity between the water table and root zone diminishing with depth — a real physical effect that the standard SSM's fixed $\beta_2$ absorbs as a depth-averaged effective parameter. The depth coupling is strongest at C1 Lake Edge ($L_E = 0.43$ m, ΔNSE = +0.148), where shallow water-table evaporation near the lake is the dominant ET pathway, and weakest at C4 Main Forest (no attenuation detected), where the gain is negligible (+0.002). The standard SSM is retained as the primary operational model because its fixed $\beta_2$ preserves the closed-form threshold equations and scenario algebra on which the operational outputs depend. The mechanistic implications — particularly the finding that C4 shows the weakest depth coupling despite having the highest $\beta_2$ — are discussed in Section 5.6.1.

### 5.2.3 The C4 Main Forest Drainage Coefficient: Identifiability and Sensitivity

The drainage coefficient $\beta_3$ is markedly low at the C4 Main Forest cluster. The cluster centroid returns $\beta_3 = 0.018$ month$^{-1}$, the lowest in the network (Section 4.2.2), and one constituent well (CEH14) returns a physically inadmissible negative value. A low coefficient of this kind can arise in two quite different ways: as a genuine hydrogeological property, or as an artefact of the atmospheric-draw term absorbing variance the drainage term would otherwise hold. Because the distinction bears on both the C4 water balance and the interpretation of the cluster's high $\beta_2$, the fit was tested directly for that degeneracy (Script 30).

The diagnostic does not support it. The drainage predictor in the SSM is the displacement of the water table above the 3.7 m drainage datum at the start of each month (Section 3.4.1),

and the question is whether that predictor is separable from PET. At C4 it is among the most cleanly separated in the network (Section 4.2.2): the variance inflation factor on the displacement term is the second lowest of the five clusters and the PET–displacement correlation the smallest in magnitude; the term is not starved of signal — C4 carries the network's largest displacement variation — and the centroid coefficient is significant. The atmospheric-draw and drainage coefficients are separately identifiable at C4, and the low drainage coefficient is a property of the data rather than an artefact of the fit.

The per-well scatter that first raised the question has a simpler explanation. Five of the nine C4 wells return a non-significant $\beta_3$ on the hundred-month per-well window, and one (CEH14) returns a negative value, but the median per-well variance inflation factor at C4 is the lowest in the network (Section 4.2.2): the scatter reflects limited statistical power in resolving a coefficient of order 0.02 month$^{-1}$ from a hundred-month record, not a $\beta_2$–$\beta_3$ trade-off. Refitting each well on its full record bears that out — seven of the nine then return a significant positive $\beta_3$, with only CEH13 and CEH14 failing — and what trade-off appears is carried almost entirely by CEH14 and CEH13, the two wells the pipeline already sets aside on independent grounds.

This has a direct — and reassuring — consequence for the cluster water balance (Section 4.2.3, Figure 11). At the fitted centroid coefficient the forest-interior loss budget is led by atmospheric draw under both forms, drainage taking 25% of losses under Model A and 29% under Model B, and the steady-state balance closes almost exactly; swept across $\beta_3$, Model A's closure residual is minimized at 0.019 month$^{-1}$ against the fitted 0.018 (Section 4.2.2), so the closure test does not admit the fitted coefficient as one of a plausible range but lands on it. The coefficient is stable under the sensitivities that bear on it: the identified per-well coefficients lie above the centroid, and excluding the two ridge-flank wells whose drainage the model does not resolve raises the centroid to 0.029 month$^{-1}$ and shortens Model A's recession time from 54 to 34 months. Across that whole range the qualitative result is unchanged: the partition remains atmospheric-draw-led. The identifiability diagnostic, the identified per-well coefficients and the water-balance closure thus converge on the fitted $\beta_3 \approx 0.018$ (Model A $t_R \approx 54$ months at the network datum): the forest interior drains genuinely more slowly than the open dune (C2 $t_R \approx 15.9$ months), a real canopy-and-substrate effect.

One consequence follows directly for the atmospheric-draw coefficient. Because $\beta_3$ is cleanly identified at C4, the cluster's high $\beta_2$ cannot be read as a collapsed drainage term reappearing in the evaporative one: C4's $\beta_2$ of 2.56 is a real feature of the cluster, standing well above the open-dune clusters (C2, 1.84; C3, 1.83) rather than resembling them. That is not, however, the same as saying that more water evaporates beneath the plantation. $\beta_2$ measures the water-table response per unit of atmospheric demand, not the evaporative flux itself: the same loss produces a larger head change where less drainable storage is available to absorb it, so the coefficient carries the substrate as well as the demand. The two forest clusters make the point directly. C4 and C5 carry broadly the same Corsican pine canopy,

yet between them they span almost the whole network range of $\beta_2$ — C4 the highest at 2.57, C5 among the lowest at 1.34, below both open-dune clusters. A canopy effect common to both cannot produce a two-fold contrast between them. What differs is position and substrate: C4 on the ridge flank over a thin, low-storage sand column, C5 on deeper coastal ground whose higher WTF-derived Sy converts a given evaporative flux into a smaller head change (Section 5.6.1). The report therefore treats the fitted C4 $\beta_2$ as real, and reads it as a storage-mediated head response rather than as evidence of elevated evaporation beneath the canopy.

CEH14 is treated separately. On the ridge flank, it is the single well where the SSM fails outright (iterative NSE = −12.87, the worst fit in the network; Section 4.4): the displacement formulation does not capture the bedrock-ridge boundary condition, and its negative $\beta_3$ is a symptom of that failure rather than a measurement. It is excluded from the drainage decay half-life map (Section 4.9.3) and the per-well mechanistic interpretation on that ground.

Across the reference network only one well returns a negative atmospheric-draw coefficient: CEH6 (C1 Lake Edge), with $\beta_2 = -0.40$. Taken at face value this is physically implausible, implying a water table that rises with evaporative demand, but the estimate is small and statistically indistinguishable from zero ($p = 0.26$, with an approximate 95% confidence interval spanning zero). The most parsimonious reading is weak identifiability of the PET term at this well rather than any real inverse sensitivity. CEH6 is a lake-buffered, strongly drainage-controlled site whose fit is dominated by recharge ($\beta_1 = 5.33$) and head-dependent drainage to Llyn Rhos-Ddu ($\beta_3 = 0.12$, $p < 10^{-20}$); consistent with this, the Lake Edge cluster carries the strongest PET–displacement collinearity in the network (Section 4.2.2), so little independent variance remains to constrain the PET coefficient and it is free to take a small, wrong-signed value under sampling noise. The anomaly is confined to the per-well estimate: the C1 cluster-centroid fit, calibrated on the pooled cluster series rather than by averaging per-well coefficients, returns a positive and highly significant $\beta_2$ of 0.94 ($p < 10^{-4}$; bootstrap 95% CI [0.55, 1.18]), and no downstream result depends on CEH6's individual value. Like CEH14's negative drainage coefficient above, CEH6's $\beta_2$ is flagged as noise-level and is not interpreted physically.

### 5.2.4 Predictive Skill and Operational Forecasting

The physical distinction between the SSM and the traditional linear model (TLM) is most directly revealed by long-range iterative simulation (Section 3.4.2; Section 4.4). In one-step diagnostic mode both models perform almost identically, because each prediction is corrected with the observed water level from the previous month; under iterative simulation the SSM returns a positive NSE at 65 of 66 wells against 55 of 66 for the TLM (Table 9; Figure 18). The explicit $\beta_3$ term in the SSM captures storage decay, preventing the cumulative drift that otherwise causes TLM divergence. The Main Forest cluster (C4) shows the smallest improvement (+0.121), consistent with two non-exclusive readings: the canopy may smooth the recharge signal into a near-

linear climate response that the TLM can handle adequately without a storage decay term, and the structurally suppressed water table at C4 carries less nonlinear storage dynamics for the SSM to capture. The two readings are not distinguishable from the NSE benchmark alone. Across the network the gain identifies where the head-dependent drainage term is indispensable — the eastern, lake-proximal aquifer, where $\beta_3$ is highest — and where atmospheric draw dominates the water balance regardless of drainage formulation, the forest interior (Section 4.2.3; Table 5). The forest clusters are also the least well fit by the cluster-level equations (Section 4.7.1), consistent with their greater storage buffering: the higher LCSC values (Table 3) attenuate the direct rainfall–water-table coupling the SSM captures.

This predictive capacity has three practical consequences for conservation management. First, the SSM can generate forward projections of water table trajectory under specified climate scenarios (Section 4.4), enabling managers to anticipate when summer minima may cross the SD15b / SD16 community-mean levels reported by Curreli et al. (2013) rather than relying solely on backward-looking trend extrapolation. The seasonal prediction equations (Table 16) and P_flood threshold equations (Table 19) are algebraic inversions of the SSM's fitted coefficients — they are not separate models but direct consequences of the SSM parameterization, and their predictive credibility rests on the SSM's iterative skill (Table 9).

Second, the spatial pattern of SSM improvement indicates where the aquifer's structural boundaries may lie, without invasive coring (Figure 18). The largest median NSE gain is at the C1 Lake Edge cluster (+0.45, against +0.38 at C2 and +0.37 at C3), where substantial seasonal head variation and head-dependent drainage to Llyn Rhos-Ddu make the storage decay term essential. The smallest gains occur in C4 Main Forest. The improvement map is itself a diagnostic: it identifies which parts of the aquifer are governed by nonlinear storage dynamics and which approximate a linear climate response.

Third, the SSM framework is operationally transferable. The model requires only three fitted parameters per well ($\beta_1$, $\beta_2$, $\beta_3$), a uniform drainage datum, and monthly rainfall and temperature from a single climate station. It does not require hydraulic conductivity measurements, aquifer thickness estimates, or lateral flow modelling — the parameter-sparsity constraints that produced calibration difficulties in the Betson et al. (2002) MODFLOW model at this site — where hydraulic conductivity rested on a single tracer test (K = 6.0 m/day) and aquifer thickness on four borehole logs across 700 ha (Bristow, 2003; Section 5.7.3). Any coastal dune aquifer with a monitoring network and a nearby climate station can be parameterized in the same framework. The displacement formulation (Section 3.4.1), which references drainage to a fixed datum rather than the ground surface, is the methodological choice that makes this possible: it captures the depth-dependence of the drainage response without requiring site-specific calibration of the drainage geometry. What the framework requires in order to be applied elsewhere, and which of its parameters do not transfer, is set out in the final section of this chapter.

### 5.2.5 Seasonal Prediction and the Intervention Window

The finding that antecedent summer minimum depth governs peak winter levels in the two forest clusters (C4, C5; $p < 0.001$, with winter rainfall not significant) and shares that role equally with winter rainfall in the Western Residual, where neither is significant on its own at the 5 % level (summer minimum $p = 0.060$, winter rainfall $p = 0.051$; Table 16) is the most consequential single finding for conservation management. It means that outside the eastern clusters managers cannot reliably forecast slack flooding from seasonal rainfall forecasts alone — how deep the aquifer drains each summer matters as much as, and in the forest more than, the winter rainfall that follows. The seasonal prediction equations — which regress peak winter water table against antecedent summer minimum depth and cumulative winter rainfall (Section 3.6.2; Table 16) — were fitted with one equation per cluster under the $k = 5$ partition: winter rainfall was significant in the two eastern clusters (Lake Edge $p = 0.002$, Dune $p = 0.011$), marginal in the Western Residual ($p = 0.051$), and non-significant in the Forest ($p = 0.903$) and Coastal Forest ($p = 0.388$). Antecedent summer minimum depth was the dominant predictor in the two forest clusters (Forest $p < 0.001$, Coastal Forest $p < 0.001$) and carried equal weight with winter rainfall in the Western Residual, both marginal ($p = 0.060$ and 0.051). The eastern pattern is consistent with the faster-draining, faster-recovering character of that aquifer, where same-season rainfall dominates the winter flooding response; the western and forest pattern reflects the multi-month carry-over of the antecedent state — a deep summer minimum requires progressively more winter rainfall simply to restore the water table to its pre-drought position before any flooding can occur — and the significant intercepts at Forest and Coastal Forest (Section 4.7.2) are consistent with the structural offset the plantation canopy imposes on water table depth. The ecological stakes of this finding are clarified by the community-mean levels reported by Curreli et al. (2013), whose four-year monitoring programme showed that wet slack vegetation (SD15b) has a mean annual maximum water level of about +0.21 m, flooded above the ground surface, and dry slack communities (SD16) persist at winter maxima around −0.25 m. Summer minimum depth sets the precondition for achieving these winter maxima: a water table that has drained deeply through summer must receive proportionally greater winter recharge to recover to flooding height. The P_flood equations quantify this precondition directly (Section 3.6.3; Table 19).

The summer minimum trajectory (Section 4.8.1, Figure 47) shows this precondition deteriorating over the full record: all five clusters show declining summer minima, significant at all but C4, the decline concentrated before 2011. Where the slack floor is surveyed the shift is measured directly: every van Willegen quadrat read in both windows is deeper than in Curreli et al.'s survey years, and none now lies above the SD15b community mean (Section 4.8.3). The post-2013 period is running 0.94°C above the pre-2013 baseline in mean summer maximum temperature (Section 4.1.1). Recent ecological field surveys provide observational support: Callaghan et al. (2021) found that despite active topographical intervention at Newborough, Petalophyllum ralfsii populations remained

restricted due to limited water availability, and recorded a complete absence of winter flooding following the severe 2018 summer drought. One mechanism operates weakly against this trajectory at C1 specifically: the depth-dependent PET diagnostic (Section 4.4; Section 5.2.2) shows C1 to have the strongest depth-coupling in the network (L_E = 0.43 m, ΔNSE = +0.148), so as the C1 water table deepens the capillary connection to the evaporating surface weakens and the atmospheric-draw term attenuates, a modest negative feedback that slows, but does not reverse, the deepening of the summer trough.

The Forest (C4) prediction equation ($R^2 = 0.872$, summer minimum dominant, winter rainfall non-significant $p = 0.903$; Table 16) is consistent with two non-exclusive interpretations. Under the canopy interception interpretation, summer drought conditions were attenuated by the canopy, producing a more compressed range of summer minima and consequently more predictable winter peaks. Under an alternative interpretation, the forest water table sits sufficiently deep that the winter recharge pulse cannot reach flooding height across the rainfall range in the record, so the regression cannot detect a winter rainfall signal. The two interpretations are not mutually exclusive — both contribute to the statistical pattern — and the available data do not allow them to be partitioned. The P_flood unreachability finding (Section 5.8.1) — that at C4 summer minima of approximately 1.7 m below ground, the required winter rainfall to achieve flooding exceeds anything in the 95-year RAF Valley record — directly supports the second interpretation. The BACI summer minima null (Section 5.5) is also consistent with a system sitting far below threshold, where neither canopy presence nor canopy removal materially affects whether winter flooding is reached.

The Eastern Block (C1, C2), by contrast, is rainfall-driven: the shallow till-floored aquifer reaches the dry slack winter level in most winters (Section 4.8.2), and flooding is driven by within-season rainfall accumulation rather than antecedent drought depth.

The primacy of summer minimum depth is reinforced by the contrasting behaviour of the two ecologically referenced metrics under the climate projection. The two ecologically referenced metrics are the summer minimum, which sets the drought precondition discussed above, and the five-year mean spring water level (MSL5), which describes the sustained baseline within which dune-slack vegetation establishes (Section 4.8.3; van Willegen et al., 2025). The two metrics diverge in their exposure to the projected climate trajectory, and the divergence is sharpest in the open dune. Under UKCP18 RCP8.5 by the 2080s, MSL5 moves by +52 and −22 mm at C1 and C2 while the summer level falls by 315 and 359 mm; at C3 and C5, where the longer mean-reversion time carries a summer deficit into the following spring, MSL5 falls by between a third and a half as much as the summer level (−150 against −440 mm at C3, and −141 against −319 mm at C5; Section 4.13.1, Figure 79). The summer minimum, by contrast, receives the full projected increase in summer atmospheric demand compounded by the projected reduction in summer rainfall. The consequence for management is that of the two ecologically referenced metrics, the spring baseline is comparatively robust to climate change in the open dune while the summer minimum is

highly exposed. This is why summer minimum depth, rather than the spring baseline, is the binding ecological constraint at Newborough: it is both the precondition for winter flooding recovery and the metric that the projected climate trajectory most directly threatens. The open-dune clusters (C1, C2) that hold a comfortable MSL5 baseline (Section 4.8.3) are nonetheless the clusters whose summer minima the Section 4.8.1 trajectory shows deepening — the divergence between the two metrics is not a contradiction but a description of a system whose long-term baseline remains viable while its drought extremes deteriorate.

## 5.3 Recharge, Loss and the Water Balance

The cluster-level water balance decomposition (Section 3.7.2; Section 4.2.3; Table 5) makes explicit a structural feature that is implicit throughout the state-space model results: under mean climate conditions, the three-term displacement-formulation SSM closes the monthly water balance at the cluster mean level, with residuals of less than 1.8% of total losses (Table 5). This near-closure indicates that recharge, atmospheric draw and head-dependent drainage together account for the principal water balance fluxes at the cluster aggregate level, with any unmeasured boundary contributions absorbed into the fitted β coefficients (Section 5.3.1) rather than appearing as systematic residuals at the cluster mean. Spatial variation in local closure across the site is examined in Section 5.2.1.

### 5.3.1 Recharge Partitioning and Atmospheric Draw

Recharge into the water table ($\beta_1 \cdot P$) declines systematically from the Lake Edge cluster to the forested clusters (Section 4.2.3, Table 5), reflecting increasing storage capacity and — in C4 and C5 — the additional suppression of throughfall by canopy interception, which is captured implicitly in the lower $\beta_1$ rather than as a separate model term. Total losses (atmospheric draw plus drainage) track recharge closely at every cluster. The drainage partition varies markedly: drainage accounts for 86% of total losses at C1, reflecting rapid lateral exchange with Llyn Rhos-Ddu, but only 24% at C4, where atmospheric draw dominates; C2, C3 and C5 lie between.

It should be noted that in open-boundary systems, lumped parameter models of this form are known to absorb unmeasured boundary fluxes into the rainfall coefficient when no intercept term is fitted (Young, 2011; Knotters and van Walsum, 1997), which means the recharge term ($\beta_1 \cdot P$) should not be read as a direct measure of local precipitation reaching the water table. At Newborough, however, there is no evidence that such absorption is occurring. The water balance residual field (Section 4.9.6, Figure 59) shows that the three-term balance closes across the site: per-well residual magnitudes are small and uncorrelated with position on either axis, and show no forest-margin or ridge-flank concentration of positive values. The field carries a small systematic negative offset (median −0.0040 m/month; 51 of 66 wells negative). The largest negative residual falls at the ridge-flank well CEH14, with CEH32 next; at both, modelled recharge marginally exceeds modelled losses. Two independent diagnostic tests reach the same conclusion by other routes: a lag-distance test

found no relationship between the lag of peak rainfall–residual cross-correlation and distance from the bedrock ridge, and a test for unmodelled evaporative demand using sunshine hours also returned a null result. The caution regarding $\beta_1$ therefore stands on general grounds — it remains a lumped coefficient carrying whatever the model does not otherwise represent — but at this site it is a statement about model form rather than a response to any observed residual structure (Section 5.2.1).

Thornthwaite PET is an empirical index of atmospheric evaporative demand derived from temperature and day length; it does not define a specific reference surface in the manner of Penman-Monteith, so a formal crop coefficient decomposition does not apply. The $\beta_2$ coefficient under the displacement-formulation SSM therefore absorbs the combined effects of vegetation transpiration, canopy interception, specific yield and depth-coupling into a single effective scaling between atmospheric demand and water table response (Section 5.6.1). C4 and C5 carry broadly similar Corsican pine canopy; the principal difference between them is topographic position and water-table depth, with C4 on the flank of the bedrock ridge (which outcrops within the forest) and C5 on the lower coastal ground. The substantial $\beta_2$ difference — C4 considerably higher than C5 — therefore tracks elevation and the rock base C4 is known to overlie rather than canopy; the depth and extent of that base, and the depth of sand beneath C5, are not constrained without coring. The WTF-derived Sy estimates (Table 8) show C5's Sy higher than C4's, supporting the direction of this interpretation: a higher Sy at C5 means a given evaporative flux produces a smaller head change at the water table, which would lower the apparent $\beta_2$. The magnitude of the $\beta_2$ ratio is larger than the WTF Sy ratio alone would predict if substrate were the only factor, indicating that depth-coupling differences (Section 5.2.2) also contribute. C4's position on the elevated ridge flank is consistent with reduced effective drainable volume per unit head change: the forest overlies the bedrock ridge, which outcrops within the plantation, and an irregular, low-storage rock base — thin sand over a variable bedrock surface — is the natural reading, though its detailed geometry beneath C4 is unconstrained without coring. The drainage decay half-life (Figure 55) is consistent with this: C4 wells show the longest t½ in the network, consistent with sluggish lateral discharge rather than free flow through a uniform sand column (Section 5.3.2). $\beta_2$ is the atmospheric-draw coefficient, and the high forest $\beta_2$ should be read as a property of the ridge-flank setting — elevation, water-table depth and the rock base beneath C4 — rather than as evidence of a boundary input.

A further question the $\beta_2$ evidence is sometimes asked to address is whether the Corsican pine canopy provides a shielding effect on the water table: the proposition that the canopy intercepts solar radiation and reduces wind speed at the ground surface, thereby suppressing evapotranspirative demand during summer, when groundwater is already at its seasonal minimum and ecological stress is highest.

One coefficient-level observation is consistent with the proposition. C5 Coastal Forest's $\beta_2$ (1.25) lies below that of the C2 and C3 open dune (C2: 1.84; C3: 1.83), though above C1

(0.91): per unit of Thornthwaite PET, the water table beneath the Coastal Forest experiences less drawdown than equivalent open-dune ground. C4 gives no comparable support. Its $\beta_2$ (2.57) stands well above the open-dune band, and the identifiability diagnostic (Section 5.2.3) indicates that this is a real property of the cluster rather than an artefact of a poorly resolved drainage term. Since the two clusters carry broadly the same canopy, the two-fold $\beta_2$ contrast between them cannot itself be a canopy signal. The C5 comparison is also not strictly like-for-like: because the SSM is fitted on gross rainfall and above-canopy PET, $\beta_2$ at the forested clusters carries the canopy's interception loss as well as sub-canopy demand (Section 3.7.2), so C5's $\beta_2$ sits below the open dune despite carrying a rainfall tax that C2 and C3 do not.

The annual $\beta_2$ coefficient is in any case a weak instrument for the question, and two diagnostics already run bear this out. The depth-dependent ET formulation (Section 3.7.1) returns the network's weakest depth coupling at C4 (no attenuation detected) alongside its highest $\beta_2$, so the elevated coefficient is not a depth-mediated evaporative effect. The residual seasonality diagnostic finds no unmodelled summer evaporation anywhere on the site: the correlation between SSM residuals and independent sunshine-hour data is negligible across all clusters (network mean $r = -0.035$; no well outside the Bartlett null band), and no well in the network carries a summer-phased residual peak, 48 of 63 peaking instead in winter or early spring. Whatever systematic bias the Thornthwaite formulation carries relative to actual forest evapotranspiration, $\beta_2$ has already absorbed it.

There is also a reason to expect any real pine–dune contrast in evaporative loss to be winter-weighted rather than summer-weighted, and it is a matter of rainfall volume rather than plant phenology. The canopy's principal hydrological cost is interception: approximately 24% of incident rainfall is caught and re-evaporated before it can reach the water table (Freeman, 2008). That loss scales with rainfall, and rainfall at Newborough is strongly winter-dominated, so the difference in water reaching the ground between forest and open dune is at its largest in winter. In summer, when rainfall is much lower, the interception difference between the two — and the evaporative loss that follows from it — is correspondingly smaller. A rainfall-scaled, winter-weighted loss represented through a coefficient on PET, a summer-weighted quantity, is represented awkwardly at best. The annual $\beta_2$ contrast between forest and open dune is therefore not a measurement of summer canopy shielding, and should not be read as one in either direction.

One confound attaches to the C5 evidence. The Coastal Forest cluster carries a long-term coastal-retreat gradient (Section 4.10.2; Section 5.7.2) that is embedded in the fitted coefficients: the lower C5 $\beta_2$ may partly reflect a secular convergence of the water table toward a lower baseline driven by shoreline retreat rather than a canopy effect on evaporative demand. Separating these mechanisms in C5 is not possible without a longer record or unforested coastal-margin control wells alongside the forested ones. The shielding

interpretation of C5's $\beta_2$ is therefore consistent with but not uniquely supported by the data; the evidence for canopy effects on atmospheric draw is suggestive rather than conclusive.

The clearfell record bears on the proposition from a different direction and is discussed in Section 5.5.1. Its seasonal asymmetry — a winter rise following felling (+128 mm over October–March, $p = 0.048$) that is not distinguishable from zero over the summer months — runs in the direction a real shielding effect would predict, but the summer estimate is too imprecise to establish it. Taken together with the C4–C5 contrast under common canopy and the interception confound above, the shielding proposition remains open on the present evidence rather than supported or refuted by it.

### 5.3.2 Drainage Decay and the Volumetric Water Balance

The drainage decay coefficient $\beta_3$ controls the rate at which the water table relaxes toward its equilibrium position, and its spatial distribution (Section 4.9.2, Figure 54d) shows the strongest contrast of any SSM parameter across the site. C1 Lake Edge drains fastest, driven by rapid lateral exchange with Llyn Rhos-Ddu. C2 and C3 show similar intermediate rates. C5 Coastal Forest is lower, and C4 Main Forest has the weakest drainage in the network (Table 3). The displacement formulation (Section 3.4.1) is essential to resolving the C4 drainage term: under a ground-surface reference C4's $\beta_3$ was borderline significant; referencing drainage to the 3.7 m datum resolves it to a clearly significant value (Table 3). The slow drainage at C4 is physically consistent with the canopy suppressing recharge pulses that would otherwise drive the water table above its equilibrium and generate a strong drainage response, combined with sluggish lateral discharge through C4's inferred irregular, low-storage base (Section 5.3.1; Section 5.6.1). The contrast between fast-draining eastern clusters and slow-draining forest clusters shapes the volumetric water balance partition between drainage and atmospheric draw.

The indicative volumetric water balance (Table 7) provides a complementary perspective. All clusters receive approximately 890 mm/yr of rainfall (889–893 mm/yr by cluster); for the forested clusters, canopy interception (24%; Freeman, 2008) reduces the throughfall reaching the ground surface to 678–679 mm/yr. The ET/drainage partition is bracketed by two independent methods — the SSM $\beta_2/\beta_3$ ratio and a seasonal recession curve analysis — with the mid-point reported in Table 7. The cluster partitions differ substantially: at C1, drainage dominates total losses, reflecting rapid lateral exchange with the lake; at C4, atmospheric draw dominates losses and drainage is weak, indicating that the Forest cluster's water balance is governed primarily by the competition between rainfall input and evaporative demand rather than by hydraulic drainage. C2 and C3 sit intermediate, with drainage-dominated partitions; C5 is drainage-dominated too, at about two-thirds of losses.

The water table fluctuation (WTF) analysis (Section 4.2.4; Table 8; Figure 13) addresses the question of whether the substrate's storage capacity differs across the cluster structure. Uncorrected forested cluster Sy medians exceed the open dune range, but this reflects the

WTF method attributing the full rainfall total to recharge when in reality a fraction is intercepted by the canopy. Once corrected for canopy interception, C4's Sy estimate falls within the open dune range and C5's sits at its upper edge, though C5's is constrained by the plausibility clip and is only weakly corroborative. The effective storage inferred from the water-table response is therefore broadly uniform across the site — consistent with, though on these data not proof of, a well-mixed aeolian depositional history — so the forest clusters' distinctive β-coefficient signature (lower $\beta_1$, higher LCSC) is read as deriving largely from canopy boundary conditions at the surface rather than from differences in subsurface storage architecture, though depth-coupling and lateral drainage geometry also contribute to within-forest β variation (Section 5.3.1).

The implication for management interpretation is that the surface-mediated canopy effects identified by the analysis are in principle responsive to silvicultural intervention, while the subsurface storage and transmission properties of the aquifer are fixed by geological history. The net hydrological response to silvicultural intervention at the cluster level is small relative to climate forcing over the same horizon (Sections 5.5.2, 5.7.1), and the spatial reach of forest management does not extend to the open dune clusters (Section 5.8); the surface-vs-subsurface distinction therefore concerns *which* of the canopy's effects are accessible to management rather than *whether* canopy management is the dominant restoration lever.

The slow drainage and thin-aquifer substrate identified above also explain a finding that bears directly on the shielding hypothesis: the inter-annual amplification field (Figure 75; Section 4.12) shows C4 Main Forest to be the most climate-sensitive zone on the site, with a mean spring water-table swing approximately 1.66 times the network mean between the canonical dry and wet year sets (range 1.31–2.32×, the upper end being CEH14, the well with the smallest drainage coefficient in the network ($\beta_3$ negative and not significant); excluding it and CEH13 the cluster mean is 1.56× and C4 remains the most sensitive zone). The forest interior also occupies the hydraulic high of the aquifer, furthest from any lateral discharge boundary — the lake to the east, the coast to the south-west, the estuary to the south-east — so the water table there rises and falls without the damping that proximity to a constant-head boundary provides to the lake-edge and coastal clusters. The lake edge damps the shared climate swing to 0.64× — the lake providing a hydraulic backstop in dry years and an overflow in wet ones — while the open dune sits near unity. The amplification coefficients correlate negatively with $\beta_3$ ($r = -0.41$, $p < 0.001$): slower-draining wells retain wet-year recharge longer and therefore show larger spring swings between dry and wet years. This is the same mechanism identified in Section 5.3.1 — low Sy over an irregular bedrock base produces a thin saturated column that amplifies any given volume of recharge or discharge into a proportionally larger head change.

This amplification result runs counter to the shielding hypothesis as a statement about dynamic climate buffering. A canopy that meaningfully shields the ground surface from climatic forcing would be expected to damp inter-annual variability, not amplify it. The

forest interior over-responds to climate swings relative to the open dune. Two positional factors account for this without invoking canopy effects, and both are consistent with the WTF Sy analysis and the slow-drainage evidence above: the thin saturated column over irregular bedrock (low effective Sy amplifying head change per unit flux), and the forest's position at the hydraulic high of the aquifer, remote from any constant-head boundary that would buffer wet-year rises or dry-year falls. The forest canopy and the bedrock substrate are spatially collinear at Newborough Warren — the plantation occupies the ridge-proximal zone where the aquifer is inherently thin — and their contributions to the amplification signal cannot be separated from the dipwell record alone. Independent substrate characterization beneath C4 (slug tests; cored transect) and a targeted seasonal $\beta_2$ decomposition comparing summer and winter months across pine and open-dune wells are identified as the analyses most likely to resolve the question in future work.

The amplification result does not, however, constitute evidence that the canopy provides no hydrological buffering and can therefore be removed without hydrological cost. The amplification field (Figure 75) measures absolute inter-annual swing — the difference between wet-year and dry-year spring water levels at each well, expressed as a ratio to the network mean. It is a structural property of the aquifer at each location: a well sitting at the hydraulic high over thin, slowly draining sand will amplify any given climatic forcing regardless of what vegetation covers it. The relevant question for buffering is not whether the forest zone swings more than the open dune in absolute terms — it does, for structural reasons — but whether the canopy modulates the response relative to what the same aquifer location would do without it. That question is answered on the relative (BACI) scale, not the absolute one. The BACI climate-sensitivity analysis (Figure 30; Section 4.6.2) operates on exactly this scale: it compares the felled well against matched controls sharing the same climate forcing, so the structural amplification signal — common to felled and unfelled wells alike — cancels. The CWB² × clearfell interaction is negative at both zones (Impact: −14.4 mm per (100 mm CWB)², p = 0.22; Edge: −8.4 mm per (100 mm CWB)², p = 0.36), the direction expected if the felled wells responded more strongly to dry-period deficits after canopy removal than before, relative to their controls — but it is not significant at either zone and the test cannot support that reading. The BACI relative scale is the only scale on which the question can be asked without confounding by the structural amplification identified above, and on that scale, on a single felling event, the test returns a null. The amplification field and the BACI curvature result therefore address different questions: the former characterizes where the aquifer is structurally sensitive to climate; the latter provides the most direct available test of whether the canopy modulates that sensitivity. They are not in conflict, and the amplification field should not be read as evidence against canopy buffering.

In summary, the cleaner and less confounded evidence for a canopy effect on the dune-slack water balance remains the recharge suppression captured in the lower $\beta_1$ of both forested clusters — a consistent finding across the WTF, LCSC and SSM analyses that is not

materially affected by substrate or coastal-retreat confounds. The summer-shielding component, assessed through $\beta_2$, is suggested by C5's coefficient alone, but the amplification result and the substrate/coastal confounds mean it cannot be stated as a confirmed finding on the present evidence.

## 5.4 Dune Scraping

Three dipwells — CEH36, CEH18 and CEH21 — were subject to ground surface lowering by mechanical scraping, with CEH36 scraped in April 2015 and re-scraped in October 2023, when CEH18 and CEH21 were scraped for the first time (Section 4.5). CEH36 is located approximately 340 m seaward and downslope of the December 2017 clearfell footprint, a spatial relationship that becomes important when the interaction between the two interventions is considered below.

### 5.4.1 Measured Benefit and Durability

At CEH36, three independent methods (raw BACI, synthetic control, SSM forward residual) all identify a scraping benefit in the same direction during the pure scraping era from April 2015 to December 2017, with the raw-BACI and synthetic-control estimates in close agreement and the SSM forward-residual estimate lower but of the same sign (Section 4.5; Figures 20, 21). The seasonal record reinforces the finding: the era-mean summer minimum shallowed substantially during the scraping era and held its position through the post-felling period, even as the CEH4 control on the eroding margin deepened, consistent with coastal retreat together with the warren-wide decline, which the record cannot separate (Section 4.5; Figure 23). Calibrated against the Curreli et al. (2013) eco-hydrological gradient (Section 5.2), the summer improvement at CEH36 represents a shift across more than half the gradient separating the SD15b and SD16 community-mean levels.

Scraping creates a topographic low that acts as a local drain, raising the water table at the scraped site by collecting lateral inflow. The whole-array step map for the scraping era shows no coherent uphill gradient in the observed level change: across the wells uphill of CEH36 the climate-corrected steps scatter in both signs, without the distance-decay a signal radiating from a point source would produce (Section 4.5; Figure 27). WMC3, the one uphill well with a full pre-scraping record and a clean forest-control comparison, does not show such a signal either. Its climate-corrected step during the pure-scraping era (−106 mm; Figure 27) is not distinctive against the array, whose steps over the same era scatter widely, and its raw displacement gap against the forest control falls by 46 mm at the April 2015 scraping and by 53 mm at the October 2023 re-scraping, neither step distinguishable from the gap's month-to-month variability (Section 4.6.3, Figure 31). No off-site level effect of the scraping is resolved, and the level evidence agrees with the drainage-coefficient evidence: the separate $\Delta\beta_3$ propagation test found no distance-decay either (Section 4.5.5). A neighbour drawdown of the size the drain-cone model predicts at these distances (Section 5.4.3) is below what the record can detect; it is not excluded. The benefit at the scraped site

does not alter the rate at which the water table responds to climate forcing, so the background drying trend continues to erode the initial improvement; at CEH36, the improvement is diminishing but the gain has been retained through the post-felling period to date. CEH18 and CEH21, scraped only in October 2023, have shown limited and inconsistent responses that do not survive Tier 1 correction for background drift — consistent with their position closer to the coast where progressive erosion outpaces the localized topographic benefit (Section 4.5).

The shape of this diminishing-but-retained trajectory is itself informative. Against the climate-corrected control the summer-minimum gap at CEH36 rose to a peak of about +422 mm roughly two years after the April 2015 scraping, then relaxed — at approximately −17 mm $yr^{-1}$ measured from the peak, or −14 mm $yr^{-1}$ from the scraping year — toward a residual of about +187 mm, a partial decline that leaves the slack well above its pre-scrape state (Section 4.5). Three mechanisms acting on different timescales can account for this shape. Removing the dune-slack vegetation eliminates a growing-season transpiration draw that would tend to raise the local table, while the newly exposed bare sand and standing water at the scrape floor add direct evaporation that would tend to lower it; which of the two dominates is not established from the present evidence. This is not the same physics as the clearfell canopy-removal step: the clearfell's benefit is attributed either to a reduced canopy draw or to a recharge gain that disperses beyond the felled compartment (Section 4.13.2), mechanisms with no counterpart at an unforested slack, and it registers only in the monthly mean — the clearfell's own summer-minimum step is close to zero (Section 4.6.4) — so the clearfell cannot serve as a validating precedent for a summer-minimum-specific effect at the scrape. Cutting headward into the landward flank creates a new low point in ground where the inland head is higher, which steepens the gradient into the scrape floor: inflow is high initially and then relaxes as the enhanced inflow draws the surrounding head down and the gradient flattens — the transient expected on relief-drainage grounds (Hooghoudt, 1940; van Schilfgaarde, 1963; Freeze and Cherry, 1979) and quantified for a comparable fine-sand dune–swale setting as largely completing within about five years (Gerla, 2019). As vegetation recolonizes the exposed floor, transpiration returns and the balance of the two competing surface losses shifts back toward the vegetated state — eroding whatever net benefit the first mechanism had contributed, while also reducing the offsetting bare-surface evaporation term as the floor is shaded and stabilized again. The observed curve is consistent with an initial overshoot that relaxes and is then slowly modified by regrowth; the residual reflects this recolonization alongside the drainage relaxation of the second mechanism, and cannot be decomposed into the two contributions with the present data. The clearfell offers a useful contrast in shape, though not in mechanism: it involves no excavation, and in the matched summer-minimum currency the felled-well response shows no comparable relaxation (a post-felling trend of about +1 mm $yr^{-1}$, indistinguishable from zero; Section 4.6.4). Because the clearfell's own summer-minimum step is itself close to zero, this flat trajectory shows only that the observed decay at CEH36 is specific to the scrape rather than a

generic post-intervention artefact — it does not isolate a transpiration-only, decay-free vegetation mechanism, since the clearfell's measured effect operates through a different pathway (a reduced draw or a dispersed recharge gain) and a different metric (the monthly mean). This excess decay is consistent with the recharge-capture transient, but it cannot be attributed to relaxation alone: the decline is present against the climate control yet absent against the CEH4-paired control (about +7 mm $yr^{-1}$, not significant), because CEH4 lies at the seaward end of the slack and is subject to the same coastal drawdown, so pairing cancels the shared coastal signal. The relaxation is therefore reported as consistent with, not diagnostic of, the drainage-theory mechanism, given the single scraped well and this currency dependence. Unlike the clearfell, whose recovery is confined to the monthly mean and does not reach the summer minimum or the five-year spring baseline (Section 4.6.4), the scraping benefit at CEH36 is expressed across all three metrics. This broader pattern is more consistent with a mechanism operating through the growing season — transpiration removal and/or the geometric drainage effect described above — than with the clearfell's pathway, though it does not on its own resolve which of the two dominates. These inflow statements concern the scrape floor, which the inflow wets — an on-site effect, distinct from the surrounding drawdown considered in Section 5.4.3.

### 5.4.2 Interaction with the Clearfell

The interaction between scraping and the subsequent clearfell is instructive. CEH36 sits approximately 340 m south of the felling footprint and down-gradient of it: kriged flow lines leaving the footprint's eastern edge pass through the well (Section 4.9.5, Figure 56). The CEH36 BACI step during the felling pulse era was smaller than during the pure scraping era — against CEH4 the scraping benefit diminished (+0.023 m, from +0.128 m) but was not reversed, whereas against the CEH21 coastal benchmark it reversed (−0.055 m) (Section 4.5.2). The CEH4 control, on the eroding margin, deepened across the same period, consistent with coastal retreat together with the warren-wide decline (the record cannot separate the two), and a substantial part of the apparent reduction in BACI step at CEH36 reflects CEH4's own deterioration rather than a loss at the scraped well. Any recharge benefit from the clearfell propagating downslope to CEH36 would operate on the multi-year to multi-decadal timescale developed in Section 5.4.3 and may not yet be detectable in the monitoring record.

The reverse interaction also warrants consideration: the April 2015 scraping at CEH36 created a topographic drain that may have drawn water away from the wells used to measure the clearfell response. WMC3 (the sole Impact well) and CEH31 (the nearest Edge well) sit closer to CEH36 than the other Edge wells. At WMC3, 262 m from CEH36, no scraping drawdown is resolved (Section 5.4.1), so the scraping drain is not observed there directly. Treating the fitted clearfell step as a residual net of any scraping drain therefore rests on the distance-weighted scraping covariate rather than on an observed drawdown. The combined footprint of the scrape drain and the coastal-retreat gradient on the pre-fell baseline is

mapped in Figure 68; it is the spatial counterpart of the distance-weighted scraping covariate and the coastal-drift term (Section 4.6), making explicit that the fitted clearfell step is a residual net of two non-felling drawdown processes. The two interventions propagate through the same continuous aquifer on overlapping timescales, and the interaction illustrates the difficulty of isolating individual management signals in a system where perturbations are entangled.

Scraping at CEH36 produced a benefit in summer minimum depth that persisted through the post-felling period. Against CEH4 the benefit diminished but was not reversed; the dominant threats to scraping durability identified by the monitoring record are the background climate drying trend and coastal erosion at the western boundary, rather than upstream canopy management.

### 5.4.3 Coastal Erosion Analogy and Cascade Risk

The mechanism by which scraping produces its measured benefit — lowering the ground surface to reduce the depth to water table — is physically identical to the mechanism of coastal erosion. Both reduce ground elevation relative to a fixed absolute water table position; both increase the displacement head (h_disp) at adjacent wells; and both consequently enhance drainage gradients away from the intervention or erosion zone. The Tier 1 CUSUM trajectories at CEH4 and CEH22 (Section 4.5; Figure 19) show this process operating naturally at the western coastline, with both control wells showing accelerating negative CUSUMs consistent with progressive coastal erosion of the dune edge. That natural process is now resolved spatially. The coastal-margin field of Section 4.10.5 (Figure 65) maps an erosion-dominated band of net water-table deepening along the dune-toe margin, coincident with the western C5 and C3 wells; its driving shoreline retreat is documented independently by Pye and Blott (2024), whose dune-toe sequence for the Twyni Penrhos frontage records up to 16.3 m of retreat at the southern end between 2013 and 2022 — of the order of 1.8 m $yr^{-1}$ at the most active location — placing a measured floor under the rate the field assumes.

This analogy carries a management implication that the SSM physics make explicit but the available monitoring record can only partly test. Scraping along the western coastline, while locally beneficial in reducing the depth to water table at the scraped location, increases the hydraulic gradient between the coast and the dune interior. The SSM drainage term operates in both directions: a scraped coastal zone with elevated h_disp draws water seaward, deepening the water table at wells uphill of the scrape. This dipole — a rise at the scraped slack, drawdown of the surrounding water table — is given spatial form in Figure 67, which renders the scrape as a steady-state drain anchored to the measured CEH36 response. The near-field level drawdown is not resolved in the record: the $\lambda \approx 220$ m decay model places the predicted equilibrium head at 262 m at −41 mm, below what the record resolves at WMC3 (Section 5.4.1). The off-site magnitudes below, expressed as a volumetric flux (mm water-equivalent per month), are modelled steady-state values the record cannot resolve. The

scenario framework puts an indicative magnitude on this neighbour drawdown: modelled from the same drain cone (Figures 28–29; Section 4.5.6), the off-site drawdown at a well 100 m from a CEH36-equivalent scrape is of order −30 mm water-equivalent per month, over the $\lambda \approx 220$ m length scale. These are steady-state figures on the same footing as the uphill drawdown of Figure 67, not values the 31-month record can confirm, but they indicate the neighbour effect is a modest fraction of the on-site benefit rather than negligible. The magnitude of this effect is governed by the same aquifer parameters that control the natural coastal erosion signal — and the results at CEH4 and CEH22 show that signal to be substantial and progressive.

The risk of a cascade follows from the mechanism. If coastal scraping deepens summer minima at interior wells, those wells may in turn be identified as candidates for scraping — lowering their ground surface, increasing their h_disp, and propagating the drainage enhancement further inland. Each successive scrape addresses the local symptom while intensifying the systemic cause. This chase-inland dynamic is the topographic analogue of the drainage acceleration that the scraping analysis identified as a fitting artefact in Section 4.5.3: the aquifer is not gaining storage decay capacity, but the systematic lowering of ground surfaces produces a head trajectory that the model — and the manager — interprets as progressive deepening requiring further intervention. The BACI-corrected $\Delta\beta_3$ values at uphill wells (Section 4.5; Script 09b) showed no distance-decay from the scraping site within the observation window. This null concerns the drainage coefficient, and is consistent with a pre-existing spatial gradient rather than a propagating $\beta_3$ signal (Section 4.5.5): the uphill wells traverse the C3 $\beta_3$ gradient and the forest zone's elevation dependence, so a split-window fit cannot separate a genuine propagating signal from the static spatial structure already present. The aquifer diffusivity timescale bounds the same conclusion physically: with $L_B^2Sy/(Kb)$ exceeding seven years at 570 m, a real drainage front could not traverse the uphill well array within the 31-month post-scraping record. The scraping benefit at CEH36 is therefore best understood as a direct geometric perturbation confined to the scraped well itself, and the SSM forward residual during the pure-scraping era, which is positive (+0.100 m) and persistently so (Figure 21c), is consistent with a permanent surface-lowering effect rather than a change in the aquifer's dynamic response properties. It is not evidence that the cascade mechanism is inactive: the level drawdown is not resolved either (Section 5.4.1), and the far-field equilibrium magnitudes remain modelled figures the record cannot confirm.

Most of the Western Residual already lies below the hydrological range for functioning dune slack habitat (Section 4.8; Figure 47), so the practical scope for this cascade is limited to the transitional margins. The principle that coastal scraping mimics coastal erosion in its effect on the inland water balance is well-founded in the SSM physics regardless of the available data's ability to test it within the observation window, and on the precautionary principle should inform the spatial planning of future scraping programmes. The most productive scraping targets identified by the analysis (Section 5.7) are the C1/C2/C3 transitional wells

in the mid-eastern sector, where structural sensitivity maximizes the retained benefit and where the scraping does not amplify an existing coastal drainage gradient. This recommendation is reinforced by the spatial asymmetry of the shoreline process itself. Pye and Blott (2024) document a north-positive (progradation) to south-negative (retreat) gradient along the frontage, with the chronic retreat concentrated at the southern end and the erosion front migrating northwards. Coastal scraping sited at the actively retreating southern margin would coincide with a location already undergoing net retreat, and would there behave less like a controlled, durable surface-lowering than like an addition to that retreat — a further argument for confining scraping to the mid-eastern transitional wells away from the eroding south-western margin.

The geometry of the scrapes reinforces this precaution. The cuts are elongated fingers projecting inland from the eroding coast, aligned with the coast-to-inland hydraulic gradient and discharging seaward at the lowest available base level; whatever their habitat intent, this is an effective relief-drainage configuration, disposed to lower the water table in the dune around each finger even while the finger floor itself is wetted by inflow (Section 5.4.1). The retreating coast does not lengthen the fingers — erosion shortens them from their seaward end — but it lowers the base level to which they drain, so the surrounding drawdown a finger imposes and the coastal-erosion drawdown act in the same direction and cannot be separated within the record. This drainage-engineering dimension of scrape design appears to be an unexamined one. Conservation guidance recognizes that creating scrapes or pools can dry adjacent dune slacks (Buglife, 2019) but frames this as a siting caution without a mechanism, and otherwise treats scraping as cut-to-the-water-table habitat creation (Grootjans et al., 2002; Denning et al., 2024); we are not aware of guidance that applies relief-drainage theory to scrape design, or that relates a scrape's geometry — its length, its orientation to the hydraulic gradient, and its discharge to a falling seaward base level — to its severity as a relief drain on the surrounding dune. The mechanism set out here offers a candidate explanation for that recognized but unexplained risk, and it reinforces the case for siting scrapes among the mid-eastern transitional wells rather than along the retreating south-western margin.

The timescale over which these processes operate is governed by the aquifer's hydraulic diffusivity. Using representative values for clean dune sand and the interception-corrected WTF Sy ranges from Section 4.2.4, the characteristic propagation timescale at 340 m is years to a decade, and at 570 m is one to several decades; the literature values for unconfined dune sand (Fetter, 2001) shorten these ranges but do not alter the order of magnitude. A direct measurement of the inland reach of the coastal-erosion gradient from the 21-year monitoring record (Section 4.10.2; Figure 60) gives a fitted strip-aquifer width of about 900 m (± 144 m on the well basis) under a Dupuit–Forchheimer steady-state interpretation, or an exponential decay length of 498 ± 78 m (the other specification; Section 4.10.2) under a transient diffusive interpretation. Both fall within the diffusivity-derived range above, providing independent network-scale corroboration of the years-to-decades timescale

regime; and the modelled coast-edge rate is itself independently corroborated by a model-free coast-to-inland transect (Section 4.10.4), whose −27.8 mm yr⁻¹ growth in the coast-minus-inland head difference is consistent with the −31.3 mm yr⁻¹ gradient $\delta_0$ from an entirely separate, collinearity-free construction. Whichever estimate is adopted, the propagation timescale is measured in years to decades. This has two consequences. First, it explains why the BACI-corrected $\Delta\beta_3$ at uphill wells showed no distance-decay: the propagation timescale exceeds the 31-month observation window. Second, the coastal erosion signal visible in the CEH4 and CEH22 CUSUMs is a lagged response to erosion that occurred years earlier; if the rate of coastal retreat is itself accelerating, consistent with regional projections under sea level rise (Pye and Saye, 2005) and the CEH22 trajectory from 2021, then the full hydrological consequences of recent erosion have not yet propagated to the interior wells. The current CUSUM trajectories are a leading indicator rather than a real-time measure of present inland conditions. The aquifer's response to any boundary perturbation — whether erosion, scraping or land cover change — is measured in years to decades, and monitoring programmes need to be designed accordingly.

Sea-level rise is the opposing process acting at the same western boundary (Figure 64). The Holyhead tide gauge records a historic mean-sea-level rise of 2.3 mm yr⁻¹ (Pye and Blott, 2024), with recent satellite altimetry suggesting in excess of 3 mm yr⁻¹ in the north-east Atlantic. The western C5 and C3 wells therefore lie under two coastal pressures of opposing sign — chronic dune-toe retreat and gradual sea-level rise — whose net effect Section 4.10.5 shows to be erosion-dominated along the dune-toe margin, reinforcing the monitoring priority for the western-margin wells.

The direction of this coastal influence — shoreline retreat narrowing the dune body and lowering the water table — is consistent with the behaviour documented at comparable Welsh and north-west European dune systems, though the present network cannot independently confirm its amplitude here (Section 4.12). The closest analogue is Whiteford Burrows, a natural dune-slack spit on the Gower coast of South Wales, where Robins, Pye and Wallace (2013) found that erosion narrowing the sand body lowered the dune water table by approximately one metre following a loss of only about 4% of the spit width — a mechanism and direction precedent for the erosion-drawdown process inferred here, not a rate comparison. Two further systems bracket the behaviour. The engineered Sand Engine nourishment in the Netherlands (Huizer et al., 2018) shows the same coupling between sand-body geometry and freshwater-lens elevation in the opposite, net-growth direction, and the natural barrier island of Dog Island, Florida (Kiflai and Whitman, 2023) demonstrates that the water-table response to shoreline change can lag the erosion itself by years to decades — a disequilibrium that is one reason the modelled coast-to-inland profile of Section 4.10.5 need not track the present-day retreat rate exactly. This erosion-drawdown mechanism is the inverse of the more commonly cited expectation that sea-level rise raises coastal water tables (Ferguson and Gleeson, 2012; Michael et al., 2013): at Newborough the dominant coastal signal at the western margin is a lowering driven by dune-toe retreat, not a rise driven by the

sea, and the two act in opposing directions as Section 4.10.5 sets out. Stating this inversion explicitly is more faithful to the evidence than presenting the sea-level-rise expectation as the default.

## 5.5 The Clearfell Experiment

The December 2017 clearfell of approximately 4.2 ha of Corsican pine plantation provides the most direct test of how canopy removal affects the water table in the dune slack system. The clearfell zone sits upslope of CEH36 (the scraped slack described in Section 5.4) and straddles the gradational boundary between C4 Main Forest and C5 Coastal Forest. The sole Impact well, WMC3, is assigned to C3 Western Residual but shows strongest Pearson affinity with C5 Coastal Forest (r = 0.961), with C3 a close second (r = 0.954), and the extended-network FE wells within the felled footprint show similarly mixed affinities across C3, C4 and C5 (Section 4.3). The felled compartment therefore cannot be treated as a clean C4-only experiment; the clearfell zone spans the transition from forest to open dune, and its hydrological effects propagate downslope into the C5 catchment via the continuous drainage pathway identified in the Darcy flow analysis (Section 4.9.5, Figure 56).

The three-counterfactual ANCOVA-BACI (Section 4.6.3) returned a clearfell step against the Forest control — the most direct test, comparing felled wells against unfelled forest sharing the same canopy and substrate — of +108 mm at the Impact well (WMC3), consistent with a recovery but not significant on its own once autocorrelation is allowed for (95% CI −21 to +237 mm, p = 0.10); over October–March the step is +128 mm (95% CI +1 to +254 mm, p = 0.048). At the Edge tier the step against the Forest control was +31 mm and not statistically distinguishable from zero (p = 0.50). The measurable clearfell recovery is therefore localized at the felled well rather than expressed across the surrounding edge of the compartment; an alternative reading — that the edge wells are undergoing a continuous decline that a step-change BACI is not well suited to detect — cannot be excluded on the present record, and is taken up in Section 5.5.1. The Climate-control comparison captures the standing forest's pre-existing hydrological footprint alongside the December 2017 clearfell; the Forest control is the appropriate test for canopy-removal-specific attribution. The step is measured net of a differential coastal drift between the felled compartment and its controls (Section 3.7). The contrast feels that field at 1.51 ± 0.45 times the amplitude fitted at network scale, which is not distinguishable from feeling it in full; constrained to exactly that amplitude the step is +77 mm rather than +108 mm. The constrained value is reported as a sensitivity rather than as the headline, because the record does not reject the unconstrained fit. Nor is the step stationary: split at January 2021, the Impact-well recovery against the Forest control falls from +152 mm in the early post-felling era to +39 mm thereafter, no longer distinguishable from zero, and the Edge-tier contrast from +66 mm to −6 mm (Section 4.6.8). The felled compartment is held permanently clear by grazing, so this relaxation is not a recovery of canopy: the +108 mm is the record-mean of a front-loaded

recovery that is not sustained even though the canopy never returns, a reading developed in Section 5.5.1.

The clearfell's date and extent are independently confirmed by the dated aerial series, from evidence the state-space model never sees. A canopy texture index measured from the imagery (Script 41; Section 4.6.8) shows the felled compartment and the felling experiment cleared between the frames of 22 April 2017 and 28 June 2018, while the unfelled conifer control holds. The index is a texture position between open ground and mature conifer rather than a canopy fraction, and it is reported here only as corroboration of when and where the canopy was removed.

The felled area is deliberately kept clear of trees and is not being restocked. The step measured here is therefore a permanent change in canopy state rather than a transient one, and should be read as such: there is no returning canopy to re-impose the effect that removal relieved. The imagery is consistent with that management — the felled compartment's texture index remains low through 2026, and what recovery it shows is of ground vegetation rather than of canopy.

Four supplementary analyses bear on the ANCOVA result: SSM residual normalization, synthetic control via donor weighting, synthetic extension of FE1 and FE2 within the felled compartment, and a clearfell transect regression (Section 4.6.7). They do not all point the same way, and the disagreement is reported here rather than reconciled. The synthetic extension carries the same Forest-control structure as the ANCOVA and returns a step consistent with it at the WMC3 + FE2 centroid; it is the one direct corroboration. The SSM residual normalization and the synthetic control agree that WMC3's gross post-felling step is of order +80 to +100 mm, but they differ on how much survives normalization against the unfelled forest controls — on the residual method the forest controls' own upward drift (+53 mm) takes up about two-thirds of the Impact well's (+81 mm), leaving an Impact-specific component of about +27 mm. None of the four carries the drift term. In the paired design that term is substantial: removing it moves the Impact step from +108 mm to +16 mm, so the four estimates and the ANCOVA are not corrected on a common basis, and the spread among them should be read with that in mind rather than as a like-for-like disagreement. The transect regression returned no significant spatial gradient, although WMC3 carries the largest post-felling step of the seven transect wells (Figure 38c) — suggesting a distance-dependent response that the transect regression lacks the power to resolve. The summer minima analysis (Section 4.6.4) returned no significant clearfell effect at the Impact tier and a small negative effect at the Edge tier, opposite in sign to the monthly ANCOVA result — summer minima at the felled Edge wells deepened slightly relative to the Forest control after felling. The clearfell therefore registers in the mean monthly water level but not in the summer extremes, where felled and unfelled wells track each other. The felled well's own coefficients do not show the partition a recharge-suppressing, shielding canopy would predict: after felling its recharge sensitivity did not rise (0.98 of its pre-felling value, net of

the climate controls) and its atmospheric draw fell, by 4.8% beyond the climate controls (Section 4.13.2). Over June–September alone the Impact step against the Forest control is −1 mm (95% CI −118 to +116 mm; Section 4.6.4). The +108 mm mean lift therefore arises either from the reduced draw or from a recharge gain that disperses beyond the 4.2 ha compartment, which is narrower than the drawdown reach, and the record does not choose between them (Section 4.13.2).

The two implementations differ in what the drift term is. In the pooled four-zone panel it is a per-well easting × time interaction, a genuine spatial term whose removal moves the Impact step by about 2 mm. In the paired design it is the fitted coastal profile differenced between target and control, and the step does rest on it: removing it costs 92 mm of the 108 mm step. The two are not interchangeable, and a sensitivity measured in one does not describe the other.

The magnitude of that paired-design correction can be stated rather than left implicit, and checked against the network gradient. The Impact well sits 553 m from the shore against a Forest-control mean of 955 m, a 402 m contrast across which the network coastal gradient predicts a differential secular fall of −10.7 mm yr⁻¹; the paired drift term absorbs −16.1 mm yr⁻¹, the 1.51 ± 0.45 amplitude noted above. The correction it applies is therefore of the rate the network gradient predicts, consistent within the per-well scatter about that gradient (a difference of 5.5 mm yr⁻¹; Section 4.10.2), and it offsets a coastal decline that would otherwise mask part of the felling signal.

The coefficient decomposition (Section 4.6.6; Table 15; Figure 37) provides additional context. Recharge sensitivity ($\beta_1$) declined at most wells of the BACI network, at the unfelled Forest Control tier as well as the treatment tiers but negligibly at the Climate Control tier, so a clearfell-specific origin is unlikely but independence of management is not established (Section 5.5.1). Atmospheric draw ($\beta_2$) increased at Forest Control wells, consistent with progressive canopy maturation raising the canopy's share of the PET budget at mature unfelled pine or, in part, with the fit trading variance between two seasonally anti-correlated regressors (winter rainfall and summer evaporative demand) at wells where the canopy lags recharge, but showed mixed responses at Climate Controls; the shift is not a climate-sampling effect, since refitting $\beta_2$ on a common PET support across the two eras leaves the Forest-Control increase positive while the Climate-Control tier remains negative (Section 4.6.6), so a change in climate forcing common to all tiers cannot account for the forest-specific direction. Storage decay ($\beta_3$) changes were small in absolute terms but negative at fourteen of the seventeen wells. The predicted clearfell effect from coefficient shifts alone was negative at the felled tiers, but because controls show comparable shifts, the net coefficient-attributable component is near zero. This sits alongside the ANCOVA's +108 mm Impact-well step. The two analyses ask different questions. The ANCOVA measures the gap between felled and control wells, so the site-wide $\beta_1$ decline — which affects both — cancels and the canopy-removal effect is visible. The per-well decomposition measures

coefficient changes at each well individually; the $\beta_1$ decline registers as a negative contribution at every well, and because controls and felled wells show comparable shifts, no clearfell-specific signal emerges. The two are reconciled rather than contradictory.

### 5.5.1 Seasonal Asymmetry, the $\beta_1$ Decline, and the Confounded Baseline

The network-wide $\beta_1$ decline is reported here separately from the clearfell result. It is a candidate mechanism for the observed summer minimum decline documented in Section 4.8.1, but its independence of management is weaker than the network mean suggests: the unfelled Climate Control tier shows almost none of it. The decline is most parsimoniously attributed to changing within-month rainfall distribution. If the same monthly total arrives in fewer, more intense events separated by longer dry spells, water held in the unsaturated zone between events is increasingly drawn back out by ET before it can percolate to the water table. The net recharge per millimetre of rainfall therefore falls even where the gross monthly total is unchanged. The mechanism is measurable at this site. Freeman (2008) recorded the open-dune profile at roughly 17.7% volumetric moisture against 6.5% beneath the plantation, with the open-dune upper 20 cm held nearly static by capillary contact with the water table. Where that contact is maintained, a lengthening inter-event interval draws the profile down before the next pulse arrives, and the recharge efficiency of a given monthly total falls accordingly. The evidence is one season at three plots and is offered as a supporting analogy rather than a test of the site-wide decline.

The magnitude of this network-wide decline is moderate. Across the seventeen wells of the BACI network (Section 4.6.6; Table 15), the mean $\beta_1$ recharge coefficient fell from 2.83 in the pre-felling era to 2.70 in the post-felling era, a network-mean reduction of 4.6%, with thirteen of the seventeen wells showing a decline. The decline is observed at the unfelled Forest Control tier (mean −4.0%, three of five wells) but is negligible at the unfelled Climate Control tier (mean −0.3%, three of five wells), sited at distance from both the clearfell footprint and the coastal margin; the larger Edge-tier and Coastal Control declines (−10.9% and −10.6%) absorb indirect contributions from post-fell recovery dynamics in the immediate clearfell vicinity and from the coastal-retreat gradient documented in Section 4.10.2 respectively. The Climate Control tier, which would give the cleanest test that the $\beta_1$ decline operates independently of canopy removal, does not show it; the decline is network-wide in the mean but is not established as a site-wide trend independent of the treatment and coastal tiers.

Against this background, the felled well does not show the seasonal trade the canopy framework predicts. Removing the canopy was expected to recover the interception loss, raising $\beta_1$, and to raise $\beta_2$, the water table's sensitivity to atmospheric demand. At WMC3 $\beta_1$ did not rise and $\beta_2$ fell (Section 4.13.2). The step appears in the mean monthly level and not in the summer months or the summer minima; whether it arises from the reduced draw or from a recharge gain that disperses beyond the compartment, the record does not decide. Its sustained effect under the scenario framework is addressed in Section 5.5.2.

The CWB × clearfell interaction — testing whether the water table's sensitivity to cumulative climate forcing changed after felling — was non-significant in all six ANCOVA models (Section 4.6.2). The climate sensitivity of the felled wells did not detectably change after canopy removal: the ANCOVA estimates an upward shift in mean water table level (+108 mm, not significant on its own as an annual mean; +128 mm over October–March, p = 0.048), while the rate at which the BACI gap responds to cumulative water balance deficit remains unchanged, with pre-felling and post-felling slopes approximately parallel (Figure 30). This is consistent with the canopy acting as a hydrological sink (transpiration, interception) and a microclimatic buffer (shading, reduced wind exposure, organic forest floor), though the felled well shows no loss of the buffer after felling (Section 4.13.2). Whatever the mechanism — a reduced draw or a dispersed recharge gain (Section 4.13.2) — the system's net sensitivity to cumulative climate forcing was preserved. The practical consequence is that the clearfell effect on the BACI gap is a fixed offset — it shifts the baseline from which the water table responds to climate — rather than a change in the rate of response to wet or dry years. The sensitivity variant of Section 4.6.2, in which the felling effect concentrates in dry periods, is the pattern a dry-season canopy-buffering mechanism would produce, but it has alternative explanations — post-felling non-stationary drift, or the coastal-retreat gradient — that the analysis does not exclude, and it rests on a single interacted term and one felling event at one site; it is a preliminary finding, not a confirmed buffering effect.

The Edge-tier result merits a specific interpretive note. The absence of a clean post-felling step at the Edge wells admits two readings: that the canopy-removal signal is genuinely confined to the felled compartment and does not extend to the surrounding edge, or that the Edge wells are undergoing a continuous decline — possibly the $\beta_1$ decline described above, which is largest at the Edge tier, or the closing replant canopy over two of those wells (Section 4.6.8) — against which a step-change BACI model has little discriminating power. A step-change design tests for a discrete offset at a fixed date; it is structurally ill-suited to a trajectory that declines gradually from before the intervention onwards, as the Edge displacement series does under the record-length-balanced pre-felling window. The present record cannot separate these two readings, and the Edge result is therefore reported as a null with respect to a discrete clearfell step rather than as positive evidence of no effect.

The developing-canopy setting resolves the Edge-tier ambiguity and reframes the Impact result (Section 4.6.8). The felled compartment is held permanently clear by grazing and does not regenerate, yet the Impact-well recovery decays from +152 mm to +39 mm across the record; the decay survives every coastal-drift treatment, so it is a real relaxation rather than a correction artefact. A mechanism already established here can account for a recovery that fades even with the canopy permanently removed: the raised water table drives faster head-dependent drainage — the $\beta_3$ term acts on displacement, so the recharge surplus that lifts the table is exported more rapidly as it rises, relaxing the system toward a lower equilibrium, the same relief-drainage relaxation invoked for the scrape (Section 5.4.1). The Edge-tier null, by

contrast, is a confound specific to those wells: two of them (CEH30, CEH16), and both Forest-control wells inside a 1998 replant block (CEH32, CEH33), sit over a canopy that was itself closing across the monitoring window (Section 4.6.8), so the Edge contrast crosses from a positive early step to a negative late one as the replant draws those wells down. The confound does not reach the headline — the untouched-forest control was stable and dense, and the step is +110 mm using only the two outside controls — so the +108 mm is best read as a genuine early recovery that is not durable, and the Edge null as that recovery masked at the edge by the maturing replant.

The summer minima record makes this seasonal partition explicit. At WMC3 — the sole Impact well with sufficient pre-felling baseline — summer minimum depth was −1.573 m (mean, n = 3 years) before the April 2015 scraping, −1.620 m during the scraping era (2015–17, n = 3), and −1.600 m post-felling (2018–25, n = 7). The Forest Control tier was stable across the same period, with a tier-mean summer minimum of −1.822 m pre-scraping, −1.753 m during the scraping era and −1.766 m post-felling (Figure 35). Both Impact and Edge tiers track the Forest Control trajectory in summer extremes, with the felled wells' summer minima moving in parallel with unfelled forest while their mean monthly levels sit above it. The climate-corrected spatial step-change mapping (Section 4.6.5; Figure 36) resolves no spatial felling footprint: the corrected steps carry an east–west trend and their most negative values lie along the south-western coastal strip, with no distance-decay from the felled compartment; the transect analysis reaches the same conclusion (Section 4.6.7). At the felled well FE2 the wet-year spring rise and the co-temporal amplification both lie below the unfelled C4 interior (Section 4.12), which is physically consistent with canopy removal: the interception-dependent recharge suppression that drives the high wet-year amplification of the unfelled interior is absent at FE2, and its slightly deeper summer drought floor is not read as drying of exposed ground, which the felled well's canopy-on draw does not show (−1.7% net of the climate controls; Section 4.13.2). Felling thus attenuates the wet-year amplification that drives the apparent C4 differential rise while not improving the summer minimum.

The ANCOVA additionally includes a coastal-drift term, which controls for progressive divergence between the felled compartment and the control wells across the felling window (Section 4.6.3). Without it, divergence of that kind — from whatever cause — is attributed to the fitted step. With the term included, the ANCOVA recovers a clearfell-specific step of +108 mm at the Impact well. The four-zone pooled-panel fit of the same contrast returns +27 mm (Section 4.6.3). The difference between the two is not this covariate: dropping it moves the pooled step by 2.0 mm (Section 4.6.3). The two estimators partition climate variance differently — the joint panel estimates climate sensitivity from the full cross-zone monthly record, so a portion of the variance the separately-fitted ANCOVA attributes to the felling step is attributed to climate in the joint fit. The coefficient of that term is not a measurement of coastal retreat and is not read as one here (Section 5.7.7).

The western-margin signal that a coastal correction would have to represent is set out in Section 4.10.5: it partitions spatially into an erosion-dominated band of net deepening along the dune-toe margin — coincident with the western C5 and C3 wells, in the western half of the BACI network — flanked by a sea-level-rise–dominated rise at the shore (Figure 65). Pye and Blott (2024) provide independent empirical corroboration of the underlying asymmetry: their dune-toe sequence records progradation in the north and chronic retreat in the south of the frontage. That asymmetry is real, and it is what made the easting reading plausible. What it does not do is make a single easting difference within a fifteen-well panel a measurement of it: within that panel easting carries no information about distance to the shore, the correlation falling to r = +0.022 (Section 5.7.7). After the clearfell the felled well rose against the Forest control over October–March (+128 mm, p = 0.048), while the annual-mean step (+108 mm) is not significant on its own (p = 0.10); in summer extremes felled and unfelled wells track each other, following the background climate trajectory.

The 1990s-era attribution of water table depression to the forest (Cottingham, 1994; Bristow and Bailey, 2001; Betson et al., 2002) was conducted during the driest decade in the 95-year RAF Valley record (mean annual rainfall 793 mm against a long-run average of 858 mm; Section 4.1.1) and without the kind of climate-partitioned analysis applied in this study. The apparent depression of the 1989–1995 water table baseline is consistent with a compound signal of forest influence and exceptional climatic dryness that the data available at the time could not separate. The present three-counterfactual design separates these contributions and additionally partitions the canopy effect by season. The forest does depress the water table — $\beta_1$ is lower under canopy, and the BACI record shows a mean water table recovery following clearfell at the felled well, significant over October–March (+128 mm, p = 0.048) though not as an annual mean (+108 mm, p = 0.10). The earlier studies would have recorded this as a uniform water table depression; the present analysis shows that the felling left summer extremes unchanged. The 1989–1995 baseline was therefore not solely a forest signal, nor solely a climate signal, but a compound of both — and a compound whose seasonal structure was not visible at the time.

The monitoring evidence therefore places the clearfell step in the mean monthly water level, not in the summer months or the summer minima — the metric that determines dune slack viability — where felled and unfelled wells track each other.

### 5.5.2 Forest Management Scenario Responses

The scenario framework (Section 4.13.2) anchors the forestry scenarios to what the December 2017 felling measured at the felled well: a fall in atmospheric draw 4.8% beyond that at the climate-control wells, and no rise in recharge. As sustained changes in level, clearfell gives +0.217 m at C4 under Model A and +0.038 to +0.066 m at C5 under Models B and A. Fifty percent thinning gives at most half of each, an upper bound, because the remaining trees and the understorey take up part of the water released (Gebhardt et al., 2014). C4 responds more than C5 because its slower drainage carries the same reduction in

draw to a higher equilibrium. These are the lower end of the clearfell range: the 4.2 ha compartment is narrower than the drawdown reach, so a recharge gain there would spread into the surrounding forest, and a wider felling could keep it (Section 4.13.2). The same canopy treatment therefore produces different responses across the forest zone, and management prescriptions cannot assume a uniform per-area benefit even where the canopy is the same.

Broadleaf conversion is reported as a range (Section 4.13.2). With the summer transpiration increase under a deciduous canopy in full leaf accounted for ($\beta_2$ summer multiplier ×1.075, reflecting comparable growing-season transpiration rates between deciduous and Corsican pine canopies; Komatsu et al., 2011) and no interception gain, it lowers the water table slightly, by 0.121 m at C4 and 0.037 m at C5 under Model A. With the lower interception of a broadleaf canopy (15% of rainfall, against 24% under pine) returned as recharge, it raises it instead, by 0.506 m at C4 and 0.308 m at C5 under Model A. Catchment modelling on sandy soils favours the gain (Sonnenborg et al., 2017); the record here cannot choose between the two ends.

At the lower ends of their ranges the three scenarios therefore differ in direction as well as size. Clearfell and thinning raise the water table by reducing the canopy’s draw; broadleaf conversion, without an interception gain, lowers it slightly, because a canopy in full leaf draws more in summer than a leafless one saves in winter. Where climate is also projected (C5), these lower ends are smaller than the climate-driven change projected over the same horizon (Section 5.7.1).

## 5.6 Hydrological Effects of the Plantation

The Corsican pine plantation intercepts approximately 24% of incident rainfall before it reaches the aquifer (Freeman, 2008). This interception is a partition of the available PET energy budget, not an additional loss on top of PET (Section 3.7.2): intercepted water evaporates from wet canopy surfaces without stomatal control, consuming a fraction of the atmospheric energy that would otherwise drive transpiration and direct soil evaporation. The $\beta_2$ coefficient for the forested clusters therefore reflects the combined effect of this unmediated interception evaporation plus the residual stomatal transpiration and direct evaporation beneath the canopy. The interception fraction is taken from Freeman's (2008) site-specific measurement. That measurement is a single plot over a single season (August 2007 to January 2008), sited deep within the stand to minimize edge effects and assessed by Freeman as representative of the majority of the plantation in canopy cover, tree density and age. It is the only site-specific interception measurement available, and the quantities that depend on it — the interception-corrected WTF specific yields (Section 4.2.4) and the volumetric water balance (Table 7) — inherit that limitation. The broader conifer plantation literature reports interception losses of 25–45% (Rutter et al., 1971; Gash, 1979; Gash et al.,

1980), with the higher end deriving from dense Sitka spruce stands in UK upland catchments (Calder and Newson, 1979; Robinson and Dupeyrat, 2005) — a different canopy structure from the partially thinned Corsican pine at Newborough. Numerical modelling of comparable coastal pine settings in northern Poland produced interception losses of approximately 27% (Gumuła-Kawęcka et al., 2021, 2024), consistent with Freeman's figure.

The canopy has further hydrological effects beyond recharge suppression. It provides microclimatic shielding — reducing direct solar radiation, insulating the sand surface through the organic forest floor, and reducing wind-driven advective evaporation — and the forest floor and root-zone macroporosity produce a smoothed vertical recharge signal relative to the rapid, pulsed infiltration of the open dune (Calder, 2007). The mechanism by which that suppression operates has been measured at this site. Freeman (2008) recorded mean volumetric soil moisture beneath the Newborough plantation at roughly half that of the open Warren (approximately 6.5% against 17.7%; one-way ANOVA $p < 0.001$, all plot pairs separated at $p < 0.01$), with the depth structure inverted: the open-dune profile is wettest at 40 cm while the forest profile is wettest in the top 5 cm and dries downward. She attributes the contrast to capillary contact between the upper soil layers and the water table, maintained in the open dune and lost beneath the canopy because the forest water table sits approximately 1.07 m deeper (ANOVA $p < 0.001$). The canopy therefore suppresses recharge twice over: once at the crown, where a fraction of rainfall is intercepted and re-evaporated, and again in the profile beneath it, where a deeper and drier unsaturated zone carries a larger storage deficit that each pulse must satisfy before the water table responds. Freeman's felled clearing plot, which carried a water table statistically indistinguishable from the open Warren, indicates that the second effect is a consequence of the canopy rather than an independent property of the forest zone. The lag diagnostic (Script 03) provides independent support for this smoothing effect. At lag-0, C5 Coastal Forest carries the lowest $R^2$ in the network (0.706, against 0.734 to 0.827 across the three open-dune clusters), indicating that the contemporaneous rainfall-to-water-table transfer is weakest beneath the coastal canopy; C4 Main Forest is second-lowest at 0.733, just below the open-dune range, and the case for canopy smoothing there rests on the lag structure rather than on the lag-0 fit. At lag-1, the pattern inverts: C4's lag-1 $R^2$ (0.421) is the highest in the network, with a significant and physically coherent $\beta_1$ of 1.564 ($p < 10^{-12}$). No other cluster retains this level of explanatory power at one-month lag — C1 Lake Edge drops to $R^2 = 0.037$ ($\beta_1$ non-significant). The canopy delays and smooths the recharge pulse so that a substantial fraction of each month's rainfall reaches the forest water table during the following month rather than contemporaneously. The lag-0 formulation is retained for all clusters because it maximizes overall network $R^2$ and preserves the operational simplicity of a single-lag framework. The strong lag-1 signal at C4 indicates that the forest's lower apparent $\beta_1$ (2.502 vs 3.808–4.577 at the open dune clusters) partly reflects temporal redistribution of recharge rather than a reduction in total recharge reaching the water table. Freeman's dipwell record provides a direct observational analogue at the plot scale. Over her 2007–08 monitoring period the

forest water table responded far less to rainfall than the clearing or open Warren, while the upper 40 cm of the forest soil profile responded more rapidly than either — the signature of a pulse arriving quickly at the surface and being absorbed within a thick unsaturated zone before it reaches the water table.

This temporal-redistribution reading carries a consequence for the management interpretation developed in Section 5.8.1. If the forest's lower apparent $\beta_1$ reflects recharge that is delayed rather than lost, then canopy removal recovers less water table gain than the open-dune-to-forest $\beta_1$ contrast would suggest — the intercepted-and-redistributed fraction was reaching the water table under canopy in any case, only later. This reinforces the P_flood finding (Section 5.8.1) that the forest clusters remain largely out of reach regardless of canopy management (of their 24 wells, 18 are marginal and 5 unreachable): the winter recharge deficit that canopy removal recovers is smaller than the raw $\beta_1$ difference implies, and is in any event insufficient to lift summer minima sitting ~1.4–1.7 m below ground into the reachable band.

These effects, and their interactions with the aquifer substrate, are examined in the subsections below.

### 5.6.1 Atmospheric Draw, Specific Yield and the $\beta_2$ Coefficient

The compound nature of $\beta_2$ and the dominance of specific yield in the C4/C5 contrast were established in Section 5.3.1. The per-well analysis within the forest zone is consistent with and extends this finding.

C4 Main Forest has the highest $\beta_2$ in the network. This does not mean the forest water table experiences the greatest atmospheric demand — it means that a given evaporative flux produces the largest head change at C4, because C4 has the lowest effective specific yield. C4's low effective specific yield reduces the drainable volume per unit head change. The WTF-derived Sy estimates (Table 8) give the measured contrast: C5's Sy is about a quarter higher than C4's. C4 overlies the bedrock ridge, which outcrops within the forest, so the nature of the underlying rock is established directly; what is not constrained without coring is the extent and depth of the ridge beneath the cluster, and hence the precise geometry of any buried ridges and the degree to which they impede lateral drainage. The contrast with C5 is consistent with C5 occupying deeper, more uniform coastal sand — essentially C3-type substrate beneath forest canopy, against C4's more C2-like setting — though the C5 sand depth is itself inferred from topography and coefficients rather than cored. Coastal proximity may also moderate the atmospheric-draw signal directly, through higher ambient humidity. The drainage decay half-life map (Section 4.9.3, Figure 55) is consistent with this: C4 wells show the longest t½ values in the network, indicating sluggish drainage that persists across years rather than free flow through a uniform sand column. The storage-drainage index $\tau = Sy/\beta_3$ provides a related perspective: where t½ captures the drainage recession at the project datum from $\beta_3$ alone, $\tau$ additionally weights by specific yield, expressing the ratio of stored

water volume to drainage flux at each well. Across the Newborough network, Sy and $1/\beta_3$ are spatially uncorrelated (r = 0.19), so the two indices carry independent information about aquifer architecture: t½ reflects how fast the aquifer drains; τ reflects how much water it holds relative to that drainage rate. The near-identical spatial patterns of the two fields (r = 0.94 across the reference network) indicate that the drainage rate $\beta_3$ is the dominant control on both — storage capacity modulates the index but does not reorganize it spatially at the scale of this network.

C5 Coastal Forest has the second-lowest $\beta_2$ in the network despite carrying the same Corsican pine canopy as C4. This reflects C5's higher Sy — the thicker coastal sand provides more drainable storage per unit depth, so the same atmospheric demand produces a smaller head change. The per-well $\beta_2$ range within C4 alone spans nearly the full site-wide range (Section 4.9.4; Table 23), indicating that within-cluster substrate variation is at least as important as between-cluster canopy differences. Cluster-level centroid values summarize a spatially variable field; they should not be taken as implying uniform conditions across the forest.

Across the forest wells, elevation alone explains the bulk of the variance in $\beta_2$ (Section 4.9.4), with higher-elevation wells on the ridge showing systematically higher atmospheric draw sensitivity. The C4/C5 boundary corresponds to a clean elevation break between the elevated ridge and plateau terrain and the low-lying coastal fringe, with no overlap. Both zones carry the same Corsican pine canopy, so the difference tracks elevation and water-table depth rather than canopy: C4 sits on the bedrock ridge (which outcrops within the forest) and C5 on the lower coastal ground, though the depth and extent of rock beneath the canopy are not constrained without coring. The implication for management interpretation is that $\beta_2$ values at any given forest well are dominated by position, not canopy: changes in $\beta_2$ following management must therefore be measured against a position-matched control if they are to be canopy-attributable. This is precisely the role of the Forest control tier in the clearfell ANCOVA (Section 4.6.6).

Although position — and the substrate inferred to underlie it — dominates the spatial pattern of $\beta_2$, the post-clearfell coefficient analysis (Section 4.6.6; Table 15) is consistent with a canopy shielding effect on top of this baseline but does not establish one. $\beta_2$ increased at the intact-pine Forest Control wells following felling, consistent with the maturing canopy moderating the atmospheric demand signal reaching the water table or, in part, with a fitting artefact (Section 5.5.1); and the direct test at the felled well finds no loss of shielding, its canopy-on draw changing by −1.7% net of the climate controls (Section 4.13.2). The response at the Edge tier was mixed, with no consistent direction across its four wells, and $\beta_2$ did not increase at the Impact well or at the Coastal Control. The shielding mechanism operates at the surface: intercepted water evaporates readily from wet canopy surfaces without stomatal control, consuming PET energy before it can act on the water table. At the one felled well the record does not show that exposure: its atmospheric draw fell after

felling, by 4.8% beyond the climate controls and by 1.7% in the canopy-on months (Section 4.13.2), so the loss of shielding that removing the canopy would imply is not observed there. This shielding effect is superimposed on a position-driven $\beta_2$ baseline — reflecting Sy and the inferred substrate — that varies by a factor of two across the forest, and the cross-cluster $\beta_2$ pattern reflects Sy differences rather than canopy differences.

The depth-dependent PET diagnostic (Section 4.4; Section 3.7.1) is consistent with this interpretation. C1 Lake Edge shows the strongest depth coupling, consistent with shallow water-table evaporation near the lake being the dominant ET pathway, while C4 Main Forest shows the weakest depth coupling despite having the highest $\beta_2$. If C4's high $\beta_2$ were driven by deep-root transpiration overriding capillary disconnection, C4 would show the strongest depth coupling, not the weakest. Together with the sunshine-hours residual diagnostic (Supplementary Note S6; Section 5.2), which returned a null result, this is consistent with the standard SSM's fixed $\beta_2$ having already absorbed the bulk of the ET signal at the cluster level.

The practical consequence connects the plantation directly to the ecological outcome of most concern. Summer minimum water table depth is the dominant predictor of winter flooding potential in the two forest clusters and shares that role equally with winter rainfall in the Western Residual (Section 5.2; Table 16). Under a warming climate with increasing summer PET, any intervention that raises $\beta_2$ would amplify the head response to that PET, deepening summer minima per unit of evaporative demand and raising the rainfall threshold required for winter flooding. The felling itself did not raise $\beta_2$: at the felled well draw fell, by 4.8% beyond the climate controls (Section 4.13.2). The scenario projections anchored to that felling (Section 4.13.2; Section 5.5.2) give small sustained rises at C4 and C5 and, at their lower ends, smaller changes than the climate-driven change projected over the same horizon where both are projected (C5). The BACI climate-sensitivity analysis (Figure 30; Section 4.6.2) provides a complementary line of evidence: the $CWB^2$ × clearfell interaction points, without reaching significance (Impact $p = 0.22$, Edge $p = 0.36$), toward the felling-attributable water table rise having been concentrated in dry conditions under the canopy — the direction a canopy partially buffering the water table against the worst dry-period deficits would produce. It is not a finding, but it is measured on the relative scale that the amplification field (Section 5.3.2) cannot provide.

### 5.6.2 Summer Minimum Depths at the Forest Margin

The relatively shallow summer minima at NW2 and CEH2 are not explained by the water balance residual field. Neither well carries a positive residual (NW2 −0.0008 m/month, CEH2 −0.0043 m/month); both sit close to the network median, and no well in the network exceeds +0.02 m/month (Section 4.9.6, Figure 59). No local departure from closure, and so no unmodelled local input, need be invoked at these wells. Clearance of the plantation in this zone would eliminate the canopy contribution to the local water balance — the coefficient decomposition (Section 4.6.6) shows that $\beta_2$

increased at the intact-pine Forest Control wells following felling (mean $\Delta\beta_2$ +0.157, excluding the broadleaf well NW10), consistent with the canopy moderating the atmospheric demand signal reaching the water table. Whether removing it at NW2/CEH2 would raise $\beta_2$ is not shown by the one felling observed, at which draw fell (Section 4.13.2). The BACI analysis (Section 5.5) found that canopy removal in the adjacent clearfell zone raised the mean monthly water table but did not improve summer minima — the ecologically decisive metric — at the felled wells. The same seasonal partition would be expected at NW2/CEH2: clearance could lift the mean level while leaving the summer minimum, following the background climate trajectory, substantially unchanged.

The summer minimum depth map identifies where water table levels are currently most actively sustained against the background climate drying trend. The favourable position at the forest-adjacent margin reflects canopy condition together with local factors the monitoring record does not resolve; the available evidence does not support clearance as a route to preserving or improving it.

### 5.6.3 Broadleaf Conversion: The NW10 Natural Experiment

The monitoring network contains a direct observational test of broadleaf conversion effects. A block of mixed broadleaf woodland was established following clearfell in 1993 and restocked in 1995 on the north-western margin of the plantation, with dipwell NW10 situated within the block and monitored continuously from 2007. CEH2 and CEH34, both classified as C4 Forest under intact Corsican pine canopy, serve as the principal comparators; NW11 and NW13 provide a paired pine-warren boundary transect approximately 90 m apart at virtually identical elevation.

The per-well SSM coefficients at NW10 are informative. Its recharge sensitivity $\beta_1 = 3.493$ substantially exceeds the C4 per-well median of 2.333 and the adjacent Corsican pine well CEH2 ($\beta_1 = 2.760$), consistent with the broadleaf canopy intercepting less rainfall than dense pine and allowing more recharge to reach the water table per millimetre of rainfall. The seasonal interception pattern expected from a deciduous canopy supports this reading: during the October–March recharge season the leafless broadleaf stand intercepts less rainfall than the year-round Corsican pine canopy. The meta-analytical review of Komatsu et al. (2011) reports that broadleaf deciduous forests in high-winter-precipitation climates consistently transmit more rainfall to the sub-canopy than coniferous forests at the same sites, with the effect most pronounced during the dormant season.

The atmospheric draw coefficient $\beta_2 = 2.740$ at NW10 is below CEH2 ($\beta_2 = 2.807$) and close to the C4 per-well median (2.581). The broadleaf stand does not show an increase in atmospheric draw above the C4 range. This is consistent with the seasonal offset of deciduous leaf-off during October–March partially compensating for higher growing-season ET demand, such that the year-round integrated atmospheric draw at the water table is

comparable to that under adjacent pine. The comparison is complicated by positional differences: NW10 carries a higher $\beta_3$ (0.039) than the C4 median (0.019), so drains faster than most C4 wells, and its ground elevation and distance from the ridge differ from CEH2 and CEH34. These positional differences mean that the $\beta_2$ comparison isolates species effects imperfectly.

NW10's behaviour relative to its pine comparators differs depending on which comparators are used and whether the comparison is climate-normalized. When normalized to the shared climate signal, NW10's summer minimum sits consistently shallower than the pine interior composite (CEH2, CEH32, CEH33, CEH34), with a mean anomaly of +249 mm across the full record and +268 mm during the post-felling period (2018–2025). The anomaly declines at −26 mm/yr over the post-felling summers, a trend that does not reach significance ($p = 0.10$, six summers; Script 10g). Its direction is the one expected if the broadleaf stand progressively deepens summer minima as canopy closure increases interception and transpiration demand — the direction of trajectory observed for the standing pine canopy as it matures — but the record does not resolve it. The direct comparison against CEH2 (NW10's nearest pine neighbour, about 350 m downslope) shows NW10 shallower than CEH2 across the monitoring record (full-record mean summer minima −1.52 m against −1.71 m); however, CEH2 itself shows a progressive deepening trend relative to the network mean that is not present in other pine interior wells, which complicates the local-pair comparison. The reconciliation of these two pictures is the principal limitation of the NW10 experiment, and is addressed below.

The scenario framework (Section 4.13.2; Section 5.5.2) provides an independent quantitative reference. With the summer transpiration increase under mature deciduous canopy accounted for ($\beta_2$ summer multiplier ×1.075) and no interception gain, broadleaf conversion lowers the water table slightly at both forest clusters (−0.121 m at C4 and −0.037 m at C5 under Model A); with a broadleaf canopy’s lower interception returned as recharge it raises it (+0.506 and +0.308 m). The scenario and the diminishing climate-normalized anomaly at NW10 agree on the seasonal pattern: the winter leaf-off advantage is real, and a mature broadleaf canopy gives part of it back in summer. The phenological winter recharge advantage at NW10 does not translate into a sustained shallower summer minimum on the available evidence.

NW10 is a single dipwell, and the positional confounders are of comparable magnitude to the observed contrasts. The spread between CEH2 (mean summer minimum −1.711 m) and CEH34 (−1.312 m), both Corsican pine wells within the same cluster, is 399 mm, exceeding the NW10 climate-normalized anomaly of +249 mm. No reliable attribution to canopy type is possible from a single dipwell, and a minimum of three to four instrumented wells distributed across the broadleaf compartment would be required before firm conclusions could be drawn. The current −26 mm/yr trend is not significant and supports no extrapolation; it rests on a single well's post-felling record.

A related observation suggests the broadleaf block's hydrological footprint may extend beyond its physical boundary. CEH2, situated approximately 350 m downslope within the adjacent Corsican pine plantation, shows the progressive deepening trend relative to the network mean noted above. This is consistent with the maturing broadleaf canopy progressively increasing winter recharge within the block, steepening the local hydraulic gradient and drawing down the water table in the adjacent downslope zone — the same mechanism that operates at the scraped slacks (Section 5.4). The observation is based on a single well comparison and cannot be attributed to canopy type with confidence, but it warrants targeted instrumentation downslope of the broadleaf block.

### 5.6.4 Stand Density, Thinning and the Dothistroma Natural Experiment

The preceding sections characterize the two alternatives the record or the model can speak to. Clearfell raises the mean water level through a reduced atmospheric draw, with no measurable gain in recharge at the felled well (Section 4.13.2). Broadleaf conversion lowers it slightly without an interception gain and raises it with one, its winter advantage set against growing-season transpiration. A third option — partial canopy retention through stand density management — has not been tested in the monitoring record but is mechanistically distinct from both alternatives.

No thinning experiment has been conducted within the study period and no thinned-stand wells exist in the network. The theoretical argument that an intermediate canopy density could retain a measure of microclimatic shielding whilst partially recovering the recharge penalty is mechanistically plausible — the β coefficient framework predicts that partial canopy retention would reduce $\beta_2$ less than complete removal while recovering a proportional fraction of the interception loss from $\beta_1$ — but this remains a prediction rather than an observed outcome. The scenario framework (Section 4.13.2; Section 5.5.2) estimates that 50% thinning would produce approximately half the annual net response of full clearfell at both forest clusters, with a similar seasonal structure. The thinning and clearfell seasonal profiles are compared in Figure 82 (Section 5.7).

On prograding Atlantic shorelines, stabilized back dunes naturally succeed to deciduous woodland — the recognized climax community of the UK psammosere (Ranwell, 1972). The geomorphological position of the Newborough plantation is therefore not unusual for back-dune terrain; what differs from natural succession at this site is the species choice and the stand density, which reflect commercial forestry management. Freeman's (2008) site-specific measurement of 24% interception sits just below the 25–45% range typically reported for UK conifer plantations, reflecting the exposed coastal setting and the partial thinning that has already occurred. The plantation's effect on the water balance reflects stand density as much as species choice.

Progressive Dothistroma needle blight infection is already driving natural canopy thinning across parts of the Newborough plantation. Monitoring the $\beta_2$ response of wells adjacent to

blight-affected stands of varying severity would provide a direct field test of the canopy shielding mechanism, which this study leaves open, and would constitute a natural experiment in progress requiring no additional silvicultural intervention. Whether the trajectory leads toward a hydrologically distinguishable equilibrium cannot be asserted from the present data, but it is a well-defined monitoring question for the 2026–2030 period.

### 5.6.5 Canopy Salt Interception and Post-Felling Geochemistry

A further geochemical consequence of canopy removal appears to have received limited attention in the dune restoration literature but follows from the LAI-deposition relationship documented in coastal forest hydrology. Pine plantation canopies intercept sea salt aerosols with greater efficiency than open dune vegetation: throughfall studies in coastal *Pinus radiata* plantations have reported chloride deposition rates up to 89% higher beneath the canopy than in adjacent open fields, attributable to the high leaf area index of the needle canopy (Deng et al., 2013), and independent soil studies in the same species have shown systematically higher exchangeable sodium and magnesium pools under pine than under adjacent pasture (Giddens et al., 1997). The Newborough plantation is a different pine species (*Pinus nigra* var. *laricio*), so the specific deposition magnitudes do not transfer directly, but the qualitative mechanism — canopy interception of marine aerosols at LAI characteristic of mature coniferous stands — applies at any site exposed to onshore marine airflow. The Newborough Warren plantation has been exposed to frequent onshore Atlantic airflow throughout its 75-year history.

If this mechanism operates at Newborough at any meaningful magnitude, the forest soils will have accumulated an elevated $Cl^-$ and $Na^+$ load relative to the open dune system over the plantation's history. Forest removal simultaneously terminates this elevated deposition input and, by increasing aquifer responsiveness following canopy removal, accelerates the downward flushing of accumulated ions through the highly transmissive dune sand. The geochemical consequence is a progressive leaching of base cations from soils that previously received sustained buffering from canopy-concentrated deposition. Curreli et al. (2013) and Sival and Grootjans (1996) identify the maintenance of carbonate-rich groundwater buffering as critical for the persistence of basiphilous slack communities; this flushing dynamic represents a potential geochemical pathway by which large-scale clearance could alter the soil chemistry of base cation buffering, operating on a timescale longer than the water table response and potentially with a longer lag time before its effects become apparent in the vegetation.

A countervailing process should be acknowledged: pine needle litter produces organic acids that acidify the soil surface over time, and removal of the canopy would terminate this acidification pathway. The net geochemical effect of clearfell — salt-leaching versus relief from acidification — is likely to be site-specific and depth-dependent, and warrants targeted investigation through paired throughfall and soil porewater chemistry monitoring across the forest–dune boundary.

## 5.7 Spatial Groundwater Modelling: Findings and Limitations

A well-level scenario framework (Section 3.8.4; Section 4.13) was applied to translate the per-well SSM parameterization into site-wide summaries of equilibrium head change under prescribed climate and forest management perturbations. Climate scenario responses are spatially coherent and physically consistent with the per-cluster β coefficient structure: UKCP18 2050s forcing lowers the site-wide mean annual level by 0.159 m, and by 0.286 m under UKCP18 2080s (sustained responses; C4 is not projected). At C5 the annual decline reflects reduced net recharge under canopy interception combined with the cluster's already-steep observed trajectory (Section 4.8.1). These outputs are presented as indicative spatial summaries of the SSM equilibrium framework rather than calibrated predictions.

### 5.7.1 Seasonal Asymmetry under Climate Projections

The annual responses conceal a pronounced seasonal asymmetry that is ecologically critical (Section 4.13.1). Under both UKCP18 trajectories the summer level falls most (site-wide 0.213 m by the 2050s, 0.387 m by the 2080s), and the winter level falls too except at C1: the extra winter rain raises the winter forcing, but the summer deficit, carried forward by the slow return of the water table to its mean, outlasts it. This summer-led signature reflects the same mechanism established in the observed record (Section 4.8.1): hydrological deterioration under climate change operates through intensified summer atmospheric demand rather than declining winter recharge, and the slow return to the mean spreads it through the year. Annual-mean Δh therefore understates the ecological consequence of summer-dominated climate forcing; the seasonal partition should be reported alongside any annual summary.

### 5.7.2 Forest Management Scenarios and Structural Isolation

For forest management scenarios, the per-well framework carries an implicit structural feature that is itself informative: because β coefficient changes under canopy removal, thinning, or broadleaf conversion are applied only at C4 and C5 wells, the IDW-interpolated Δh field produces a response concentrated in the forest zone. The drawdown propagation analysis (Section 4.11, Figure 66) estimates a characteristic e-folding distance of 220 m, with the modelled perturbation falling from 72 mm at the nearest C1 well (CEH27) to 8 mm at CEH6 across the six eastern C1 wells, and to 2 mm or less at CEH24, CEH10 and CEH11. Site-wide annual means under forest management are correspondingly small (Section 4.13.2; Section 5.5.2), because only 14 of 66 reference wells are in the two forest clusters and the propagation signal does not reach the eastern dune and lake-edge wells CEH24, CEH10 and CEH11 at measurable magnitude.

Clearfell is predicted to raise C5 by +0.038 m (Model B) to +0.066 m (Model A) as a sustained change in level, less than C4 (+0.217 m, Model A), whose slower drainage carries

the same reduction in draw to a higher equilibrium (Section 4.13.2). C5 is also the cluster with the steepest observed decline in the network — 2.5 times the summer rate of C1, a winter decline that falls just short of significance, and an ongoing collapse in seasonal amplitude (Section 4.8.1). Over the full record the magnitude of that divergence is consistent with a contribution beyond the climate forcing all clusters share; but the decline is concentrated before 2011, and refitted from 2011 C5's summer trend is +0.003 m $yr^{-1}$ ($p = 0.79$), so the divergence is a feature of the early record rather than a continuing departure. Three candidate explanations, not mutually exclusive, present themselves: amplified head response per unit climate forcing due to thin saturated thickness at the coastal margin, where a given recharge deficit produces a proportionally larger head change at lower Sy; progressive coastal boundary retreat, reducing aquifer volume at the seaward edge and producing an ongoing decline with no equilibrium; and progressive canopy maturation, increasing interception and transpiration demand as the Corsican pine stand ages. The post-2018 amplitude collapse — recharge peaks no longer reaching their pre-2018 heights even after correcting for drought summers (Section 4.2.1) — points to at least one site-specific process beyond the common climate signal, and the non-stabilization of the C5 CUSUM trajectory to the end of the record favours an ongoing process over a one-off step change. The network-scale partition of Section 4.10.2 then makes coastal retreat the largest quantified contributor: applied to C5's wells, each at its own distance from the shore, the fitted gradient accounts for about half of the cluster's decline, leaving −17.6 mm $yr^{-1}$ unexplained, so the other two candidates are not excluded and retain room for a quantitative role at C5. That attribution is sensitive to cluster composition (Section 4.10.2). Whether the predicted benefit could arrest the observed decline depends on the mechanism driving the decline: if the decline is primarily climatic, the scenario gain offsets several months of trajectory per year; if a structural process such as coastal boundary retreat contributes substantially, the gain offsets only the climatic component, with the structural component continuing. Two independent strands of the change analysis below indicate that the structural component is real and spatially concentrated at exactly these wells. The differential-movement map (Figure 73; Section 4.12), which removes the common climate signal and fits a per-well secular trend, identifies the south-western coastal and western margin — including the C5 coastal members — as the robust, window-independent sinkers on the site, consistent with the coastal-retreat gradient of Section 4.10.2. Independently, the two-window MSL5 change correlates with the long-run summer-minimum slope (14 to 21 years per well) at $r = 0.43$ ($p = 0.003$, $n = 46$; Section 4.12), indicating that the coastal decline is a secular trend rather than a window artefact. The C5 decline therefore carries a structural coastal-retreat component that clearfell cannot offset, so the predicted scenario benefit should be read as addressing the climatic component only. A seasonal robustness check (Supplementary Note S8) finds the C5 decline present in both the summer minimum and the spring mean (−29 and −24 mm $yr^{-1}$), and a full-panel season × distance-decay interaction test finds the coastal-retreat gradient about 31% steeper in spring than over the rest of the

year ($\gamma = +0.31$, $p < 0.001$) but acting year-round — consistent with a coastal-boundary process rather than a drought-season artefact.

The spatial coefficient atlas (Section 4.9.2, Figure 54) provides independent characterization of the forest zone's hydrological position. The $\beta_1$ surface shows a sharp step at the forest boundary — recharge sensitivity drops from open-dune values to substantially lower values beneath the canopy — and the $\beta_3$ surface shows an order-of-magnitude contrast between the fast-draining lake edge and the near-stagnant forest interior. These are not gradual transitions; they are consistent with a zone that runs its own local water balance with limited coupling to the open dune clusters. The bedrock ridge forms the northern boundary of the entire study area (Section 2), and most groundwater flows southward off the ridge toward the coast; only the slacks beside Llyn Rhos-Ddu drain north to the lake. On the kriged water table (Section 4.9.5, Figure 56) the forest zone drains south and south-west toward Caernarfon Bay, while the eastern open dune drains east and south-east toward the Afon Braint. The flow arrows describe direction and relative gradient only; they are not evidence of a lateral input at the ridge margin, and the water-balance residual field does not support one — the Main Forest cluster carries the most negative residual in the network (C4 mean −0.0066 m/month), not a positive anomaly. A head divide in the central dune field separates these two drainage paths. A canopy-management perturbation in the C4/C5 zone therefore propagates south-westward toward the coast, not south-eastward into the C1 and C2 clusters. Hydrological connectivity along the coastward gradient is consistent with the April 2015 scraping response at CEH36 (Section 5.4.2); this connectivity operates between the forest zone and the south-western coastal margin rather than between the forest zone and the eastern slacks.

### 5.7.3 Framework Limitations and Continuous-Flow Modelling

The conclusion that forest management perturbations do not reach the eastern slacks requires qualification. The forest drawdown propagation analysis (Section 4.11, Figure 66) estimates the steady-state head perturbation attributable to the existing Corsican pine canopy: the signal exceeds 50 mm out to about 240 m of the forest edge and falls below 10 mm beyond about 600 m, reaching the six nearer lake-edge wells at 8–72 mm and negligible (≤2 mm) at CEH24, CEH10 and CEH11. This quantifies the spatial reach of the canopy's current influence on the water table. It does not set the reach of a felling's recovery: a compartment narrower than that reach does not keep a local recharge gain, which spreads into the surrounding forest (Section 4.13.2).

The dune-scrape and clearfell-baseline fields (Figures 67, 68) carry the same single-mechanism, steady-state caveats as the forest and coastal fields: the scrape edge magnitude is anchored to the one measured response, but the surrounding drawdown is inferred through the drain model rather than measured well-by-well, and the combined baseline field is a first-order superposition, not a closed budget. However, the kriged flow field (Section 4.9.5, Figure 56) and the Pearson affinity analysis (Section 4.3) show that C4 and C5 are

hydrologically connected along a continuous downslope drainage pathway. The clearfell's downslope effects on CEH36 (Section 5.4) indicate that perturbations in the C4 zone can propagate into C5. Whether they propagate further into C3 and C1/C2 at magnitudes below the propagation model's resolution cannot be resolved by the current framework, and would require a calibrated continuous-flow model.

The practical limits of the spatial framework lie in what it deliberately does not attempt. The per-well IDW framework is a geometric aggregator of SSM responses rather than a physical model of groundwater flow, so it cannot represent cluster-to-cluster hydraulic coupling, lateral redistribution of localized recharge perturbations, or the diffusive response of the aquifer to boundary-flux changes. For one question this study addresses — how climate perturbations partition across the cluster structure — the framework is adequate. For the other — whether forest management in the C4/C5 zone affects summer minima in C1 and C2 — it cannot show propagation, and the answer rests on the BACI record, which found no felling-specific response outside the forest zone. Answering a different class of question, such as how a hypothetical intervention along the northern forest margin would propagate through the aquifer, would require a calibrated continuous-flow model, and for that the critical unknowns are hydraulic conductivity — currently constrained by a single tracer test (K = 6.0 m/day; Betson et al., 2002) — and aquifer thickness, inferred from four borehole logs across 700 ha (Bristow, 2003). Slug tests at two to three representative wells per cluster and a ground-penetrating radar survey of aquifer thickness are identified as the priority field measurements required to enable such a model. These data would also support a HYDRUS-1D vadose zone analysis (Šimůnek et al., 2008) at the cluster level, providing an independent test of the vegetation-driven recharge suppression signal identified in C4 and C5 (Gumuła-Kawęcka et al., 2021). The coastal transect of Section 4.10.4 bears on one of these unknowns. A static substrate geometry — the thin-saturated-thickness mechanism — would hold the coast-to-inland head offset constant whatever the interior did, whereas the observed offset trends downward across 2010–2023 while the spring climate water balance does not trend: the signature of a coastal boundary that fails to rise with the interior, not a disguised climate trend. Acting against retreat, sea-level rise raises the head at the seaward boundary rather than lowering the ground beside it (Section 4.10.6); the commonly cited expectation that sea-level rise raises coastal water tables is therefore not wrong here, merely subordinate at the margin.

### 5.7.4 Forest Scenario Predictions Against the Observed Record

The scenario framework’s predicted responses can be set against the observed post-felling record at WMC3 (Figure 82). Applied to the observed C4 mean seasonal cycle on the cluster-centroid coefficients under Model A, with the felled well’s measured change in draw and no interception recharge (Section 4.13.2), the three management options separate in every month rather than in summer alone, because the responses are sustained. Full clearfell raises the water table by 0.374 m at the September trough and thinning by at most half that

(0.187 m). These centroid figures are larger than the mean of the per-well responses at C4 (+0.217 m; Section 4.13.2), because the C4 centroid drains slowly under Model A, and larger again than the sustained rise at the felled well itself (+0.060 m under Model B, +0.135 m under Model A), which is a C3 well.

Broadleaf conversion without an interception gain lowers the water table in every month, by 0.313 m at the September trough on the same coefficients: the higher growing-season draw of a deciduous canopy in full leaf outweighs its lower draw in the leafless months (Section 5.5.2). This is the lower end of the broadleaf range; returning a broadleaf canopy's lower interception as recharge reverses it (Section 4.13.2). At WMC3 the felling changed the summer minimum by only −7 mm against the Forest control (Section 4.6.4) while the mean monthly level rose +108 mm (Section 4.6.3): the sustained responses describe the mean level, and the summer minimum at the felled well did not follow it.

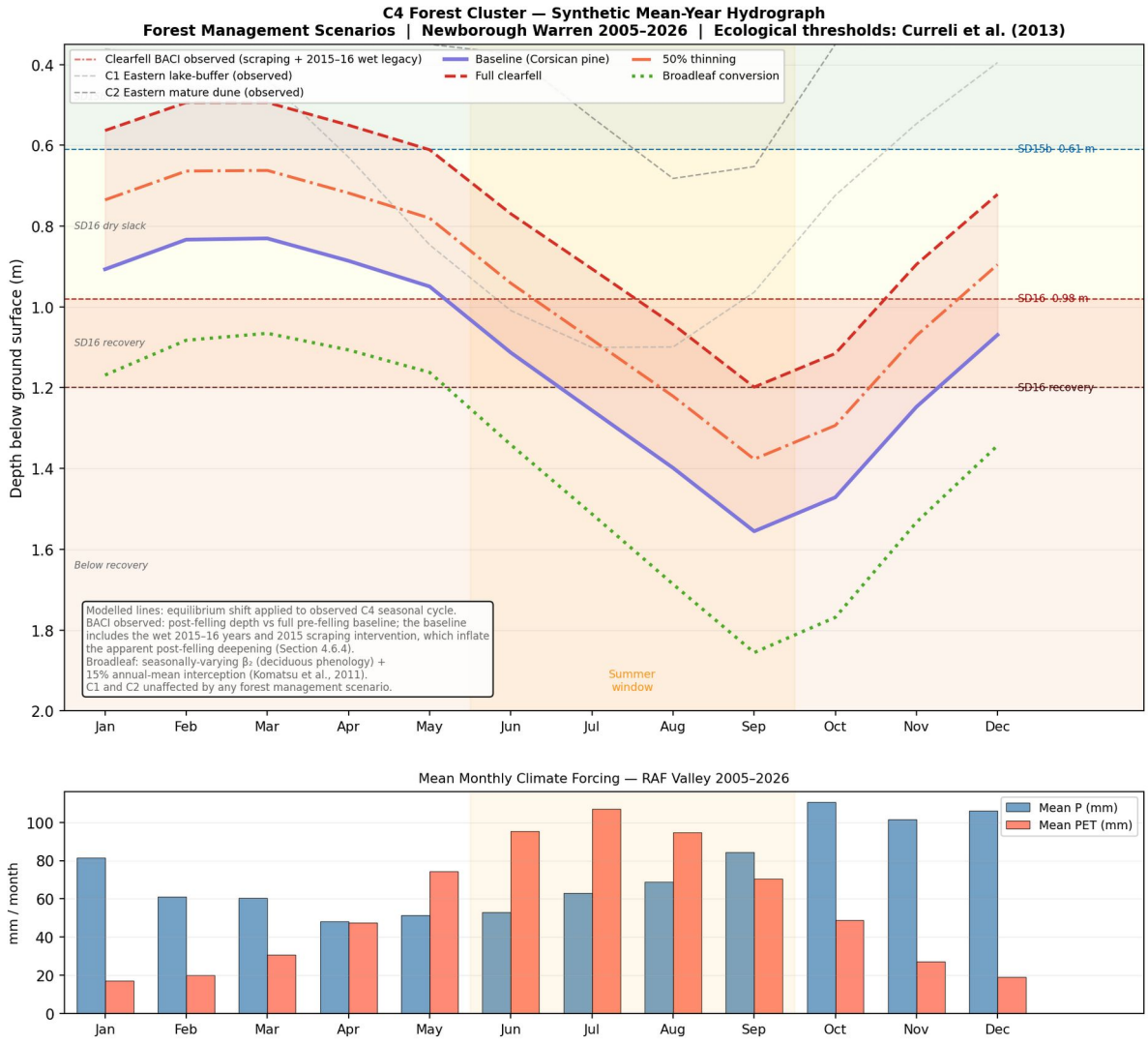



***Figure 82:*** *Synthetic mean-year hydrograph for the C4 Forest cluster under four management scenarios:*

*baseline Corsican pine, full clearfell, 50% thinning, and broadleaf conversion. Lines show equilibrium head shifts applied to the observed 2005–2026 C4 seasonal cycle, derived from the felled well's measured change in $\beta_2$ (ET draw) for clearfell and thinning, with no interception recharge term, on the C4 cluster-centroid coefficients under Model A. The BACI-observed WMC3 summer minimum is shown for context: its era-mean summer minimum was −1.573 m before scraping, −1.62 m through 2015–17, and −1.60 m post-felling; controlling for climate against the Forest control, the felling itself changed the WMC3 summer minimum by only −7 mm (p = 0.91, ns; Section 4.6.4). The modelled clearfell line describes the mean level; the summer minimum at the felled well did not follow it (Section 4.6.4), while the mean monthly level rose +108 mm (Impact; not significant on its own, p = 0.10, with a winter rise of +128 mm, p = 0.048; Section 4.6.3). C1 and C2 observed mean seasonal cycles are shown for ecological context; no management scenario produces any detectable effect on these clusters. Broadleaf conversion: seasonally-varying $\beta_2$ applied to reflect deciduous phenology (lower winter ET, higher summer ET, $\beta_2$ summer multiplier ×1.075). Lower panel: mean monthly rainfall and PET from RAF Valley 2005–2026. Ecological zone colouring follows Curreli et al. (2013). Generated from 03_master_data.csv and 03_regional_averages_maod.csv. (Source: 21_forestry_01_hydrograph.png).*

### 5.7.5 Parametric Prediction and Observed Network Change

The parametric net state map (Figure 70, Section 4.12) and the observed MSL5 change map (Figure 72, Section 4.12) present the same aquifer through different lenses: one through the single-mechanism annual-mean driver fields of Sections 4.10.5 and 4.11, the other through the directly measured change in five-year mean spring water level between the pre-clearfell baseline (MSL5 window end 2017, springs 2013–2017, network mean −467 mm) and the current period (window end 2023, springs 2019–2023, network mean −572 mm). The comparison window is discussed below.

**The standing canopy effect is invisible to a change map**

The parametric map predicts a large net loss across the forest interior (about −50 to nearly −300 mm, median about −150 mm) driven by the standing Corsican pine canopy drawdown. The observed MSL5 map shows a site-wide deepening with no positive zone in the forest interior. This is not a contradiction. The forest canopy drawdown represents a steady-state penalty relative to a hypothetical no-forest aquifer — a penalty equally present in both MSL5 windows. A before–after comparison cannot detect a driver whose effect is constant across both periods. The MSL5 comparison therefore cannot see the canopy drawdown at all; its magnitude is constrained by the parametric model (Section 4.11) and by the per-well $\beta_2$ coefficients of Section 4.9.2. The observed site-wide deepening reflects a change in the aquifer state between the two windows, not a response to the standing canopy.

**What the MSL5 change map shows**

The 2017–2023 MSL5 comparison (Figure 72; raw difference, n = 59 wells) shows the aquifer deeper across the site in the current window. The network mean MSL5 has deepened by 105 mm, from −467 mm (2017) to −572 mm (2023) below ground; 58 of the 59 wells exceed the ±25 mm threshold and every one of them deepens — no well shallows. Because the figure differences two five-year windows, every time-invariant driver cancels — in particular the constant drawdown of the standing forest — so the surviving pattern reflects only drivers that progressed across the study period, and the deepening intensifies from the eastern dune interior toward the south-western coastal margin. The eastern C1/C2 dune wells deepen least (CEH25 −11, D10 −29, L7 −31, D7 −34 mm), CEH25 alone within the ±25 mm near-neutral band. The clearfell Impact well WMC3 (−90 mm) and the scrape well CEH36 (−97 mm) deepen broadly in line with the −105 mm site-wide mean, showing no intervention-specific departure from the surrounding network. The most strongly negative zone is the south-western coastal margin: CEH22 (−233 mm), CEH21 (−213 mm), CEH18 (−145 mm), and the C5 coastal members CEH19, CEH17 and NW9 (−164 to −172 mm). The five Forest Control wells deepen by −140 mm on average (−138 mm for the C4 cluster mean), more than the site-wide average; as the standing-canopy drawdown cancels in a window-to-window difference and these are unfelled controls, this excess is neither a

standing-canopy nor a clearfell effect but the C4 cluster's stronger differential response to the between-window climate anomaly, expressed in full because the map applies no climate correction (Section 4.12).

**Window selection and the spring composition of the two windows**

The choice of the 2023 current window requires justification. An alternative current window ending in 2025 (springs 2021–2025) was initially examined but not adopted: it contains the wettest spring of the 2007–2025 well record (2024, by RAF Valley spring rainfall), and the forested clusters respond more strongly to wet conditions than the open dune. In the 2025 window the Forest Control wells (CEH32, CEH34, CEH33, NW10, CEH2) show a large gain over their 2013–2017 baseline of +133 mm, compared with +29 mm for the open-dune Climate Controls — an apparent Forest Control excess of +103 mm. Three checks indicate this excess arises from the spring composition of the two windows and from the limited resolution of a five-well group contrast, rather than from a real management or structural signal.

Check 1 — spring gap stability. The spring gap between the Forest Control group mean and the Climate Control group mean was −567 mm pre-clearfell (2013–2017) and −565 mm post-clearfell (2018–2025, excluding 2024, the wettest post-felling spring by RAF Valley spring rainfall): a change of +3 mm, Mann–Whitney $p = 0.876$; with 2024 included the change is +17 mm, $p = 0.943$. The Forest Controls are structurally deeper than the Climate Controls in spring — reflecting the canopy drawdown — and this gap has not changed. In the 2025 MSL5 window the Forest Control excess over Climate Controls is +103 mm; in the 2023 window it collapses to −15 mm. The switch reflects which springs each window contains, not a change at the Forest Control wells. Measured against the 2013–2017 baseline gap, the five springs of the 2021–2025 window contribute +27, +43, −17, +24 and +26 mm respectively to the +103 mm excess. No single spring accounts for it: the largest contribution comes from 2022, and the anomalously wet 2024 spring contributes 24 mm, less than a quarter of the total. The 2019–2023 window contains 2019 and 2020, the two widest-gap springs of the 2013–2025 comparison period, which is why the same comparison there returns −15 mm.

The gap varies systematically with how wet the network is. Regressed on the site-mean spring level of the same year and of the preceding year, both terms are significant (contemporaneous +0.31 mm per mm, $p = 0.013$; one-year carry-over +0.28, $p = 0.007$; $R^2 = 0.60$, $n = 13$). This is consistent with the larger climate response of the C4 Main Forest (Section 4.12) acting together with the far slower decay of a fluctuation in the forest. C4's free-intercept drainage coefficient is not identified on the record (Section 3.4.1), so the forest's own mean-reversion time cannot be read from the model; the one-year carry-over term above is the measured statement, and it belongs to the forest-minus-control gap, not to a single well's spring series. The head-space recession times at the datum ($1/\beta_3$, about 58

months at the Forest Controls against 21 at the Climate Controls, per-well group means; Section 4.9.3) order the two groups the same way.

This mechanism accounts for part of the excess but not all of it. Carrying the observed window wetness through the fitted relation predicts about +28 mm of the +103 mm. The remainder lies within the resolution of the contrast: the gap has an interannual standard deviation of 121 mm over 2013–2025, so a five-year window mean carries a standard error of about 54 mm and a difference between two such windows about 76 mm. On that footing the +103 mm excess is 1.4 standard errors and the −15 mm of the 2023 window is 0.2, neither separable from zero. This resolution limit is a property of a contrast between two five-well groups and does not bear on the network-mean deepening of Figure 72, which rests on 59 wells of which 58 exceed the threshold and none reverse. Both group figures are consistent with the unchanged pre- and post-clearfell gap established above, and neither supports a management or structural interpretation.

Check 2 — independent metric agreement. The MSL5 2017–2023 raw change correlates significantly with the long-run summer minimum slope from Script 25 (Section 4.12; Pearson $r = 0.43$, $p = 0.003$; Spearman $r = 0.40$, $p = 0.006$; $n = 46$ wells). These are entirely independent metrics: the summer minimum slope is a per-well linear regression over 14 to 21 years (median 16) of annual April–September minima (hydrological-year indexed) (Section 4.10.2, Script 25); the MSL5 comparison uses five-year means of the readings dated March–May from two specific windows. The convergence ($r \approx 0.4$ on both measures) indicates that the 2017–2023 MSL5 map is showing the same spatial signal as the long-run secular trend established across the full monitoring record, not a window-specific artefact.

Check 3 — climate seasonality. Examination of monthly PET trends across the 95-year RAF Valley record (1931–2025) reveals significant increases in May (slope +0.068 mm month$^{-1}$ yr$^{-1}$, $p = 0.002$), June (+0.048, $p = 0.033$), and July (+0.063, $p = 0.010$). Total June–September PET has also risen over the same record (+0.148 mm yr$^{-1}$, $p = 0.007$), while summer rainfall shows no trend ($p = 0.81$); the intensification of early-summer atmospheric demand — occurring precisely when the water table is falling from its spring peak toward the summer minimum — is consistent with an independent physical mechanism for the secular deepening visible in the MSL5 and summer minimum metrics; over the well record alone (2006–2025, $n = 20$ years) none of the three monthly slopes is distinguishable from zero. The rising early-summer PET draw is consistent with regional temperature increases during the growing season and applies uniformly across the site, producing the observed broad deepening with strongest expression at wells where the water table is most sensitive to atmospheric demand (C4, C5) and weakest at the lake-buffered eastern fringe (C1, eastern C2).

Together, these three checks indicate that the 2023 window is the defensible choice: its spring composition is not dominated by the wettest springs of the record, it matches the independent long-run secular trend, and it is physically explained by the documented intensification of early-summer PET. The 2025 window was not selected because its dominant spatial feature, the Forest Control excess, is not separable from the spring composition of the window or from the resolution of a five-well group contrast.

**Independence of spring readings within a window**

The two-window comparison assumes the spring readings within each window are approximately independent, and the wide spread of drainage recession times across the network invites the expectation that this fails at the slow-draining wells. It does not: spring-level autocorrelation is approximately zero throughout (Section 4.12), as the mean-reversion time predicts: its datum-free e-folding time is about half a year (Section 3.4.1), so little of one spring survives to the next. The expectation from the published form's drainage coefficient is larger, but that coefficient carries the drainage datum and is not the mean-reversion time (Section 3.4.1). The five-year average therefore behaves as intended, with an effective sample size of approximately five at every cluster. Window sensitivity is nonetheless real and graded across the network, but as a matter of amplitude rather than persistence: C4 is the most window-sensitive cluster because its spring levels swing furthest, not because its springs are serially correlated, and what governs that amplitude is $\beta_2$ rather than $\beta_3$ (Section 4.12). Because the two parameters are correlated across wells, the recession time appears diagnostic when examined alone. This is the same higher $\beta_2$ that produces the C4 cluster's stronger climate response, and it accounts for both the larger deepening at C4 and the greater sensitivity of any window comparison there.

**Matched-window sensitivity and the secular trend**

The three checks above establish that the 2017 → 2023 deepening is internally robust: it is not an artefact of window length or of gaps in the spring record, and it agrees with the independent long-run summer-minimum trend. They do not, however, establish that the −105 mm headline is a measure of secular change. Because the amplitude of interannual variation differs so widely across the network, the magnitude of a two-window MSL5 difference — and at most wells its sign — depends on which pair of windows is differenced (Script 34; Supplementary Material Note S11). Holding the well panel fixed across each comparison and admitting every well-sampled window pair (common panels of ≥ 40 wells, which excludes the thin early baselines (every window ending 2014 or earlier), so the spread reflects the window choice alone), the site-mean change spans −0.16 to +0.23 m (−161 to +229 mm) across the 55 admissible pairs and changes sign — 17 falling, 38 rising. The −105 mm 2017/2023 comparison is one interior cell; the most negative pairing is 2017 → 2019 (−161 mm) and the most positive 2015 → 2024 (+229 mm), the latter resting on the exceptionally wet 2024 spring, the wettest of the 2007–2025 well record by RAF Valley spring rainfall, whose effect on window composition the window-selection checks above

already flag. That a single methodologically admissible change of window pair can move the site-mean estimate by more than a third of a metre, and reverse its sign, is why absolute site-wide change is not rested on the two-window comparison (Script 34; Supplementary Material Note S11, Figure S11.1). Consistent with this, a trend fitted to the site-mean spring level across the full record is −4.8 mm yr⁻¹ (AR-corrected, p = 0.63), not significantly different from zero. The two-window MSL5 comparison is therefore best read as a snapshot of where the aquifer stood in one window relative to another, not as a secular rate; the window-independent secular signal is recovered instead by the relative and absolute differential-movement analyses that follow (Figures 73 and 74).

**Absence of a detectable management signal in the MSL5 map**

The MSL5 2017–2023 map shows no spatially distinct signal attributable to either the 2015 dune scraping or the 2017 clearfell. WMC3 (clearfell Impact) shows −90 mm and CEH36 (scrape well) shows −97 mm — both deepening broadly in line with the −105 mm site-wide mean and indistinguishable from the surrounding network. The Forest Controls show −140 mm (mean of the five) — deeper than baseline, consistent with the site-wide secular trend and not exhibiting any departure from it that could be attributed to management. This is consistent with the BACI findings that the clearfell response is expressed in the monthly-mean water level, chiefly over October–March (Section 4.6.3), and not in the summer minimum (Section 4.6.4). The scraping benefit at CEH36 (+128 mm, Section 4.5.2) is expressed in the monthly-mean era comparison but is largely absorbed by the construction of the five-year spring means. The 2013–2017 baseline window already contains two fully post-scraping springs (2016 and 2017), with the 2015 spring straddling the April 2015 works, so much of the benefit is present in the baseline as well as in the current window and does not survive the difference. CEH36's −97 mm against the −105 mm site-wide mean lies well inside the ±25 mm display threshold, so no residual of the benefit is resolved in the difference. This is consistent with a window-overlap effect rather than a drainage one: the scrape lowered the ground surface, which is consistent with a new equilibrium rather than a transient anomaly subject to head-space recession.

**The coastal decline signal**

The coastal margin decline in Figure 72 is the spatially distinctive element of the MSL5 map that is not explained by the site-wide secular deepening alone. The south-western coastal wells (CEH22 −233 mm, CEH21 −213 mm, CEH18 −145 mm; C5 coastal members CEH19, CEH17 and NW9 −164 to −172 mm) show substantially larger deepening than the network mean, consistent with the coastal-retreat gradient documented in Section 4.10.2. The parametric map places these wells in a modest net-gain zone (SLR marginally exceeding the Storm Brendan erosion drawdown), which does not match the observed decline. This discrepancy is consistent with the known limitations of the one-dimensional erfc SLR model at a finite bounded dune body and with the residual unexplained drift absorbed by the coastal-drift term (Section 4.6.3, Table 12).

**Differential movement with the climate signal removed (Figure 73)**

The differential-movement map (Figure 73, Section 4.12) recovers the secular component that a two-window comparison cannot isolate. By removing the common climate signal shared across the network and fitting a per-well trend in spring (MAM) level, it shows which parts of the aquifer are moving relative to the site as a whole, independently of how favourable any individual spring happened to be. Over 2011–2025, 13 of 74 mapped wells move significantly: eight sinkers on the coastal margins — CEH22 (−27.8 mm yr⁻¹), CEH21 (−17.0 mm yr⁻¹), CEH19 (−14.0 mm yr⁻¹) on the south-western frontage and CEH11 (−11.7 mm yr⁻¹) on the southern shore, opposite the rest of its cluster, among them — and five risers in the western residual block (NW1, NW2, NW11, NW13 and WMC4, +10 to +12 mm yr⁻¹). Extending the window to 2005–2025 gives 12 of 77 wells, with CEH17, CEH3 and NW9 strengthening, CEH22 little changed (−27.8 to −26.8 mm yr⁻¹) and NW8, which has no 2011–2025 slope, entering at −30.3 mm yr⁻¹, so that the coastal and western sinkers remain the dominant significant signal. The coastal-margin decline identified above is thus supported as the robust, window-independent secular movement on the site, consistent with the coastal-retreat gradient of Section 4.10.2 and the coastal-drift absorption of Section 4.6.3.

**Reconciling the parametric forest loss (Figure 70) with the observed forest holding (Figure 73 and Figure 74)**

The forest interior behaves very differently in Figure 73 from the way the parametric net-state map (Figure 70) would lead one to expect. Figure 70 predicts a large net loss across the forest interior under the standing-canopy drawdown; in Figure 73 the forest, far from declining, holds its position relative to the rest of the Warren, and the forest control wells (CEH32 +18.9, CEH34 +15.8, CEH2 +15.3 mm yr⁻¹) are among the stronger upward drifts, though 38 of the 74 mapped wells have positive slopes and none of these three is individually significant. This is not a contradiction between the two figures; they describe different quantities. Figure 70 is a steady-state penalty — the depression the standing canopy maintains relative to a hypothetical no-forest aquifer, a constant offset present throughout the record — whereas Figure 73 is a rate of change with the common climate signal removed, and a constant offset, however large, contributes nothing to a secular trend.

The absolute climate-removed map (Figure 74) supports this reading from the other direction. Where Figure 73 shows the forest holding its position *relative to the network*, Figure 74 — which removes the climate signal without re-referencing to the site mean — shows the same wells close to zero in *absolute* terms: the main-forest cluster sits at C4 ≈ −0.6 mm yr⁻¹, alongside the lake-edge and dune clusters (C1 ≈ −0.3, C2 ≈ +0.2 mm yr⁻¹; cluster means), none of them trending materially in either direction once the referencing is removed. The upward drift of the forest wells in Figure 73 is therefore a property of the reference frame — the forest declining more slowly than a wet-spring-lifted network mean — and not a secular gain. Read together, the relative and absolute maps agree: the forest

interior is stable, neither recovering nor declining, while the genuine secular losses are confined to the western and coastal margins (C3 ≈ −6.9, C5 ≈ −16.7 mm yr⁻¹).

The forest's predicted sensitivity is nonetheless real; it is expressed in two ways that both leave its relative position intact rather than as secular decline. First, as climate volatility: the amplification field (Figure 75) shows the forest interior magnifying the dry-to-wet spring swing more strongly than any other zone — the C4 forest at ≈ 1.7× the network-mean swing of 0.77 m, against ≈ 0.6× at the lake edge and ≈ 1× on the open dune — so the forest moves furthest within each climate cycle while not trending relative to the site across cycles. Second, through its position in the recession geometry: the forest occupies a high, slow-draining mound (C4 Model A recession time $1/\beta_3$ ≈ 54 months at the network datum, against ≈ 16 months at open-dune C2), and a slow-draining high holds its relative elevation as the whole network rises and falls. The amplification field correlates with the per-well $\beta_2$ coefficient (r ≈ 0.8), a marker of this deep, slow-recession store rather than a literal evaporation rate, since $\beta_2$ measures head response per unit atmospheric demand and is therefore raised where drainable storage is small (Section 4.9.2). The parametric forest loss and the observed forest holding are therefore two consistent readings of the same aquifer — annual-mean steady-state drawdown versus climate-removed secular drift — not a discrepancy to be resolved in favour of one figure.

**The dry-year spring envelope and the SD15b / SD16 community-mean levels (Figure 76)**

The envelope analysis also bears directly on ecological condition. Figure 76 maps the spring water-table depth in the dry extreme years (2011, 2012, 2019) against the Curreli et al. (2013) SD15b / SD16 community-mean levels (SD15b 0.61 m; SD16 0.98 m). A well's depth indicates a community only where the well stands on the slack floor (Section 4.8.3). Across the forest interior and the western residual block the dry-year spring table already sits below the SD16 community-mean level. Because spring is the seasonal high, a spring level already below a level defined on the annual minimum guarantees that the summer minimum is deeper still: the contour is a strict, conservative lower bound on the area below the SD16 community mean, and the true footprint below it in late summer of a dry year is larger. This is the dry-year complement to the secular picture — the coastal margin is where the table is declining fastest (Figures 72 and 74, the latter showing the coastal decline in absolute terms, though no coastal well's absolute trend is individually significant), while the forest and western dune are where, in dry years, the table is already deepest relative to the SD15b / SD16 community-mean levels (Figure 76). The single raised inter-slack well CEH10 is excluded from the interpolated surface, as it measures slack-edge rather than slack-floor ground; it is retained as a distinct slack-edge marker.

**Robustness of the BACI clearfell estimates**

The finding that no management signal is detectable in the MSL5 comparison bears on the interpretation of the BACI estimates, and it is important to distinguish which estimates are robust and which require qualification.

The primary ANCOVA-BACI monthly-mean step change at WMC3 relative to the Forest Control counterfactual (+108 mm; Section 4.6.3) is consistent with a recovery against its own control but is not significant on its own once autocorrelation is allowed for (95% CI −21 to +237 mm, p = 0.10); over October–March it is +128 mm (95% CI +1 to +254 mm, p = 0.048). It is derived from the continuous monthly record using a common-climate covariate structure, controls for the shared climate trajectory of Impact and Control tiers through the ANCOVA design, and both its magnitude and its significance depend on the drift correction (without it the step is +16 mm, p = 0.51 with ordinary errors; Section 5.5), as its magnitude does on which counterfactual is used. The directional reading — a rise at WMC3 after felling, carried by the October–March months — is consistent with the record and is not affected by the MSL5 window analysis.

The era-split β coefficient shifts ($\Delta\beta_2$, $\Delta\beta_3$) at the Forest Control tier (Section 4.6.6) are less robust. The network-wide α_B decline post-felling, visible even at the open-dune Climate Controls, indicates that the two eras have different climate-adjusted baselines driven by the changed antecedent storage state that the monthly SSM does not capture. The post-felling $\beta_3$ decline at the Forest Controls may therefore partly reflect this era-level climate confound rather than a real physical change in the aquifer, and should be treated as indicative pending corroboration from direct hydraulic measurements.

The summer minimum analysis (Section 4.6.4), the four-zone BACI corroboration (Section 4.6.7), and the felled well's $\beta_2$ change used in the scenario framework (Section 4.13.2) are derived from the full monthly record with the pre-specified BACI design and are unaffected by the MSL5 window issue; the $\beta_2$ change is a single-well estimate whose caveats are stated in Section 4.13.2.

**Priority measurements to reduce uncertainty**

The driver scale-factor test of Section 4.12.1 is a bounded confirmation rather than a measurement of amplitude. Each driver field is detectable in the window where it acts, at a scale consistent with the modelled amplitude, but the intervals are wide, for the reasons that limit the test. The coastal signature is collinear with easting (r = +0.67 to +0.73 across the three windows) and with the underlying clay-dip substrate gradient; the per-well residual scatter (of order ±150–200 mm) is of the same size as the driver amplitudes being tested; and only around fifteen of the wells entering the fit carry a modelled coastal amplitude large enough to constrain the regression at all. The coastal shape ($\delta_0$, L_cg) is moreover fitted from this same network via the gradient analysis of Section 4.10.2, so the scale-factor test is in part self-confirming; the one genuinely independent check is an expanding-window trajectory of the implied coastal decline rate, which is itself too contaminated by the 2006–

2012 drought signal to be conclusive. The regression detects the coastal field without resolving its amplitude; the background common-mode movement is the larger term in every window.

Three field measurements would materially resolve the remaining uncertainties. First, continued monitoring through a dry spring sequence equivalent to 2013–2017 would test whether the MSL5 pattern under neutral climate conditions shows any management signal above the secular deepening trend. Second, slug tests at two or three representative C4 wells (CEH32, CEH34, CEH13) would provide hydraulic conductivity estimates independent of the SSM coefficient fits, allowing the post-felling $\beta_3$ decline to be evaluated against measured drainage properties. Third, a ground-penetrating radar survey of aquifer thickness, with coring, would replace the indirect $\beta_1$-proxy constraints with direct measurements of thickness and specific yield, and would establish whether a ridge boundary flux exists at all rather than assuming one.

### 5.7.6 External Validation Against Dune-Slack Vegetation, and a Structural Wetness Index

MSL5 (Section 4.8.3) follows van Willegen et al. (2025), the strongest single hydrological correlate of dune-slack vegetation moisture at Newborough. As implemented here it reproduces their published spring levels to within 2 mm and predicts their co-located Ellenberg-F between wells ($r = +0.83$, $n = 18$), supporting the view that the metric carries the ecohydrological signal attributed to it (Section 4.8.3).

The state-space coefficients afford a structurally-derived estimate of the same quantity — an equilibrium water-table level obtained by inverting the head-dependent drainage term under long-term mean climate. Tested directly against Ellenberg-F it is statistically indistinguishable from MSL5 (RMSE 0.37 vs 0.34 Ellenberg-F units; overlapping confidence intervals; Williams $p = 0.49$), and the two distribute almost identically across per-well accuracy bands. That coefficients calibrated purely on dynamics predict vegetation moisture as well as the direct spring-level metric is the substantive result; on $n = 18$ it is equivalence within the data, not proven identity.

Calibrated onto the MSL5 scale, the index also reconstructs observed MSL5 across the open dune (clusters C1–C3) to 73 mm RMSE, generalizing, with some loss of accuracy, to the wells van Willegen never studied (82 mm, against 62 mm at the 20 wells of van Willegen's seventeen stations) — but it is scoped to the open dune, as MSL5 itself is (Section 4.8.3): in the forest the coefficients are least constrained (Section 4.9.2) and prediction degrades to ~176 mm, too coarse to classify a well across a Curreli community-mean level.

Its practical value should be stated precisely. The index is not simpler to construct than MSL5 — it additionally requires climate data and a fitted model, and it is fragile where the fit is weak. Its one genuine advantage is record length: an SSM fit needs of order thirty months, whereas MSL5 needs five clean consecutive springs, so the index is available earlier

in a monitoring programme. At Newborough, with a 21-year record, that advantage is immaterial; its utility is transferability to other dune systems or to early monitoring elsewhere, where MSL5 cannot yet be formed. A distinction matters here. The agreement with vegetation is independent of MSL5: the equilibrium index is built from the state-space coefficients and climate alone, so its prediction of Ellenberg-F is an external ecological validation of the model, not a restatement of the spring-level metric. The reconstruction of MSL5 from the index, by contrast, is calibrated on MSL5 and is a diagnostic of usability rather than an independent finding. The vegetation result rests on eighteen wells at a single site, so its statistical power is limited, but it stands on its own — coefficients fitted purely on month-to-month dynamics carry an ecological signal not borrowed from MSL5. The slack-floor wet-area check (Section 4.8.5) is the report's only spatial, out-of-sample validation of the modelled water table, and its recovery of the satellite-observed extent — most strongly for the wet-floor class that carries the dune-slack habitat signal — is independent corroboration that the SSM level is spatially, not merely temporally, faithful. Two of its features are read here: the failure of a digital elevation model plus a uniformly rising water table at cell scale is interpolation error, not evidence of lateral or surface supply; and the floor's lag on the falling limb is water standing in the hollows after the water table has fallen beneath them.

### 5.7.7 What the Record Cannot Resolve: Rates Through Time, and the Length of Record They Would Require

This record does not resolve rates through time, and the scale of that limitation is worth stating once in numbers, because it governs more of this analysis than any single quantity does.

Over the full record the site-mean level moves about its own fitted trend with an interannual residual standard deviation of 270 mm on the spring basis and 211 mm on the annual basis. Against variability of that order, and over the 21 years available, the smallest site-wide rate of change that this record could distinguish from zero at 80 % power is about ±28 mm $yr^{-1}$ (27 on the spring basis, 29 on the annual). Every site-mean trend the record yields falls inside that threshold: a decline of 4.77 mm $yr^{-1}$ over 2005–2025 and a rise of 19.78 mm $yr^{-1}$ over 2011–2025 on the spring basis, and declines and rises of similar size on the annual basis. None is significant, all have confidence intervals spanning zero, and the two periods return opposite signs on overlapping data. Neither is evidence of a trend, and their disagreement is not a contradiction to be resolved but the expected behaviour of estimates below the detection threshold. Bringing that threshold down to the order of 7 mm $yr^{-1}$ would take a record of roughly five decades at the present variability.

That threshold explains which of this study's results stand and which do not, and the division is not arbitrary. Quantities estimated as a difference between groups, or as a contrast across space at one time, are not exposed to interannual variability in the same way, because the variability is common to both sides of the contrast and cancels. The clearfell step is a level

discontinuity between felled and unfelled wells sharing the same weather, so interannual variability cancels from it; what does not cancel is the persistence of the felled well's own departures from month to month. It is positive across three impact-well specifications against the Forest control (WMC3 alone, WMC3 with FE2, and WMC3 with FE1 and FE2: +108, +94 and +78 mm), yet with autocorrelation-robust errors none of the three annual-mean steps is significant on its own ($p = 0.10$ to 0.12), and only the October–March step is (+128 mm, $p = 0.048$). A contrast removes the shared weather, not the persistence of a single well's record; the coastal gradient, a contrast across many wells at a common time, is not limited in that way. At −26.4 mm yr$^{-1}$ the coastal gradient is estimated with a well-basis standard error of 3.5 mm yr$^{-1}$ (95 % CI −33.2 to −19.5), more than seven standard errors from zero — a resolution no site-wide rate in this dataset approaches. Quantities estimated as a rate through time from the site mean are all unresolved: the far-field constant, the secular trend, the post-felling trajectory and the pre-felling parallel-trends comparison alike. The distinction is one of estimator, not of importance. The far-field constant deserves one more word, because a constant near zero invites the reading that the climate is not changing. It is not that. The summer and winter water-balance trends are of opposite sign and comparable size (Section 4.10.3), so their sum — which is what an annual term sees — is flat; the far-field constant is fitted against a covariate built from that annual balance, so a redistribution that cancels within the year cannot express itself in it. A near-zero far-field term is therefore not evidence of an unchanging climate; it is evidence that the change here is seasonal in character, and that this term, being an annual-rate quantity, is blind to it.

A resolved coastal gradient and an unresolved site-wide rate can appear to sit oddly together, and the relationship between them is worth stating, because it is the same distinction seen from the other side. What the gradient resolves is how the rate of change varies with distance from the shore, pooled over sixty-one wells. It does not follow that the gradient can be turned around and applied to individual wells as a correction, and it is not so applied here.

Three things prevent it. The gradient is estimated with the entire BACI clearfell zone excluded — the impact well, the four edge wells, the five forest controls and the two coastal controls are all withheld, so that the clearfell signal cannot enter the gradient estimate. Of the six BACI tiers only the climate and far-field controls appear in the coastal panel at all. Correcting the clearfell wells with the gradient would return an estimate to the population it was built to exclude. Second, the relationship is a population one with substantial scatter about it: distance to the shore accounts for 18 % of the variance in per-well trend, and individual wells depart from the fitted profile with a standard deviation of 14.9 mm yr$^{-1}$, against a predicted impact-to-forest differential of 10.2 mm yr$^{-1}$. A per-well correction would carry more scatter than signal, however well determined the pooled relationship is. Third, the gradient is a rate averaged over the full record, whereas a BACI correction needs the differential over the specific pre- and post-intervention windows — and rates in this record move with the window, which is the finding of the two paragraphs above.

The direction of the omission is worth stating, because it is not neutral. The felled compartment sits about 550 m from the shore and the forest controls average about 950 m, so the coastal gradient acts against the felled wells. The step reported here is corrected for it, at a fitted amplitude larger than one unit rather than smaller; because the felled compartment sits nearer the eroding shore than its controls, the correction adds to the step rather than subtracting from it (+108 mm corrected, +16 mm without the term; Section 5.5). The clearfell result reported here is therefore measured net of this confound rather than bounded by it, and so is the scraping term fitted alongside it in the same model at the felled well (WMC3; Section 4.6.3). Quantifying the differential explicitly, in distance-to-coast rather than in easting, was identified as the way to test it, and is what the drift covariate of Section 3.5.4 now does.

The practical consequence is that record length, rather than network density, is the binding constraint on the questions that remain open, and the requirements this places on the continuity of the network are taken up in Section 5.9.

### 5.7.8 Out-of-Sample Test Against the 1989–96 Record

The 1989–96 hindcast (Section 4.14.1) is the only out-of-sample test this study can perform, and the only observation of this aquifer from outside the window the model was calibrated on. The model reproduces the shape of the historic record well — the seasonal filling and draining of the aquifer sixteen years before the calibration window opens is described by coefficients that never saw it — but not the level as reliably: a negative Nash–Sutcliffe efficiency beside a positive bias-removed efficiency is the signature of a model that has the dynamics and the datum only approximately, and the level error is small and falls either side of the record.

The three wells that have changed canopy state or sat under canopy throughout fit worse and shift more than the six open-ground wells; the well that has been forest across both epochs is simultaneously the poorest fit in the set and the largest shift, which is what a model carrying a modern canopy state would produce.

The contrast between epochs is very largely climate. The comparison window is among the driest sustained stretches in the ninety-five-year RAF Valley record, with a rainfall surplus over potential evapotranspiration barely half the modern one (Section 4.14.1). A water table standing lower in those years is what climate alone predicts, and the hindcast, which knows nothing else, produces it; what is left over at the six open-ground wells is +0.109 m against a contrast of −0.347 m.

The epoch shift is a level contrast between a dry epoch and a wetter one, and it should be read as one. Its direction is a recovery rather than a decline: the table stood lower in 1989–96 and stands higher now. Dividing the six-well median by the twenty-three years separating the midpoints of the two records would give a figure near 15.2 mm yr⁻¹, but the division assumes a straight line across years that hold no observations, and two windows cannot say

what happened between them. Nor is the contrast in tension with the declines reported inside the modern record — a recovery from the drought of the early 1990s followed by a decline within 2005–2026 is consistent with both — but it does not corroborate them, and it should not be quoted as though it did.

The test therefore does what this record does well and stops where it stops. It indicates that the model transfers across three decades, reproducing the shape of a record from sixteen years before its calibration window opens; and it indicates that the water table of the early 1990s stood materially lower than today’s, for reasons the climate of those years is sufficient to explain. It resolves no rate, and it is reported as a validation of the model and a contrast between epochs rather than as a measurement of change per year.

### 5.7.9 The 1951–53 Record: Fifty-Four Years Before Calibration

Ranwell's 1951–53 readings (Section 4.14.2) take the same test thirty-eight years further back, into the last years before the forest closed, and add a second test of a different kind: whether the level has moved. Coefficients fitted to 2005–2026 reproduce the seasonal filling and draining of these slacks fifty-four years before the calibration window opens (Figure 80); the level comparison is not point to point, and its error budget is the measure of what it can say.

The record therefore supports the reading the introduction gives it. Inland of the coastal reach, at the open slacks Ranwell measured, the water table today stands within about a decimetre of where it stood in 1951–53 once climate is allowed for (−0.09 ± 0.10 m at six sites); a fall larger than about 0.3 m is excluded there, and so is a rise larger than about 0.1 m. Only at the seaward end of the coastal slack has the level fallen by more than its uncertainty: Site 8 is about a metre lower, the size the modern coastal-retreat gradient would produce there over the interval if the rate had held, and its reading belongs with the coastal record (Section 5.4.3) rather than with the warren as a whole. Taken over all eight sites, the change is −0.15 ± 0.09 m. That bound is what “near-stationary” means here, and it is a bound, not a rate: two windows sixty-six years apart cannot say what happened between them, and the 1989–96 record shows that the intervening decades contained at least one excursion of several decimetres that has since reversed. Nor does the comparison reach the forest. No pipe stood under the canopy, and at the Clwt Gwlyb sites, which lie 124–432 m from the nearest forest wells, the drawdown the interception model predicts (Section 4.11, Figure 66) is of order 60–80 mm — inside this comparison’s error, and so neither confirmed nor excluded by it. What the century does not contain, on the only measurements that exist, is the progressive regional decline the site’s literature has repeated since the 1970s without a measurement to rest on.

The floors themselves carry the same message, and carry it further back than any dipwell. A deflation floor is cut down to the wet sand and no further (Ranwell, 1959), so its height above the water table records the level at which cutting stopped. On the open warren that

level is the spring water table: the floors stand about 0.2 m above the spring level and about 0.9 m above the annual minimum today, and Ranwell’s own floors stood in the same relation to his readings in 1951–53 (Section 4.14.2, Figure 81), which is consistent with floors cut to the spring level plus a capillary margin rather than to the summer minimum, and with a spring level that has not moved in seventy years. Inland, therefore, the floors date nothing: they sit where the water table still is. At the coast the picture is mixed rather than uniform. Of the fourteen open-dune wells within the coastal reach, four stand 0.3–0.7 m further above the annual minimum than the inland baseline and ten do not, so a coastal floor left high by a retreating shoreline is a feature of particular slacks, not of the frontage. Site 8 shows why. Its level has fallen by about a metre in Ordnance Datum terms since 1951–53, the amount the coastal-retreat gradient would produce there, and the fall divides about equally between the floor and the water beneath it: the lowest ground near the site now lies 0.53 m below his floor, and the water table stands a further 0.52 m deeper below that floor than his readings stood below his. At the six inland sites the same two measures scatter by about ±0.1 and ±0.2 m around zero (Section 4.14.2), so the division at Site 8 is consistent with a floor cut down again as the water table fell, though one site cannot show it. If that is the mechanism, the floor at the coast is a current surface that follows the level, and a former water table is preserved only where cutting has stopped.

The forest floors are the exception, and the sand's own history says why. The eastern warren was still mobile in the 1940s, and its hollows, under the trees and in the open alike, were being cut down to the wet sand in the years the plantation went in, between 1947 and 1965; when the surface fixed, deflation stopped everywhere at much the same time, so both sets of floors record the water table of the last mobile years. The open-dune floors sit where a modern floor would (Section 4.14.2), and Ranwell's 1951–53 pipes at Clwt Gwlyb, 124–432 m from the nearest forest wells, show why: the open-dune water table has not moved since they were cut. The met record says the same for both sets: driven by RAF Valley rainfall and evaporation alone, the water table of the planting years stood no higher than a decimetre above its modern level, so neither set was cut under a wetter climate than the present one. The 0.4–0.7 m by which the forest floors stand further above the modern water table than the open-dune floors, once the shoreline retreat since the 1950s and the litter are allowed for (Section 4.14.2), is then the change under the plantation since the surface fixed, and it is confined to under the trees. It is four times the modelled canopy drawdown, and it has not reversed at the wells felled in 1993 and replanted in 1998, which is what a floor that could not re-cut under a young crop would show; neither the drainage channel nor the ploughing that preceded planting accounts for it. What the record cannot do is divide that change between a canopy drawdown larger than the interception model allows and a water table that already stood higher under the inland warren than at its edge before the trees: the clearfell step (Section 4.6) measures the recovery of a young clearfell, not the full effect of seventy years of canopy, the satellite cannot read a floor under canopy or the regrowth on a felled one (Section 4.8.5), and the hindcast is fitted under the canopy it is asked to look behind.

The reading is consistent with the clearfell and drawdown results: a forest effect confined to the ground under the trees, whose magnitude the interception model, carrying interception alone, bounds from below, as it bounds from above the reach of the effect beyond the canopy.

## 5.8 The Reach of Forest Management Interventions

The preceding sections have characterized the clearfell signal as concentrated in the mean monthly water level rather than in summer minima at the felled wells (Section 5.5), with the felling acting either through a reduced atmospheric draw or through a recharge gain that disperses beyond the compartment (Section 4.13.2). A separate question is whether forest management in the C4/C5 zone could in principle improve water table conditions in the open dune clusters — C1 Lake Edge, C2 Dune and C3 Western Residual — where the remaining wet-slack resource is concentrated.

The available evidence indicates that forest management in the C4/C5 zone does not measurably affect water table conditions in C1, C2 or C3. The spatial scenario framework (Section 4.13.2) evaluates the forest management scenarios well by well, with no flow between wells, so it cannot show a response at C1, C2 or C3; the drawdown the canopy imposes has an e-folding length of about 220 m beyond the forest edge (Section 3.8.2). The climate-corrected spatial step-change maps (Section 4.6.5; Figure 36) resolve no spatial felling footprint: the corrected steps carry an east–west trend and a south-western coastal strip, not a response centred on the felled compartment, and the felled-well signal is the winter rise at WMC3 (+128 mm over October–March, $p = 0.048$; annual +108 mm, $p = 0.10$; Section 4.6.3). Site-wide annual means under forest management are correspondingly small (Section 5.7.2). Broadleaf conversion ranges from a slight fall to a gain at the forest clusters themselves (Section 5.5.2) and produces no meaningful site-wide signal. The BACI monitoring record is consistent with this structural isolation: wells outside the forest zone showed no detectable felling-specific response once common-mode climate variability was accounted for (Section 4.6.7). The transect from uncleared plantation interior to clearfell core showed no significant distance gradient in the post-felling trajectories (Section 4.6.5; Figure 38), so no spatial footprint of the felling is resolved beyond the winter rise at the felled well.

This structural isolation is not a framework artefact. The per-well analysis (Section 5.6.1) shows that the C4/C5 boundary corresponds to an elevation transition (and inferred substrate change), and the drainage geography indicates that forest-zone perturbations propagate south-westward toward Caernarfon Bay rather than eastward into the C1 and C2 clusters (Section 5.7.2). The forest drawdown propagation analysis (Section 4.11, Figure 66) provides a quantitative upper bound on the reach of the forest effect beyond the canopy ($\lambda$), though, carrying interception alone, only a lower bound on its magnitude: the steady-state head perturbation from the existing 75-year-old plantation is 2 mm or less at CEH24, CEH10

and CEH11 and falls to 8 mm at CEH6, the furthest of the six eastern C1 wells, despite the plantation having had its full lifetime to reach equilibrium. A felling narrower than the drawdown reach does not keep a local recharge gain, which spreads into the surrounding forest (Section 4.13.2); the scenarios carry no flow between wells and cannot show how far a recovery would travel. The P_flood equations (Table 19; Section 4.7.5) characterize the consequence at the forest clusters themselves: at forest-cluster summer minima of approximately 1.4–1.7 m below ground, the required winter rainfall to achieve flooding exceeds anything in the 95-year RAF Valley record regardless of canopy management.

The clusters where the ecological stakes are highest — C1 and C2, the open-dune clusters with the shallowest summer minima — show no detectable response to forest management in the C4/C5 zone, and their summer minimum trajectory is driven by the site-wide climate signal. Peak winter level is predicted by winter rainfall at C1 and C2 ($p = 0.002$ and 0.011), where the antecedent summer minimum is not significant; at the Western Residual the two carry equal weight ($p = 0.051$ and 0.060); and at C4 and C5 antecedent summer depth dominates ($p < 0.001$), with winter rainfall not significant (Table 16; Section 5.2.5). Summer minimum depth is therefore the restoration lever in the forest clusters and shares that role with winter rainfall at C3; at C1 and C2 winter flooding follows winter rainfall, and over their summer minima forest management in the C4/C5 zone has no detectable influence.

### 5.8.1 P_flood Recoverability and the Operational Boundary

The P_flood threshold mapping (Section 4.7.4, Figures 41–43) characterizes the recoverability of each well in the network under realistic rainfall conditions. Of the 88 classified wells, 59 (67%) are reachable under mildly wet winters ($m_P < 1.5$) — these lie almost all in the open-dune clusters C1, C2 and C3 (58 of the 59) and represent the zone where targeted scraping can shift a marginal well into reach of a mildly wet winter.

A further 24 wells (27%) require wet winters of varying rarity ($1.5 \leq m_P < 2.5$), and the remaining 5 wells (6%) are effectively unreachable under the present climate ($m_P \geq 2.5$). The unreachable wells are concentrated in the forest zone — three in C5 Coastal Forest (CEH17, FE3 and CEH31) and two in C4 Main Forest (CEH30 and CEH33) — where summer minima of 1.9 to 2.3 m below ground place the required winter rainfall beyond anything in the 95-year RAF Valley record (the wettest observed October–February, 701 mm in 1959/60, delivered only 1.52× the climatological mean over the same months and 61% of the $m_P = 2.5$ demand).

C3 Western Residual occupies the middle ground — most C3 wells sit within the achievable band (median $m_P = 1.25$) but four deeper-end wells are marginal, from $m_P = 1.61$ (with the deepest at $m_P = 2.00$), and C3's near-zero winter flooding record (Section 5.1) indicates that even nominally reachable thresholds are rarely achieved in practice.

The recoverability map delineates a clear operational boundary: the target zone for direct intervention is C1/C2 where m_P values cluster close to 1.0 and a single scraping event can bring a well within reach of a mildly wet winter.

The recession time $t_R = 1/\beta_3$ adds a further dimension to the achievability picture that m_P alone does not capture. Two wells with identical m_P can differ substantially in their sensitivity to the distribution of rainfall within the winter. A well with a short recession time (fast-draining, typical of C1) has $t_R = 1/\beta_3 \approx 7–11$ months (range over the seven C1 wells on the comparison window): significant drainage losses occur between monthly recharge pulses, so an early wet October contributes relatively little more to the February peak than a wet February. A well with a long recession time (sluggish, typical of C4) has t_R of about 50 months (median well 48 months, cluster centroid 54 months): drainage losses within a single winter are negligible, and the aquifer integrates rainfall contributions almost equally regardless of when in the winter they arrive. The operational implication is that at long-recession wells sitting near a m_P threshold, the achievability classification is more stable — a borderline m_P = 1.4 well in C4 will flood given any sufficiently wet winter, with the exact timing of rainfall within the season mattering little. At short-recession wells the same m_P carries greater sensitivity to whether the wet months fall early or late in the recharge window. This distinction is invisible in the m_P map but is encoded in the recession-time field (Section 4.9.3) and should inform operational interpretation of achievability classifications near the m_P = 1.0 and m_P = 1.5 boundaries.

### 5.8.2 Climate Trajectory as the Binding Constraint

After the clearfell the felled well rose against the Forest control over October–March (+128 mm, p = 0.048) and not over June–September, and the annual-mean step (+108 mm) is not significant on its own (p = 0.10; Section 5.5); the rise is in the winter months rather than in summer extremes, and the $\beta_1$ decline, a candidate mechanism for summer minimum deterioration, appears at the unfelled Forest Control tier as well as the felled tiers but barely at the Climate Control tier (Section 5.5.1). The seasonal pattern at Newborough places the measured effect of canopy management in the mean level rather than the summer months, while the ecologically critical constraint — summer minimum depth — is set primarily by the climate trajectory operating across all clusters regardless of canopy cover.

Whether this seasonal asymmetry generalizes to other coastal dune sites depends on the local water balance. The seasonal prediction equations at Newborough show that summer minimum depth dominates winter flooding prediction in the forest clusters (C4, C5) and carries equal weight with winter rainfall in the Western Residual (C3), while winter rainfall is the dominant predictor at the eastern lake-buffered and shallow till clusters (C1, C2; Table 16). Where this pattern holds at other sites, canopy management would address a different season from the ecologically critical one. Where the local water balance is winter-rainfall-limited rather than summer-minimum-limited, the calculation would differ. The

Newborough finding cannot be directly transferred without site-specific seasonal partitioning of the flooding-control mechanism.

At Newborough specifically, the progressive redistribution of rainfall toward wetter winters and drier summers under a changing climate is visible in the declining summer minimum trends (Section 4.8.1). The scenario projections anchored to the 2017 felling predict small sustained rises at the forest clusters, at their lower ends smaller than the projected climate-driven change, and cannot show propagation to the C1/C2 zone where the ecological need is greatest (Section 5.9). The operational domain for direct hydrological intervention at Newborough therefore lies in topographic scraping in the C1/C2 zone where the P_flood threshold remains achievable under realistic winter rainfall (Section 5.8.1).

These reaches can be consolidated on a common footing by expressing each intervention and driver as an equilibrium head change against distance from its source (Figure 83). The scrape dipole and the forest drawdown, already discussed, decay exponentially over λ ≈ 220 m and are spatially confined; the figure adds the comparison the preceding analysis has not made — coastal retreat. A single Storm-Brendan-class 6 m retreat imposes only a shallow edge drawdown (≈ −81 mm), but because the coastal signal decays linearly across the whole western dune field rather than exponentially, it reaches far further inland (to L_cg ≈ 900 m) than any management effect. Sustained over five years, the fitted coast-edge retreat trend accumulates to a drawdown comparable in magnitude to the standing forest yet spread across the entire western system. The climate term, by contrast, is represented with no reach decay: it is applied uniformly across the site by assumption, modest at any one point but everywhere, and on the shared five-year footing coastal retreat is the larger driver along the western fringe, while the climate decline, small but ubiquitous, carries the larger site-wide mean.

WMC3's raw scraping step at 262 m is drawn with its standard error (−46 ± 63 mm): the bar spans zero and contains the modelled head at that distance, so the record neither confirms nor contradicts the dipole, and the whole off-site cone is a modelled projection (Section 4.6.3).

Read with the caveat that every curve is a single-mechanism steady-state construction anchored at one or two measured points, and that the coastal curves are forward projections of a fitted trend rather than observations, the figure reinforces the section's central finding: the management interventions most often proposed are spatially localized, whereas the shoreline retreat and the climate decline are basin-wide. The mechanisms behind the four drivers are drawn schematically in Figure 84.

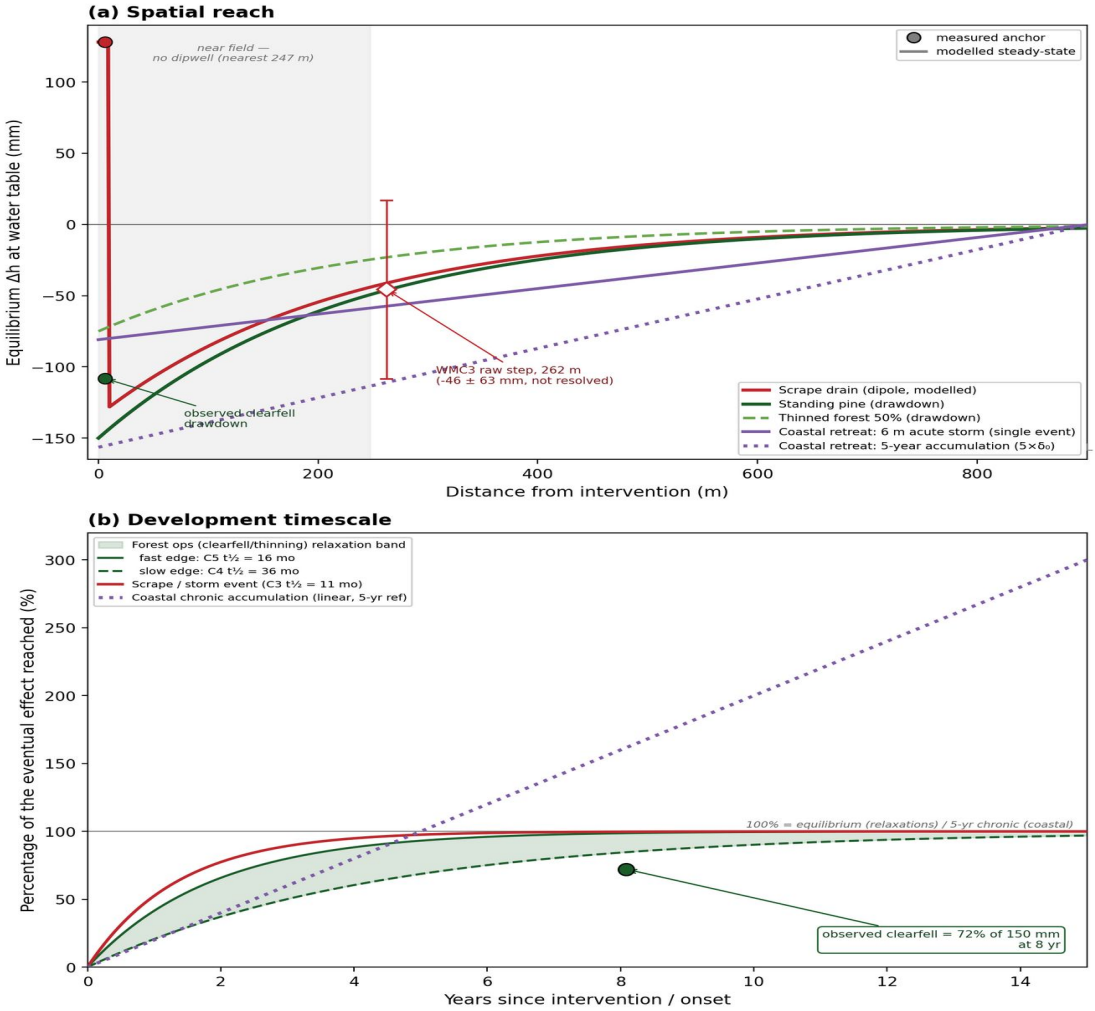



***Figure 83:*** *Spatial reach and development timescale of the management interventions and coastal retreat.* ***(a) Modelled equilibrium water-table displacement change (Δh)*** *against distance from each source. The scrape dipole and the standing/50%-thinned forest drawdowns decay over the open-dune e-folding length (λ ≈ 220 m); the two coastal curves show a single 6 m storm event and a five-year accumulation of the fitted coast-edge trend (5 × $\delta_0$ ≈ 156 mm at the toe, $\delta_0$ = 31 mm yr$^{-1}$, reach L_cg ≈ 900 m); no spatially uniform climate line is drawn: the fitted far-field level is a window statistic, not a site-wide rate. Measured anchors are the CEH36 scrape and clearfell BACI responses at the source (r ≈ 0); WMC3's raw 2015 scraping step at 262 m is shown hollow with its ±1 standard error (−46 ± 63 mm), not resolved from month-to-month variability (Section 4.6.3), so no off-site point is evidenced. The scrape cone and the coastal curves are modelled. The shaded band marks the near field the network cannot resolve (nearest well 247 m).* ***(b) Time to develop****, as a percentage of the eventual effect; being dimensionless it composes with panel (a) — the head at a given time and distance is the panel-(b) fraction times the panel-(a) magnitude. Forest operations relax as a saturating exponential spanning the forest-cluster Model A half-lives at the network datum (C5 ≈ 16, C4 ≈ 36 months); a scrape or single storm-retreat step relaxes on the faster Western Residual (C3) half-life of about 11 months, sharing one curve; the coastal decline instead accumulates roughly linearly toward no equilibrium, the saturating-versus-linear contrast distinguishing a management relaxation from ongoing accumulation. The observed clearfell response (≈ 108 mm as the record mean) is not a point on the forest band: the step is front-loaded, +152 mm to December 2020 and +39 mm thereafter (Section 4.6.8), and the 150 mm interception equilibrium drawn here is the upper end of the clearfell range rather than its expected value (Section 4.13.2). Drainage coefficients and m_P are live pipeline values; the coastal, storm and thinning curves are modelled scenarios flagged as such. (Source: 09f_management_effects.png).*

The four drivers act on the water table by distinct mechanisms, drawn schematically in Figure 84. A **dune scrape** cuts down to meet the water table, so the excavation becomes a pool at the unchanged local level; cutting headward into the landward edge of a slack extends the low seaward control inland and draws the adjacent slack below its floor. Only the first is measured: the cut wets by +128 mm (CEH36 against its control) — the pool sitting marginally higher because the headward cut reaches ground where the inland head is naturally higher, not because excavation lifted the table — while the drawdown of the neighbouring ground is not resolved: the near-field well WMC3 shows raw steps of −46 mm in 2015 and −53 mm in 2023, neither distinguishable from month-to-month variability (Section 4.6.3), and the uphill wells show no decay with distance from CEH36 in their change of drainage coefficient (Section 4.5.5), so no drawdown cone is resolved. The scrape is locally wetting but redistributive, with a small net site-wide loss, and it relaxes to a new level over a few years.

A **clearfell** lifts part of the canopy's local suppression of the water table (modelled at about 150 mm under standing Corsican pine, whose canopy intercepts roughly a quarter of incident rainfall). After felling the felled well rose +128 mm over October–March (95% CI +1 to +254 mm, p = 0.048) and not at all over June–September, the ecologically limiting season (−1 mm, 95% CI −118 to +116 mm), so felling does not restore the summer minimum; the annual-mean step, +108 mm, is not significant on its own once autocorrelation is allowed for (95% CI −21 to +237 mm, p = 0.10). The recovery is front-loaded rather than settling, +152 mm to December 2020 and +39 mm thereafter (Section 4.6.8), and felling has not undone the plantation's long-run effect (Section 4.13.2).

**Coastal retreat** lowers the seaward boundary as the shoreline recedes (about 2.3 m per year over 2006–2026; Section 3.8.2), drawing the water table down over a coastal fringe. Unlike the other three, this drawdown is not directly measured: the retreat rate is known but its effect on the water table is modelled at about −31 mm per year at the shore (≈ −626 mm over twenty years), tapering inland over a reach of about 900 m. The independent observational check is the coast-to-inland head transect (−28 mm per year), consistent with a moving boundary; the mechanism matches the Whiteford Burrows analogue. This is an accumulating process — it does not settle to an equilibrium.

**Climate** is represented here as a spatially uniform decline in the recharge/PET balance, the direction UKCP18 RCP8.5 projections indicate, which would lower the whole water table by the same amount everywhere and dry both slacks across the board; uniformity is the model's assumption, not an observation. The warren-wide fall in levels is observed across the well network, but the part of it that coastal retreat does not account for is not spatially uniform on this decomposition: it is largest at C2 and close to zero at C3. Across the three open-dune clusters — seven, twenty-two and twenty-two wells, on different substrates and at different distances from the sea — the balanced observed decline less the modelled coastal gradient averages −8.5 mm $yr^{-1}$ with a spread of 14.1 mm $yr^{-1}$ (C1 −10.4, C2 −14.6, C3 −0.5). An

independent measure, the site-mean trend, gives −13.0 mm yr⁻¹ over the same record, of the same order.

That magnitude is a central estimate and not a resolved rate, and the reason matters. The clusters' year-to-year swings are common-mode — the same weather over the same aquifer — so averaging them does not lower the uncertainty on its size, and at a 14.1 mm yr⁻¹ spread, larger than the mean, the decomposition indicates that the decline is not uniform: −8.5 sits inside the ±28.5 mm yr⁻¹ minimum detectable rate the site-mean trend carries. The panel regression's own fitted constant is a different quantity and is not used here, being not separately identified from the cumulative-water-balance covariate.

The term is carried on the common footing of Table 25 (Section 4.12.1) as an unexplained residual decline, applied uniformly by assumption, and it is named that rather than named for a mechanism. Climate is the leading candidate — the site-wide decline in recharge sensitivity, the PET trend and the summer temperature step all point the same way — but what the decomposition isolates is what the coastal gradient leaves behind, not a driver it has identified.

Expressed as an equivalent depth averaged over the 949 ha mapped site (the site outline on the 50 m map grid, not the 865 ha study area of Section 2) at a common twenty-year horizon and referenced to coastal retreat, the largest term the record resolves, the drivers rank as in Section 4.12.1 (Table 25).

That site-wide ranking differs from the drivers' local intensity, and the two orderings invert. Coastal retreat has by far the largest *peak* — about −626 mm at the shore over twenty years — but it decays inland over the ~900 m reach, so integrated across the site it is second. The unexplained residual decline is far smaller anywhere in particular, but it is applied across the whole 949 ha, and that is what places it first on the *site-wide volume* — on its central estimate, which the record does not resolve.

The two statements are complementary: coastal retreat is the largest thing happening at the shore, and the unexplained residual decline, on its central estimate and applied uniformly by assumption, is the largest thing happening to the site.

The figure shows only terms that vary with distance. A flat climate line and the distance at which the coastal curve would cross it are not drawn, because the level they would rest on is not separately identified. Of the footings ranked, the coastal drawdown, the off-site scrape drawdown and the broadleaf restock are modelled; the clearfell step and the on-site scrape gain are observed.

The management challenge at Newborough is hydrological adaptation rather than restoration to a prior state — maintaining flooding capacity in an increasingly seasonal climate, while the summer drought trajectory approaches the SD15b / SD16 community-mean levels of Curreli et al. (2013) within the next one to two decades.

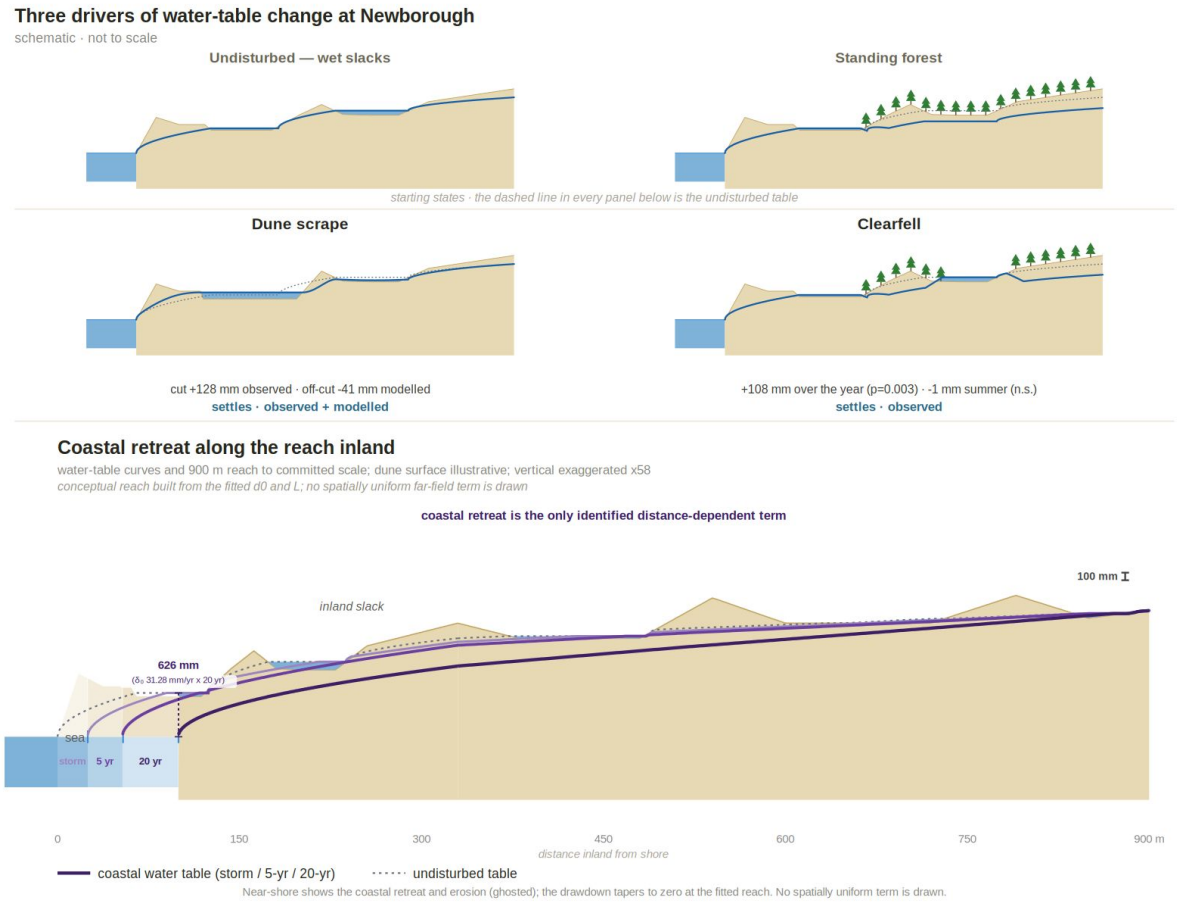



*Figure 84:* ***Three drivers of water-table change and the undisturbed state — schematic cross-sections (not to scale; vertical scale exaggerated). Climate, the fourth driver, is treated as acting uniformly across the site by assumption and has no reach to draw.*** *Conceptual before/after sections for the site's water-table drivers — dune scrape and clearfell (local, settling interventions) and coastal retreat (the basin-wide, accumulating driver) — drawn on one shared profile and a common vertical amplitude scale so their relative magnitudes can be read directly. The panels show mechanism and direction, not measured section geometry. Solid line: water table; dashed line: the undisturbed (reference) table. The full-width lower panel places the coastal-retreat drawdown on one continuous reach inland from the shore, largest at the shore and tapering to zero at the fitted reach. No spatially uniform climate line or crossing is drawn. Magnitudes and the site-wide footing of each driver are developed in the text above. (Source: 09g_mechanism_grid.png.)*

**The dune-slack community water levels.** Curreli et al. (2013) give community means of the four-year mean annual minimum water level at this site: −0.61 m for the SD15b wet slack and −0.98 m for the SD16 dry slack (their Table 4). The Dune Managers Handbook v2 (Denning et al., 2024) gives nine-year mean spring levels for the same communities, −0.13 and −0.48 m. Both are levels relative to the ground at slack-floor quadrats. Where that ground is surveyed, at van Willegen's quadrats, the record shows a shift: every quadrat read in both windows is deeper than in Curreli's survey years, and none now lies above the SD15b mean (Section 4.8.3). Away from the quadrats this report does not place its wells against the communities. The dipwells stand at every position from slack floor to dune flank, and placing a well's level against a community needs the height of the slack floor at that well; the

2 m lidar recovers Ranwell's levelled floors to 0.34 m at best (Section 4.14.2), close to the 0.37 m that separates the two communities.

What the network shows there is direction, because a change in level needs no floor height. The summer trough fell in four of the five clusters over the record, mostly before 2011. The spring baseline, MSL5, shows no decline that the window choice can resolve (Section 5.7.5). Curreli et al., applying their community means to the Ainsdale water-table projection of Clarke and Sanitwong Na Ayutthaya (2010), concluded that conditions supporting wet slack vegetation there may become non-viable by the 2030s. A differential GPS survey of the slack floor at each dipwell would extend the quadrat comparison to the whole network, and is the one addition that would turn the cluster trajectories into statements about the communities.

## 5.9 Implications for Restoration and Monitoring

**Summer minima** are the restoration lever in the forest clusters and share it with winter rainfall in the Western Residual. Antecedent summer depth is the dominant predictor of peak winter water table at C4 and C5 ($p < 0.001$), where winter rainfall is not significant, and carries equal weight with winter rainfall at C3 ($p = 0.060$ and 0.051); at C1 and C2 winter rainfall dominates (Section 5.2.5; Table 16). Where summer depth governs, restoration success depends on raising summer water table floors through topographic modification and reduced evaporative demand rather than on maximizing winter recharge. This has direct implications for monitoring design: in those clusters summer dipwell readings are more diagnostically informative than winter peak measurements for assessing restoration trajectory.

**Topographic scraping** is the most directly effective intervention identified in the monitoring record, with a measured improvement in summer minimum depth at CEH36 during the pure scraping era representing more than half the gradient separating the SD15b and SD16 community-mean levels (Section 5.4). The benefit operates on summer minima and is achievable within the C1/C2 zone where the P_flood threshold remains within reach of mildly wet winters (Section 5.8.1). Scraping steepens the local hydraulic gradient and draws the water table downward upslope (Section 5.4.3), so the spatial planning of any scraping programme must account for the cascade risk and avoid amplifying the existing coastal drainage gradient.

Scraping is therefore not a hydrologically neutral intervention. The same drain-cone physics that quantify the standing forest's drawdown (Section 4.11) put an indicative magnitude on the scrape's own neighbour effect: a well 100 m upslope of a CEH36-equivalent scrape sees a modelled drawdown of order −30 mm water-equivalent per month, decaying to roughly −16 at 250 m. These are steady-state modelled figures the 31-month record cannot yet confirm, and they are modest beside the on-site benefit — but they are not zero, and they act in the same direction as, and add to, the coastal-retreat drawdown already deepening the western margin. On the eroding south-western frontage in particular, coastal scraping compounds a drainage gradient the shoreline is already steepening; the intervention is best confined to the mid-eastern C1/C2 transitional zone (Section 5.4.3), where it raises a slack without reinforcing an existing coastal cascade.

The background climate drying trend continues to operate at the scraped site as elsewhere, and the structural benefit at CEH36 has diminished — though, against CEH4, not reversed — over the post-felling period. Re-scraping intervals cannot yet be estimated from the available record.

Per-well categorization against the P_flood multiplier (Figure 85) operationalizes the priority criterion identified in Conclusion 6. Each well is binned on its m_P value into three bands: achievable ($m_P < 1.5$, reachable in normal-to-mildly-wet winters), marginal ($1.5 \leq m_P <$

2.5, reachable only in wet winters), and unreachable (m_P ≥ 2.5, effectively unreachable under current climate). Across the sixty-four wells in the open-dune clusters C1, C2 and C3, fifty-eight are achievable, six are marginal, and none are unreachable. By contrast, of the twenty-four forest-zone wells in C4 and C5, only one sits in the achievable band and five are unreachable — three of the unreachable wells are in C5 Coastal Forest and two in C4 Main Forest. The achievable wells in the C1/C2/C3 transitional zone are the operationally feasible scrape-targeting candidates; the six marginal open-dune wells define the upper edge of the operational envelope under current climate.

**Plantation clearance** as implemented in 2017 was followed by a winter rise at the felled well against the Forest control (+128 mm over October–March, p = 0.048) and none in summer; the annual-mean step, +108 mm, is consistent with a recovery but not significant on its own once autocorrelation is allowed for (p = 0.10; Section 4.6.3), and synthetic extension within the felled compartment returns a step of similar size (Section 4.6.7). The clearfell signal is concentrated in the mean monthly water level rather than in summer minima, where felled and unfelled wells track each other (Section 5.5).

The evidence does not support the view that the plantation is the principal driver of the warren's summer drying, nor that its removal would reverse it. The metric that governs dune-slack viability — the summer minimum — is deteriorating across felled and unfelled wells alike, to which a decline in recharge sensitivity ($\beta_1$) across the BACI network is a candidate contributor, though its independence of management is not established (Section 5.5.1); clearfell raised the mean monthly water level but left the summer minimum unchanged. Where the canopy does shape the climate response, the effect is two-sided rather than simply drying: the standing pine amplifies the wet-year water-table swing through interception-driven recharge suppression (attenuated at the felled well FE2; Section 5.5.1), while the post-felling record shows the felling-attributable rise concentrated in dry conditions (the $CWB^2$ × clearfell curvature, Section 4.6, which does not reach significance) — the direction a canopy buffering the water table against drought extremes would produce.

Removing the forest would thus trade an unconfirmed dry-season buffering effect and a wet-year amplification for a mean-level gain that does not reach the ecologically binding summer minimum, on the same multi-decadal propagation timescale as the plantation took to develop its influence.

The scenario projections, anchored to the 2017 felling (Section 4.13.2; Section 5.5.2), predict small rises at the forest clusters under clearfell and thinning, with a felling wider than the drawdown reach as their upper end, and a range from a slight fall to a gain under broadleaf conversion. At their lower ends these are smaller than the climate-driven change projected over the same horizon where both are projected (C5; Section 5.7.1). The scenarios are evaluated well by well and cannot show propagation: the BACI record found no felling-specific response outside the forest zone, and the canopy’s modelled drawdown has an e-folding length of about 220 m beyond its edge, falling to 8–72 mm at the lake-edge wells

nearest the forest and 2 mm or less at the eastern dune and lake-edge wells furthest from it, in the C1 and C2 zone where the ecological need is greatest (Sections 4.11, 5.7, 5.8).

**The prediction equations, P_flood thresholds and scenario framework are available as interactive tools** hosted on the Newborough Warren Groundwater Research project page (Hollingham, 2026; https://newbroman.github.io/Newborough_Hydrology/): a per-well flood probability forecaster driven by the P_flood threshold equations, a scenario viewer implementing the SSM equilibrium framework for exploring climate and management scenarios, and a network scatter plot of seasonal extremes referenced against the Curreli et al. (2013) SD15b / SD16 community-mean levels. The three tools operate on different aspects of the analytical framework and are complementary.

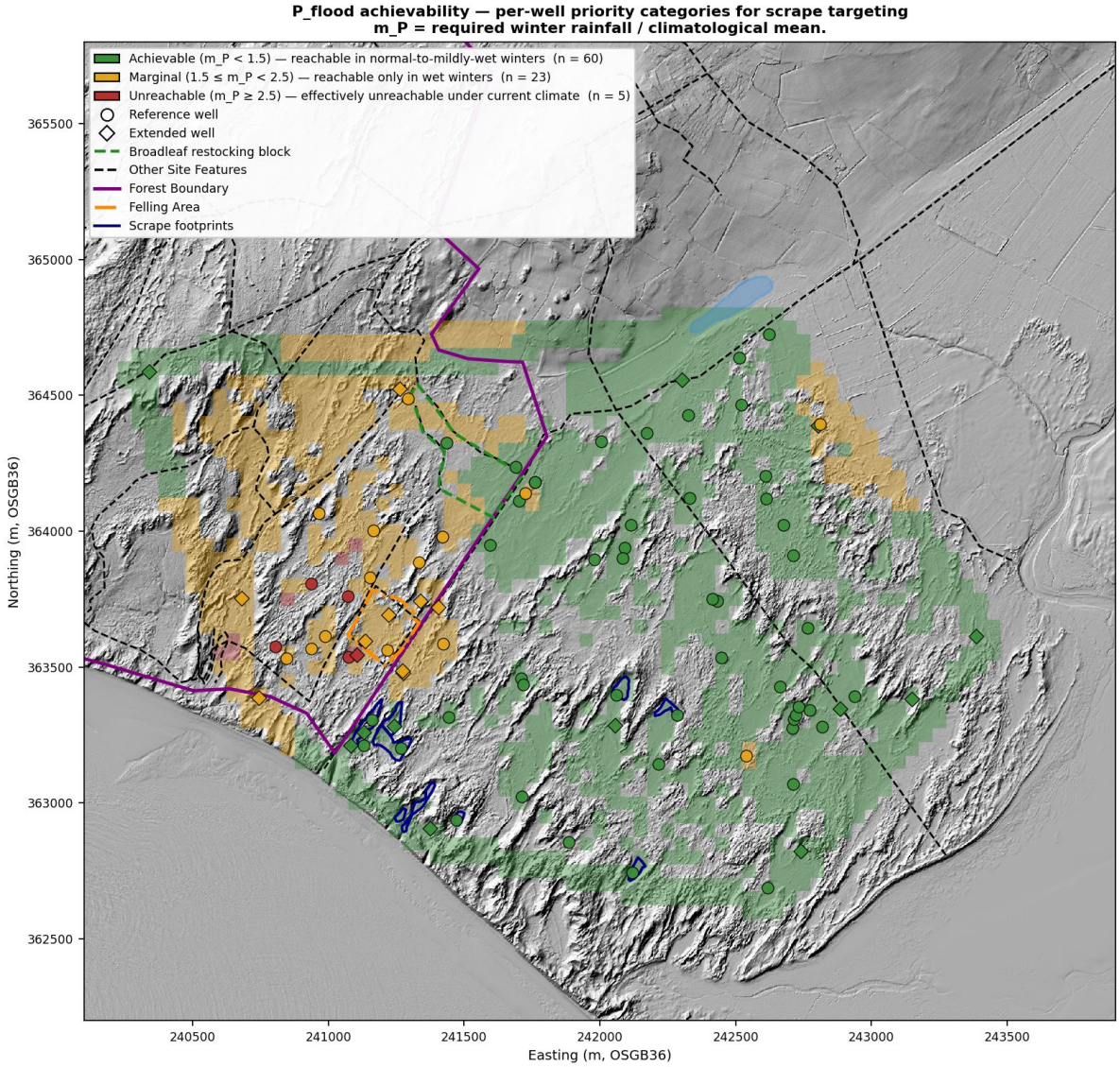



***Figure 85:*** *Per-well achievability categorization against the P_flood rainfall multiplier (m_P), the cumulative winter rainfall required to raise each well's water table from its mean summer minimum to the local slack floor (h_target = 0 m) by the end of the recharge season, expressed as a multiple of climatological winter mean. Wells in the achievable category (m_P < 1.5, green) are reachable in normal-to-mildly-wet winters; marginal*

*wells (1.5 ≤ m_P < 2.5, amber) only in wet winters; unreachable wells (m_P ≥ 2.5, red) are effectively unreachable under current climate. The shading is m_P interpolated between the wells on the piecewise-linear surface, banded at the same limits and masked across dune ridges. The cluster pattern (open-dune C1/C2/C3 dominated by achievable; forest C4/C5 dominated by marginal-to-unreachable) operationalizes Conclusion 6's priority criterion for scrape-target identification. (Source: 11c_pflood_achievability.png; per-well lookup table in 11c_pflood_achievability_per_well.csv).*

**Four priorities for future research are identified.** First, slug tests at two to three representative wells per cluster and a ground-penetrating radar survey of aquifer thickness would enable a calibrated continuous-flow spatial model and directly test the interpretation that substrate thickness, inferred here from elevation and the SSM coefficients, is the dominant control on within-forest coefficient variation (Section 5.6.1). These measurements would also lift the principal limitation on the spatial framework identified in Section 5.7.3, where the absence of calibrated hydraulic conductivity and aquifer thickness data prevents direct testing of cluster-to-cluster propagation.

Second, the C5 Coastal Forest trajectory — the steepest decline in the network in both seasons, with three unresolved candidate explanations (Section 4.8.1; Section 5.7.2) — requires continued monitoring alongside targeted measurement of coastal retreat rates, since coastal boundary retreat is one of the candidate explanations and is the one most directly measurable in the field. Pye and Blott (2024) provide a citable empirical anchor for this priority: dune-toe retreat at the southern Twyni Penrhos frontage of up to 1.8 m $yr^{-1}$ over 2013–2022, projected to continue, with the erosion front migrating northwards toward the western C5 and C3 wells.

Third, the relationship between canopy density and microclimatic shielding could be characterized through monitoring of $\beta_2$ responses at wells adjacent to blight-affected stands as the Dothistroma needle blight trajectory progresses (Section 5.6.4); this would be an opportunistic rather than designed study, but the natural experiment is already in progress and would require only sustained monitoring at existing or modestly extended sites.

Fourth, the internal structure of the bedrock ridge forming the northern boundary warrants direct investigation. Field observation indicates folds and faults running along the ridge, crossed by step faults, such that parts of it terminate in a stepped fashion — discernible in plan view in the DEM — with one termination immediately north-north-east of CEH14. Such a structure could in principle support an along-ridge sand aquifer, bounded below by the ridge-valley floor, conveying water along the ridge and discharging into the dune system at the structural terminations. The monitoring record does not test this and provides no positive evidence for it: the water balance closes at every cluster without requiring a lateral input (Section 4.9.6), and the diagnostic tests in Supplementary Notes S5 and S6 find neither distance-dependent transport from the ridge nor a residual signature attributable to it. Any such conveyance is therefore either absent, or small and steady enough to be indistinguishable from the fitted coefficients at monthly resolution. Shallow boreholes or geophysical profiling along the ridge, with direct head-gradient measurement at the candidate discharge points, would settle a question the dipwell network is not positioned to answer. It is placed fourth because the hydrological case for it is now the weakest of the four; the structural observation nonetheless stands on its own field evidence, and the northern dune margin's long-term water supply is the interest at stake.

## 5.10 Transferability: Requirements and Limits

The framework needs four things: a network of dipwells read manually at monthly intervals, a single climate station supplying monthly rainfall and mean temperature, surveyed well elevations, and a site boundary. Everything reported here follows from those — the behavioural partition, the state-space coefficients, the specific-yield estimates, the water balance and the intervention analyses. Nothing at any stage requires telemetry, borehole logging, tracer testing or laboratory analysis, and that is the substantive sense in which the approach is portable. It has not, however, been applied at a second site: what follows is therefore a requirement schedule rather than a demonstration — the inputs, the record lengths, and the parameters that must be re-derived locally rather than carried across.

The monitoring requirement is modest but not trivial. The network must be dense enough for the partition to have something to partition and for the interpolated coefficient surfaces to be supported between wells, and the climate station must be close enough that its monthly totals stand for conditions at the site. Both are met comfortably at Newborough; neither is guaranteed elsewhere, and both are cheaper to check before the monitoring starts than after.

What transfers unchanged is the method rather than its calibration: the state-space specification and its displacement formulation, the fit–benchmark–map sequence, the water-table-fluctuation estimation of specific yield, the before-after-control-impact design used for the interventions, and the accompanying identifiability and stability diagnostics.

The state-space model is the portable core of that list, and its requirements are specific. It needs a monthly rainfall total, a monthly potential evapotranspiration estimate — Thornthwaite here, derived from mean temperature and latitude, so it asks nothing of the site that a standard climate station does not already provide — and a water-level series in a consistent vertical convention. The displacement formulation is what makes the drainage term portable: referencing drainage to a fixed datum rather than to the ground surface removes the dependence on local topography, at the cost of requiring that the datum itself be established for each site. The three coefficients then carry the same meaning anywhere the model fits, which is what allows the benchmark against a traditional linear model to test the approach rather than this site.

The network need not be uniformly long-recorded, and this is the practical answer to the record-length constraint. The partition is established from the reference-network wells, which carry the long records; wells with shorter series are then placed against the resulting cluster centroids by Pearson correlation of their z-scored hydrographs (Section 3.3), requiring only twenty-four months of overlap — enough to resolve one annual cycle — and are labelled by the margin between their assigned centroid and the next best. Twenty-two of the eighty-eight wells here reach the cluster maps by that route rather than by clustering. A new site therefore needs a core of long-record wells to define the behavioural groups, not a

whole network of them: the difference between applying this framework after two decades of monitoring and applying it after two years against a legacy core.

The derived products transfer as methods and not as numbers. The seasonal prediction equations and the critical rainfall thresholds (Section 3.6) are regressions calibrated on this site's own history: their form is general, their coefficients are not, and carrying the coefficients elsewhere without refitting would be a plain error. The equilibrium wetness index is a further step removed — it is built from the fitted coefficients rather than measured, so it inherits their uncertainty and is weakest exactly where the fit is weakest. Where a site has a usable spring-level record the five-year mean spring level remains the simpler and more direct instrument; the index earns its place where the model fits well and the spring record does not exist.

What must be re-derived is everything that encodes local conditions. The drainage datum is a site-specific choice, justified here by the depth at which the drainage response becomes Darcy-like (Section 3.4.1), and it should be established at each site rather than adopted from this one. The cluster partition and the anchor wells that fix its labels are properties of this network. The canopy interception fraction applies to mature Corsican pine. The vegetation depth levels are Curreli et al.'s (2013) community means for the UK dune-slack communities SD15b and SD16, and transfer to comparable dune systems as they stand; a different habitat, with different communities, would need its own community means established before such levels mean anything. The BACI tier assignments follow this site's felling geometry.

Record length governs which conclusions are available, and the three scales differ. The 100-month admission threshold is what a well needs to enter the network, and it is sufficient for that well's individual coefficients. That threshold governs entry to the reference network only; the affinity route above brings shorter-record wells onto the cluster maps at twenty-four months. The cluster-scale coefficients reported here are estimated on each cluster's full record, which is substantially longer. The hydrogeological partition — the clustering of the well network into behavioural groups, on which every cluster-scale result rests — is the element least well supported at short record lengths, and its two stability diagnostics say different things about it. Individual memberships are robust to the period of record: under a twelve-month block resample of months the median well keeps its cluster-mates as reliably (0.989) as it does under a resample of network membership (0.949). The overall structure is not: two disjoint halves of the record produce partitions agreeing at an adjusted Rand index of 0.53 (5th to 95th percentile 0.35 to 0.73), because a merge or a split moves that index sharply while leaving most wells with most of their neighbours. A site with only a few years of record should therefore treat its classification as provisional, and report both diagnostics rather than presenting the grouping as settled.

# 6. Limitations

## 6.1 Temporal resolution and conservative volatility estimates

The reliance on monthly-averaged groundwater levels mathematically dampens the event-scale fluctuations inherent to highly conductive unconfined aquifers. Because daily peaks and troughs of individual storm events are smoothed over a 30-day window, the derived water balance parameters are optimized for seasonal trend analysis rather than event-based flood routing. This is a standard constraint in historic groundwater modelling (Taylor and Alley, 2001), and means the model captures the overall volumetric water balance while inherently underestimating daily transient spikes. The assessment of seasonal amplitude presented here is therefore conservative: the actual event-scale fluctuation — and any periodic severing of the capillary fringe from the root zone — is likely more pronounced than the monthly averages depict.

## 6.2 Record length — the difference between a level and a rate

The binding constraint on this study is the length of the record rather than the density of the network, and it falls unevenly across the results. Over the full record the site-mean level moves about its own fitted trend with an interannual residual standard deviation of 270 mm on the spring basis and 211 mm on the annual basis. Against variability of that order, and over the twenty-one years available, the smallest site-wide rate of change this record could distinguish from zero at 80 % power is about ±28 mm yr$^{-1}$ (27 on the spring basis, 29 on the annual). Every site-mean trend the record yields falls inside that threshold, and the two longest windows return opposite signs on overlapping data. Bringing the threshold down to the order of 7 mm yr$^{-1}$ would take a record of roughly five decades at the present variability.

The consequence is a division by estimator rather than by importance, and it is set out in full in Section 5.7.7. A quantity estimated as a difference between groups, or as a contrast across space at one time, is not exposed to interannual variability in the same way, because the variability is common to both sides of the contrast and cancels; a quantity estimated as a rate through time from the site mean has no such protection. The clearfell step, the scraping steps, the coastal gradient and the epoch contrast against the 1989–96 record are of the first kind and are resolved. The secular trend, the far-field constant and the post-felling trajectory are of the second kind and are not. Where a rate would have to be obtained by dividing a resolved contrast by an interval containing no observations — as it would for the 1989–96 comparison — the division is not performed and no rate is quoted. Readers should treat every level and every contrast in this work as supported by the record, and treat the absence of quoted site-wide rates as deliberate.

## 6.3 Parallel trends — unverifiable rather than violated

The difference-in-differences design underlying the BACI analyses assumes that, absent the intervention, the felled and control wells would have continued to diverge at whatever rate they diverged at beforehand. That assumption is conventionally supported by demonstrating that the pre-intervention trends were parallel. At this record length it cannot be supported in either direction. The pre-felling comparison is a rate estimated over a window of a few years from a quantity with the variability described above, and it sits far below the detection threshold: the pre-felling trends are neither demonstrably parallel nor demonstrably divergent, and the test has no power to distinguish the two. This is a limitation of the record, not a finding about the design, and it should not be read as a violated assumption.

The step estimates are reported accordingly. Each is a discontinuity at the intervention date, net of a linear drift control between treatment and control wells, which is the conservative construction because it charges any accumulating divergence to the control rather than to the intervention. Their support comes from methods that rest on different assumptions rather than from a verified parallel-trends premise. The ANCOVA and the within-compartment synthetic extension agree; the synthetic control and SSM residual normalization agree on the gross step at the felled well but leave no clearfell-specific component distinguishable from zero once the unfelled controls' own drift is removed (Section 4.6.7). A longer pre-intervention baseline is the only thing that would convert the assumption from untestable to tested, and the requirement is taken up with the other record-length constraints in Section 5.7.7.

## 6.4 Thornthwaite PET and sensitivity of absolute thresholds

The Thornthwaite method was selected for its compatibility with the low-cost replication objective (Section 3.10). Because the same PET series is applied to every cluster and every period, its bias is shared by all cluster-to-cluster and before-versus-after comparisons (Section 3.1.2). The absolute values produced by the prediction equations and P_flood thresholds do depend on the PET input, but not in the direction usually assumed. Thornthwaite is commonly taken to overestimate evaporative demand by about 10–15% in humid maritime climates (Donohue et al., 2010), which would make thresholds built on it conservative upper bounds. At RAF Valley it does not: its annual total agrees with an FAO-56 Penman–Monteith series from the same station within about 1% (Section 4.1.1). Its bias is seasonal instead, about 30% low from January to April and about 16% high from July to October. $\beta_2$ is fitted to Thornthwaite's seasonal shape and absorbs much of that difference, and the model fits the water table better on Thornthwaite than on Penman–Monteith: refitted on Penman–Monteith, its forward forecasts were poorer and its seasonal residual larger (Script 50, E13 and E13b). The absolute thresholds and scenario magnitudes are therefore the model's best estimates on the forcing it was fitted to, not bounds in either direction. The relative ranking of scenarios, the spatial pattern of responses, and the BACI finding that no

spatial felling footprint is resolved are independent of the PET method, because they depend on coefficient ratios and spatial structure rather than absolute magnitudes.

Thornthwaite's bias is also in its response. Thornthwaite carries the heat index $I$ in its denominator and builds $I$ from the same monthly temperatures, so a warming climate raises numerator and denominator together and the formula partly cancels its own response. Over the RAF Valley record, mean annual temperature rose 0.73 °C between the first and last thirty complete years (1931–1960 against 1996–2025), a rise of 7.2%, while mean annual Thornthwaite PET rose only 3.0%, from 631 to 650 mm — an elasticity of 0.42. Penman–Monteith, which carries radiation, wind and humidity, rose by about 25 mm per decade over 2002–2023 against about 7 mm for Thornthwaite over the same years. That is consistent with Thornthwaite damping the warming signal, although the earlier Penman–Monteith record is not homogeneous enough for a thirty-year comparison (Section 4.1.1). The PET series the models are fitted to therefore carries roughly two-fifths of the warming signal the temperature record shows. This does not affect the cluster-to-cluster and before-versus-after comparisons discussed above, in which the same series is applied throughout. It does mean that where a result is interpreted as a response to rising evaporative demand, the demand implied by the temperature record is larger than the forcing the model was given, so the atmospheric-draw route is understated rather than overstated.

A second limit on the climate input is spatial. Every rainfall and PET value in this study comes from one Met Office station, RAF Valley, approximately 16 km north-west of the site (Section 3.1.2); there was no gauge on the warren during the monitoring period, so any difference between the exposed dune front and the sheltered eastern slacks, and any gradient in rainfall across the 4 km of the site, is unmeasured. A uniform error in the station's rainfall behaves like the PET bias above: it scales the absolute magnitude of $\beta_1$ and the recharge fluxes built on it, and cancels in every cluster-to-cluster and before-versus-after comparison. A spatial gradient would not cancel — it would be absorbed into the between-cluster differences in $\beta_1$ — and the record cannot rule one out, though a gradient across a site this size would have to be large to approach the between-cluster range. The only on-site rainfall record is Ranwell's Parc Mawr nursery gauge of 1951–53, which over his reading span totals 0.97 of the RAF Valley figure with a monthly correlation of 0.92 (Section 4.14.2): in those years the station tracked the site closely in timing and to within a few per cent in amount. The coefficient rankings and the spatial structure of the results do not depend on the station's absolute values; the absolute thresholds and fluxes carry its error, in the same direction for every cluster.

## 6.5 The NW10 broadleaf comparison — a single uncontrolled well

The comparison of NW10 (broadleaf block, established 1993–1995) with adjacent Corsican pine wells represents the only direct observational evidence of species conversion effects within the monitoring network, but it is subject to several limitations that preclude a fully

controlled comparison. NW10 is a single well in a single block; replication across multiple broadleaf wells and blocks would be required to establish a statistically robust species effect independent of local variation. The comparison wells (CEH2, CEH34) differ from NW10 in ground elevation, distance from the ridge, and drainage rate (NW10 $\beta_3$ = 0.040 vs C4 median 0.019), and whilst the direction of the summer minimum contrast is consistent across all comparators the absolute magnitude is sensitive to these positional differences. Most significantly, the broadleaf stand was characterized by a dense bramble understorey throughout most of the monitoring period, which maintained high ground-level evapotranspiration demand and confounded the tree canopy signal. The stand has only begun to approach a closed canopy configuration from approximately 2022, leaving fewer than four years of post-closure record. The conclusions drawn from NW10 should therefore be treated as indicative observations from a natural experiment rather than confirmed causal attributions, and will require validation against a longer post-closure record and, ideally, against additional broadleaf wells installed within the block.

## 6.6 Spatial analysis — absence of calibrated continuous-flow modelling

The spatial framework used in this study (Sections 3.8, 4.9) is a per-well SSM scenario aggregator interpolated as a piecewise-linear surface, not a calibrated continuous-flow model. It is well-suited to the questions the paper addresses — how the per-well β parameterization aggregates spatially under scenarios, and whether the BACI-observed forest response propagates beyond the forest zone — but cannot represent cluster-to-cluster hydraulic coupling or the diffusive redistribution of localized recharge perturbations. Hydraulic conductivity is constrained by a single tracer test (K = 6.0 m/day; Betson et al., 2002) and aquifer thickness in the western interior is inferred from four borehole logs across 700 ha (Bristow, 2003). These are the same parameter-sparsity constraints that produced calibration difficulties in the Betson et al. (2002) MODFLOW model at this site. The priority field measurements required to enable a calibrated continuous-flow model are identified in Section 5.9.

## 6.7 BACI experiment — network design limitations

The clearfell zone and scraping sites were selected on different grounds: scraping was a targeted conservation intervention, while the clearfell was an experimental felling recommended by previous researchers and implemented as an operational forestry action. Neither was randomly assigned, which introduces the possibility that pre-existing site characteristics could confound the intervention signal. The monitoring network was not designed as a purpose-built propagation transect. The transect analysis in Section 4.6.7 (Figure 38) uses the available surveillance wells to test for an inland gradient in the post-felling signal and finds none, with post-felling step changes distributed uniformly from the plantation interior to the felling core. All 17 wells in the five-tier network have a minimum of seven years' pre-felling baseline (Section 3.5.4). Wells with shorter records (FE1–4, LIS1,

NW8B), and CEH42, whose record begins in July 2014, were excluded from the clearfell BACI design. The network permits a test of whether clearfell-driven effects propagate into the adjacent warren, but does not provide fine-grained spatial resolution of any residual propagation signal at distances under 100 m from the clearfell edge, nor a purpose-built warren transect perpendicular to the clearfell boundary.

## 6.8 Coastal-erosion confound — first-order spatial control rather than time-lagged propagation

Both intervention analyses treat the coastal-erosion confound through covariates and control selection rather than through explicit time-lagged modelling. The clearfell ANCOVA includes a coastal-drift covariate that absorbs progressive divergence between the felled compartment and its controls over the observation window, and the scraping analysis tracks coastal-driven drift through CEH4's CUSUM trajectory and through the synthetic-control and SSM-forward-residual robustness checks. Neither approach explicitly represents the multi-year to multi-decadal propagation timescale (L_B²Sy/(Kb), Section 5.4.3) over which coastal erosion signals travel inland through the aquifer. The result is that both intervention analyses estimate the felling-attributable and scraping-attributable steps cleanly within the available analysis windows, but cannot fully separate the contemporaneous management signal from the time-integrated coastal-erosion signal arriving from earlier historical retreat.

A direct test of the spatial component of this confound is provided by the network-scale gradient regression of Section 4.10.2, which fits the strip-aquifer reach and coast-edge deepening rate directly to the 21-year monthly panel. The clearfell ANCOVA's drift covariate is a partial test of it. Written as the fitted coastal profile differenced between target and control (Section 3.7), its coefficient ξ is the fraction of that field the contrast feels, and against the Forest and Combined controls it returns 1.51 ± 0.45 and 1.13 ± 0.63, neither distinguishable from unity. That is agreement with the network-scale fit rather than an independent measurement of it, since the profile is imported from Section 4.10.2 rather than re-estimated; and the far-field contrast disagrees, returning −1.52 ± 0.61. The remaining limitation is the absence of an explicit transient boundary-condition flow model, which would refine the time-lagged propagation estimates and resolve the residual covariate absorption beyond what the steady-state gradient model captures. Direct measurement of present-day coastal retreat rates at the south-western dune front remains a monitoring priority.

## 6.9 Water balance closure as weak validation

The cluster-level water balance closes to within 1.8% of total losses at all five clusters (Table 5), and the residual field is spatially uniform in magnitude, with 65 of the 66 reference wells within ±0.01 m/month and no gradient in residual magnitude across the site (Section 4.9.6).

Closure of this quality is a necessary condition for the model to be trusted, but it is a weak one.

The residual is a single scalar per well formed from the long-term means of all three fitted terms, so compensating errors cancel within it: a $\beta_1$ biased high and a $\beta_2$ biased high in the same proportion close the balance as well as two correct coefficients. Good closure at the long-term mean therefore constrains the terms jointly and says little about any one of them individually, and nothing about month-to-month adequacy, which is addressed separately by the fit statistics and the residual diagnostics of Supplementary Notes S5 and S6. The residual also absorbs nonlinear soil-moisture storage dynamics, coefficient sensitivity and any systematic bias in the Thornthwaite PET estimate indistinguishably, so a small residual cannot be read as evidence that none of these is present.

Interpretation of the field is correspondingly limited. It supports the statement that no large unmodelled flux is required anywhere in the network; it does not support inferences about the magnitude of any individual term, and management conclusions drawn in Section 5 do not rest on it.

## 6.10 Topographic proxies, aeolian deflation, and bedrock causation

At the scale of the dune field the water table follows the ground: against the DEM smoothed over 300 m it has a slope of 1.01 and r = 0.92, against r = 0.76 at native resolution (Section 4.9.5, Figure 57). The DEM-derived drainage lines are a different quantity — surface routing, which spills each closed slack over its saddle — and are not a guide to the direction of groundwater flow. The large-scale correspondence does not imply direct causation: surface sand deposition patterns do not dictate groundwater divides. Both are co-determined by the underlying architecture of glacial till and bedrock ridges, where present. In coastal dune systems this structural relationship is mediated by aeolian deflation — wind erosion is physically limited by the capillary fringe, as cohesive damp sand resists aeolian transport (Bristow and Bailey, 2001; Hesp and Thom, 1990), and because the regional water table is governed by subsurface geology, that geology indirectly controls the baseline limits of surface erosion (Haitjema and Mitchell-Bruker, 2005). Surface topography at Newborough Warren is therefore a physical expression of subsurface hydrological structure, supporting the smoothed DEM as a proxy for the regional form of the water table in the absence of high-resolution bedrock mapping, though not for local flow paths, which the dipwells alone resolve.

## 6.11 MSL5 metric: application beyond its intended domain

The five-year mean spring water level (MSL5; van Willegen et al. 2025) was validated as an ecohydrological descriptor for dune-slack vegetation communities at lake-edge, open-dune and western-residual wells (C1, C2 and C3). Its use in Section 4.12 as a network-wide change-detection tool across all clusters reflects a deliberate methodological choice: shorter comparison windows, one to three years either side of each intervention, proved too unstable to yield interpretable results because the expected management signal is smaller than the interannual variability of spring water levels at this site. Following involvement in the van Willegen et al. (2025) research on the same network, MSL5 was recognized as the most stable available metric for a longer-window empirical comparison. The spring season was selected because it is the only season in which readings are reliably obtainable across the full network: summer levels frequently fall below dipwell reach at the deeper wells, and winter readings are subject to surface ponding that truncates the measurable maximum.

Because this extends the metric beyond its original design, the properties governing its window sensitivity were characterized directly (Section 4.12). Two findings bear on its use elsewhere. First, the assumption of independence between spring readings holds: the expectation that slow-draining wells would carry a head anomaly from one spring to the next is not borne out, because spring is the seasonal maximum and each winter's recharge resets the water table. Second, window sensitivity is nonetheless graded across the network, but it is driven by the amplitude of interannual variation rather than by storage decay, and that amplitude tracks the atmospheric-draw coefficient $\beta_2$ rather than the drainage coefficient $\beta_3$. Researchers applying the metric at sites with heterogeneous aquifer properties should therefore triage wells on the interannual standard deviation of spring level, or on $\beta_2$ as its structural proxy, and not on the recession time; the standard deviation has the further advantage of requiring no state-space fit. The equilibrium wetness index of Section 4.8.4 is not a substitute for change detection, being a steady-state descriptor with no before and after, and it is in any case the less precise of the two indices at every cluster.

The check that remains generally applicable is the external one. Testing whether the chosen window's spatial pattern correlates with an independent long-run trend metric, such as the per-well summer-minimum slope, distinguishes a genuine secular signal from a window-specific artefact; at this site the 2017 → 2023 window converges with that metric at $r = 0.43$ (Section 5.7.5). A worked illustration of why the internal checks require care is provided by the anomalously wet 2024 spring. It lifts both groups: in the window ending 2025 the Forest Control wells stand +133 mm above their 2013–2017 baseline, but the open-dune Climate Controls also stand +29 mm above theirs, and of the +103 mm excess between the two, 2024 contributes 24 mm (Section 5.7.5). A wet spring changes both the signal and its comparator, and does not by itself isolate a cluster-specific effect.

One consequence of the metric's provenance is a limit that should be stated. Van Willegen et al. (2025) related MSL5 to community moisture as a continuous quantity, mean Ellenberg F,

and did not derive community reference values for it; the values of Curreli et al. (2013) describe the annual minimum, and the report applies them to that series only (Section 4.8.3). MSL5 therefore has no published community reference values, and none are constructed here by transferring the annual-minimum values across metrics. The table that would supply them — community means, ranges and counts of MSL5 at the classified permanent quadrats, with the four-year mean annual minimum at the same quadrats reproduced against Curreli's Table 4 as the admission test — is specified in Supplementary Note S12. Its one missing input is the National Vegetation Classification community of each permanent quadrat by survey year, which the deposited dataset does not carry. Until it is built, statements about clusters relative to the SD15b and SD16 values rest on the annual minimum, and MSL5 is read for its ordering of the clusters and its change through time.

## 6.12 Reconstructed records — borrowed from a cluster-mate

A reconstructed month is the donor's level carried across by a line fitted over the months both wells were read. It is right only to the extent that the target shared its donor's departures outside that overlap, which the overlap cannot show; that is why the test hides months that exist (Section 3.3.4). The test has three weaknesses. The early window is thin: few wells were read through it, so its errors rest on 17 targets against 68 in the late window. The rules are not scored on identical targets, because each is scored on the targets it can serve, and Model A could be fitted outside the early window at only 10. And the reference partition is used as committed, so the rules that use no cluster are the clean comparison, and they come close to the cluster-mate.

Some gaps cannot be checked by the test at all. A target that experienced something its donor did not, such as a local felling, a scrape or a drain, is filled wrongly, and the version that accepts donors read through half the gap was not tested. The reconstructed values are therefore kept apart from the measured record, flagged, and not used in any coefficient, threshold or projection in this report; using them in the MSL5 comparison or the vegetation references would be a separate decision.

# 7. Conclusions

This study provides the first network-wide, multi-method quantitative characterization of groundwater behaviour at Newborough Warren, integrating 88 wells, 21 years of monitoring and an analytical pipeline of 64 steps across 22 phases. The principal conclusions are:

1. **Five hydrogeological clusters govern site hydrology.** The eastern open dune (C1 Lake Edge, C2 Dune) responds rapidly to rainfall with high recharge sensitivity ($\beta_1$ = 4.58, 3.90); drainage is fastest at the lake-coupled C1 ($\beta_3$ = 0.089) and slower at the archetypal C2 Dune ($\beta_3$ = 0.063, longer turnover). The Western Residual (C3) acts as a deep buffer (LCSC = 26.6%). The two Forest clusters (C4 Main Forest, C5 Coastal Forest) share elevated LCSC values (40.4%, 41.5%) reflecting canopy interception, but differ in setting: C4 occupies the elevated ridge flank (high $\beta_2$ = 2.56, weak $\beta_3$ = 0.018) and C5 the lower coastal ground (lower $\beta_2$ = 1.34, faster $\beta_3$ = 0.045). Within the forest zone, elevation alone explains 98% of the variance in $\beta_2$, identifying topographic position — and the substrate inferred to accompany it — as the major control on the atmospheric draw coefficient, with depth-coupling and other factors also acting. That the equally forested C5 (Coastal Forest) carries the same Corsican pine canopy yet returns a low $\beta_2$ indicates that the elevated C4 value reflects position rather than the canopy: were the high atmospheric draw a canopy effect, both forested clusters would show it. The plantation subsurface is not cored, so the C4/C5 substrate contrast is an interpretation of the elevation and coefficient evidence rather than a direct observation. The co-temporal amplification field (Section 4.12; Figure 75) is consistent with this substrate reading: C4 amplifies the shared dry-to-wet spring climate swing to 1.66 times the network mean (range 1.31–2.32×), while the lake-edge cluster damps it to 0.64×. This over-response is the opposite of what a canopy-shielding mechanism would predict and is more consistently explained by two positional factors: thin saturated thickness over irregular bedrock, and the forest's occupation of the hydraulic high of the aquifer, furthest from any lateral discharge boundary — lake, coast or estuary — and therefore without the constant-head damping that buffers the lake-edge and coastal clusters in both wet and dry years.
2. **Position and substrate, rather than canopy, best explain the forest atmospheric-draw coefficient.** Of the two forested clusters, C5 ($\beta_2$ = 1.25) lies below C2 (1.84) and C3 (1.83) but within the open-dune range, which extends down to C1 (0.91); C4's $\beta_2$ (2.57) is genuinely elevated. An identifiability diagnostic finds the C4 drainage coefficient cleanly resolved, on a near-orthogonal drainage predictor (among the least collinear in the network), so the elevated $\beta_2$ is not an artefact of a degenerate $\beta_2$–$\beta_3$ pair. It is attributed instead to position and substrate: $\beta_2$ measures head response per unit atmospheric demand, and therefore rises where drainable storage is small.

Whether the canopy suppresses summer atmospheric draw cannot be stated as a finding on the present evidence — the amplification result, the coastal-retreat confound at C5, and the imprecision of the seasonal clearfell comparison all leave it open.

3. **Summer minimum depth is the binding ecological constraint, and in the forest clusters the dominant predictor of winter flooding.** Winter rainfall is the dominant predictor of winter flooding potential in the eastern open dune (C1, C2; $p = 0.002$, 0.010); in the Western Residual (C3) rainfall and antecedent summer minimum carry equal weight ($p = 0.047$, 0.045); antecedent summer minimum dominates at the two Forest clusters ($p < 0.001$), where rainfall is not significant (Table 16). The summer level is therefore the restoration lever in the forest clusters and shares that role with rainfall at C3. The entire ecological gradient from SD15b wet slack to SD16 dry slack spans only 37 cm of summer minimum depth (Curreli et al., 2013). Any management intervention must be evaluated on its summer minimum effect.
4. **State-space modelling substantially improves long-range forecast skill.** The median per-well iterative NSE improvement of +0.37 over the traditional linear model — with only 1 of 66 SSM wells producing a negative iterative NSE compared with 11 of 66 for the TLM — indicates that explicit parameterization of the head-dependent drainage term is essential for long-range prediction in this aquifer.
5. **The spring-level metric is supported externally by vegetation and by satellite-observed inundation extent, and the state-space coefficients carry ecohydrological signal.** MSL5 predicts the co-located Ellenberg-F of van Willegen et al. (2025) between wells ($r = 0.83$, $n = 18$). An equilibrium wetness index obtained by inverting the SSM drainage term is statistically indistinguishable from MSL5 as an Ellenberg-F predictor (Williams $p = 0.49$) and, calibrated onto the MSL5 scale, reconstructs open-dune spring levels to 73 mm RMSE, generalizing beyond the vegetation-calibrated set — indicating that dynamics-only coefficients carry ecohydrological signal. Its prediction of vegetation moisture is independent of MSL5 — computed from the coefficients and climate alone — and so provides external ecological support for the state-space model; its reconstruction of MSL5, being calibrated on that metric, is a usability diagnostic rather than an independent result. It is a supporting diagnostic scoped to the open dune (unreliable in the forest), advantageous mainly in needing a shorter record, so its value is transferability to other sites rather than local gap-filling. The modelled level is independently tested in space as well as at points: across 43 winter Sentinel-2 scenes (2016–2026) the wetted area of the open slack floor is a monotonic function of the modelled median level (wet floor $R^2 = 0.77$, open water $R^2 = 0.71$), and driving the SSM level through these curves reproduces the observed inundation out of sample ($n = 238$ months; wet-floor $R^2 = 0.80$, median modelled:observed ratio 0.96) — the report's only spatial, out-of-sample validation of the water table, strongest for the wet-floor class that carries the dune-slack habitat signal.

6. **Dune scraping delivers a measurable but spatially and temporally limited benefit whose placement within the drainage system is critical.** Three independent methods identify a scraping benefit at CEH36 in the same direction (raw BACI +0.128 m, synthetic control +0.141 m, SSM forward residual +0.100 m), with a paired summer minimum shift of +195 mm (p = 0.004) — more than half the SD15b–SD16 ecological gradient. The benefit is consistent with a geometric effect (permanent surface lowering) rather than a change in aquifer dynamics: +0.128 m from a cut inferred at about 0.43 m. Scraping operates through the same physical mechanism as coastal erosion — lowering ground elevation to increase the displacement head and drainage gradient — and its spatial placement determines whether it complements or exacerbates the existing coastal drainage regime. CEH21's limited response to the October 2023 intervention indicates that position within the coastal drainage pathway is a significant determinant of scraping efficacy. Priority targets are the C1/C2/C3 transitional wells where the aquifer base is stable and P_flood thresholds remain achievable (rainfall multiplier m_P < 1.5); the per-well achievability categorization is mapped in Figure 85 (Section 5.9).
7. **The clearfell experiment was followed by a winter rise at the felled well and none in summer.** Against the Forest control the felled well (WMC3) rose +128 mm over October–March (95% CI +1 to +254 mm, p = 0.048) and not at all over June–September (−1 mm); the annual-mean step of the three-counterfactual ANCOVA-BACI, +108 mm, is not significant on its own with autocorrelation-robust (Newey–West) errors (95% CI −21 to +237 mm, p = 0.10). At the Edge tier the step against the Forest control was +31 mm and not statistically distinguishable from zero (p = 0.50); the measurable clearfell recovery is localized at the felled well rather than expressed across the surrounding edge of the compartment. The recovery is concentrated in the mean monthly water level rather than in summer minima: the summer minima analysis returned no significant effect at the felled wells, whose summer minima moved in parallel with the unfelled Forest controls. Synthetic extension to the WMC3 + FE2 centroid within the felled compartment returns a clearfell step of +94 mm (95% CI −24 to +212 mm, p = 0.12) against the Forest control, consistent in direction and magnitude with the Impact-well result. Scenario projections anchored to this felling, a fall in atmospheric draw and no rise in recharge, reproduce the step at the felled well (+0.060 m under Model B, +0.135 m under Model A). They predict sustained rises under clearfell of +0.038 to +0.066 m at C5 and +0.217 m at C4 (Model A), at most half of each under thinning, and a range from a slight fall to a gain under broadleaf conversion. The interception-recharge form used previously overstated the felled well’s response three- to tenfold; the 4.2 ha compartment is narrower than the drawdown reach, so a recharge gain there would disperse into the surrounding forest, and that form remains the upper end for a wider felling. The +108 mm Impact-well clearfell step rests on the paired ANCOVA against a physically matched Forest control (Section 4.6.3). It is the record mean of a front-loaded recovery: +152 mm to

December 2020 and +39 mm thereafter, no longer distinguishable from zero (Section 4.6.8). An empirical MSL5 comparison (window end 2017 vs window end 2023) shows no spatially distinct clearfell signal in five-year spring means, against a five-year window standard error of 82–181 mm by cluster, consistent with the BACI finding that the recovery is concentrated in the monthly-mean water level rather than in spring peaks. The monthly-mean ANCOVA result (+108 mm) is the primary clearfell estimate; the MSL5 comparison reflects the secular deepening trend rather than the management response.

8. **Recharge sensitivity ($\beta_1$) has declined across the BACI network, but its independence of management is not established.** The network-mean $\beta_1$ fell from 2.83 to 2.70 between the pre-fell and post-fell eras (Section 4.6.6; Table 15), a reduction of 4.6%, with thirteen of seventeen wells declining; but the unfelled Climate Control tier — five wells distant from both the clearfell footprint and the coastal margin — declined by only 0.3% (three of five wells), so the tier that would demonstrate management-independence does not. The decline remains a candidate mechanism for the observed summer minimum deterioration across the network, consistent with, but not shown to be, a site-wide trend. The network-wide decline in the SSM water balance residual α_B between eras, visible even at open-dune Climate Control wells distant from the forest and coast, indicates that changed antecedent storage state between the two eras is a confound in the era-split coefficient analysis; the post-felling $\beta_3$ decline at Forest Control wells should be treated as indicative pending direct hydraulic corroboration. The monitoring evidence therefore places the clearfell rise in the winter months at the felled well (+128 mm October–March, $p = 0.048$), not in the summer months or the summer minimum, where felled and unfelled wells track each other.
9. **Modelled forest drawdown reaches the lake edge; felling left no spatial gradient.** The drawdown propagation analysis (Figure 66) places the steady-state head deficit from the existing 75-year-old plantation at 8–72 mm at the lake-edge wells nearest the forest (CEH6 8 mm, CEH23 24 mm, CEH26 43 mm, CEH27 72 mm), falling to 2 mm or less only at wells furthest from it along the flow path, among them CEH7, CEH8, CEH24, the T41 transect and D43. These are modelled steady-state values, not observed ones. The model represents canopy interception alone, so its heads are a lower bound on the magnitude of the forest effect, while its e-folding length λ is an upper bound on how far beyond the canopy that effect reaches. The BACI transect analysis shows no felling-specific spatial gradient. The operational domain for direct hydrological intervention is the C1/C2 zone where P_flood thresholds remain achievable.
10. WTF-derived specific yields are broadly convergent once canopy interception is corrected. C4's corrected estimate (0.261) falls within the open dune range (0.210–0.325) and the interpolated yield surface shows no discontinuity at the cluster margins; C5's (0.316) sits just above the upper edge but is constrained by the plausibility clip

and is only weakly corroborative. The plantation's distinctive β-coefficient signature therefore derives largely from canopy boundary conditions at the surface rather than from differences in subsurface storage architecture.

11. **The water balance closes to within 1.8% under the displacement formulation, and the per-well residual field shows no coherent spatial structure.** Residual magnitudes are small and uncorrelated with position on either East or North axis, and show no forest-margin or ridge-flank concentration; no well exceeds +0.02 m/month, and the largest positive values — NW3 at +0.0080 m/month, with CEH23 and NW6 close behind — fall in the open dune rather than in the forest. The field carries a small systematic negative offset (median −0.0040 m/month, 51 of 66 wells negative). Two independent diagnostic tests for a ridge-derived lateral input — a rainfall–residual lag-distance test and a sunshine-hours test for unmodelled evaporative demand — returned null results, consistent with the residual field itself.
12. **C5 Coastal Forest shows the steepest full-record decline in the network in both summer (−0.029 m yr⁻¹, p = 0.004) and winter (−0.027 m yr⁻¹, p = 0.056).** At 2.5 times the summer decline rate of C1 — one of the three other clusters (C1, C2, C3) with a statistically significant summer trend over the full record — the magnitude of C5's divergence is consistent with a contribution beyond the climate forcing all clusters share. The divergence belongs to the early record: refitted from 2011, C5's summer trend is +0.003 m yr⁻¹ (p = 0.79), as flat as the other clusters'. Three candidate explanations — amplified head response per unit climate forcing due to thin saturated thickness at the coastal margin, coastal boundary retreat, and progressive canopy maturation — are not mutually exclusive, and the scale of the divergence is reinforced by a parallel post-2018 collapse in seasonal amplitude (Section 4.2.1) indicating that recharge peaks are no longer reaching their pre-2018 heights. A network-scale non-linear panel regression on perpendicular distance to the eroding Caernarfon Bay shoreline (Section 4.10.2; Table 24 and Figure 60) attributes about half of C5's exceptional summer-minimum decline (four C5 wells in the per-well basis, three reference members and the extended well NW8B; the direction of the result is robust, the magnitude sensitive to which wells enter the filter), within uncertainty, to a coastal-retreat gradient (coast-edge trend −26.4 mm yr⁻¹ at the 150 m reference distance, from a fitted shoreline amplitude of −31.3 mm yr⁻¹ decaying linearly to the far-field asymptote by about 900 m inland; AIC favours the exponential form, and the linear-capped form is retained on physical grounds). Progressive coastal boundary retreat is therefore quantified as the largest single contributor to the C5 anomaly, leaving −17.6 mm yr⁻¹ unexplained; the attribution is sensitive to cluster composition, since the same decomposition left −0.7 mm yr⁻¹ before NW8B was assigned to C5. The other two candidate mechanisms are not excluded and, on the present decomposition, retain room for a quantitative role at C5. Direct measurement of present-day coastal retreat rates and continued monitoring of the CUSUM trajectory remain a priority. Monthly decomposition of the RAF Valley PET record shows significant increases in May

(+0.068 mm month$^{-1}$ yr$^{-1}$, p = 0.002), June (+0.048, p = 0.033) and July (+0.063, p = 0.010), providing an independent physical mechanism for the site-wide secular deepening operating through intensified early-summer atmospheric demand.

13. **The wet-slack summer regime has narrowed where it can be measured.** At van Willegen's quadrats, the one place the slack floor is surveyed, every quadrat read in both windows is deeper than in Curreli et al.'s survey years, by a median of 0.19 m, and none now lies above the SD15b community mean. Across the network, summer minima fell in four of the five clusters over 2005–2025, the decline concentrated before 2011, while the five-year spring baseline shows no decline that the window choice can resolve. A differential GPS survey of the slack floor at each dipwell would let the whole 21-year record be read against the communities (Section 5.8.2).
14. **Progressive coastal boundary retreat is an independent and potentially accelerating threat whose full consequences are lagged.** The convergent deterioration of two spatially separated control wells (CEH4 and CEH22), independently of any management action, indicates that the aquifer is losing volume at its seaward edge. The aquifer's hydraulic diffusivity means that the coastal erosion signal visible in the current CUSUM trajectories represents erosion that occurred years earlier; if the rate of retreat is accelerating, as CEH22's trajectory from 2021 suggests, the full hydrological consequences have not yet propagated to interior wells. The network-scale gradient regression of Section 4.10.2 quantifies the steady-state coastal-retreat signal directly from 21 years of monitoring data: the coast-edge water-table anomaly above climate is approximately −26.4 mm yr$^{-1}$ at the 150 m reference distance, decaying to zero over an inland reach of about 900 m. C5 Coastal Forest sits closest to the shore, all four of its wells within the gradient zone; three of the four C4 wells and half of the C3 wells also lie inside the fitted reach, one C1 well (CEH11) just inside it, and no C2 well. Direct measurement of present-day coastal retreat rates and continued monitoring of inland propagation through the CEH4 and CEH22 CUSUM trajectories remain a priority.

## 8. Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Anthropic's Claude (successive Claude Opus and Claude Fable models, accessed through the Claude desktop application) for substantial drafting and editing of the manuscript text: composing prose from the author's analytical outputs and decisions, structuring and restructuring sections, and correcting the text against the study's committed analytical outputs. The conception and design of the study, the twenty-one-year manual dipwell record on which it rests, the analytical decisions, and the interpretation of the results are the author's own. Numerical values in the manuscript were verified against the committed outputs of the analysis pipeline, and a decision log records the methodological and editorial calls made by the author during the work. After using this tool the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

# 9. References

Arribas-Bel, D., Fleischmann, M., & others. (2020) *contextily: Context geo tiles in Python*. GitHub repository. https://github.com/geopandas/contextily

Bakker, M. and Schaars, F. (2019) Solving groundwater flow problems with time series analysis: you may not even need another model. *Groundwater*, 57(6), pp. 826–833. https://doi.org/10.1111/gwat.12927

Bear, J. (1972) Dynamics of Fluids in Porous Media. New York: American Elsevier.

Betson, M., Connell, L. and Bristow, C. (2002) Groundwater Modelling of Newborough Warren. In: Bristow, C. (ed.) The Impact of Forestry on Coastal Geomorphology at Newborough Warren/Ynys Llanddwyn NNR, SSSI, pSAC. Volume 5: Hydrogeology. Report to the Countryside Council for Wales, Contract FC 73-05-18. Department of Geological Sciences, UCL and School of Earth Sciences, Birkbeck University of London. July 2002.

Betson, M. and Scholefield, P. (2004) Implications for the Water Balance of Newborough Warren NNR of the Evapotranspiration of Herbaceous and Afforested Duneland Ecosystems. Unpublished report to Countryside Council for Wales. ADAS.

Beven, K. (2012) Rainfall-Runoff Modelling: The Primer. 2nd edn. Chichester: Wiley-Blackwell.

Bristow, C.S. (2003) The Impact of Forestry on Coastal Geomorphology at Newborough Warren / Ynys Llanddwyn NNR, SSSI, pSAC. Volumes 1–5. Contract FC 73-05-18, Final Report. Countryside Council for Wales. London: Birkbeck University of London.

Bristow, C.S. and Bailey, S.D. (2001) Non-invasive investigation of water table and structures in coastal dunes using ground-penetrating radar (GPR): implications for dune management. In: Houston, J.A., Edmondson, S.E. and Rooney, P.J. (eds) *Coastal Dune Management: Shared Experience of European Conservation Practice.* Liverpool University Press, pp. 408–417.

Buglife (2019) *Coastal sand dunes.* Habitat management guidance. Buglife – The Invertebrate Conservation Trust, Peterborough.

Burnham, K.P. and Anderson, D.R. (2002) Model Selection and Multimodel Inference: A Practical Information-Theoretic Approach. 2nd edn. New York: Springer.

Calder, I.R. (2007) "Forests and water — Ensuring forest benefits outweigh water costs." *Forest Ecology and Management* 251, 110–120. doi:10.1016/j.foreco.2007.06.015

Calder, I.R. and Newson, M.D. (1979) Land-use and upland water resources in Britain — a strategic look. Water Resources Bulletin, 15(6), pp. 1628–1639.

Caliński, T. and Harabasz, J. (1974) A dendrite method for cluster analysis. *Communications in Statistics*, 3(1), pp. 1–27.

Callaghan, D. A., van Willegen, L., Williams, G., Hollingham, M., & Jones, L. (2021) Pony trails, hydrology and habitat restoration: aspects of the ecology of *Petalophyllum ralfsii* in a Welsh oceanic dune system. *Journal of Bryology*, 43(2), 150-161. https://doi.org/10.1080/03736687.2020.1819719

Clarke, D. and Sanitwong Na Ayutthaya, S. (2010) Predicted effects of climate change, vegetation and tree cover on dune slack habitats at Ainsdale on the Sefton Coast, UK. Journal of Coastal Conservation, 14(2), pp. 115–125. doi:10.1007/s11852-009-0066-7.

**Collenteur, R.A., Bakker, M., Caljé, R., Klop, S.A. and Schaars, F. (2019)** Pastas: open source software for the analysis of groundwater time series. *Groundwater*, 57(6), pp. 877–885. https://doi.org/10.1111/gwat.12925

Cottingham, P. (1994) The effects of afforestation on the hydrology of Newborough Warren dune system. Unpublished MSc dissertation, University of Wales, Bangor.

Crosbie, R.S., Binning, P. and Kalma, J.D. (2005) A time series approach to inferring groundwater recharge using the water table fluctuation method. *Water Resources Research* 41(1), W01008. doi:10.1029/2004WR003077.

Curreli, A., Wallace, H., Freeman, C., Hollingham, M., Stratford, C., Johnson, H. and Jones, L. (2013) Eco-hydrological requirements of dune slack vegetation and the implications of climate change. Science of the Total Environment, 443, pp. 910–919. doi:10.1016/j.scitotenv.2012.11.035.

Dargie, T.C.D. (1995) Sand dune vegetation survey of Great Britain: a national inventory. Part 3: Wales. Peterborough: Joint Nature Conservation Committee.

Davy, A.J., Grootjans, A.P., Hiscock, K. and Petersen, J. (2006) Development of Eco-Hydrological Guidelines for Dune Habitats — Phase 1. English Nature Research Reports, No. 696. Peterborough: English Nature.

Davy, A.J., Hiscock, K.M., Jones, M.L.M., Low, R., Robins, N.S. and Stratford, C. (2010) Protecting the plant communities and rare species of dune wetland systems: ecohydrological guidelines for wet dune habitats — Phase 2. Bristol: Environment Agency. Science Report GEHO0310BSGV-E-E.

Deng, Z., Priestley, S.C., Guan, H., Love, A.J. and Simmons, C.T. (2013). Canopy enhanced chloride deposition in coastal South Australia and its application for the chloride mass balance method. *Journal of Hydrology*, 497, 62–70. https://doi.org/10.1016/j.jhydrol.2013.05.040

Denning, L., Pugh, M., Pitts, V. and Dynamic Dunescapes partners (2024) *The Sand Dune Managers Handbook*, Version 2, September 2024. Produced for the Dynamic Dunescapes project. https://dynamicdunescapes.co.uk/wp-content/uploads/2024/09/Dynamic-Dunescapes-Sand-Dune-Managers-Handbook-v2.pdf

Desbarats, A.J., Logan, C.E., Hinton, M.J. and Sharpe, D.R. (2002) On the kriging of water table elevations using collateral information from a digital elevation model. *Journal of Hydrology*, 255(1–4), 25–38. https://doi.org/10.1016/S0022-1694(01)00504-2

Donohue, R.J., McVicar, T.R. and Roderick, M.L. (2010) Assessing the ability of potential evaporation formulations to capture the dynamics in evaporative demand within a changing climate. Journal of Hydrology, 386(1–4), pp. 186–197. https://doi.org/10.1016/j.jhydrol.2010.03.020

Ferguson, G. and Gleeson, T. (2012) Vulnerability of coastal aquifers to groundwater use and climate change. *Nature Climate Change*, 2, 342–345.

Fetter, C. W. (2001) *Applied Hydrogeology* (4th ed.). Prentice Hall.

Flyamer, I. M., Illingworth, R. S., & others. (2020) *adjustText: Automatically adjust text position in matplotlib plots to minimize overlaps*. GitHub repository. https://github.com/Phlya/adjustText

Forgrave, A. (2020) Dramatic pictures show the beautiful little corner of North Wales that's fast being lost to the sea. *North Wales Live / Daily Post*, 22 January 2020.

Freeman, S. (2008) *The hydrological impact of Corsican pine plantation on the dune system at Newborough Warren, Anglesey*. Unpublished BSc Honours dissertation, University of Wales, Bangor.

Freeze, R.A. and Cherry, J.A. (1979) Groundwater. Englewood Cliffs, NJ: Prentice-Hall.

Gash, J.H.C. (1979) An analytical model of rainfall interception by forests. Quarterly Journal of the Royal Meteorological Society, 105(443), pp. 43–55.

Gash, J.H.C., Wright, I.R. and Lloyd, C.R. (1980) Comparative estimates of interception loss from three coniferous forests in Great Britain. Journal of Hydrology, 48, pp. 89–105.

Gebhardt, T., Häberle, K.-H., Matyssek, R., Schulz, C. and Ammer, C. (2014) The more, the better? Water relations of Norway spruce stands after progressive thinning. Agricultural and Forest Meteorology, 197, pp. 235–243. https://doi.org/10.1016/j.agrformet.2014.05.013

Gerla, P.J. (2019) Monitoring and Modeling the Effect of Agricultural Drainage and Recent Channel Incision on Adjacent Groundwater-Dependent Ecosystems. *Water* 11(4): 863. https://doi.org/10.3390/w11040863

Giddens, K.M., Parfitt, R.L. and Percival, H.J. (1997) Comparison of some soil properties under *Pinus radiata* and pasture. *New Zealand Journal of Agricultural Research*, 40(4), pp. 443–450.

Gillies, S., & others. (2013) *Rasterio: Geospatial raster I/O for Python programmers*. GitHub repository. https://github.com/rasterio/rasterio

Gotelli, N.J. and Ellison, A.M. (2004) A Primer of Ecological Statistics. Sunderland, MA: Sinauer Associates.

Grootjans, A.P., Geelen, H.W.T., Jansen, A.J.M. and Lammerts, E.J. (2002) Restoration of coastal dune slacks in the Netherlands. Hydrobiologia, 478(1–3), pp. 181–203.

Grootjans, A. P., Adema, E. B., Bekker, R. M., & Lammerts, E. J. (2004) Why young coastal dune slacks sustain a high biodiversity. In M. L. Martínez & N. P. Psuty (Eds.), *Coastal Dunes: Ecology and Conservation* (Ecological Studies, Vol. 171, pp. 85-101). Springer-Verlag, Berlin, Heidelberg. https://doi.org/10.1007/978-3-662-06259-5_6

Grootjans, A.P., Shahrudin, R., van de Craats, A., Kooijman, A., Oostermeijer, G., Petersen, J., Amatirsat, D., Bland, C. and Stuyfzand, P.J. (2017) Window of opportunity of *Liparis loeselii* populations during vegetation succession on the Wadden Sea islands. *Journal of Coastal Conservation*, 21(5), 631–641. https://doi.org/10.1007/s11852-016-0448-6

Gumuła-Kawęcka, A., Jaworska-Szulc, B., Szymkiewicz, A., Gorczewska-Langner, W., Pruszkowska-Caceres, M., Angulo-Jaramillo, R. and Šimůnek, J. (2021) Estimation of groundwater recharge in a shallow sandy aquifer using unsaturated zone modeling and water table fluctuation method. Journal of Hydrology, 605, 127283. https://doi.org/10.1016/j.jhydrol.2021.127283

Gumuła-Kawęcka, A., Jaworska-Szulc, B. and Jefimow, M. (2024) Climate change impact on groundwater resources in sandbar aquifers in southern Baltic coast. *Scientific Reports* 14, 11828. https://doi.org/10.1038/s41598-024-62522-0

Gupta, H. V., Wagener, T., & Liu, Y. (2008) Reconciling theory with observations: elements of a diagnostic approach to model evaluation. *Hydrological Processes*, 22(18), 3802–3813. https://doi.org/10.1002/hyp.6989

Haitjema, H.M. and Mitchell-Bruker, S. (2005) Are water tables a subdued replica of the topography? Groundwater, 43(6), pp. 781–786.

Hannah, D. M., Smith, B. P. G., Gurnell, A. M., & McGregor, G. R. (2000) An approach to characterize data streams in hydrology. *Journal of Hydrology*, 232(1-4), 71-91. https://doi.org/10.1016/S0022-1694(00)00214-9

Harris, C. R., Millman, K. J., van der Walt, S. J., Gommers, R., Virtanen, P., Cournapeau, D., ... & Oliphant, T. E. (2020) Array programming with NumPy. *Nature*, 585(7825), 357-362. https://doi.org/10.1038/s41586-020-2649-2

Healy, R.W. and Cook, P.G. (2002) Using groundwater levels to estimate recharge. *Hydrogeology Journal* 10, 91–109. doi:10.1007/s10040-001-0178-0

Hennig, C. (2007) Cluster-wise assessment of cluster stability. *Computational Statistics and Data Analysis*, 52(1), pp. 258–271.

Hesp, P.A. and Thom, B.G. (1990) Geomorphology and evolution of active transgressive dunefields. In: Nordstrom, K.F., Psuty, N. and Carter, R.W.G. (eds.) Coastal Dunes: Form and Process. Chichester: John Wiley and Sons, pp. 253–288.

Hill, M.O., Mountford, J.O., Roy, D.B. & Bunce, R.G.H. (1999). Ellenberg's indicator values for British plants. ECOFACT Volume 2, Technical Annex. Institute of Terrestrial Ecology, Huntingdon, 46 pp.

Hollingham, M. (2026) Newborough Warren Groundwater Research: interactive modelling suite — scenario viewer, per-well flood-probability forecaster, and seasonal-extremes explorer. Available at: https://newbroman.github.io/Newborough_Hydrology/ [Accessed August 2026].

Hooghoudt, S.B. (1940) Bijdragen tot de kennis van eenige natuurkundige grootheden van den grond. 7. Algemeene beschouwing van het probleem van de detailontwatering en de infiltratie door middel van parallel loopende drains, greppels, slooten en kanalen [General consideration of the problem of field drainage by parallel drains, ditches, watercourses and channels]. *Verslagen van Landbouwkundige Onderzoekingen* 46(14)B, 515–707. Algemeene Landsdrukkerij, 's-Gravenhage.

Huizer, S., Radermacher, M., de Vries, S., Oude Essink, G. H. P. and Bierkens, M. F. P. (2018) Impact of coastal forcing and groundwater recharge on the growth of a fresh groundwater lens in a mega-scale beach nourishment. *Hydrology and Earth System Sciences*, 22, 1065–1080. https://doi.org/10.5194/hess-22-1065-2018

Hunter, J. D. (2007). Matplotlib: A 2D graphics environment. *Computing in Science & Engineering*, 9(3), 90-95. https://doi.org/10.1109/MCSE.2007.55

Hypolite, J., Arvor, D., Bertrand, G., & Fovet, O. (2021) A state-space approach to groundwater level forecasting: Comparisons with lumped and machine learning models. *Journal of Hydrology*, 603, 127115. https://doi.org/10.1016/j.jhydrol.2021.127115

Jennings, T.A.H. (1990) The changing hydrology of the Newborough Dune System due to afforestation. Unpublished M.Sc. dissertation, University of North Wales, Bangor.

JNCC (2026) Y Twyni o Abermenai i Aberffraw / Abermenai to Aberffraw Dunes, Special Area of Conservation UK0020021: site page. Peterborough: Joint Nature Conservation Committee. https://sac.jncc.gov.uk/site/UK0020021 (accessed 28 September 2026).

Jordahl, K., Van den Bossche, J., Fleischmann, M., Wasserman, J., McBride, P., Gerard, J., ... & Ward, B. (2020). *geopandas/geopandas: v0.8.1*. Zenodo. https://doi.org/10.5281/zenodo.3946761

Kiflai, M. E. and Whitman, D. (2023) Morphologic, atmospheric, and oceanic drivers cause multi-temporal saltwater intrusion on a remote, sand island. *Water Resources Research*, 59. https://doi.org/10.1029/2022WR033582

Knotters, M. and Bierkens, M.F.P. (2000) Physical basis of time series models for water table depths. Water Resources Research, 36(1), pp. 181–188. https://doi.org/10.1029/1999WR900288

Knotters, M. and van Walsum, P.E.V. (1997) Estimating fluctuation quantities from time series of water-table depths using models with a stochastic component. Journal of Hydrology, 197(1–4), pp. 25–46.

Komatsu, H., Kume, T. and Otsuki, K. (2011) Increasing annual runoff — broadleaf or coniferous forests? *Hydrological Processes*, 25(2), 302–318. https://doi.org/10.1002/hyp.7898

Krause, P., Boyle, D. P., & Bäse, F. (2005) Comparison of different efficiency criteria for hydrological model assessment. *Advances in Geosciences*, 5, 89-97. https://doi.org/10.5194/adgeo-5-89-2005

Manis, J., Shanafield, M., Cook, P. G., Andersen, M. S., & McCallum, J. L. (2022) Quantifying internal drainage and vertical leakage in multi-layered aquifer systems using state-space representations. *Groundwater*, 60(4), 512–525. https://doi.org/10.1111/gwat.13171

McKinney, W. (2010) Data structures for statistical computing in Python. In *Proceedings of the 9th Python in Science Conference* (Vol. 445, pp. 51-56). https://doi.org/10.25080/Majora-92bf1922-00a

Met Office (2018) UKCP18 Regional Projections on a 12 km grid over the UK for 1980–2080. Centre for Environmental Data Analysis. Met Office Hadley Centre. Available at: https://catalogue.ceda.ac.uk/uuid/b4d24b3df3754b9d9028447eb3cbd878

Michael, H. A., Russoniello, C. J. and Byron, L. A. (2013) Global assessment of vulnerability to sea-level rise in topography-limited and recharge-limited coastal groundwater systems. *Water Resources Research*, 49, 2228–2240.

Nash, J.E. and Sutcliffe, J.V. (1970) River flow forecasting through conceptual models. Part I — A discussion of principles. Journal of Hydrology, 10(3), pp. 282–290.

Obergfell, C., Bakker, M. and Maas, K. (2019) Identification and explanation of a change in the groundwater regime using time series analysis. *Groundwater*, 57(6), pp. 886–894. https://doi.org/10.1111/gwat.12891

Osenberg, C.W., Schmitt, R.J., Holbrook, S.J., Abu-Saba, K.E. and Flegal, A.R. (1994) Detection of environmental impacts: natural variability, effect size, and power analysis. *Ecological Applications*, 4(1), 16–30.

Page, E.S. (1954) Continuous inspection schemes. Biometrika, 41(1/2), pp. 100–115.

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M. and Duchesnay, É. (2011) Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research* 12: 2825–2830.

Peters, E., Torfs, P.J.J.F., van Lanen, H.A.J. and Bier, G. (2003) Propagation of drought through groundwater — a new approach using linear reservoir theory. Hydrological Processes, 17(15), pp. 3023–3040.

Peterson, T.J. and Western, A.W. (2014) Nonlinear time-series modeling of unconfined groundwater head. *Water Resources Research*, 50(10), pp. 8330–8355. https://doi.org/10.1002/2013WR014800

Pye, K. and Blott, S.J. (2024) *Dune rejuvenation AfterLIFE options for Twyni Penrhos, Newborough.* NRW Evidence Report No. 787, 73 pp. Natural Resources Wales, Bangor.

Pye, K. and Saye, S. (2005) The Geomorphological Response of Welsh Sand Dunes to Sea Level Rise Over the Next 100 Years and the Management Implications for SAC and SSSI Sites. CCW Contract Science Report No. 670. Bangor: Countryside Council for Wales.

Ranwell, D.S. (1959) Newborough Warren, Anglesey 1. The dune system and dune slack habitat. Journal of Ecology, 47, pp. 571–601.

Ranwell, D.S. (1972) *Ecology of Salt Marshes and Sand Dunes*. Chapman and Hall, London.

Rao, A. R., & Srinivas, V. V. (2006) Regionalization of watersheds by fuzzy cluster analysis. *Journal of Hydrology*, 318(1-4), 57-79. https://doi.org/10.1016/j.jhydrol.2005.06.004

Rao, P., Wang, Y., Liu, Y., Wang, X., Hou, Y., Pan, S., Wang, F. and Zhu, D. (2022) A comparison of multiple methods for mapping groundwater levels in the Mu Us Sandy Land, China. *Journal of Hydrology: Regional Studies*, 43, 101189. https://doi.org/10.1016/j.ejrh.2022.101189

Robins, N. S., Jones, M. L. M., & Farr, G. (2013) Shallow groundwater in the dune and slack environment: The implications for management. *Quarterly Journal of Engineering Geology and Hydrogeology*, 46(3), 261–265. https://doi.org/10.1144/qjegh2012-045

Robins, N. S., Pye, K. and Wallace, H. (2013) Dynamic coastal dune spit: the impact of morphological change on dune slacks at Whiteford Burrows, South Wales, UK. *Journal of Coastal Conservation*, 17(3), 473–482. https://doi.org/10.1007/s11852-013-0245-4

Robinson, E.L., Huntingford, C., Semeena, V.S. and Bullock, J.M. (2023) CHESS-SCAPE: high-resolution future projections of multiple climate scenarios for the United Kingdom derived from downscaled United Kingdom Climate Projections 2018 regional climate model output. Earth System Science Data, 15(12), pp. 5371–5401. https://doi.org/10.5194/essd-15-5371-2023

Robinson, M. and Dupeyrat, A. (2005) Effects of commercial timber harvesting on streamflow regimes in the Plynlimon catchments, mid-Wales. Hydrological Processes, 19(6), pp. 1213–1226.

Roy, V., Plamondon, A.P. and Bernier, P.-Y. (2000) Influence of vegetation removal and regrowth on interception and water table level on wetlands. International Peat Journal, 10, pp. 3–12.

Rutter, A.J., Kershaw, K.A., Robins, P.C. and Morton, A.J. (1971) A predictive model of rainfall interception in forests. I: Derivation of the model from observations in a plantation of Corsican pine. Agricultural Meteorology, 9, pp. 367–384.

Scanlon, B.R., Healy, R.W. and Cook, P.G. (2002) Choosing appropriate techniques for quantifying groundwater recharge. *Hydrogeology Journal*, 10(1), 18–39. https://doi.org/10.1007/s10040-001-0176-2

Seabold, S., & Perktold, J. (2010). statsmodels: Econometric and statistical modeling with python. In *Proceedings of the 9th Python in Science Conference* (Vol. 57, pp. 61-66). https://doi.org/10.25080/Majora-92bf1922-011

Šimůnek, J., Šejna, M., Saito, H., Sakai, M. and van Genuchten, M.T. (2008) *The HYDRUS-1D Software Package for Simulating the One-Dimensional Movement of Water, Heat, and Multiple Solutes in Variably Saturated Media, Version 4.0*. HYDRUS Software Series 3, Department of Environmental Sciences, University of California Riverside, Riverside, California, USA.

Sival, F.P. and Grootjans, A.P. (1996). Dynamics of seasonal bicarbonate supply in a dune slack: effects on organic matter, nitrogen pool and vegetation succession. *Vegetatio*, 126, 39–50.

Smokorowski, K.E. and Randall, R.G. (2017). Cautions on using the Before-After-Control-Impact design in environmental effects monitoring programs. *FACETS* 2: 212–232. https://doi.org/10.1139/facets-2016-0058

Sonnenborg, T.O., Christiansen, J.R., Pang, B., Bruge, A., Stisen, S. and Gundersen, P. (2017) Analyzing the hydrological impact of afforestation and tree species in two catchments with contrasting soil properties using the spatially distributed model MIKE SHE SWET. Agricultural and Forest Meteorology, 239, pp. 118–133.

Stewart-Oaten, A., Murdoch, W.W. and Parker, K.R. (1986) Environmental impact assessment: 'pseudoreplication' in time? *Ecology* 67(4): 929–940. https://doi.org/10.2307/1939815

Stratford, C. (2006) Review of Hydrological Reports for Newborough Warren, Anglesey. Unpublished report. Wallingford: Centre for Ecology and Hydrology.

Stratford, C., Ratcliffe, J., Hughes, A.G., Roberts, J. and Robins, N.S. (2007) Complex interaction between shallow groundwater and changing woodland, surface water, grazing and other influences in partly wooded duneland in Anglesey, Wales. In: Ribeiro, L., Chambel, A. and Condesso de Melo, M.T. (eds) IAH 35th Congress, Groundwater and Ecosystems, Lisbon.

Stuyfzand, P. J. (1993) *Hydrochemistry and hydrology of the coastal dune area of the Western Netherlands*. (Doctoral dissertation, Vrije Universiteit Amsterdam)

Sun, G., Riekerk, H. and Kornhak, L.V. (2000) Ground-water-table rise after forest harvesting on cypress-pine flatwoods in Florida. Wetlands, 20(1), pp. 101–112.

Sun, G., Caldwell, P.V. and McNulty, S.G. (2015) Modelling the potential role of forest thinning in maintaining water supplies under a changing climate across the conterminous United States. Hydrological Processes, 29, pp. 5016–5030. https://doi.org/10.1002/hyp.10469

Taylor, C.J. and Alley, W.M. (2001) Ground-Water-Level Monitoring and the Importance of Long-Term Water-Level Data. U.S. Geological Survey Circular 1217. Denver, CO: U.S. Geological Survey.

Thornthwaite, C.W. and Mather, J.R. (1957) Instructions and tables for computing potential evapotranspiration and the water balance. Publications in Climatology, 10(3). Centerton, NJ: Drexel Institute of Technology, Laboratory of Climatology.

UKHO (2026) *Admiralty Tide Tables, Volume 1B* (NP201B-26). Taunton: United Kingdom Hydrographic Office.

Underwood, A.J. (1992). Beyond BACI: the detection of environmental impacts on populations in the real, but variable, world. *Journal of Experimental Marine Biology and Ecology*, 161(2), 145-178.

van Schilfgaarde, J. (1963) Design of tile drainage for falling water tables. Journal of the Irrigation and Drainage Division, ASCE 89(2), 1–12.

van Willegen, L., Wallace, H., Curreli, A., Dwyer, C., Ratcliffe, J., Jones, D.L., Williams, G., Hollingham, M. & Jones, L. (2024). Dune slack Ecohydrology Dataset. Mendeley Data, V1. https://doi.org/10.17632/p4xvb6xxp9.1

van Willegen, L., Wallace, H., Curreli, A., Dwyer, C., Ratcliffe, J., Jones, D.L., Williams, G., Hollingham, M. and Jones, L. (2025) Five-year carry-over effects in dune slack vegetation response to hydrology. *Ecological Indicators*, 170, 113016. https://doi.org/10.1016/j.ecolind.2024.113016

Varouchakis, E.A. and Hristopulos, D.T. (2013) Comparison of stochastic and deterministic methods for mapping groundwater level spatial variability in sparsely monitored basins. *Environmental Monitoring and Assessment*, 185(1), 1–19. https://doi.org/10.1007/s10661-012-2527-y

Virtanen, P., Gommers, R., Oliphant, T.E., Haberland, M., Reddy, T., Cournapeau, D., ... & van der Walt, S.J. (2020) SciPy 1.0: Fundamental algorithms for scientific computing in Python. Nature Methods, 17(3), 261–272. https://doi.org/10.1038/s41592-020-0772-5

von Asmuth, J.R., Bierkens, M.F.P. and Maas, K. (2002) Transfer function-noise modelling in continuous time using predefined impulse response functions. Water Resources Research, 38(12), 1287. doi:10.1029/2001WR001136.

Ward, J.H. (1963) Hierarchical grouping to optimize an objective function. Journal of the American Statistical Association, 58(301), pp. 236–244.

Waskom, M.L. (2021) seaborn: statistical data visualization. *Journal of Open Source Software* 6(60): 3021. https://doi.org/10.21105/joss.03021

Williams, E.J. (1959). The comparison of regression variables. Journal of the Royal Statistical Society, Series B (Methodological), 21(2), 396–399.

Wu, X., Koch, M. and Xu, Y. (2021) Unsupervised learning for groundwater level clustering and pattern recognition in data-sparse aquifer systems. Hydrogeology Journal, 29(4), pp. 1507–1528.

Young, P.C. (2011) Recursive Estimation and Time-Series Analysis: An Introduction for the Student and Practitioner. Berlin: Springer.

Yuan, C. and Yang, H. (2019) Research on K-value selection method of K-means clustering algorithm. J, 2(2), pp. 226–235.

# 10. Acknowledgements

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# 11. Data Availability

## 11.1 Data Availability and Software

The complete monitoring network across Newborough Warren and the adjacent dune systems is mapped at: https://www.google.com/maps/d/viewer?mid=1hXLAauiMeaVsXhBR_IoUTziAtjk (M. Hollingham, unpublished).

The analytical pipeline underpinning this study is documented at https://newbroman.github.io/Newborough_Hydrology/ and comprises 64 steps across 22 phases covering data preparation, state-space modelling, and intervention analysis and mapping. A Methods Supplement, a per-script record of the analytical rationale, implementation decisions and limitations of each pipeline step, is deposited as a document in the project repository (https://github.com/newbroman/Newborough_Hydrology). All analyses were implemented in Python using NumPy (Harris et al., 2020), pandas (McKinney, 2010), SciPy (Virtanen et al., 2020), statsmodels (Seabold and Perktold, 2010), scikit-learn (Pedregosa et al., 2011) and seaborn (Waskom, 2021), with figures produced in matplotlib (Hunter, 2007). Spatial analysis used GeoPandas (Jordahl et al., 2020), rasterio (Gillies, 2013), contextily (Arribas-Bel et al., 2020) and adjustText (Flyamer et al., 2020). Geographic boundaries and analysis regions are held as KML. Driver availability for that format is a property of the local GDAL build rather than of the code — LIBKML is a separate build option that several packaged distributions do not enable, and the read engine differs between GeoPandas installations — so every KML read in the pipeline is routed through a single helper (src/utils/kml_io.py) that attempts the KML driver, then LIBKML, then a pure-XML parse requiring no GDAL driver at all. A reviewer whose GDAL provides neither driver can therefore reproduce every spatial result: the absence changes the speed of a read, not its outcome. For the same reason the pipeline runner re-executes itself under the project virtual environment, so that the analysis and the verification gates always run on the same interpreter.

Full diagnostic outputs for all 66 reference network wells and all 17 clearfell BACI wells are reproducible by running the published pipeline. Groundwater monitoring data supporting this study are available from the author on request.

**Source Data Credits:**

- **Climate Data:** Meteorological records for RAF Valley were obtained from the UK Met Office (MIDAS Open Government Licence).
- **Topographic Data:** Terrain analysis was performed using the 2 m resolution LiDAR digital terrain model provided by Natural Resources Wales (NRW) and accessed via DataMapWales (the Welsh Government geospatial portal, successor to the

decommissioned Lle Geo-Portal); 2 m DTM, captured March 2023. © Natural Resources Wales and Ordnance Survey.

- **Groundwater Data:** The primary dipwell monitoring record was independently maintained by the author.
- **Historic Mapping:** The nineteenth-century shoreline and dune edge were digitized from Ordnance Survey, Anglesey Sheet XXV.NW, revised 1899, published 1901, reproduced with the permission of the National Library of Scotland (https://maps.nls.uk) under CC-BY-NC-SA 4.0. The sheet was georeferenced to OSGB36 / British National Grid from 27 ground control points (RMSE 5.95 m; 3.27 m excluding two outliers). The source raster is not redistributed; the digitized vectors are original work.
- **Shoreline Data:** The mean-high-water shoreline from which every well-to-coast distance in this study is measured was derived from OpenStreetMap (**natural=coastline**), retrieved via the Overpass API and clipped to the west-facing Caernarfon Bay frontage. Map data © OpenStreetMap contributors, available under the Open Database Licence (ODbL, https://www.openstreetmap.org/copyright).
- **Methods Supplement:** Newborough Warren Hydrogeological Analysis — Methods Supplement (M. Hollingham), deposited in the project repository, https://github.com/newbroman/Newborough_Hydrology.

# 12. Notation

This chapter lists the mathematical symbols, abbreviations and statistics used in the report, with the units of each and where it is defined. Quantities that appear only in the pipeline's supplementary outputs are named in the scripts that define them.

## 12.1 Mathematical symbols

| Symbol | Meaning | Units | Defined in |
|---|---|---|---|
| Δh | Change in water-table level during month *t*; the SSM response variable | m/month | Report §3.4; `build_ssm_frame()` |
| h | Water-table level, negative-below-ground convention | m | Report §3.4; `build_ssm_frame()` col `h` |
| h_prev | Level at the end of the preceding month, `h.shift(1)` | m | `build_ssm_frame()` col `h_prev` |
| h_disp | Displacement above the | m | Report §3.4 |

| Symbol | Meaning | Units | Defined in |
|---|---|---|---|
| | drainage datum, $z_0 + h$ | | |
| h_disp,eq | Steady-state displacement, $(\beta_1 \cdot \bar{P} - \beta_2 \cdot \bar{PET})/\beta_3$; the quantity the EWI is built from | m | Report §3.7.6 |
| $z_0$ | Drainage datum: reference depth below ground against which displacement is measured | m | `config.DRAINAGE_DATUM` = 3.7 m |
| P | Monthly rainfall total | m/month (reported in mm) | Report §3.4; climate col `P_m` |
| PET | Monthly Thornthwaite potential evapotranspiration total | m/month (reported in mm) | Report §3.4; climate col `PET` |
| $\beta_1$ | Recharge coefficient (`beta_1_recharge`) | dimensionless | Report §3.4; `03_master_data.csv` |
| $\beta_2$ | Atmospheric-draw coefficient (`beta_2_atmospheric_draw`) | dimensionless | Report §3.4; `03_master_data.csv` |
| $\beta_3$ | Drainage coefficient (`beta_3_drainage`) | per month | Report §3.4; `03_master_data.csv` |
| $\Delta\beta_1$, $\Delta\beta_2$, $\Delta\beta_3$ | Era-to-era change in the coefficient, before vs after an intervention | as the coefficient | Report §3.5.4; Script 10e |
| Sy | Specific yield (drainable porosity) | dimensionless | Report §3.7.3; `17_wtf_01_sy_estimates.csv`, `18_wtf_01_well_sy_estimates.csv` |
| $\lambda$ | Drawdown e-folding reach, $\sqrt{(Kb/(Sy \cdot \beta_3))}$; distance over which a forest/clearfell/scrape drawdown decays | m | Report §3.8.2; `20_report_numbers.csv` `drawdown_lambda` |
| L_E | Attenuation length of evaporative draw in the depth-dependent PET term $\beta_2 \cdot \exp(-d_w/L_E)$; larger = PET draw persists deeper. Fitted as its | m | Report §3.7.1; `15_04_best_params.csv` col `Best_inv_L_E` |

| Symbol | Meaning | Units | Defined in |
|---|---|---|---|
| | reciprocal 1/L_E ($m^{-1}$); formerly κ | | |
| m_P | Rainfall multiplier in the P_flood calculation: multiple of climatological winter rainfall to lift a well to target | dimensionless | Report §3.6.3; `model_utils.pflood_lambda` |
| $\delta_0$ | Amplitude of the coastal-retreat drift gradient at the shoreline ($d = 0$); an extrapolation beyond the network | mm/yr | Report §4.10; `25_01_panel_fit_parameters.csv` |
| L_cg | Inland reach of the coastal-retreat gradient (distance beyond which the shoreline exerts no fitted influence). | m | Report §4.10; `25_01_panel_fit_parameters.csv` |
| c_far | Fitted far-field asymptote of the coastal decay model. | mm/yr | Report §4.10; `25_01_panel_fit_parameters.csv` |
| d | Cross-shore distance from the shoreline, or cost-distance from a drawdown source | m | Report §3.8.2 and §4.10 |
| d_w | Depth of the water table below ground surface, the argument of `exp(−d_w/L_E)`. Distinct from displacement (referenced to the datum). | m | Report §3.7.1; Script 15 |
| τ | Storage–drainage index, $S_y/\beta_3$. **Retired as a timescale**: a diagnostic weighting recession time by storage, not a residence time; not to be called a drainage timescale | months | Script 18; `18_wtf_05_storage_drainage_index.csv` |
| t½ | Drainage half-life, $\ln(2)/\beta_3$, at the project datum | months | Report §3.4.5; from `beta_3_drainage` |
| t_R | Recession e-folding time, $1/\beta_3$ (Sy-free) | months | Report §3.4.5 |

| Symbol | Meaning | Units | Defined in |
|---|---|---|---|
| $H_0$ | Drawdown deficit at the source edge (forest / clearfell / scrape lip) | mm | `config.DRAWDOWN_H0_MM` = 150.0 |
| K | Saturated hydraulic conductivity of the dune sand | m/day | `config.DRAWDOWN_K_MDAY` = 6.0 (Betson et al., 2002) |
| b | Saturated aquifer thickness in the reach calculation | m | `config.DRAWDOWN_B_M` = 5.0 |
| D | Hydraulic diffusivity, `K·b/Sy`, governing how fast a boundary perturbation spreads inland | m²/day | Report §3.8.2, sea-level-rise model |
| P_flood | Cumulative winter rainfall needed to lift a well's summer minimum above its ecological target | mm | Report §3.6.3; `11b_03_pflood_per_well.csv` |
| α | Two-sided significance level for a hypothesis test | dimensionless | `config.DETECTABILITY_ALPHA` = 0.05 |
| φ | First-order autoregressive coefficient of a series or residual series | dimensionless | Scripts 22, 24 |
| ε(t) | Monthly SSM time-series residual, observed minus modelled Δh | m | Report §3.4.1; Script 22 |
| σ, μ | Standard deviation, arithmetic mean of the named quantity | as the quantity | Standard usage |
| ρ | Spearman rank-correlation coefficient | dimensionless, −1 to +1 | See Section 12.4 |
| Σ | Summation operator | — | Standard usage |
| $\bar{P}$, $\overline{PET}$, $\bar{h}$ | Long-term monthly means of rainfall, PET and level | as the quantity | Report §3.4 |
| *n*, *N* | Sample size (wells, months or panel observations, as stated at use) | count | Standard usage |

## 12.2 Abbreviations and acronyms

| Abbreviation | Expansion | Units where it carries any | Defined in |
|---|---|---|---|
| SSM | State-Space Model — the | — | Report §3.4; |

| Abbreviation | Expansion | Units where it carries any | Defined in |
|---|---|---|---|
| | three-term monthly water-balance model, $\Delta h = \beta_1 \cdot P + \beta_2 \cdot (-PET) + \beta_3 \cdot (-h_disp_prev)$ | | `model_utils.fit_ssm()` |
| TLM | Traditional Linear Model — the counterfactual benchmark, an intercept model with no drainage term | — | Report §3.4; Script 08 |
| LCSC | Lumped Catchment Storage Coefficient, $100/\beta_1$ | dimensionless | Report §3.4.1; `config.LCSC_DATA_LIMIT` |
| BACI | Before-After-Control-Impact — the intervention design comparing treated and control wells before and after an event | — | Report §3.5 |
| ANCOVA | Analysis of covariance — the BACI panel regression carrying climate covariates alongside the treatment indicators | — | Report §3.5.4 |
| CUSUM | Cumulative Sum — the running sum of departures from a reference mean, used to date the onset and test the permanence of a shift | m, cumulative | Report §3.5 |
| CWB | Cumulative water balance — the running rainfall-minus-PET total used as the BACI climate covariate | mm | Report §3.5.4 |
| WTF | Water Table Fluctuation — the independent method for estimating specific yield from recharge episodes | — | Report §3.7.3 |
| MSL5 | Five-Year Mean Spring Water Level — the five-year running mean of the spring levels (the readings dated March–May) | m; changes in mm | Report §3.7.5 (van Willegen framework) |
| EWI | Equilibrium Wetness Index | m | Report §3.7.6; Script 26 |

| Abbreviation | Expansion | Units where it carries any | Defined in |
|---|---|---|---|
| | — the steady-state displacement expressed on the MSL5 scale | | |
| MCA | Multi-Cluster Affinity — a well whose hydrograph correlates above the affinity threshold with three or more cluster centroids | — | Report §4.3 |
| ESS | Error Sum of Squares — the within-cluster variance criterion Ward's linkage minimizes | m², as the clustered variable | Report §3.2.2 |
| SD15b, SD16 | National Vegetation Classification dune-slack communities: wet slack (SD15b) and dry slack (SD16). Curreli et al. (2013) community means of the four-year mean annual minimum, used as the ecological reference levels | levels in m relative to ground surface | Report §3.6; `config.SD15b`, `config.SD16` |
| PET | Potential evapotranspiration, computed by the Thornthwaite method | m per month; reported in mm | Report §3.1.2 |
| ET | Evapotranspiration, the realized flux as distinct from the potential demand | mm per month | Report §3.1.2 |
| P_flood | Critical rainfall threshold; see Section 12.1 | mm | Report §3.6.3 |
| SLR | Sea-level rise | m | Report §3.8.2; `config.SLR_RISE_M` |
| UKCP18 | UK Climate Projections 2018 — the source of the seasonal rainfall and PET change factors | multipliers, dimensionless | `config.UKCP18_SCENARIOS` |
| RCP8.5 | Representative Concentration Pathway 8.5 — the high-emissions UKCP18 scenario used for the projections | — | Report §3.7.5 |

| Abbreviation | Expansion | Units where it carries any | Defined in |
|---|---|---|---|
| IDW | Inverse-distance weighting — used only for the MSL5-change map; other per-well statistics are mapped as a piecewise-linear surface and the water table is kriged | — | Report §3.4.3; `map_utils.interpolate_surface()` |
| DEM | Digital Elevation Model | m | Report §3.1.1 |
| LiDAR | Light detection and ranging; the 2 m Natural Resources Wales DTM, captured March 2023, from which the DEM is built | m | Report §3.1.1 |
| DGPS | Differential Global Positioning System — the independently surveyed ground elevations used to validate the DEM | m | Report §3.1.1 |
| AOD | Above Ordnance Datum — the national vertical datum for absolute elevations | m | Report §3.1.1 |
| OS | Ordnance Survey | — | Report §3.1.1 |
| NRW | Natural Resources Wales — the data provider and site manager | — | Report §2 |
| CCW | Countryside Council for Wales — the predecessor body whose 1989–96 dipwell record the hindcast tests against | — | Report §5.7.8; Script 39 |
| SAC | Special Area of Conservation | — | Report §2 |
| SSSI | Site of Special Scientific Interest | — | Report §2 |
| NNR | National Nature Reserve | — | Report §2 |
| MAM, JJA, DJF | Seasonal three-month groupings: March-to-May (spring), June-to-August (summer), December-to-February (winter) | — | `config.MSL_SPRING_MONTHS` |
| OLS | Ordinary least squares | — | Report §3.4 |

| Abbreviation | Expansion | Units where it carries any | Defined in |
|---|---|---|---|
| AR(1) | First-order autoregressive; the lag-one dependence structure of a series | — | See Section 12.4 |
| VIF | Variance inflation factor | dimensionless | See Section 12.4; `30_c4_identifiability_by_cluster.csv` column `VIF` |
| NSE | Nash–Sutcliffe Efficiency | dimensionless, ≤ 1 | See Section 12.4 |
| RMSE | Root mean square error | as the quantity, usually m or mm | See Section 12.4 |
| AIC | Akaike Information Criterion | dimensionless | See Section 12.4 |
| CI | Confidence interval | as the quantity | See Section 12.4 |
| SE | Standard error | as the quantity | See Section 12.4 |
| SD | Standard deviation | as the quantity | See Section 12.4 |
| IQR | Interquartile range, the 25th-to-75th-percentile span | as the quantity | See Section 12.4 |
| C1–C5 | Cluster labels: C1 Lake Edge, C2 Dune, C3 Western Residual, C4 Main Forest, C5 Coastal Forest | — | `config.CLUSTER_LABELS` |

## 12.3 Identifier conventions

| Convention | What it names | Defined in |
|---|---|---|
| `cehNN` | Reference-network dipwell, the CEH monitoring series | `well_metadata.csv`; `02_07_cluster_membership_k5.csv` |
| `nwNN`, `feN`, `lisN`, `wmcN` | Extended-network dipwells, classified by Pearson affinity rather than by fitted membership | Report §4.3 |
| `llynrhos` | Llyn Rhos-Ddu lake gauge. Not a classified dipwell; excluded from the 88-dipwell network | `config.LAKE_GAUGE_KEYS`; `config.LAKE_GAUGE_REASON` |
| `YYYY-MM-01` | Monthly timestamp. The `-01` is a pandas formatting convention, not the first of the month; the row is the month as a whole | Report §3.1 |

## 12.4 Statistics, and how to read them

| Statistic | What it measures | Range | Good / warning | How to read it in this study |
|---|---|---|---|---|
| **NSE** (Nash–Sutcliffe Efficiency) | How much of the observed variance the simulated series reproduces, relative to simply predicting the record mean | −∞ to 1 | A value of 1 is a perfect forecast; a value of 0 indicates the model does no better than the mean; negative values indicate it does worse | Iterative NSE is the study's hardest test, because the model is run forward from one initial condition and its errors compound. The SSM-over-TLM gain, ΔNSE, is what quantifies the value of the explicit drainage term, and it is one of the three axes on which the five clusters separate |
| **RMSE** (root mean square error) | Typical size of the model's error, in the units of the thing being modelled | ≥ 0 | Lower is better; the meaningful comparison is against the observed range, not against zero | Reported in metres of head. It is the metric that makes the EWI's calibration onto the MSL5 scale legible, because it says how far a predicted level typically sits from the measured one |
| **Bias** | Mean signed error: whether the model sits systematically high or low, as opposed to merely scattered | any sign | A value near zero is good; a bias comparable to the RMSE indicates a systematic offset rather than noise | Bias is the statistic that separates the EWI's known systematic depth offset from random error. A model can have acceptable RMSE and still be unusable as a level if the bias is large |
| **$R^2$** (coefficient of determination) | Fraction of variance in the response explained by the fitted model | 0 to 1 for a fitted OLS model | Higher is better, but a high value on a one-step fit is a weak claim | Applied to the one-step SSM fit, $R^2$ is generous, because last month's level does most of the work. Prefer iterative NSE for any claim about predictive skill |
| **Pearson r** | Strength of the straight-line relationship between | −1 to +1 | Values near ±1 indicate a tight linear relationship; values | Pearson r is the affinity measure that classifies extended- |

| Statistic | What it measures | Range | Good / warning | How to read it in this study |
|---|---|---|---|---|
| | two variables | | near 0 indicate none | network wells against cluster centroids, and the threshold for Multi-Cluster Affinity is set on it. It is sensitive to shape, not to amplitude, which is why a well can be affine to several centroids at once |
| **Spearman ρ** | Strength of the monotone relationship between the *ranks* of two variables | −1 to +1 | Values near ±1 indicate a consistent ordering; values near 0 indicate none | Used where the ordering matters and the functional form does not — the coastal-end ordering of anchor-referenced levels, and peak rainfall lag against ridge distance. Being rank-based, it is not distorted by one extreme well |
| **VIF** (variance inflation factor) | How much wider a coefficient's error bar is because that predictor overlaps with the other predictors | ≥ 1 | A value of 1 indicates no overlap. Values above about 5 warrant attention and above about 10 are conventionally treated as serious overlap; both are rules of thumb, not pipeline outputs | Consider what happens if rainfall, PET and antecedent displacement all rise and fall together: the fit cannot tell which of them is doing the work, so it spreads the credit arbitrarily and every coefficient's error bar widens. The VIF is exactly how much wider — a VIF of 4 indicates the variance is four times what it would be with independent predictors, so the error bar is twice as wide. In this study, it answers one specific worry: that C4 Main |

| Statistic | What it measures | Range | Good / warning | How to read it in this study |
|---|---|---|---|---|
| | | | | Forest's small $\beta_3$ is an artefact of PET and displacement moving together at a deep-water-table cluster rather than a real property of the cluster. The per-cluster values sit in `30_c4_identifiability_by_cluster.csv`, column `VIF` |
| **Condition number** | How well-shaped the $\geq 1$ predictor data are as a whole, rather than one predictor at a time | | A value near 1 indicates a well-shaped design. Large values indicate an ill-conditioned one; the conventional flag is above about 30, which is a rule of thumb, not a pipeline output | Picture every month of data as a point in a space with one axis per predictor. If those points form a roughly round cloud, the fit has seen every combination and stands on firm ground. If they form a long thin cigar, there is one direction the data barely explores, and the coefficients along it swing about wildly for tiny changes in the input. The condition number is how elongated that cigar is — the longest axis divided by the shortest. Where VIF interrogates one predictor, the condition number interrogates the design as a whole, which is why the two are reported side by side; the value sits in `30_c4_identifiability_by_clu` |

| Statistic | What it measures | Range | Good / warning | How to read it in this study |
|---|---|---|---|---|
| | | | | `ster.csv`, column `cond` |
| **t-statistic** | Coefficient divided by its own standard error: how many error bars the estimate sits away from zero | any sign | Magnitudes above about 2 correspond to significance at the conventional level for a reasonable sample; magnitudes below 1 indicate an estimate the data cannot separate from zero | Reported alongside every SSM and BACI coefficient. Because it is a ratio, a large t can come from a large effect or from a small standard error, so read it beside the coefficient itself and never on its own |
| **p-value** | Probability of seeing an effect at least this large if the true effect were zero | 0 to 1 | Values below `DETECTABILITY_ALPHA` = 0.05 are treated as significant; values just either side of it are treated as inconclusive rather than as decisive | The study's convention is that a non-significant result indicates the record cannot resolve the effect, not that the effect is absent. Language is kept to "consistent with" and "indicates" for that reason |
| **AIC and ΔAIC** | Model fit penalized for the number of parameters, so that a more complex model has to earn its extra terms | AIC on an arbitrary scale; ΔAIC is the difference between two models | Only differences mean anything. A ΔAIC below −2 indicates the more complex model is worth keeping; a value near zero indicates the two forms are indistinguishable | AIC is what decides whether the depth-dependent PET term is retained over fixed $\beta_2$, and whether the curvature terms are retained in the BACI specification. Where the linear-capped and exponential coastal decay forms sit within a couple of AIC units of each other, the choice between them is a methods footnote rather than a headline decision |
| **Confidence interval** | Range of values the data are consistent with, at a stated level | as the quantity | An interval excluding zero indicates a resolved effect; a wide interval | The study prefers the interval to the point estimate throughout. Two intervals that |

| Statistic | What it measures | Range | Good / warning | How to read it in this study |
|---|---|---|---|---|
| | of confidence | | indicates a poorly constrained one even where the point estimate looks large | overlap do not by themselves indicate two quantities are equal, which is why the seasonal gradient question is settled by a single interaction test rather than by comparing two subset fits |
| **Bootstrap CI** | Confidence interval built by resampling the data many times and reading the spread of the answers, rather than by assuming a distribution | as the quantity | Read exactly as any other interval; a bootstrap interval much wider than the analytic one indicates the analytic assumptions were optimistic | Used where the analytic form is not trustworthy: a moving-block resample preserves the seasonal structure of a monthly series, and a well-level resample answers the different question of how much the answer depends on which wells are in the cluster. The block length, replicate count and seed are fixed in `config.py`, so an interval reproduces exactly |
| **AR(1) correction and effective N** | Adjustment for the fact that consecutive observations in a time series carry overlapping information, so the record holds fewer independent data points than it has rows | effective N ≤ N | An effective N far below N indicates that an uncorrected p-value would be badly overconfident | Trend significance is quoted on the AR(1)-corrected footing throughout — `p_ar` in `32_site_mean_trend.csv`, not `p_ols` — because an annual spring series of two decades can carry an effective sample size of only about five. This is the single correction that most changes which trends the study is |

| Statistic | What it measures | Range | Good / warning | How to read it in this study |
|---|---|---|---|---|
| | | | | willing to call resolved |
| **Statistical power and minimum detectable effect** | The size of effect the record could have detected, had one been present | as the quantity | A minimum detectable effect larger than the effect of interest indicates the analysis cannot settle the question either way | Reported alongside every site-mean trend as `min_detectable_mm_yr` in `32_site_mean_trend.csv`, computed at `DETECTABILITY_ALPHA` = 0.05 and `DIFF_POWER_TARGET` = 0.80. It is what turns a null result into a statement about the record rather than about the site: where the minimum detectable rate exceeds the rate being looked for, the correct reading is that the record is too short, not that nothing is happening |